

Gerhard Bohm, Günter Zech

Introduction to Statistics and Data Analysis for Physicists

Verlag Deutsches Elektronen-Synchrotron

Prof. Dr. Gerhard Bohm

Deutsches Elektronen-Synchrotron
Platanenallee 6
D-15738 Zeuthen
e-mail: bohm@ifh.de

Univ.-Prof. Dr. Günter Zech

Universität Siegen
Fachbereich Physik
Walter-Flex-Str. 3
D-57068 Siegen
e-mail: zech@physik.uni-siegen.de

Bibliografische Information der Deutschen Bibliothek

Die Deutsche Bibliothek verzeichnet diese Publikation in der Deutschen Nationalbibliografie; detaillierte bibliografische Daten sind im Internet über [<http://dnb.ddb.de>](http://dnb.ddb.de) abrufbar.

**Herausgeber
und Vertrieb:**

Deutsches Elektronen-Synchrotron
in der Helmholtz-Gemeinschaft
Zentralbibliothek
D-22603 Hamburg
Telefon (040) 8998-3602
Telefax (040) 8998-4440

Umschlaggestaltung:

Christine Iezzi
Deutsches Elektronen-Synchrotron, Zeuthen

Druck:

Druckerei, Deutsches Elektronen-Synchrotron

Copyright:

Gerhard Bohm, Günter Zech

ISBN 978-3-935702-41-6

DOI 10.3204/DESY-BOOK/statistics (e-book)

[<http://www-library.desy.de/elbook.html>](http://www-library.desy.de/elbook.html)

Das Buch darf nicht ohne schriftliche Genehmigung der Autoren durch Druck, Fotokopie oder andere Verfahren reproduziert oder unter Verwendung elektronischer Systeme verarbeitet, vervielfältigt oder verbreitet werden.
Der ganze oder teilweise Download des e-Buchs zum persönlichen Gebrauch ist erlaubt.

Preface

There is a large number of excellent statistic books. Nevertheless, we think that it is justified to complement them by another textbook with the focus on modern applications in nuclear and particle physics. To this end we have included a large number of related examples and figures in the text. We emphasize less the mathematical foundations but appeal to the intuition of the reader.

Data analysis in modern experiments is unthinkable without simulation techniques. We discuss in some detail how to apply Monte Carlo simulation to parameter estimation, deconvolution, goodness-of-fit tests. We sketch also modern developments like artificial neural nets, bootstrap methods, boosted decision trees and support vector machines.

Likelihood is a central concept of statistical analysis and its foundation is the likelihood principle. We discuss this concept in more detail than usually done in textbooks and base the treatment of inference problems as far as possible on the likelihood function only, as is common in the majority of the nuclear and particle physics community. In this way point and interval estimation, error propagation, combining results, inference of discrete and continuous parameters are consistently treated. We apply Bayesian methods where the likelihood function is not sufficient to proceed to sensible results, for instance in handling systematic errors, deconvolution problems and in some cases when nuisance parameters have to be eliminated, but we avoid improper prior densities. Goodness-of-fit and significance tests, where no likelihood function exists, are based on standard frequentist methods.

Our textbook is based on lecture notes from a course given to master physics students at the University of Siegen, Germany, a few years ago. The content has been considerably extended since then. A preliminary German version is published as an electronic book at the DESY library. The present book is addressed mainly to master and Ph.D. students but also to physicists who are interested to get an introduction into recent developments in statistical methods of data analysis in particle physics. When reading the book, some parts can be skipped, especially in the first five chapters. Where necessary, back references are included.

We welcome comments, suggestions and indications of mistakes and typing errors. We are prepared to discuss or answer questions to specific statistical problems.

We acknowledge the technical support provided by DESY and the University of Siegen.

February 2010,

Gerhard Bohm, Günter Zech

Contents

1	Introduction: Probability and Statistics	1
1.1	The Purpose of Statistics	1
1.2	Event, Observation and Measurement	2
1.3	How to Define Probability?	3
1.4	Assignment of Probabilities to Events	4
1.5	Outline of this Book	6
2	Basic Probability Relations	9
2.1	Random Events and Variables	9
2.2	Probability Axioms and Theorems	10
2.2.1	Axioms	10
2.2.2	Conditional Probability, Independence, and Bayes' Theorem	11
3	Probability Distributions and their Properties	15
3.1	Definition of Probability Distributions	16
3.1.1	Discrete Distributions	16
3.1.2	Continuous Distributions	16
3.1.3	Empirical Distributions	20
3.2	Expected Values	20
3.2.1	Definition and Properties of the Expected Value	21
3.2.2	Mean Value	22
3.2.3	Variance	23
3.2.4	Skewness	26
3.2.5	Kurtosis (Excess)	26
3.2.6	Discussion	27
3.2.7	Examples	28

II Contents

3.3	Moments and Characteristic Functions	32
3.3.1	Moments	32
3.3.2	Characteristic Function	33
3.3.3	Examples	36
3.4	Transformation of Variables.....	38
3.4.1	Calculation of the Transformed Density	39
3.4.2	Determination of the Transformation Relating two Distributions	41
3.5	Multivariate Probability Densities	42
3.5.1	Probability Density of two Variables	43
3.5.2	Moments	44
3.5.3	Transformation of Variables	46
3.5.4	Reduction of the Number of Variables.....	47
3.5.5	Determination of the Transformation between two Distributions	50
3.5.6	Distributions of more than two Variables	51
3.5.7	Independent, Identically Distributed Variables	52
3.5.8	Angular Distributions.....	53
3.6	Some Important Distributions.....	55
3.6.1	The Binomial Distribution.....	55
3.6.2	The Multinomial Distribution.....	58
3.6.3	The Poisson Distribution	58
3.6.4	The Uniform Distribution	65
3.6.5	The Normal Distribution	65
3.6.6	The Exponential Distribution	69
3.6.7	The χ^2 Distribution	70
3.6.8	The Gamma Distribution.....	72
3.6.9	The Lorentz and the Cauchy Distributions.....	74
3.6.10	The Log-normal Distribution	75
3.6.11	Student's t Distribution	75
3.6.12	The Extreme Value Distributions.....	77
4	Measurement errors	81
4.1	General Considerations.....	81
4.1.1	Importance of Error Assignments.....	81
4.1.2	Verification of Assigned Errors	82
4.1.3	The Declaration of Errors	82

4.1.4	Definition of Measurement and its Error	83
4.2	Different Types of Measurement Uncertainty	84
4.2.1	Statistical Errors	84
4.2.2	Systematic Errors (G. Bohm)	88
4.2.3	Systematic Errors (G. Zech)	90
4.2.4	Controversial Examples	94
4.3	Linear Propagation of Errors	94
4.3.1	Error Propagation	94
4.3.2	Error of a Function of Several Measured Quantities	95
4.3.3	Averaging Uncorrelated Measurements	98
4.3.4	Averaging Correlated Measurements	98
4.3.5	Several Functions of Several Measured Quantities	100
4.3.6	Examples	101
4.4	Biased Measurements	103
4.5	Confidence Intervals	104
5	Monte Carlo Simulation	107
5.1	Introduction	107
5.2	Generation of Statistical Distributions	109
5.2.1	Computer Generated Pseudo Random Numbers	109
5.2.2	Generation of Distributions by Variable Transformation	110
5.2.3	Simple Rejection Sampling	115
5.2.4	Importance Sampling	116
5.2.5	Treatment of Additive Probability Densities	119
5.2.6	Weighting Events	120
5.2.7	Markov Chain Monte Carlo	120
5.3	Solution of Integrals	123
5.3.1	Simple Random Selection Method	123
5.3.2	Improved Selection Method	126
5.3.3	Weighting Method	127
5.3.4	Reduction to Expected Values	128
5.3.5	Stratified Sampling	129
5.4	General Remarks	129
6	Parameter Inference I	131
6.1	Introduction	131

6.2	Inference with Given Prior	133
6.2.1	Discrete Hypotheses	133
6.2.2	Continuous Parameters	135
6.3	Definition and Visualization of the Likelihood	137
6.4	The Likelihood Ratio	140
6.5	The Maximum Likelihood Method for Parameter Inference	142
6.5.1	The Recipe for a Single Parameter.....	143
6.5.2	Examples	144
6.5.3	Likelihood Inference for Several Parameters.....	148
6.5.4	Combining Measurements	151
6.5.5	Normally Distributed Variates and χ^2	151
6.5.6	Likelihood of Histograms	152
6.5.7	Extended Likelihood	154
6.5.8	Complicated Likelihood Functions	155
6.5.9	Comparison of Observations with a Monte Carlo Simulation ..	155
6.5.10	Parameter Estimate of a Signal Contaminated by Background	160
6.6	Inclusion of Constraints	163
6.6.1	Introduction	163
6.6.2	Eliminating Redundant Parameters	164
6.6.3	Gaussian Approximation of Constraints	166
6.6.4	The Method of Lagrange Multipliers	167
6.6.5	Conclusion	168
6.7	Reduction of the Number of Variates.....	168
6.7.1	The Problem	168
6.7.2	Two Variables and a Single Linear Parameter	169
6.7.3	Generalization to Several Variables and Parameters	169
6.7.4	Non-linear Parameters	171
6.8	Method of Approximated Likelihood Estimator.....	171
6.9	Nuisance Parameters.....	174
6.9.1	Nuisance Parameters with Given Prior	175
6.9.2	Factorizing the Likelihood Function.....	176
6.9.3	Parameter Transformation, Restructuring	177
6.9.4	Profile Likelihood	179
6.9.5	Integrating out the Nuisance Parameter	181
6.9.6	Explicit Declaration of the Parameter Dependence.....	181

6.9.7	Advice	181
7	Parameter Inference II	183
7.1	Likelihood and Information	183
7.1.1	Sufficiency	183
7.1.2	The Conditionality Principle	185
7.1.3	The Likelihood Principle	186
7.1.4	Bias of Maximum Likelihood Results	187
7.1.5	Stopping Rules	190
7.2	Further Methods of Parameter Inference	191
7.2.1	The Moments Method	191
7.2.2	The Least Square Method	195
7.2.3	Linear Regression	198
7.3	Comparison of Estimation Methods	199
8	Interval Estimation	201
8.1	Introduction	201
8.2	Error Intervals	202
8.2.1	Parabolic Approximation	203
8.2.2	General Situation	204
8.3	Error Propagation	205
8.3.1	Averaging Measurements	205
8.3.2	Approximating the Likelihood Function	208
8.3.3	Incompatible Measurements	209
8.3.4	Error Propagation for a Scalar Function of a Single Parameter	210
8.3.5	Error Propagation for a Function of Several Parameters	210
8.4	One-sided Confidence Limits	214
8.4.1	General Case	214
8.4.2	Upper Poisson Limits, Simple Case	215
8.4.3	Poisson Limit for Data with Background	216
8.4.4	Unphysical Parameter Values	219
8.5	Summary	219
9	Deconvolution	221
9.1	Introduction	221
9.1.1	The Problem	221

9.1.2	Deconvolution by Matrix Inversion	224
9.1.3	The Transfer Matrix	226
9.1.4	Regularization Methods	226
9.2	Deconvolution of Histograms	227
9.2.1	Fitting the Bin Content	227
9.2.2	Iterative Deconvolution	231
9.2.3	Regularization of the Transfer Matrix	232
9.3	Binning-free Methods	234
9.3.1	Iterative Deconvolution	234
9.3.2	The Satellite Method	235
9.3.3	The Maximum Likelihood Method	237
9.4	Comparison of the Methods	239
9.5	Error Estimation for the Deconvoluted Distribution	241
10	Hypothesis Tests	245
10.1	Introduction	245
10.2	Some Definitions	246
10.2.1	Single and Composite Hypotheses	246
10.2.2	Test Statistic, Critical Region and Significance Level	246
10.2.3	Errors of the First and Second Kind, Power of a Test	247
10.2.4	P-Values	248
10.2.5	Consistency and Bias of Tests	248
10.3	Goodness-of-Fit Tests	250
10.3.1	General Remarks	250
10.3.2	P-Values	252
10.3.3	The χ^2 Test in Generalized Form	254
10.3.4	The Likelihood Ratio Test	261
10.3.5	The Kolmogorov–Smirnov Test	263
10.3.6	Tests of the Kolmogorov–Smirnov – and Cramer–von Mises Families	265
10.3.7	Neyman’s Smooth Test	266
10.3.8	The L_2 Test	268
10.3.9	Comparing a Data Sample to a Monte Carlo Sample and the Metric	269
10.3.10	The k-Nearest Neighbor Test	270
10.3.11	The Energy Test	270

10.3.12 Tests Designed for Specific Problems	272
10.3.13 Comparison of Tests	273
10.4 Two-Sample Tests	275
10.4.1 The Problem	275
10.4.2 The χ^2 Test	276
10.4.3 The Likelihood Ratio Test	276
10.4.4 The Kolmogorov–Smirnov Test	277
10.4.5 The Energy Test	277
10.4.6 The k-Nearest Neighbor Test	278
10.5 Significance of Signals	279
10.5.1 Introduction	279
10.5.2 The Likelihood Ratio Test	281
10.5.3 Tests Based on the Signal Strength	286
11 Statistical Learning	289
11.1 Introduction	289
11.2 Smoothing of Measurements and Approximation by Analytic Functions	291
11.2.1 Smoothing Methods	292
11.2.2 Approximation by Orthogonal Functions	294
11.2.3 Wavelets	298
11.2.4 Spline Approximation	300
11.2.5 Approximation by a Combination of Simple Functions	302
11.2.6 Example	302
11.3 Linear Factor Analysis and Principal Components	303
11.4 Classification	309
11.4.1 The Discriminant Analysis	311
11.4.2 Artificial Neural Networks	312
11.4.3 Weighting Methods	319
11.4.4 Decision Trees	322
11.4.5 Bagging and Random Forest	325
11.4.6 Comparison of the Methods	326
12 Auxiliary Methods	329
12.1 Probability Density Estimation	329
12.1.1 Introduction	329

12.1.2	Fixed Interval Methods	330
12.1.3	Fixed Number and Fixed Volume Methods	333
12.1.4	Kernel Methods	333
12.1.5	Problems and Discussion	334
12.2	Resampling Techniques	336
12.2.1	Introduction	336
12.2.2	Definition of Bootstrap and Simple Examples	337
12.2.3	Precision of the Error Estimate	339
12.2.4	Confidence Limits	339
12.2.5	Precision of Classifiers	340
12.2.6	Random Permutations	340
12.2.7	Jackknife and Bias Correction	340
13	Appendix	343
13.1	Large Number Theorems	343
13.1.1	The Chebyshev Inequality and the Law of Large Numbers	343
13.1.2	The Central Limit Theorem	344
13.2	Consistency, Bias and Efficiency of Estimators	345
13.2.1	Consistency	345
13.2.2	Bias of Estimates	345
13.2.3	Efficiency	346
13.3	Properties of the Maximum Likelihood Estimator	347
13.3.1	Consistency	347
13.3.2	Efficiency	348
13.3.3	Asymptotic Form of the Likelihood Function	349
13.3.4	Properties of the Maximum Likelihood Estimate for Small Samples	350
13.4	Error of Background-Contaminated Parameter Estimates	350
13.5	Frequentist Confidence Intervals	353
13.6	Comparison of Different Inference Methods	355
13.6.1	A Few Examples	355
13.6.2	The Frequentist approach	358
13.6.3	The Bayesian Approach	358
13.6.4	The Likelihood Ratio Approach	359
13.6.5	Conclusion	359
13.7	P-Values for EDF-Statistics	359

13.8 Comparison of two Histograms, Goodness-of-Fit and Parameter Estimation	361
13.8.1 Comparison of two Poisson Numbers with Different Normalization.....	361
13.8.2 Comparison of Weighted Sums	362
13.8.3 χ^2 of Histograms	362
13.8.4 Parameter Estimation.....	363
13.9 Extremum Search	364
13.9.1 Monte Carlo Search	365
13.9.2 The Simplex Algorithm	365
13.9.3 The Parabola Method.....	366
13.9.4 The Method of Steepest Descent	366
13.9.5 Stochastic Elements in Minimum Search.....	367
13.10 Linear Regression with Constraints	368
13.11 Formulas Related to the Polynomial Approximation	369
13.12 Formulas for B-Spline Functions	370
13.12.1 Linear B-Splines	370
13.12.2 Quadratic B-Splines.....	370
13.12.3 Cubic B-Splines	371
13.13 Support Vector Classifiers.....	371
13.13.1 Linear Classifiers	371
13.13.2 General Kernel Classifiers.....	373
13.14 The Bayes Factor	374
13.15 Robust Fitting Methods	375
13.15.1 Introduction	375
13.15.2 Robust Methods	376
References	381
Index	387

Introduction: Probability and Statistics

Though it is exaggerated to pretend that in our life only the taxes and the death are certain, it is true that the majority of all predictions suffer from uncertainties. Thus the occupation with probabilities and statistics is useful for everybody, for scientists of experimental and empirical sciences it is indispensable.

1.1 The Purpose of Statistics

Whenever we perform an experiment and want to interpret the collected data, we need statistical tools. The accuracy of measurements is limited by the precision of the equipment which we use, and thus the results emerge from a random process. In many cases also the processes under investigation are of stochastic nature, i.e. not predictable with arbitrary precision, such that we are forced to present the results in form of estimates with error intervals. Estimates accompanied by an uncertainty interval allow us to test scientific hypotheses and by averaging the results of different experiments to improve continuously the accuracy. It is by this procedure that a constant progress in experimental sciences and their applications was made possible.

Inferential statistics provides mathematical methods to infer the properties of a population from a randomly selected sample taken from it. A population is an arbitrary collection of elements, a sample just a subset of it.

A trivial, qualitative case of an application of statistics in every day life is the following: To test whether a soup is too salted, we taste it with a spoon. To obtain a reliable result, we have to stir the soup thoroughly and the sample contained in the spoon has to be large enough: Samples have to be representative and large enough to achieve a sufficiently precise estimate of the properties of the population.

Scientific measurements are subject to the same scheme. Let us look at a few statistical problems:

1. From the results of an exit poll the allocation of seats among the different parties in the parliament is predicted. The population is the total of the votes of all electors, the sample a representative selection from it. It is relatively simple to compute the distribution of the seats from the results of the poll, but one wants to know in addition the accuracy of the prognosis, respectively how many electors have to be asked in order to issue a reasonably precise statement.

2. In an experiment we record the lifetimes of 100 decays of an unstable nucleus. To determine the mean life of the nucleus, we take the average of the observed times. Here the uncertainty has its origin in the quantum mechanical random process. The laws of physics tell us, that the lifetimes follow a random exponential distribution. The sample is assumed to be representative of the total of the infinitely many decay times that could have occurred.
3. From 10 observations the period of a pendulum is to be determined. We will take as estimate the mean value of the replicates. Its uncertainty has to be evaluated from the dispersion of the individual observations. The actual observations form a sample from the infinite number of all possible observations.

These examples are related to *parameter inference*. Further statistical topics are *testing*, *deconvolution*, and *classification*.

4. A bump is observed in a mass distribution. Is it a resonance or just a background fluctuation?
5. An angular distribution is predicted to be linear in the cosine of the polar angle. Are the observed data compatible with this hypothesis?
6. It is to be tested whether two experimental setups perform identically. To this end, measurement samples from both are compared to each other. It is tested whether the samples belong to the same population, while the populations themselves are not identified explicitly.
7. A frequency spectrum is distorted by the finite resolution of the detector. We want to reconstruct the true distribution.
8. In a test beam the development of shower cascades produced by electrons and pions is investigated. The test samples are characterized by several variables like penetration depth and shower width. The test samples are used to develop procedures which predict the identity of unknown particles from their shower parameters.

A further, very important part of statistics is *decision theory*. We shall not cover this topic.

1.2 Event, Observation and Measurement

Each discipline has its own terminology. The notations used in statistics are somewhat different from those used by physicists, and even physicists from different fields do not always use the same language. To avoid confusion, we will fix the meaning of some terms which we need in the following.

We will distinguish between *event*, *observation* and *measurement*. A random event¹ or in short an *event* is the possible outcome of an experiment governed by some stochastic process. This could be the decay of a nucleus in a certain time interval or the coincidence that two pupils of a class have their birthdays the same day. An *observation* is the element of a sample, the realization of an event.

¹At some occasions we will use the term *event* in the way physicists do. This will become clear from the context.

As indicated above, a *population* is the set of all possible events, i.e. all potential observations. In the natural sciences, ideally experiments can be repeated infinitely often, thus we usually deal with infinite populations.

When we infer properties, i.e. parameters characterizing the population, from the sample, we talk about an *estimate* or a *measurement*. The decay times of 10 pion decays correspond to a sample of observations from the population of all possible events, the decay times. The estimation of the mean life of pions from the sample is a measurement. An observation as such – the reading of a meter, a decay time, the number of detected cosmic muons – has no error associated with it. Its value is fixed by a random process. On the contrary, the measurement which corresponds to parameter inference is afflicted with an uncertainty. In many simple situations, observation and measurement coincide numerically, in other cases the measurement is the result of an extensive analysis based on a large amount of observations.

1.3 How to Define Probability?

Statistics is at least partially based on experience which is manifest in fields like deconvolution and pattern recognition. It applies probability theory but should not be confounded with it. Probability theory, contrary to statistics, is a purely mathematical discipline and based on simple axioms. On the other hand, all statistical methods use probability theory. Therefore, we will deal in the first part of this book with simple concepts and computational rules of this field.

In statistics, there exist several different notions on what probability means. In the *Dictionary of Statistical Terms* of Kendall and Buckland [1] we find the following definition:

“**probability**, a basic concept which may be taken as undefinable, expressing in some way a *degree of belief*, or as the limiting frequency in an infinite random series. Both approaches have their difficulties and the most convenient axiomatization of probability theory is a matter of personal taste. Fortunately both lead to much the same calculus of probability.”

We will try to extend this short explanation:

In the *frequentist statistics*, sometimes also called *classical statistics*, the probability of an event, the possible outcome of an experiment, is defined as the frequency with which it occurs in the limit of an infinite number of repetitions of the experiment. If in throwing dice the result *five* occurs with frequency $1/6$ in an infinite number of trials, the probability to obtain *five* is defined to be $1/6$.

In the more modern, so-called *Bayesian statistics*² this narrow notion of probability is extended. Probability is also ascribed to fixed but incompletely known facts and to processes that cannot be repeated. It may be assigned to deterministic physical phenomena when we lack sufficient information. We may roll a dice and before looking at the result, state that the result is “5” with probability $1/6$. Similarly, a probability can be attributed to the fact that the electron mass is located within a certain mass interval. That in the context of a constant like the electron mass probability statements are applied, is due to our limited knowledge of the true facts.

²Thomas Bayes was a mathematician and theologian who lived in the 18th century.

It would be more correct, but rather clumsy, to formulate: “The probability that we are right with the proposition that the electron mass is located in that error interval is such and such.” The assignment of probabilities sometimes relies on assumptions which cannot be proved but usually they are well founded on symmetry arguments, physical laws or on experience³. The results obviously depend on these assumptions and can be interpreted only together with those.

The frequentist concept as compared to the Bayesian one has the advantage that additional not provable assumptions are obsolete but the disadvantages that its field of application is rather restricted. Important parts of statistics, like deconvolution, pattern recognition and decision theory are outside its reach. The Bayesian statistics exists in different variants. Its extreme version permits very subjective assignments of probabilities and thus its results are sometimes vulnerable and useless for scientific applications. Anyway, these very speculative probabilities do not play a significant role in the scientific practice.

Both schools, the classical frequentist oriented and the modern Bayesian have developed important statistical concepts. In most applications the results are quite similar. A short comparison of the two approaches will be presented in the appendix.

For completeness we mention a third *classical* interpretation of probability which is appreciated by mathematicians⁴: If an experiment has N equally likely and mutually exclusive outcomes, and if the event A can occur in P of them, then the probability of event A is equal to P/N . It has the difficulty that it can hardly be translated into real situations and a slight logical problem in that the term *equally likely* already presumes some idea of what probability means.

Independent of the statistical approach, in order to be able to apply the results of probability theory, it is necessary that the statistical probability follows the axioms of the mathematical probability theory, i.e. it has to obey Kolmogorov’s axioms. For example, probabilities have to be positive and smaller or equal to one. We will discuss these axioms below.

In this book we will adopt a moderately Bayesian point of view. This means that in some cases we will introduce sensible assumptions without being able to prove their validity. However, we will establish fixed, simple rules that have to be applied in data analysis. In this way we achieve an objective parametrization of the data. This does not exclude that in some occasions as in goodness-of-fit tests we favor methods of frequentist statistics.

1.4 Assignment of Probabilities to Events

The mathematician assumes that the assignment of probabilities to events exists. To achieve practical, useful results in the natural sciences, in sociology, economics or medicine, statistical methods are required and a sensible assignment has to be made.

There are various possibilities to do so:

³Remark, also probability assignments based on experience have a frequency background.

⁴For two reasons: The proof that the Kolmogorov’s axioms are fulfilled is rather easy, and the calculation of the probability for complex events is possible by straight forward combinatorics.

- *Symmetry properties* are frequently used to assign equal probabilities to events. This is done in gambling, examples are rolling dice, roulette and card games. The isotropy of space predicts equal probabilities for radiation from a point source into different directions.
- *Laws of nature* like the Boltzmann's law of thermodynamics, the exponential decay law of quantum mechanics or Mendel's laws allow us to calculate the probabilities for certain events.
- From the observed frequencies of certain events in empirical studies we can estimate their probabilities, like those of female and male births, of muons in cosmic rays, or of measurement errors in certain repeatable experiments. Here we derive frequencies from a large sample of observations from which we then derive with sufficient accuracy the probability of future events.
- In some situations we are left with educated guesses or we rely on the opinion of experts, when for example the weather is to be predicted or the risk of an accident of a new oil-ship has to be evaluated.
- In case of absolute ignorance often a *uniform probability distribution* is assumed. This is known as *Bayes' postulate*. When we watch a tennis game and do not know the players, we will assign equal probabilities of winning to both players.

To illustrate the last point in a more scientific situation, let us look at a common example in particle physics:

Example 1. Uniform prior for a particle mass

Before a precise measurement of a particle mass is performed, we only know that a particle mass m lies between the values m_1 and m_2 . We may assume initially that all values of the mass inside the interval are equally likely. Then the a priori probability $P\{m_0 \leq m < m_2\}$ (or prior probability) that m is larger than m_0 , with m_0 located between m_1 and m_2 , is equal to:

$$P\{m_0 \leq m < m_2\} = \frac{m_2 - m_0}{m_2 - m_1} .$$

This assertion relies on the assumption of a uniform distribution of the mass within the limits and is obviously assailable, because, had we assumed – with equal right – a uniform distribution for the mass squared, we had obtained a different result:

$$P\{m_0^2 \leq m^2 < m_2^2\} = \frac{m_2^2 - m_0^2}{m_2^2 - m_1^2} \neq P\{m_0 \leq m < m_2\} .$$

Of course, the difference is small, if the interval is small, $m_2 - m_1 \ll m$, for then we have:

$$\begin{aligned} \frac{m_2^2 - m_0^2}{m_2^2 - m_1^2} &= \frac{m_2 - m_0}{m_2 - m_1} \times \frac{m_2 + m_0}{m_2 + m_1} \\ &= \frac{m_2 - m_0}{m_2 - m_1} \left(1 + \frac{m_0 - m_1}{m_2 + m_1} \right) \\ &\approx \frac{m_2 - m_0}{m_2 - m_1} . \end{aligned}$$

When the Z_0 mass and its error were determined, a uniform prior probability in the mass was assumed. If instead a uniform probability in the mass squared had been used, the result had changed only by about 10^{-3} times the uncertainty of the mass determination. This means that applying Bayes' assumption to either the mass or the mass squared makes no difference within the precision of the measurement in this specific case.

In other situations prior probabilities which we will discuss in detail in Chap. 6 can have a considerable influence on a result.

1.5 Outline of this Book

After a short chapter on probability axioms and theorems we present properties of probability distributions in Chapter 3, and its application to simple error calculus and Monte Carlo simulation in Chapters 4 and 5.

The statistics part starts in Chapters 6 and 7 with point estimation followed by interval estimation, Chapter 8

Chapter 9 deals with deconvolution problems.

In Chapter 10 significance and goodness-of-fit tests are discussed.

Chapter 11 with the title *Statistical Learning* summarizes some approximation and classification techniques.

In Chapter 12 a short introduction into probability density estimation and bootstrap techniques is given.

Finally, the Appendix contains some useful mathematical or technical objects, introduces important frequentist concepts and theorems and presents a short comparison of the different statistical approaches.

Recommendations for Ancillary Literature

- The standard book of Kendall and Stuart "The Advanced Theory of Statistics" [2], consisting of several volumes provides a excellent and rather complete presentation of classical statistics with all necessary proofs and many references. It is a sort of Bible of conservative statistics, well suited to look up specific topics. Modern techniques, like Monte Carlo methods are not included.

- The books of Brandt "Data Analysis" [3] and Frodesen et. al. "Probability and Statistics in Particle Physics" [4] give a pretty complete overview of the standard statistical methods as used by physicists.

- Less formal, very intuitive and also well suited for beginners is the thinner book by Lyons [5], "Statistics for Nuclear and Particle Physicists". It reflects the large practical experience of the author.

- Larger, very professional and more ambitious is the book of Eadie et al. "Statistical Methods in Experimental Physics" [6], also intended mainly for particle and nuclear physicists and written by particle physicists and statisticians. A new edition has appeared recently [7]. Modern techniques of data analysis are not discussed.

- Very useful especially for the solution of numerical problems is a book by Blobel and Lohrman “Statistische und numerische Methoden der Datenanalyse” [8] written in German.

- Other useful books written by particle physicists are found in Refs. [9, 10, 11, 12]. The books by Roe and Barlow are more conventional while Cowan and D’Agostini favor a moderate Bayesian view.

- Modern techniques of statistical data analysis are presented in a book written by professional statisticians for non-professional users, Hastie et al. “The Elements of Statistical Learning” [13]

- A modern professional treatment of Bayesian statistics is the textbook by Box and Tiao “Bayesian Inference in Statistical Analysis” [14].

The interested reader will find work on the foundations of statistics, on basic principles and on the standard theory in the following books:

- Fisher’s book [15] “Statistical Method, Experimental Design and Scientific Inference” provides an interesting overview of his complete work.

- Edward’s book “Likelihood” [16] stresses the importance of the likelihood function, contains many useful references and the history of the Likelihood Principle.

- Many basic considerations and a collection of personal contributions from a moderate Bayesian view are contained in the book “Good Thinking” by Good [17]. A collection of work by Savage [18], presents a more extreme Bayesian point of view.

- Somewhat old fashioned textbooks of Bayesian statistic which are of historical interest are the books of Jeffreys [19] and Savage [20].

Recent statistical work by particle physicists and astrophysicists can be found in the proceedings of the PHYSTAT Conferences [21] held during the past few years. Many interesting and well written articles can be found also in the internet.

This personal selection of literature is obviously in no way exhaustive.

Basic Probability Relations

2.1 Random Events and Variables

Events are processes or facts that are characterized by a specific property, like obtaining a “3” with a dice. A goal in a soccer play or the existence of fish in a lake are events. Events can also be complex facts like rolling two times a dice with the results *three* and *five* or the occurrence of a certain mean value in a series of measurements or the estimate of the parameter of a theory. There are elementary events, which mutually exclude each other but also events that correspond to a class of elementary events, like the result *greater than three* when throwing a dice. We are concerned with random events which emerge from a stochastic process as already introduced above.

When we consider several events, then there are events which exclude each other and events which are compatible. We stick to our standard example *dice*. The elementary events *three* and *five* exclude each other, the events *greater than two* and *five* are of course compatible. An other common example: We select an object from a bag containing blue and red cubes and spheres. Here the events *sphere* and *cube* exclude each other, the events *sphere* and *red* may be compatible.

The event $\overline{A}$ is called the complement of event A if either event A or event $\overline{A}$ applies, but *not* both at the same time (exclusive *or*). Complementary to the event *three* in the dice example is the event *less than three or larger than three* (inclusive *or*). Complementary to the event *red sphere* is the event *cube or blue sphere*.

The event consisting of the fact that an arbitrary event out of all possible events applies, is called the certain event. We denote it with Ω . The complementary event is the impossible event, that none of all considered events applies: It is denoted with $\emptyset$, thus $\emptyset = \overline{\Omega}$.

Some further definitions are useful:

Definition 1: $A \cup B$ means *A or B*.

The event $A \cup B$ has the attributes of event A or event B or those of both A and B (inclusive *or*). (The attribute *cube* $\cup$ *red* corresponds to the complementary event *blue sphere*.)

Definition 2: $A \cap B$ means *A and B*.

The event $A \cap B$ has the attributes of event A as well as those of event B . (The attribute *cube* $\cap$ *red* corresponds to *red cube*.) If $A \cap B = \emptyset$, then A, B mutually exclude each other.

Definition 3: $A \subset B$ means that A implies B .

It is equivalent to both $A \cup B = B$ and $A \cap B = A$.

From these definitions follow the trivial relations

$$A \cup \overline{A} = \Omega, \quad A \cap \overline{A} = \emptyset,$$

and

$$\emptyset \subset A \subset \Omega. \quad (2.1)$$

For any A, B we have

$$\overline{A \cup B} = \overline{A} \cap \overline{B}, \quad \overline{A \cap B} = \overline{A} \cup \overline{B}.$$

To the random event A we associate the probability $P\{A\}$ as discussed above. In all practical cases random events can be identified with a variable, the *random variable* or *variate*. Examples for variates are the decay time in particle decay, the number of cosmic muons penetrating a body in a fixed time interval and measurement errors. When the random events involve values that cannot be ordered, like shapes or colors, then they can be associated with classes or categorical variates.

2.2 Probability Axioms and Theorems

2.2.1 Axioms

The assignment of probabilities $P\{A\}$ to members $A, B, C, \dots$ of a set of events has to satisfy the following axioms¹. Only then the rules of probability theory are applicable.

- **Axiom 1** $0 \leq P\{A\}$
The probability of an event is a positive real number.
- **Axiom 2** $P\{\Omega\} = 1$
The probability of the certain event is *one*.
- **Axiom 3** $P\{A \cup B\} = P\{A\} + P\{B\}$ if $A \cap B = \emptyset$
The probability that *A or B* applies is equal to the sum of the probabilities that *A* or that *B* applies, if the events *A* and *B* are mutually exclusive.

These axioms and definitions imply the following theorems whose validity is rather obvious. They can be illustrated with so-called Venn diagrams, Fig. 2.1. There the areas of the ellipses and their intersection are proportional to the probabilities.

$$\begin{aligned} P\{\overline{A}\} &= 1 - P\{A\}, \quad P\{\emptyset\} = 0, \\ P\{A \cup B\} &= P\{A\} + P\{B\} - P\{A \cap B\}, \\ \text{if } A \subset B &\Rightarrow P\{A\} \leq P\{B\}. \end{aligned} \quad (2.2)$$

¹They are called *Kolmogorov* axioms, after the Russian mathematician A. N. Kolmogorov (1903-1987).

Relation (2.1) together with (2.2) and axioms 1, 2 imply $0 \leq P\{A\} \leq 1$. For arbitrary events we have

$$P\{A \cup B\} \geq P\{A\}, P\{B\} ; P\{A \cap B\} \leq P\{A\}, P\{B\} .$$

If all events with the attribute A possess also the attribute B , $A \subset B$, then we have $P\{A \cap B\} = P\{A\}$, and $P\{A \cup B\} = P\{B\}$.

2.2.2 Conditional Probability, Independence, and Bayes' Theorem

In the following we need two further definitions:

Definition: $P\{A | B\}$ is the *conditional probability* of event A under the condition that B applies. It is given, as is obvious from Fig. 2.1, by:

$$P\{A | B\} = \frac{P\{A \cap B\}}{P\{B\}} , \quad P\{B\} \neq 0 . \quad (2.3)$$

A conditional probability is, for example, the probability to find a *sphere* among the *red* objects. The notation $A | B$ expresses that B is considered as fixed, while A is the random event to which the probability refers. Contrary to $P\{A\}$, which refers to arbitrary events A , we require that also B is valid and therefore $P\{A \cap B\}$ is normalized to $P\{B\}$.

Among the events $A | B$ the event $A = B$ is the certain event, thus $P\{B | B\} = 1$. More generally, from definition 3 of the last section and (2.3) follows $P\{A | B\} = 1$ if B implies A :

$$B \subset A \Rightarrow A \cap B = B \Rightarrow P\{A | B\} = 1 .$$

Definition: If $P\{A \cap B\} = P\{A\} \times P\{B\}$, the events A and B (more precisely: the probabilities for their occurrence) are *independent*.

From (2.3) then follows $P\{A | B\} = P\{A\}$, i.e. the conditioning on B is irrelevant for the probability of A . Likewise $P\{B | A\} = P\{B\}$.

In Relation (2.3) we can exchange A and B and thus $P\{A | B\}P\{B\} = P\{A \cap B\} = P\{B | A\}P\{A\}$ and we obtain the famous Bayes' theorem:

$$P\{A | B\}P\{B\} = P\{B | A\}P\{A\} . \quad (2.4)$$

Bayes' theorem is frequently used to relate the conditional probabilities $P\{A | B\}$ and $P\{B | A\}$, and, as we will see, is of some relevance in parameter inference.

The following simple example illustrates some of our definitions. It assumes that each of the considered events is composed of a certain number of elementary events which mutually exclude each other and which because of symmetry arguments all have the same probability.

Example 2. Card game, independent events

The following table summarizes some probabilities for randomly selected cards from a card set consisting of 32 cards and 4 colors.

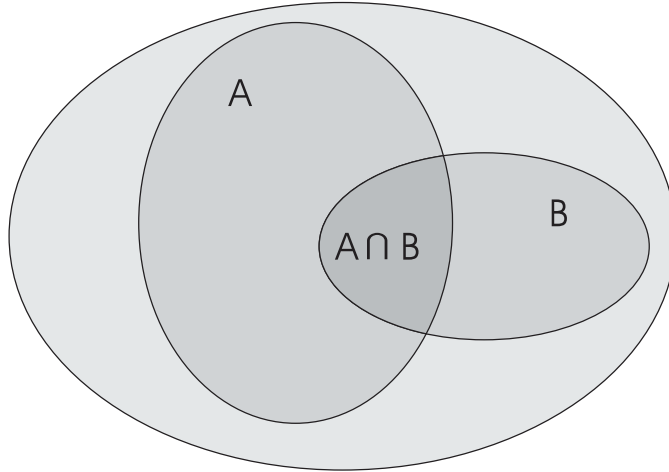


Fig. 2.1. Venn diagram.

$P\{\text{king}\}:$	$4/32 = 1/8$	(prob. for king)
$P\{\text{heart}\}:$	$1/4$	(prob. for heart)
$P\{\text{heart} \cap \text{king}\}:$	$1/8 \cdot 1/4 = 1/32$	(prob. for heart king)
$P\{\text{heart} \cup \text{king}\}:$	$1/8 + 1/4 - 1/32 = 11/32$	(prob. for heart or king)
$P\{\text{heart} \mid \text{king}\}:$	$1/4$	(prob. for heart if king)

The probabilities $P\{\text{heart}\}$ and $P\{\text{heart} \mid \text{king}\}$ are equal as required from the independence of the events A and B .

The following example illustrates how we make use of independence.

Example 3. Random coincidences, measuring the efficiency of a counter

When we want to measure the efficiency of a particle counter (1), we combine it with a second counter (2) in such a way that a particle beam crosses both detectors. We record the number of events n_1, n_2 in the two counters and in addition the number of coincidences $n_{1 \cap 2}$. The corresponding efficiencies relate these numbers² to the unknown number of particles n crossing the detectors.

$$n_1 = \varepsilon_1 n, \quad n_2 = \varepsilon_2 n, \quad n_{1 \cap 2} = \varepsilon_{1 \cap 2} n.$$

For *independent* counting efficiencies we have $\varepsilon_{1 \cap 2} = \varepsilon_1 \varepsilon_2$ and we get

$$\varepsilon_1 = \frac{n_{1 \cap 2}}{n_2}, \quad \varepsilon_2 = \frac{n_{1 \cap 2}}{n_1}, \quad n = \frac{n_1 n_2}{n_{1 \cap 2}}.$$

This scheme is used in many analog situations.

Bayes' theorem is applied in the next two examples, where the attributes are not independent.

Example 4. Bayes' theorem, fraction of women among students

²Here we ignore the statistical fluctuations of the observations.

From the proportion of students and women in the population and the fraction of students among women we compute the fraction of women among students:

$$\begin{aligned} P\{A\} &= 0.02 && \text{(fraction of students in the population)} \\ P\{B\} &= 0.5 && \text{(fraction of women in the population)} \\ P\{A \mid B\} &= 0.018 && \text{(fraction of students among women)} \\ P\{B \mid A\} &=? && \text{(fraction of women among students)} \end{aligned}$$

The dependence of the events A and B manifests itself in the difference of $P\{A\}$ and $P\{A \mid B\}$. Applying Bayes' theorem we obtain

$$\begin{aligned} P\{B \mid A\} &= \frac{P\{A \mid B\}P\{B\}}{P\{A\}} \\ &= \frac{0.018 \cdot 0.5}{0.02} = 0.45 . \end{aligned}$$

About 45% of the students are women.

Example 5. Bayes' theorem, beauty filter

The probability $P\{A\}$ that beauty quark production occurs in a colliding beam reaction be 0.0001. A filter program selects beauty reactions A with efficiency $P\{b \mid A\} = 0.98$ and the probability that it falsely assumes that beauty is present if it is not, be $P\{b \mid \bar{A}\} = 0.01$. What is the probability $P\{A \mid b\}$ to have genuine beauty production in a selected event? To solve the problem, first the probability $P\{b\}$ that a random event is selected has to be evaluated,

$$\begin{aligned} P\{b\} &= P\{b\} [P\{A \mid b\} + P\{\bar{A} \mid b\}] \\ &= P\{b \mid A\}P\{A\} + P\{b \mid \bar{A}\}P\{\bar{A}\} \end{aligned}$$

where the bracket in the first line is equal to 1. In the second line Bayes' theorem is applied. Applying it once more, we get

$$\begin{aligned} P\{A \mid b\} &= \frac{P\{b \mid A\}P\{A\}}{P\{b\}} \\ &= \frac{P\{b \mid A\}P\{A\}}{P\{b \mid A\}P\{A\} + P\{b \mid \bar{A}\}P\{\bar{A}\}} \\ &= \frac{0.98 \cdot 0.0001}{0.98 \cdot 0.0001 + 0.01 \cdot 0.9999} = 0.0097 . \end{aligned}$$

About 1% of the selected events corresponds to b quark production.

Bayes' theorem is rather trivial, thus the results of the last two examples could have easily been written down without referring to it.

Probability Distributions and their Properties

A probability distribution assigns probabilities to random variables. As an example we show in Fig. 3.1 the distribution of the sum s of the points obtained by throwing three ideal dice. Altogether there are $6^3 = 216$ different combinations. The random variable s takes values between 3 and 18 with different probabilities. The sum $s = 6$, for instance, can be realized in 10 different ways, all of which are equally probable. Therefore the probability for $s = 6$ is $P\{s = 6\} = 10/216 \approx 4.6\%$. The distribution is symmetric with respect to its mean value 10.5. It is restricted to discrete values of s , namely natural numbers.

In our example the variate is discrete. In other cases the random variables are continuous. Then the probability for any fixed value is zero, we have to describe the distribution by a probability density and we obtain a finite probability when we integrate the density over a certain interval.

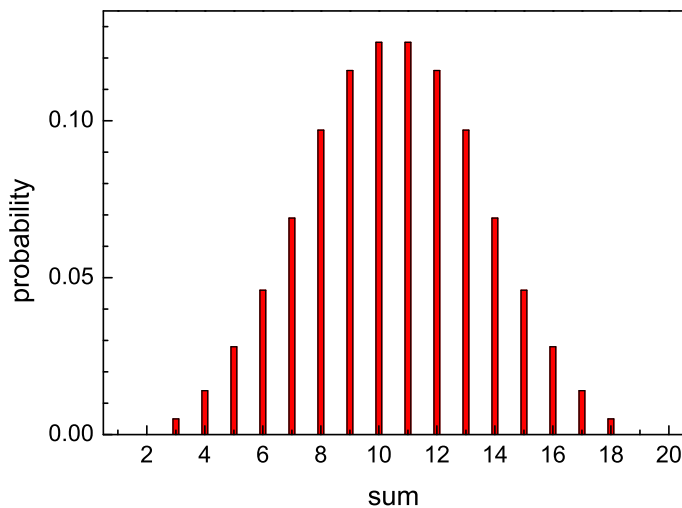


Fig. 3.1. Probability distribution of the sum of the points of three dice.

3.1 Definition of Probability Distributions

We define a distribution function, also called cumulative or integral distribution function, $F(t)$, which specifies the probability P to find a value of x smaller than t :

$$F(t) = P\{x < t\} \quad , \quad \text{with} \quad -\infty < t < \infty .$$

The probability axioms require the following properties of the distribution function:

- a) $F(t)$ is a non-decreasing function of t ,
- b) $F(-\infty) = 0$,
- c) $F(\infty) = 1$.

We distinguish between

- Discrete distributions (Fig. 3.2)
- Continuous distributions (Fig. 3.3)

3.1.1 Discrete Distributions

If not specified differently, we assume in the following that discrete distributions assign probabilities to an enumerable set of different events, which are characterized by an ordered, real variate x_i , with $i = 1, \dots, N$, where N may be finite or infinite. The probabilities $p(x_i)$ to observe the values x_i satisfy the normalization condition:

$$\sum_{i=1}^N p(x_i) = 1 .$$

It is defined by

$$p(x_i) = P\{x = x_i\} = F(x_i + \epsilon) - F(x_i - \epsilon) ,$$

with ϵ positive and smaller than the distance to neighboring variate values.

Example 6. Discrete probability distribution (dice)

For a fair die, the probability to throw a certain number k is just one-sixth:

$$p(k) = 1/6 \text{ for } k = 1, 2, 3, 4, 5, 6.$$

It is possible to treat discrete distributions with the help of Dirac's δ -function like continuous ones. Therefore we will often consider only the case of continuous variates.

3.1.2 Continuous Distributions

We replace the discrete probability distribution by a *probability density*¹ $f(x)$, abbreviated as *p.d.f.* (*probability density function*). It is defined as follows:

¹We will, however, use the notations *probability distribution* and *distribution* for discrete as well as for continuous distributions.

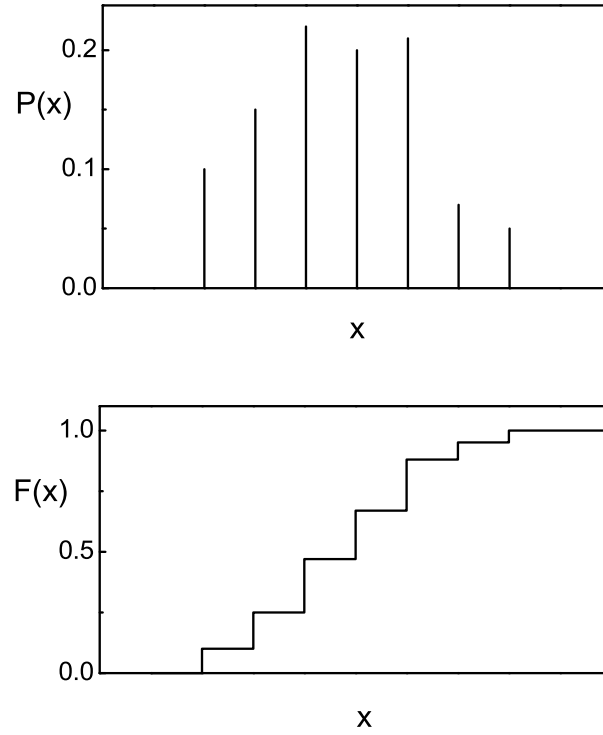


Fig. 3.2. Discrete probability distribution and distribution function.

$$f(x) = \frac{dF(x)}{dx} . \quad (3.1)$$

Remark that the p.d.f. is defined in the full range $-\infty < x < \infty$. It may be zero in certain regions.

It has the following properties:

- a) $f(-\infty) = f(+\infty) = 0$,
- b) $\int_{-\infty}^{\infty} f(x)dx = 1$.

The probability $P\{x_1 \leq x \leq x_2\}$ to find the random variable x in the interval $[x_1, x_2]$ is given by

$$P\{x_1 \leq x \leq x_2\} = F(x_2) - F(x_1) = \int_{x_1}^{x_2} f(x)dx .$$

We will discuss specific distributions in Sect. 3.6 but we introduce two common distributions already here. They will serve us as examples in the following sections.

Example 7. Probability density of an exponential distribution

The decay time t of an instable particles follows an exponential distribution with the p.d.f.

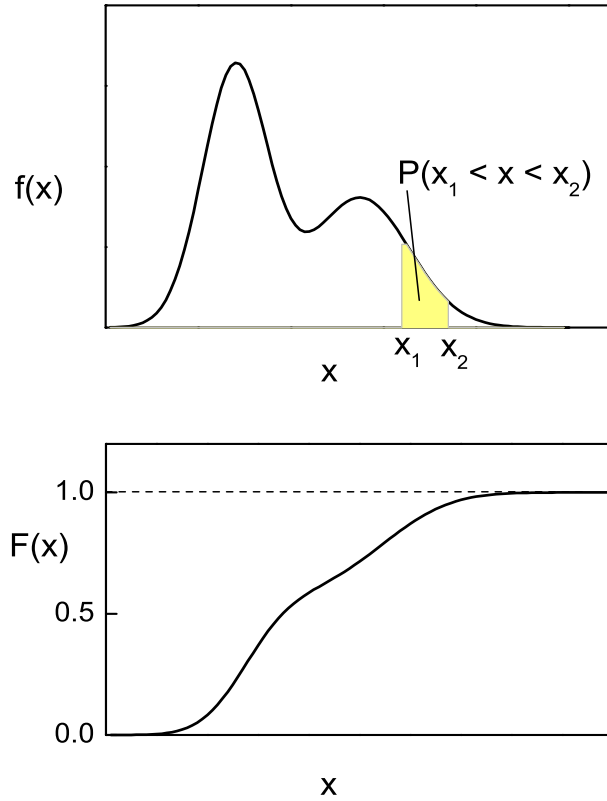


Fig. 3.3. Probability density and distribution function of a continuous distribution.

$$f(t) \equiv f(t|\lambda) = \lambda e^{-\lambda t} \text{ for } t \geq 0, \quad (3.2)$$

where the parameter² $\lambda > 0$, the decay constant, is the inverse of the mean lifetime $\tau = 1/\lambda$. The probability density and the distribution function

$$F(t) = \int_{-\infty}^t f(t') dt' = 1 - e^{-\lambda t}$$

are shown in Fig. 3.4. The probability of observing a lifetime longer than τ is

$$P\{t > \tau\} = F(\infty) - F(\tau) = e^{-1}.$$

Example 8. Probability density of the normal distribution

An oxygen atom is drifting in argon gas, driven by thermal scattering. It starts at the origin. After a certain time its position is (x, y, z) . Each projec-

²We use the bar | to separate random variables (x, t) from parameters (λ) which specify the distribution. This notation will be discussed below in Chap. 6, Sect. 6.3.

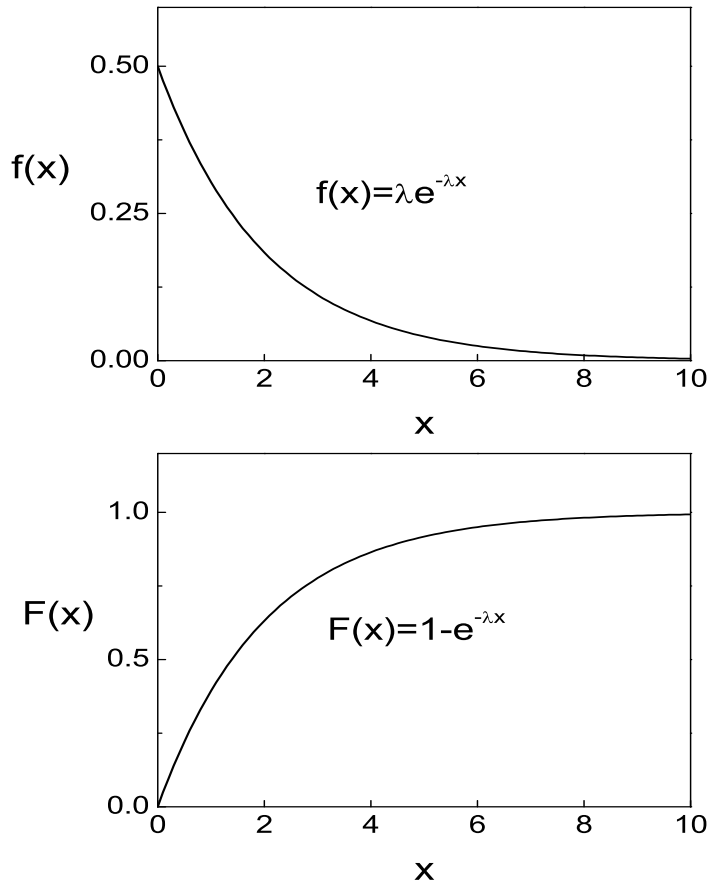


Fig. 3.4. Probability density and distribution function of an exponential distribution.

tion, for instance x , has approximately a normal distribution (see Fig. 3.5), also called Gauss distribution³

$$f(x) = N(x|0, s) ,$$

$$N(x|x_0, s) = \frac{1}{\sqrt{2\pi}s} e^{-(x-x_0)^2/(2s^2)} . \quad (3.3)$$

The width constant s is, as will be shown later, proportional to the square root of the number of scattering processes or the square root of time. When we descent by the factor $1/\sqrt{e}$ from the maximum, the full width is just $2s$. A statistical drift motion, or more generally a random walk, is met frequently in science and also in every day life. The normal distribution also describes approximately the motion of snow flakes or the erratic movements of a drunkard in the streets.

³In the formula we separate parameters from variates by a bar.

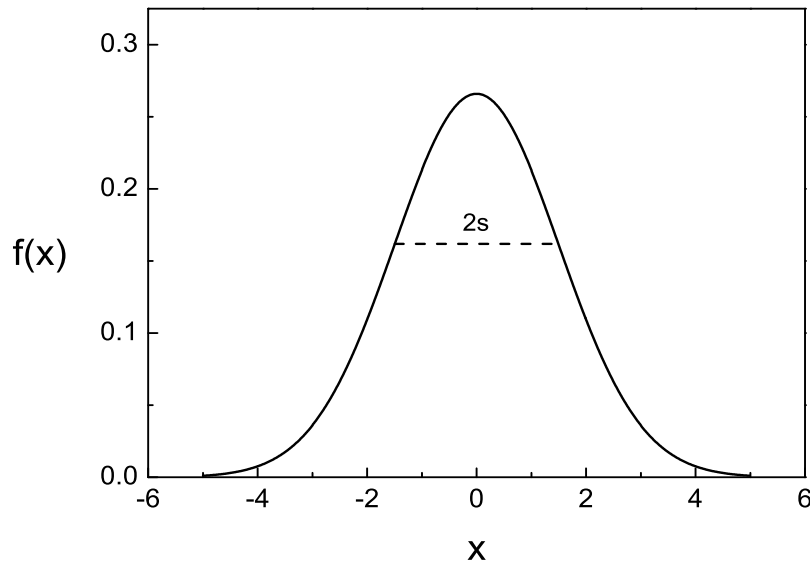


Fig. 3.5. Normal distribution.

3.1.3 Empirical Distributions

Many processes are too complex or not well enough understood to be described by a distribution in form of a simple algebraic formula. In these cases it may be useful to approximate the underlying distribution using an experimental data sample. The simplest way to do this, is to histogram the observations and to normalize the frequency histogram. More sophisticated methods of probability density estimation will be sketched in Chap. 12. The quality of the approximation depends of course on the available number of observations.

3.2 Expected Values

In this section we will consider some general characteristic quantities of distributions, like mean value, width, and asymmetry or skewness. Before introducing the calculation methods, we turn to the general concept of the *expected value*.

The expected value $E(u)$ of a quantity $u(x)$, which depends on the random variable x , can be obtained by collecting infinitely many random values x_i from the distribution $f(x)$, calculating $u_i = u(x_i)$, and then averaging over these values. Obviously, we have to assume the existence of such a limiting value.

In quantum mechanics, expected values of physical quantities are the main results of theoretical calculations and experimental investigations, and provide the connection to classical mechanics. Also in statistical mechanics and thermodynamics the calculation of expected values is frequently needed. We can, for instance, calculate

from the velocity distribution of gas molecules the expected value of their kinetic energy, that means essentially their temperature. In probability theory and statistics expected values play a central role.

3.2.1 Definition and Properties of the Expected Value

Definition:

$$E(u(x)) = \sum_{i=1}^{\infty} u(x_i)p(x_i) \quad (\text{discrete distribution}) , \quad (3.4)$$

$$E(u(x)) = \int_{-\infty}^{\infty} u(x)f(x) dx \quad (\text{continuous distribution}) . \quad (3.5)$$

Here and in what follows, we assume the existence of integrals and sums. This condition restricts the choice of the allowed functions u , p , f .

From the definition of the expected value follow the relations (c is a constant, u , v are functions of x):

$$E(c) = c, \quad (3.6)$$

$$E(E(u)) = E(u), \quad (3.7)$$

$$E(u + v) = E(u) + E(v), \quad (3.8)$$

$$E(cu) = cE(u) . \quad (3.9)$$

They characterize E as a linear functional.

For *independent* (see also Chap. 2 and Sect. 3.5) variates x , y the following important relation holds:

$$E(u(x)v(y)) = E(u)E(v) . \quad (3.10)$$

Often expected values are denoted by angular brackets:

$$E(u) \equiv \langle u \rangle .$$

Sometimes this simplifies the appearance of the formulas. We will use both notations.

In case of a nonlinear function $u(x)$, its expected value differs, in general, from its value at the argument $\langle x \rangle$:

$$\langle u \rangle \neq u(\langle x \rangle) .$$

Example 9. Relation between the expected values of the track momentum and of its curvature

In tracking detectors the momentum p of a charged particle is usually determined from the curvature ρ in a magnetic field, $p(\rho) = c/\rho$. Here $\langle p(\rho) \rangle > p(\langle \rho \rangle) = c/\langle \rho \rangle$. In fact the momentum determined from the curvature is biased to large values. More generally, if $u(x)$ is the inverse of a positive function $v(x)$, its expected value is always *greater* than (or equal to) the inverse expected value of v :

$$\langle u \rangle = \langle 1/v \rangle \geq 1/\langle v \rangle ,$$

where equality is only possible in the degenerate case of a one-point distribution $f(x) = \delta(x - c)$.

To show this assertion, we define an inner product of two positive functions $g(x)$, $h(x)$ of a random variable x with p.d.f. $f(x)$:

$$(g, h) = \int g(x)h(x)f(x)dx$$

and apply the Cauchy-Schwarz inequality

$$(g, h)^2 \leq (g, g)(h, h)$$

to $g(x) = \sqrt{u(x)}$ and $h(x) = 1/\sqrt{u(x)}$, giving

$$\langle u \rangle \langle 1/u \rangle \geq 1 .$$

The equality requires $g \propto h$, or $f(x) = \delta(x - c)$.

3.2.2 Mean Value

The expected value of the variate x is also called the *mean value*. It can be visualized as the center of gravity of the distribution. Usually it is denoted by the Greek letter μ . Both names, mean value, and expected value⁴ of the corresponding distribution are used synonymously.

Definition:

$$\begin{aligned} E(x) &\equiv \langle x \rangle = \mu = \sum_{i=1}^{\infty} x_i p(x_i) \text{ (discrete distribution) ,} \\ E(x) &\equiv \langle x \rangle = \mu = \int_{-\infty}^{\infty} x f(x) dx \text{ (continuous distribution) .} \end{aligned}$$

The mean value of the exponential distribution (3.2) is

$$\langle t \rangle = \int_0^{\infty} \lambda t e^{-\lambda t} dt = 1/\lambda = \tau .$$

We will distinguish $\langle x \rangle$ from the average value of a sample, consisting of a *finite* number N of variate values, $x_1, \dots, x_N$, which will be denoted by $\bar{x}$:

$$\bar{x} = \frac{1}{N} \sum_i x_i .$$

It is called sample mean. It is a random variable and has the expected value

$$\langle \bar{x} \rangle = \frac{1}{N} \sum_i \langle x_i \rangle = \langle x \rangle ,$$

as follows from (3.8), (3.9).

⁴The notation *expected value* may be somewhat misleading, as the probability to obtain it can be zero (see the example “dice” in Sect. 3.2.7).

3.2.3 Variance

The variance σ^2 measures the spread of a distribution, defined as the mean quadratic deviation of the variate from its mean value. Usually, we want to know not only the mean value of a stochastic quantity, but require also information on the dispersion of the individual random values relative to it. When we buy a laser, we are of course interested in its mean energy per pulse, but also in the variation of the single energies around that mean value. The mean value alone does not provide information about the shape of a distribution. The mean height with respect to sea level of Switzerland is about 700 m, but this alone does not say much about the beauty of that country, which, to a large degree, depends on the spread of the height distribution.

The square root σ of the variance is called *standard deviation* and is the standard measure of stochastic uncertainties.

A mechanical analogy to the variance is the moment of inertia for a mass distribution along the x -axis for a total mass equal to unity.

Definition:

$$\text{var}(x) = \sigma^2 = E[(x - \mu)^2] .$$

From this definition follows immediately

$$\text{var}(cx) = c^2 \text{var}(x) ,$$

and σ/μ is independent of the scale of x .

Very useful is the following expression for the variance which is easily derived from its definition and (3.8), (3.9):

$$\begin{aligned} \sigma^2 &= E(x^2 - 2x\mu + \mu^2) \\ &= E(x^2) - 2\mu^2 + \mu^2 \\ &= E(x^2) - \mu^2 . \end{aligned}$$

Sometimes this is written more conveniently as

$$\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2 = \langle x^2 \rangle - \mu^2 . \quad (3.11)$$

In analogy to Steiner's theorem for moments of inertia, we have

$$\begin{aligned} \langle (x - a)^2 \rangle &= \langle (x - \mu)^2 \rangle + \langle (\mu - a)^2 \rangle \\ &= \sigma^2 + (\mu - a)^2 , \end{aligned}$$

implying (3.11) for $a = 0$.

The variance is invariant against a translation of the distribution by a :

$$x \rightarrow x + a , \mu \rightarrow \mu + a \Rightarrow \sigma^2 \rightarrow \sigma^2 .$$

Variance of a Sum of Random Numbers

Let us calculate the variance σ^2 for the distribution of the sum x of two *independent* random numbers x_1 and x_2 , which follow different distributions with mean values μ_1, μ_2 and variances σ_1^2, σ_2^2 :

$$x = x_1 + x_2,$$

$$\begin{aligned}\sigma^2 &= \langle (x - \langle x \rangle)^2 \rangle \\ &= \langle ((x_1 - \mu_1) + (x_2 - \mu_2))^2 \rangle \\ &= \langle (x_1 - \mu_1)^2 + (x_2 - \mu_2)^2 + 2(x_1 - \mu_1)(x_2 - \mu_2) \rangle \\ &= \langle (x_1 - \mu_1)^2 \rangle + \langle (x_2 - \mu_2)^2 \rangle + 2\langle x_1 - \mu_1 \rangle \langle x_2 - \mu_2 \rangle \\ &= \sigma_1^2 + \sigma_2^2.\end{aligned}$$

In the fourth step the independence of the variates (3.10) was used.

This result is important for all kinds of error estimation. For a sum of two independent measurements, their standard deviations add quadratically. We can generalize the last relation to a sum $x = \sum x_i$ of N variates or measurements:

$$\sigma^2 = \sum_{i=1}^N \sigma_i^2. \quad (3.12)$$

Example 10. Variance of the convolution of two distributions

We consider a quantity x with the p.d.f. $g(x)$ with variance σ_g^2 which is measured with a device which produces a smearing with a p.d.f. $h(y)$ with variance σ_h^2 . We want to know the variance of the “smeared” value $x' = x + y$. According to 3.12, this is the sum of the variances of the two p.d.f.s:

$$\sigma^2 = \sigma_g^2 + \sigma_h^2.$$

Variance of the Sample Mean of Independent Identically Distributed Variates

From the last relation we obtain the variance $\sigma_{\bar{x}}^2$ of the sample mean $\bar{x}$ from N independent random numbers x_i , which all follow the same distribution⁵ $f(x)$, with expected value μ and variance σ^2 :

$$\begin{aligned}\bar{x} &= \sum_{i=1}^N x_i / N, \\ \text{var}(N\bar{x}) &= N^2 \text{var}(\bar{x}) = N\sigma^2, \\ \sigma_{\bar{x}} &= \frac{\sigma}{\sqrt{N}}.\end{aligned} \quad (3.13)$$

The last two relations (3.12), (3.13) have many applications, for instance in random walk, diffusion, and error propagation. The root mean square distance reached by a diffusing molecule after N scatters is proportional to $\sqrt{N}$ and therefore also to $\sqrt{t}$, t being the diffusion time. The total length of 100 aligned objects, all having the same standard deviation σ of their nominal length, will have a standard deviation of only 10σ . To a certain degree, random fluctuations compensate each other.

⁵The usual abbreviation is i.i.d. variates for independent identically distributed.

Width v of a Sample and Variance of the Distribution

Often, as we will see in Chap. 6, a sample is used to estimate the variance σ^2 of the underlying distribution. In case the mean value μ is known, we calculate the quantity

$$v_\mu^2 = \frac{1}{N} \sum_i (x_i - \mu)^2$$

which has the correct expected value $\langle v_\mu^2 \rangle = \sigma^2$. Usually, however, the true mean value μ is unknown – except perhaps in calibration measurements – and must be estimated from the same sample as is used to derive v_μ^2 . We then are obliged to use the sample mean $\bar{x}$ instead of μ and calculate the mean quadratic deviation v^2 of the sample values relative to $\bar{x}$. In this case the expected value of v^2 will depend not only on σ , but also on N . In a first step we find

$$\begin{aligned} v^2 &= \frac{1}{N} \sum_i (x_i - \bar{x})^2 \\ &= \frac{1}{N} \sum_i (x_i^2 - 2x_i\bar{x} + \bar{x}^2) \\ &= \frac{1}{N} \sum_i x_i^2 - \bar{x}^2. \end{aligned} \quad (3.14)$$

To calculate the expected value, we use (3.11) and (3.13),

$$\begin{aligned} \langle x^2 \rangle &= \sigma^2 + \mu^2, \\ \langle \bar{x}^2 \rangle &= \text{var}(\bar{x}) + \langle \bar{x} \rangle^2 \\ &= \frac{\sigma^2}{N} + \mu^2 \end{aligned}$$

and get with (3.14)

$$\begin{aligned} \langle v^2 \rangle &= \langle x^2 \rangle - \langle \bar{x}^2 \rangle = \sigma^2 \left(1 - \frac{1}{N} \right), \\ \sigma^2 &= \frac{N}{N-1} \langle v^2 \rangle = \frac{\langle \sum_i (x_i - \bar{x})^2 \rangle}{N-1}. \end{aligned} \quad (3.15)$$

The expected value of the mean squared deviation is smaller than the variance of the distribution by a factor of $(N-1)/N$.

The relation (3.15) is widely used for the estimation of measurement errors, when several independent measurements are available. The variance σ_x^2 of the sample mean $\bar{x}$ itself is approximated, according to (3.13), by

$$\frac{v^2}{N-1} = \frac{\sum_i (x_i - \bar{x})^2}{N(N-1)}.$$

Mean Value and Variance of a Superposition of two Distributions

Frequently a distribution consists of a superposition of elementary distributions. Let us compute the mean μ and variance σ^2 of a linear superposition of two distributions

$$f(x) = \alpha f_1(x) + \beta f_2(x), \quad \alpha + \beta = 1,$$

where f_1, f_2 may have different mean values μ_1, μ_2 and variances σ_1^2, σ_2^2 :

$$\mu = \alpha\mu_1 + \beta\mu_2,$$

$$\begin{aligned} \sigma^2 &= E\left((x - E(x))^2\right) \\ &= E(x^2) - \mu^2 \\ &= \alpha E_1(x^2) + \beta E_2(x^2) - \mu^2 \\ &= \alpha(\mu_1^2 + \sigma_1^2) + \beta(\mu_2^2 + \sigma_2^2) - \mu^2 \\ &= \alpha\sigma_1^2 + \beta\sigma_2^2 + \alpha\beta(\mu_1 - \mu_2)^2. \end{aligned}$$

Here, E, E_1, E_2 denote expected values related to the p.d.f.s f, f_1, f_2 . In the last step the relation $\alpha + \beta = 1$ has been used. Of course, the width increases with the distance of the mean values. The result for σ^2 could have been guessed by considering the limiting cases, $\mu_1 = \mu_2, \sigma_1 = \sigma_2 = 0$.

3.2.4 Skewness

The skewness coefficient γ_1 measures the asymmetry of a distribution with respect to its mean. It is zero for the normal distribution, but quite sizable for the exponential distribution. There it has the value $\gamma_1 = 2$, see Sect. 3.3.3 below.

Definition:

$$\gamma_1 = E\left[(x - \mu)^3\right] / \sigma^3.$$

Similarly to the variance, γ_1 can be expressed by expected values of powers of the variate x :

$$\begin{aligned} \gamma_1 &= E\left[(x - \mu)^3\right] / \sigma^3 \\ &= E\left[x^3 - 3\mu x^2 + 3\mu^2 x - \mu^3\right] / \sigma^3 \\ &= \{E(x^3) - 3\mu[E(x^2) - \mu E(x)] - \mu^3\} / \sigma^3 \\ &= \frac{E(x^3) - 3\mu\sigma^2 - \mu^3}{\sigma^3}. \end{aligned}$$

The skewness coefficient is defined in such a way that it satisfies the requirement of invariance under translation and dilatation of the distribution. Its square is usually denoted by $\beta_1 = \gamma_1^2$.

3.2.5 Kurtosis (Excess)

A fourth parameter, the kurtosis β_2 , measures the tails of a distribution.

Definition:

$$\beta_2 = E\left[(x - \mu)^4\right] / \sigma^4.$$

A kurtosis coefficient or excess γ_2 ,

$$\gamma_2 = \beta_2 - 3,$$

is defined such that it is equal to zero for the normal distribution which is used as a reference. (see Sect. 3.6.5).

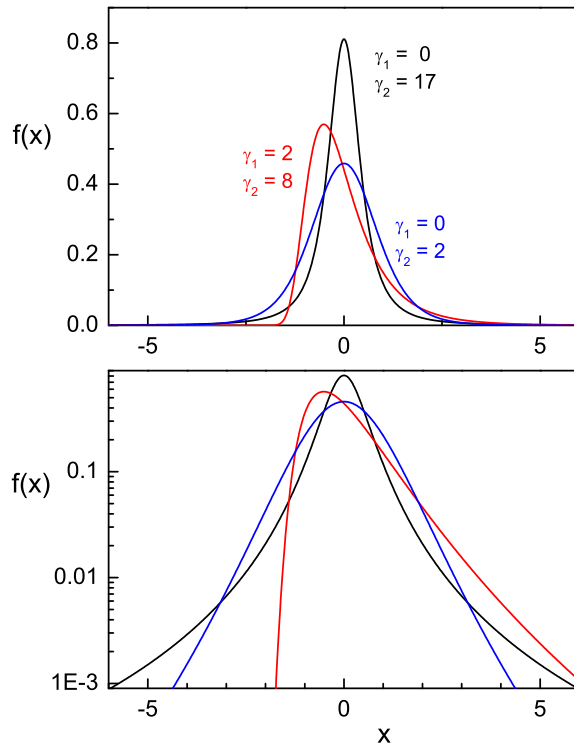


Fig. 3.6. Three distribution with equal mean and variance but different skewness and kurtosis.

3.2.6 Discussion

The mean value of a distribution is a so-called *position* or *location* parameter, the standard deviation is a *scale parameter*. A translation of the variate $x \rightarrow y = x + a$ changes the mean value correspondingly, $\langle y \rangle = \langle x \rangle + a$. This parameter is therefore sensitive to the location of the distribution (like the center of gravity for a mass distribution). The variance (corresponding to the moment of inertia for a mass distribution), respectively the standard deviation remain unchanged. A change of the scale (dilatation) $x \rightarrow y = cx$ entails, besides $\langle y \rangle = c\langle x \rangle$ also $\sigma(y) = c\sigma(x)$. Skewness and kurtosis remain unchanged under both transformations. They are *shape parameters*.

The four parameters mean, variance, skewness, and kurtosis, or equivalently the expected values of x , x^2 , x^3 and x^4 , fix a distribution quite well if in addition the range of the variates and the behavior of the distribution at the limits is given. Then the distribution can be reconstructed quite accurately [74].

Fig. 3.6 shows three probability densities, all with the same mean $\mu = 0$ and standard deviation $\sigma = 1$, but different skewness and kurtosis. The apparently narrower curve has clearly longer tails, as seen in the lower graph with logarithmic scale.

Mainly in cases, where the type of the distribution is not well known, i.e. for empirical distributions, other location and scale parameters are common. These are the *mode* x_{mod} , the variate value, at which the distribution has its maximum, and the *median*, defined as the variate value $x_{0.5}$, at which $P\{x < x_{0.5}\} = F(x_{0.5}) = 0.5$, i.e.

the median subdivides the domain of the variate into two regions with equal probability of 50%. More generally, we define a *quantile* x_a of order a by the requirement $F(x_a) = a$.

A well known example for a median is the half-life $t_{0.5}$ which is the time at which 50% of the nuclei of an unstable isotope have decayed. From the exponential distribution (3.2) follows the relation between the half-life and the mean lifetime τ

$$t_{0.5} = \tau \ln 2 \approx 0.693 \tau .$$

The median is invariant under non-linear transformations $y = y(x)$ of the variate, $y_{0.5} = y(x_{0.5})$ while for the mean value μ and the mode x_{mod} this is usually not the case, $\mu_y \neq y(\mu_x)$, $y_{mod} \neq y(x_{mod})$. The reason for these properties is that probabilities but not probability densities are invariant under variate transformations. Thus the mode should not be considered as the “most probable value”. The probability to obtain exactly the mode value is zero. To obtain finite probabilities, we have to integrate the p.d.f. over some range of the variate as is the case for the median.

In statistical analyses of data contaminated by background the sample median is more “robust” than the sample mean as estimator of the distribution mean. (see Appendix, Sect. 13.15). Instead of the sample width v , often the *full width at half maximum* (f.w.h.m.) is used to characterize the spread of a distribution. It ignores the tails of the distribution. This makes sense for empirical distributions, e.g. in the investigation of spectral lines over a sizable background. For a normal distribution the f.w.h.m. is related to the standard deviation by

$$f.w.h.m._{gauss} \approx 2.36 \sigma_{gauss} .$$

This relation is often used to estimate quickly the standard deviation σ for an empirical distribution given in form of a histogram. As seen from the examples in Fig.3.6, which, for the same variance, differ widely in their f.w.h.m., this procedure may lead to wrong results for non-Gaussian distributions.

3.2.7 Examples

In this section we compute expected values of some quantities for different distributions.

Example 11. Expected values, dice

We have $p(k) = 1/6$, $k = 1, \dots, 6$.

$$\begin{aligned} \langle x \rangle &= (1 + 2 + 3 + 4 + 5 + 6) 1/6 = 7/2 , \\ \langle x^2 \rangle &= (1 + 4 + 9 + 16 + 25 + 36) 1/6 = 91/6 , \\ \sigma^2 &= 91/6 - (7/2)^2 = 35/12 , \\ \sigma &\approx 1.71 , \\ \gamma_1 &= 0 . \end{aligned}$$

The expected value has probability zero.

Example 12. Expected values, lifetime distribution

$$f(t) = \frac{1}{\tau} e^{-t/\tau}, t \geq 0,$$

$$\langle t^n \rangle = \int_0^\infty \frac{t^n}{\tau} e^{-t/\tau} dt = n! \tau^n,$$

$$\begin{aligned}\langle t \rangle &= \tau, \\ \langle t^2 \rangle &= 2\tau^2, \\ \langle t^3 \rangle &= 6\tau^3, \\ \sigma &= \tau, \\ \gamma_1 &= 2.\end{aligned}$$

Example 13. Mean value of the volume of a sphere with a normally distributed radius

The normal distribution is given by

$$f(x) = \frac{1}{\sqrt{2\pi}s} e^{-(x-x_0)^2/(2s^2)}.$$

It is symmetric with respect to x_0 . Thus the mean value is $\mu = x_0$, and the skewness is zero. For the variance we obtain

$$\begin{aligned}\sigma^2 &= \frac{1}{\sqrt{2\pi}s} \int_{-\infty}^{\infty} dx (x - x_0)^2 e^{-(x-x_0)^2/(2s^2)} \\ &= s^2.\end{aligned}$$

The parameters x_0 , s of the normal distribution are simply the mean value and the standard deviation μ , σ , and the p.d.f. with these parameters is abbreviated as $N(x|\mu, \sigma)$.

We now assume that the radius r_0 of a sphere is smeared according to a normal distribution around the mean value r_0 with standard deviation s . This assumption is certainly only approximately valid for $r_0 \gg s$, since negative radii are of course impossible. Let us calculate the expected value of the volume $V(r) = 4/3 \pi r^3$:

$$\begin{aligned}\langle V \rangle &= \int_{-\infty}^{\infty} dr V(r) f(r) \\ &= \frac{4}{3} \frac{\pi}{\sqrt{2\pi}s} \int_{-\infty}^{\infty} dr r^3 e^{-\frac{(r-r_0)^2}{2s^2}} \\ &= \frac{4}{3} \frac{\pi}{\sqrt{2\pi}s} \int_{-\infty}^{\infty} dz (z + r_0)^3 e^{-\frac{z^2}{2s^2}} \\ &= \frac{4}{3} \frac{\pi}{\sqrt{2\pi}s} \int_{-\infty}^{\infty} dz (z^3 + 3z^2 r_0 + 3z r_0^2 + r_0^3) e^{-\frac{z^2}{2s^2}} \\ &= \frac{4}{3} \pi (r_0^3 + 3s^2 r_0).\end{aligned}$$

The mean volume is larger than the volume calculated using the mean radius.

Example 14. Playing poker until the bitter end

Two players are equally clever, but dispose of different capital K_1 , respectively K_2 . They play, until one of the players is left without money. We denote the probabilities for player 1, (2) to win finally with w_1 (w_2). The probability, that one of the two players wins, is unity⁶:

⁶That the game ends if the time is not restricted can be proven in game theory and is supported by experience.

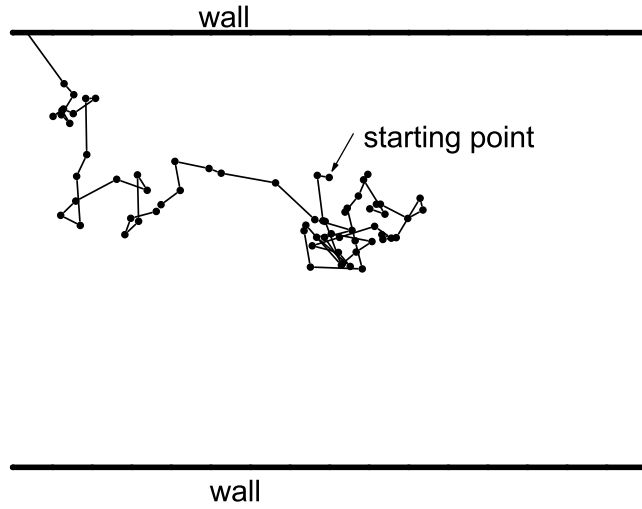


Fig. 3.7. Brownian motion.

$$w_1 + w_2 = 1 .$$

Player 1 gains the capital K_2 with probability w_1 and loses K_1 with probability w_2 . Thus his mean gain is $w_1 K_2 - w_2 K_1$. The same is valid for player two, only with reversed sign. As both players play equally well, the expected gain should be zero for both

$$w_1 K_2 - w_2 K_1 = 0 .$$

From the two relation follows:

$$w_1 = \frac{K_1}{K_1 + K_2}; \quad w_2 = \frac{K_2}{K_1 + K_2} .$$

The probability to win is proportional to the capital disposed of. However, the greater risk of the player with the smaller capital comes along with the possibility of a higher gain.

Example 15. Diffusion (random walk)

A particle is moving stochastically according to the Brownian motion, where every step is independent of the previous ones (Fig. 3.7). The starting point has a distance d_1 from the wall 1 and d_2 from the opposite wall 2. We want to know the probabilities w_1 , w_2 to hit wall 1 or 2. The direct calculation of w_1 and w_2 is a quite involved problem. However, using the properties of expected values, it can be solved quite simply, without even knowing the probability density. The problem here is completely analogous to the previous one:

$$w_1 = \frac{d_2}{d_1 + d_2} , \quad w_2 = \frac{d_1}{d_1 + d_2} .$$

Example 16. Mean kinetic energy of a gas molecule

The velocity of a particle in x -direction v_x is given by a normal distribution

$$f(v_x) = \frac{1}{s\sqrt{2\pi}} e^{-v_x^2/(2s^2)},$$

with

$$s^2 = \frac{kT}{m},$$

where k , T , m are the Boltzmann constant, the temperature, and the mass of the molecule. The kinetic energy is

$$\epsilon_{kin} = \frac{m}{2}(v_x^2 + v_y^2 + v_z^2)$$

with the expected value

$$E(\epsilon_{kin}) = \frac{m}{2} (E(v_x^2) + E(v_y^2) + E(v_z^2)) = \frac{3m}{2} E(v_x^2),$$

where in the last step the velocity distribution was assumed to be isotropic. It follows:

$$E(v_x^2) = \frac{1}{s\sqrt{2\pi}} \int_{-\infty}^{\infty} dv_x v_x^2 e^{-v_x^2/(2s^2)} = s^2 = kT/m,$$

$$E(\epsilon_{kin}) = \frac{3}{2} kT.$$

Example 17. Reading accuracy of a digital clock

For an accurate digital clock which displays the time in seconds, the deviation of the reading from the true time is maximally ± 0.5 seconds. After the reading, we may associate to the true time a uniform distribution with the actual reading as its central value. To simplify the calculation of the variance, we set the reading equal to zero. We thus have

$$f(t) = \begin{cases} 1 & \text{if } -0.5 < t < 0.5 \\ 0 & \text{else} \end{cases}$$

and

$$\sigma^2 = \int_{-0.5}^{0.5} t^2 dt = \frac{1}{12}. \quad (3.16)$$

The root mean square measurement uncertainty (standard deviation) is $\sigma = 1 \text{ s}/\sqrt{12} \approx 0.29 \text{ s}$.

The variance of a uniform distribution, which covers a range of a , is accordingly $\sigma^2 = a^2/12$. This result is widely used for the error estimation of digital measurements. A typical example from particle physics is the coordinate measurement with a wire chamber.

Example 18. Efficiency fluctuations of a detector

A counter registers on average the fraction $\varepsilon = 0.9$ of all traversing electrons. How large is the relative fluctuation σ of the registered number N_1 for N particles passing the detector? The exact solution of this problem requires

the knowledge of the probability distribution, in this case the binomial distribution. But also without this knowledge we can derive the dependence on N with the help of relation (3.13):

$$\sigma\left(\frac{N_1}{N}\right) \sim \frac{1}{\sqrt{N}}.$$

The whole process can be split into single processes, each being associated with the passing of a single particle. Averaging over all processes leads to the above result. (The binomial distribution gives $\sigma(N_1/N) = \sqrt{\varepsilon(1-\varepsilon)/N}$, see Sect. 3.6.1).

All stochastic processes, which can be split into N identical, independent elementary processes, show the typical $1/\sqrt{N}$ behavior of their relative fluctuations.

3.3 Moments and Characteristic Functions

The characteristic quantities of distributions considered up to now, mean value, variance, skewness, and kurtosis, have been calculated from expected values of the lower four powers of the variate. Now we will investigate the expected value of arbitrary powers of the random variable x for discrete and continuous probability distributions $p(x)$, $f(x)$, respectively. They are called *moments* of the distribution. Their calculation is particularly simple, if the characteristic function of the distribution is known. The latter is just the Fourier transform of the distribution.

3.3.1 Moments

Definition: The n -th moments of $f(x)$, respectively $p(x)$ are

$$\mu_n = E(x^n) = \int_{-\infty}^{\infty} x^n f(x) dx,$$

and

$$\mu_n = E(x^n) = \sum_{k=1}^{\infty} x_k^n p(x_k)$$

where n is a natural number⁷.

Apart from these moments, called *moments about the origin*, we consider also the moments about an arbitrary point a where x^n is replaced by $(x - a)^n$. Of special importance are the moments about the expected value of the distribution. They are called *central moments*.

Definition: The n -th central moment about $\mu = \mu_1$ of $f(x)$, $p(x)$ is:

$$\mu'_n = E((x - \mu)^n) = \int_{-\infty}^{\infty} (x - \mu)^n f(x) dx,$$

respectively

⁷In one dimension the zeroth moment is irrelevant. Formally, it is equal to *one*.

$$\mu'_n = E((x - \mu)^n) = \sum_{k=1}^{\infty} (x_k - \mu)^n p(x_k) .$$

Accordingly, the first central moment is zero: $\mu'_1 = 0$. Generally, the moments are related to the expected values introduced before as follows:

$$\begin{aligned} \text{First central moment: } \mu'_1 &= 0 \\ \text{Second central moment: } \mu'_2 &= \sigma^2 \\ \text{Third central moment: } \mu'_3 &= \gamma_1 \sigma^3 \\ \text{Fourth central moment: } \mu'_4 &= \beta_2 \sigma^4 \end{aligned}$$

Under conditions usually met in practise, a distribution is uniquely fixed by its moments. This means, if two distributions have the same moments in all orders, they are identical. We will present below plausibility arguments for the validity of this important assertion.

3.3.2 Characteristic Function

We define the characteristic function $\phi(t)$ of a distribution as follows:

Definition: The characteristic function $\phi(t)$ of a probability density $f(x)$ is

$$\phi(t) = E(e^{itx}) = \int_{-\infty}^{\infty} e^{itx} f(x) dx , \quad (3.17)$$

and, respectively for a discrete distribution $p(x_k)$

$$\phi(t) = E(e^{itx}) = \sum_{k=1}^{\infty} e^{itx_k} p(x_k) . \quad (3.18)$$

For continuous distributions, $\phi(t)$ is the Fourier transform of the p.d.f..

From the definition of the characteristic function follow several useful properties.

$\phi(t)$ is a continuous, in general complex-valued function of t , $-\infty < t < \infty$ with $|\phi(t)| \leq 1$, $\phi(0) = 1$ and $\phi(-t) = \phi^*(t)$. $\phi(t)$ is a real function, if and only if the distribution is symmetric, $f(x) = f(-x)$. Especially for continuous distributions there is $\lim_{t \rightarrow \infty} \phi(t) = 0$. A linear transformation of the variate $x \rightarrow y = ax + b$ induces a transformation of the characteristic function of the form

$$\phi_x(t) \rightarrow \phi_y(t) = e^{ibt} \phi_x(at) . \quad (3.19)$$

Further properties are found in handbooks on the Fourier transform.

The transformation is invertible: With (3.17) it is

$$\begin{aligned} \int_{-\infty}^{\infty} \phi(t) e^{-itx} dt &= \int_{-\infty}^{\infty} e^{-itx} \int_{-\infty}^{\infty} e^{itx'} f(x') dx' dt \\ &= \int_{-\infty}^{\infty} f(x') \left(\int_{-\infty}^{\infty} e^{it(x'-x)} dt \right) dx' \\ &= 2\pi \int_{-\infty}^{\infty} f(x') \delta(x' - x) dx' \\ &= 2\pi f(x) , \end{aligned}$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \phi(t) e^{-itx} dt .$$

The same is true for discrete distributions, as may be verified by substituting (3.18):

$$p(x_k) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \phi(t) e^{-itx_k} dt .$$

In all cases of practical relevance, the probability distribution is uniquely determined by its characteristic function.

Knowing the characteristic functions simplifies considerably the calculation of moments and of the distributions of sums or linear combinations of variates. For continuous distributions moments are found by n -fold derivation of $\phi(t)$:

$$\frac{d^n \phi(t)}{dt^n} = \int_{-\infty}^{\infty} (ix)^n e^{itx} f(x) dx .$$

With $t = 0$ follows

$$\frac{d^n \phi(0)}{dt^n} = \int_{-\infty}^{\infty} (ix)^n f(x) dx = i^n \mu_n . \quad (3.20)$$

The Taylor expansion of $\phi(t)$,

$$\phi(t) = \sum_{n=0}^{\infty} \frac{1}{n!} t^n \frac{d^n \phi(0)}{dt^n} = \sum_{n=0}^{\infty} \frac{1}{n!} (it)^n \mu_n , \quad (3.21)$$

generates the moments of the distribution.

The characteristic function $\phi(t)$ is closely related to the *moment generating function* which is defined through $M(t) = E(e^{tx})$. In some textbooks M is used instead of ϕ for the evaluation of the moments.

We realize that the moments determine ϕ uniquely, and, since the Fourier transform is uniquely invertible, the moments also determine the probability density, as stated above.

In the same way we obtain the central moments:

$$\phi'(t) = E(e^{it(x-\mu)}) = \int_{-\infty}^{\infty} e^{it(x-\mu)} f(x) dx = e^{-it\mu} \phi(t) , \quad (3.22)$$

$$\frac{d^n \phi'(0)}{dt^n} = i^n \mu'_n . \quad (3.23)$$

The Taylor expansion is

$$\phi'(t) = \sum_{n=0}^{\infty} \frac{1}{n!} (it)^n \mu'_n . \quad (3.24)$$

The results (3.20), (3.21), (3.23), (3.24) remain valid also for discrete distributions. The expansion of the right hand side of relation (3.22) allows us to calculate the central moments from the moments about the origin and vice versa:

$$\mu'_n = \sum_{k=0}^n (-1)^k \binom{n}{k} \mu_{n-k} \mu^k , \quad \mu_n = \sum_{k=0}^n \binom{n}{k} \mu'_{n-k} \mu^k .$$

Note, that for $n = 0$, $\mu_0 = \mu'_0 = 1$.

In some applications we have to compute the distribution $f(z)$ where z is the sum $z = x + y$ of two *independent* random variables x and y with the probability densities $g(x)$ and $h(y)$. The result is given by the convolution integral, see Sect. 3.5.4,

$$f(z) = \int g(x)h(z-x) dx = \int h(y)g(z-y) dy$$

which often is difficult to evaluate analytically. It is simpler in most situations to proceed indirectly via the characteristic functions $\phi_g(t)$, $\phi_h(t)$ and $\phi_f(t)$ of the three p.d.f.s which obey the simple relation

$$\phi_f(t) = \phi_g(t)\phi_h(t) . \quad (3.25)$$

Proof:

According to (3.10) we get

$$\begin{aligned} \phi_f(t) &= E(e^{it(x+y)}) \\ &= E(e^{itx}e^{ity}) \\ &= E(e^{itx})E(e^{ity}) \\ &= \phi_g(t)\phi_h(t) . \end{aligned}$$

The third step requires the independence of the two variates. Applying the inverse Fourier transform to $\phi_f(t)$, we get

$$f(z) = \frac{1}{2\pi} \int e^{-itz} \phi_f(t) dt .$$

The solution of this integral is not always simple. For some functions it can be found in tables of the Fourier transform.

In the general case where x is a linear combination of independent random variables, $x = \sum c_j x_j$, we find in an analogous way:

$$\phi(t) = \prod \phi_j(c_j t) .$$

Cumulants

As we have seen, the characteristic function simplifies in many cases the calculation of moments and the convolution of two distributions. Interesting relations between the moments of the three distributions $g(x)$, $h(y)$ and $f(z)$ with $z = x + y$ are obtained from the expansion of the logarithm $K(t)$ of the characteristic functions into powers of it :

$$K(t) = \ln \phi(t) = \ln \langle \exp(itx) \rangle = \kappa_1(it) + \kappa_2 \frac{(it)^2}{2!} + \kappa_3 \frac{(it)^3}{3!} + \dots .$$

Since $\phi(0) = 1$ there is no constant term. The coefficients κ_i , defined in this way, are called *cumulants* or *semiinvariants*. The second notation was chosen, since the cumulants κ_i , with the exception of κ_1 , remain invariant under the translations $x \rightarrow x+b$ of the variate x . Of course, the cumulant of order i can be expressed by moments about the origin or by central moments μ_k , μ'_k up to the order i . We do not present the general analytic expressions for the cumulants which can be derived from the power expansion of $\exp K(t)$ and give only the remarkably simple relations for $i \leq 6$ as a function of the central moments:

$$\begin{aligned}
\kappa_1 &= \mu_1 \equiv \mu = \langle x \rangle, \\
\kappa_2 &= \mu'_2 \equiv \sigma^2 = \text{var}(x), \\
\kappa_3 &= \mu'_3, \\
\kappa_4 &= \mu'_4 - 3\mu'^2_2, \\
\kappa_5 &= \mu'_5 - 10\mu'_2\mu'_3, \\
\kappa_6 &= \mu'_6 - 15\mu'_2\mu'_4 - 10\mu'^2_3 + 30\mu'^3_2.
\end{aligned} \tag{3.26}$$

Besides expected value and variance, also skewness and excess are easily expressed by cumulants:

$$\gamma_1 = \frac{\kappa_3}{\kappa_2^{3/2}}, \quad \gamma_2 = \frac{\kappa_4}{\kappa_2^2}. \tag{3.27}$$

Since the product of the characteristic functions $\phi(t) = \phi^{(1)}(t)\phi^{(2)}(t)$ turns into the sum $K(t) = K^{(1)}(t) + K^{(2)}(t)$, the cumulants are additive, $\kappa_i = \kappa_i^{(1)} + \kappa_i^{(2)}$. In the general case, where x is a linear combination of independent variates, $x = \sum c_j x^{(j)}$, the cumulants of the resulting x -distribution, κ_i , are derived from those of the various $x^{(j)}$ distributions according to

$$\kappa_i = \sum_j c_j^i \kappa_i^{(j)}. \tag{3.28}$$

We have met examples for this relation already in Sect. 3.2.3 where we have computed the variance of the distribution of a sum of variates. We will use it again in the discussion of the Poisson distribution in the following example and in Sect. 3.6.3.

3.3.3 Examples

Example 19. Characteristic function of the Poisson distribution

The Poisson distribution

$$P_\lambda(k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

has the characteristic function

$$\phi(t) = \sum_{k=0}^{\infty} e^{itk} \frac{\lambda^k}{k!} e^{-\lambda}$$

which can be simplified to

$$\begin{aligned}
\phi(t) &= \sum_{k=0}^{\infty} \frac{1}{k!} (e^{it}\lambda)^k e^{-\lambda} \\
&= \exp(e^{it}\lambda) e^{-\lambda} \\
&= \exp(\lambda(e^{it} - 1)),
\end{aligned}$$

from which we derive the moments:

$$\begin{aligned}
\frac{d\phi}{dt} &= \exp(\lambda(e^{it} - 1)) \lambda i e^{it}, \\
\frac{d\phi(0)}{dt} &= i\lambda,
\end{aligned}$$

$$\begin{aligned}\frac{d^2\phi}{dt^2} &= \exp(\lambda(e^{it} - 1)) ((\lambda ie^{it})^2 - \lambda e^{it}) , \\ \frac{d^2\phi(0)}{dt^2} &= -(\lambda^2 + \lambda) .\end{aligned}$$

Thus, the two lowest moments are

$$\begin{aligned}\mu &= \langle k \rangle = \lambda , \\ \mu_2 &= \langle k^2 \rangle = \lambda^2 + \lambda\end{aligned}$$

and the mean value and the standard deviation are given by

$$\begin{aligned}\langle k \rangle &= \lambda , \\ \sigma &= \sqrt{\lambda} .\end{aligned}$$

Expanding

$$K(t) = \ln \phi(t) = \lambda(e^{it} - 1) = \lambda[(it) + \frac{1}{2!}(it)^2 + \frac{1}{3!}(it)^3 + \dots] ,$$

we find for the cumulants the simple result

$$\kappa_1 = \kappa_2 = \kappa_3 = \dots = \lambda .$$

The calculation of the lower central moments is then trivial. For example, skewness and excess are simply given by

$$\gamma_1 = \kappa_3/\kappa_2^{3/2} = 1/\sqrt{\lambda} , \quad \gamma_2 = \kappa_4/\kappa_2^2 = 1/\lambda .$$

Example 20. Distribution of a sum of independent, Poisson distributed variates

We start from the distributions

$$\begin{aligned}P_1(k_1) &= P_{\lambda_1}(k_1), \\ P_2(k_2) &= P_{\lambda_2}(k_2)\end{aligned}$$

and calculate the probability distribution $P(k)$ for $k = k_1 + k_2$. When we write down the characteristic function for $P(k)$,

$$\begin{aligned}\phi(t) &= \phi_1(t)\phi_2(t) \\ &= \exp(\lambda_1(e^{it} - 1)) \exp(\lambda_2(e^{it} - 1)) \\ &= \exp((\lambda_1 + \lambda_2)(e^{it} - 1)) ,\end{aligned}$$

we observe that $\phi(t)$ is just the characteristic function of the Poisson distribution $P_{\lambda_1+\lambda_2}(k)$. The sum of two Poisson distributed variates is again Poisson distributed, the mean value being the sum of the mean values of the two original distributions. This property is sometimes called *stability*.

Example 21. Characteristic function and moments of the exponential distribution

For the p.d.f.

$$f(x) = \lambda e^{-\lambda x}$$

we obtain the characteristic function

$$\begin{aligned}\phi(t) &= \int_0^\infty e^{itx} \lambda e^{-\lambda x} dx \\ &= \frac{\lambda}{-\lambda + it} e^{(-\lambda + it)x} \Big|_0^\infty \\ &= \frac{\lambda}{\lambda - it}\end{aligned}$$

and deriving it with respect to t , we get from

$$\begin{aligned}\frac{d\phi(t)}{dt} &= \frac{i\lambda}{(\lambda - it)^2} , \\ \frac{d^n \phi(t)}{dt^n} &= \frac{n! i^n \lambda}{(\lambda - it)^{n+1}} , \\ \frac{d^n \phi(0)}{dt^n} &= \frac{n! i^n}{\lambda^n}\end{aligned}$$

the moments of the distribution:

$$\mu_n = n! \lambda^{-n} .$$

From these we obtain the mean value

$$\mu = 1/\lambda ,$$

the standard deviation

$$\sigma = \sqrt{\mu_2 - \mu^2} = 1/\lambda ,$$

and the skewness

$$\gamma_1 = (\mu_3 - 3\sigma^2\mu - \mu^3)/\sigma^3 = 2 .$$

Contrary to the Poisson example, here we do not gain in using the characteristic function, since the moments can be calculated directly:

$$\int_0^\infty x^n \lambda e^{-\lambda x} dx = n! \lambda^{-n} .$$

3.4 Transformation of Variables

In one of the examples of Sect. 3.2.7 we had calculated the expected value of the energy from the distribution of velocity. Of course, for certain applications it may be necessary to know not only the mean value of the energy but its complete distribution. To derive it, we have to perform a variable transformation.

For discrete distributions, this is a trivial exercise: The probability that the event “ u has the value $u(x_k)$ ” occurs, where u is an uniquely invertible function of x , is of course the same as for “ x has the value x_k ”:

$$P\{u = u(x_k)\} = P\{x = x_k\} .$$

For continuous distributions, the probability densities are transformed according to the usual rules as applied for example for mass or charge densities.

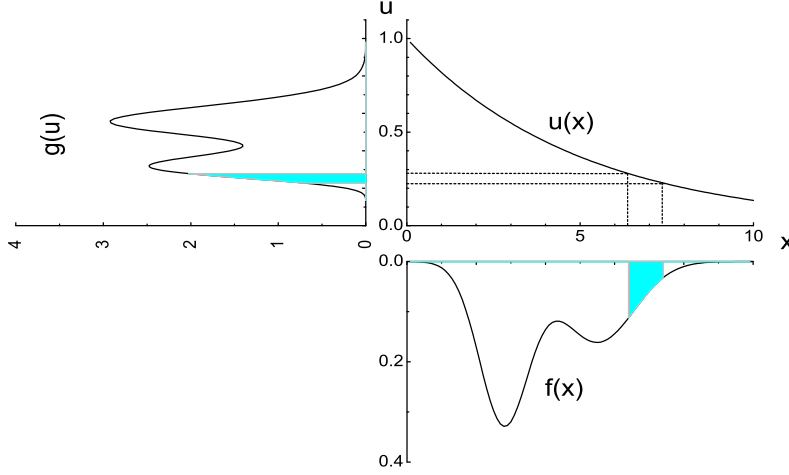


Fig. 3.8. Transformation of a probability density $f(x)$ into $g(u)$ via $u(x)$. The shaded areas are equal.

3.4.1 Calculation of the Transformed Density

We consider a probability density $f(x)$ and a monotone, i.e. uniquely invertible function $u(x)$. We are interested in the p.d.f. of u , $g(u)$ (Fig. 3.8).

The relation $P\{x_1 < x < x_2\} = P\{u_1 < u < u_2\}$ with $u_1 = u(x_1)$, $u_2 = u(x_2)$ has to hold, and therefore

$$\begin{aligned} P\{x_1 < x < x_2\} &= \int_{x_1}^{x_2} f(x') dx' \\ &= \int_{u_1}^{u_2} g(u') du'. \end{aligned}$$

This may be written in differential form as

$$\begin{aligned} |g(u)du| &= |f(x)dx|, \\ g(u) &= f(x) \left| \frac{dx}{du} \right|. \end{aligned} \quad (3.29)$$

Taking the absolute value guarantees the positivity of the probability density. Integrating (3.29), we find numerical equality of the cumulative distribution functions, $F(x) = G(u(x))$.

If $u(x)$ is not a monotone function, then, contrary to the above assumption, $x(u)$ is not a unique function (Fig. 3.9) and we have to sum over the contributions of the various branches of the inverse function:

$$g(u) = \left\{ f(x) \left| \frac{dx}{du} \right| \right\}_{branch1} + \left\{ f(x) \left| \frac{dx}{du} \right| \right\}_{branch2} + \dots. \quad (3.30)$$

Example 22. Calculation of the p.d.f. for the volume of a sphere from the p.d.f. of the radius

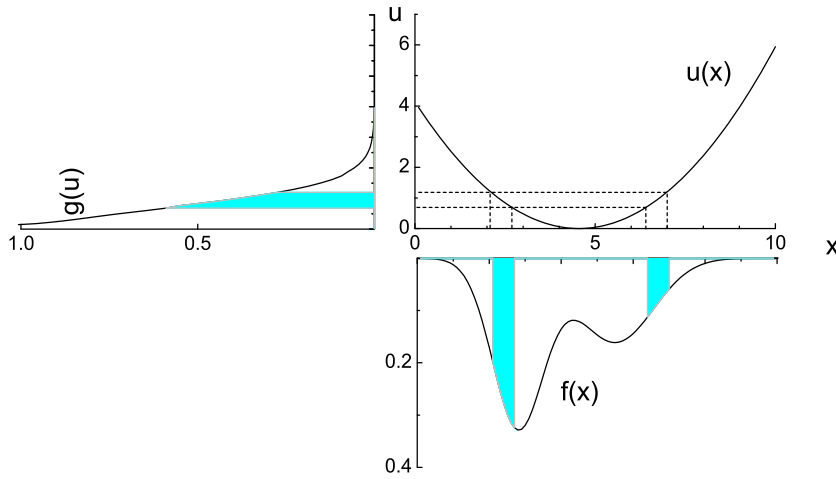


Fig. 3.9. Transformation of a p.d.f. $f(x)$ into $g(u)$ with $u = x^2$. The sum of the shaded areas below $f(x)$ is equal to the shaded area below $g(u)$.

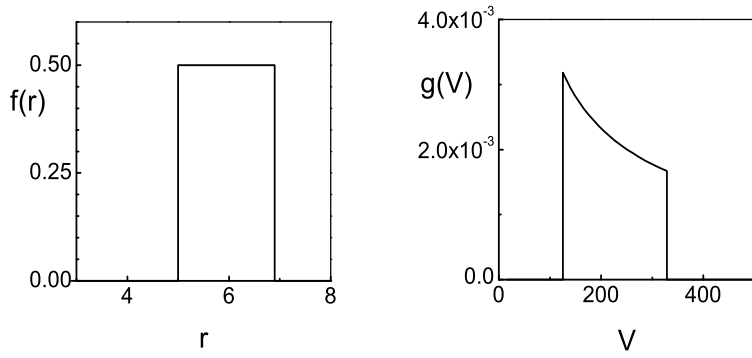


Fig. 3.10. Transformation of a uniform distribution of the radius into the distribution of the volume of a sphere.

Given a uniform distribution for the radius r

$$f(r) = \begin{cases} 1/(r_2 - r_1) & \text{if } r_1 < r < r_2 \\ 0 & \text{else} \end{cases}.$$

we ask for the distribution $g(V)$ of the volume $V(r)$. With

$$g(V) = f(r) \left| \frac{dr}{dV} \right|, \quad \frac{dV}{dr} = 4\pi r^2$$

we get

$$g(V) = \frac{1}{r_2 - r_1} \frac{1}{4\pi r^2} = \frac{1}{V_2^{1/3} - V_1^{1/3}} \frac{1}{3} V^{-2/3}.$$

Example 23. Distribution of the quadratic deviation

For a normal distributed variate x with mean value x_0 and variance s^2 we ask for the distribution $g(u)$, where

$$u = (x - x_0)^2 / s^2$$

is the normalized quadratic deviation. The expected value of u is unity, since the expected value of $(x - \mu)^2$ per definition equals σ^2 for any distribution. The function $x(u)$ has two branches. With

$$f(x) = \frac{1}{s\sqrt{2\pi}} e^{-(x-x_0)^2/(2s^2)}$$

and

$$\frac{dx}{du} = \frac{s}{2\sqrt{u}}$$

we find

$$g(u) = \left\{ \frac{1}{2\sqrt{2\pi u}} e^{-u/2} \right\}_{branch1} + \left\{ \dots \right\}_{branch2}.$$

The contributions from both branches are the same, thus

$$g(u) = \frac{1}{\sqrt{2\pi u}} e^{-u/2}. \quad (3.31)$$

The function $g(u)$ is the so-called χ^2 - *distribution* (chi-square distribution) for one degree of freedom.

Example 24. Distribution of kinetic energy in the one-dimensional ideal gas

Be v the velocity of a particle in x direction with probability density

$$f(v) = \sqrt{\frac{m}{2\pi kT}} e^{-mv^2/(2kT)}.$$

Its kinetic energy is $E = v^2/(2m)$, for which we want to know the distribution $g(E)$. The function $v(E)$ has again two branches. We get, in complete analogy to the example above,

$$\frac{dv}{dE} = \frac{1}{\sqrt{2mE}},$$

$$g(E) = \left\{ \frac{1}{2\sqrt{\pi kTE}} e^{-E/kT} \right\}_{branch1} + \left\{ \dots \right\}_{branch2}.$$

The contributions of both branches are the same, hence

$$g(E) = \frac{1}{\sqrt{\pi kTE}} e^{-E/kT}.$$

3.4.2 Determination of the Transformation Relating two Distributions

In the computer simulation of stochastic processes we are frequently confronted with the problem that we have to transform the uniform distribution of a random number generator into a desired distribution, e.g. a normal or exponential distribution.

More generally, we want to obtain for two given distributions $f(x)$ and $g(u)$ the transformation $u(x)$ connecting them.

We have

$$\int_{-\infty}^x f(x') dx' = \int_{-\infty}^u g(u') du' .$$

Integrating, we get $F(x)$ and $G(u)$:

$$F(x) = G(u) ,$$

$$u(x) = G^{-1}(F(x)) .$$

G^{-1} is the inverse function of G . The problem can be solved analytically, only if f and g can be integrated analytically and if the inverse function of G can be derived.

Let us consider now the special case mentioned above, where the primary distribution $f(x)$ is uniform, $f(x) = 1$ for $0 \leq x \leq 1$. This implies $F(x) = x$ and

$$\begin{aligned} G(u) &= x, \\ u &= G^{-1}(x) . \end{aligned} \tag{3.32}$$

Example 25. Generation of an exponential distribution starting from a uniform distribution

Given are the p.d.f.s

$$\begin{aligned} f(x) &= \begin{cases} 1 & \text{for } 0 < x < 1 \\ 0 & \text{else} , \end{cases} \\ g(u) &= \begin{cases} \lambda e^{-\lambda u} & \text{for } 0 < u \\ 0 & \text{else} . \end{cases} \end{aligned}$$

The desired transformation $u(x)$, as demonstrated above in the general case, is obtained by integration and inversion:

$$\begin{aligned} \int_0^u g(u') du' &= \int_0^x f(x') dx' , \\ \int_0^u \lambda e^{-\lambda u'} du' &= \int_0^x f(x') dx' , \\ 1 - e^{-\lambda u} &= x , \\ u &= -\ln(1 - x)/\lambda . \end{aligned}$$

We could have used, of course, also the relation (3.32) directly. Obviously in the last relation we could substitute $1 - x$ by x , since both quantities are uniformly distributed.

When we transform the uniformly distributed random numbers x delivered by our computer according to the last relation into the variable u , the latter will be exponentially distributed. This is the usual way to simulate the lifetime distribution of instable particles and other decay processes (see Chap. 5).

3.5 Multivariate Probability Densities

The results of the last sections are easily extended to multivariate distributions. We restrict ourself here to the case of continuous distributions⁸.

⁸An example of a multivariate discrete distribution will be presented in Sect. 3.6.2.

3.5.1 Probability Density of two Variables

Definitions

As in the one-dimensional case we define an integral distribution function $F(x, y)$, now taken to be the probability to find values of the variates x' , y' smaller than x , respectively y

$$F(x, y) = P\{(x' < x) \cap (y' < y)\} . \quad (3.33)$$

The following properties of this distribution function are satisfied:

$$\begin{aligned} F(\infty, \infty) &= 1, \\ F(-\infty, y) &= F(x, -\infty) = 0 . \end{aligned}$$

In addition, F has to be a monotone increasing function of both variables. We define a two-dimensional probability density, the so-called joined probability density, as the partial derivation of f with respect to the variables x , y :

$$f(x, y) = \frac{\partial^2 F}{\partial x \partial y} .$$

From these definitions follows the normalization condition

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \, dx \, dy = 1 .$$

The projections $f_x(x)$ respectively $f_y(y)$ of the joined probability density onto the coordinate axes are called marginal distributions :

$$\begin{aligned} f_x(x) &= \int_{-\infty}^{\infty} f(x, y) \, dy , \\ f_y(y) &= \int_{-\infty}^{\infty} f(x, y) \, dx . \end{aligned}$$

The marginal distributions are one-dimensional (univariate) probability densities.

The conditional probability densities for fixed values of the second variate and normalized with respect to the first one are denoted by $f_x(x|y)$ and $f_y(y|x)$ for given values of y or x , respectively. We have the following relations:

$$\begin{aligned} f_x(x|y) &= \frac{f(x, y)}{\int_{-\infty}^{\infty} f(x, y) \, dx} \\ &= \frac{f(x, y)}{f_y(y)} , \end{aligned} \quad (3.34)$$

$$\begin{aligned} f_y(y|x) &= \frac{f(x, y)}{\int_{-\infty}^{\infty} f(x, y) \, dy} \\ &= \frac{f(x, y)}{f_x(x)} . \end{aligned} \quad (3.35)$$

Together, (3.34) and (3.35) express again Bayes' theorem:

$$f_x(x|y)f_y(y) = f_y(y|x)f_x(x) = f(x, y) . \quad (3.36)$$

Example 26. Superposition of two two-dimensional normal distributions⁹

The marginal distributions $f_x(x)$, $f_y(y)$ and the conditional p.d.f.

$$f_y(y|x=1)$$

for the joined two-dimensional p.d.f.

$$f(x, y) = \frac{1}{2\pi} \left[0.6 \exp \left(-\frac{x^2}{2} - \frac{y^2}{2} \right) + \frac{0.4}{\sqrt{3}} \exp \left(-\frac{(x-2)^2}{3} - \frac{(y-2.5)^2}{4} \right) \right]$$

are

$$\begin{aligned} f_x(x) &= \frac{1}{\sqrt{2\pi}} \left[0.6 \exp \left(-\frac{x^2}{2} \right) + \frac{0.4}{\sqrt{1.5}} \exp \left(-\frac{(x-2)^2}{3} \right) \right], \\ f_y(y) &= \frac{1}{\sqrt{2\pi}} \left[0.6 \exp \left(-\frac{y^2}{2} \right) + \frac{0.4}{\sqrt{2}} \exp \left(-\frac{(y-2.5)^2}{4} \right) \right], \\ f(y, x=1) &= \frac{1}{2\pi} \left[0.6 \exp \left(-\frac{1}{2} - \frac{y^2}{2} \right) + \frac{0.4}{\sqrt{3}} \exp \left(-\frac{1}{3} - \frac{(y-2.5)^2}{4} \right) \right], \\ f_y(y|x=1) &= 0.667 \left[0.6 \exp \left(-\frac{1}{2} - \frac{y^2}{2} \right) + \frac{0.4}{\sqrt{3}} \exp \left(-\frac{1}{3} - \frac{(y-2.5)^2}{4} \right) \right]. \end{aligned}$$

$f_y(y|1)$ and $f(y, 1)$ differ in the normalization factor, which results from the requirement $\int f_y(y|1) dy = 1$.

Graphical Presentation

Fig. 3.11 shows a similar superposition of two Gaussians together with its marginal distributions and one conditional distribution. The chosen form of the graphical representation as a contour plot for two-dimensional distributions is usually to be favored over three-dimensional surface plots.

3.5.2 Moments

Analogously to the one-dimensional case we define moments of two-dimensional distributions:

$$\begin{aligned} \mu_x &= E(x), \\ \mu_y &= E(y), \\ \sigma_x^2 &= E[(x - \mu_x)^2], \\ \sigma_y^2 &= E[(y - \mu_y)^2], \\ \sigma_{xy} &= E[(x - \mu_x)(y - \mu_y)], \\ \mu_{lm} &= E(x^l y^m), \\ \mu'_{lm} &= E[(x - \mu_x)^l (y - \mu_y)^m]. \end{aligned}$$

Explicitly,

⁹The two-dimensional normal distribution will be discussed in Sect. 3.6.5.

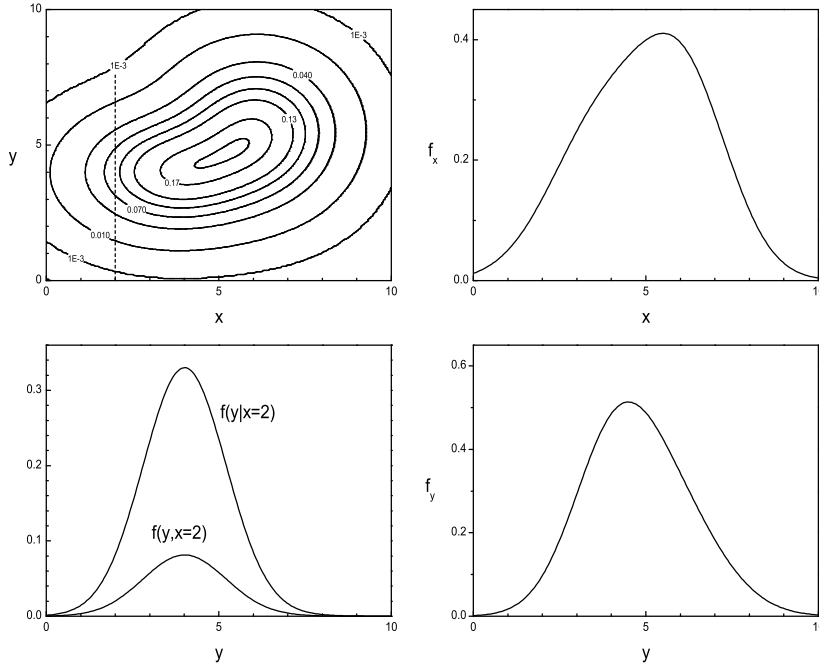


Fig. 3.11. Two-dimensional probability density. The lower left-hand plot shows the conditional p.d.f. of y for $x = 2$. The lower curve is the p.d.f. $f(y, 2)$. It corresponds to the dashed line in the upper plot. The right-hand side displays the marginal distributions.

$$\mu_x = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f(x, y) dx dy = \int_{-\infty}^{\infty} x f_x(x) dx ,$$

$$\mu_y = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dx dy = \int_{-\infty}^{\infty} y f_y(y) dy ,$$

$$\mu_{lm} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^l y^m f(x, y) dx dy ,$$

$$\mu'_{lm} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_x)^l (y - \mu_y)^m f(x, y) dx dy .$$

Obviously, μ'_x, μ'_y ($= \mu'_{10}, \mu'_{01}$) are zero.

Correlations, Covariance, Independence

The mixed moment σ_{xy} is called *covariance* of x and y , and sometimes also denoted as $\text{cov}(x, y)$. If σ_{xy} is different from zero, the variables x and y are said to be *correlated*. The mean value of y depends on the value chosen for x and vice versa. Thus, for instance, the weight of a man is positively correlated with its height.

The degree of correlation is quantified by the dimensionless quantity

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y} ,$$

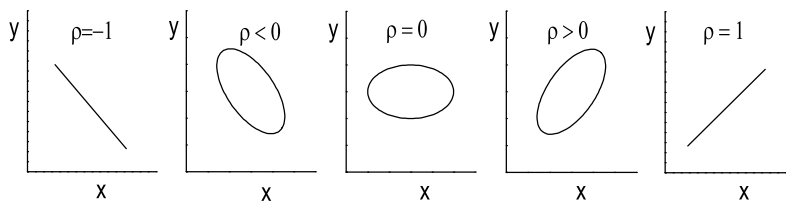


Fig. 3.12. Curves $f(x, y) = \text{const.}$ with different correlation coefficients.

the *correlation coefficient*. Schwarz' inequality insures $|\rho_{xy}| \leq 1$.

Figure 3.12 shows lines of constant probability for various kinds of correlated distributions. In the extreme case $|\rho| = 1$ the variates are linearly related.

If the correlation coefficient is zero, this does not necessarily mean statistical *independence* of the variates. The dependence may be more subtle, as we will see shortly. As defined in Chap. 2, two random variables x, y are called *independent* or *orthogonal*, if the probability to observe one of the two variates x, y is independent from the value of the other one, i.e. the conditional distributions are equal to the marginal distributions, $f_x(x|y) = f_x(x)$, $f_y(y|x) = f_y(y)$. Independence is realized only if the joined distribution $f(x, y)$ factorizes into its marginal distributions (see Chap. 2):

$$f(x, y) = f_x(x)f_y(y) .$$

Clearly, correlated variates cannot be independent.

Example 27. Correlated variates

A measurement uncertainty of a point in the xy -plane follows independent normal distributions in the polar coordinates r, φ (the errors are assumed small enough to neglect the regions $r < 0$ and $|\varphi| > \pi$). A line of constant probability in the xy -plane would look similar to the second graph of Fig. 3.12. The cartesian coordinates are negatively correlated, although the original polar coordinates have been chosen as uncorrelated, in fact they are even independent.

Example 28. Dependent variates with correlation coefficient zero

For the probability density

$$f(x, y) = \frac{1}{2\pi\sqrt{x^2 + y^2}} e^{-\sqrt{x^2 + y^2}}$$

we find $\sigma_{xy} = 0$. The curves $f = \text{const.}$ are circles, but x and y are *not* independent, the conditional distribution $f_y(y|x)$ of y depends on x .

3.5.3 Transformation of Variables

The probability densities $f(x, y)$ and $g(u, v)$ are transformed via the transformation functions $u(x, y)$, $v(x, y)$, analogously to the univariate case

$$g(u, v) \, du \, dv = f(x, y) \, dx \, dy ,$$

$$g(u, v) = f(x, y) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| ,$$

with the Jacobian determinant replacing the differential quotient dx/du .

Example 29. Transformation of a normal distribution from cartesian into polar coordinates

A two-dimensional normal distribution

$$f(x, y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2}$$

is to be transformed into polar coordinates

$$\begin{aligned} x &= r \cos \varphi , \\ y &= r \sin \varphi . \end{aligned}$$

The Jacobian is

$$\frac{\partial(x, y)}{\partial(r, \varphi)} = r .$$

We get

$$g(r, \varphi) = \frac{1}{2\pi} r e^{-r^2/2}$$

with the marginal distributions

$$\begin{aligned} g_r &= \int_0^{2\pi} g(r, \varphi) \, d\varphi = r e^{-r^2/2} , \\ g_\varphi &= \int_0^\infty g(r, \varphi) \, dr = \frac{1}{2\pi} . \end{aligned}$$

The joined distribution factorizes into its marginal distributions (Fig. 3.13). Not only x, y , but also r, φ are independent.

3.5.4 Reduction of the Number of Variables

Frequently, we are faced with the problem to find from a given joined distribution $f(x, y)$ the distribution $g(u)$ of a dependent random variable $u(x, y)$. We can reduce it to that of a usual transformation, by inventing a second variable $v = v(x, y)$, performing the transformation $f(x, y) \rightarrow h(u, v)$ and, finally, by calculating the marginal distribution in u ,

$$g(u) = \int_{-\infty}^{\infty} h(u, v) \, dv .$$

In most cases, the choice $v = x$ is suitable. More formally, we might use the equivalent reduction formula

$$g(u) = \int_{-\infty}^{\infty} f(x, y) \delta(u - u(x, y)) \, dx \, dy . \quad (3.37)$$

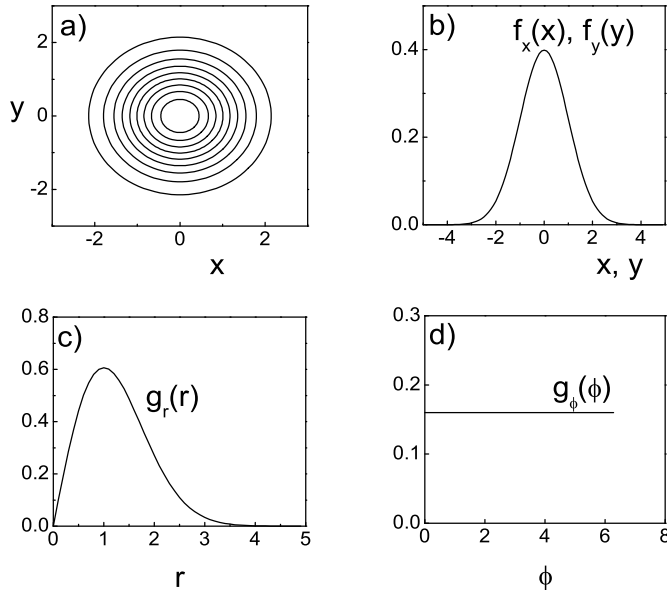


Fig. 3.13. Transformation of a two-dimensional normal distribution of cartesian coordinates into the distribution of polar coordinates: (a) lines of constant probability; b) cartesian marginal distributions; c), d) marginal distributions of the polar coordinates.

For the distribution of a sum $u = x + y$ of two independent variates x, y , i.e. $f(x, y) = f_x(x)f_y(y)$, after integration over y follows

$$g(u) = \int f(x, u - x) dx = \int f_x(x)f_y(u - x) dx .$$

This is called the convolution integral or convolution product of f_x and f_y .

Example 30. Distribution of the difference of two digitally measured times

The true times t_1, t_2 are taken to follow a uniform distribution

$$f(t_1, t_2) = \begin{cases} 1/\Delta^2 & \text{for } |t_1 - T_1|, |t_2 - T_2| < \Delta/2 \\ 0 & \text{else} \end{cases}$$

around the readings T_1, T_2 . We are interested in the probability density of the difference $t = t_1 - t_2$. To simplify the notation, we choose the case $T_1 = T_2 = 0$ and $\Delta = 2$ (Fig. 3.14). First we transform the variables according to

$$\begin{aligned} t &= t_1 - t_2 , \\ t_1 &= t_1 \end{aligned}$$

with the Jacobian

$$\frac{\partial(t_1, t_2)}{\partial(t_1, t)} = 1 .$$

The new distribution is also uniform:

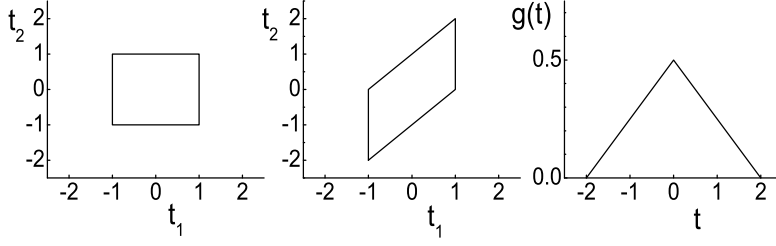


Fig. 3.14. Distribution of the difference t between two times t_1 and t_2 which have both clock readings equal to zero.

$$h(t_1, t) = f(t_1, t_2) = 1/\Delta^2 ,$$

and has the boundaries shown in Fig. 3.14. The form of the marginal distribution is found by integration over t_1 , or directly by reading it off from the figure:

$$g(t) = \begin{cases} (t - T + \Delta)/\Delta^2 & \text{for } t - T < 0 \\ (-t + T + \Delta)/\Delta^2 & \text{for } 0 < t - T \\ 0 & \text{else .} \end{cases}$$

where $T = T_1 - T_2$ now for arbitrary values of T_1 and T_2 .

Example 31. Distribution of the transverse momentum squared of particle tracks

The projections of the momenta are assumed to be independently normally distributed,

$$f(p_x, p_y) = \frac{1}{2\pi s^2} e^{-(p_x^2 + p_y^2)/(2s^2)} ,$$

with equal variances $\langle p_x^2 \rangle = \langle p_y^2 \rangle = s^2$. For the transverse momentum squared we set $q = p^2$ and calculate its distribution. We transform the distributions into polar coordinates

$$\begin{aligned} p_x &= \sqrt{q} \cos \varphi , \\ p_y &= \sqrt{q} \sin \varphi \end{aligned}$$

with

$$\frac{\partial(p_x, p_y)}{\partial(q, \varphi)} = \frac{1}{2}$$

and obtain

$$h(q, \varphi) = \frac{1}{4\pi s^2} e^{-q/(2s^2)}$$

with the marginal distribution

$$\begin{aligned} h_q(q) &= \int_0^{2\pi} \frac{1}{4\pi s^2} e^{-q/(2s^2)} d\varphi \\ &= \frac{1}{2s^2} e^{-q/(2s^2)} , \\ g(p^2) &= \frac{1}{\langle p^2 \rangle} e^{-p^2/\langle p^2 \rangle} . \end{aligned}$$

The result is an exponential distribution in p^2 with mean $\langle p^2 \rangle = \langle p_x^2 \rangle + \langle p_y^2 \rangle$.

Example 32. Quotient of two normally distributed variates

For variates x, y , independently and identically normally distributed, i.e.

$$f(x, y) = f(x)f(y) = \frac{1}{2\pi} \exp\left(-\frac{x^2 + y^2}{2}\right),$$

we want to find the distribution $g(u)$ of the quotient $u = y/x$. Again, we transform first into new variates $u = y/x$, $v = x$, or, inverted, $x = v$, $y = uv$ and get

$$h(u, v) = f(x(u, v), y(u, v)) \frac{\partial(x, y)}{\partial(u, v)},$$

with the Jacobian

$$\frac{\partial(x, y)}{\partial(u, v)} = -v,$$

hence

$$\begin{aligned} g(u) &= \int h(u, v) dv \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp\left(-\frac{v^2 + u^2 v^2}{2}\right) |v| dv \\ &= \frac{1}{\pi} \int_0^{\infty} e^{-(1+u^2)z} dz \\ &= \frac{1}{\pi} \frac{1}{1+u^2}, \end{aligned}$$

where the substitution $z = v^2/2$ has been used. The result $g(u)$ is the Cauchy distribution (see Sect. 3.6.9). Its long tails are caused here by the finite probability of arbitrary small values in the denominator. This effect is quite important in experimental situations when we estimate the uncertainty of quantities which are the quotients of normally distributed variates in cases, where the p.d.f. in the denominator is not negligible at the value zero.

The few examples given above should not lead to the impression that transformations of variates always yield more or less simple analytical expressions for the resulting distributions. This is not the rule, but rather the exception. However, as we will learn in Chap. 5, a simple, straight forward numerical solution is provided by Monte Carlo methods.

3.5.5 Determination of the Transformation between two Distributions

As in the one-dimensional case, for the purpose of simulation, we frequently need to generate some required distribution from the uniformly distributed random numbers delivered by the computer. The general method of integration and inversion of the cumulative distribution can be used directly, only if we deal with independent variates. Often, a transformation of the variates is helpful. We consider here a special example, which we need later in Chap. 5.

Example 33. Generation of a two-dimensional normal distribution starting from uniform distributions

We use the result from example 29 and start with the representation of the two-dimensional Gaussian in polar coordinates

$$g(\rho, \varphi) d\rho d\varphi = \frac{1}{2\pi} d\varphi \rho e^{-\rho^2/2} d\rho ,$$

which factorizes in φ and ρ . With two in the interval $[0, 1]$ uniformly distributed variates r_1, r_2 , we obtain the function $\rho(r_1)$:

$$\begin{aligned} \int_0^\rho \rho' e^{-\rho'^2/2} d\rho' &= r_1 , \\ -e^{-\rho'^2/2} \Big|_0^\rho &= r_1 , \\ 1 - e^{-\rho^2/2} &= r_1 , \\ \rho &= \sqrt{-2 \ln(1 - r_1)} . \end{aligned}$$

In the same way we get $\varphi(r_2)$:

$$\varphi = 2\pi r_2 .$$

Finally we find x and y :

$$x = \rho \cos \varphi = \sqrt{-2 \ln(1 - r_1)} \cos(2\pi r_2) , \quad (3.38)$$

$$y = \rho \sin \varphi = \sqrt{-2 \ln(1 - r_1)} \sin(2\pi r_2) . \quad (3.39)$$

These variables are independent and distributed normally about the origin with variance unity:

$$f(x, y) = \frac{1}{2\pi} e^{-(x^2+y^2)/2} .$$

(We could replace $1 - r_1$ by r_1 , since $1 - r_1$ is uniformly distributed as well.)

3.5.6 Distributions of more than two Variables

It is not difficult to generalize the relations just derived for two variables to multivariate distributions, of N variables. We define the distribution function $F(x_1, \dots, x_N)$ as the probability to find values of the variates smaller than $x_1, \dots, x_N$,

$$F(x_1, \dots, x_N) = P\{(x'_1 < x) \cap \dots \cap (x'_N < x_N)\} ,$$

and the p.d.f.

$$f(x_1, \dots, x_N) = \frac{\partial^N F}{\partial x_1, \dots, \partial x_N} .$$

Often it is convenient to use the vector notation, $F(\mathbf{x})$, $f(\mathbf{x})$ with

$$\mathbf{x} = \{x_1, x_2, \dots, x_N\} .$$

These variate vectors can be represented as points in an N -dimensional space.

The p.d.f. $f(\mathbf{x})$ can also be defined directly, without reference to the distribution function $F(\mathbf{x})$, as the density of points at the location $\mathbf{x}$, by setting

$$\begin{aligned} f(x_1, \dots, x_N) dx_1 \cdots dx_N &= dP\{(x_1 - \frac{dx_1}{2} \leq x'_1 \leq x_1 + \frac{dx_1}{2}) \cap \dots \\ &\quad \cdots \cap (x_N - \frac{dx_N}{2} \leq x'_N \leq x_N + \frac{dx_N}{2})\} \end{aligned}$$

Expected Values and Correlation Matrix

The expected value of a function $u(\mathbf{x})$ is

$$E(u) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} u(\mathbf{x}) f(\mathbf{x}) \prod_{i=1}^N dx_i .$$

Because of the additivity of expected values this relation also holds for vector functions $\mathbf{u}(\mathbf{x})$.

The dispersion of multivariate distributions is now described by the so-called covariance matrix $\mathbf{C}$:

$$C_{ij} = \langle (x_i - \langle x_i \rangle)(x_j - \langle x_j \rangle) \rangle = \langle x_i x_j \rangle - \langle x_i \rangle \langle x_j \rangle .$$

The correlation matrix is given by

$$\rho_{ij} = \frac{C_{ij}}{\sqrt{C_{ii}C_{jj}}} .$$

Transformation of Variables

We multiply with the absolute value of the N -dimensional Jacobian

$$g(\mathbf{y}) = f(\mathbf{x}) \left| \frac{\partial(x_1, \dots, x_N)}{\partial(y_1, \dots, y_N)} \right| .$$

Correlation and Independence

As in the two-dimensional case two variables x_i, x_j are called uncorrelated if their correlation coefficient ρ_{ij} is equal to zero. The two variates x_i, x_j are independent if the conditional p.d.f. of x_i conditioned on all other variates does not depend on x_j . The combined density f then has to factorize into two factors where one of them is independent of x_i and the other one is independent of x_j ¹⁰. All variates are independent of each other, if

$$f(x_1, x_2, \dots, x_N) = \prod_{i=1}^N f_{x_i}(x_i) .$$

3.5.7 Independent, Identically Distributed Variables

One of the main topics of statistics is the estimation of free parameters of a distribution from a random sample of observations all drawn from the same population. For example, we might want to estimate the mean lifetime τ of a particle from N independent measurements t_i where t follows an exponential distribution depending on τ . The probability density $\tilde{f}$ for N *independent* and *identically distributed* variates (abbreviated as i.i.d. variates) x_i , each distributed according to $f(x)$, is, according to the definition of independence,

$$\tilde{f}(x_1, \dots, x_N) = \prod_{i=1}^N f(x_i) .$$

The covariance matrix of i.i.d. variables is diagonal, with $C_{ii} = \text{var}(x_i) = \text{var}(x_1)$.

¹⁰we omit the formulas because they are very clumsy.

3.5.8 Angular Distributions

In physics applications we are often interested in spatial distributions. Fortunately our problems often exhibit certain symmetries which facilitate the description of the phenomena. Depending on the kind of symmetry of the physical process or the detector, we choose appropriate coordinates, spherical, cylindrical or polar. These coordinates are especially well suited to describe processes where radiation is emitted by a local source or where the detector has a spherical or cylindrical symmetry. Then the distance, i.e. the radius vector, is not the most interesting parameter and we often describe the process solely by angular distributions. In other situations, only directions enter, for example in particle scattering, when we investigate the polarization of light crossing an optically active medium, or of a particle decaying in flight into a pair of secondaries where the orientation of the normal of the decay plane contains relevant information. Similarly, distributions of physical parameters on the surface of the earth are expressed as functions of the angular coordinates.

Distribution of the Polar Angle

As already explained above, the expressions

$$\begin{aligned}x &= r \cos \varphi, \\y &= r \sin \varphi\end{aligned}$$

relate the polar coordinates r, φ to the cartesian coordinates x, y . Since we have periodic functions, we restrict the angle φ to the interval $[-\pi, \pi]$. This choice is arbitrary to a certain extent.

For an isotropic distribution all angles are equally likely and we obtain the uniform distribution of φ

$$g(\varphi) = \frac{1}{2\pi}.$$

Since we have to deal with periodic functions, we have to be careful when we compute moments and in general expected values. For example the mean of the two angles $\varphi_1 = \pi/2$, $\varphi_2 = -\pi$ is not $(\varphi_1 + \varphi_2)/2 = -\pi/4$, but $3\pi/4$. To avoid this kind of mistake it is advisable to go back to the unit vectors $\{x_i, y_i\} = \{\cos \varphi_i, \sin \varphi_i\}$, to average those and to extract the resulting angle.

Example 34. The v. Mises distribution

We consider the Brownian motion of a particle on the surface of a liquid. Starting from a point $\mathbf{r}_0$ its position $\mathbf{r}$ after some time will be given by the expression

$$f(\mathbf{r}) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{|\mathbf{r} - \mathbf{r}_0|^2}{2\sigma^2}\right).$$

Taking into account the Jacobian $\partial(x, y)/\partial(r, \varphi) = r$, the distribution in polar coordinates is:

$$g(r, \varphi) = \frac{r}{2\pi\sigma^2} \exp\left(-\frac{r^2 + r_0^2 - 2rr_0 \cos \varphi}{2\sigma^2}\right).$$

For convenience we have chosen the origin of φ such that $\varphi_0 = 0$. For fixed r we obtain the conditional distribution

$$\tilde{g}(\varphi) = g(\varphi|r) = c_N(\kappa) \exp(\kappa \cos \varphi)$$

with $\kappa = rr_0/\sigma^2$ and $c_N(\kappa)$ the normalization constant. This is the v. Mises distribution. It is symmetric in φ , unimodal with its maximum at $\varphi = 0$. The normalization

$$c_N(\kappa) = \frac{1}{2\pi I_0(\kappa)}$$

contains I_0 , the modified Bessel function of order zero [22].

For large values of κ the distribution approaches a Gaussian with variance $1/\kappa$. To demonstrate this feature, we rewrite the distribution in a slightly modified way,

$$\tilde{g}(\varphi) = c_N(\kappa) e^{\kappa} e^{[-\kappa(1-\cos \varphi)]} ,$$

and make use of the asymptotic form $\lim_{x \rightarrow \infty} I_0(x) \sim e^x / \sqrt{2\pi x}$ (see [22]). The exponential function is suppressed for large values of $(1 - \cos \varphi)$, and small values can be approximated by $\varphi^2/2$. Thus the asymptotic form of the distribution is

$$\tilde{g} = \sqrt{\frac{\kappa}{2\pi}} e^{-\kappa \varphi^2/2} . \quad (3.40)$$

In the limit $\kappa = 0$, which is the case for $r_0 = 0$ or $\sigma \rightarrow \infty$, the distribution becomes uniform, as it should.

Distribution of Spherical Angles

Spatial directions are described by the polar angle θ and the azimuthal angle φ which we define through the transformation relations from the cartesian coordinates:

$$\begin{aligned} x &= r \sin \theta \cos \varphi , & -\pi \leq \varphi \leq \pi \\ y &= r \sin \theta \sin \varphi , & 0 \leq \theta \leq \pi \\ z &= r \cos \theta . \end{aligned}$$

The Jacobian is $\partial(x, y, z)/\partial(r, \theta, \varphi) = r^2 \sin \theta$. A uniform distribution inside a sphere of radius R in cartesian coordinates

$$f_u(x, y, z) = \begin{cases} 3/(4\pi R^3) & \text{if } x^2 + y^2 + z^2 \leq R^2 , \\ 0 & \text{else} \end{cases}$$

thus transforms into

$$h_u(r, \theta, \varphi) = \frac{3r^2}{4\pi R^3} \sin \theta \quad \text{if } r \leq R .$$

We obtain the isotropic angular distribution by marginalizing or conditioning on r :

$$h_u(\theta, \varphi) = \frac{1}{4\pi} \sin \theta . \quad (3.41)$$

Spatial distributions are usually expressed in the coordinates $\tilde{z} = \cos \theta$ and φ , because then the uniform distribution simplifies further to

$$g_u(\tilde{z}, \varphi) = \frac{1}{4\pi}$$

with $|\tilde{z}| \leq 1$.

The p.d.f. $g(\tilde{z}, \varphi)$ of an arbitrary distribution of $\tilde{z}, \varphi$ is defined in the standard way through the probability $d^2P = g(\tilde{z}, \varphi)d\tilde{z}d\varphi$. The product $d\tilde{z}d\varphi = \sin\theta d\theta d\varphi = d^2\Omega$ is called solid angle element and corresponds to an infinitesimal area at the surface of the unit sphere. A solid angle Ω defines a certain area at this surface and contains all directions pointing into this area.

Example 35. Fisher's spherical distribution

Instead of the uniform distribution considered in the previous example we now investigate the angular distribution generated by a three-dimensional rotationally symmetric Gaussian distribution with variances $\sigma^2 = \sigma_x^2 = \sigma_y^2 = \sigma_z^2$. We put the center of the Gaussian at the z-axis, $\mathbf{r}_0 = \{0, 0, 1\}$. In spherical coordinates we then obtain the p.d.f.

$$f(r, \theta, \varphi) = \frac{1}{(2\pi)^{3/2}\sigma^3} r^2 \sin\theta \exp\left(-\frac{r^2 + r_0^2 - 2rr_0 \cos\theta}{2\sigma^2}\right).$$

For fixed distance r we obtain a function of θ and φ only which for our choice of $\mathbf{r}_0$ is also independent of φ :

$$g(\theta, \varphi) = c_N(\kappa) \sin\theta \exp(\kappa \cos\theta).$$

The parameter κ is again given by $\kappa = rr_0/\sigma^2$. Applying the normalization condition $\int g d\theta d\varphi = 1$ we find $c_N(\kappa) = \kappa/(4\pi \sinh \kappa)$ and

$$g(\theta, \varphi) = \frac{\kappa}{4\pi \sinh \kappa} e^{\kappa \cos\theta} \sin\theta \quad (3.42)$$

a two-dimensional, unimodal distribution, known as Fisher's spherical distribution. As in the previous example we get in the limit $\kappa \rightarrow 0$ the uniform distribution (3.41) and for large κ the asymptotic distribution

$$\tilde{g}(\theta, \varphi) \approx \frac{1}{4\pi} \kappa \theta e^{-\kappa\theta^2/2},$$

which is an exponential distribution of θ^2 . As a function of $\tilde{z} = \cos\theta$ the distribution (3.42) simplifies to

$$h(\tilde{z}, \varphi) = \frac{\kappa}{4\pi \sinh \kappa} e^{\kappa \tilde{z}}.$$

which illustrates the spatial shape of the distribution much better than (3.42).

3.6 Some Important Distributions

3.6.1 The Binomial Distribution

What is the probability to get with ten dice just two times a *six*? The answer is given by the binomial distribution:

$$B_{1/6}^{10}(2) = \binom{10}{2} \left(\frac{1}{6}\right)^2 \left(1 - \frac{1}{6}\right)^8.$$

The probability to get with 2 particular dice *six*, and with the remaining 8 dice *not* the number *six*, is given by the product of the two power factors. The binomial coefficient

$$\binom{10}{2} = \frac{10!}{2!8!}$$

counts the number of possibilities to distribute the 2 *sixes* over the 10 dice. This are just 45. With the above formula we obtain a probability of about 0.29.

Considering, more generally, n randomly chosen objects (or a sequence of n independent trials), which have with probability p the property A , which we will call *success*, the probability to find k out of these n objects with property A is $B_p^n(k)$,

$$B_p^n(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad , \quad k = 0, \dots, n \quad .$$

Since this is just the term of order p^k in the power expansion of $[p + (1-p)]^n$, we have the normalization condition

$$\begin{aligned} [p + (1-p)]^n &= 1 \quad , \\ \sum_{k=0}^n B_p^n(k) &= 1 \quad . \end{aligned} \tag{3.43}$$

Since the mean number of successes in one trial is given by p , we obtain, following the rules for expected values, for n independent trials

$$E(k) = np \quad .$$

With a similar argument we can find the variance: For $n = 1$, we can directly compute the expected quadratic difference, i.e. the variance σ_1^2 . Using $\langle k \rangle = p$ and that $k = 1$ is found with probability¹¹ $P\{1\} = p$ and $k = 0$ with $P\{0\} = 1 - p$, we find:

$$\begin{aligned} \sigma_1^2 &= \langle (k - \langle k \rangle)^2 \rangle \\ &= p(1-p)^2 + (1-p)(0-p)^2 \\ &= p(1-p) \quad . \end{aligned}$$

According to (3.12) the variance of the sum of n i.i.d. random numbers is

$$\sigma^2 = n\sigma_1^2 = np(1-p) \quad .$$

The characteristic function has the form:

$$\phi(t) = [1 + p(e^{it} - 1)]^n \quad . \tag{3.44}$$

It is easily derived by substituting in the expansion of (3.43) in the k_{th} term p^k with $(p e^{it})^k$. From (3.25) follows the property of stability, which is also convincing intuitively:

The distribution of a sum of numbers $k = k_1 + \dots + k_N$ obeying binomial distributions $B_p^{n_i}(k_i)$, is again a binomial distribution $B_p^n(k)$ with $n = n_1 + \dots + n_N$.

¹¹For $n = 1$ the binomial distribution is also called two-point or Bernoulli distribution.

There is no particularly simple expression for higher moments; they can of course be calculated from the Taylor expansion of $\phi(t)$, as explained in Sect. 3.3.2. We give only the results for the coefficients of skewness and excess:

$$\gamma_1 = \frac{1-2p}{\sqrt{np(1-p)}} \quad , \quad \gamma_2 = \frac{1-6p(1-p)}{np(1-p)} \quad .$$

Example 36. Efficiency fluctuations of a Geiger counter

A Geiger counter with a registration probability of 90% ($p = 0.9$) detects n' out of $n = 1000$ particles crossing it. On average this will be $\langle n' \rangle = np = 900$. The mean fluctuation (standard deviation) of this number is $\sigma = \sqrt{np(1-p)} = \sqrt{90} \approx 9.5$. The observed efficiency $\varepsilon = n'/n$ will fluctuate by $\sigma_\varepsilon = \sigma/n = \sqrt{p(1-p)/n} \approx 0.0095$

Example 37. Accuracy of a Monte Carlo integration

We want to estimate the value of π by a Monte Carlo integration. We distribute randomly n points in a square of area 4 cm^2 , centered at the origin. The number of points with a distance less than 1 cm from the origin is $k = np$ with $p = \pi/4$. To reach an accuracy of 1% requires

$$\begin{aligned} \frac{\sigma}{np} &= 0.01 \quad , \\ \frac{\sqrt{np(1-p)}}{np} &= 0.01 \quad , \\ n &= \frac{(1-p)}{0.01^2 p} = \frac{(4-\pi)}{0.01^2 \pi} \approx 2732 \quad , \end{aligned}$$

i.e. we have to generate $n = 2732$ pairs of random numbers.

Example 38. Acceptance fluctuations for weighted events

The acceptance of a complex detector is determined by Monte Carlo simulation which depends on a probability density $f_0(\mathbf{x})$ where $\mathbf{x}$ denotes all relevant kinematical variables. In order to avoid the repetition of the simulation for a different physical situation (e.g. a different cross section) described by a p.d.f. $f(\mathbf{x})$, it is customary to weight the individual events with $w_i = f(\mathbf{x})/f_0(\mathbf{x})$, $i = 1, \dots, N$ for N generated events. The acceptance ε_i for event i is either 1 or 0. Hence the overall acceptance is

$$\varepsilon_T = \frac{\sum w_i \varepsilon_i}{\sum w_i} \quad .$$

The variance for each single term in the numerator is $w_i^2 \varepsilon_i (1 - \varepsilon_i)$. Then the variance σ_T^2 of ε_T becomes

$$\sigma_T^2 = \frac{\sum w_i^2 \varepsilon_i (1 - \varepsilon_i)}{(\sum w_i)^2} \quad .$$

3.6.2 The Multinomial Distribution

When a single experiment or trial has not only two, but N possible outcomes with probabilities $p_1, p_2, \dots, p_N$, the probability to observe in n experiments $k_1, k_2, \dots, k_N$ trials belonging to the outcomes $1, \dots, N$ is equal to

$$M_{p_1, p_2, \dots, p_N}^n(k_1, k_2, \dots, k_N) = \frac{n!}{\prod_{i=1}^N k_i!} \prod_{i=1}^N p_i^{k_i},$$

where $\sum_{i=1}^N p_i = 1$ and $\sum_{i=1}^N k_i = n$ are satisfied. Hence we have $N - 1$ independent variates. The value $N = 2$ reproduces the binomial distribution.

In complete analogy to the binomial distribution, the multinomial distribution may be generated by expanding the multinom

$$(p_1 + p_2 + \dots + p_N)^n = 1$$

in powers of p_i , see (3.43). The binomial coefficients are replaced by multinomial coefficients which count the number of ways in which n distinguishable objects can be distributed into N classes which contain $k_1, \dots, k_N$ objects.

The expected values are

$$E(k_i) = np_i$$

and the covariance matrix is given by

$$C_{ij} = np_i(\delta_{ij} - p_j).$$

They can be derived from the characteristic function

$$\phi(t_1, \dots, t_{N-1}) = \left(1 + \sum_{i=1}^{N-1} p_i (e^{it_i} - 1) \right)^n$$

which is a straight forward generalization of the 1-dimensional case (3.44). The correlations are negative: If, for instance, more events k_i as expected fall into class i , the mean number of k_j for any other class will tend to be smaller than its expected value $E(k_j)$.

The multinomial distribution applies for the distribution of events into histogram bins. For total a number n of events with the probability p_i to collect an event in bin i , the expected number of events in that bin will be $n_i = np_i$ and the variance $C_{ii} = np_i(1 - p_i)$. Normally a histogram has many bins and $p_i \ll 1$ for all i . Then we approximate $C_{ij} \approx n_i \delta_{ij}$. The correlation between the bin entries can be neglected and the fluctuation of the entries in a bin is described by the Poisson distribution which we will discuss in the following section.

3.6.3 The Poisson Distribution

When a certain reaction happens randomly in time with an average frequency λ in a given time interval, then the number k of reactions in that time interval will follow a Poisson distribution (Fig. 3.15)

$$P_\lambda(k) = e^{-\lambda} \frac{\lambda^k}{k!}.$$

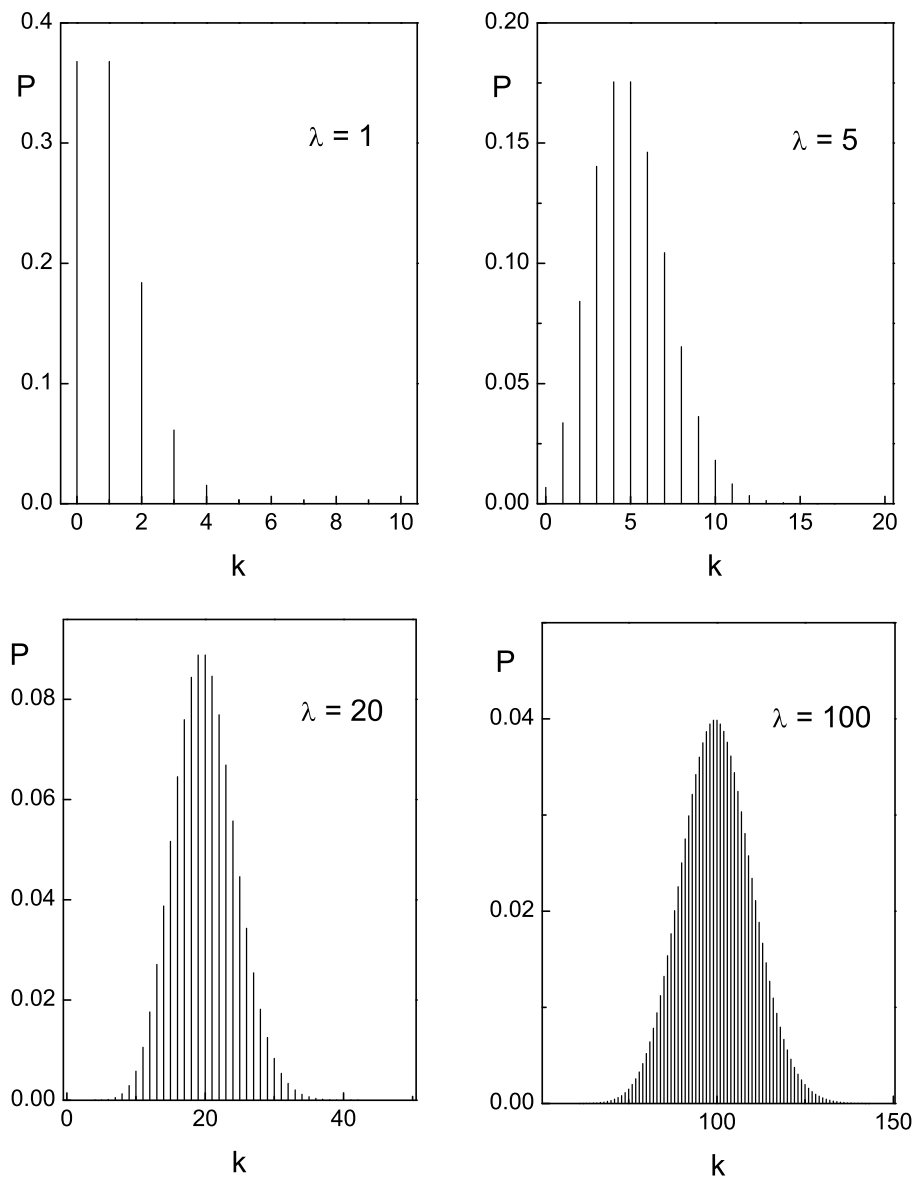


Fig. 3.15. Poisson distributions with different expected values.

Its expected value and variance have already been calculated above (see p. 36):

$$E(k) = \lambda, \text{ var}(k) = \lambda.$$

The characteristic function and cumulants have also been derived in Sect. 3.3.2 :

$$\phi(t) = \exp(\lambda(e^{it} - 1)), \quad (3.45)$$

$$\kappa_i = \lambda, i = 1, 2, \dots$$

Skewness and excess,

$$\gamma_1 = \frac{1}{\sqrt{\lambda}}, \gamma_2 = \frac{1}{\lambda}$$

indicate that the distribution becomes more Gaussian-like with increasing λ (see Fig. 3.15).

The Poisson distribution itself can be considered as the limiting case of a binomial distribution with $np = \lambda$, where n approaches infinity ($n \rightarrow \infty$) and, at the same time, p approaches zero, $p \rightarrow 0$. The corresponding limit of the characteristic function of the binomial distribution (3.44) produces the characteristic function of the Poisson distribution (3.45): With $p = \lambda/n$ we then obtain

$$\lim_{n \rightarrow \infty} \left[1 + \frac{\lambda}{n}(e^{it} - 1) \right]^n = \exp(\lambda(e^{it} - 1)).$$

For the Poisson distribution, the supply of potential events or number of trials is supposed to be infinite while the chance of a success, p , tends to zero. It is often used in cases where in principle the binomial distribution applies, but where the number of trials is very large.

Example 39. Poisson limit of the binomial distribution

A volume of 1 l contains 10^{16} hydrogen ions. The mean number of ions in a sub-volume of $1 \mu\text{m}^3$ is then $\lambda = 10$ and its standard deviation for a Poisson distribution is $\sigma = \sqrt{10} \approx 3$. The exact calculation of the standard deviation with the binomial distribution would change σ only by a factor $\sqrt{1 - 10^{-15}}$.

Also the number of radioactive decays in a given time interval follows a Poisson distribution, if the number of nuclei is big and the decay probability for a single nucleus is small.

The Poisson distribution is of exceptional importance in nuclear and particle physics, but also in the fields of microelectronics (noise), optics, and gas discharges it describes the statistical fluctuations.

Specific Properties of the Poisson Distribution

The sum $k = k_1 + k_2$ of Poisson distributed numbers k_1, k_2 with expected values λ_1, λ_2 is again a Poisson distributed number with expected value $\lambda = \lambda_1 + \lambda_2$. This property, which we called stability in connection with the binomial distribution follows formally from the structure of the characteristic function, or from the additivity of the cumulants given above. It is also intuitively obvious.

Example 40. Fluctuation of a counting rate minus background

Expected are S signal events with a mean background B . The mean fluctuation (standard deviation) of the observed number k is $\sqrt{S+B}$. This is also the fluctuation of $k-B$, because B is a constant. For a mean signal $S = 100$ and an expected background $B = 50$ we will observe on average 150 events with a fluctuation of $\sqrt{150}$. After subtracting the background, this fluctuation will remain. Hence, the background corrected signal is expected to be 100 with the standard deviation $\sigma = \sqrt{150}$. The uncertainty would even be larger, if also the mean value B was not known exactly.

If from a Poisson-distributed number n with expected value λ_0 on the average only a fraction ε is registered, for instance when the size of a detector is reduced by a factor of ε , then the expected rate is $\lambda = \lambda_0 \varepsilon$ and the number of observed events k follows the Poisson distribution $P_\lambda(k)$. This intuitive result is also obtained analytically: The number k follows a binomial distribution $B_\varepsilon^n(k)$ where n is a Poisson-distributed number. The probability $p(k)$ is:

$$\begin{aligned} p(k) &= \sum_{n=k}^{\infty} B_\varepsilon^n(k) P_{\lambda_0}(n) \\ &= \sum_{n=k}^{\infty} \frac{n!}{k!(n-k)!} \varepsilon^k (1-\varepsilon)^{n-k} e^{-\lambda_0} \frac{\lambda_0^n}{n!} \\ &= e^{-\lambda_0} \frac{(\varepsilon \lambda_0)^k}{k!} \sum_{n=k}^{\infty} \frac{1}{(n-k)!} (\lambda_0 - \lambda_0 \varepsilon)^{n-k} \\ &= e^{-\varepsilon \lambda_0} \frac{(\varepsilon \lambda_0)^k}{k!} \\ &= P_\lambda(k) . \end{aligned}$$

Of interest is also the following mathematical identity

$$\begin{aligned} \sum_{i=0}^k P_\lambda(i) &= \int_{\lambda}^{\infty} d\lambda' P_{\lambda'}(k) , \\ \sum_{i=0}^k \frac{\lambda^i}{i!} e^{-\lambda} &= \int_{\lambda}^{\infty} \frac{(\lambda')^k}{k!} e^{-\lambda'} d\lambda' , \end{aligned}$$

which allows us to calculate the probability $P\{i \leq k\}$ to find a number i less or equal k via a well known integral (described by the incomplete gamma function). It is applied in the estimation of upper and lower interval limits in Chap. 8.

Weighted Poisson Distributed Events

Let us assume that we measure the activity of a β -source with a Geiger counter. The probability that it fires, the detection probability, depends on the electron energy which varies from event to event. We can estimate the true number of decays by weighting each observation with the inverse of its detection probability. The statistics of their weighted sum is rather complex.

The statistics of weighted events plays also a role in some Monte Carlo integration methods, and sometimes also in parameter inference, if weighted observations

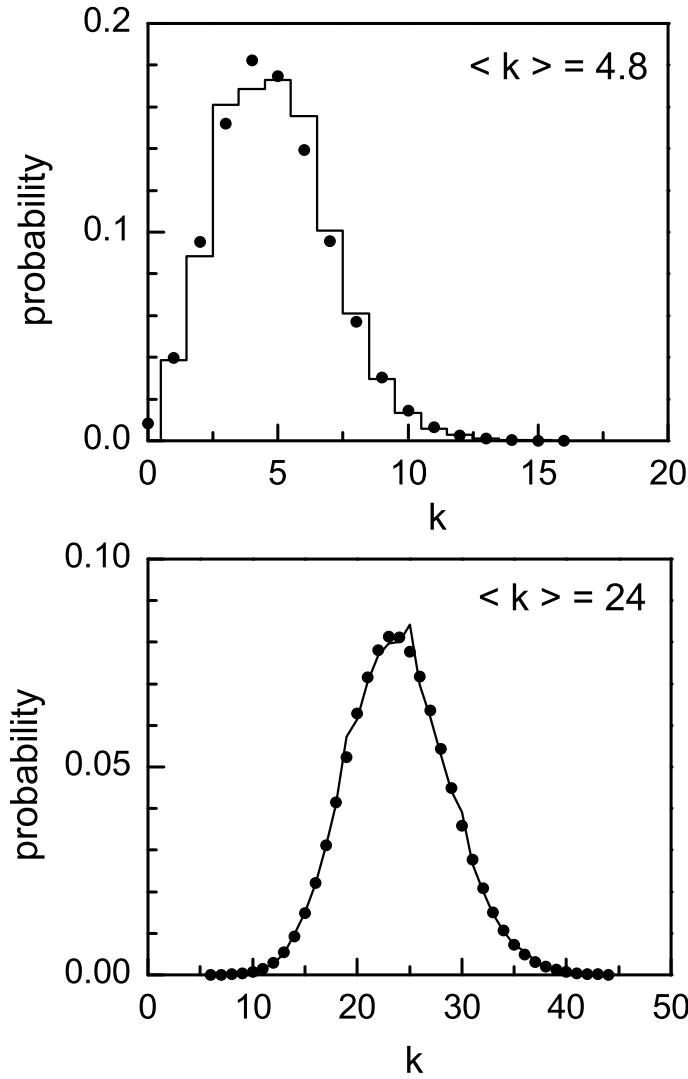


Fig. 3.16. Comparison of Poisson distributions (histograms) and scaled distribution of weighted events (dots) with a mixture of weights of one and ten.

are given in the form of histograms. The probability distribution of the corrected numbers does not have a simple analytical expression. It can, as we will see, be approximately described by a Poisson distribution. However, the cumulants and thus also the moments of the distribution can be calculated exactly.

Let us consider the definite case that on average λ_1 observations are obtained with probability ε_1 and λ_2 observations with probability ε_2 . We correct the losses by weighting the observed numbers with $w_1 = 1/\varepsilon_1$ and $w_2 = 1/\varepsilon_2$. For the Poisson-distributed numbers k_1, k_2

$$\begin{aligned}
P_{\lambda_1}(k_1) &= \frac{\lambda_1^{k_1}}{k_1!} e^{-\lambda_1} , \\
P_{\lambda_2}(k_2) &= \frac{\lambda_2^{k_2}}{k_2!} e^{-\lambda_2} , \\
k &= w_1 k_1 + w_2 k_2 ,
\end{aligned}$$

we get the mean value λ of the variate k and its variance with $\text{var}(cx) = c^2 \text{var}(x)$:

$$\begin{aligned}
\lambda &= w_1 \lambda_1 + w_2 \lambda_2 , \\
\sigma^2 &= w_1^2 \lambda_1 + w_2^2 \lambda_2 .
\end{aligned}$$

According to (3.28), the cumulant κ_i of order i of the distribution of k is related to the cumulants $\kappa_i^{(1)}, \kappa_i^{(2)}$ of the corresponding distributions of k_1, k_2 through

$$\kappa_i = w_1^i \kappa_i^{(1)} + w_2^i \kappa_i^{(2)} . \quad (3.46)$$

As mentioned above, there is no simple closed expression for the distribution of k . It can be constructed approximately using the lower moments, but this is rather tedious. Therefore we try to describe it approximately by a scaled Poisson distribution

$$P_{\tilde{\lambda}}(\tilde{k}) = \frac{\tilde{\lambda}^{\tilde{k}}}{\tilde{k}!} e^{-\tilde{\lambda}} ,$$

such that the ratio of standard deviation to mean value is the same as for the weighted sum. We obtain

$$\begin{aligned}
\frac{\sqrt{\tilde{\lambda}}}{\tilde{\lambda}} &= \frac{\sigma}{\lambda} , \\
\tilde{\lambda} &= \frac{(w_1 \lambda_1 + w_2 \lambda_2)^2}{w_1^2 \lambda_1 + w_2^2 \lambda_2} ,
\end{aligned}$$

where we have to scale the random variable $\tilde{k}$ to obtain the acceptance corrected number k :

$$\begin{aligned}
k &= \tilde{k} \frac{\lambda}{\tilde{\lambda}} \\
&= \tilde{k} \frac{w_1^2 \lambda_1 + w_2^2 \lambda_2}{w_1 \lambda_1 + w_2 \lambda_2} .
\end{aligned}$$

The approximate distribution of k that we have obtained in this way has by construction the right mean value and width, but differs from the exact distribution in details. The quantity $\tilde{k}$ is called *equivalent number of events*. A number k of observations which has been corrected for acceptance losses has the same statistical significance as $\tilde{k}$ un-weighted observations. (The statistical significance is the ratio of mean value to standard deviation μ/σ). The accuracy of our approximation can be checked by comparing the skewness and excess $\gamma_{1,2}$ as obtained from the cumulants (3.46) according to (3.27) with the results of a Poisson distribution, namely $\tilde{\gamma}_1 = 1/\sqrt{\tilde{\lambda}}$, $\gamma_2 = 1/\tilde{\lambda}$.

Example 41. Distribution of weighted, Poisson distributed observations

We expect on average $\lambda_1 = 20$ observations registered with the probability $\varepsilon_1 = 0.1$ ($w_1 = 10$), and $\lambda_2 = \lambda_1 = 20$ observations with acceptance

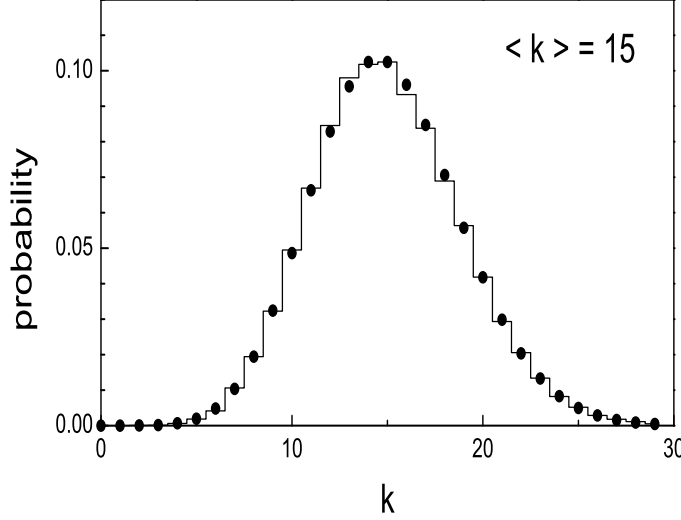


Fig. 3.17. Comparison of a Poisson distribution (histogram) with a scaled distribution of weighted events (dots). The weights are uniformly distributed between zero and one.

$\varepsilon_1 = 1$. The expected value of the acceptance corrected number is $\lambda = 220$. We approximate the distribution of the observed numbers k by the Poisson distribution with the mean value $\tilde{\lambda}$,

$$\tilde{\lambda} = \frac{(10 \cdot 20 + 1 \cdot 20)^2}{10^2 \cdot 20 + 1^2 \cdot 20} = 24.0 ,$$

in order to describe the distribution of $\tilde{k}$. The statistical significance of the 40 expected observations is the same as that of 24 unweighted observations. We obtain the approximate distribution of $k = \tilde{k}\lambda/\tilde{\lambda} = \tilde{k} \cdot 220/24$ by generating the distribution of $\tilde{k}$ and equating the probabilities of k and $\tilde{k}$. Fig. 3.16 shows a comparison between the equivalent Poisson distribution and the true distribution of $k\tilde{\lambda}/\lambda$. By construction, mean value and variance are the same; also skewness and excess are similar in both cases: The approximate (exact) values are, respectively, $\gamma_1 = 0.204$ (0.221), $\gamma_2 = 0.042$ (0.049). The differences are partially due to the fact that we compare two different discrete distributions which have discrete parameters γ_1, γ_2 .

The generalization of the above procedure to the addition of more than two weighted Poisson distributions is trivial:

$$\tilde{\lambda} = \frac{(\sum w_i \lambda_i)^2}{\sum w_i^2 \lambda_i} ,$$

$$k \approx \tilde{k} \frac{\sum w_i^2 \lambda_i}{\sum w_i \lambda_i} .$$

For the sum of N accidental weights which follow a continuous distribution $g(w)$, we obtain the expected value $\lambda = \langle k \rangle = N \int g(w)w \, dw$ and with $\lambda_i = 1$

$$\tilde{\lambda} = N \frac{(\int g(w)w \, dw)^2}{\int g(w)w^2 \, dw} ,$$

$$k \approx \tilde{k} \frac{\int g(w) w^2 dw}{\int g(w) w dw}.$$

In Fig. 3.17, a scaled distribution with uniformly distributed weights between zero and one is compared to the corresponding approximating Poisson distribution. The agreement is quite good.

3.6.4 The Uniform Distribution

The uniform distribution is the simplest continuous distribution. It describes, for instance, digital measurements where the random variable is tied to a given interval and where inside the interval all its values are equally probable.

Given an interval of length α centered at the mean value ξ the p.d.f. reads

$$f(x|\xi, \alpha) = \begin{cases} 1/\alpha & \text{if } |x - \xi| < \alpha/2 \\ 0 & \text{else.} \end{cases} \quad (3.47)$$

Mean value and variance are $\langle x \rangle = \xi$ and $\sigma^2 = \alpha^2/12$, respectively. The characteristic function is

$$\phi(t) = \frac{1}{\alpha} \int_{\xi-\alpha/2}^{\xi+\alpha/2} e^{itx} dx = \frac{2}{\alpha t} \sin \frac{\alpha t}{2} e^{i\xi t}. \quad (3.48)$$

Using the power expansion of the sinus function we find from (3.48) for $\xi = 0$ the even moments (the odd moments vanish):

$$\mu'_{2k} = \frac{1}{2k+1} \left(\frac{\alpha}{2} \right)^{2k}, \quad \mu'_{2k-1} = 0.$$

The uniform distribution is the basis for the computer simulation of all other distributions because random number generators for numbers uniformly distributed between 0 and 1 are implemented on all computers used for scientific purposes. We will discuss simulations in some detail in Chap. 5.

3.6.5 The Normal Distribution

The normal or Gauss distribution which we introduced already in Sect. 3.2.7,

$$N(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)},$$

enjoys great popularity among statisticians. This has several reasons which, however, are not independent from each other.

1. The sum of normally distributed quantities is again normally distributed (stability), with $\mu = \sum \mu_i$, $\sigma^2 = \sum \sigma_i^2$, in obvious notation.
2. The discrete binomial- and Poisson distributions and also the χ^2 -distribution, in the limit of a large number, a large mean value and many degrees of freedom, respectively, approach the normal distribution.
3. Many distributions met in natural sciences are well approximated by normal distributions. We have already mentioned some examples: velocity components of gas

molecules, diffusion, Brownian motion and many measurement errors obey normal distributions to good accuracy.

4. Certain analytically simple statistical procedures for parameter estimation and propagation of errors are valid exactly only for normally distributed errors.

The deeper reason for point 2 and 3 is explained by the *central limit theorem*: The mean value of a large number N of independent random variables, obeying the same distribution with variance σ_0^2 , approaches a normal distribution with variance $\sigma^2 = \sigma_0^2/N$. The important point is that this theorem is valid for quite arbitrary distributions, provided they have a finite variance, a condition which practically always can be fulfilled, if necessary by cutting off large absolute values of the variates. Instead of a formal proof¹², we show in Fig. 3.18, how with increasing number of variates the distribution of their mean value approaches the normal distribution better and better.

As example we have chosen the mean values for uniformly resp. exponentially distributed numbers. For the very asymmetrical exponential distribution on the left hand side of the figure the convergence to a normal distribution is not as fast as for the uniform distribution, where already the distribution of the mean of five random numbers is in good agreement with the normal distribution. The central limit theorem applies also when the individual variates follow different distributions provided that the variances are of the same order of magnitude.

The characteristic function of the normal distribution is

$$\phi(t) = \exp\left(-\frac{1}{2}\sigma^2 t^2 + i\mu t\right).$$

It is real and also of Gaussian shape for $\mu = 0$. The stability (see point 1 above) is easily proven, using the convolution theorem (3.25) and the exponential form of $\phi(t)$.

Differentiating the characteristic function, setting $\mu = 0$, we obtain the central moments of the normal distribution:

$$\mu'_{2j} = \frac{(2j)!}{2^j j!} \sigma^{2j}.$$

Cumulants, with the exception of $\kappa_1 = \mu$ and $\kappa_2 = \sigma^2$, vanish. Also the odd central moments are zero.

The Normal Distribution in Higher Dimensions

The normal distribution in two dimensions with its maximum at the origin has the general form

$$N_0(x, y) = \frac{1}{\sqrt{1 - \rho^2} 2\pi s_x s_y} \exp\left[-\frac{1}{2(1 - \rho^2)} \left(\frac{x^2}{s_x^2} - 2\rho \frac{xy}{s_x s_y} + \frac{y^2}{s_y^2}\right)\right]. \quad (3.49)$$

The notation has been chosen such that it indicates the moments:

$$\begin{aligned} \langle x^2 \rangle &= s_x^2, \\ \langle y^2 \rangle &= s_y^2, \\ \langle xy \rangle &= \rho s_x s_y. \end{aligned}$$

¹²A simplified proof is presented in the Appendix 13.1.

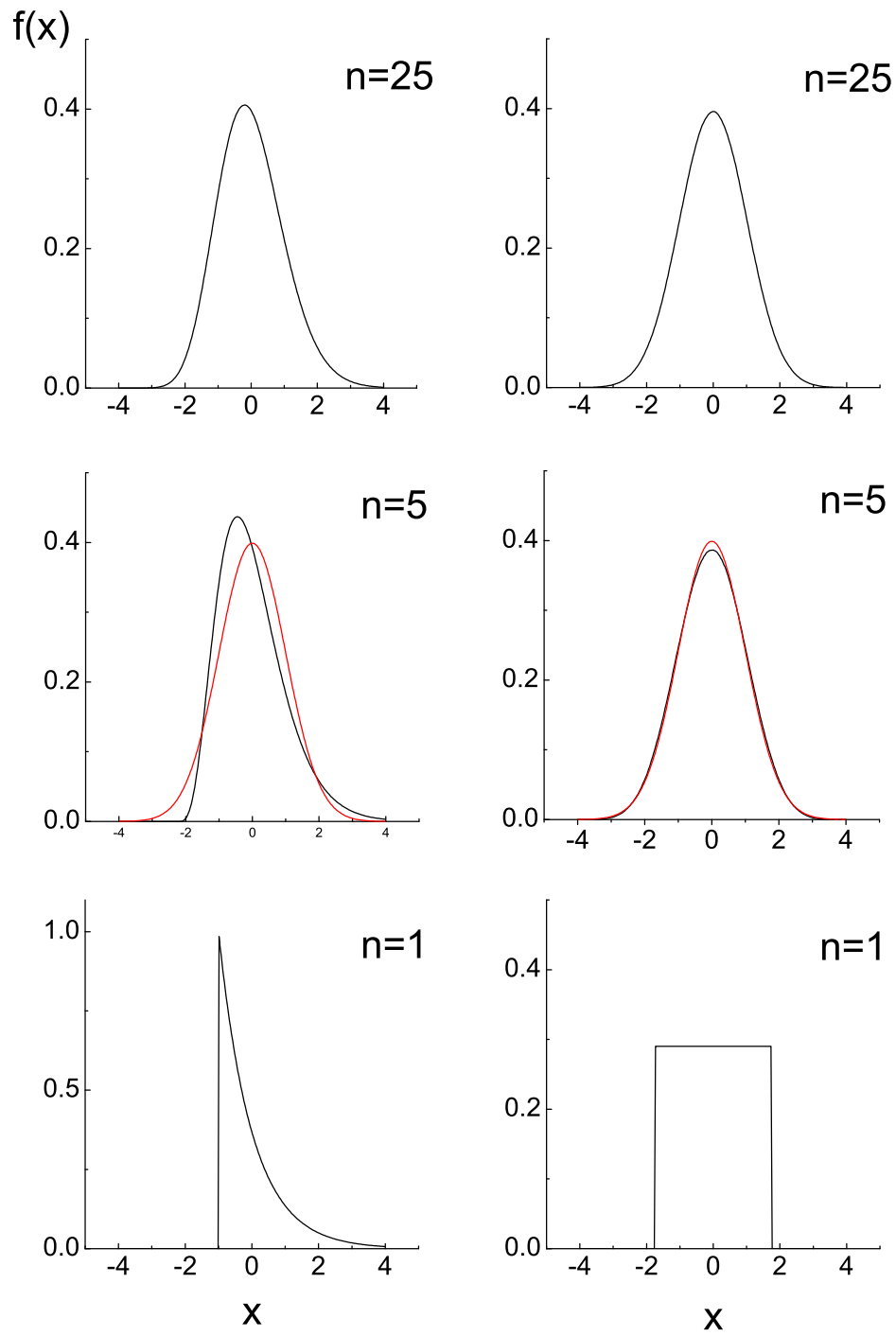


Fig. 3.18. Illustration of the central limit theorem. The mean values of n exponential or uniformly distributed variates approach with increasing n the normal distribution.

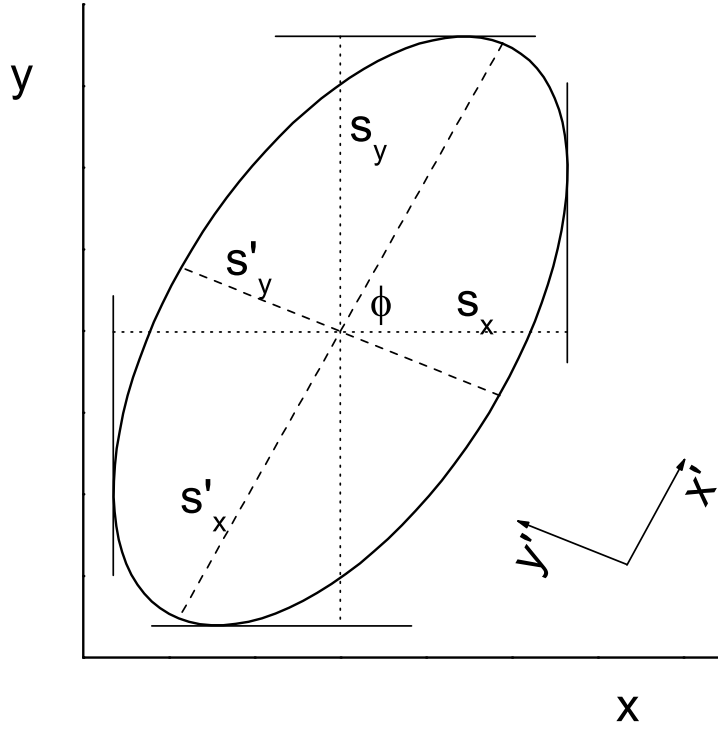


Fig. 3.19. Transformation of the error ellipse.

We skip the explicit calculation. Integrating (3.49) over $y, (x)$, we obtain the marginal distributions of $x, (y)$. They are again normal distributions with widths s_x and s_y . A characteristic feature of the normal distribution is that for a vanishing correlation $\rho = 0$ the two variables are independent, since in this case the p.d.f. $N_0(x, y)$ factorizes into normal distributions of x and y .

Curves of equal probability are obtained by equating the exponent to a constant. The equations

$$\frac{1}{1 - \rho^2} \left(\frac{x^2}{s_x^2} - 2\rho \frac{xy}{s_x s_y} + \frac{y^2}{s_y^2} \right) = \text{const}$$

describe concentric ellipses. For the special choice $\text{const} = 1$ we show the ellipse in Fig. 3.19. At this so-called error ellipse the value of the p.d.f. is just $N_0(0, 0)/\sqrt{e}$, i.e. reduced with respect to the maximum by a factor $1/\sqrt{e}$.

By a simple rotation we achieve uncorrelated variables x' and y' :

$$\begin{aligned} x' &= x \cos \phi + y \sin \phi, \\ y' &= -x \sin \phi + y \cos \phi, \\ \tan 2\phi &= \frac{2\rho s_x s_y}{s_x^2 - s_y^2}. \end{aligned}$$

The half-axes, i.e. the variances s'^2_x and s'^2_y of the uncorrelated variables x' and y' are

$$s_x'^2 = \frac{s_x^2 + s_y^2}{2} + \frac{s_x^2 - s_y^2}{2 \cos 2\phi},$$

$$s_y'^2 = \frac{s_x^2 + s_y^2}{2} - \frac{s_x^2 - s_y^2}{2 \cos 2\phi}.$$

In the new variables, the normal distribution has then the simple form

$$N'_0(x', y') = \frac{1}{2\pi s'_x s'_y} \exp\left(-\frac{1}{2}\left(\frac{x'^2}{s_x'^2} + \frac{y'^2}{s_y'^2}\right)\right) = f(x')g(y').$$

The two-dimensional normal distribution with its maximum at (x_0, y_0) is obtained from (3.49) with the substitution $x \rightarrow x - x_0$, $y \rightarrow y - y_0$.

We now generalize the normal distribution to n dimensions. We skip again the simple algebra and present directly the result. The variables are written in vector form $\mathbf{x}$ and with the symmetric and positive definite covariance matrix $\mathbf{C}$, the p.d.f. is given by

$$N(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n \det(\mathbf{C})}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T \mathbf{C}^{-1}(\mathbf{x} - \mathbf{x}_0)\right).$$

Frequently we need the inverse of the covariance matrix

$$\mathbf{V} = \mathbf{C}^{-1}$$

which is called weight matrix. Small variances C_{ii} of components x_i lead to large weights V_{ii} . The normal distribution in n dimensions has then the form

$$N(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n \det(\mathbf{C})}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T \mathbf{V}(\mathbf{x} - \mathbf{x}_0)\right).$$

In the two-dimensional case the matrices $\mathbf{C}$ and $\mathbf{V}$ are

$$\mathbf{C} = \begin{pmatrix} s_x^2 & \rho s_x s_y \\ \rho s_x s_y & s_y^2 \end{pmatrix},$$

$$\mathbf{V} = \frac{1}{1 - \rho^2} \begin{pmatrix} \frac{1}{s_x^2} & -\frac{\rho}{s_x s_y} \\ -\frac{\rho}{s_x s_y} & \frac{1}{s_y^2} \end{pmatrix}$$

with the determinant $\det(\mathbf{C}) = s_x^2 s_y^2 (1 - \rho^2) = 1 / \det(\mathbf{V})$.

3.6.6 The Exponential Distribution

Also the exponential distribution appears in many physical phenomena. Besides life time distributions (decay of instable particles, nuclei or excited states), it describes the distributions of intervals between Poisson distributed events like time intervals between decays or gap lengths in track chambers, and of the penetration depth of particles in absorbing materials.

The main characteristics of processes described by the exponential distribution is lack of memory, i.e. processes which are not influenced by their history. For instance, the decay probability of an instable particle is independent of its age, or the scattering probability for a gas molecule at the time t is independent of t and of the time that

has passed since the last scattering event. The probability density for the decay of a particle at the time $t_1 + t_2$ must be equal to the probability density $f(t_2)$ multiplied with the probability $1 - F(t_1)$ to survive until t_1 :

$$f(t_1 + t_2) = (1 - F(t_1)) f(t_2) .$$

Since $f(t_1 + t_2)$ must be symmetric under exchanges of t_1 and t_2 , the first factor has to be proportional to $f(t_1)$,

$$1 - F(t_1) = cf(t_1) , \quad (3.50)$$

$$f(t_1 + t_2) = cf(t_1)f(t_2) \quad (3.51)$$

with constant c . The property (3.51) is found only for the exponential function: $f(t) = ae^{bt}$. If we require that the probability density is normalized, we get

$$f(t) = \lambda e^{-\lambda t} .$$

This result could also have been derived by differentiating (3.50) and solving the corresponding differential equation $f = -c df/dt$.

The characteristic function

$$\phi(t) = \frac{\lambda}{\lambda - it}$$

and the moments

$$\mu_n = n! \lambda^{-n}$$

have already been derived in Example 21 in Sect. 3.3.3.

3.6.7 The χ^2 Distribution

The chi-square distribution (χ^2 distribution) plays an important role in the comparison of measurements with theoretical distributions (see Chap. 10). The corresponding tests allow us to discover systematic measurement errors and to check the validity of theoretical models. The variable χ^2 which we will define below, is certainly the quantity which is most frequently used to quantify the quality of the agreement of experimental data with the theory.

The variate χ^2 is defined as the sum

$$\chi^2 = \sum_{i=1}^f \frac{x_i^2}{\sigma_i^2} ,$$

where x_i are independent, normally distributed variates with zero mean and variance σ_i^2 .

We have already come across the simplest case with $f = 1$ in Sect. 3.4.1: The transformation of a normally distributed variate x with expected value zero to $u = x^2/s^2$, where s^2 is the variance, yields

$$g_1(u) = \frac{1}{\sqrt{2\pi u}} e^{-u/2} \quad (f = 1) .$$

(We have replaced the variable χ^2 by $u = \chi^2$ to simplify the writing.) Mean value and variance of this distribution are $E(u) = 1$ and $\text{var}(u) = 2$.

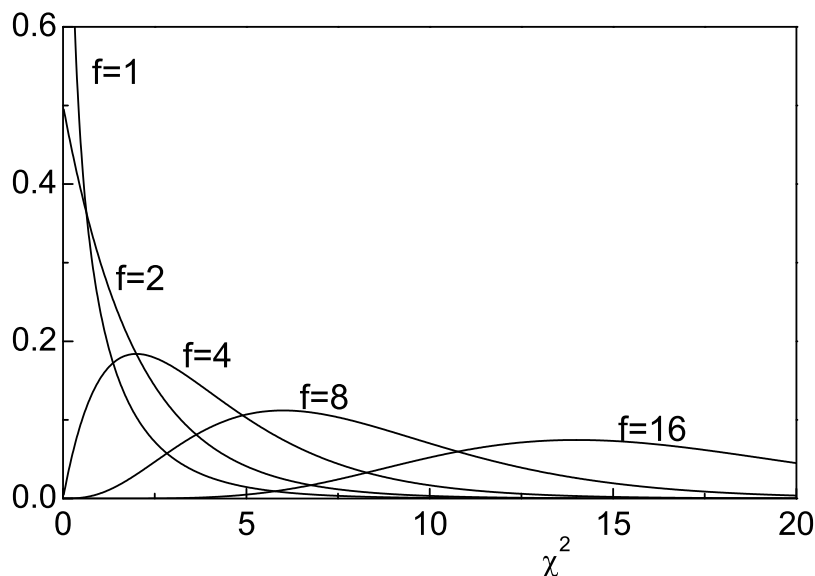


Fig. 3.20. χ^2 distribution for different degrees of freedom.

When we now add f independent summands we obtain

$$g_f(u) = \frac{1}{\Gamma(f/2)2^{f/2}} u^{f/2-1} e^{-u/2} . \quad (3.52)$$

The only parameter of the χ^2 distribution is the number of degrees of freedom f , the meaning of which will become clear later. We will prove (3.52) when we discuss the gamma distribution, which includes the χ^2 distribution as a special case. Fig. 3.20 shows the χ^2 distribution for some values of f . The value $f = 2$ corresponds to an exponential distribution. As follows from the central limit theorem, for large values of f the χ^2 distribution approaches a normal distribution.

By differentiation of the p.d.f. we find for $f > 2$ the maximum at the mode $u_{mod} = f - 2$. The expected value of the variate u is equal to f and its variance is $2f$. These relations follow immediately from the definition of u .

$$\begin{aligned} u_{mod} &= f - 2 \quad \text{for } f > 2 , \\ E(u) &= f , \\ \text{var}(u) &= 2f . \end{aligned}$$

Distribution of the Sample Width

We define the width v of a sample of N elements x_i as follows (see 3.2.3):

$$\begin{aligned} v^2 &= \frac{1}{N} \sum_{i=1}^N x_i^2 - \bar{x}^2 \\ &= \overline{x^2} - \bar{x}^2 . \end{aligned}$$

If the variates x_i of the sample are distributed normally with mean x_0 and variance σ^2 , then Nv^2/σ^2 follows a χ^2 distribution with $f = N - 1$ degrees of freedom. We omit the formal proof; the result is plausible, however, from the expected value derived in Sect. 3.2.3:

$$\begin{aligned}\langle v^2 \rangle &= \frac{N-1}{N} \sigma^2, \\ \left\langle \frac{Nv^2}{\sigma^2} \right\rangle &= N-1.\end{aligned}$$

Degrees of Freedom and Constraints

In Sect. 7.2 we will discuss the method of least squares for parameter estimation. To adjust a curve to measured points x_i with Gaussian errors σ_i we minimize the quantity

$$\chi^2 = \sum_{i=1}^N \frac{\left(x_i - x_i^{(t)}(\lambda_1, \dots, \lambda_Z) \right)^2}{\sigma_i^2},$$

where $x_i^{(t)}$ are the ordinates of the curve depending on the Z free parameters λ_k . Large values of χ^2 signal a bad agreement between measured values and the fitted curve. If the predictions $x_i^{(t)}$ depend linearly on the parameters, the sum χ^2 obeys a χ^2 distribution with $f = N - Z$ degrees of freedom. The reduction of f accounts for the fact that the expected value of χ^2 is reduced when we allow for free parameters. Indeed, for $Z = N$ we could adjust the parameters such that χ^2 would vanish.

Generally, in statistics the term degrees of freedom¹³ f denotes the number of independent predictions. For $N = Z$ we have no prediction for the observations x_i . For $Z = 0$ we predict all N observations, $f = N$. When we fit a straight line through 3 points with given abscissa and observed ordinate, we have $N = 3$ and $Z = 2$ because the line contains 2 parameters. The corresponding χ^2 distribution has 1 degree of freedom. The quantity Z is called the *number of constraints*, a somewhat misleading term. In the case of the sample width discussed above, one quantity, the mean, is adjusted. Consequently, we have $Z = 1$ and the sample width follows a χ^2 distribution of $f = N - 1$ degrees of freedom.

3.6.8 The Gamma Distribution

The distributions considered in the last two sections, the exponential- and the chi-square distribution, are special cases of the gamma distribution

$$G(x|\nu, \lambda) = \frac{\lambda^\nu}{\Gamma(\nu)} x^{\nu-1} e^{-\lambda x}, \quad x > 0.$$

The parameter $\lambda > 0$ is a scale parameter, while the parameter $\nu > 0$ determines the shape of the distribution. With $\nu = 1$ we obtain the exponential distribution. The parameter ν is not restricted to natural numbers. With the special choice $\nu = f/2$ and $\lambda = 1/2$ we get the χ^2 -distribution with f degrees of freedom (see Sect. 3.6.7).

¹³Often the notation *number of degrees of freedom*, abbreviated by *n.d.f.* or *NDF* is used in the literature.

The gamma distribution is used typically for the description of random variables that are restricted to positive values, as in the two cases just mentioned. The characteristic function is very simple:

$$\phi(t) = \left(1 - \frac{it}{\lambda}\right)^{-\nu}. \quad (3.53)$$

As usual, we obtain expected value, variance, moments about the origin, skewness and excess by differentiating $\phi(t)$:

$$\langle x \rangle = \frac{\nu}{\lambda}, \quad \text{var}(x) = \frac{\nu}{\lambda^2}, \quad \mu_i = \frac{\Gamma(i + \nu)}{\lambda^i \Gamma(\nu)}, \quad \gamma_1 = \frac{2}{\sqrt{\nu}}, \quad \gamma_2 = \frac{6}{\nu}.$$

The maximum of the distribution is at $x_{\text{mod}} = (\nu - 1)/\lambda$, ($\nu > 1$).

The gamma distribution has the property of stability in the following sense: The sum of variates following gamma distributions with the *same* scaling parameter λ , but different shape parameters ν_i is again gamma distributed, with the shape parameter ν ,

$$\nu = \sum \nu_i.$$

This result is obtained by multiplying the characteristic functions (3.53). It proves also the corresponding result (3.52) for the χ^2 -distribution.

Example 42. Distribution of the mean value of decay times

Let us consider the sample mean $\bar{x} = \sum x_i/N$, of exponentially distributed variates x_i . The characteristic function is (see 3.3.3)

$$\phi_x(t) = \frac{1}{1 - it/\lambda}.$$

Forming the N -fold product, and using the scaling rule for Fourier transforms (3.19), $\phi_{x/N}(t) = \phi_x(t/N)$, we arrive at the characteristic function of a gamma distribution with scaling parameter $N\lambda$ and shape parameter N :

$$\phi_{\bar{x}}(t) = \left(1 - \frac{it}{N\lambda}\right)^{-N}. \quad (3.54)$$

Thus the p.d.f. $f(\bar{x})$ is equal to $G(\bar{x}|N, N\lambda)$. Considering the limit for large N , we convince ourself of the validity of the law of large numbers and the central limit theorem. From (3.54) we derive

$$\begin{aligned} \ln \phi_{\bar{x}}(t) &= -N \ln \left(1 - \frac{it}{N\lambda}\right) \\ &= -N \left[\left(-\frac{it}{N\lambda}\right) - \frac{1}{2} \left(-\frac{it}{N\lambda}\right)^2 + O(N^{-3}) \right], \\ \phi_{\bar{x}}(t) &= \exp \left[i \frac{1}{\lambda} t - \frac{1}{2} \frac{1}{N\lambda^2} t^2 + O(N^{-2}) \right]. \end{aligned}$$

When N is large, the term of order N^{-2} can be neglected and with the two remaining terms in the exponent we get the characteristic function of a normal distribution with mean $\mu = 1/\lambda = \langle x \rangle$ and variance $\sigma^2 = 1/(N\lambda^2) =$

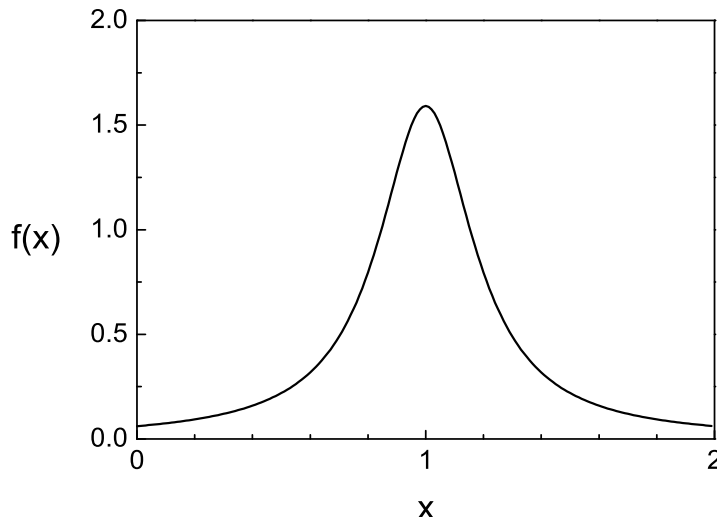


Fig. 3.21. Lorentz distribution with mean equal to 1 and halfwidth $\Gamma/2 = \sqrt{2}$.

$\text{var}(x)/N$, (see 3.3.3), in agreement with the central limit theorem. If N approaches infinity, only the first term remains and we obtain the characteristic function of a delta distribution $\delta(1/\lambda - \bar{x})$. This result is predicted by the law of large numbers (see Appendix 13.1). This law states that, under certain conditions, with increasing sample size, the difference between the sample mean and the population mean approaches zero.

3.6.9 The Lorentz and the Cauchy Distributions

The Lorentz distribution (Fig. 3.21)

$$f(x) = \frac{1}{\pi} \frac{\Gamma/2}{(x - a)^2 + (\Gamma/2)^2}$$

is symmetric with respect to $x = a$. Although it is bell-shaped like a Gaussian, it has, because of its long tails, no finite variance. This means that we cannot infer the location parameter a of the distribution¹⁴ from the sample mean, even for arbitrary large samples. The Lorentz distribution describes resonance effects, where Γ represents the width of the resonance. In particle or nuclear physics, mass distributions of short-lived particles follow this p.d.f. which then is called *Breit-Wigner distribution*.

The Cauchy distribution corresponds to the special choice of the scale parameter $\Gamma = 2$.¹⁵ For the location parameter $a = 0$ it has the characteristic function $\phi(t) = \exp(-|t|)$, which obviously has no derivatives at $t = 0$, an other consequence of the nonexistence of moments. The characteristic function for the sample mean of N measurements, $\bar{x} = \sum_1^N x_i/N$, is found with the help of (3.19), (3.25) as

$$\phi_{\bar{x}}(t) = (\phi(t/N))^N = \phi(t) .$$

The sample mean has the same distribution as the original population. It is therefore, as already stated above, not suited for the estimation of the location parameter.

¹⁴The first moment exists only as a Cauchy principal value and equals a .

¹⁵In the literature also the more general definition with two parameters is met.

3.6.10 The Log-normal Distribution

The distribution of a variable $x > 0$ whose logarithm u is normally distributed

$$g(u) = \frac{1}{\sqrt{2\pi}s} e^{-(u-u_0)^2/2s^2}$$

with mean u_0 and variance s^2 follows the log-normal distribution, see Fig. 3.22:

$$f(x) = \frac{1}{xs\sqrt{2\pi}} e^{-(\ln x - u_0)^2/2s^2}.$$

This is, like the normal distribution, a two-parameter distribution where the parameters u_0 , s^2 , however, are not identical with the mean μ and variance σ^2 , but the latter are given by

$$\begin{aligned} \mu &= e^{u_0 + s^2/2}, \\ \sigma^2 &= (e^{s^2} - 1)e^{2u_0 + s^2}. \end{aligned} \quad (3.55)$$

Note that the distribution is declared only for positive x , while u_0 can also be negative.

The characteristic function cannot be written in closed form, but only as a power expansion. This means, the moments of order k about the origin are

$$\mu_k = e^{ku_0 + \frac{1}{2}k^2s^2}.$$

Other characteristic parameters are

$$\begin{aligned} \text{median : } x_{0.5} &= e^{u_0}, \\ \text{mode : } x_{mod} &= e^{u_0 - s^2}, \\ \text{skewness : } \gamma_1 &= (e^{s^2} + 2)\sqrt{e^{s^2} - 1}, \\ \text{kurtosis : } \gamma_2 &= e^{4s^2} + 2e^{3s^2} + 3e^{2s^2} - 6. \end{aligned} \quad (3.56)$$

The distribution of a variate $x = \prod x_i$ which is the product of many variates x_i , each of which is positive and has a small variance, σ_i^2 compared to its mean squared μ_i^2 , $\sigma_i^2 \ll \mu_i^2$, can be approximated by a log-normal distribution. This is a consequence of the central limit theorem (see 3.6.5). Writing

$$\ln x = \sum_{i=1}^N \ln x_i$$

we realize that $\ln x$ is normally distributed in the limit $N \rightarrow \infty$ if the summands fulfil the conditions required by the central limit theorem. Accordingly, x will be distributed by the log-normal distribution.

3.6.11 Student's t Distribution

This distribution, introduced by W. S. Gosset (pseudonym “Student”) is frequently used to test the compatibility of a sample with a normal distribution with given mean

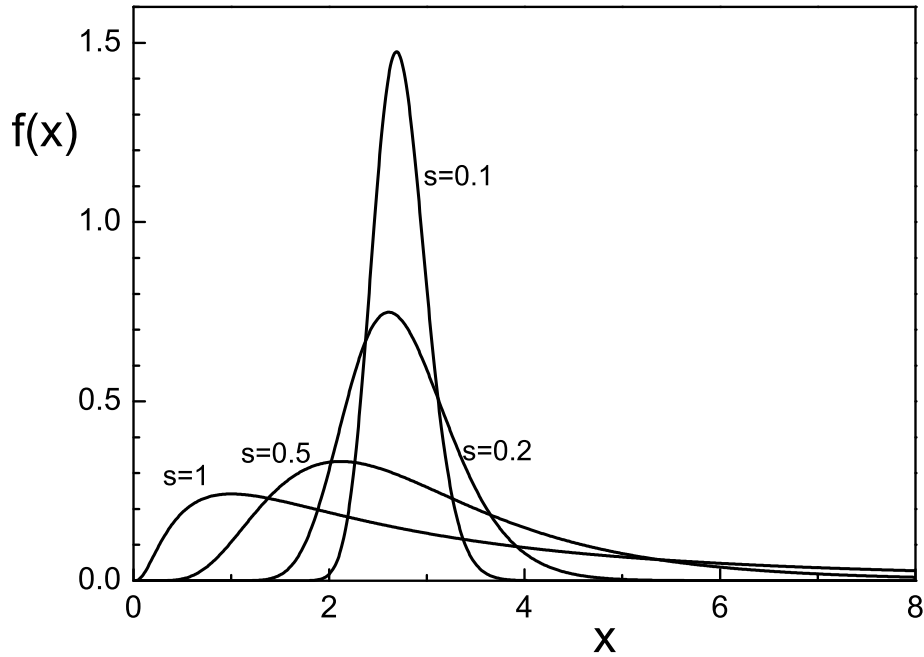


Fig. 3.22. Log-normal distribution with $u_0 = 1$ and different values of s .

but unknown variance. It describes the distribution of the so-called “studentized” variate t , defined as

$$t = \frac{\bar{x} - \mu}{s}. \quad (3.57)$$

The numerator is the difference between a sample mean and the mean of the Gaussian from which the sample of size N is drawn. It follows a normal distribution centered at zero. The denominator s is an estimate of the standard deviation of the numerator derived from the sample. It is defined by (3.15).

$$s^2 = \frac{1}{N(N-1)} \sum_{i=1}^N (x_i - \bar{x})^2.$$

The sum on the right-hand side, after division by the variance σ^2 of the Gaussian, follows a χ^2 distribution with $f = N - 1$ degrees of freedom, see (3.52). Dividing also the numerator of (3.57) by its standard deviation $\sigma/\sqrt{N}$, it follows a normal distribution of variance unity. Thus the variable t of the t distribution is the quotient of a normal variate and the square root of a χ^2 variate.

The analytical form of the p.d.f. can be found by the standard method used in Sect. 3.5.4. The result is

$$h(t|f) = \frac{\Gamma((f+1)/2)}{\Gamma(f/2)\sqrt{\pi f}} \left(1 + \frac{t^2}{f}\right)^{-\frac{f+1}{2}}.$$

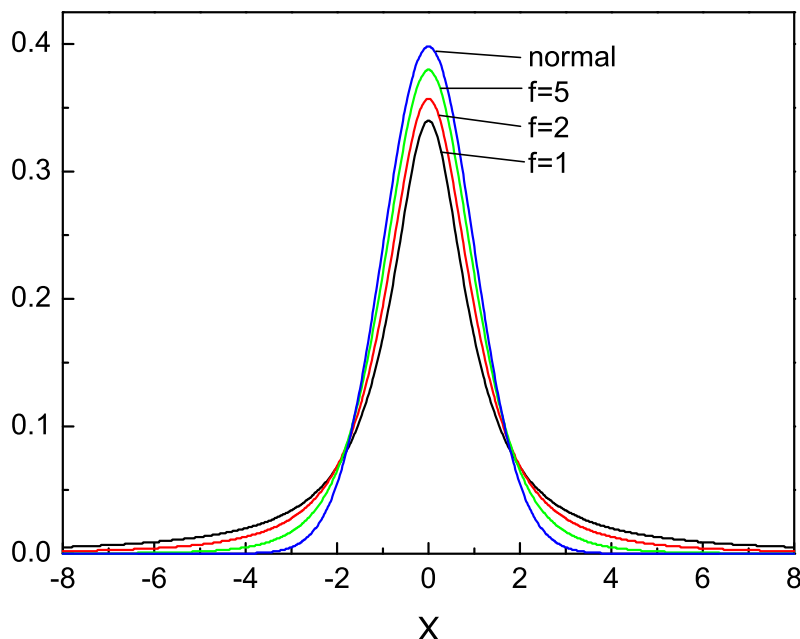


Fig. 3.23. Student's distributions for 1, 2, 5 degrees of freedom and normal distribution.

The only parameter is f , the number of degrees of freedom. For $f = 1$ we recover the Cauchy distribution. For large f it approaches the normal distribution $N(0, 1)$ with variance equal to one. The distribution is symmetric, centered at zero, and bell shaped, but with longer tails than $N(0, 1)$. The even moments are

$$\mu_i = f^{\frac{i}{2}} \frac{1 \cdot 3 \cdots (i-1)}{(f-2)(f-4) \cdots (f-i)}.$$

They exist only for $i \leq f-1$. The variance for $f \geq 3$ is $\sigma^2 = f/(f-2)$, the excess for $f \geq 5$ is $\gamma_2 = 6/(f-4)$, disappearing for large f , in agreement with the fact that the distribution approaches the normal distribution.

The typical field of application for the t distribution is the derivation of tests or confidence intervals in cases where a sample is supposed to be taken from a normal distribution of unknown variance but known mean μ . Qualitatively, very large absolute values of t indicate that the sample mean is incompatible with μ . Sometimes the t distribution is used to approximate experimental distributions which differ from Gaussians because they have longer tails. In a way, the t distribution interpolates between the Cauchy (for $f = 1$) and the Gauss distribution (for $f \rightarrow \infty$).

3.6.12 The Extreme Value Distributions

The family of extreme value distributions is relevant for the following type of problem: Given a sample taken from a certain distribution, what can be said about the distribution of its maximal or minimal value? It is found that these distributions converge with increasing sample size to distributions of the types given below.

The Weibull Distribution

This distribution has been studied in connection with the lifetime of complex aggregates. It is a limiting distribution for the minimal member of a sample taken from a distribution limited from below. The p.d.f. is

$$f(x|a, p) = \frac{p}{a} \left(\frac{x}{a}\right)^{p-1} \exp\left(-\left(\frac{x}{a}\right)^p\right), \quad x > 0 \quad (3.58)$$

with the positive scale and shape parameters a and p . The mode is

$$x_m = a \left(\frac{p-1}{p}\right)^{1/p} \quad \text{for } p \geq 1,$$

mean value and variance are

$$\begin{aligned} \mu &= a\Gamma(1 + 1/p), \\ \sigma^2 &= a^2 \left(\Gamma(1 + 2/p) - \Gamma^2(1 + 1/p) \right). \end{aligned}$$

The moments are

$$\mu_i = a^i \Gamma(1 + i/p).$$

For $p = 1$ we get an exponential distribution with decay constant $1/a$.

The Fisher–Tippett Distribution

Also this distribution with the p.d.f.

$$f_{\pm}(x|x_0, s) = \frac{1}{s} \exp\left(\pm \frac{x - x_0}{s} - e^{\pm(x - x_0)/s}\right)$$

belongs to the family of extreme value distributions. It is sometimes called extreme value distribution (without further specification) or log-Weibull distribution.

If y is Weibull-distributed (3.58) with parameters a, p , the transformation to $x = -\ln y$ leads for x to a log-Weibull distribution with parameters $x_0 = -\ln a$ and $s = 1/p$. The first of these, the location parameter x_0 , gives the position of the maximum, i.e. $x_{mod} = x_0$, and the parameter $s > 0$ is a scale parameter. Mean value μ and variance σ^2 depend on these parameters through

$$\begin{aligned} \mu &= x_0 \mp Cs, \quad \text{with Euler's constant } C = 0.5772\dots, \\ \sigma^2 &= s^2 \frac{\pi^2}{6}. \end{aligned}$$

Mostly, the negative sign in the exponent is realized. Its normal form

$$f(x|0, 1) = \exp(-x - e^{-x})$$

is also known as Gumbel's distribution and shown in Fig. 3.24.

Using mathematical properties of Euler's Γ function [22] one can derive the characteristic function in closed form:

$$\phi(t) = \Gamma(1 \pm ist) e^{ix_0 t},$$

whose logarithmic derivatives give in turn the cumulants for this distribution:

$$\kappa_1 = x_0 \mp Cs, \quad \kappa_{i \geq 2} = (\mp 1)^i (i-1)! s^i \zeta(i),$$

with Riemann's zeta function $\zeta(z) = \sum_{n=1}^{\infty} 1/n^z$. (see [22]). Skewness and excess are given by $\gamma_1 \approx 1.14$ and $\gamma_2 = 12/5$.

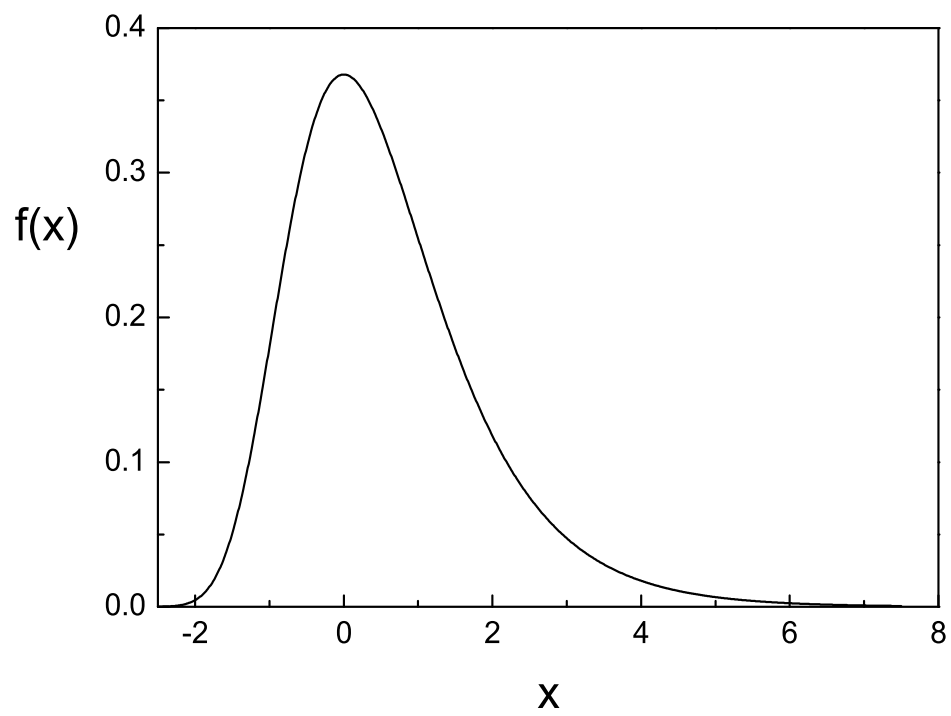


Fig. 3.24. Log-Weibull distribution.

Measurement errors

4.1 General Considerations

When we talk about measurement errors, we do not mean mistakes caused by the experimenter, but the unavoidable random dispersion of measurements. Therefore, a better name would be measurement uncertainties. We will use the terms *uncertainty* and *error* synonymously.

The correct determination and treatment of measurement errors is not always trivial. In principle, the evaluation of parameters and their uncertainties are part of the statistical problem of parameter inference, which we will treat in Chaps. 6, 7 and 8. There we will come back to this problem and look at it from a more general point of view. In the present chapter we will introduce certain, in practice often well justified approximations.

Official recommendations are given in “Guide to the Expression of Uncertainty of Measurement”, published in 1993 and updated in 1995 in the name of many relevant organizations like ISO and BIMP (Guide to the Expression of Uncertainty of Measurement, International Organization for Standardization, Geneva, Switzerland) [23]. More recently, a task force of the European cooperation for Accreditation of Laboratories (EAL) with members of all western European countries has issued a document (EAL-R2) with the aim to harmonize the evaluation of measurement uncertainties. It follows the rules of the document mentioned above but is more specific in some fields, especially in calibration issues which are important when measurements are exchanged between different laboratories. The two reports essentially recommend to estimate the expected value and the standard deviation of the quantity to be measured. Our treatment of measurement uncertainty will basically be in agreement with the recommendations of the two cited documents which deal mainly with systematic uncertainties and follow the Bayesian philosophy, but we will extend their concept in Sect. 8.2 where we introduce asymmetric error limits.

4.1.1 Importance of Error Assignments

The natural sciences owe their success to the possibility to compare quantitative hypotheses to experimental facts. However, we are able to check theoretical predictions only if we have an idea about the accuracy of the measurements. If this is not the case, our measurements are completely useless.

Of course, we also want to compare the results of different experiments to each other and to combine them. Measurement errors must be defined in such a way that this is possible without knowing details of the measurement procedure. Only then, important parameters, like constants of nature, can be determined more and more accurately and possible variations with time, like it was hypothesized for the gravitational constant, can be tested.

Finally, it is indispensable for the utilization of measured data in other scientific or engineering applications to know their accuracy and reliability. An overestimated error can lead to a waste of resources and, even worse, an underestimated error may lead to wrong conclusions.

4.1.2 Verification of Assigned Errors

In some situations a system of variables is overconstrained and thus allows us to check whether our measurements are consistent within their error limits. An example is the comparison of the sum of measured angles of a triangle with the value 180° which is common in surveying. In the experiments of particle physics we can apply among other laws the constraints provided by energy and momentum conservation. When we adjust curves, e.g. a straight line to measured points, the deviations of the points from the line permit us to check the goodness of the fit, and if the fit is poor, we might reject the presumed parametrization or revise the error assignment. A systematic treatment of the corresponding goodness-of-fit tests will be presented in Chap. 10.

4.1.3 The Declaration of Errors

There are several ways to present measurements with their uncertainties. Some of the more frequent ones are given in the following examples:

$$\begin{aligned} t &= (34.5 \pm 0.7) 10^{-3} \text{ s} \\ t &= 34.5 10^{-3} \text{ s} \pm 2\% \\ x &= 10.3^{+0.7}_{-0.3} \\ m_e &= (0.510\,999\,06 \pm 0.000\,000\,15) \text{ MeV}/c^2 \\ m_e &= 0.510\,999\,06 (15) \text{ MeV}/c^2 \\ m_e &= 9.109\,389\,7 10^{-31} \text{ kg} \pm 0.3 \text{ ppm} \end{aligned}$$

The abbreviation *ppm* means *parts per million*. The treatment of asymmetric errors will be postponed to Chap. 8. The measurement and its error must have the same number of significant digits. Declarations like $x = 3.2 \pm 0.01$ or $x = 3.02 \pm 0.1$ are inconsistent.

A relatively crude declaration of the uncertainty is sufficient, one or two significant digits are adequate in any case, keeping in mind that often we do not know all sources of errors or are unable to estimate their influence on the result with high accuracy¹. This fact also justifies in most cases the approximations which we have to apply in the following.

We denote the error of x with δx or δ_x . Sometimes it is convenient, to quote dimensionless relative errors $\delta x/x$ that are useful in error propagation – see below.

¹There are exceptions to this rule in hypothesis testing (see Chap. 10).

4.1.4 Definition of Measurement and its Error

Measurements are either quantities read from a measurement device or simply an instrument – we call them input quantities – or derived quantities, like the average of two or more input quantities, the slope of a street, or a rate which are computed from several input quantities. Let us first restrict ourselves to input quantities. An input quantity can be regarded as an *observation*, i.e. a random variable x drawn from a distribution centered around the true value x_t of the quantity which we want to determine. The measurement process, including the experimental setup, determines the type of this distribution (Gauss, Poisson, etc.) For the experimenter the true value is an unknown parameter of the distribution. The measurement and its error are estimates of the true value and of the standard deviation of the distribution². This definition allows us to apply relations which we have derived in the previous chapter for the standard deviation to calculations of the uncertainty, e.g. the error δ of a sum of independent measurements with individual errors δ_i is given by $\delta^2 = \sum \delta_i^2$.

In an ideal situation the following conditions are fulfilled:

1. The mean value of infinitely often repeated measurements coincides with the true value, i.e. the true value is equal to the expectation value $\langle x \rangle$ of the measurement distribution, see Sect. 3.2. The measurement is then called unbiased.
2. The assigned measurement error is independent of the measured value.

These properties can not always be realized exactly but often they are valid to a sufficiently good approximation. The following two examples refer to asymmetric errors where in the first but not in the second the asymmetry can be neglected.

Example 43. Scaling error

A tape measure is slightly elastic. The absolute measurement error increases with the measured length. Assuming a scaling error of 1% also the estimate of the error of a measured length would in average be wrong by 1% and asymmetric by the same proportion. This, however, is completely unimportant.

Example 44. Low decay rate

We want to measure the decay rate of a radioactive compound. After one hour we have recorded one decay. Given such small rates, it is not correct to compute the error from a Poisson distribution (see Sect. 3.6.3) in which we replace the mean value by the observed measurement. The declaration $R = 1 \pm 1$ does not reflect the result correctly because $R = 0$ is excluded by the observation while $R = 2.5$ on the other hand is consistent with it.

In Sect. 8.2 we will, as mentioned above, also discuss more complex cases, including asymmetric errors due to low event rates or other sources.

Apart from the definition of a measurement and its error by the estimated mean and standard deviation of the related distribution there exist other conventions: Distribution median, maximal errors, width at half maximum and confidence intervals. They are useful in specific situations but suffer from the crucial disadvantage that

²Remark that we do not need to know the full error distribution but only its standard deviation.

they are not suited for the combination of measurement or the determination of the errors of depending variables, i.e. error propagation.

4.2 Different Types of Measurement Uncertainty

There are uncertainties of different nature: *statistical or random errors* and *systematic errors*. Their definitions are not unambiguous, disagree from author to author and depend somewhat on the scientific discipline in which they are treated. Also the authors of this book have different opinions regarding the meaning of systematic errors. Hence, we have decided to present two parallel subsections for this topic.

4.2.1 Statistical Errors

Errors Following a Known Statistical Distribution

Relatively simple is the interpretation of measurements if *the distributions of the errors follow known statistical laws*³. The corresponding uncertainties are called *statistical errors*. Examples are the measurement of counting rates (Poisson distribution), counter efficiency (binomial distribution) or of the lifetime of unstable particles (exponential distribution). Characteristic for statistical errors is that sequential measurements are uncorrelated and thus the precision of the combined results is improved by the repetition of the measurement. In these cases the distribution is known up to a parameter – its expected value. We then associate the actually observed value to that parameter and declare as measurement error the standard deviation belonging to that distribution.

Example 45. Poisson distributed rate

Recorded have been $N = 150$ decays. We set the rate and its error equal to $Z = N \pm \sqrt{N} = (150 \pm \sqrt{150}) \approx 150 \pm 12$.

Example 46. Digital measurement (uniform distribution)

With a digital clock the time $t = 237$ s has been recorded. The error is $\delta t = 1/\sqrt{12}$ s ≈ 0.3 s, thus $t = (237.0 \pm 0.3)$ s.

Example 47. Efficiency of a detector (binomial distribution)

From $N_0 = 60$ particles which traverse a detector, 45 are registered. The efficiency is $\varepsilon = N/N_0 = 0.75$. The error derived from the binomial distribution is

$$\delta\varepsilon = \delta N/N_0 = \sqrt{\varepsilon(1-\varepsilon)/N_0} = \sqrt{0.75 \cdot 0.25/60} = 0.06 .$$

Example 48. Calorimetric energy measurement (normal distribution)

The energy of an high energy electron is measured by a scintillating fiber calorimeter by collecting light produced by the electromagnetic cascade in

³G. Bohm's definition is slightly different, i.e. more restrictive, see examples at the end of this section.

the scintillator of the device. From the calibration of the calorimeter with electrons of known energies E we know that the calorimeter response is well described by a Gaussian with mean proportional to E and variance proportional to $\sqrt{E}$.

Many experimental signals follow to a very good approximation a normal distribution. This is due to the fact that they consist of the sum of many contributions and a consequence of the central limit theorem.

A typical property of statistical errors is that the relative error is proportional to $1/\sqrt{N}$, the inverse of the square root of the number of events. This is obvious in the first and third example, but also in the second example the relative error would show the $1/\sqrt{N}$ behavior if we were able to repeat the measurement N times. If the relative error is sufficiently small, we usually can treat it as independent of the observed value.

Errors Determined from a Sample of Measurements

An often used method for the estimation of errors is to repeat a measurement several times and to estimate the error from the fluctuation of the results. The results presented below will be justified in subsequent chapters but are also intuitively plausible.

In the simplest case, for instance in calibration procedures, the true value x_t of the measured quantity x is known, and the measurement is just done to get information about the accuracy of the measurement. An estimate of the average error δx of x from N measurements is in this case

$$(\delta x)^2 = \frac{1}{N} \sum_{i=1}^N (x_i - x_t)^2.$$

We have to require that the fluctuations are purely statistical and that correlated systematic variations are absent, i.e. the data have to be independent from each other. The relative uncertainty of the error estimate follows the $1/\sqrt{N}$ law. It will be studied below. For example with 100 repetitions of the measurement, the uncertainty of the error itself is reasonably small, i.e. about 10 % but depends on the distribution of x .

When the true value is unknown, we can approximate it by the sample mean $\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$ and use the following recipe:

$$(\delta x)^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2. \quad (4.1)$$

In the denominator of the formula used to determine the mean quadratic deviation $(\delta x)^2$ of a single measurement figures $N-1$ instead of N . This is plausible because, when we compute the empirical mean value $\bar{x}$, the measurements x_i enter and thus they are expected to be in average nearer to their mean value than to the true value. In particular the division by N would produce the absurd value $\delta x = 0$ for $N = 1$, while the division by $N-1$ yields an indefinite result. The derivation of (4.1) follows from (3.15). The quantity $(\delta x)^2$ in (4.1) is sometimes called *empirical variance*. We have met it already in Sect. 3.2.3 of the previous chapter.

Table 4.1. Error estimate for a mean value.

measurements x_i	quadratic deviations $(x_i - \bar{x})^2$
2.22	0.0009
2.25	0.0000
2.30	0.0025
2.21	0.0016
2.27	0.0004
$\sum x_i = 11.25$ $\bar{x} = 2.25$	$\sum (x_i - \bar{x})^2 = 0.0054$ $(\delta x)^2 = \sum (x_i - \bar{x})^2 / 4 = 0.0013$

Frequently, we want to find the error for measurements x_i which are constrained by physical or mathematical laws and where the true values are estimated by a parameter fit (to be explained in subsequent chapters). The expression (4.1) then is generalized to

$$(\delta x)^2 = \frac{1}{N - Z} \sum_{i=1}^N (x_i - \hat{x}_i)^2 . \quad (4.2)$$

where $\hat{x}_i$ are the estimates of the true values corresponding to the measurements x_i and Z is the number of parameters that have been adjusted using the data. When we compare the data of a sample to the sample mean we have $Z = 1$ parameter, namely $\bar{x}$, when we compare coordinates to the values of a straight line fit then we have $Z = 2$ free parameters to be adjusted from the data, for instance, the slope and the intercept of the line with the ordinate axis. Again, the denominator $N - Z$ is intuitively plausible, since for $N = Z$ we have 2 points lying exactly on the straight line which is determined by them, so also the numerator is zero and the result then is indefinite.

Relation (4.2) is frequently used in particle physics to estimate momentum or coordinate errors from empirical distributions (of course, all errors are assumed to be the same). For example, the spatial resolution of tracking devices is estimated from the distribution of the residuals $(x_i - \hat{x}_i)$. The individual measurement error δx as computed from a M tracks and N points per track is then estimated quite reliably to

$$(\delta x)^2 = \frac{1}{(N - Z)M} \sum_{i=1}^{M \times N} (x_i - \hat{x}_i)^2 .$$

Not only the precision of the error estimate, but also the precision of a measurements can be increased by repetition. The error $\delta \bar{x}$ of a corresponding sample mean is, following the results of the previous section, given by

$$\begin{aligned} (\delta \bar{x})^2 &= (\delta x)^2 / N , \\ &= \frac{1}{N(N - 1)} \sum_{i=1}^N (x_i - \bar{x})^2 . \end{aligned} \quad (4.3)$$

Example 49. Average from 5 measurements

In Table 4.1 five measurements are displayed. The resulting mean value is $\bar{x} = 2.25 \pm 0.02$. We have used that the error of the mean value is smaller by

the factor $\sqrt{5}$ than that of a single measurement, $\delta\bar{x} = \delta x/\sqrt{5}$. With only 5 repetitions the precision of the error estimate is rather poor.

Our recipe yields $\delta\bar{x} \sim 1/\sqrt{N}$, i.e. the error becomes arbitrarily small if the number of the measurements approaches infinity. The validity of the $1/\sqrt{N}$ behavior relies on the assumption that the fluctuations are purely statistical and that correlated systematic variations are absent, i.e. the data have to be independent of each other. When we measure repeatedly the period of a pendulum, then the accuracy of the measurements can be deduced from the variations of the results only if the clock is not stopped systematically too early or too late and if the clock is not running too fast or too slow. Our experience tells us that some correlation between the different measurements usually cannot be avoided completely and thus there is a lower limit for $\delta\bar{x}$. To obtain a reliable estimate of the uncertainty, we have to take care that the systematic uncertainties are small compared to the statistical error $\delta\bar{x}$.

Error of the Empirical Variance

Sometimes we are interested in the variance of an empirical distribution and in its uncertainty. In the same category falls the problem to estimate *the error of the error* of a parameter which is determined from a series of measurements. For example, we may need to know the resolution of a meter or the width of a spectral line and the related accuracy. It is also of interest to know how often a calibration measurement has to be performed to estimate the corresponding error with sufficient accuracy. In these situations the variance s^2 itself is the result of the investigation to which we would like to associate an uncertainty.

The variance of $(x - \mu)^2$ for a given distribution is easily calculated using the formulas of Sect. 3.2.3. We omit the details of the calculation and quote the result which is related of the second and fourth central moments.

$$\begin{aligned}\text{var}[(x - \mu)^2] &= \left\langle [(x - \mu)^2 - \sigma^2]^2 \right\rangle \\ &= (\mu'_4 - \sigma^4) .\end{aligned}$$

We now assume that our sample is large and replace the distribution moments μ'_n by the empirical central moments m'_n ,

$$m'_n = \frac{1}{N} \sum (x_i - \bar{x})^n .$$

The moment $s^2 = m'_2$ is an estimate for σ^2 . For N events in the sample, we get for the uncertainty δs^2 of s^2

$$(\delta s^2)^2 = \frac{m'_4 - m'^2_2}{N}$$

and from error propagation (see next section 4.3) we derive the uncertainty of s itself

$$\begin{aligned}\frac{\delta s}{s} &= \frac{1}{2} \frac{\delta s^2}{s^2} , \\ &= \frac{1}{2\sqrt{N}} \frac{\sqrt{m'_4 - s^4}}{s^2} .\end{aligned}$$

If the type of distribution is known, we can use relations between moments. Thus, for the normal distribution we have $\mu'_4 = 3\sigma^4$ (see Sect. 3.6.5), and it follows

$\delta s/s = 1/\sqrt{2N}$ which also follows from the variance of the χ^2 distribution. This relation sometimes is applied to arbitrary distributions. It then often underestimates the uncertainty.

4.2.2 Systematic Errors (G. Bohm)

A stringent requirement on scientific results is the possibility of verification or refutation. In case of experiments, the most direct way, indispensable in case of crucial experiments, is their *independent* repetition. Experimental results have to be presented in a way which allows this: First of all, any experimental number has to be given together with its statistical error. Only then we can state “agreement within errors” and combine the results to a more accurate one, or, not so seldom as one might wish, we find the result “unconfirmed” and the experimental situation “still unclear”.

A cornerstone in such comparisons is the assumption of statistical independence: Statistical fluctuations in the two experiments “have nothing to do with each other”. Anticipating later developments on error propagation for independent variates, we compare the squared difference of the two results, $|x_1 - x_2|^2$ with the variance of $x_1 - x_2$, given by

$$\text{var}(x_1 - x_2) = \text{var}(x_1) + \text{var}(x_2) - 2\text{cov}(x_1, x_2)$$

or in terms of the covariance matrix, see Chap. 3, Sect. 3.5.6

$$\text{var}(x_1 - x_2) = C_{11} + C_{22} - 2C_{12} .$$

Using independence of the x_i , we have $C_{12} = 0$ and the diagonal elements of $\mathbf{C}$ are the squared errors $\delta x_{1,2}^2$; so, confirmation means

$$|x_1 - x_2|^2 \leq k(C_{11} + C_{22}) \quad (4.4)$$

with k of order unity⁴.

The next step is usually the combination of the two results as a weighted mean. This will be explained in the next section; here we anticipate that the error of the combined result will be smaller than the minimum of δx_1 and δx_2 . So, under the assumption of independence, there is no lower limit to the statistical error of repeatable experiments.

Of course, in reality there is always a limit on statistical accuracy, besides the trivial one that time for repetitions is limited: The assumption of independence is violated in a very specific way by so-called *systematic errors* which appear in any realistic experiment. Postponing an abstract definition we first give some examples to illustrate the usage of this term in physical applications.

1. Reading error of an analog instrument: The readings performed by an observer will not average to the true value, but show a personal bias. This effect may be estimated by comparing the results of several observers.

⁴This is of course a rough statement, to be qualified by probabilistic statements if required.

2. Calibration errors of the measurement apparatus: The art of measuring with analog instruments is merely history, since digital displays replaced them. Corrections, like zero shifts, non-linearities etc. are usually taken care of already in the displayed value, but the applier should know, besides the resolution, also the limit of accuracy of the average result for a large sample.
3. Errors of auxiliary quantities: Besides calibration constants, many other physical quantities will influence the final result, all being subject to measurement errors (particle characteristics, former experimental results...)
4. Errors of auxiliary quantities of theoretical origin: Sometimes it is not possible to reach the required accuracy in theoretical calculations of such quantities. Also then an uncertainty should be given, corresponding to a variance in a Bayesian sense.
5. Uncertainties in the definition of the quantity to be measured: A measurement result usually depends, besides from the quantity one wants to measure, on surrounding conditions like temperature or pressure. The size of a body is well-defined only for certain values of these parameters.
6. Errors of parameters determining the statistical model describing the data: Often the parameter of interest is found by adapting a model which depends on this parameter, to the data. Usually such model depends also on other, uninteresting, parameters, called nuisance parameters in statistics, which often cannot be derived from the data alone, but need external input with unavoidable uncertainties.

The above items are neither non-overlapping nor complete. The following is an attempt to give a more formal characterization of systematic errors, meeting, hopefully, the general usage of this denotation.

Let us consider a measured quantity x with statistical error δx . To describe a realistic situation, the final, corrected result $\tilde{x}$ is a function u of x and other, generally also uncertain, auxiliary parameters $a, b, \dots$. Then a first measurement of x gives

$$\tilde{x}_1 = u(x_1, a, b, \dots) .$$

A repeated measurement, presumably under the same conditions, of x yields x_2 and

$$\tilde{x}_2 = u(x_2, a, b, \dots)$$

with *the same unchanged values*⁵ of $a, b, \dots$. Taylor-expanding u around the expected values $\langle x \rangle, \langle a \rangle, \langle b \rangle \dots$ and writing $\Delta x_i = x_i - \langle x \rangle$, $u_x = \partial u / \partial x$ etc., we get (in linear approximation)

$$\Delta \tilde{x}_i = u_x \Delta x_i + u_a \Delta a + u_b \Delta b + \dots .$$

Assuming, for simplicity, all variates on the right hand side as uncorrelated, we find, after squaring and taking expectation values

$$\begin{aligned} \text{var}(\tilde{x}_i) &= u_x^2 \text{var}(x_i) + u_a^2 \text{var}(a) + u_b^2 \text{var}(b) + \dots , \\ \text{cov}(\tilde{x}_1, \tilde{x}_2) &= u_a^2 \text{var}(a) + u_b^2 \text{var}(b) + \dots . \end{aligned}$$

The corrected covariance matrix is no longer diagonal. The results of repetitions are not independent, just because they are subject to the *same* “systematic” effects from fluctuations of $a, b, \dots$:

⁵Besides x_1, x_2 also their statistical errors may differ.

$$\tilde{C} = u_x^2 \begin{pmatrix} \text{var}(x_1) & 0 \\ 0 & \text{var}(x_2) \end{pmatrix} + [u_a^2 \text{var}(a) + u_b^2 \text{var}(b) + \dots] \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

This covariance matrix describes the uncertainty of a doubled measurement completely. The second matrix (of rank unity) expresses the fact that the two measurements are totally positively correlated with respect to the fluctuations of the auxiliary parameters $a, b, \dots$. The square root of the term in square brackets is called systematic error and customarily given separately from the “statistical” error $\delta x_i^{\text{stat}} = \sqrt{\text{var}(x_i)}$. Even more recommendable is the practice to give the list of the systematic errors taken into account, i.e. the values $a \pm \delta a$, $u_a \delta a$, the same for b , and so on. The main advantage of this separation is that it gives the information on how far the accuracy can be enhanced by repetitions of the experiment. A second argument against mixing up statistical and systematic errors is the fact that in practice the latter are often less well known.

Coming back to the comparison of two independent measurements of x under “apparently the same conditions” (4.4), we see that this criterium remains the same, whether or not the identical systematic errors are taken into account. When combining the results, however, the systematic error stays constant. A nasty consequence from this is that systematic errors not taken into account do not signal their existence by larger fluctuations of the data. It is an important part of experimental work to check carefully for any source of systematic errors, and to document every parameter of potential influence.

4.2.3 Systematic Errors (G. Zech)

Introduction and Definition

Experiments in nuclear and particle physics usually extract the information from a statistical data sample. The precision of the results then is mainly determined by the number N of collected reactions. Besides the corresponding well defined *statistical errors*, nearly every measurement is subject to further uncertainties, the *systematic errors*, typically associated with auxiliary parameters related to the measuring apparatus, or with model assumptions. The result is typically presented in the form $x = 2.34 \pm 0.06 = 2.34 \pm 0.05(\text{stat.}) \pm 0.03(\text{syst.})$.

The only reason for the separate quotation of the two uncertainties is that the size of the systematic uncertainties is less well known than that of the purely statistical error. Thus, for example, excluding a prediction by say a 4 standard deviation measurement where the errors are dominantly of systematic type is certainly less convincing than if the result is purely statistical.

To illustrate the different quality of purely statistical errors and systematic errors, imagine we read a pico-ampere meter. The needle is quivering and sometimes large amplitudes occur due to pick-up. Observing the movement of the needle for some time, we will be able to estimate the current and to guess the uncertainty. Repeating the measurement many times will not sizably reduce the error. A tape-measure may have expanded or shrunk due to temperature effects. The corresponding uncertainty can be estimated roughly from the estimated range of temperature variations and the known expansion coefficient of the tape material if it is made out of metal. It may also be guessed from previous experience. Characteristic of most systematic errors is that repeated measurements are generally correlated and thus the accuracy

is not improved by a repetition. However, there are other systematic errors where a repetition of a measurement is impossible, for instance when an auxiliary parameter is taken from a technical data sheet or when the measurement has happened in the past. It may happen that we have to derive a parameter from two or three independent and thus uncorrelated observations following an unknown distribution. For instance, the current of a magnet may have been measured at the beginning and at the end of an experiment. The variation of the current introduces an error for the momentum measurement of charged particles. The estimate of the uncertainty from only two measurements will be rather vague and thus the error is of type *systematic*.

From these considerations follow the definitions:

Statistical errors are determined by objective statistical procedures. They follow a known distribution like a Poisson rate or are determined empirically from the distribution of an unbiased, sufficiently large sample. They show the typical reduction of the error with $1/\sqrt{N}$ where N is the sample size.

On the contrary, *systematic errors* cannot be calculated solely from sampling fluctuations and *are at least partially based on assumptions made by the experimenter, are model dependent or follow unknown distributions.* Thus, the systematic error and / or its distribution are only vaguely known.

Most systematic errors do not share the $1/\sqrt{N}$ law of the statistical errors, but there are exceptions and many systematic errors can be reduced with increasing statistics of an experiment.

Systematic errors arise in most experiments. They are especially important in high precision measurements like those of the magnetic dipole moment of the muon or of the CP violation constants in the neutral kaon system.

Examples of Systematic Errors

Systematic uncertainties arise typically when instruments are poorly calibrated, or are influenced, in a way not known exactly, by the environment (temperature, pressure. . .), or if thresholds or base levels are not constant in time. Besides uncertainties regarding the measuring instruments, often the knowledge or the understanding of parts of experimental setup, its acceptance and the sensitivity of detector components is incomplete. These effects and only partially known background contributions lead to systematic errors. Sometimes the data analysis requires additional experimental input from other experiments or theoretical results have to be used, where especially the latter may have uncertainties of unclear nature.

In some of these examples there is no “ N ”, i.e. there is no possibility to produce several independent guesses, or a repetition of the measurement would not produce an independent result. It does not make sense to repeat the estimate of theoretical uncertainties arising for instance from neglecting higher order terms in an expansion. Sometimes the reason of the uncertainty lies in the past, the failure of some electronic part or the physicist has forgotten to record the atmospheric pressure which influences some calibration constant. Other uncertainties like those of background subtraction can be reduced with increasing statistics, but nevertheless there remains some systematic uncertainty from an extrapolation. Often the measurement of an interesting parameter is correlated with the value of another uninteresting parameter. We call it a nuisance parameter. The uncertainty of the nuisance parameter can

induce a systematic uncertainty on the interesting parameter which in first order decreases like $1/\sqrt{N}$ with the number of observations.

How to Detect and Avoid Systematic Errors

In order to detect and to estimate systematic errors, experience, common sense, and intuition is needed. A general advice is to try to suppress them as far as possible already by an appropriate design of the experiment and to include the possibility of control measurements, like regular calibration. Since correlation of repeated measurement is characteristic for the presence of systematic errors, observed correlations of results with parameters related to the systematic effects provide the possibility to estimate and reduce the latter. In the pendulum example discussed above the systematic contribution to the error due to a possible unknown bias in the stopping procedure can be estimated by studying the result as a function of the number of periods and reduced by increasing the measurement time. In particle physics experiments where usually only a fraction of events is accepted by some filtering procedure, it is advisable to record also a fraction of those events that are normally rejected (down-scaling) and to try to understand their nature. Some systematic effects are related to the beam intensity, thus a variation of the beam intensity helps to study them.

How can we detect systematic errors caused for instance by background subtraction or efficiency corrections at the stage of data analysis? A widely used method is the investigation of the results as a function of the selection criteria. A correlation of the interesting parameter with the value of a cut-off parameter in a certain variable is a clear indication for the presence of systematic errors. It is evident though that the systematic errors then have to be much larger than the normal statistical fluctuations in order to be detected. Obviously, we want to discriminate also systematic errors which are of the same order of magnitude as the statistical ones, preferably much smaller. Therefore we have to investigate samples, where the systematic effects are artificially enhanced. If we suspect rate dependent distortion effects as those connected with dead times, it is recommended to analyze a control sample with considerably enhanced rate. When we eliminate a background reaction by a selection criterion, we should investigate its importance in the region which has been excluded, where it is supposed to be abundant.

Frequently made mistakes are: 1. From the fact that the data are consistent with the absence of systematic errors, it is supposed that they do not exist. This leads always to underestimation of systematic errors. 2. The changes of the results found by changing the selection criteria are directly converted into systematic errors. This in most cases leads to overestimates, because the variations are partially due to the normal statistical fluctuations.

Treatment of Systematic Errors

Because of the difficulty to specify systematic errors accurately, scientists are rather sceptical about results based on measurements which are dominated by systematic errors. In any case, it is indispensable to declare the systematic and the statistical contributions to the measurement error separately.

In many experiments there appears a quite large number – typically a dozen or so – of such systematic uncertainties. When we combine systematic errors (see Sect.

8.2), we can often profit from the central limit theorem (see Sect. 3.6.5) provided that they are all of the same order of magnitude. The distribution of the sum of variables suffering from systematic uncertainties approaches a normal distribution, with variance equal to the sum of variances of the contributing distributions. If the variables enter multiplicatively, the logarithms are additive and approach a normal distribution (see Sect. 8.2)

Sometimes systematic errors are combined linearly. There is no justification for such a procedure.

Interesting studies of systematic error are presented in [25, 26].

Remarks: The notations *statistical* and *systematic* errors as used by us are somewhat unfortunate⁶ but they are common in particle physics. In many other fields of physics errors which follow a precisely known distribution are absent and thus all errors are more or less of the same quality. In the corresponding literature a systematic error is associated with a faulty measurement, leading to measured values being systematically too high or too low with respect to the true value. This definition is different from ours where *error* always means *uncertainty*. To avoid, and if this is not possible, to detect systematic biases is an important part of experimentation but systematic biases are not subject to a statistical treatment. If they are known they are corrected for and need not be discussed. If they are unknown, we cannot account for them. Some authors [26] implicitly define systematic errors as a systematic shift of unknown direction to be opposed to a random error. This denotation is due to a frequentist view, where probabilities of constant parameters do not exist. Nevertheless, in order to handle systematic errors also in this scheme a variance is attributed to the systematics and the errors are added in quadrature. Thus the result is the same as in our treatment. Systematic errors can only be treated in the Bayesian approach. But contrary to the pure Bayesian philosophy which does not distinguish between the two kind of errors, we stress the importance to differentiate between uncertainties which are due to a random process with known distribution and errors which are partially based on vague information or subjective input. We have to admit though that there are cases where a clear classification is difficult.

In part of the particle physics literature all error contributions from secondary parameters which influence the parameter of interest, independent of their origin are called systematic errors. This classification has the difficulty that in most cases the parameter of interest is a complex function of many input variables and then the error would be solely of the systematic type and a reason to distinguish between statistical and systematic errors would no longer exist.

In Ref. [25] purely statistical uncertainties related to detector effects or secondary measurements are called *class 1 systematic errors*, but the author states that a classification of these uncertainties as statistical errors would be more informative. He subdivides further the real systematic errors following his definition of systematic errors (which coincides with ours), into systematic errors related to experimental effects (*class 2*) and those depending on theoretical models (*class 3*). This distinction makes sense, because our possibilities to reduce, detect and estimate class 2 and class 3 errors are very different.

⁶All quantities which are not exactly known are random variables and thus have statistical uncertainties.

4.2.4 Controversial Examples

The main difference in our points of view are illustrated in the two following examples:

1. An experiment measures a parameter A with a purely statistical error and publishes a ratio $R = A/B$ where B is a published number, also with a purely statistical error. The result has a statistical error only because it follows a known statistical law (G. Zech); has also a systematic error component due to the fact that B is an external parameter which, in case it might be used also in a second experiment, would lead to correlations (G. Bohm).
2. A cross section is measured as a function of energy, where the different rates A_i/B have a common normalization. The numerators A_i and the normalization B have statistical errors only. The result has correlated statistical errors and no systematic error (G. Zech); has a systematic error component (G. Bohm).

We both agree, that parameters where the errors are not well known or depend on assumptions introduce systematic errors. While G. Bohm stresses the correlation aspect as characteristic of systematic errors, G. Zech emphasizes the reliability of the error estimation as relevant for the distinction of the two types of errors.

4.3 Linear Propagation of Errors

4.3.1 Error Propagation

We now want to investigate how a measurement error propagates into quantities which are functions of the measurement. We consider a function $y(x)$, a measurement value $x_m \pm \delta x$, with the standard deviation δx , and are interested in y_m , the corresponding measurement of y and its error δy . If the p.d.f. $f(x)$ is known, we can determine the p.d.f. of y , its expectation value y_m and the standard deviation δy by an analytic or numerical transformation of the variables, as introduced above in Chap. 3. We will assume, however, that the measurement error is small enough to justify the approximation of the function by a linear expression within the error limits. Then we need not know the p.d.f. $f(x)$.

We use the Taylor expansion of y around x_m :

$$y = y(x_m) + y'(x_m)\Delta x + \frac{1}{2!}y''(x_m)(\Delta x)^2 + \dots$$

We neglect quadratic and higher order terms, set y_m equal to the expected value of y , and $(\delta y)^2$ equal to the expected value of the squared deviation. According to the definition, the expected value of $\Delta x = x - x_m$ is zero, and that of $(\Delta x)^2$ equals $(\delta x)^2$. (In our notation quantities denoted by δ are expected values, i.e. fixed positive parameters, while Δx is a random variable). We get

$$\begin{aligned} y_m &= \langle y(x) \rangle \\ &\approx \langle y(x_m) \rangle + \langle y'(x_m)\Delta x \rangle = y(x_m), \end{aligned}$$

and

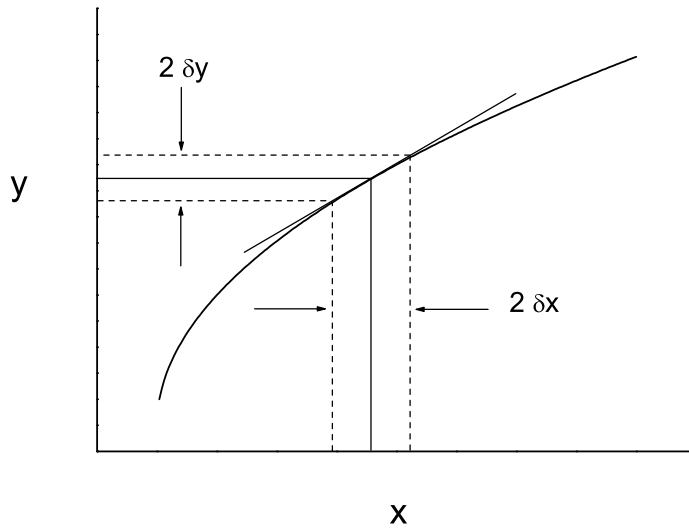


Fig. 4.1. Linear error propagation.

$$\begin{aligned}
 (\delta y)^2 &= \langle (y - y_m)^2 \rangle \\
 &\approx \langle (y(x_m) + y'(x_m)\Delta x - y_m)^2 \rangle \\
 &= y'^2(x_m) \langle (\Delta x)^2 \rangle \\
 &= y'^2(x_m) (\delta x)^2, \\
 \delta y &= |y'(x_m)| \delta x.
 \end{aligned}$$

This result also could have been read off directly from Fig. 4.1.

Examples of the linear propagation of errors for some simple functions are compiled below:

Function :	Relation between errors :
$y = ax^n \Rightarrow$	$\frac{\delta y}{ y } = \frac{ n \delta x}{ x }$
$y = a \ln(bx) \Rightarrow$	$\delta y = \frac{ a \delta x}{ x }$
$y = ae^{bx} \Rightarrow$	$\frac{\delta y}{ y } = b \delta x$
$y = \tan x \Rightarrow$	$\frac{\delta y}{ y } = \frac{\delta x}{ \cos x \sin x }$

4.3.2 Error of a Function of Several Measured Quantities

Most physical measurements depend on several input quantities and their uncertainties. For example, a velocity measurement $v = s/t$ based on the measurements of length and time has an associated error which obviously depends on the errors of both s and t .

Let us first consider a function $y(x_1, x_2)$ of only two measured quantities with values x_{1m}, x_{2m} and errors $\delta x_1, \delta x_2$. With the Taylor expansion

$$y = y(x_{1m}, x_{2m}) + \frac{\partial y}{\partial x_1}(x_{1m}, x_{2m})\Delta x_1 + \frac{\partial y}{\partial x_2}(x_{1m}, x_{2m})\Delta x_2 + \dots$$

we get as above to lowest order:

$$\begin{aligned} y_m &= \langle y(x_1, x_2) \rangle \\ &= y(x_{1m}, x_{2m}) \end{aligned}$$

and

$$\begin{aligned} (\delta y)^2 &= \langle (\Delta y)^2 \rangle \\ &= \left(\frac{\partial y}{\partial x_1}\right)^2 \langle (\Delta x_1)^2 \rangle + \left(\frac{\partial y}{\partial x_2}\right)^2 \langle (\Delta x_2)^2 \rangle + 2\left(\frac{\partial y}{\partial x_1}\right)\left(\frac{\partial y}{\partial x_2}\right) \langle \Delta x_1 \Delta x_2 \rangle \\ &= \left(\frac{\partial y}{\partial x_1}\right)^2 (\delta x_1)^2 + \left(\frac{\partial y}{\partial x_2}\right)^2 (\delta x_2)^2 + 2\left(\frac{\partial y}{\partial x_1}\right)\left(\frac{\partial y}{\partial x_2}\right) R_{12} \delta x_1 \delta x_2, \end{aligned} \quad (4.5)$$

with the correlation coefficient

$$R_{12} = \frac{\langle \Delta x_1 \Delta x_2 \rangle}{\delta x_1 \delta x_2}.$$

In most cases the quantities x_1 and x_2 are uncorrelated. Then the relation (4.5) simplifies with $R_{12} = 0$ to

$$(\delta y)^2 = \left(\frac{\partial y}{\partial x_1}\right)^2 (\delta x_1)^2 + \left(\frac{\partial y}{\partial x_2}\right)^2 (\delta x_2)^2.$$

If the function is a product of independent quantities, it is convenient to use relative errors as indicated in the following example:

$$\begin{aligned} z &= x^n y^m, \\ \left(\frac{\delta z}{z}\right)^2 &= \left(n \frac{\delta x}{x}\right)^2 + \left(m \frac{\delta y}{y}\right)^2. \end{aligned}$$

It is not difficult to generalize our results to functions $y(x_1, \dots, x_N)$ of N measured quantities. We obtain

$$\begin{aligned} (\delta y)^2 &= \sum_{i,j=1}^N \left(\frac{\partial y}{\partial x_i} \frac{\partial y}{\partial x_j} R_{ij} \delta x_i \delta x_j \right) \\ &= \sum_{i=1}^N \left(\frac{\partial y}{\partial x_i} \right)^2 (\delta x_i)^2 + \sum_{i \neq j=1}^N \left(\frac{\partial y}{\partial x_i} \frac{\partial y}{\partial x_j} R_{ij} \delta x_i \delta x_j \right) \end{aligned}$$

with the correlation coefficient

$$\begin{aligned} R_{ij} &= \frac{\langle \Delta x_i \Delta x_j \rangle}{\delta x_i \delta x_j}, \\ R_{ij} &= R_{ji}, \\ R_{ii} &= 1. \end{aligned}$$

Covariance Matrix

To simplify the notation, we introduce the covariance matrix $\mathbf{C}$

$$\mathbf{C} = \begin{pmatrix} \langle \Delta x_1 \Delta x_1 \rangle, & \langle \Delta x_1 \Delta x_2 \rangle, & \dots & \langle \Delta x_1 \Delta x_n \rangle \\ \langle \Delta x_2 \Delta x_1 \rangle, & \langle \Delta x_2 \Delta x_2 \rangle, & \dots & \langle \Delta x_2 \Delta x_n \rangle \\ \vdots & \vdots & & \vdots \\ \langle \Delta x_n \Delta x_1 \rangle, & \langle \Delta x_n \Delta x_2 \rangle, & \dots & \langle \Delta x_n \Delta x_n \rangle \end{pmatrix},$$

$$C_{ij} = R_{ij} \delta x_i \delta x_j$$

which, in this context, is also called error matrix. The covariance matrix by definition is positive definite and symmetric. The error δy of the dependent variable y is then given in linear approximation by

$$(\delta y)^2 = \sum_{i,j=1}^N \left(\frac{\partial y}{\partial x_i} \frac{\partial y}{\partial x_j} C_{ij} \right)$$

which can also be written in matrix notation as

$$(\delta y)^2 = \nabla \mathbf{y}^T \mathbf{C} \nabla \mathbf{y}.$$

For two variables with normally distributed errors following (3.49)

$$N(\Delta x_1, \Delta x_2) = \frac{1}{2\pi\delta_1\delta_2\sqrt{1-\rho^2}} \exp \left(-\frac{1}{2} \frac{\frac{(\Delta x_1)^2}{\delta_1^2} - 2\rho\frac{\Delta x_1\Delta x_2}{\delta_1\delta_2} + \frac{(\Delta x_2)^2}{\delta_2^2}}{1-\rho^2} \right) \quad (4.6)$$

we get

$$\mathbf{C} = \begin{pmatrix} \delta_1^2, & \rho\delta_1\delta_2 \\ \rho\delta_1\delta_2, & \delta_2^2 \end{pmatrix}.$$

Error Ellipsoids

Two-dimensional Gaussian error distributions like (4.6) (see Sect. 3.6.5) have the property that the curves of constant probability density are ellipses. Instead of $n\sigma$ error intervals in one dimension, we define $n\sigma$ error ellipses. The curve of constant probability density with density down by a factor of $\exp(-n^2/2)$ relative to the maximal density is the $n\sigma$ error ellipse.

For the error distribution in the form of (4.6) the error ellipse is

$$\frac{\frac{(\Delta x_1)^2}{\delta_1^2} - 2\rho\frac{\Delta x_1\Delta x_2}{\delta_1\delta_2} + \frac{(\Delta x_2)^2}{\delta_2^2}}{1-\rho^2} = n^2.$$

For uncorrelated errors the one standard deviation error ellipse is simply

$$\frac{(\Delta x_1)^2}{\delta_1^2} + \frac{(\Delta x_2)^2}{\delta_2^2} = 1.$$

In higher dimensions, we obtain ellipsoids which we better write in vector notation:

$$\nabla \mathbf{y}^T \mathbf{C} \nabla \mathbf{y} = n^2.$$

4.3.3 Averaging Uncorrelated Measurements

Important measurements are usually performed by various experiments in parallel, or are repeated several times. The combination of the results from various measurements should be performed in such a way that it leads to optimal accuracy. Under these conditions we can calculate a so-called *weighted* mean, with an error smaller than that of any of the contributing measurements. As usual, we assume that the individual measurements are independent.

As an example let us consider two measurements with measured values x_1, x_2 and errors δ_1, δ_2 . With the relations given in Sect. 3.2.3, we find for the error squared δ^2 of a weighted sum

$$\begin{aligned} x &= w_1 x_1 + w_2 x_2, \\ \delta^2 &= w_1^2 \delta_1^2 + w_2^2 \delta_2^2. \end{aligned}$$

Now we chose the weights in such a way that the error of the weighted sum is minimal, i.e. we seek for the minimum of δ^2 under the condition $w_1 + w_2 = 1$. The result is

$$w_i = \frac{1/\delta_i^2}{1/\delta_1^2 + 1/\delta_2^2}$$

and for the combined error we get

$$\frac{1}{\delta^2} = \frac{1}{\delta_1^2} + \frac{1}{\delta_2^2}.$$

Generally, for N measurements we find

$$x = \sum_{i=1}^N \frac{x_i}{\delta_i^2} / \sum_{i=1}^N \frac{1}{\delta_i^2}, \quad (4.7)$$

$$\frac{1}{\delta^2} = \sum_{i=1}^N \frac{1}{\delta_i^2}. \quad (4.8)$$

When all measurements have the same error, all the weights are equal to $w_i = 1/N$, and we get the normal arithmetic mean, with the corresponding reduction of the error by the factor $1/\sqrt{N}$.

Remark: If the original raw data of different experiments are available, then we have the possibility to improve the averaging process compared to the simple use of the relations 4.7 and 4.8. When, for example, in two rate measurements of 1 and 2 hours duration, 2, respectively 12 events are observed, then the combined rate is $(2 + 12)/(1 \text{ h} + 2 \text{ h}) = 3.5 \text{ h}^{-1}$, with an error $\pm 0.9 \text{ h}^{-1}$. Averaging according to (4.7) would lead to too low a value of $(3.2 \pm 1.2) \text{ h}^{-1}$, due to the above mentioned problem of small rates and asymmetric errors. The optimal procedure is in any case the addition of the log-likelihoods which will be discussed in Chap. 8. It will correspond to the addition of the original data, as done here.

4.3.4 Averaging Correlated Measurements

In Sect. 4.3.3 we derived the expression for the weighted mean of *independent* measurements of one and the same quantity. This is a special case of a more general

result for a sample of N measurements of the same quantity which differ not only in their variances, but are also correlated, and therefore not statistically independent. Consequently, they have to be described by a complete $N \times N$ covariance or error matrix C .

We choose the weights for a weighted mean such that the variance of the combined value is minimal, in much the same way as in Sect.4.3.3 for uncorrelated measurements. For simplicity, we restrict ourselves to two measurements $x_{1,2}$. The weighted sum x is

$$x = w_1 x_1 + w_2 x_2, \text{ with } w_1 + w_2 = 1.$$

To calculate $\text{var}(x)$ we have to take into account the correlation terms:

$$\delta_x^2 \equiv \text{var}(x) = w_1^2 C_{11} + w_2^2 C_{22} + 2w_1 w_2 C_{12}.$$

The minimum of δ_x^2 is achieved for

$$\begin{aligned} w_1 &= \frac{C_{22} - C_{12}}{C_{11} + C_{22} - 2C_{12}}, \\ w_2 &= \frac{C_{11} - C_{12}}{C_{11} + C_{22} - 2C_{12}}. \end{aligned} \quad (4.9)$$

The uncorrelated weighted mean corresponds to $C_{12} = 0$. Contrary to this case, where the expression for the minimal value of δ_x^2 is particularly simple, it is not as transparent in the correlated case.

The case of N correlated measurements leads to the following expression for the weights:

$$w_i = \frac{\sum_{j=1}^N V_{ij}}{\sum_{i,j=1}^N V_{ij}},$$

where V is the inverse matrix of C which we called the weight matrix in Sect. 3.6.5.

Example 50. Average of measurements with common off-set error

Several experiments (i) determine the energy E_i^* of an excited nuclear state by measuring its transition energy E_i with the uncertainty δ_i to the ground state with energy E_0 . They take the value of E_0 from the same table which quotes an uncertainty of δ_0 of the ground state energy. Thus the results $E_i^* = E_i + E_0$ are correlated. The covariance matrix is

$$C_{ij} = \langle (\Delta_i + \Delta_0)(\Delta_j + \Delta_0) \rangle = \delta_i^2 \delta_{ij} + \delta_0^2.$$

C is the sum of a diagonal matrix and a matrix where all elements are identical, namely equal to δ_0^2 . In this special situation the variance $\text{var}(x) \equiv \delta_x^2$ of the combined result $x = \sum w_i x_i$ is

$$\begin{aligned} \delta^2 &= \sum_i w_i^2 C_{ii} + \sum_{i \neq j} w_i w_j C_{ij} \\ &= \sum w_i^2 \delta_i^2 + \left(\sum w_i \right)^2 \delta_0^2. \end{aligned}$$

Since the second sum is unity, the second term is unimportant when we minimize δ^2 , and we get the same result (4.7) for the weighted mean E^* as in the uncorrelated case. For its error we find, as could have been expected,

$$\delta^2 = \left(\sum \frac{1}{\delta_i^2} \right)^{-1} + \delta_0^2.$$

It is interesting that in some rare cases the weighted mean of two correlated measurements x_1 and x_2 is not located between the individual measurement, the so-called “mean value” is not contained in the interval $[x_1, x_2]$.

Example 51. Average outside the range defined by the individual measurements

The matrix

$$\mathbf{C} = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$$

with eigenvalues

$$\lambda_{1,2} = 3 \pm \sqrt{8} > 0$$

is symmetric and positive definite and thus a possible covariance matrix. But following (4.9) it leads to weights $w_1 = \frac{3}{2}$, $w_2 = -\frac{1}{2}$. Thus the weighted mean $x = \frac{3}{2}x_1 - \frac{1}{2}x_2$ with $x_1 = 0$, $x_2 = 1$ will lead to $x = -\frac{1}{2}$ which is less than both input values. The reason for this sensible but at first sight unexpected result can be understood intuitively in the following way: Due to the strong correlation, x_1 and x_2 , both will usually be either too large or too low. An indication, that x_2 is too large is the fact that it is larger than x_1 which is the more precise measurement. Thus the true value x then is expected to be located below both x_1 and x_2 .

4.3.5 Several Functions of Several Measured Quantities

When we fix a straight line by two measured points in the plane, we are normally interested in its slope and its intercept with a given axis. The errors of these two quantities are usually correlated. The correlations often have to be known in subsequent calculations, e.g. of the crossing point with a second straight line.

In the general case we are dealing with K functions $y_k(x_1, \dots, x_N)$ of N variables with given measurement values x_i and error matrix $\mathbf{C}$. The symmetric error matrix $\mathbf{E}$ related to the values y_k is

$$\begin{aligned} \langle \Delta y_k \Delta y_l \rangle &= \sum_{i,j=1}^N \left(\frac{\partial y_k}{\partial x_i} \frac{\partial y_l}{\partial x_j} \langle \Delta x_i \Delta x_j \rangle \right), \\ E_{kl} &= \sum_{i,j=1}^N \frac{\partial y_k}{\partial x_i} \frac{\partial y_l}{\partial x_j} C_{ij}. \end{aligned} \quad (4.10)$$

Defining a matrix

$$D_{ki} = \frac{\partial y_k}{\partial x_i},$$

we can write more compactly

$$E_{kl} = \sum_{i,j=1}^N D_{ki} D_{lj} C_{ij}.$$

4.3.6 Examples

The following examples represent some standard cases of error propagation.

Example 52. Error propagation: velocity of a sprinter

Given are $s = (100.0 \pm 0.1) \text{ m}$, $t = (10.00 \pm 0.02) \text{ s}$, searched for is δv :

$$\begin{aligned} \left(\frac{\delta v}{v}\right)^2 &= \left(\frac{\delta t}{t}\right)^2 + \left(\frac{\delta s}{s}\right)^2, \\ \frac{\delta v}{v} &= \sqrt{\left(\frac{0.02}{10}\right)^2 + \left(\frac{0.1}{100}\right)^2} = 2.2 \cdot 10^{-3}. \end{aligned}$$

Example 53. Error propagation: area of a rectangular table

Given are the sides a , b with a reading error δ_1 and a relative scaling error δ_2 , caused by a possible extension or shrinkage of the measuring tape. We want to calculate the error δF of the area $F = ab$. We find

$$\begin{aligned} (\delta_a)^2 &= (\delta_1)^2 + (a\delta_2)^2, \\ (\delta_b)^2 &= (\delta_1)^2 + (b\delta_2)^2, \\ C_{ab} &= ab(\delta_2)^2, \\ (\delta F)^2 &= b^2(\delta_a)^2 + a^2(\delta_b)^2 + 2abC_{ab}, \\ \left(\frac{\delta F}{F}\right)^2 &= (\delta_1)^2 \left(\frac{1}{a^2} + \frac{1}{b^2}\right) + 2(\delta_2)^2. \end{aligned}$$

For large areas, the contribution of the reading error is negligible compared to that of the scaling error.

Example 54. Straight line through two measured points

Given are two measured points $(x_1, y_1 \pm \delta y_1)$, $(x_2, y_2 \pm \delta y_2)$ of the straight line $y = mx + b$, where only the ordinate y possesses an error. We want to find the error matrix for the intercept

$$b = (x_2 y_1 - x_1 y_2) / (x_2 - x_1)$$

and the slope

$$m = (y_2 - y_1) / (x_2 - x_1).$$

According to (4.10) we calculate the errors

$$\begin{aligned} (\delta m)^2 &= \frac{(\delta y_2)^2 + (\delta y_1)^2}{(x_2 - x_1)^2}, \\ (\delta b)^2 &= \frac{x_2^2 (\delta y_1)^2 + x_1^2 (\delta y_2)^2}{(x_2 - x_1)^2}, \\ E_{12} = \langle \Delta m \Delta b \rangle &= -\frac{x_2 (\delta y_1)^2 + x_1 (\delta y_2)^2}{(x_2 - x_1)^2}. \end{aligned}$$

The error matrix $\mathbf{E}$ for m and b is therefore

$$E = \frac{1}{(x_2 - x_1)^2} \begin{pmatrix} (\delta y_1)^2 + (\delta y_2)^2, & -x_2(\delta y_1)^2 - x_1(\delta y_2)^2 \\ -x_2(\delta y_1)^2 - x_1(\delta y_2)^2, & x_2^2(\delta y_1)^2 + x_1^2(\delta y_2)^2 \end{pmatrix}.$$

The correlation matrix element R_{12} is

$$\begin{aligned} R_{12} &= \frac{E_{12}}{\delta m \delta b}, \\ &= -\frac{x_2(\delta y_1)^2 + x_1(\delta y_2)^2}{\{[(\delta y_2)^2 + (\delta y_1)^2][x_2^2(\delta y_1)^2 + x_1^2(\delta y_2)^2]\}^{1/2}}. \end{aligned} \quad (4.11)$$

For the special case $\delta y_1 = \delta y_2 = \delta y$ the results simplify to

$$\begin{aligned} (\delta m)^2 &= \frac{2}{(x_1 - x_2)^2} (\delta y)^2, \\ (\delta b)^2 &= \frac{(x_1^2 + x_2^2)}{(x_1 - x_2)^2} (\delta y)^2, \\ E_{12} &= -\frac{(x_1 + x_2)}{(x_1 - x_2)^2} (\delta y)^2, \\ R_{12} &= -\frac{x_1 + x_2}{\sqrt{2(x_1^2 + x_2^2)}}. \end{aligned}$$

Remark: As seen from (4.11), for a suitable choice of the abscissa the correlation disappears. To achieve this, we take as the origin the “center of gravity” x_s of the x -values x_i , weighted with the inverse squared errors of the ordinates, $1/(\delta y_i)^2$:

$$x_s = \sum \frac{x_i}{(\delta y_i)^2} / \sum \frac{1}{(\delta y_i)^2}.$$

Example 55. Error of a sum of weighted measurements

In the evaluation of event numbers, the events are often counted with different weights, in order to take into account, for instance, a varying acceptance of the detector. Weighting is also important in Monte Carlo simulations (see 5.2.6) especially when combined with parameter estimation Sect. 6.5.9 An event with weight 10 stands for 10 events with weight 1. For N events with weights $w_1 \dots w_N$ the weighted number of events is

$$s = \sum_{i=1}^N w_i.$$

This sum s stands for N measurements with result $x_i = 1$ in each of them and where the results are added after weighting with w_i : $s = \sum w_i x_i$. The error of this sum is $\delta_s^2 = \sum w_i^2 \delta_i^2$. Since each individual result corresponds to a Poisson distribution with mean equal to 1, it has also variance of 1 and thus $\delta_i^2 = 1$. We thus obtain

$$\delta_s^2 = \sum w_i^2.$$

which corresponds to the variance of the sum of weighted events as derived in Sect. 3.6.3.

4.4 Biased Measurements

We have required that our measurement values x_i are undistorted (unbiased). We have used this property in the discussion of error propagation. Anyway, it is rather plausible that we should always avoid biased measurements, because averaging measurements with a common bias would produce a result with the same bias. The average from infinitely many measurements would thus be different from the true parameter value but the associated error would be infinitely small. However a closer look at the problem reveals that the requirement of unbiasedness has also its problems: When we average measurements, the measurements x_i are weighted with $1/\delta_i^2$, their inverse squared errors, as we have seen above. *To be consistent, it is therefore required that the quantities x_i/δ_i^2 are unbiased!* Of course, we explicitly excluded the possibility of errors which depend on the measurement values, but since this requirement is violated so often in reality and since a bias which is small compared to the uncertainty in an individual experiment can become important in the average, we stress this point here and present an example.

Example 56. Bias in averaging measurements

Let us assume that several measurements of a constant x_0 produce unbiased results x_i with errors $\delta_i \sim x_i$ which are proportional to the measurements. This could be, for instance, measurements of particle lifetimes, where the relative error is determined by the number of recorded decays and thus the absolute error is set proportional to the observed mean life. When we compute the weighted mean x over many such measurements

$$\begin{aligned} x &= \sum \frac{x_i}{\delta_i^2} / \sum \frac{1}{\delta_i^2} \\ &= \sum \frac{1}{x_i} / \sum \frac{1}{x_i^2} \\ &\approx \langle 1/x \rangle / \langle 1/x^2 \rangle \end{aligned}$$

the expected value is shifted systematically to lower values. This is easily seen from a Taylor expansion of the expected values:

$$\begin{aligned} \langle x - x_0 \rangle &= \frac{\langle 1/x \rangle}{\langle 1/x^2 \rangle} - x_0, \\ \left\langle \frac{1}{x} \right\rangle &= \frac{1}{x_0} \left(1 - \left\langle \frac{\Delta x}{x_0} \right\rangle + \left\langle \frac{\Delta x^2}{x_0^2} \right\rangle + \dots \right) \\ &\approx \frac{1}{x_0} \left(1 + \frac{\delta^2}{x_0^2} \right), \\ \left\langle \frac{1}{x^2} \right\rangle &= \frac{1}{x_0^2} \left(1 - 2 \left\langle \frac{\Delta x}{x_0} \right\rangle + 3 \left\langle \frac{\Delta x^2}{x_0^2} \right\rangle + \dots \right) \\ &\approx \frac{1}{x_0^2} \left(1 + 3 \frac{\delta^2}{x_0^2} \right), \\ \langle x - x_0 \rangle &\approx x_0 \frac{1 + \delta^2/x_0^2}{1 + 3\delta^2/x_0^2} - x_0 \\ &\approx x_0 (1 - 2\delta^2/x_0^2) - x_0, \\ \frac{\langle x - x_0 \rangle}{x_0} &\approx -2 \frac{\delta^2}{x_0^2}. \end{aligned}$$

Here δ^2 is the expectation of the error squared in an individual measurement. For a measurement error δ/x_0 of 20% we obtain a sizable final bias of 8% for the asymptotic result of infinitely many contributions.

The revised requirement of unbiasedness of measurements divided by the error squared does not alter the other results which we have derived for the general error propagation in the linear approximation.

4.5 Confidence Intervals

Under the condition that the error distribution is a one-dimensional Gaussian, with a width independent of the expected value, the error intervals of many repeated measurements will cover the true parameter value in 68.3 % of the cases, because for any true value μ the probability to observe x inside one standard deviation interval is

$$\frac{1}{\sqrt{2\pi}\delta} \int_{-\delta}^{\delta} \exp \left[-\frac{(x-\mu)^2}{2\delta^2} \right] dx \approx 0.683$$

The region $[x-\delta, x+\delta]$ is called a confidence interval⁷ with the confidence level (*CL*) of 68.3 %, or, in physicists' jargon, a 1σ confidence interval. Thus in about one third of the cases our standard error intervals, under the above assumption of normality, will not contain the true value. Often a higher safety is desired, for instance 90 %, 95 %, or even 99 %. The respective limits can be calculated, provided the probability distribution is known with sufficient accuracy. For the normal distribution we present some limits in units of the standard deviation in Table 4.2. The numerical values can be taken from tables of the χ^2 -distribution function.

Example 57. Confidence level for the mean of normally distributed variates

Let us consider a sample of N measurements $x_1, \dots, x_N$ which are supposed to be normally distributed with *unknown* mean μ but *known* variance σ^2 . The sample mean $\bar{x}$ is also normally distributed with variance $\delta_N = \sigma/\sqrt{N}$. The 1σ confidence interval $[\bar{x}-\delta_N, \bar{x}+\delta_N]$ covers, as we have discussed above, the true value μ in 68.3 % of the cases. We can, with the help of Table 4.2, also find a 99 % confidence level, i.e. $[\bar{x}-2.58\delta_N, \bar{x}+2.58\delta_N]$.

We have to keep in mind that the Gaussian confidence limits do not or only approximately apply to other distributions. Error distributions often have tails which are not well understood. Then it is impossible to derive reliable confidence limits with high confidence levels. The same is true when systematic errors play a role, for example due to background and acceptance which usually are not known with great accuracy. Then for a given confidence level much wider intervals than in the above case are required.

We come back to our previous example but now we assume that the error has to be estimated from the sample itself, according to (4.1), (4.3):

$$\delta_N^2 = \sum_{i=1}^N (x_i - \bar{x})^2 / [N(N-1)] .$$

⁷We will discuss confidence intervals in more detail in Chap. 8 and in Appendix 13.5.

Table 4.2. Left hand side: confidence levels for several error limits and dimensions, right hand side: error limits in units of σ for several confidence levels and dimensions.

Deviation	Dimension				Confidence level	Dimension			
	1	2	3	4		1	2	3	4
1 σ	0.683	0.393	0.199	0.090	0.50	0.67	1.18	1.54	1.83
2 σ	0.954	0.865	0.739	0.594	0.90	1.65	2.14	2.50	2.79
3 σ	0.997	0.989	0.971	0.939	0.95	1.96	2.45	2.79	3.08
4 σ	1.	1.	0.999	0.997	0.99	2.58	3.03	3.37	3.64

Table 4.3. Values of the factor k for the Student's t-distribution as a function of the confidence levels CL and sample size N .

N	68.3%	99%
3	1.32	3.85
10	1.06	1.26
20	1.03	1.11
∞	1.00	1.00

To compute the confidence level for a given interval in units of the standard deviation, we now have to switch to Student's distribution (see Sect. 3.6.11). The variate t , given by $(\bar{x} - \mu)/\delta_N$, can be shown to be distributed according to $h_f(t)$ with $f = N - 1$ degrees of freedom. The confidence level for a given number of standard deviations will now be lower, because of the tails of Student's distribution. Instead of quoting this number, we give in Table 4.3 the factor k by which we have to increase the interval length to get the same confidence level as in the Gaussian case. To clarify its meaning, let us look at two special cases: For 68.3% confidence and $N = 3$ we require a 1.32 standard deviation interval and for 99% confidence and $N = 10$ a $1.26 \times 2.58 = 3.25$ standard deviation interval. As expected, the discrepancies are largest for small samples and high confidence levels. In the limit when N approaches infinity the factor k has to become equal to one.

Often it is overlooked that for distributions of several variates, the probability to find *all* variables inside their error limits is strongly decreasing with the number of variables. Some probabilities for Gaussian errors are given in Table 4.2. In three dimensions only 20 % of the observations are found in the 1σ ellipsoid. Fig. 4.2 shows confidence ellipses and probabilities for two variables.

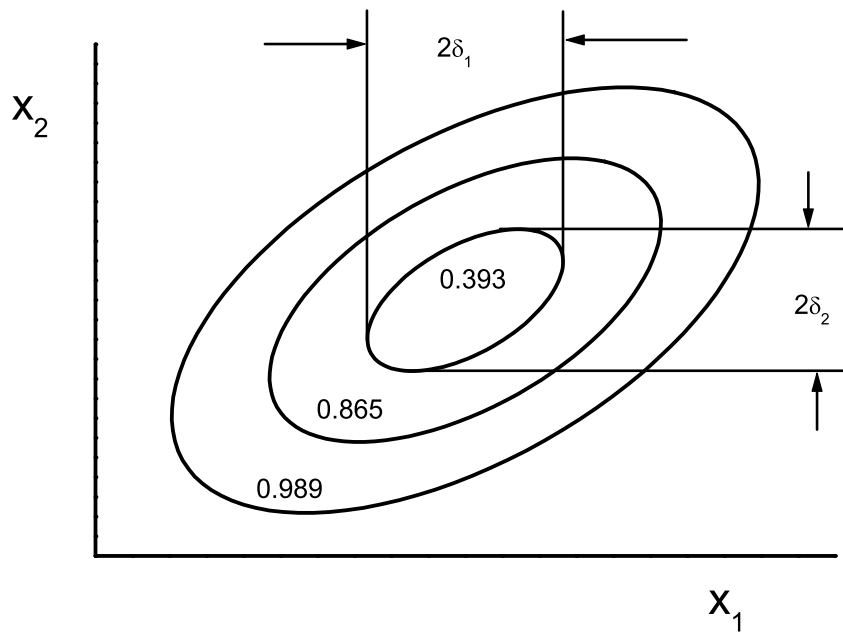


Fig. 4.2. Confidence ellipses for 1, 2 and 3 standard deviations and corresponding probabilities.

Monte Carlo Simulation

5.1 Introduction

The possibility to simulate stochastic processes and of numerical modeling on the computer simplifies extraordinarily the solution of many problems in science and engineering. The deeper reason for this is characterized quite aptly by the German saying “Probieren geht über studieren” (Trying beats studying). Monte Carlo methods replace intellectual by computational effort which, however, is realized by the computer.

A few simple examples will demonstrate the advantages, but also the limits of this method. The first two of them are purely mathematical integration problems which could be solved also by classical numerical methods, but show the conceptual simplicity of the statistical approach.

Example 58. Area of a circle of diameter d

We should keep in mind that without the knowledge of the quantity π the problem requires quite some mathematics but even a child can solve this problem experimentally. It may inscribe a circle into a square with edge length d , and sprinkles confetti with uniform density over it. The fraction of confetti confined inside the circle provides the area of the circle in units of the square area. Digital computers have no problem in “sprinkling confetti” homogeneously over given regions.

Example 59. Volume of the intersection of a cone and a torus

We solve the problem simply by scattering points homogeneously inside a cuboid containing the intersect. The fraction of points inside both bodies is a measure for the ratio of the intersection volume to that of the cuboid.

In the following three examples we consider the influence of the measurement process on the quantity to be determined.

Example 60. Correction of decay times

The decay time of instable particles is measured with a digital clock which is stopped at a certain maximal time. How can we determine the mean lifetime of the particles? The measured decay times are distorted by both the limited

resolution as well as by the finite measurement time, and have to be corrected. The correction can be determined by a simulation of the whole measurement process. (We will come back to details below.)

Example 61. Efficiency of particle detection

Charged particles passing a scintillating fiber produce photons. A fraction of the photons is reflected at the surface of the fiber, and, after many reflections, eventually produces a signal in a photomultiplier. The photon yield per crossing particle has to be known as a function of several parameters like track length of the particle inside the fiber, its angle of incidence, fiber length and curvature, surface parameters of the fiber etc.. Here a numerical solution using classical integration methods would be extremely involved and an experimental calibration would require a large number of measurements. Here, and in many similar situations, a Monte Carlo simulation is the only sensible approach.

Example 62. Measurement of a cross section in a collider experiment

Particle experiments often consist of millions of detector elements which have to measure the trajectories of sometimes thousands of particles and the energies deposited in an enormous number of calorimeter cells. To measure a specific cross section, the corresponding events have to be selected, acceptance losses have to be corrected, and unavoidable background has to be estimated. This can only be achieved by sophisticated Monte Carlo simulations which require a huge amount of computing time. These simulations consist of two distinct parts, namely the generation of the particle reaction (event generation) which contains the interesting physics, and the simulation of the detector response. The computing time needed for the event generation is negligible compared to that required for the detector simulation. As a consequence one tries to avoid the repetition of the detector simulation and takes, if possible, modifications of the physical process into account by re-weighting events.

Example 63. Reaction rates of gas mixtures

A vessel contains different molecules with translational and rotational movements according to the given temperature. The molecules scatter on the walls, with each other and transform into other molecules by chemical processes depending on their energy. To be determined is the composition of the gas after a certain time. The process can be simulated for a limited number of particles. The particle trajectories and the reactions have to be computed.

All examples lead finally to integration problems. In the first three examples also numerical integration, even exact analytical methods, could have been used. For the Examples 61 and 63, however, this is hardly possible, since the number of variables is too large. Furthermore, the mathematical formulation of the problems becomes rather involved.

Monte Carlo simulation does not require a profound mathematical expertise. Due to its simplicity and transparency mistakes can be avoided. It is true, though, that the results are subject to statistical fluctuations which, however, may be kept small

enough in most cases thanks to the fast computers available nowadays. For the simulation of chemical reactions, however, (Example 63) we reach the limits of computing power quite soon, even with super computers. The treatment of macroscopic quantities (one mole, say) is impossible. Most questions can be answered, however, by simulating small samples.

Nowadays, even statistical problems are often solved through Monte Carlo simulations. In some big experiments the error estimation for parameters determined in a complex analysis is so involved that it is easier to simulate the experiment, including the analysis, several times, and to derive the errors quasi experimentally from the distribution of the resulting parameter values. The relative statistical fluctuations can be computed for small samples and then scaled down with the square root of the sample size.

In the following section we will treat the simulation of the basic univariate distributions which are needed for the generation of more complex processes. The generalization to several dimensions is not difficult. Then we continue with a short summary on Monte Carlo integration methods.

5.2 Generation of Statistical Distributions

The simplest distribution is the uniform distribution which serves as the basis for the generation of all other distributions. In the following we will introduce some frequently used methods to generate random numbers with desired distributions.

Some of the simpler methods have been introduced already in Chap. 3, Sect. 3.6.4, 3.6.5: By a linear transformation we can generate uniform distributions of any location and width. The sum of two uniformly distributed random numbers follows a triangular distribution. The addition of only five such numbers produces a quite good approximation of a Gaussian variate.

Since our computers work deterministically, they cannot produce numbers that are really random, but they can be programmed to deliver for practically any application sufficiently unordered numbers, *pseudo random numbers* which approximate random numbers to a very good accuracy.

5.2.1 Computer Generated Pseudo Random Numbers

The computer delivers pseudo random numbers in the interval between zero and one. Because of the finite number of digits used to represent data in a computer, these are discrete, rational numbers which due to the usual floating point accuracy can take only $2^{18} \approx 8 \cdot 10^6$ different values, and follow a fixed, reproducible sequence which, however, appears as stochastic to the user. More refined algorithms can avoid, though, the repetition of the same sequence after 2^{18} calls. The *Mersenne twister*, one of the fastest reasonable random number generators, invented in 1997 by M. Matsumoto and T. Nishimura has the enormous period of 2^{19937} which never can be exhausted and is shown to be uniformly distributed in 623 dimensions. In all generators, the user has the possibility to set some starting value, called *seed*, and thus to repeat exactly the same sequence or to interrupt a simulation and to continue with the sequence in order to generate statistically independent samples.

In the following we will speak of random numbers also when we mean pseudo random numbers.

There are many algorithms for the generation of random numbers. The principle is quite simple: One performs an arithmetic operation and uses only the insignificant digits of the resulting number. How this works is shown by the prescription

$$x_{i+1} = n^{-1} \text{mod}(\lambda x_i; n) ,$$

producing from the old random number x_i a new one between zero and one. The parameters λ and n fulfil the condition $\lambda \gg n$. With the values $x_1 = 0.7123$, $\lambda = 4158$, $n = 1$ we get, for instance, the number

$$x_2 = \text{mod}(2961.7434; 1) = 0.7434 .$$

The apparent “randomness” is due to the cutting off the significant digits by the *mod* operation.

This random number generator is far from being perfect, as can be shown experimentally by investigation of the correlations of consecutive random numbers. The generators installed in the commonly used program libraries are almost always sufficiently good. Nevertheless it is advisable to check their quality before starting important calculations. Possible problems with random number generators are that they have a shorter than expected repetition period, correlations of successive values and lack of uniformity. For simulations which require a high accuracy, we should remember that with the standard generators only a limited number of random numbers is available. Though intuitively attractive, randomly mixing the results of different random number generators does not improve the overall quality.

In Fig. 5.1 the values of two consecutive random numbers from a PC routine are plotted against each other. Obvious correlations and clustering cannot be detected. The histogram of a projection is well compatible with a uniform distribution. A quantitative judgment of the quality of random number generators can be derived with goodness-of-fit tests (see Chap. 10).

In principle, one could of course integrate random number generators into the computers which indeed work stochastically and replace the deterministic generators. As physical processes, the photo effect or, even simpler, the thermal noise could be used. Each bit of a computer word could be set by a dual oscillator which is stopped by the stochastic process. Unfortunately, such hardware random number generators are presently not used, although they could be produced quite economically, presumably $\approx 10^3$ in a single chip. They would make obsolete some discussions, which come up from time to time, on the reliability of software generators. On the other hand, the reproducibility of the random number sequence is quite useful when we want to compare different program versions, or to debug them.

5.2.2 Generation of Distributions by Variable Transformation

Continuous Variables

With the restrictions discussed above, we can generate with the computer random numbers obeying the uniform distribution

$$u(r) = 1 \text{ for } 0 \leq r \leq 1.$$

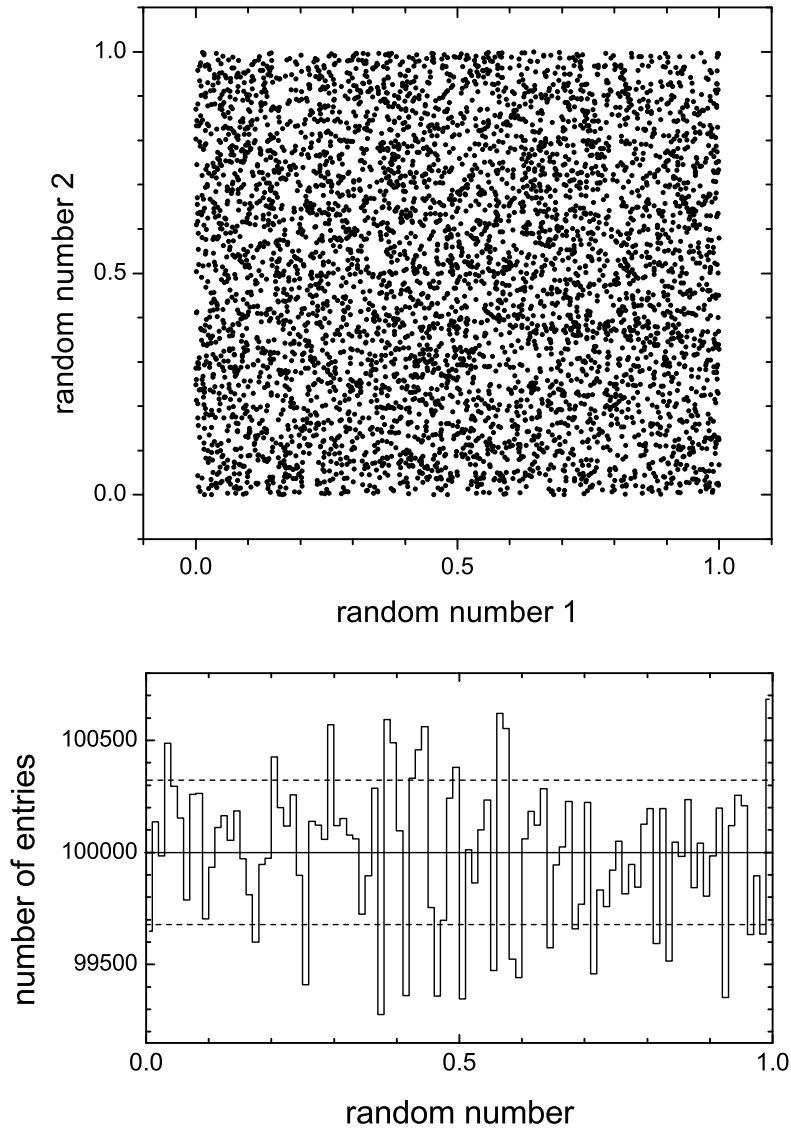


Fig. 5.1. Correlation plot of consecutive random numbers (top) and frequency of random numbers (bottom).

In the following we use the notations u for the uniform distribution and r for a uniformly distributed variate in the interval $[0, 1]$. Other univariate distributions $f(x)$ are obtained by variable transformations $r(x)$ with r a monotone function of x (see Chap. 3):

$$f(x)dx = u(r)dr,$$

$$\int_{-\infty}^x f(x')dx' = \int_0^{r(x)} u(r')dr' = r(x),$$

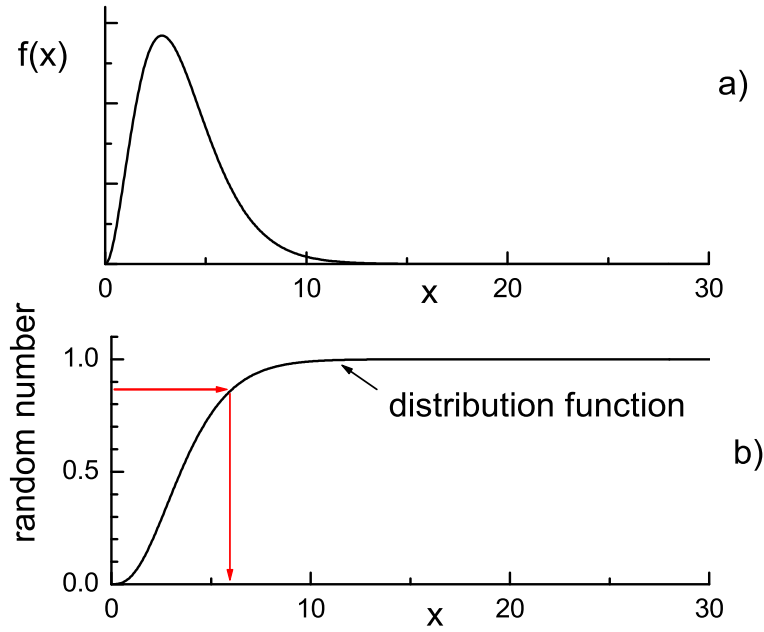


Fig. 5.2. The p.d.f (top) follows from the distribution function as indicated by the arrows.

$$F(x) = r,$$

$$x(r) = F^{-1}(r) .$$

The variable x is calculated from the inverse function F^{-1} where $F(x)$ is the distribution function which is set equal to r . For an analytic solution the p.d.f. has to be analytically integrable and the distribution function must have an inverse in analytic form.

The procedure is explained graphically in Fig. 5.2: A random number r between zero and one is chosen on the ordinate. The distribution function (or rather its inverse) then delivers the respective value of the random variable x .

In this way it is possible to generate the following distributions by simple variable transformation from the uniform distribution:

- Linear distribution:

$$f(x) = 2x \quad 0 \leq x \leq 1 ,$$

$$x(r) = \sqrt{r} .$$

- Power-law distribution:

$$f(x) = (n+1)x^n \quad 0 \leq x \leq 1, \quad n > -1 ,$$

$$x(r) = r^{1/(n+1)} .$$

- Exponential distribution (Sect. 3.6.6) :

$$f(x) = \gamma e^{-\gamma x},$$

$$x(r) = -\frac{1}{\gamma} \ln(1-r).$$

- Normal distribution (Sect. 3.6.5) : Two independent normally distributed random numbers x, y are obtained from two uniformly distributed random numbers r_1, r_2 , see (3.38), (3.39).

$$f(x, y) = \frac{1}{2\pi} \exp \left[-\frac{x^2 + y^2}{2} \right],$$

$$x(r_1, r_2) = \sqrt{-2 \ln(1-r_1)} \cos(2\pi r_2),$$

$$y(r_1, r_2) = \sqrt{-2 \ln(1-r_1)} \sin(2\pi r_2).$$

- Breit-Wigner distribution (Sect. 3.6.9) :

$$f(x) = \frac{1}{\pi \Gamma/2} \frac{(\Gamma/2)^2}{x^2 + (\Gamma/2)^2},$$

$$x(r) = \frac{\Gamma}{2} \tan \left[\pi \left(r - \frac{1}{2} \right) \right].$$

- Log-Weibull (Fisher-Tippett) distribution (3.6.12)

$$f(x) = \exp(-x - e^{-x}),$$

$$x(r) = -\ln(-\ln r).$$

The expression $1-r$ can be replaced by r in the formulas. More general versions of these distributions are obtained by translation and/or scaling operations. A triangular distribution can be constructed as a superposition of two linear distributions. Correlated normal distributed random numbers are obtained by scaling x and y differently and subsequently rotating the coordinate frame. How to generate superpositions of distributions will be explained in Sect. 5.2.5.

Uniform Angular, Circular and Spherical Distributions

Very often the generation of a uniform angular distribution is required. The azimuthal angle φ is given by

$$\varphi = 2\pi r.$$

To obtain a spatially isotropic distribution, we have also to generate the polar angle θ . As we have discussed in Sect. 3.5.8, its cosine is uniformly distributed in the interval $[-1, 1]$. Therefore

$$\cos \theta = (2r_1 - 1),$$

$$\theta = \arccos(2r_1 - 1),$$

$$\varphi = 2\pi r_2.$$

A uniform distribution inside a circle of radius R_0 is generated by

$$R = R_0 \sqrt{r_1},$$

$$\varphi = 2\pi r_2.$$

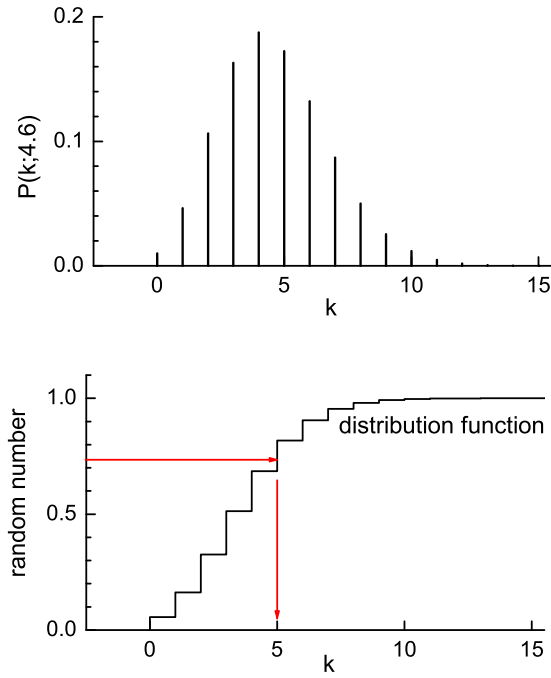


Fig. 5.3. Generation of a Poisson distributed random number.

Because the differential area element is $R dR d\varphi$, we have a linear distribution in R .

A uniform distribution inside a sphere of radius R_0 is obtained similarly from

$$\begin{aligned} R &= R_0 r_1^{1/3}, \\ \theta &= \arccos(2r_2 - 1), \\ \varphi &= 2\pi r_3, \end{aligned}$$

with a quadratic distribution in R .

Discrete Distributions

The generation of random numbers drawn from discrete distributions is performed in a completely analogous fashion. We demonstrate the method with a simple example: We generate random numbers k following a Poisson distribution $P(k|4.6)$ with expected value 4.6 which is displayed in Fig. 5.3. By summation of the bins starting from the left (integration), we obtain the distribution function $S(k) = \sum_{i=0}^k P(i|4.6)$ shown in the figure. To a uniformly distributed random number r we attach the value k which corresponds to the minimal $S(k)$ fulfilling $S > r$. The numbers k follow the desired distribution.

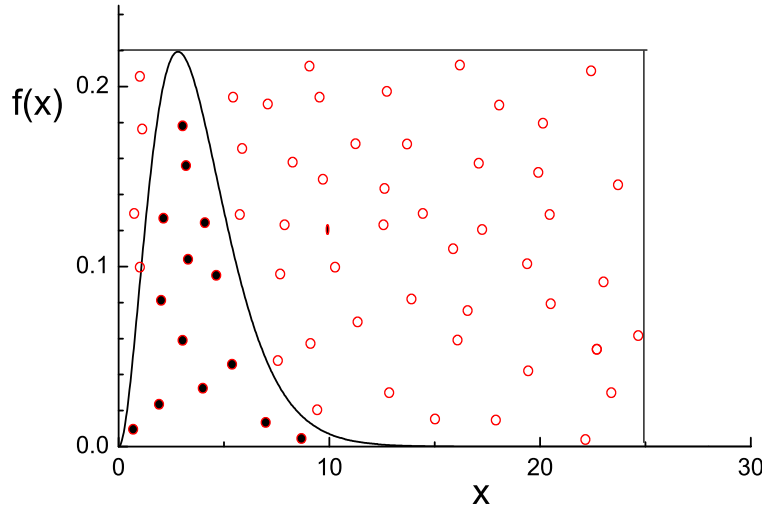


Fig. 5.4. Random selection method. The projection of the points located below the curve follow the desired distribution.

Histograms

A similar method is applied when an empirical distribution given in the form of a histogram has to be simulated. The random number r determines the bin j . The remainder $r - S(j - 1)$ is used for the interpolation inside the bin interval. Often the bins are small enough to justify a uniform distribution for this interpolation. A linear approximation does not require much additional effort.

For two-dimensional histograms h_{ij} we first produce a projection,

$$g_i = \sum_j h_{ij} ,$$

normalize it to one, and generate at first i , and then for given i in the same way j . That means that we need for each value of i the distribution summed over j .

5.2.3 Simple Rejection Sampling

In the majority of cases it is not possible to find and invert the distribution function analytically. As an example for a non-analytic approach, we consider the generation of photons following the Planck black-body radiation law. The appropriately scaled frequency x obeys the distribution

$$f(x) = c \frac{x^3}{e^x - 1} \quad (5.1)$$

with the normalization constant c . This function is shown in Fig. 5.4 for $c = 1$, i.e. not normalized. We restrict ourselves to frequencies below a given maximal frequency x_{max} .

A simple method to generate this distribution $f(x)$ is to choose two uniformly distributed random numbers, where r_1 is restricted to the interval (x_{min}, x_{max}) and

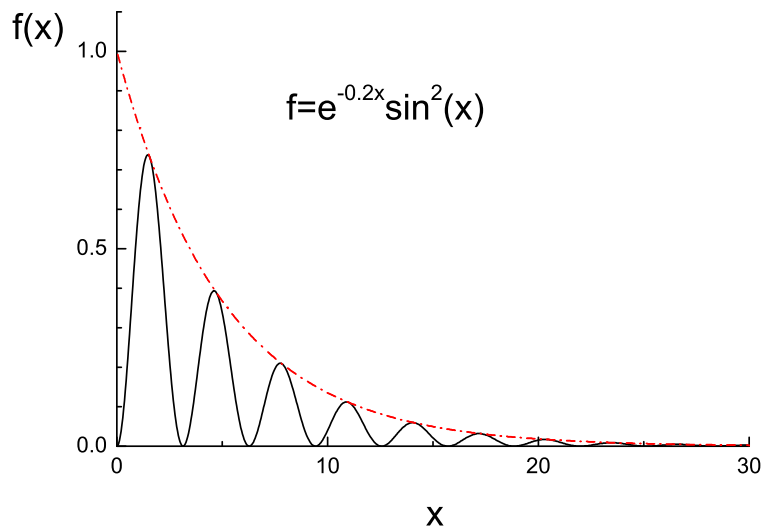


Fig. 5.5. Majorant (dashed) used for importance sampling.

r_2 to $(0, f_{max})$. This pair of numbers $P(r_1, r_2)$ corresponds to a point inside the rectangle shown in the figure. We generate points and those lying above the curve $f(x)$ are rejected. The density of the remaining r_1 values follows the desired p.d.f. $f(x)$.

A disadvantage of this method is that it requires several randomly distributed pairs to select one random number following the distribution. In our example the ratio of successes to trials is about 1:10. For generating photons up to arbitrary large frequencies the method cannot be applied at all.

5.2.4 Importance Sampling

An improved selection method, called importance sampling, is the following: We look for an appropriate function $m(x)$, called majorant, with the properties

- $m \geq f$ for all x ,
- $x = M^{-1}(r)$, i.e. the indefinite integral $M(x) = \int_{-\infty}^x m(x')dx'$ is invertible,

If it exists (see Fig. 5.5), we generate x according to $m(x)$ and, in a second step, drop stochastically for given x the fraction $[m(x) - f(x)]/f(x)$ of the events. This means, for each event (i.e. each generated x) a second, this time uniform random number between zero and $m(x)$ is generated, and if it is larger than $f(x)$, the event is abandoned. The advantage is, that for $m(x)$ being not much different from $f(x)$ in most of the cases, the generation of one event requires only two random numbers. Moreover, in this way it is possible to generate also distributions which extend to infinity, as for instance the Planck distribution, and many other distributions.

We illustrate the method with a simple example (Fig. 5.5):

Example 64. Importance sampling

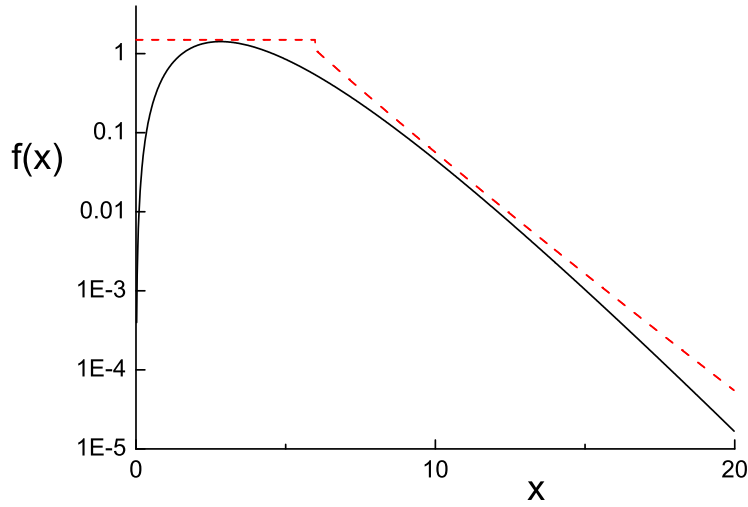


Fig. 5.6. Planck spectrum with majorant.

To generate

$$f(x) = c(e^{-0.2x} \sin^2 x) \quad \text{for } 0 < x < \infty$$

with the majorant

$$m(x) = c e^{-0.2x} ,$$

we normalize $m(x)$ and calculate its distribution function

$$\begin{aligned} r &= \int_0^x 0.2 e^{-0.2x'} dx' \\ &= 1 - e^{-0.2x} . \end{aligned}$$

Thus the variate transformation from the uniformly distributed random number r_1 to x is

$$x = -\frac{1}{0.2} \ln(1 - r_1) .$$

We draw a second uniform random number r_2 , also between zero and one, and test whether $r_2 m(x)$ exceeds the desired p.d.f. $f(x)$. If this is the case, the event is rejected:

$$\begin{aligned} \text{for } r_2 m(x) < \sin^2 x &\rightarrow \text{keep } x, \\ \text{for } r_2 m(x) > \sin^2 x &\rightarrow \text{drop } x . \end{aligned}$$

With this method a uniform distribution of random points below the majorant curve is generated, while only those points are kept which lie below the p.d.f. to be generated. On average about 4 random numbers per event are needed in this example, since the test has a positive result in about half of the cases.

If an appropriate continuous, analytical majorant function cannot be found, often a piecewise constant function (step function) is chosen.

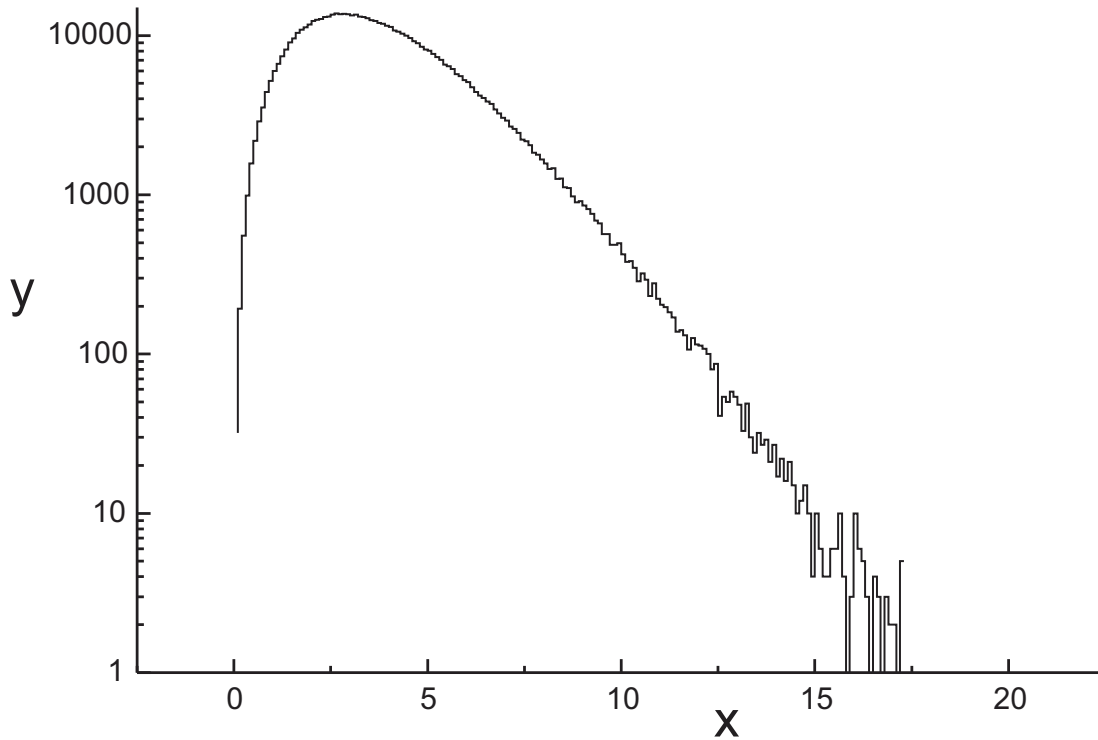


Fig. 5.7. Generated Planck spectrum.

Example 65. Generation of the Planck distribution

Here a piecewise defined majorant is useful. We consider again the Planck distribution (5.1), and define the majorant in the following way: For small values $x < x_1$ we chose a constant majorant $m_1(x) = 6c$. For larger values $x > x_1$ the second majorant $m_2(x)$ should be integrable with invertible integral function. Due to the x^3 -term, the Planck distribution decreases somewhat more slowly than e^{-x} . Therefore we chose for m_2 an exponential factor with x substituted by $x^{1-\varepsilon}$. With the arbitrary choice $\varepsilon = 0.1$ we take

$$m_2(x) = 200cx^{-0.1}e^{-x^{0.9}}.$$

The factor $x^{-0.1}$ does not influence the asymptotic behavior significantly but permits the analytical integration:

$$\begin{aligned} M_2(x) &= \int_{x_1}^x m_2(x')dx', \\ &= \frac{200c}{0.9} \left[e^{-x_1^{0.9}} - e^{-x^{0.9}} \right]. \end{aligned}$$

This function can be easily solved for x , therefore it is possible to generate m_2 via a uniformly distributed random number. Omitting further details of the calculation, we show in Fig. 5.6 the Planck distribution with the two majorant pieces in logarithmic scale, and in Fig. 5.7 the generated spectrum.

5.2.5 Treatment of Additive Probability Densities

Quite often the p.d.f. to be considered is a sum of several terms. Let us restrict ourselves to the simplest case with two terms,

$$f(x) = f_1(x) + f_2(x) ,$$

with

$$\begin{aligned} S_1 &= \int_{-\infty}^{\infty} f_1(x) dx , \\ S_2 &= \int_{-\infty}^{\infty} f_2(x) dx , \\ S_1 + S_2 &= 1 . \end{aligned}$$

Now we chose with probability S_1 (S_2) a random number distributed according to f_1 (f_2). If the integral functions

$$\begin{aligned} F_1(x) &= \int_{-\infty}^x f_1(x') dx' , \\ F_2(x) &= \int_{-\infty}^x f_2(x') dx' \end{aligned}$$

are invertible, we obtain with a uniformly distributed random number r the variate x distributed according to $f(x)$:

$$x = F_1^{-1}(r) \quad \text{for } r < S_1 ,$$

respectively

$$x = F_2^{-1}(r - S_1) \quad \text{for } r > S_1 .$$

The generalization to more than two terms is trivial.

Example 66. Generation of an exponential distribution with constant background

In order to generate the p.d.f.

$$f(x) = \varepsilon \frac{\lambda e^{-\lambda x}}{1 - e^{-\lambda a}} + (1 - \varepsilon) \frac{1}{a} \quad \text{für } 0 < x < a ,$$

we chose for $r < \varepsilon$

$$x = \frac{-1}{\lambda} \ln \left(1 - \frac{1 - e^{-\lambda a}}{\varepsilon} r \right) ,$$

and for $r > \varepsilon$

$$x = a \frac{r - \varepsilon}{1 - \varepsilon} .$$

We need only one random number per event. The direct way to use the inverse of the distribution function $F(x)$ would not have been successful, since it cannot be given in analytic form.

The separation into additive terms is always recommended, even when the individual terms cannot be handled by simple variate transformations as in the example above.

5.2.6 Weighting Events

In Sect. 3.6.3 we have discussed some statistical properties of weighted events and realized that the relative statistical error of a sum of N weighted events can be much larger than the Poisson value $1/\sqrt{N}$, especially when the individual weights are very different. Thus we will usually refrain from weighting. However, there are situations where it is not only convenient but essential to work with weighted events. If a large sample of events has already been generated and stored and the p.d.f. has to be changed afterwards, it is of course much more economical to re-weight the stored events than to generate new ones because the simulation of high energy reactions in highly complex detectors is quite expensive. Furthermore, for small changes the weights are close to one and will not much increase the errors. As we will see later, parameter inference based on a comparison of data with a Monte Carlo simulation usually requires re-weighting anyway.

An event with weight w stands for w identical events with weight 1. When interpreting the results of a simulation, i.e. calculating errors, one has to take into account the distribution of a sum of weights, see last example in Sect. 4.3.6. There we showed that

$$\text{var}\left(\sum w_i\right) = \sum w_i^2.$$

Relevant is the relative error of a sum of weights:

$$\frac{\delta(\sum w_i)}{\sum w_i} = \frac{\sqrt{\sum w_i^2}}{\sum w_i}.$$

Strongly varying weights lead to large statistical fluctuations and should therefore be avoided.

To simulate a distribution

$$f(x) : \text{ with } x_a < x < x_b$$

with weighted events is especially simple: We generate events x_i that are uniformly distributed in the interval $[x_a, x_b]$ and weight each event with $w_i = f(x_i)$.

In the Example 64 we could have generated events following the majorant distribution, weighting them with $\sin^2 x$. The weights would then be $w_i = f(x_i)/m(x_i)$.

When we have generated events following a p.d.f. $f(x|\theta)$ depending on a parameter θ and are interested in the distribution $f'(x|\theta')$ we have only to re-weight the events by f'/f .

5.2.7 Markov Chain Monte Carlo

Introduction

The generation of distributions of high dimensional distributions is difficult with the methods that we have described above. Markov chain Monte Carlo (MCMC) is able to generate samples of distributions with hundred or thousand dimensions. It has become popular in thermodynamics where statistical distributions are simulated to compute macroscopic mean values and especially to study phase transitions. It has also been applied for the approximation of functions on discrete lattices. The method

is used mainly in theoretical physics to sample multi-dimensional distributions, so far we know of no applications in experimental particle physics. However, MCMC is used in artificial neural networks to optimize the weights of perceptron nets.

Characteristic of a Markov chain is that a random variable x is modified stochastically in discrete steps, its value at step i depending only on its value at the previous step $i - 1$. Values of older steps are forgotten: $P(x_i|x_{i-1}, x_{i-2}, \dots, x_1) = P(x_i|x_{i-1})$. A typical example of a Markov chain is random walk. Of interest are Markov chains that converge to an equilibrium distribution, like random walk in a fixed volume. MCMC generates a Markov chain that has as its equilibrium distribution the desired distribution. Continuing with the chain once the equilibrium has been reached produces further variates of the distribution. To satisfy this requirement, the chain has to satisfy certain conditions which are fulfilled for instance for the so-called Metropolis algorithm, which we will use below. There exist also several other sampling methods. Here we will only sketch this subject and refer the interested reader to the extensive literature which is nicely summarized in [27].

Thermodynamical Model, Metropolis Algorithm

In thermodynamics the molecules of an arbitrary initial state always approach – if there is no external intervention – a stationary equilibrium distribution. Transitions then obey the *principle of detailed balance*. In a simple model with atoms or molecules in only two possible states in the stationary case, the rate of transitions from state 1 to state 2 has to be equal to the reverse rate from 2 to 1. For occupation numbers N_1 , N_2 of the respective states and transition rates per molecule and time W_{12} , respectively W_{21} , we have the equation of balance

$$N_1 W_{12} = N_2 W_{21} .$$

For instance, for atoms with an excited state, where the occupation numbers are very different, the equilibrium corresponds to a Boltzmann distribution, $N_1/N_2 = e^{-\Delta E/kT}$, with ΔE being the excitation energy, k the Boltzmann constant and T the absolute temperature. When the stationary state is not yet reached, e.g. the number N_1 is smaller than in the equilibrium, there will be less transitions to state 2 and more to state 1 on average than in equilibrium. The occupation number of state 1 will therefore increase until equilibrium is reached. Since transitions are performed stochastically, even in equilibrium the occupation numbers will fluctuate around their nominal values.

If now, instead of discrete states, we consider systems that are characterized by a continuous variable x , the occupation numbers are to be replaced by a density distribution $f(x)$ where x is multidimensional. It represents the total of all energies of all molecules. As above, for a stationary system we have

$$f(x)W(x \rightarrow x') = f(x')W(x' \rightarrow x) .$$

As probability $P(x \rightarrow x')$ for a transition from state x to a state x' we choose the Boltzmann acceptance function

$$\begin{aligned} P(x \rightarrow x') &= \frac{W(x \rightarrow x')}{W(x \rightarrow x') + W(x' \rightarrow x)} \\ &= \frac{f(x')}{f(x) + f(x')} . \end{aligned}$$

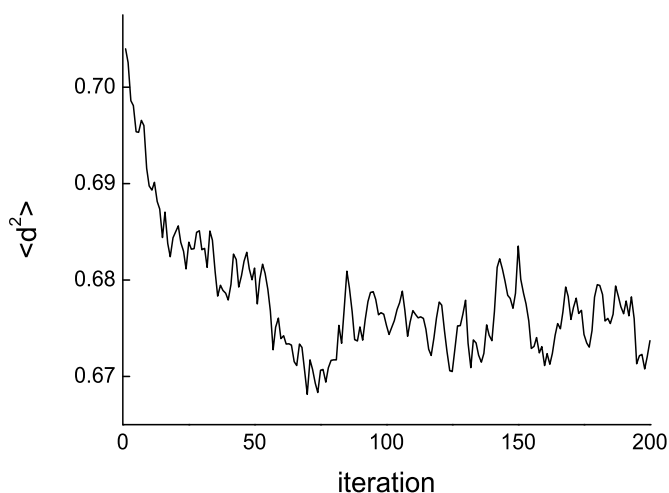


Fig. 5.8. Mean distance of spheres as a function of the number of iterations.

In an ideal gas and in many other systems the transition regards only one or two molecules and we need only consider the effect of the change of those. Then the evaluation of the transition probability is rather simple. Now we simulate the stochastic changes of the states with the computer, by choosing a molecule at random and change its state with the probability $P(x \rightarrow x')$ into a also randomly chosen state x' ($x \rightarrow x'$). The choice of the initial distribution for x is relevant for the speed of convergence but not for the asymptotic result.

This mechanism has been introduced by Metropolis et al. [28] with a different acceptance function in 1953. It is well suited for the calculation of mean values and fluctuations of parameters of thermodynamical or quantum statistical distributions. The process continues after the equilibrium is reached and the desired quantity is computed periodically. This process simulates a periodic measurement, for instance of the energy of a gas with small number of molecules in a heat bath. Measurements performed shortly one after the other will be correlated. The same is true for sequentially probed quantities of the MCMC sampling. For the calculation of statistical fluctuations the effect of correlations has to be taken into account. It can be estimated by varying the number of moves between subsequent measurements.

Example 67. Mean distance of gas molecules

We consider an atomic gas enclosed in a cubic box located in the gravitational field of the earth. The N atoms are treated as hard balls with given radius R . Initially the atoms are arranged on a regular lattice. The p.d.f. is zero for overlapping atoms, and proportional to $e^{-\alpha z}$, where z is the vertical coordinate of a given atom. The exponential factor corresponds to the Boltzmann distribution for the potential energy in the gravitational field. An atom is chosen randomly. Its position may be (x, y, z) . A second position inside the box is randomly selected by means of three uniformly distributed random numbers. If within a distance of less than $2R$ another atom is found, the move is rejected and we repeat the selection of a possible new location. If the position search with the coordinates (x', y', z') is successful, we form the ratio

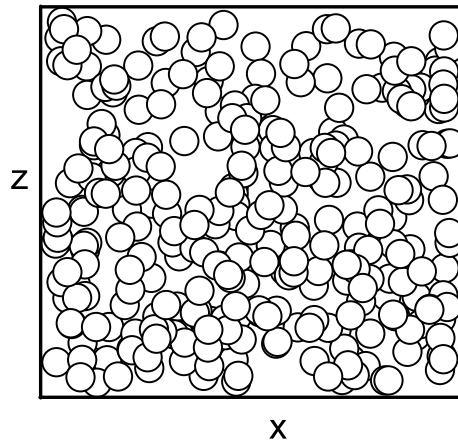


Fig. 5.9. Solid spheres in a box. The plot is a projection onto the x-z plane.

$w = e^{-az'} / (e^{-az'} + e^{-az})$. The position is changed if the condition $r < w$ is fulfilled, with a further random number r . Periodically, the quantity being studied, here the mean distance between atoms, is calculated. It is displayed in Fig. 5.8 as a function of the iteration number. Its mean value converges to an asymptotic value after a number of moves which is large compared to the number of atoms. Fig. 5.9 shows the position of atoms projected to the x-z plane, for 300 out of 1000 considered atoms, after 20000 moves. Also the statistical fluctuations can be found and, eventually, re-calculated for a modified number of atoms according to the $1/\sqrt{N}$ -factor.

5.3 Solution of Integrals

The generation of distributions has always the aim, finally, to evaluate integrals. There the integration consists in simply counting the sample elements (the events), for instance, when we determine the acceptance or efficiency of a detector.

The integration methods follow very closely those treated above for the generation of distributions. To simplify the discussion, we will consider mainly one-dimensional integrals. The generalization to higher dimensions, where the advantages of the Monte Carlo method become even more pronounced than for one-dimensional integration, does not impose difficulties.

Monte Carlo integration is especially simple and has the additional advantage that the accuracy of the integrals can be determined by the usual methods of statistical error estimation. Error estimation is often quite involved with the conventional numerical integration methods.

5.3.1 Simple Random Selection Method

Integrals with the integrand changing sign are subdivided into integrals over intervals with only positive or only negative integrand. Hence it is sufficient to consider only

the case

$$I = \int_{x_a}^{x_b} y(x) dx \quad \text{with } y > 0. \quad (5.2)$$

As in the analogous case when we generate a distribution, we produce points which are distributed randomly and uniformly in a rectangle covering the integrand function. An estimate $\hat{I}$ for the area I is obtained from the ratio of successes – this are the points falling below the function $y(x)$ – to the number of trials N_0 , multiplied by the area I_0 of the rectangle:

$$\hat{I} = I_0 \frac{N}{N_0}.$$

To evaluate the uncertainty of this estimate, we refer to the binomial distribution in which we approximate the probability of success ε by the experimental value $\varepsilon = N/N_0$:

$$\begin{aligned} \delta N &= \sqrt{N_0 \varepsilon (1 - \varepsilon)}, \\ \frac{\delta \hat{I}}{\hat{I}} &= \frac{\delta N}{N} = \sqrt{\frac{1 - \varepsilon}{N}}. \end{aligned} \quad (5.3)$$

As expected, the accuracy raises with the square root of the number of successes and with ε . The smaller the deviation of the curve from the rectangle, the less will be the uncertainty.

Example 68. Photon-yield for a particle crossing a scintillating fiber

Ionizing particles are crossing a scintillating fiber with circular cross section perpendicular to the fiber axis which is parallel to the z-axis (Fig. 5.10), and generate photons with spatially isotropic angular distribution (see 5.2.2). Photons hitting the fiber surface will be reflected if the angle with respect to the surface normal is larger than $\beta_0 = 60^\circ$. For smaller angles they will be lost. We want to know, how the number of captured photons depends on the location where the particle intersects the fiber. The particle traverses the fiber in y direction at a distance x from the fiber axis. To evaluate the acceptance, we perform the following steps:

- Set the fiber radius $R = 1$, create a photon at x, y uniformly distributed in the square $0 < x, y < 1$,
- calculate $r^2 = x^2 + y^2$, if $r^2 > 1$ reject the event,
- chose azimuth angle φ for the photon direction, with respect to an axis parallel to the fiber direction in the point x, y , $0 < \varphi < 2\pi$, φ uniformly distributed,
- calculate the projected angle α ($\sin \alpha = r \sin \varphi$),
- choose a polar angle ϑ for the photon direction, $0 < \cos(\vartheta) < 1$, $\cos(\vartheta)$ uniformly distributed,
- calculate the angle β of the photon with respect to the (inner) surface normal of the fiber, $\cos \beta = \sin \vartheta \cos \alpha$,
- for $\beta < \beta_0$ reject the event,
- store x for the successful trials in a histogram and normalize to the total number of trials.

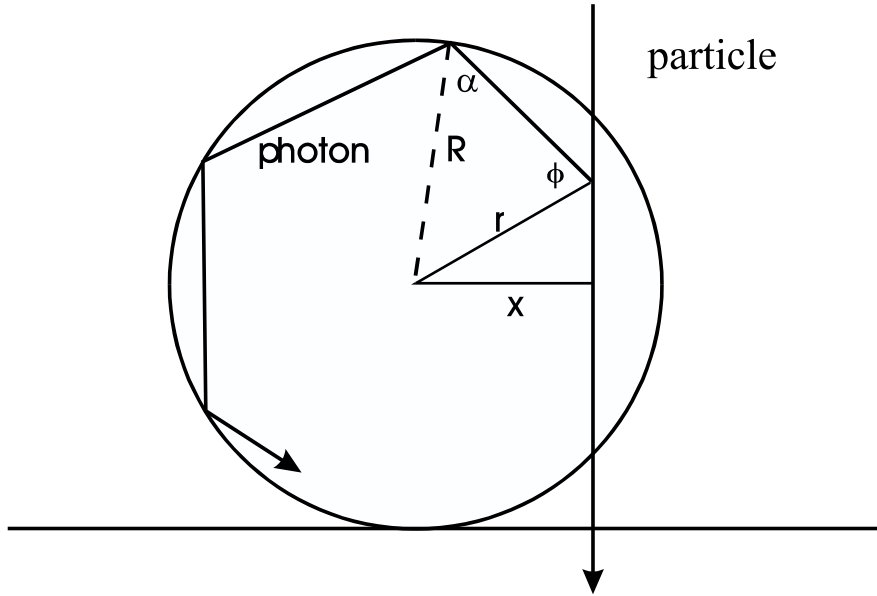


Fig. 5.10. Geometry of photon radiation in a scintillating fiber.

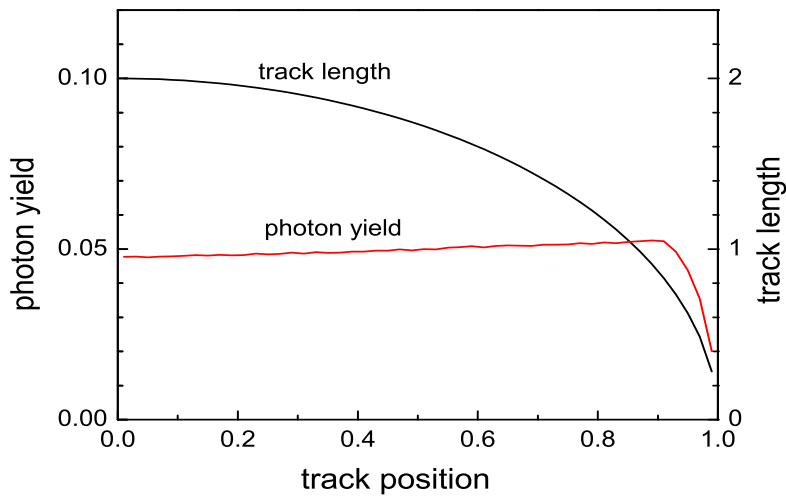


Fig. 5.11. Photon yield as a function of track position.

The efficiency is normalized such that particles crossing the fiber at $x = 0$ produce exactly 1 photon. Fig. 5.11 shows the result of our simulation. For large values of x the track length is small, but the photon capture efficiency is large, therefore the yield increases with x almost until the fiber edge.

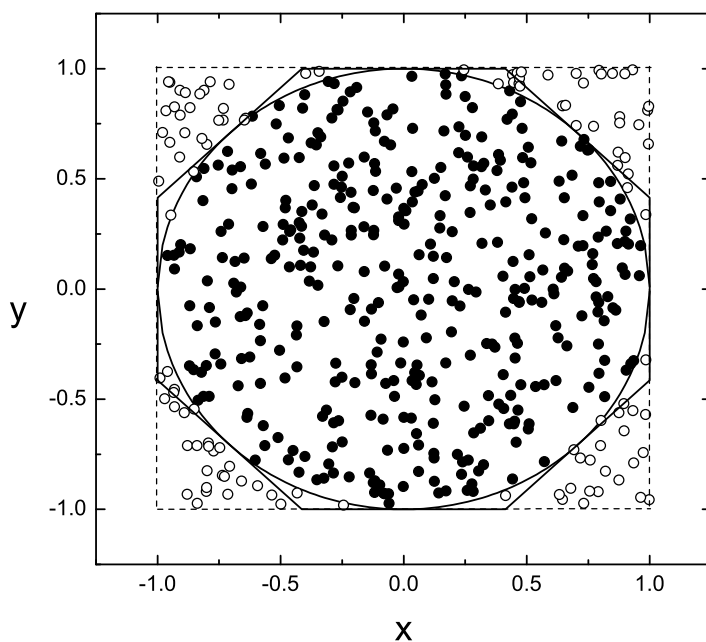


Fig. 5.12. Estimation of the number π .

5.3.2 Improved Selection Method

a) Reducing the Reference Area

We can gain in accuracy by reducing the area in which the points are distributed, as above by introduction of a majorant function, Fig. 5.5. As seen from (5.3), the relative error is proportional to the square root of the inefficiency.

We come back to the first example of this chapter:

Example 69. Determination of π

The area of a circle with radius 1 is π . For N_0 uniformly distributed trials in a circumscribed square of area 4 (Fig. 5.12) the number of successes N is on average

$$\langle N \rangle = \frac{\pi}{4} N_0 .$$

An estimate $\hat{\pi}$ for π is

$$\begin{aligned} \hat{\pi} &= \frac{4N}{N_0} , \\ \frac{\delta \hat{\pi}}{\pi} &= \frac{\sqrt{1 - \pi/4}}{\sqrt{N_0 \pi/4}} , \\ &\approx 0.52 \frac{1}{\sqrt{N_0}} . \end{aligned}$$

Choosing a circumscribed octagon as the reference area, the error is reduced by about a factor two. A further improvement is possible by inscribing an-

other polygon in the circle and considering only the area between the polygons.

b) Importance Sampling

If there exists a majorant $m(x)$ for the function $y(x)$ to be integrated,

$$I = \int_{x_a}^{x_b} y(x) dx, \quad (5.4)$$

with the property that the indefinite integral $M(x)$

$$M(x) = \int_{x_a}^x m(x') dx'$$

can be inverted, we generate N_0 x -values according to the distribution $m(x)$. For each x_i a further random number y_i in the interval $0 < y < m(x_i)$ is generated. Again, as for the simulation of distributions, points lying above $y(x_i)$ are rejected. The number N of the remaining events provides the integral

$$\hat{I} = M(x_b) \frac{N}{N_0}.$$

5.3.3 Weighting Method

a) Simple Weighting

We generate N random numbers x_i in the interval $x_a < x < x_b$ and average over the function values:

$$\bar{y} = \sum_{i=1}^N y(x_i) / N.$$

An estimate for the integral (5.4) is given by

$$\hat{I} = (x_b - x_a) \bar{y}.$$

This method corresponds to the usual numerical integration, with the peculiarity that the supporting points on the abscissa are not chosen regularly but are distributed at random. This alone cannot be an advantage, and indeed the Monte Carlo integration in one and two dimensions for a given number of supporting points is less efficient than conventional methods. It is, however, superior to other methods for multi-dimensional integrations. Already in three dimensions it competes favorably in many cases.

To estimate the accuracy, we apply the usual statistical error estimation. We consider the numbers $y_i = y(x_i)$ as N stochastic measurements of $\bar{y}$. The expected mean squared error of $\bar{y}$ is then given by (4.3):

$$(\delta \bar{y})^2 = \frac{1}{N(N-1)} \sum (y_i - \bar{y})^2.$$

The relative errors of $\bar{y}$ and $\hat{I}$ are the same,

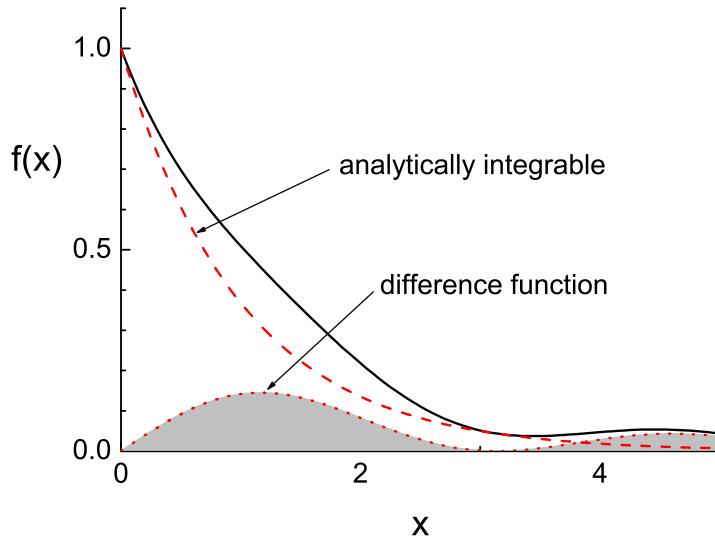


Fig. 5.13. Monte Carlo integration of the difference between the function to be integrated and an integrable function.

$$\begin{aligned} \left(\frac{\delta \hat{I}}{\hat{I}} \right)^2 &= \left(\frac{\delta \bar{y}}{\bar{y}} \right)^2, \\ &= \frac{\sum (y_i - \bar{y})^2}{N(N-1)\bar{y}^2}. \end{aligned} \quad (5.5)$$

The numerator is an estimate of the variance of the y distribution. The accuracy is the better, the smaller the fluctuations of the function around its mean value are.

b) Subtraction method

The accuracy can be improved through a reduction of the fluctuations of the integrand.

If we find a function $\tilde{y}(x)$ which is integrable analytically and does not differ too much from the original integrand $y(x)$ we cast the integral into the form

$$\int_{x_a}^{x_b} y(x) dx = \int_{x_a}^{x_b} \tilde{y}(x) dx + \int_{x_a}^{x_b} (y(x) - \tilde{y}(x)) dx.$$

We now have to evaluate by Monte Carlo only the second term with relatively small fluctuations (Fig. 5.13).

5.3.4 Reduction to Expected Values

In many cases it makes sense to factorize the integrand $y(x) = f(x)y_1(x)$ into a factor $f(x)$ corresponding to a p.d.f. normalized inside the integration interval which

is easy to generate, and a second factor $y_1(x)$. To be effective, the method requires that f is close to y . Our integral has now the form of an expected value:

$$\begin{aligned}\int_{x_a}^{x_b} y(x) dx &= \int_{x_a}^{x_b} f(x) y_1(x) dx \\ &= \langle y_1 \rangle .\end{aligned}$$

We generate values x_i distributed according to $f(x)$ and obtain from these an estimate for the integral I :

$$\begin{aligned}\hat{I} &= \frac{\sum_i y_1(x_i)}{N} , \\ \left(\frac{\delta \hat{I}}{\hat{I}} \right)^2 &= \frac{\sum [y_1(x_i) - \bar{y}_1]^2}{N(N-1)\bar{y}_1^2} .\end{aligned}$$

The estimate is again the better, the less the y_1 -values are fluctuating, i.e. the more similar the functions y and f are. The error estimate is analogous to (5.5).

5.3.5 Stratified Sampling

In stratified sampling the domain of integration is partitioned into sub-domains. Over each of these we integrate separately. The advantage is that the distribution in each sub-domain is more uniform and thus the fluctuations of the random variables are smaller and the statistical error is reduced. This method is somewhat antithetical to the basic idea of the simple Monte Carlo method, since it produces a more uniform (equidistant) distribution of the supporting points and requires some effort to combine the errors from the different contributions. Thus we recommend it only if the integrand shows very strong variations.

5.4 General Remarks

Often we need to solve integrals over different domains but always with the same integrand. In these cases the Monte Carlo approach is particularly advantageous. We store all single simulated values (usually called “events”) and are able to select events afterwards according to the chosen domain, and obtain the integral with relatively small computing expense by summation. Similarly a change of event weights is possible without repeating the generation of the events.

Let us illustrate this feature with a mechanical example: If, for instance, we want to obtain the tensor of inertia for a complex mass distribution like a car, we distribute points stochastically within the body and store their coordinates together with the respective mass densities. With these data it is easy to calculate by summations the mass, the center of mass and the moments of inertia with respect to arbitrary axes. If desired, parts of the body can be eliminated simply by rejecting the corresponding points in the sums and different materials can be considered by changing the density.

In thermodynamic systems we are often interested in several mean values, like the mean free path length, mean kinetic or potential energy, velocities etc.. Once a

statistical ensemble has been generated, all these quantities are easily obtained, while with the usual integration methods, one has to repeat each time the full integration.

Even more obvious are these advantages in acceptance calculations. Big experiments in particle physics and other areas have to be simulated as completely and realistically as allowed by the available computing power. The acceptance of a given system of particle detectors for a certain class of events is found in two steps: first, a sample of interesting events is generated and the particles produced are traced through the detecting apparatus. The hits in various detectors together with other relevant information (momenta, particle identities) are stored in data banks. In a second step the desired acceptance for a class of events is found by simulating the selection procedure and counting the fraction of events which are retained. Arbitrary changes in the selection procedure are readily implemented without the need to simulate large event samples more than once.

Finally, we want to stress again how easy it is to estimate the errors of Monte Carlo integration. It is almost identical¹ to the error estimation for the experimental data. We usually will generate a number of Monte Carlo reactions which is large enough to neglect their statistical error compared to the experimental error. In other words, the number of Monte Carlo events should be large compared to the number of experimental events. Usually a factor of ten is sufficient, a higher factor is reasonable if enough computing power is available.

¹The Monte Carlo errors are usually described by the binomial distribution, those of the experimental data by the Poisson distribution.

Parameter Inference I

6.1 Introduction

We now leave the probability calculus and its simple applications and turn to the field of statistics. More precisely we are concerned with inferential statistics.

While the probability calculus, starting from distributions, predicts properties of random samples, in statistics, given a data sample, we look for a theoretical description of the population from which it has been derived by some random process. In the simplest case, the sample consists of independent observations, randomly drawn from a parent population. If not specified differently, we assume that the population is a collection of elements which all follow the same discrete or continuous distribution. Frequently, the sample consists of data collected in a measurement sequence.

Usually we either want to check whether our sample is compatible with a specific theory, or we decide between several theories, or we infer unknown parameters of a given theory.

To introduce the problem, we discuss three simple examples:

1. At a table we find eight playing cards: two *kings*, three *ladies*, one *ten*, one *eight* and one *seven*. Do the cards belong to a set of Canasta cards or to a set of Skat cards?
2. A college is attended by 523 boys and 490 girls. Are these numbers compatible with the assumption that in average the tendency to attend a college is equal for boys and girls?
3. The lifetimes of five instable particles of a certain species have been measured. How large is the mean life of that particle and how large is the corresponding uncertainty?

In our first example we would favor the Skat game because none of the cards *two* to *six* is present which, however, are part of Canasta card sets. Assuming that the cards have been taken at random from a complete card set, we can summarize the available information in the following way: The probability to observe no card with value below *seven* in eight cards of a Canasta game is $L_C = (5/13)^8 = 4.8 \times 10^{-4}$ whereas it is $L_S = 1$ for a Skat game. We call these quantities *likelihoods*¹. The likelihood indicates how well a given hypothesis is supported by a given observation

¹The term *likelihood* was first used by the British biologist and statistician Sir Ronald Aylmer Fisher (1890-1962).

but the likelihood alone is not sufficient for a decision in favor of one or the other hypothesis. Additional considerations may play an important role. When the cards are located in a Swiss youth hostel we would consider the hypothesis *Skat* more sceptically than when the cards are found in a pub at Hamburg. We therefore would weight our hypotheses with prior probabilities (short: priors) which quantify this additional piece of information. Prior probabilities are often hard to estimate, often they are completely unknown. As a consequence, results depending on priors are model dependent.

We usually will avoid to introduce prior probabilities and stay with likelihoods but sometimes this is not possible. Then the results have to be interpreted conditional to the validity of the applied prior probabilities.

Similar situations as described in our trivial example also occur in empirical sciences. Whenever an observation is more or less compatible with two alternative theories, we cannot simply derive probabilities for the validities of theories based solely on the experimental data. Other criteria like the attractiveness of the theory, the compatibility with previous measurements will enter into our judgement. These additional attributes again can be taken into account by introducing prior probabilities which of course will depend to some extent on subjective prejudices. The cases where well founded quantitative priors are available are rare.

Some years ago, in an experiment a significant deviation from the gravitational law had been observed. From the two alternatives: H_1 : *Newton's law is correct* and H_2 : *A $1/r^2$ term has to be added to the $1/r$ potential*, the latter was much more compatible with the experimental data. In spite of this experimental evidence, hardly any serious physicist doubted the validity of the classical gravitational law. The reason is that our experience shows that the laws of nature are basically simple. In the mean time, as a consequence of the availability of more precise data, the $1/r^2$ hypothesis has been rejected.

Nevertheless it is correct and necessary to publish the observed data without weighting them with a prior, i.e. to restrict the presentation to the purely statistical result and to leave the subjective part to the reader of the publication.

The scientific method requires the possibility to compare and combine results from independent measurements. This is impossible when different authors apply varying priors. We will see that it is almost always possible to avoid the introduction of this kind of subjective assumptions into the statistical analysis.

In our second example the situation is even more complex because we are confronted with only one hypothesis and no well specified alternative. The validity of the alternative, e.g. a deviation from the equality of the distribution of the sexes is hardly measurable since an arbitrarily small deviation from the equality is present in any case. There is no other possibility as to quantify the deviation of the data with the prediction in some proper way. We will accept the hypothesis if the deviation is not abnormally high. We will treat this problem in the chapter *goodness-of-fit tests*.

In our third example the number of hypotheses is infinite. To each value of the unknown parameter, i.e. to each different mean life, corresponds a different prediction. The difficulties are very similar to those in case one. If we want to quote probabilities, we are forced to introduce a priori probabilities – here for the parameter under investigation. Again, in most cases no reliable prior information will be available but we can avoid the subjective part by documenting the results in a sensible way. We

will quote the parameter best supported by the data and define an error interval based on the likelihood of the parameter values.

The following table summarizes the cases which we have discussed.

<u>case 1</u>	given: N alternative hypotheses H_i wanted: relative probabilities for the validity of H_i
<u>case 2</u>	given: one hypothesis H_0 wanted: a quantitative statement about the validity of H_0
<u>case 3</u>	given: one valid hypothesis $H(\lambda)$ where λ is a single parameter or a set of unknown continuous parameters wanted: “best” value of λ and its uncertainty

In practice we often will compare observations with a theory which contains free parameters. In this case we have to infer parameters and to test the compatibility of the hypothesis with the data, i.e. case 2 and case 3 apply.

We now address the different problems one after the other.

Remark 1: In the following chapters we consider parameters as random variables as is common in Bayesian statistics. Sometimes, we assign probabilities to the possible values or intervals of parameters, probabilities which reflect our knowledge of these parameters.

Remark 2: We are primarily interested in the estimation of constants of nature. A different problem is the estimation of the individual parameters of an ensemble of events, like those of particle tracks, where the individual tracks have different parameter values. An analogue situation is common in social sciences and commercial applications but is of little importance in particle physics and astronomy. It will be treated shortly in the appendix.

6.2 Inference with Given Prior

We now try to derive from a given sample probabilities for hypotheses or parameters. If prior information is available this is possible by means of Bayes’ theorem.

6.2.1 Discrete Hypotheses

In Chap. 1 we had shown that conditional probabilities fulfil the following relation (*Bayes’ theorem*):

$$P\{A \cap B\} = P\{A|B\}P\{B\} = P\{B|A\}P\{A\}. \quad (6.1)$$

The probability $P\{A \cap B\}$, that both the properties A and B apply is equal to the probability $P\{B\}$, to find property B multiplied by the conditional probability $P\{A|B\}$ to find A , when B is realized. This is the first part of the relation above. The second part is analogous.

We apply this relation to a discrete random variable k and hypotheses H_i . The index denoting the hypothesis is interpreted as a random variable².

²In this case this is a categorical variable which denotes a certain class.

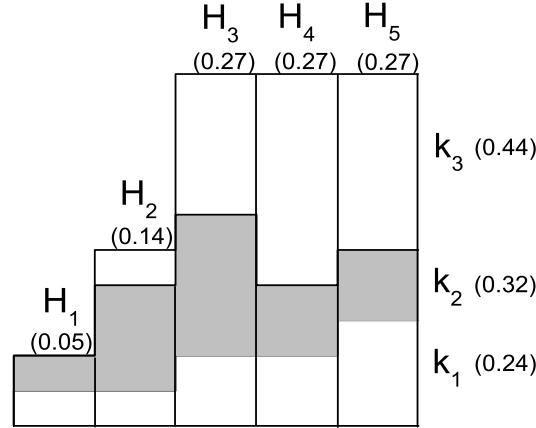


Fig. 6.1. Quantitative Venn diagram. The areas indicate the probabilities for certain combinations of hypotheses H_i and discrete events of type k_j .

We assume that the probability $P\{k|H_i\}$ to observe k is given for a finite number of alternatively exclusive hypotheses H_i . Then we have

$$P\{k|H_i\}P\{H_i\} = P\{H_i|k\}P\{k\},$$

$$P\{H_i|k\} = \frac{P\{k|H_i\}P\{H_i\}}{P\{k\}}. \quad (6.2)$$

Here $P\{H_i\}$ is the assumed probability for the validity of hypothesis i before the observation happens, it is the a priori probability, in short the prior.

In Fig. 6.1 we illustrate relation (6.2) in form of a so called Venn diagram where in the present example 3 out of the 5 hypotheses have the same prior. Each hypothesis bin is divided into 3 regions with areas proportional to the probabilities to observe $k = k_1$, $k = k_2$ and $k = k_3$, respectively. For example when the observation is $k = k_2$ (shadowed in gray) then the gray areas provide the relative probabilities of the validity of the corresponding hypotheses. In our example hypothesis H_3 is the most probable, H_1 the most unlikely.

The computation of $P\{k\}$ which is the marginal distribution of k , i.e. the probability of a certain observation, summed over all hypotheses, yields:

$$P\{k\} = \sum_i P\{k|H_i\}P\{H_i\}.$$

As required, $P\{H_i|k\}$ is normalized in such a way that the probability that any of the hypotheses is fulfilled is equal to one. We get

$$P\{H_i|k\} = \frac{P\{k|H_i\}P\{H_i\}}{\sum_j P\{k|H_j\}P\{H_j\}}. \quad (6.3)$$

In words: The probability for the validity of hypothesis H_i after the measurement k is equal to the prior $P\{H_i\}$ of H_i multiplied with the probability to observe k if H_i applies and divided by a normalization factor. When we are only interested in the

relative probabilities of two different hypotheses H_i and H_j for an observation k , the denominator $P\{k\}$ cancels:

$$\frac{P\{H_i|k\}}{P\{H_j|k\}} = \frac{P\{k|H_i\}P\{H_i\}}{P\{k|H_j\}P\{H_j\}}.$$

Example 70. Bayes' theorem: pion or kaon decay?

A muon has been detected. Does it originate from a pion or from a kaon decay? The decay probabilities inside the detector are known and are $P\{\mu|\pi\} = 0.02$ and $P\{\mu|K\} = 0.10$, respectively. The ratio of pions and kaons in the beam is $P\{\pi\} : P\{K\} = 3 : 1$. With these numbers we obtain:

$$\begin{aligned} \frac{P\{K|\mu\}}{P\{\pi|\mu\}} &= \frac{0.10 \times 1}{0.02 \times 3} = \frac{5}{3}, \\ \frac{P\{K|\mu\}}{P\{K|\mu\} + P\{\pi|\mu\}} &= \frac{0.10 \times 1}{0.02 \times 3 + 0.10 \times 1} = 0.625. \end{aligned}$$

The kaon hypothesis is more likely than the pion hypothesis. Its probability is 0.625.

6.2.2 Continuous Parameters

Now we extend our considerations to the case where the hypothesis index is replaced by a continuous parameter θ , i.e. we have an infinite number of hypotheses. Instead of probabilities we obtain probability densities. Bayes' theorem now reads

$$f(x, \theta) = f_x(x|\theta)\pi_\theta(\theta) = f_\theta(\theta|x)\pi_x(x) \quad (6.4)$$

which is just the relation 3.36 of Sect. 3.5, where f_x, f_θ are conditional distribution densities and $\pi_x(x), \pi_\theta(\theta)$ are the marginal distributions of $f(x, \theta)$. The joint probability density $f(x, \theta)$ of the two random variables x, θ is equal to the conditional probability density $f_x(x|\theta)$ of x , where θ is fixed, multiplied by the probability density $\pi_\theta(\theta)$, the marginal distribution of θ . For an observation x we obtain analogously to our previous relations

$$f_\theta(\theta|x) = \frac{f_x(x|\theta)\pi_\theta(\theta)}{\pi_x(x)},$$

and

$$f_\theta(\theta|x) = \frac{f_x(x|\theta)\pi_\theta(\theta)}{\int_{-\infty}^{\infty} f_x(x|\theta)\pi_\theta(\theta)d\theta}. \quad (6.5)$$

In words: For a measurement with the result x , we compute the probability density for the parameter θ from the value of the probability density $f_x(x|\theta)$ for x , multiplied by the probability density (prior) $\pi_\theta(\theta)$ of θ before the measurement, divided by a normalization integral. Again, the quantity $f_x(x|\theta)$ determines how strongly various parameter values θ are supported by the given observation x and is called – in this context – likelihood of θ .

From the probability density $f_\theta(\theta|x)$ of the interesting parameter we can derive a best estimate $\hat{\theta}$ and an error interval. An obvious choice is the expectation value and the standard deviation. Thus the estimate is a function of the observations³, $\hat{\theta} = \hat{\theta}(x)$.

³A function of the observations is called a *statistic*, to be distinguished from the discipline *statistics*.

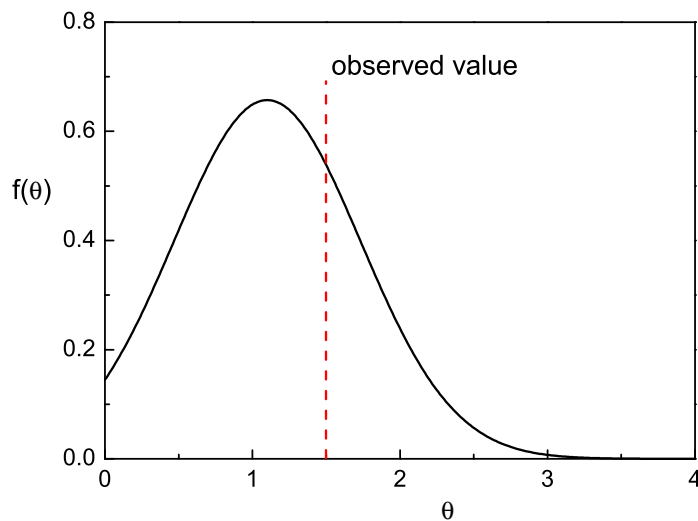


Fig. 6.2. Probability density for the true decay time. The mean decay time is 1, the observed value is 1.5.

Example 71. Time of a decay with exponential prior

A detector with finite resolution registers at time t the decay of a K meson. The time resolution corresponds to a Gaussian with variance σ^2 . We are interested in the time θ at which the decay occurred. The mean lifetime τ of kaons is known. The probability density for the parameter θ before the measurement, the prior, is $\pi(\theta) = e^{-\theta/\tau}/\tau$, $\theta \geq 0$. The probability density for t with θ fixed is the Gaussian. Applying (6.5) we obtain the probability density $f(\theta) = f(\theta|t)$ of the parameter θ ,

$$f(\theta) = \frac{e^{-(t-\theta)^2/(2\sigma^2)} e^{-\theta/\tau}}{\int_0^\infty e^{-(t-\theta)^2/(2\sigma^2)} e^{-\theta/\tau} d\theta},$$

which is displayed in Fig. 6.2. As a consequence of the exponential prior it is visibly shifted to the left with respect to the observation.

If the value of the probability density $f_x(x|\theta)$ in (6.5) varies much more rapidly with θ than the prior – this is the case when the observation restricts the parameter drastically – then to a good approximation the prior can be regarded as constant in the interesting region. We then have

$$f_\theta(\theta|x) \approx \frac{f_x(x|\theta)}{\int_{-\infty}^{\infty} f_x(x|\theta) d\theta}.$$

In this approximation the probability density f_θ of the parameter corresponds to the normalized likelihood function.

In practice, f_θ often follows to a good approximation a normal distribution. The value $\hat{\theta}$ where f_θ is maximal then is the estimate of θ and the values where f_θ has decreased by the factor $e^{1/2}$ define a standard deviation error interval and thus fix the uncertainty of the estimate $\hat{\theta}$.

6.3 Definition and Visualization of the Likelihood

Usually we do not know the prior or our ideas about it are rather vague.

Example 72. Likelihood ratio: $V + A$ or $V - A$ reaction?

An experiment is performed to measure the energy E of muons produced in the decay of the tau lepton, $\tau^- \rightarrow \mu^- \nu_\tau \bar{\nu}_\mu$, to determine whether the decay corresponds to a $V - A$ or a $V + A$ matrix element. We know the corresponding normalized decay distributions $f_-(E)$, $f_+(E)$ and we can derive the ratio $R_L = f_-(E)/f_+(E)$. But how should we choose the prior densities for the two alternative hypotheses? In this example it would not make sense to quantify our prejudices for one or the other hypothesis and to publish the resulting probabilities. We restrict the information to the ratio

$$R_L = \frac{f_-(E)}{f_+(E)} .$$

The quantity R_L is called likelihood ratio.

In the absence of prior information the likelihood ratio is the only element which we have, to judge the relative virtues of alternative hypotheses.

Definition: The likelihood L_i of a hypothesis H_i , to which corresponds a probability density $f_i(x) \equiv f(x|H_i)$ or a discrete probability distribution $W_i(k) \equiv P\{k|H_i\}$, when the observation x, k , respectively, has been realized, is equal to

$$L_i \equiv L(i|x) = f_i(x)$$

and

$$L_i \equiv L(i|k) = W_i(k) ,$$

respectively. Here the index i denoting the hypothesis is treated as an independent random variable. When we replace it by a continuous parameter θ and consider a parameter dependent p.d.f. $f(x|\theta)$ or a discrete probability distribution $W(k|\theta)$ and observations x, k , the corresponding likelihoods are

$$\begin{aligned} L(\theta) &\equiv L(\theta|x) = f(x|\theta) , \\ L(\theta) &\equiv L(\theta|k) = W(k|\theta) . \end{aligned}$$

While the likelihood is related to the validity of a hypothesis given an observation, the p.d.f. is related to the probability to observe a variate for a given hypothesis. In our notation, the quantity which is considered as fixed is placed behind the bar while the variable quantity is located left of it. When both quantities are fixed the function values of both the likelihood and the p.d.f. are equal. To attribute a likelihood makes sense only if alternative hypotheses, either discrete or differing by parameters, can apply. If the likelihood depends on one or several continuous parameters, we talk of a *likelihood function*

Remark: The likelihood function is not a probability density of the parameter. It has no differential element like $d\theta$ involved and does not obey the laws of probability. To distinguish it from probability, R.A. Fisher had invented the name likelihood. Multiplied by a prior and normalized, a probability density of the parameter is obtained. Statisticians call this *inverse probability* or *probability of causes* to emphasize

that compared to the *direct probability* where the parameter is known and the chances of an event are described, we are in the inverse position where we have observed the event and want to associate probabilities to the various causes that could have led to the observation.

As already stated above, the likelihood of a certain hypothesis is large if the observation is probable for this hypothesis. It measures how strongly a hypothesis is supported by the data. If an observation is very unlikely the validity of the hypothesis is doubtful – however this classification applies only when there is an alternative hypothesis with larger likelihood. Only relations between likelihoods make sense.

Usually experiments provide a sample of N independent observations x_i which all follow independently the same p.d.f. $f(x|\theta)$ which depends on the unknown parameter θ (i.i.d. variates). The combined p.d.f. $\tilde{f}$ then is equal to the product of the N simple p.d.f.s

$$\tilde{f}(x_1, \dots, x_N|\theta) = \prod_{i=1}^N f(x_i|\theta) .$$

For discrete variates we have the corresponding relation

$$\tilde{W}(k_1, \dots, k_N|\theta) = \prod_{i=1}^N W(k_i|\theta) .$$

For all values of θ the function $\tilde{f}$ evaluated for the sample $x_1, \dots, x_N$ is equal to the *likelihood* $\tilde{L}$

$$\begin{aligned} \tilde{L}(\theta) &\equiv \tilde{L}(\theta|x_1, x_2, \dots, x_N) \\ &= \tilde{f}(x_1, x_2, \dots, x_N|\theta) \\ &= \prod_{i=1}^N f(x_i|\theta) \\ &= \prod_{i=1}^N L(\theta|x_i) . \end{aligned}$$

The same relation also holds for discrete variates:

$$\begin{aligned} \tilde{L}(\theta) &\equiv \tilde{L}(\theta|k_1, \dots, k_N) \\ &= \prod_{i=1}^N W(k_i|\theta) \\ &= \prod_{i=1}^N L(\theta|k_i) . \end{aligned}$$

When we have a sample of independent observations, it is convenient to consider the logarithm of the likelihood. It is called *log-likelihood*. It is equal to

$$\ln \tilde{L}(\theta) = \sum_{i=1}^N \ln [f(x_i|\theta)]$$

for continuous variates. A corresponding relation holds for discrete variates.

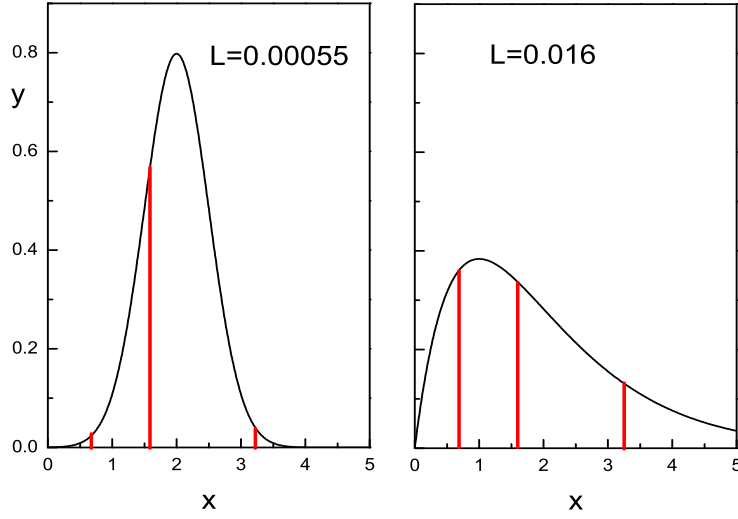


Fig. 6.3. Likelihood of three observations and two hypotheses with different p.d.f.s.

Fig. 6.3 illustrates the notion of likelihood in a concrete case of two hypotheses. For two given hypotheses and a sample of three observations we present the values of the likelihood, i.e. the products of the three corresponding p.d.f. values. The broad p.d.f. in the right hand picture matches better. Its likelihood is about thirty times higher than that of the left hand hypothesis.

So far we have considered the likelihood of samples of i.i.d. variates. Also the case where two independent experiments A , B measure the same quantity x is of considerable interest. The combined likelihood L is just the product of the individual likelihoods $L_A(\theta|x_1) = f_A(x_1|\theta)$ and $L_B(\theta|x_2) = f_B(x_2|\theta)$ as is obvious from the definition:

$$\begin{aligned} f(x_1, x_2|\theta) &= f_A(x_1|\theta)f_B(x_2|\theta) , \\ L(\theta) &= f(x_1, x_2|\theta) , \end{aligned}$$

hence

$$\begin{aligned} L &= L_A L_B , \\ \ln L &= \ln L_A + \ln L_B . \end{aligned}$$

We state: *The likelihood of several independent observations or experiments is equal to the product of the individual likelihoods. Correspondingly, the log-likelihoods add up.*

$$\begin{aligned} L &= \prod L_i , \\ \ln L &= \sum \ln L_i . \end{aligned}$$

6.4 The Likelihood Ratio

To discriminate between hypotheses, we use the likelihood ratio. According to a lemma of Neyman and Pearson there is no other more powerful quantity. This means that classifying according to the likelihood ratio, we can obtain the smallest number of false decisions (see Chap. 10). When we have to choose between more than two hypotheses, there are of course several independent ratios.

Example 73. Likelihood ratio of Poisson frequencies

We observe 5 decays and want to compute the relative probabilities for three hypotheses. Prediction H_1 assumes a Poisson distribution with expectation value 2, H_2 and H_3 have expectation values 9 and 20, respectively. The likelihoods following from the Poisson distribution $P_\lambda(k)$ are:

$$\begin{aligned} L_1 &= P_2(5) \approx 0.036 , \\ L_2 &= P_9(5) \approx 0.061 , \\ L_3 &= P_{20}(5) \approx 0.00005 . \end{aligned}$$

We can form different likelihood ratios. If we are interested for example in hypothesis 2, then the quotient $L_2/(L_1 + L_2 + L_3) \approx 0.63$ is relevant⁴. If we observe in a second measurement in the same time interval 8 decays, we obtain:

$$\begin{aligned} L_1 &= P_2(5)P_2(8) = P_4(13) \approx 6.4 \cdot 10^{-3} , \\ L_2 &= P_9(5)P_9(8) = P_{18}(13) \approx 5.1 \cdot 10^{-2} , \\ L_3 &= P_{20}(5)P_{20}(8) = P_{40}(13) \approx 6.1 \cdot 10^{-7} . \end{aligned}$$

The likelihood ratio $L_2/(L_1 + L_2 + L_3) \approx 0.89$ (for H_1 and H_3 correspondingly 0.11 and 10^{-5}) now is much more significant. The fact that all values L_i are small is unimportant because one of the three hypotheses has to be valid.

We now apply the same procedure to hypotheses with probability densities.

Example 74. Likelihood ratio of normal distributions

We compare samples drawn from one out of two alternative normal distributions with different expectation values and variances (Fig. 6.4)

$$\begin{aligned} f_1 &= \frac{1}{\sqrt{2\pi}1} e^{-(x-1)^2/2} , \\ f_2 &= \frac{1}{\sqrt{2\pi}2} e^{-(x-2)^2/8} . \end{aligned}$$

a) Initially the sample consists of a single observation at $x = 0$, for both cases one standard deviation off the mean values of the two distributions (Fig. 6.4a):

$$\frac{L_1}{L_2} = 2 \frac{e^{-1/2}}{e^{-4/8}} = 2 .$$

⁴We remember that the likelihood ratio is not the ratio of the probabilities. The latter depends on prior probabilities of the hypotheses.

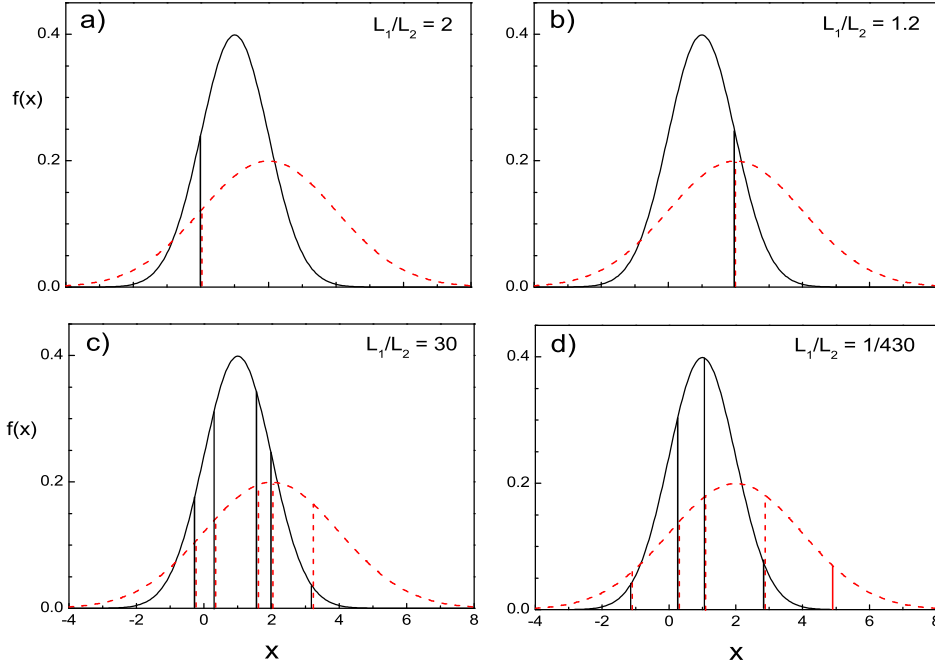


Fig. 6.4. Likelihood ratio for two normal distributions. Top: 1 observation, bottom: 5 observations.

b) Now we place the observation at $x = 2$, the maximum of the second distribution (Fig. 6.4b):

$$\frac{L_1}{L_2} = 2 \frac{e^{-1/2}}{e^{-0}} = 1.2 .$$

c) We now consider five observations which have been taken from distribution f_1 (Fig. 6.4c) and distribution f_2 , respectively (Fig. 6.4d). We obtain the likelihood ratios

$$\begin{aligned} L_1/L_2 &= 30 \quad (\text{Fig. 5.3c}) , \\ L_1/L_2 &= 1/430 \quad (\text{Fig. 5.3d}) . \end{aligned}$$

It turns out that small distributions are easier to exclude than broad ones. On the other hand we get in case b) a preference for distribution 1 even though the observation is located right at the center of distribution 2.

Example 75. Likelihood ratio for two decay time distributions

A sample of N decay times t_i has been recorded in the time interval $t_{\min} < t < t_{\max}$. The times are expected to follow either an exponential distribution $f_1(t) \sim e^{-t/\tau}$ (hypothesis 1), or an uniform distribution $f_2(t) = \text{const.}$ (hypothesis 2). How likely are H_1 , H_2 ? First we have to normalize the p.d.f.s:

$$f_1(t) = \frac{1}{\tau} \frac{e^{-t/\tau}}{e^{-t_{\min}/\tau} - e^{-t_{\max}/\tau}} ,$$

$$f_2(t) = \frac{1}{t_{\max} - t_{\min}} .$$

The likelihoods are equal to the product of the p.d.f.s at the observations:

$$L_1 = \left[\tau \left(e^{-t_{\min}/\tau} - e^{-t_{\max}/\tau} \right) \right]^{-N} \exp \left(- \sum_{i=1}^N t_i / \tau \right) ,$$

$$L_2 = 1 / (t_{\max} - t_{\min})^N .$$

With $\bar{t} = \sum t_i / N$ the mean value of the times, we obtain the likelihood ratio

$$\frac{L_1}{L_2} = \left(\frac{t_{\max} - t_{\min}}{\tau (e^{-t_{\min}/\tau} - e^{-t_{\max}/\tau})} \right)^N e^{-N\bar{t}/\tau} .$$

6.5 The Maximum Likelihood Method for Parameter Inference

In the previous examples we have compared a sample with different hypotheses which differed only in the value of a parameter but corresponded to the same distribution. We now allow for an infinite number of hypotheses by varying the value of a parameter. As in the discrete case, in the absence of a given prior probability, *the only available piece of information which allows us to judge different parameter values is the likelihood function*. A formal justification for this assertion is given by the *likelihood principle* (LP) which states that the likelihood function exhausts all the information contained in the observations related to the parameters and which we will discuss in the following chapter. It is then plausible to choose the parameter such that the likelihood is as large as possible. This is the *maximum likelihood estimate* (MLE). When we are interested in a parameter range, we will choose the interval such that the likelihood outside is always less than inside.

Remark that the MLE as well as likelihood intervals are invariant against transformations of the parameter. The likelihood is not a p.d.f. but a function of the parameter and therefore $L(\theta) = L'(\theta')$ for $\theta'(\theta)$. Thus a likelihood analysis estimating, for example, the mass of a particle will give the same result as that inferring the mass squared, and estimates of the decay rate γ and mean life $\tau = 1/\gamma$ will be consistent.

Here and in the following sections we assume that the likelihood function is continuous and differentiable and has exactly one maximum inside the valid range of the parameter. This condition is fulfilled in the majority of all cases.

Besides the maximum likelihood (ML) method, invented by Fisher, there exist a number of other methods of parameter estimation. Popular is especially the method of *least squares* (LS) which was first proposed by Gauß⁵. It is used to adjust parameters of curves which are fixed by some measured points and will be discussed in the next chapter. It can be traced back to the ML method if the measurement errors are normally distributed.

⁵Carl Friedrich Gauß (1777-1855), German mathematician, astronomer and physicist.

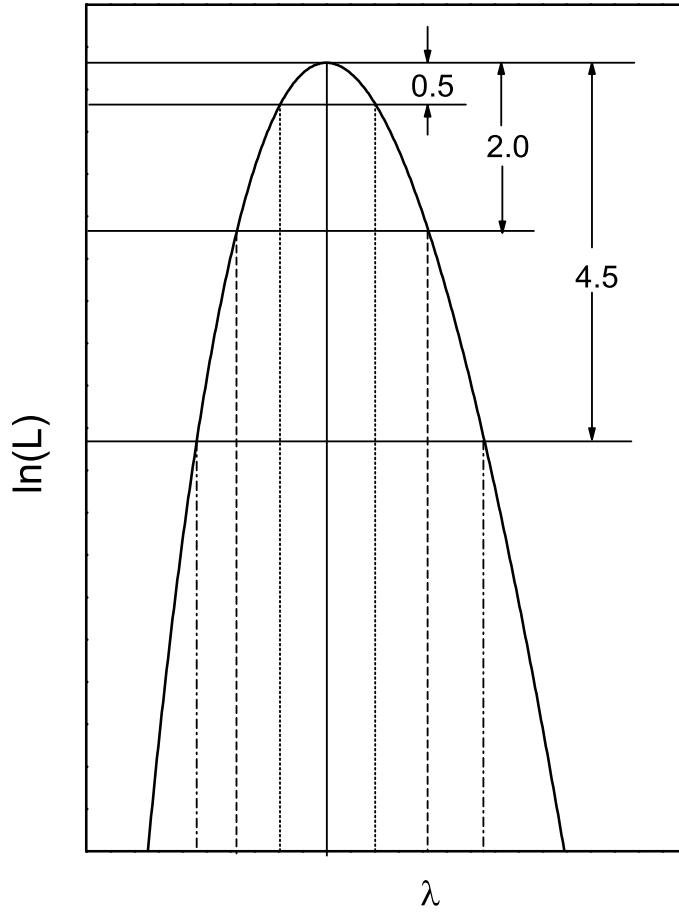


Fig. 6.5. Log-likelihood function and uncertainty limits for 1, 2, 3 standard deviations.

In most cases we are not able to compute analytically the location of the maximum of the likelihood. To simplify the numerical computation, still linear approximations (e.g. linear regression) are used quite frequently. These methods find the solution by matrix operations and iteration. They are dispensable nowadays. With common PCs and maximum searching programs the maximum of a function of some hundred parameters can be determined without problem, given enough observations to fix it.

6.5.1 The Recipe for a Single Parameter

We proceed according to the following recipe. Given a sample of N i.i.d. observations $\{x_1, \dots, x_N\}$ from a p.d.f. $f(x|\theta)$ with unknown parameter θ , we form the likelihood or its logarithm, respectively, in the following way:

$$L(\theta) = \prod_{i=1}^N f(x_i|\theta), \quad (6.6)$$

$$\ln L(\theta) = \sum_{i=1}^N \ln f(x_i|\theta). \quad (6.7)$$

In most cases the likelihood function resembles a bell shaped Gaussian and $\ln L(\theta)$ approximately a downwards open parabola (see Fig. 6.5). This approximation is especially good for large samples.

To find the maximum of L ($\ln L$ and L have their maxima at the same location), we derive the log-likelihood⁶ with respect to the parameter and set the derivative equal to zero. The value $\hat{\theta}$, that satisfies the equation which we obtain in this way is the MLE of θ .

$$\left. \frac{d \ln L}{d\theta} \right|_{\hat{\theta}} = 0 \quad (6.8)$$

Since only the derivative of the likelihood function is of importance, factors in the likelihood or summands in the log-likelihood which are independent of θ can be omitted.

The estimate $\hat{\theta}$ is a function of the sample values x_i , and consequently a statistic.

The point estimate has to be accompanied by an error interval. Point estimate and error interval form an ensemble and cannot be discussed separately. Choosing as point estimate the value that maximizes the likelihood function it is natural to include inside the error limits parameter values with higher likelihood than all parameters that are excluded. This prescription leads to so-called likelihood ratio error intervals.

We will discuss the error interval estimation in a separate chapter, but fix the error limit already now by definition:

Definition: The limits of a standard error interval are located at the parameter values where the likelihood function has decreased from its maximum by a factor $e^{1/2}$. For two and three standard deviations the factors are e^2 and $e^{4.5}$. This choice corresponds to differences for the log-likelihood of 0.5 for one, of 2 for two and of 4.5 for three standard error intervals as illustrated in Fig. 6.5. For the time being we assume that these limits exist inside the parameter range.

The reason for this definition is the following: As already mentioned, asymptotically, when the sample size N tends to infinity, under very general conditions the likelihood function approaches a Gaussian and becomes proportional to the probability density of the parameter (for a proof, see Appendix 13.3). Then our error limit corresponds exactly to the standard deviation of the p.d.f., i.e. the square root of the variance of the Gaussian. We keep the definition also for non normally shaped likelihood functions and small sample sizes. Then we usually get asymmetric error limits.

6.5.2 Examples

Example 76. Maximum likelihood estimate (MLE) of the mean life of an unstable particle

Given be N decay times t_i of an unstable particle with unknown mean life τ . For an exponential decay time distribution

$$f(t|\gamma) = \gamma e^{-\gamma t}$$

with $\gamma = 1/\tau$ the likelihood is

⁶The advantage of using the log-likelihood compared to the normal likelihood is that we do not need to derive a product but a sum which is much more convenient.

$$\begin{aligned}
L &= \gamma^N \prod_{i=1}^N e^{-\gamma t_i} \\
&= \gamma^N e^{-\sum_{i=1}^N \gamma t_i} , \\
\ln L &= N \ln \gamma - \gamma \sum_{i=1}^N t_i .
\end{aligned}$$

The estimate $\hat{\gamma}$ satisfies

$$\begin{aligned}
\frac{d \ln L}{d \gamma} \Big|_{\hat{\gamma}} &= 0 , \\
0 &= \frac{N}{\hat{\gamma}} - \sum_{i=1}^N t_i , \\
\hat{\tau} = \hat{\gamma}^{-1} &= \sum_{i=1}^N t_i / N = \bar{t} .
\end{aligned}$$

Thus the estimate is just equal to the mean value $\bar{t}$ of the observed decay times. In practice, the full range up to infinitely large decay times is not always observable. If the measurement is restricted to an interval $0 < t < t_{\max}$, the p.d.f. changes, it has to be renormalized:

$$\begin{aligned}
f(t|\gamma) &= \frac{\gamma e^{-\gamma t}}{1 - e^{-\gamma t_{\max}}} , \\
\ln L &= N [\ln \gamma - \ln(1 - e^{-\gamma t_{\max}})] - \gamma \sum_{i=1}^N t_i .
\end{aligned}$$

The maximum is now located at the estimate $\hat{\gamma}$, which fulfils the relation

$$\begin{aligned}
0 &= N \left(\frac{1}{\hat{\gamma}} - \frac{t_{\max} e^{-\hat{\gamma} t_{\max}}}{1 - e^{-\hat{\gamma} t_{\max}}} \right) - \sum_{i=1}^N t_i , \\
\hat{\tau} &= \bar{t} + \frac{t_{\max} e^{-t_{\max}/\hat{\tau}}}{1 - e^{-t_{\max}/\hat{\tau}}} ,
\end{aligned}$$

which has to be evaluated numerically. If the time interval is not too short, $t_{\max} > \tau$, an iterative computation lends itself: The correction term at the right hand side is neglected in zeroth order. At the subsequent iterations we insert in this term the value τ of the previous iteration. We notice that the estimate again depends solely on the mean value $\bar{t}$ of the observed decay times. The quantity $\bar{t}$ is a *sufficient statistic*. We will explain this notion in more detail later. The case with also a lower bound $t_{\min}$ of t can be reduced easily to the previous one by transforming the variable to $t' = t - t_{\min}$.

In the following examples we discuss the likelihood functions and the MLEs of the parameters of the normal distribution in four different situations:

Example 77. MLE of the mean value of a normal distribution with known width (case Ia)

Given are N observation x_i drawn from a normal distribution of known width σ and mean value μ to be estimated:

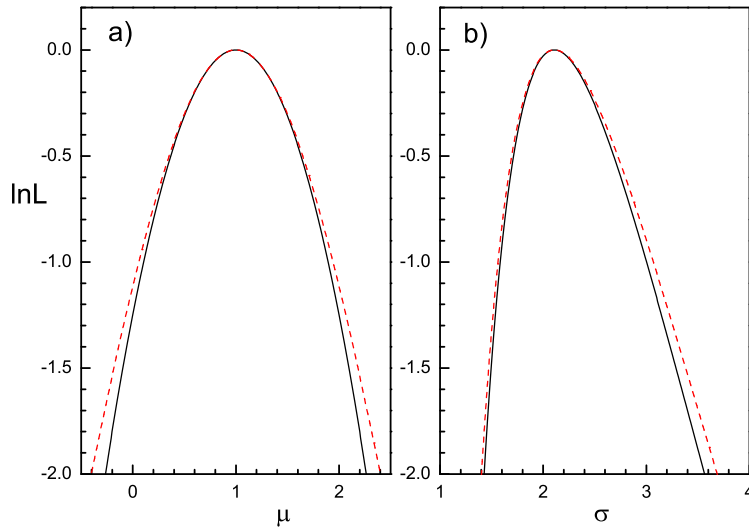


Fig. 6.6. Log-likelihood functions for the parameters of a normal distribution: a) for the mean μ with known width (solid curve) and unknown width (dashed curve), b) for the width σ with known mean (solid curve) and unknown mean (dashed curve). The position of the maximum is $\hat{\sigma} = \sigma \sqrt{N/(N-1)} = 2.108$ for both curves.

$$\begin{aligned}
 f(x|\mu) &= \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{(x-\mu)^2}{2\sigma^2} \right], \\
 L(\mu) &= \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{(x_i-\mu)^2}{2\sigma^2} \right], \\
 \ln L(\mu) &= -\sum_{i=1}^N \frac{(x_i-\mu)^2}{2\sigma^2} + \text{const} \\
 &= -N \frac{\overline{x^2} - 2\bar{x}\mu + \mu^2}{2\sigma^2} + \text{const}.
 \end{aligned} \tag{6.9}$$

The log-likelihood function is a parabola. It is shown in Fig. 6.6a for $\sigma = 2$. Deriving it with respect to the unknown parameter μ and setting the result equal to zero, we get

$$\begin{aligned}
 N \frac{(\bar{x} - \hat{\mu})}{\sigma^2} &= 0, \\
 \hat{\mu} &= \bar{x}.
 \end{aligned}$$

The likelihood estimate $\hat{\mu}$ for the expectation value of the normal distribution is equal to the arithmetic mean $\bar{x}$ of the sample. It is independent of σ , but σ determines the width of the likelihood function and the standard error $\delta_\mu = \sigma/\sqrt{N}$.

Example 78. MLE of the width of a normal distribution with given mean (case Ib)

Given are now N observations x_i which follow a normal distribution of unknown width σ to be estimated for known mean $\mu = 5/3$. The reason for this – in principle here arbitrary – choice will become clear below.

$$\begin{aligned} L(\sigma) &= \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right), \\ \ln L(\sigma) &= -N\left(\frac{1}{2} \ln 2\pi + \ln \sigma\right) - \sum_{i=1}^N \frac{(x_i - \mu)^2}{2\sigma^2} \\ &= -N \left[\ln \sigma + \frac{\overline{(x - \mu)^2}}{2\sigma^2} \right] + \text{const}. \end{aligned}$$

The log-likelihood function for our numerical values is presented in Fig. 6.6b. Deriving with respect to the parameter of interest and setting the result equal to zero we find

$$\begin{aligned} 0 &= \frac{1}{\hat{\sigma}} - \frac{\overline{(x - \mu)^2}}{\hat{\sigma}^3}, \\ \hat{\sigma} &= \sqrt{\overline{(x - \mu)^2}}. \end{aligned}$$

Again we obtain a well known result. The mean square deviation of the sample values provides an estimate for the width of the normal distribution. This relation is the usual distribution-free estimate of the standard deviation if the expected value is known. The error bounds from the drop of the log-likelihood function by $1/2$ become asymmetric. Solving the respective transcendental equation, thereby neglecting higher orders in $1/N$, one finds

$$\delta_{\sigma}^{\pm} = \frac{\hat{\sigma} \sqrt{\frac{1}{2N}}}{1 \mp \sqrt{\frac{1}{2N}}}.$$

Example 79. MLE of the mean of a normal distribution with unknown width (case IIa)

The solution of this problem can be taken from Sect. 3.6.11 where we found that $t = (\bar{x} - \mu)/s$ with $s^2 = \sum (x_i - \bar{x})^2 / [N(N-1)] = v^2 / (N-1)$ follows the Student's distribution with $N-1$ degrees of freedom.

$$h(t|N-1) = \frac{\Gamma(N/2)}{\Gamma((N-1)/2)\sqrt{\pi(N-1)}} \left(1 + \frac{t^2}{N-1}\right)^{-\frac{N}{2}}$$

The corresponding log-likelihood is

$$\ln L(\mu) = -\frac{N}{2} \ln \left[1 + \frac{(\bar{x} - \mu)^2}{v^2} \right]$$

with the maximum $\mu = \bar{x}$. It corresponds to the dashed curve in Fig. 6.6a. From the drop of $\ln L$ by $1/2$ we get now for the standard error squared the expression

$$\delta_{\mu}^2 = (e^{1/N} - 1)v^2.$$

This becomes for large N , after expanding the exponential function, very similar to the expression for the standard error in case Ia, whereby σ is exchanged by v .

Example 80. MLE of the width of a normal distribution with unknown mean (case IIb)

Obviously, shifting a normally distributed distribution or a corresponding sample changes the mean value but not the true or the empirical variance $v^2 = \overline{(x - \bar{x})^2}$. Thus the empirical variance v^2 can only depend on σ and not on μ . Without going into the details of the calculation, we state that Nv^2/σ^2 follows a χ^2 distribution of $N - 1$ degrees of freedom.

$$f(v^2|\sigma) = \frac{N}{\Gamma[(N-1)/2] 2\sigma^2} \left(\frac{Nv^2}{2\sigma^2} \right)^{(N-3)/2} \exp\left(-\frac{Nv^2}{2\sigma^2}\right)$$

with the log-likelihood

$$\ln L(\sigma) = -(N-1) \ln \sigma - \frac{Nv^2}{2\sigma^2}$$

corresponding to the dashed curve in Fig. 6.6b. (The numerical value of the true value of μ was chosen such that the maxima of the two curves are located at the same value in order to simplify the comparison.) The MLE is

$$\hat{\sigma}^2 = \frac{N}{N-1} v^2,$$

in agreement with our result (3.15). For the asymmetric error limits we find in analogy to example 78

$$\delta_\sigma^\pm = \frac{\hat{\sigma} \sqrt{\frac{1}{2(N-1)}}}{1 \mp \sqrt{\frac{1}{2(N-1)}}}.$$

6.5.3 Likelihood Inference for Several Parameters

We can extend our concept easily to several parameters λ_k which we combine to a vector $\boldsymbol{\lambda} = \{\lambda_1, \dots, \lambda_K\}$.

$$L(\boldsymbol{\lambda}) = \prod_{i=1}^N f(x_i|\boldsymbol{\lambda}), \quad (6.10)$$

$$\ln L(\boldsymbol{\lambda}) = \sum_{i=1}^N \ln f(x_i|\boldsymbol{\lambda}). \quad (6.11)$$

To find the maximum of the likelihood function, we set the partial derivatives equal to zero. Those values $\hat{\lambda}_k$ which satisfy the system of equations obtained this way, are the MLEs $\hat{\lambda}_k$ of the parameters λ_k :

$$\frac{\partial \ln L}{\partial \lambda_k} \Big|_{\hat{\lambda}_1, \dots, \hat{\lambda}_K} = 0. \quad (6.12)$$

The error interval is now to be replaced by an error volume with its surface defined again by the drop of $\ln L$ by $1/2$:

$$\ln L(\hat{\boldsymbol{\lambda}}) - \ln L(\boldsymbol{\lambda}) = 1/2.$$

We have to assume, that this defines a closed surface in the parameter space, in two dimensions just a closed contour, as shown in the next example.

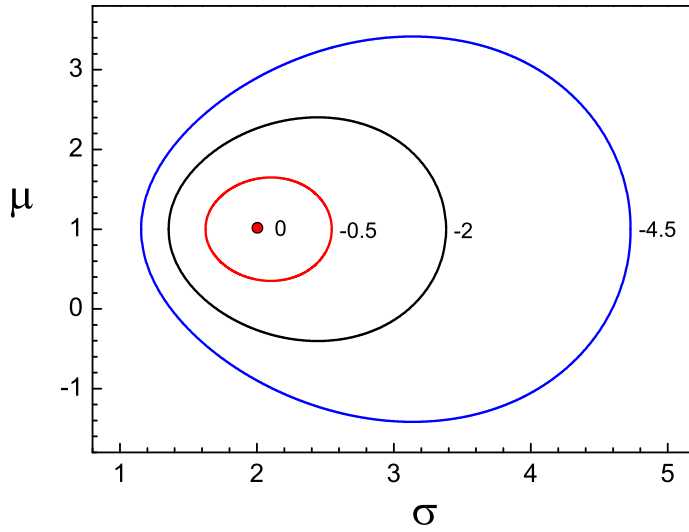


Fig. 6.7. MLE of the parameters of a normal distribution and lines of constant log-likelihood. The numbers indicate the values of log-likelihood relative to the maximum.

Example 81. MLEs of the mean value and the width of a normal distribution

Given are N observations x_i which follow a normal distribution where now both the width and the mean value μ are unknown. As above, the log-likelihood is

$$\ln L(\mu, \sigma) = -N \left[\ln \sigma + \frac{(x - \mu)^2}{2\sigma^2} \right] + \text{const} .$$

The derivation with respect to the parameters leads to the results:

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i = \bar{x} ,$$

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})^2 = \overline{(x - \bar{x})^2} = v^2 .$$

The MLE and log-likelihood contours for a sample of 10 events with empirical mean values $\bar{x} = 1$ and $\overline{x^2} = 5$ are depicted in Fig. 6.7. The innermost line encloses the standard error area. If one of the parameters, for instance $\mu = \mu_1$ is given, the log-likelihood of the other parameter, here σ , is obtained by the cross section of the likelihood function at $\mu = \mu_1$.

Similarly any other relation between μ and σ defines a curve in Fig. 6.7 along which a one-dimensional likelihood function is defined.

Remark: Frequently, we are interested only in one of the parameters, and we want to eliminate the others, the nuisance parameters. How to achieve this, will be discussed in Sect. 6.9. Generally, it is not allowed to use the MLE of a single parameter

in the multi-parameter case separately, ignoring the other parameters. While in the previous example $\hat{\sigma}$ is the correct estimate of σ if $\hat{\mu}$ applies, the solution for the estimate and its likelihood function independent of μ has been given in example 80 and that of μ independent of σ in example 79.

Example 82. Determination of the axis of a given distribution of directions⁷

Given are the directions of N tracks by the unit vectors $\mathbf{e}_k$. The distribution of the direction cosines $\cos \alpha_i$ with respect to an axis $\mathbf{u}$ corresponds to

$$f(\cos \alpha) = \frac{3}{8}(1 + \cos^2 \alpha) .$$

We search for the direction of the axis. The axis $\mathbf{u}(u_1, u_2, u_3)$ is parameterized by its components, the direction cosines u_k . (There are only two independent parameters u_1, u_2 because $u_3 = \sqrt{1 - u_1^2 - u_2^2}$ depends on u_1 and u_2 .) The log-likelihood function is

$$\ln L = \sum_{i=1}^n \ln(1 + \cos^2 \alpha_i) ,$$

where the values $\cos \alpha_i = \mathbf{u} \cdot \mathbf{e}_i$ depend on the parameters of interest, the direction cosines. Maximizing $\ln L$, yields the parameters u_1, u_2 . We omit the details of the calculation.

Example 83. Likelihood analysis for a signal with background

We want to fit a normal distribution with a linear background to a given sample. (The procedure for a background described by a higher order polynomial is analogous.) The p.d.f. is

$$f(x) = \theta_1 x + \theta_2 + \theta_3 N(x|\mu, \sigma) .$$

Here N is the normal distribution with unknown mean μ and standard deviation σ . The other parameters are not independent because f has to be normalized in the given interval $x_{\min} < x < x_{\max}$. Thus we can eliminate one parameter. Assuming that the normal distribution is negligible outside the interval, the norm D is:

$$D = \frac{1}{2}\theta_1(x_{\max}^2 - x_{\min}^2) + \theta_2(x_{\max} - x_{\min}) + \theta_3 .$$

The normalized p.d.f. is therefore

$$f(x) = \frac{\theta'_1 x + \theta'_2 + N(x|\mu, \sigma)}{\frac{1}{2}\theta'_1(x_{\max}^2 - x_{\min}^2) + \theta'_2(x_{\max} - x_{\min}) + 1} ,$$

with the new parameters $\theta'_1 = \theta_1/\theta_3$ and $\theta'_2 = \theta_2/\theta_3$. The likelihood function is obtained in the usual way by inserting the observations of the sample into $\ln L = \sum \ln f(x_i|\theta'_1, \theta'_2, \mu, \sigma)$. Maximizing this expression, we obtain the four parameters and from those the fraction of signal events $S = \theta_3/D$:

$$S = \left[1 + \frac{1}{2}\theta'_1(x_{\max}^2 - x_{\min}^2) + \theta'_2(x_{\max} - x_{\min}) \right]^{-1} .$$

⁷This example has been borrowed from the book of L. Lyons [5].

6.5.4 Combining Measurements

When parameters are determined in independent experiments, we obtain according to the definition of the likelihood the combined likelihood by multiplication of the likelihoods of the individual experiments.

$$L(\lambda) = \prod L_i(\lambda) , \\ \ln L = \sum \ln L_i .$$

The likelihood method makes it possible to combine experimental results in a extremely simple and at the same time optimal way. Thereby experimental data can originate from completely heterogeneous experiments because no assumptions about the p.d.f.s of the individual experiments enter, except that they are independent of each other.

For the combination of experimental results it is convenient to use the logarithmic presentation. In case the log-likelihoods can be approximated by quadratic parabolas, the addition again produces a parabola.

6.5.5 Normally Distributed Variates and χ^2

Frequently we encounter the situation that we have to compare measurements with normally distributed errors to a parameter dependent prediction, for instance when we fit a curve to measurements. (Remark that so far we had considered i.i.d variates, now each observation follows a different distribution.) We will come back to this problem below. For the moment let us assume that N observations x_i each following a normal distribution with variance δ_i^2

$$f(x_1, \dots, x_N) = \prod_i \frac{1}{\sqrt{2\pi\delta_i^2}} \exp \left[-\frac{(x_i - t_i(\boldsymbol{\theta}))^2}{2\delta_i^2} \right]$$

are to be compared to a function $t_i(\boldsymbol{\theta})$. The log-likelihood is

$$\ln L = -\frac{1}{2} \sum \left[\frac{(x_i - t_i(\boldsymbol{\theta}))^2}{\delta_i^2} + \ln(2\pi) + \ln \delta_i^2 \right] . \quad (6.13)$$

The first term of the sum corresponds to the expression (3.6.7) and has been denoted by χ^2 ,

$$\chi^2 = \sum \left[\frac{(x_i - t_i(\boldsymbol{\theta}))^2}{\delta_i^2} \right] .$$

For parameter inference we can omit the constant terms in (6.13) and thus have

$$\ln L = -\frac{1}{2} \chi^2 . \quad (6.14)$$

Minimizing χ^2 is equivalent to maximizing the log-likelihood. The MLE of $\boldsymbol{\theta}$ is obtained from a so-called *least square* or χ^2 fit. Since we obtain the error of the estimates $\hat{\boldsymbol{\theta}}$ from the change of the likelihood function by 1/2, χ^2 increases in the range by one unit, $\Delta\chi^2 = 1$.

6.5.6 Likelihood of Histograms

For large samples it is more efficient to analyze the data in form of histograms than to compute the likelihood for many single observations. The individual observations are classified and collected into bins where all events of a bin have approximately the same likelihood. We then compare the number of entries of a bin with the parameter dependent prediction. Often the prediction is available only as a Monte Carlo simulation in form of a histogram.

The number of entries d_i in a bin i follows a multinomial distribution where for each bin the probability $p_i(\boldsymbol{\theta})$ is computed as a function of the parameters $\boldsymbol{\theta}$. For a total N of observations the expectation t_i of the number of observations in bin i is

$$t_i(\boldsymbol{\theta}) = Np_i(\boldsymbol{\theta}) .$$

Usually the number of entries of an individual bin is small compared to the total number N . We then are allowed to apply the more comfortable Poisson distribution. The likelihood for t_i expected and d_i observed entries according to the Poisson distribution is given by

$$L_i(\boldsymbol{\theta}) = \frac{e^{-t_i} t_i^{d_i}}{d_i!} ,$$

$$\ln L_i(\boldsymbol{\theta}) = -t_i + d_i \ln t_i - \ln(d_i!) .$$

Since factors not depending on $\boldsymbol{\theta}$ are irrelevant for the likelihood inference (see Sect. 6.5.1), we are allowed to omit the term with the factorial. The log-likelihood of the complete histogram with B bins is then

$$\ln L(\boldsymbol{\theta}) = \sum_{i=1}^B (-t_i + d_i \ln t_i) . \quad (6.15)$$

The parameter dependence is hidden in the quantities t_i . The maximum of this function is determined by numerical methods.

For the numerical determination of the maximum, the sum (6.15) has to be re-computed after each modification of the parameters. Since the sum runs only over the bins but not over all individual observations as in the normal likelihood method, the computation for histograms is relatively fast.

Example 84. Adjustment of a linear distribution to a histogram

The cosine $u = \cos \alpha$ of an angle α be linearly distributed according to

$$f(u|\lambda) = \frac{1}{2}(1 + \lambda u) , \quad -1 \leq u \leq 1 , \quad |\lambda| < 1 .$$

We want to determine the parameter λ which best describes the observed distribution of 500 entries d_i into 20 bins (Fig. 6.8). In the Poisson approximation we expect t_i entries for the bin i corresponding to the average value $u_i = -1 + (i - 0.5)/10$ of the cosine,

$$t_i = \frac{500}{20}(1 + \lambda u_i) .$$

We obtain the likelihood function by inserting this expression into (6.15). The likelihood function and the MLE are indicated in the Figure 6.8.

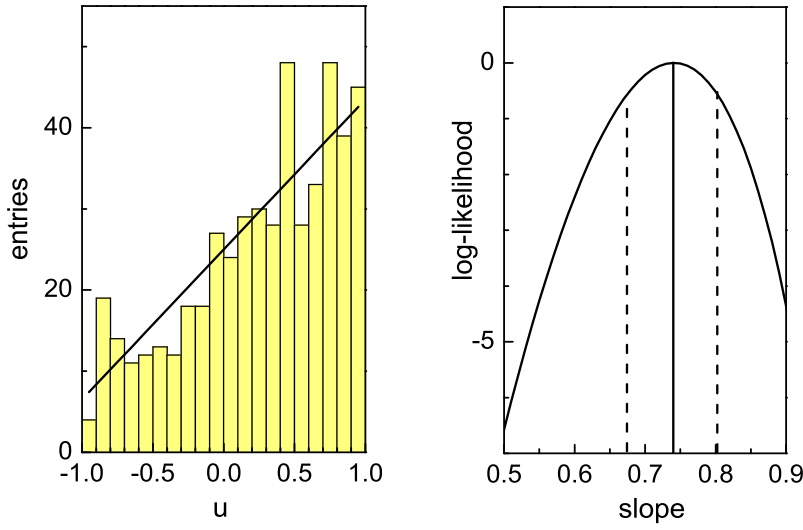


Fig. 6.8. Linear distribution with adjusted straight line (left) and likelihood function (right).

Histograms with Background

If the measurement is contaminated by background which follows the Poisson statistics with mean b , we have to modify the expression (6.15) for the log-likelihood to

$$\ln L = \sum_{i=1}^B [-(t_i + b_i) + d_i \ln(t_i + b_i)] .$$

Of course, a simple subtraction of the average background would have underestimated the uncertainties. A further modification of the above expression is necessary if the expectation value b_i of the background itself is subject to uncertainties.

χ^2 Approximation

We have seen in Sect. 3.6.3 that the Poisson distribution asymptotically with increasing mean value t approaches a normal distribution with variance t . Thus for high statistics histograms the number of events d in a bin with prediction $t(\theta)$ is described by

$$f(d) = \frac{1}{\sqrt{2\pi t}} \exp \left[-\frac{(d - t)^2}{2t} \right] .$$

The corresponding log-likelihood is

$$\ln L = -\frac{(d - t)^2}{2t} - \frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln t .$$

For large t , the logarithmic term is an extremely slowly varying function of t . Neglecting it and the constant term we find for the whole histogram

$$\begin{aligned}\ln L &= -\frac{1}{2} \sum_{i=1}^B \frac{(d_i - t_i)^2}{t_i} \\ &= -\frac{1}{2} \chi^2.\end{aligned}$$

The sum corresponds to the expression (3.6.7) and has been denoted there as χ^2 :

$$\chi^2 = \sum_{i=1}^B \frac{(d_i - t_i)^2}{t_i}.$$

If the approximation of the Poisson distribution by a normal distribution is justified, the likelihood estimation of the parameters may be replaced by a χ^2 fit and the standard errors are given by an increase of χ^2 by one unit.

The width of the χ^2 distribution for B degrees of freedom is $\sigma = \sqrt{2B}$ which, for example, is equal to 10 for 50 histogram bins. With such large fluctuations of the value of χ^2 from one sample to the other, it appears paradoxical at first sight that a parameter error of one standard deviation corresponds to such a small change of χ^2 as one unit, while a variation of χ^2 by 10 is compatible with the prediction. The obvious reason for the good resolution is that the large fluctuations from sample to sample are unrelated to the value of the parameter. In case we would compare the prediction after each parameter change to a new measurement sample, we would not be able to obtain a precise result for the estimate of the parameter λ .

Usually, histograms contain some bins with few entries. Then a binned likelihood fit is to be preferred to a χ^2 fit, since the above condition of large t_i (e.g. $t_i > 5$) is violated. We recommend to perform always a likelihood adjustment.

6.5.7 Extended Likelihood

When we record N independent multi-dimensional observations, $\{\mathbf{x}_i\}$, $i = 1, \dots, N$, of a distribution depending on a set of parameters $\boldsymbol{\theta}$, then it may happen that these parameters also determine the rate, i.e. the expected rate $\lambda(\boldsymbol{\theta})$ is a function of $\boldsymbol{\theta}$. In this situation N is no longer an ancillary statistic but a random variable like the $\mathbf{x}_i$. This means that we have to multiply two probabilities, the probability to find N observations which follow the Poisson statistics and the probability to observe a certain distribution of the variates $\mathbf{x}_i$. Given a p.d.f. $f(\mathbf{x}|\boldsymbol{\theta})$ for a single observation, we obtain the likelihood function

$$L(\boldsymbol{\theta}) = \frac{e^{-\lambda(\boldsymbol{\theta})} \lambda(\boldsymbol{\theta})^N}{N!} \prod_{i=1}^N f(\mathbf{x}_i|\boldsymbol{\theta})$$

and its logarithm

$$\ln L(\boldsymbol{\theta}) = -\lambda(\boldsymbol{\theta}) + N \ln(\lambda(\boldsymbol{\theta})) + \sum_{i=1}^N \ln f(\mathbf{x}_i|\boldsymbol{\theta}) - \ln N!.$$

Again we can omit the last term in the likelihood analysis, because it does not depend on $\boldsymbol{\theta}$.

As a special case, let us assume that the cross section for a certain reaction is equal to $g(\mathbf{x}|\boldsymbol{\theta})$. Then we get the p.d.f. by normalization of g :

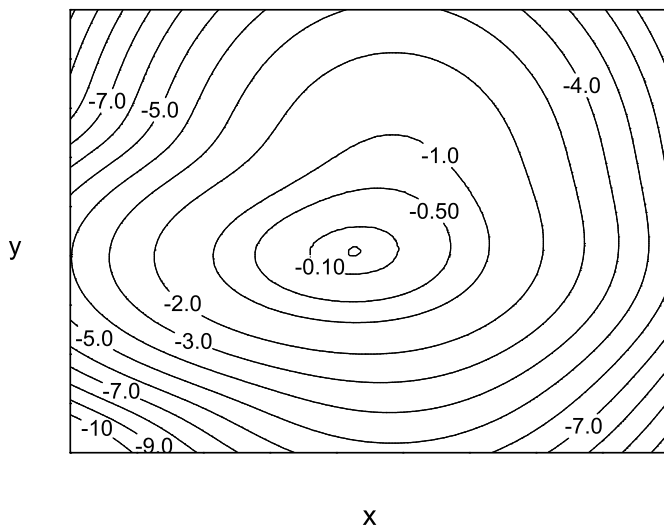


Fig. 6.9. Likelihood contours.

$$f(\mathbf{x}|\boldsymbol{\theta}) = \frac{g(\mathbf{x}|\boldsymbol{\theta})}{\int g(\mathbf{x}|\boldsymbol{\theta})d\mathbf{x}} .$$

The production rate λ is equal to the normalization factor multiplied with the luminosity S which is a constant:

$$\lambda(\boldsymbol{\theta}) = S \int g(\mathbf{x}|\boldsymbol{\theta})d\mathbf{x} .$$

The extended likelihood method is discussed in some detail in Ref. [29].

6.5.8 Complicated Likelihood Functions

If the likelihood function deviates considerably from a normal distribution in the vicinity of its maximum, e.g. contains several significant maxima, then it is not appropriate to parametrize it by the maximum and error limits. In this situation the full function or a likelihood map should be presented. Such a map is shown in Fig. 6.9. The presentation reflects very well which combinations of the parameters are supported by the data. Under certain conditions, with more than two parameters, several projections have to be considered.

6.5.9 Comparison of Observations with a Monte Carlo Simulation

Motivation

Modern research in natural sciences requires more and more complex and expensive experimental setups which nevertheless cannot be perfect. Non-linearity, limited acceptance, finite resolution, and dead time affect the measurements. These effects can

be described by an acceptance function $\alpha(x, x')$ which specifies with which probability a physical quantity x is detected as x' . The acceptance function thus describes at the same time acceptance losses and resolution effects. A frequency distribution $h(x)$ is convoluted with the acceptance function and transformed to a distribution $h'(x')$,

$$h'(x') = \int_{x_{\min}}^{x_{\max}} \alpha(x, x') h(x) dx .$$

In many cases the detection effects can only be corrected by means of very elaborate Monte Carlo simulations. This is especially true for experiments in nuclear and particle physics.

There are two fundamentally different ways to extract interesting parameters from the data: i) The function α is generated by a Monte Carlo simulation and is subsequently used to deconvolute the observed distribution. We will devote Chap. 9 to deconvolution. ii) The observed distribution is compared to the simulation and the parameters used in the simulation are adjusted. The second method is preferable because the deconvolution always requires assumptions about the shape of the distribution which are hard to justify, and introduce additional uncertainties.

The χ^2 Approximation

The theoretical models are represented by Monte Carlo samples and the parameter inference is carried out by a comparison of experimental and simulated histograms. The number M of the simulated observations has to be normalized to the number N of experimental events. For $c_m = N/M$ and m_i Monte Carlo events in bin i we get for the likelihood instead of (6.15).

$$\ln L = \sum_{i=1}^B (-c_m m_i + d_i \ln(c_m m_i)) \quad (6.16)$$

assuming that the statistical error of the simulation can be neglected, i.e. $M \gg N$ applies. If the number of the entries in all bins is large enough to approximate the Poisson distribution by the normal distribution, we can as well minimize the corresponding χ^2 expression

$$\chi^2 = \sum_{i=1}^B \frac{(d_i - c_m m_i)^2}{c_m m_i}. \quad (6.17)$$

Remark that M and thus also c_m depends on the parameters which we estimate. Alternatively, we can leave c_m as a free parameter in the fit. This is more convenient, but reduces slightly the precision of the estimates of the parameters of interest.

If also the statistical fluctuations of the simulation have to be taken into account and/or the observations are weighted, the expressions for (6.16), (6.17) become more complex. They can be found in Ref. [48], and are for convenience included in Appendix 13.6.

The simulation programs usually consist of two different parts. The first part describes the physical process which depends on the parameters of interest. The second models the detection process. Both parts often require large program packages, the so called event generators and the detector simulators. The latter usually consume considerable computing power. Limitations in the available computing time then may result in non-negligible statistical fluctuations of the simulation.

Weighting the Monte Carlo Observations

When we fit parameters, every parameter change obviously entails a modification of the Monte Carlo prediction. Now we do not want to repeat the full simulation with every fitting step. Apart from the fact that we want to avoid the computational effort there is another more important reason: With the χ^2 fit we find the standard error interval by letting vary χ^2 by one unit. On the other hand when we compare experimental data with an optimal simulation, we expect a contribution to χ^2 from the simulation of the order of $\sqrt{2BN/M}$ for B histogram bins. Even with a simulation sample which is a hundred times larger than the data sample this value is of the order of one. This means that a repetition of the simulation causes considerable fluctuations of the χ^2 value which have nothing to do with parameter changes. These fluctuations can only be reduced if the same Monte Carlo sample is used for all parameter values. We have to adjust the simulation to the modified parameters by weighting its observations.

Also re-weighting produces additional fluctuations. These, however, should be tolerable if the weights do not vary too much. If we are not sure that this assumption is justified, we can verify it: We reduce the number of Monte Carlo observations and check whether the result remains stable. We know that the contribution of the simulation to the parameter errors scales with the inverse square root of the number of simulated events. Alternatively, we can also estimate the Monte Carlo contribution to the error by repeating the simulation and the full estimation process.

The weights are computed in the following way: For each Monte Carlo observation $\mathbf{x}'$ we know the true values $\mathbf{x}$ of the variates and the corresponding p.d.f. $f(\mathbf{x}|\theta_0)$ for the parameter θ_0 , which had been used at the generation. When we modify the parameter we weight each observation by $w(\theta) = f(\mathbf{x}|\theta)/f(\mathbf{x}|\theta_0)$. The weighted distribution of $\mathbf{x}'$ then describes the modified prediction.

The weighting of single observations requires a repetition of the histogramming after each parameter change. This may require too much computing time. A more efficient method is based on a Taylor expansion of the p.d.f. in powers of the parameters. We illustrate it with two examples.

Example 85. Fit of the slope of a linear distribution with Monte Carlo correction

The p.d.f. be

$$f(x|\theta) = \frac{1 + \theta x}{1 + \theta/2}, \quad 0 \leq x \leq 1.$$

We generate observations x uniformly distributed in the interval $0 \leq x \leq 1$, simulate the experimental resolution and the acceptance, and histogram the distorted variable x' into bins i and obtain contents m_{0i} . The same observations are weighted by x and summed up to the histogram m_{1i} . These two distributions are shown in Fig. 6.10 a, b. The dotted histograms correspond to the distributions before the distortion by the measurement process. In Fig. 6.10 c is also depicted the experimental distribution. It should be possible to describe it by a superposition m_i of the two Monte Carlo distributions:

$$d_i \sim m_i = m_{0i} + \theta m_{1i}. \quad (6.18)$$

We optimize the parameter θ such that the histogram d_i is described up to a normalization constant as well as possible by a superposition of the two

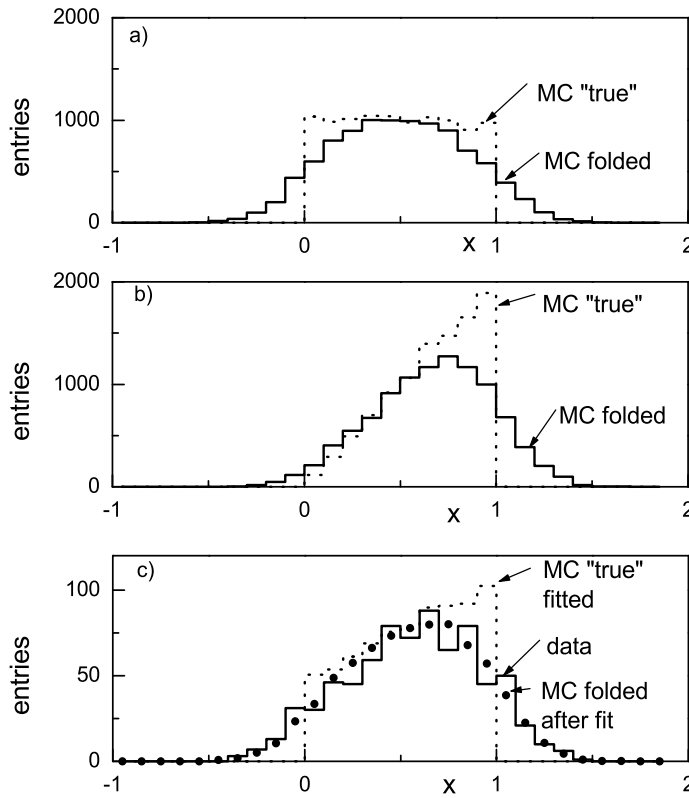


Fig. 6.10. The superposition of two Monte Carlo distributions, a) flat and b) triangular is adjusted to the experimental data.

Monte Carlo histograms. We have to insert m_i from (6.18) into (6.16) and set $c_n = N / \sum_i m_i$.

Example 86. Fit of a lifetime with Monte Carlo correction

We expand the p.d.f.

$$f(t|\gamma) = \gamma e^{-\gamma t}$$

into a Taylor expansion at γ_0 which is a first guess of the decay rate γ :

$$f(t|\gamma) = \gamma_0 e^{-\gamma_0 t} \left\{ 1 + \frac{\Delta\gamma}{\gamma_0} (1 - \gamma_0 t) + \left(\frac{\Delta\gamma}{\gamma_0} \right)^2 \left(-\gamma_0 t + \frac{\gamma_0^2 t^2}{2} \right) + \dots \right\}.$$

The Monte Carlo simulation follows the distribution $f_0 = \gamma_0 e^{-\gamma_0 t}$. Weighting the events by $(1/\gamma_0 - t)$ and $(-t/\gamma_0 + t^2/2)$, we obtain the distributions $f_1 = (1 - \gamma_0 t) e^{-\gamma_0 t}$, $f_2 = (-t + \gamma_0 t^2/2) e^{-\gamma_0 t}$ and

$$f(t|\gamma) = f_0(t) + \Delta\gamma f_1(t) + (\Delta\gamma)^2 f_2(t) + \dots.$$

If it is justified to neglect the higher powers of $\Delta\gamma/\gamma_0$, we can again describe our experimental distribution this time by a superposition of three distributions $f'_0(t)$, $f'_1(t)$, $f'_2(t)$ which are the distorted versions of f_0, f_1, f_2 . The

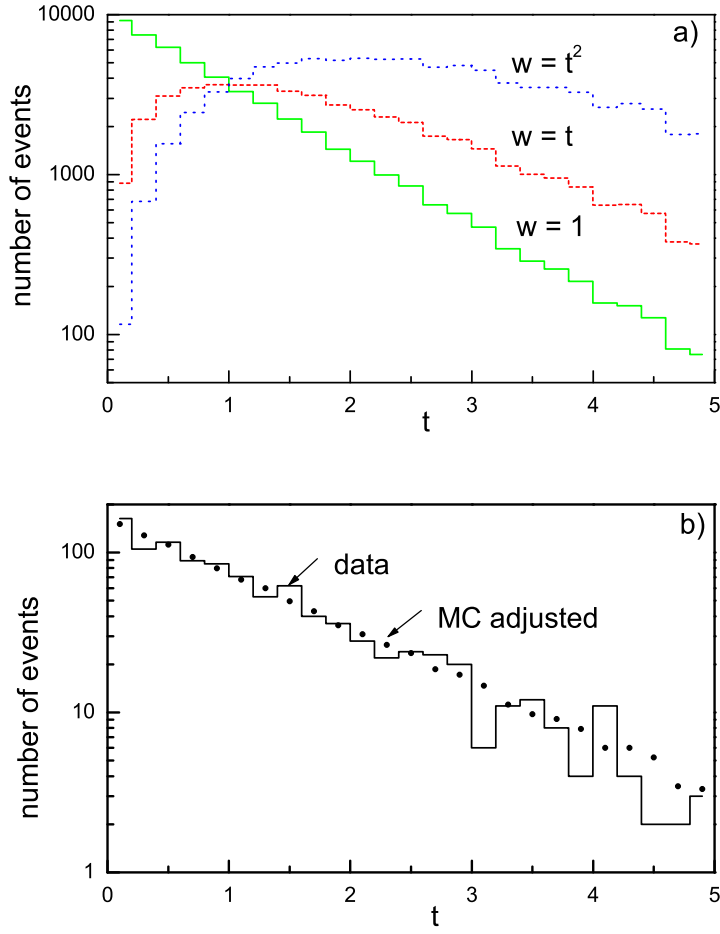


Fig. 6.11. Lifetime fit. The dotted histogram in b) is the superposition of the three histograms of a) with weights depending on $\Delta\gamma$.

parameter $\Delta\gamma$ is determined by a χ^2 or likelihood fit. In our special case it is even simpler to weight f_0 by t , and t^2 , respectively, and to superpose the corresponding distributions f_0 , $g_1 = tf_0$, $g_2 = t^2f_0$ with the factors given in the following expression:

$$f(t|\gamma) = f_0(t) \left(1 + \frac{\Delta\gamma}{\gamma_0} \right) + \gamma_0 g_1(t) \left(\frac{\Delta\gamma}{\gamma_0} + \left(\frac{\Delta\gamma}{\gamma_0} \right)^2 \right) + \frac{1}{2} g_2(t) \gamma_0^2 \left(\frac{\Delta\gamma}{\gamma_0} \right)^2.$$

The parameter $\Delta\gamma$ is then modified until the correspondingly weighted sum of the distorted histograms agrees optimally with the data. Figure 6.11 shows an example. In case the quadratic term can be neglected, two histograms are sufficient.

The general case is treated in an analogous manner. The Taylor expansion is:

$$f(\theta) = f(\theta_0) + \Delta\theta \frac{df}{d\theta}(\theta_0) + \frac{(\Delta\theta)^2}{2!} \frac{d^2f}{d\theta^2}(\theta_0) + \dots$$

$$= f(\theta_0) \left\{ 1 + \Delta\theta \frac{1}{f_0} \frac{df}{d\theta}(\theta_0) + \frac{(\Delta\theta)^2}{2!} \frac{1}{f_0} \frac{d^2f}{d\theta^2}(\theta_0) + \dots \right\} .$$

The observations x' of the distribution $f_0(x|\theta_0)$ provide the histogram m_0 . Weighting with w_1 and w_2 , where

$$w_1 = \frac{1}{f_0} \frac{df}{d\theta}(x|\theta_0) ,$$

$$w_2 = \frac{1}{2f_0} \frac{d^2f}{d\theta^2}(x|\theta_0) ,$$

we obtain two further histograms m_{1i}, m_{2i} . The parameter inference of $\Delta\theta$ is performed by comparing $m_i = (m_{0i} + \Delta\theta m_{1i} + \Delta\theta^2 m_{2i})$ with the experimental histogram d_i .

In many cases the quadratic term can be omitted. In other situations it might be necessary to iterate the procedure.

6.5.10 Parameter Estimate of a Signal Contaminated by Background

In this section we will discuss point inference in the presence of background in a specific situation, where we have the chance to record independently from a signal sample also a reference sample containing pure background. The measuring times or fluxes, i.e. the relative normalization, of the two experiments are supposed to be known. The parameters searched for could be the position and width of a Breit-Wigner bump but also the slope of an angular distribution or the lifetime of a particle. The interest in this method rests upon the fact that we do not need to parameterize the background distribution and thus are independent of assumptions about its shape. This feature has to be paid for by a certain loss of precision.

The idea behind the method is simple: The log-likelihood of the wanted signal parameter as derived for the full signal sample is a superposition of the log-likelihood of the genuine signal events and the log-likelihood of the background events. The latter can be estimated from the reference sample and subtracted from the full log-likelihood.

To illustrate the procedure, imagine we want to measure the signal response of a radiation detector by recording a sample of signal heights $x_1, \dots, x_N$ from a mono-energetic source. For a pure signal, the x_i would follow a normal distribution with resolution σ :

$$f(x|\mu) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} .$$

The unknown parameter μ is to be estimated. After removing the source, we can – under identical conditions – take a reference sample $x'_1, \dots, x'_M$ of background events. They follow a distribution which is of no interest to us.

If we knew, which observations x_i in our signal sample were signal ($x_i^{(S)}$), respectively background ($x_i^{(B)}$) events, we could take only the S signal events and calculate the correct log-likelihood function

$$\begin{aligned} \ln L &= \sum_{i=1}^S \ln f(x_i^{(S)}|\mu) = - \sum_{i=1}^S \frac{(x_i^{(S)} - \mu)^2}{2\sigma^2} + \text{const.} \\ &= - \sum_{i=1}^N \frac{(x_i - \mu)^2}{2\sigma^2} + \sum_{i=1}^B \frac{(x_i^{(B)} - \mu)^2}{2\sigma^2} + \text{const.} , \end{aligned}$$

with $S + B = N$. The second unknown term can be estimated from the control sample

$$\sum_{i=1}^B \frac{(x_i^{(B)} - \mu)^2}{2\sigma^2} \approx \sum_{i=1}^M \frac{(x'_i - \mu)^2}{2\sigma^2}.$$

The logarithm of our corrected log-likelihood becomes:

$$\ln \tilde{L} = -\sum_{i=1}^N \frac{(x_i - \mu)^2}{2\sigma^2} + \sum_{i=1}^M \frac{(x'_i - \mu)^2}{2\sigma^2}.$$

We call it pseudo log-likelihood, $\ln \tilde{L}$, to distinguish it from a genuine log-likelihood. To obtain the estimate $\hat{\mu}$ of our parameter, we look for the parameter $\hat{\mu}$ which maximizes $\ln \tilde{L}$ and find the expected simple function of the mean values $\bar{x}$, $\bar{x}'$:

$$\begin{aligned} \hat{\mu} &= \frac{\sum_{i=1}^N x_i - \sum_{i=1}^M x'}{N - M} \\ &= \frac{N\bar{x} - M\bar{x}'}{N - M}. \end{aligned} \quad (6.19)$$

The general problem where the sample and parameter spaces could be multi-dimensional and with different fluxes of the signal and the reference sample, is solved in complete analogy to our example: Given a contaminated signal distribution of size N and a reference distribution of size M and flux $1/r$ times larger than that of the signal sample, we put

$$\ln \tilde{L} = \sum_{i=1}^N \ln f(\mathbf{x}_i | \boldsymbol{\theta}) - r \sum_{i=1}^M \ln f(\mathbf{x}'_i | \boldsymbol{\theta}). \quad (6.20)$$

For small event numbers, for example if the flux-corrected number of background events in the reference sample exceeds the total number of events in the main sample, $rM > N$, it may happen that $\ln \tilde{L}$ becomes un-bounded from above (for instance asymptotically a parabola opening to the upper side), rendering a maximum undefined.

The formula (6.20) is completely general and does not depend on the shape of the background distribution⁸. It avoids histogramming which is problematic for low event counts. Especially, those methods which subtract a background histogram from the signal histogram often fail in such a situation. A different method, where the shape of the background distribution is approximated by probability density estimation (PDE) will be given in Sect. 12.1.1.

To get a feeling of the uncertainty Δ of the corrected estimate, we return to our example and look at the difference of the estimate $\hat{\mu}$ from the uncontaminated estimate $\bar{x}^{(S)}$ which we can set without loss of generality equal to zero. From simple error propagation we find:

$$\Delta = \frac{\overline{Bx^{(B)}} - M\bar{x}'}{S + B - M}.$$

⁸The method which we present in this section has been taken from the Russian translation of the book by Eadie et al. [6] and has been introduced probably by the Russian editors [32].

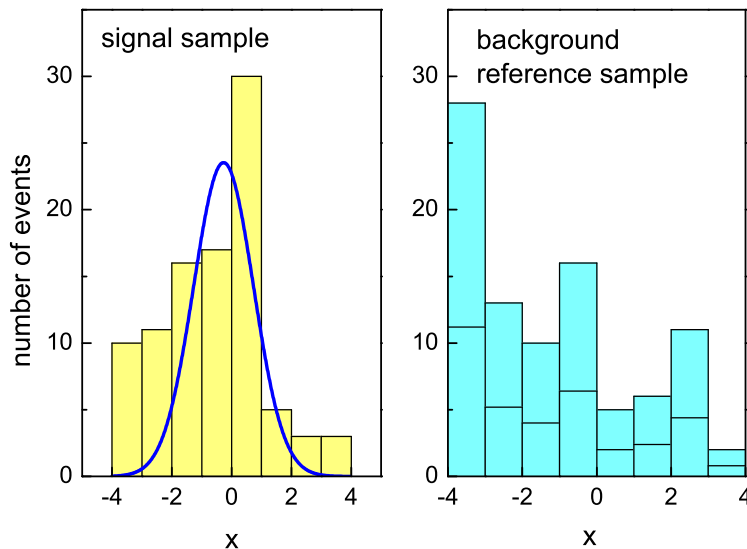


Fig. 6.12. Experimental distribution of a normally distributed signal over background (left) and background reference sample (right). The lower histogram is scaled to the signal flux.

Without going into the details, we realize that the error increases i) with the amount of background, ii) with the difference of the expected value of the reference distribution from that of the genuine signal distribution, and iii) with the variance of the background distribution. Correspondingly, we have to require $\sqrt{2M} \ll S$, more specifically $\sqrt{2M}|\mu_B - \mu_S| \ll S\sigma$ and $\sqrt{2M}\sigma_B \ll S\sigma$. Here $\sqrt{2M}$ is an estimate of the Poisson error of $B - M$, $S \approx N - M$, $\mu_B \approx \bar{x}'$, $\mu_S \approx \hat{\mu}$, $\sigma_B \approx (\bar{x'^2} - \bar{x'}^2)^{1/2}$.

Also this consideration can be generalized: Stated qualitatively, the contribution to the parameter uncertainty is small, if background and signal lead to similar parameter estimates, if the background estimate has a small uncertainty, and, trivially, if the amount of background is small. This applies also, for instance, when we estimate the asymmetry parameter of an angular distribution contaminated by background.

The shape itself of the pseudo likelihood cannot be used directly to estimate the parameter errors. The calculation of the error in the asymptotic approximation is given in Appendix 13.4.

An alternative procedure for the error estimation is the bootstrap method, where we take a large number of bootstrap samples from the experimental distributions of both the signal- and the control experiment and calculate the background corrected parameter estimate for each pair of samples, see Chap. 12.2.

Example 87. Signal over background with background reference sample

Fig. 6.12 shows an experimental histogram of a normally distributed signal of width $\sigma = 1$ contaminated by background, together 95 events with mean $\bar{x} = 0.61$ and empirical variance $v^2 = 3.00$. The right hand side is the distribution of a background reference sample with $1/r = 2.5$ times the flux of the signal sample, containing 91 events with mean $\bar{x}' = -1.17$ and variance $v'^2 = 4.79$.

The mean of the signal is obtained from the flux corrected version of (6.19):

$$\begin{aligned}\hat{\mu} &= \frac{N\bar{x} - rM\bar{x}'}{N - rM} \\ &= \frac{95 \cdot 0.61 - 0.4 \cdot 91 \cdot 1.17}{95 - 0.4 \cdot 91} = -0.26 \pm 0.33 .\end{aligned}$$

The error is estimated by linear error propagation. The result is indicated in Fig. 6.12. The distributions were generated with nominally 60 pure signal plus 40 background events and 100 background reference events. The signal corresponds to a normal distribution, $N(x|0,1)$, and the background to an exponential, $\sim \exp(-0.2x)$.

6.6 Inclusion of Constraints

6.6.1 Introduction

The interesting parameters are not always independent of each other but are often connected by constraint equations like physical or geometrical laws.

As an example let us look at the decay of a Λ particle into a proton and a pion, $\Lambda \rightarrow p + \pi$, where the direction of flight of the Λ hyperon and the momentum vectors of the decay products are measured. The momentum vectors of the three particles which participate in the reaction are related through the conservation laws of energy and momentum. Taking into account the conservation laws, we add information and can improve the precision of the momentum determination.

In the following we assume that we have N direct observations x_i which are predicted by functions $t_i(\boldsymbol{\theta})$ of a parameter vector $\boldsymbol{\theta}$ with P components as well as K constraints of the form $h_k(\theta_1, \dots, \theta_P) = 0$. Let us assume further that the uncertainties Δ_i of the observations are normally distributed and that the constraints are fulfilled with the precision δ_k ,

$$\begin{aligned}\langle (t_i(\boldsymbol{\theta}) - x_i)^2 \rangle &= \Delta_i^2 , \\ \langle h_k^2(\theta_1, \dots, \theta_P) \rangle &= \delta_k^2 ,\end{aligned}\tag{6.21}$$

then χ^2 can be written in the form:

$$\chi^2 = \sum_{i=1}^N \frac{[x_i - t_i(\boldsymbol{\theta})]^2}{\Delta_i^2} + \sum_{k=1}^K \frac{h_k^2(\boldsymbol{\theta})}{\delta_k^2} .\tag{6.22}$$

We minimize χ^2 by varying the parameters and obtain their best estimates at the minimum of χ^2 . This procedure works also when the constraints contain more than N parameters, as long as their number P does not exceed the sum $N + K$.

A corresponding likelihood fit would maximize

$$\ln L = \sum_{i=1}^N \ln f(x_i|\boldsymbol{\theta}) - \frac{1}{2} \sum_{k=1}^K \frac{h_k^2(\boldsymbol{\theta})}{\delta_k^2} .$$

In most cases the constraints are obeyed exactly, $\delta_k = 0$, and the second term in (6.22) diverges. This difficulty is avoided in the following three procedures:

1. The constraints are used to reduce the number of parameters.
2. The constraints are approximated by narrow Gaussians.
3. Lagrange multipliers are adjusted to satisfy the constraints.

We will discuss the problem in terms of a χ^2 minimization. The transition to a likelihood fit is trivial.

6.6.2 Eliminating Redundant Parameters

Sometimes it is possible to eliminate parameters by expressing them by an unconstrained subset.

Example 88. Fit with constraint: two pieces of a rope

A rope of exactly 1 m length is cut into two pieces. A measurement of both pieces yields $l_1 = 35.3$ cm and $l_2 = 64.3$ cm, both with the same Gaussian uncertainty of $\delta = 0.3$. We have to find the estimates $\hat{\lambda}_1, \hat{\lambda}_2$ of the lengths. We minimize

$$\chi^2 = \frac{(l_1 - \lambda_1)^2}{\delta^2} + \frac{(l_2 - \lambda_2)^2}{\delta^2}$$

including the constraint $\lambda_1 + \lambda_2 = L = 100$ cm. We simply replace λ_2 by $L - \lambda_1$ and adjust λ_1 , minimizing

$$\chi^2 = \frac{(l_1 - \lambda_1)^2}{\delta^2} + \frac{(L - l_2 - \lambda_1)^2}{\delta^2}.$$

The minimization relative to λ_1 leads to the result:

$$\hat{\lambda}_1 = \frac{L}{2} + \frac{l_1 - l_2}{2} = 35.5 \pm 0.2 \text{ cm}$$

and the corresponding estimate of λ_2 is just the complement of $\hat{\lambda}_1$ with respect to the full length. Note that due to the constraint the error of λ_i is reduced by a factor $\sqrt{2}$, as can easily be seen from error propagation. The constraint has the same effect as a double measurement, but with the modification that now the results are (maximally) anti-correlated: one finds $\text{cov}(\lambda_1, \lambda_2) = -\text{var}(\lambda_i)$.

Example 89. Fit of the particle composition of an event sample

A particle identification variable x has different distributions $f_i(x)$ for different particles. The p.d.f. given the relative particle abundance λ_m for particle species m out of M different particles is

$$f(x|\lambda_1, \dots, \lambda_M) = \sum_{m=1}^M \lambda_m f_m(x),$$

$$\sum \lambda_m = 1. \quad (6.23)$$

As the constraint relation is linear, we can easily eliminate the parameter λ_M to get rid of the constraint $\sum \lambda_m = 1$:

$$f_{M-1}(x|\lambda_1, \dots, \lambda_{M-1}) = \sum_{m=1}^{M-1} \lambda_m f_m(x) + (1 - \sum_{m=1}^{M-1} \lambda_m) f_M(x) .$$

The log-likelihood for N particles is

$$\ln L = \sum_{i=1}^N \ln \left[\sum_{m=1}^{M-1} \lambda_m f_m(x_i) + (1 - \sum_{m=1}^{M-1} \lambda_m) f_M(x_i) \right] .$$

From the MLE we obtain in the usual way the first $M-1$ parameters and their error matrix $\mathbf{E}$. The remaining parameter λ_M and the related error matrix elements e_{Mj} are derived from the constraint (6.23) and the corresponding relation $\sum \Delta \lambda_m = 0$. The diagonal error is the expected value of $(\Delta \lambda_M)^2$:

$$\begin{aligned} \Delta \lambda_M &= - \sum_{m=1}^{M-1} \Delta \lambda_m , \\ (\Delta \lambda_M)^2 &= \left[\sum_{m=1}^{M-1} \Delta \lambda_m \right]^2 , \\ E_{MM} &= \sum_{m=1}^{M-1} E_{mm} + \sum_m \sum_{l \neq m}^{M-1} E_{ml} . \end{aligned}$$

The remaining elements are computed analogously:

$$E_{Mj} = E_{jM} = -E_{jj} - \sum_{m \neq j}^{M-1} E_{mj} .$$

An iterative method, called *channel likelihood*, to find the particle contributions, is given in [34].

These trivial examples are not really representative for the typical problems we have to solve in particle- or astrophysics. Indeed, it is often complicated or even impossible to reduce the parameter set analytically to an unconstrained subset. But we can introduce a new unconstrained parameter set which then predicts the measured quantities. To find such a set is straight forward in the majority of problems: We just have to think how we would simulate the corresponding experimental process. A simulation is always based on a minimum set of parameters. Constraints are obeyed automatically.

Example 90. Kinematical fit with constraints: eliminating parameters

A neutral particle c is decaying into two charged particles a and b , for instance $\Lambda \rightarrow p + \pi^-$. The masses m_c, m_a, m_b are known. Measured are the momenta $\boldsymbol{\pi}_a, \boldsymbol{\pi}_b$ of the decay products and the decay vertex $\boldsymbol{\rho}$. The measurements of the components of the momentum vectors are correlated. The inverse error matrices be V_a and V_b . The origin of the decaying particle be at the origin of the coordinate system. Thus we have 9 measurements $(\mathbf{r}, \mathbf{p}_a, \mathbf{p}_b)$, 10 parameters, namely the 3 momentum vectors and the distance $(\boldsymbol{\pi}_c, \boldsymbol{\pi}_a, \boldsymbol{\pi}_b, \rho)$, and 4 constraints from momentum and energy conservation:

$$\begin{aligned}\pi(\pi_c, \pi_a, \pi_b) &\equiv \pi_c - \pi_a - \pi_b = 0, \\ \varepsilon(\pi_c, \pi_a, \pi_b) &\equiv \sqrt{\pi_c^2 + m_c^2} - \sqrt{\pi_a^2 + m_a^2} - \sqrt{\pi_b^2 + m_b^2} = 0.\end{aligned}$$

The χ^2 expression is

$$\begin{aligned}\chi^2 &= \sum_{i=1}^3 \left(\frac{r_i - \rho_i}{\delta r_i} \right)^2 + \sum_{i,j=1}^3 (p_{ai} - \pi_{ai}) V_{aij} (p_{aj} - \pi_{aj}) + \\ &\quad + \sum_{i,j=1}^3 (p_{bi} - \pi_{bi}) V_{bij} (p_{bj} - \pi_{bj}).\end{aligned}$$

The vertex parameters ρ_i are fixed by the vector relation $\boldsymbol{\rho} = \rho \boldsymbol{\pi}_c / |\boldsymbol{\pi}_c|$. Now we would like to remove 4 out of the 10 parameters using the 4 constraints.

A Monte Carlo simulation of the Λ decay would proceed as follows: First we would select the Λ momentum vector (3 parameters). Next the decay length would be generated (1 parameter). The decay of the Λ hyperon into proton and pion is fully determined when we choose the proton direction in the lambda center of mass system (2 parameters). All measured laboratory quantities and thus also χ^2 can then be expressed analytically by these 6 unconstrained quantities (we omit here the corresponding relations) which are varied in the fitting procedure until χ^2 is minimal. Of course in the fit we would not select random starting values for the parameters but the values which we compute from the experimental decay length and the measured momentum vectors. Once the reduced parameter set has been adjusted, it is easy to compute also the remaining laboratory momenta and their errors which, obviously, are strongly correlated.

Often the reduced parameter set is more relevant than the set corresponding to the measurement, because a simulation usually is based on parameters which are of scientific interest. For example, the investigation of the Λ decay might have the goal to determine the Λ decay parameter which depends on the center of mass direction of the proton relative to the Λ polarization, i.e. on one of the directly fitted quantities.

6.6.3 Gaussian Approximation of Constraints

As suggested by formula (6.22) to fulfil the constraints within the precision of our measurement, we just have to choose the uncertainty δ_k sufficiently small. We will give a precise meaning to the expression *sufficiently small* below.

Example 91. Example 88 continued

Minimizing

$$\chi^2 = \frac{(l_1 - \lambda_1)^2}{\delta^2} + \frac{(l_2 - \lambda_2)^2}{\delta^2} + \frac{(\lambda_1 + \lambda_2 - L)^2}{(10^{-5}\delta)^2}.$$

produces the same result as the fit presented above. The value $\delta_k^2 = 10^{-10}\delta^2$ is chosen small compared to δ .

Example 92. Example 90 continued

The modified χ^2 expression is

$$\begin{aligned} \chi^2 = & \sum_{i=1}^3 \left(\frac{r_i - \rho_i}{\delta r_i} \right)^2 + \sum_{i,j=1}^3 (p_{ai} - \pi_{ai}) V_{aij} (p_{aj} - \pi_{aj}) + \\ & + \sum_{i,j=1}^3 (p_{bi} - \pi_{bi}) V_{bij} (p_{bj} - \pi_{bj}) + \sum_{i=1}^3 \left(\frac{\pi_i}{\delta_\pi} \right)^2 + \left(\frac{\varepsilon}{\delta_\varepsilon} \right)^2 \end{aligned}$$

with tiny values for δ_π and δ_ε .

This direct inclusion of the constraint through a penalty term in the fit is technically very simple and efficient.

As demonstrated in example 88 the term *sufficiently small* means that the uncertainty δh_k of a constraint k as derived from the fitted values of the parameters and their errors is large compared to δ_k :

$$(\delta h)_k^2 = \sum_{i,j} \frac{\partial h_k}{\partial \theta_i} \frac{\partial h_k}{\partial \theta_j} \delta \theta_i \delta \theta_j \gg \delta_k^2. \quad (6.24)$$

The quantities $\delta \theta_i$ in (6.24) are not known precisely before the fit is performed but can be estimated sufficiently well beforehand. The precise choice of the constraint precision δ_k is not at all critical, variations by many orders of magnitude make no difference but a too small values of δ_k could lead to numerical problems.

6.6.4 The Method of Lagrange Multipliers

This time we choose the likelihood presentation of the problem. The likelihood function is extended to

$$\ln L = \sum_{i=1}^N \ln f(x_i | \boldsymbol{\theta}) + \sum_k \alpha_k h_k(\boldsymbol{\theta}). \quad (6.25)$$

We have appended an expressions that in the end should be equal to zero, the constraint functions multiplied by the so-called *Lagrange multipliers*. The MLE as obtained by setting $\partial \ln L / \partial \theta_j = 0$ yields parameters that depend on the Lagrange multipliers α . We can now use the free parameters α_k to fulfil the constraints, or in other words, we use the constraints to eliminate the Lagrange multiplier dependence of the MLE.

Example 93. Example 88 continued

Our full likelihood function is now

$$\ln L = -\frac{(l_1 - \lambda_1)^2}{2\delta^2} - \frac{(l_2 - \lambda_2)^2}{2\delta^2} + \alpha(\lambda_1 + \lambda_2 - L)$$

with the MLE $\hat{\lambda}_{1,2} = l_{1,2} - \delta^2 \alpha$. Using $\hat{\lambda}_1 + \hat{\lambda}_2 = L$ we find $\delta^2 \alpha = (l_1 + l_2 - L)/2$ and, as before, $\hat{\lambda}_1 = (L + l_1 - l_2)/2$, $\hat{\lambda}_2 = (L + l_2 - l_1)/2$.

Of course, the general situation is much more complicated than that of our trivial example. An analytic solution will hardly be possible. Instead we can set the derivative of the log-likelihood not only with respect to the parameters θ but also with respect to the multipliers α_k equal to zero, $\partial \ln L / \partial \alpha_k = 0$, which automatically implies, see (6.25), that the constraints are satisfied. Unfortunately, the zero of the derivative corresponds to a saddle point and cannot be found by a maximum searching routine. More subtle numerical methods have to be applied.

Most methods avoid this complication and limit themselves to linear regression models which require a linear dependence of the observations on the parameters and linear constraint relations. Non-linear problems are then solved iteratively. The solution then is obtained by a simple matrix calculus.

Linear regression will be sketched in Sect. 7.2.3 and the inclusion of constraints in Appendix 13.10.

6.6.5 Conclusion

By far the simplest method is the one where the constraint is directly included and approximated by a narrow Gaussian. With conventional minimizing programs the full error matrix is produced automatically.

The approach using a reduced parameter set is especially interesting when we are primarily interested in the parameters of the reduced set. Due to the reduced dimension of the parameter space, it is faster than the other methods. The determination of the errors of the original parameters through error propagation is sometimes tedious.

It is recommended to either eliminate redundant parameters or to use the simple method where we represent constraints by narrow Gaussians. The application of Lagrange multipliers is unnecessarily complicated and the linear approximation requires additional assumptions and iterations.

6.7 Reduction of the Number of Variates

6.7.1 The Problem

A statistical analysis of an univariate sample is obviously much simpler than that of a multidimensional one. This is not only true for the qualitative comparison of a sample with a parameter dependent p.d.f. but also for the quantitative parameter inference. Especially when the p.d.f. is distorted by the measurement process and a Monte Carlo simulation is required, the direct ML method cannot be applied as we have seen above. The parameter inference then happens by comparing histograms with the problem that in multidimensional spaces the number of entries can be quite small in some bins. Therefore, we have an interest to reduce the dimensionality of the variable space by appropriate transformations, of course, if possible, without loss of information. However, it is not always easy to find out which variable or which variable combination is especially important for the parameter estimation.

6.7.2 Two Variables and a Single Linear Parameter

A p.d.f. $f(x, y|\theta)$ of two variates with a linear parameter dependence can always be written in the form

$$f(x, y|\theta) = v(x, y)[1 + u(x, y)\theta] .$$

The log-likelihood for observations $u = u(x, y)$, $v = v(x, y)$,

$$\ln L(\theta) = \ln v + \ln(1 + u\theta) ,$$

is essentially a function of only one significant variate $u(x, y)$ because $\ln v$ does not depend on θ and can be omitted. A MLE of θ for a given sample $\{(x_1, y_1), \dots, (x_N, y_N)\}$ with

$$\ln L(\theta) = \sum \ln(1 + u_i\theta) , \quad (6.26)$$

and $u_i = u(x_i, y_i)$ depends only on the observations $\{u_1, \dots, u_N\}$. The analysis can be based on the individual quantities or a histogram.

The simple form of the relation (6.26) suggests that the analytic form $g(u|\theta)$ of the p.d.f. of u is not needed for the parameter inference. Only the experimental observations u_i enter into the likelihood function.

Unfortunately this nice property is lost when acceptance and resolution effects are present – and this is usually the case. In this situation, the linearity in θ is lost because we are forced to renormalize the p.d.f.. Nevertheless we gain by the reduction to one variate u . If the detector effects are not too large, the distribution of u still contains almost the complete information relative to θ .

The analytic variable transformation and reduction is possible only in rare cases, but it is not necessary because it is performed implicitly by the Monte Carlo simulation. We generate according to $f(x, y|\theta)$ and for each observation x_i, y_i we calculate the corresponding quantity $u_i = u(x_i, y_i)$. The parameter θ is determined by a comparison of the experimental sample with the Monte Carlo distribution of u by means of a likelihood or a χ^2 fit. The distribution of u lends itself also for a goodness-of-fit test (see Chap. 10).

6.7.3 Generalization to Several Variables and Parameters

The generalization to N variates which we combine to a vector $\mathbf{x}$ is trivial:

$$f(\mathbf{x}|\theta) = v(\mathbf{x}) [1 + u(\mathbf{x})\theta] .$$

Again we can reduce the variate space to a single significant variate u without losing relevant information. If simultaneously P parameters have to be determined, we usually will need also P new variates $u_1, \dots, u_P$:

$$f(\mathbf{x}|\boldsymbol{\theta}) = v(\mathbf{x}) \left[1 + \sum_p u_p(\mathbf{x})\theta_p \right] .$$

Our procedure thus makes sense only if the number of parameters is smaller than the dimension of the variate space.

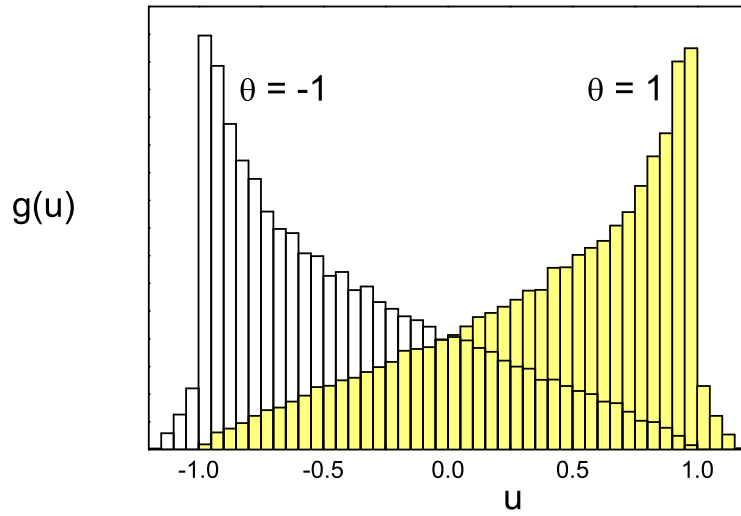


Fig. 6.13. Simulated p.d.f.s of the reduced variable u for the values ± 1 of the parameter.

Example 94. Reduction of the variate space

We consider the p.d.f.

$$f(x, y, z|\theta) = \frac{1}{\pi} \left[(x^2 + y^2 + z^2)^{1/2} + (x + y^3)\theta \right], \quad x^2 + y^2 + z^2 \leq 1, \quad (6.27)$$

which depends on three variates and one parameter. For a given sample of observations in the three dimensional cartesian space we want to determine the parameter θ . The substitutions

$$\begin{aligned} u &= \frac{x + y^3}{(x^2 + y^2 + z^2)^{1/2}}, \quad |u| \leq \sqrt{2}, \\ v &= (x^2 + y^2 + z^2)^{1/2}, \quad 0 \leq v \leq 1, \\ z &= z \end{aligned}$$

lead to the new p.d.f. $g'(u, v, z)$

$$g'(u, v, z|\theta) = \frac{v}{\pi} [1 + u\theta] \frac{\partial(x, y, z)}{\partial(u, v, z)},$$

which after integrating out v and z yields the p.d.f. $g(u|\theta)$:

$$g(u|\theta) = \int dz dv g'(u, v, z|\theta).$$

This operation is not possible analytically but we do not need to compute g explicitly. We are able to determine the MLE and its error from the simple log-likelihood function of θ

$$\ln L(\theta) = \sum_i \ln(1 + u_i \theta).$$

In case we have to account for acceptance effects, we have to simulate the u distribution. For a Monte Carlo simulation of (6.27) we compute for each observation x_i, y_i, z_i the value of u_i and histogram it. The simulated histograms g_+ and g_- of u for the two parameter values $\theta = \pm 1$ are shown in Fig. 6.13. (The figure does not include experimental effects. This is irrelevant for the illustration of the method.) The superposition $t_i = (1 - \theta)g_{-i} + (1 + \theta)g_{+i}$ has then to be inserted into the likelihood function (6.15).

6.7.4 Non-linear Parameters

The example which we just investigated is especially simple because the p.d.f. depends linearly on a single parameter. Linear dependencies are quite frequent because distributions often consist of a superposition of several processes, and the interesting parameters are the relative weights of those.

For the general, non-linear case we restrict ourselves to a single parameter to simplify the notation. We expand the p.d.f. into a Taylor series at a first rough estimate θ_0 :

$$\begin{aligned} f(\mathbf{x}|\theta) &= f(\mathbf{x}|\theta_0) + \frac{\partial f}{\partial \theta} \Big|_{\theta_0} \Delta\theta + \frac{1}{2} \frac{\partial^2 f}{\partial \theta^2} \Big|_{\theta_0} \Delta\theta^2 + \dots \\ &= V [1 + u_1 \Delta\theta + u_2 \Delta\theta^2 + \dots] . \end{aligned} \quad (6.28)$$

As before, we choose the coefficients u_i as new variates. Neglecting quadratic and higher terms, the estimate $\hat{\theta} = \theta_0 + \widehat{\Delta\theta}$ depends only on the new variate u_1 ,

$$u_1(\mathbf{x}) = \frac{\partial f(\mathbf{x}|\theta)/\partial \theta \Big|_{\theta_0}}{f(\mathbf{x}|\theta_0)}$$

which is a simple function of $\mathbf{x}$.

If the linear approximation is insufficient, a second variate u_2 should be added. Alternatively, the solution can be iterated. The generalization to several parameters is straight forward.

A more detailed description of the method with application to a physics process can be found in Refs. [30, 31]. The corresponding choice of the variate is also known under the name *optimal variable method* [42].

6.8 Method of Approximated Likelihood Estimator

As in the previous section we investigate the situation where we have to estimate parameters in presence of acceptance and resolution effects. The idea of the method is the following: We try to find a statistic $\hat{\theta}'$ of the distorted data sample which summarizes the information relative to the parameter of interest. Then we perform a Monte Carlo simulation to infer the relation $\theta(\theta')$ between the parameter of interest θ and the observed quantity θ' . Ideally, we can find an approximately sufficient statistic⁹. If we are not successful, we can use the likelihood estimate which we

⁹A sufficient statistic is a function of the sample and replaces it for the parameter estimation without loss in precision. We will define sufficiency in Chap. 7.

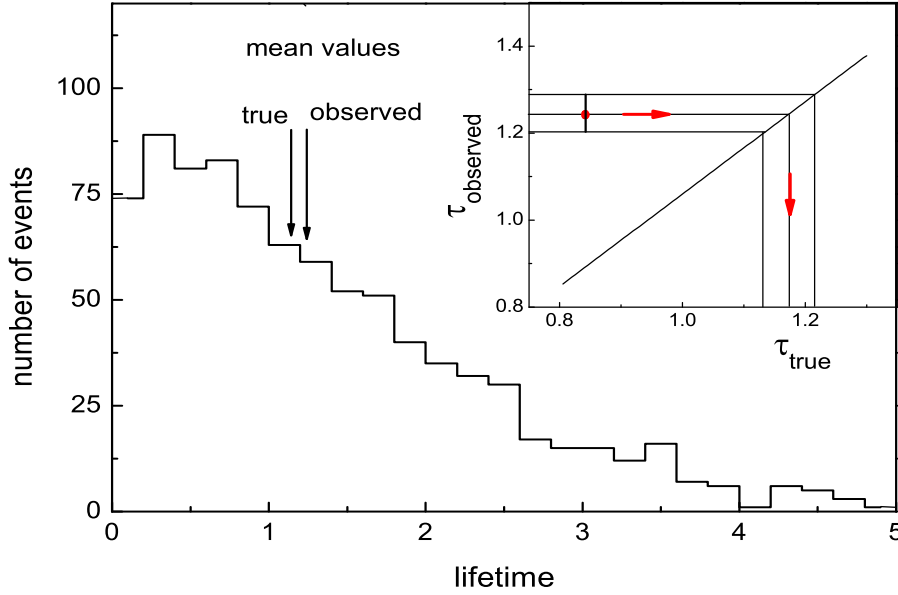


Fig. 6.14. Observed lifetime distribution. The insert indicates the transformation of the observed lifetime to the corrected one.

obtain when we insert the data into the undistorted p.d.f.. In both cases we find the relation between the experimental statistic and the estimate of the parameter by a Monte Carlo simulation. The method should become clear in the following example.

Example 95. Approximated likelihood estimator: Lifetime fit from a distorted distribution

The sample mean $\bar{t}$ of a sample of N undistorted exponentially distributed lifetimes t_i is a sufficient estimator: It contains the full information related to the parameter τ , the mean lifetime (see Sect. 7.1.1). In case the distribution is distorted by resolution and acceptance effects (Fig. 6.14), the mean value

$$\bar{t}' = \sum t'_i / N$$

of the distorted sample t'_i will usually still contain almost the full information relative to the mean life τ . The relation $\tau(\bar{t}')$ between τ and its approximation $\bar{t}'$ (see insert of Fig. 6.14) is generated by a Monte Carlo simulation. The uncertainty $\delta\tau$ is obtained by error propagation from the uncertainty $\delta\bar{t}'$ of $\bar{t}'$,

$$(\delta\bar{t}')^2 = \frac{(\overline{t'^2} - \bar{t}'^2)}{N-1},$$

$$\text{with } \overline{t'^2} = \frac{1}{N} \sum t_i'^2$$

using the Monte Carlo relation $\tau(\bar{t}')$.

This approach has several advantages:

- We do not need to histogram the observations.
- Problems due to small event numbers for bins in a multivariate space are avoided.
- It is robust, simple and requires little computing time.

For these reasons the method is especially suited for online applications, provided that we find an efficient estimator.

If the distortions are not too large, we can use the likelihood estimator extracted from the observed sample $\{x'_1, \dots, x'_N\}$ and the undistorted distribution $f(x|\lambda)$:

$$L(\lambda) = \prod f(x'_i|\lambda) ,$$

$$\frac{dL}{d\lambda}|_{\hat{\lambda}'} = 0 . \quad (6.29)$$

This means concretely that we perform the usual likelihood analysis where we ignore the distortion. We obtain $\hat{\lambda}'$. Then we correct the bias by a Monte Carlo simulation which provides the relation $\hat{\lambda}(\hat{\lambda}')$.

It may happen in rare cases where the experimental resolution is very bad that $f(x|\lambda)$ is undefined for some extremely distorted observations. This problem can be cured by scaling $\hat{\lambda}'$ or by eliminating particular observations.

Acceptance losses $\alpha(x)$ alone without resolution effects do not necessarily entail a reduction in the precision of our approach. For example, as has been shown in Sect. 6.5.2, cutting an exponential distribution at some maximum value of the variate, the mean value of the observations is still a sufficient statistic. But there are cases where sizable acceptance losses have the consequence that our method deteriorates. In these cases we have to take the losses into account. We only sketch a suitable method. The acceptance corrected p.d.f. $f'(x|\lambda)$ for the variate x is

$$f'(x|\lambda) = \frac{\alpha(x)f(x|\lambda)}{\int \alpha(x)f(x|\lambda)dx} ,$$

where the denominator is the global acceptance and provides the correct normalization. We abbreviate it by $A(\lambda)$. The log-likelihood of N observations is

$$\ln L(\lambda) = \sum \ln \alpha(x_i) + \sum \ln f(x_i|\lambda) - NA(\lambda) .$$

The first term can be omitted. The acceptance $A(\lambda)$ can be determined by a Monte Carlo simulation. Again a rough estimation is sufficient, at most it reduces the precision but does not introduce a bias, since all approximations are automatically corrected with the transformation $\lambda(\lambda')$.

Frequently, the relation (6.29) can only be solved numerically, i.e. we find the maximum of the likelihood function in the usual manner. We are also allowed to approximate this relation such that an analytic solution is possible. The resulting error is compensated in the simulation.

Example 96. Approximated likelihood estimator: linear and quadratic distributions

A sample of events x_i is distributed linearly inside the interval $[-1, 1]$, i.e. the p.d.f. is $f(x|b) = 0.5 + bx$. The slope b , $|b| < 1/2$, is to be fitted. It is located in the vicinity of b_0 . We expand the likelihood function

$$\ln L = \sum \ln(0.5 + bx_i)$$

at b_0 with

$$b = b_0 + \beta$$

and derive it with respect to β to find the value $\hat{\beta}$ at the maximum:

$$\sum \frac{x_i}{0.5 + (b_0 + \hat{\beta})x_i} = 0 .$$

Neglecting quadratic and higher order terms in $\hat{\beta}$ we can solve this equation for $\hat{\beta}$ and obtain

$$\hat{\beta} \approx \frac{\sum x_i / f_{0i}}{\sum x_i^2 / f_{0i}^2} \quad (6.30)$$

where we have set $f_{0i} = f(x_i | b_0)$.

If we allow also for a quadratic term

$$f(x|a, b) = a + bx + (1.5 - 3a)x^2 ,$$

we write, in obvious notation,

$$f(x|a, b) = f_0 + \alpha(1 - 3x^2) + \beta x$$

and get, after deriving $\ln L$ with respect to α and β and linearizing, two linear equations for $\hat{\alpha}$ and $\hat{\beta}$:

$$\begin{aligned} \hat{\alpha} \sum A_i^2 + \hat{\beta} \sum A_i B_i &= \sum A_i , \\ \hat{\alpha} \sum A_i B_i + \hat{\beta} \sum B_i^2 &= \sum B_i , \end{aligned} \quad (6.31)$$

with the abbreviations $A_i = (1 - 3x_i^2)/f_{0i}$, $B_i = x_i/f_{0i}$.

From the observed data using (6.31) we get $\hat{\beta}'(x')$, $\hat{\alpha}'(x')$, and the simulation provides the parameter estimates $\hat{b}(\hat{\beta}')$, $\hat{a}(\hat{\alpha}')$ and their uncertainties.

The calculation is much faster than a numerical minimum search and almost as precise. If $\hat{\alpha}, \hat{\beta}$ are large we have to iterate.

6.9 Nuisance Parameters

Frequently a p.d.f. $f(\mathbf{x}|\boldsymbol{\theta}, \boldsymbol{\nu})$ contains several parameters from which only some, namely $\boldsymbol{\theta}$, are of interest, whereas the other parameters $\boldsymbol{\nu}$ are unwanted, but influence the estimate of the former. Those are called *nuisance parameters*. A typical example is the following.

Example 97. Nuisance parameter: decay distribution with background

We want to infer the decay rate γ of a certain particle from the decay times t_i of a sample of M events. Unfortunately, the sample contains an unknown amount of background. The decay rate γ_b of the background particles be known. The nuisance parameter is the number of background events N . For

a fraction of background events of N/M , the p.d.f. for a single event with lifetime t is

$$f(t|\gamma, N) = \left(1 - \frac{N}{M}\right) \gamma e^{-\gamma t} + \frac{N}{M} \gamma_b e^{-\gamma_b t},$$

from which we derive the likelihood for the sample:

$$L(\gamma, N) = \prod_{i=1}^M \left[\left(1 - \frac{N}{M}\right) \gamma e^{-\gamma t_i} + \frac{N}{M} \gamma_b e^{-\gamma_b t_i} \right].$$

A contour plot of the log-likelihood of a specific data sample of 20 events and $\gamma_b = 0.2$ is depicted in Fig. 6.15. The two parameters γ and N are correlated. The question is then: What do we learn about γ , what is a sensible point estimate of γ and how should we determine its uncertainty?

We will re-discuss this example in the next subsection and present in the following some approaches which permit to eliminate the nuisance parameters. First we will investigate exact methods and then we will turn to the more problematic part where we have to apply approximations.

6.9.1 Nuisance Parameters with Given Prior

If we know the p.d.f. $\pi(\boldsymbol{\nu})$ of a nuisance parameter vector $\boldsymbol{\nu}$, the prior of $\boldsymbol{\nu}$, then we can eliminate $\boldsymbol{\nu}$ simply by integrating it out, thereby weighting $\boldsymbol{\nu}$ with its probability $\pi(\boldsymbol{\nu})$ to occur.

$$f_{\theta}(\mathbf{x}|\boldsymbol{\theta}) = \int f(\mathbf{x}|\boldsymbol{\theta}, \boldsymbol{\nu}) \pi(\boldsymbol{\nu}) d\boldsymbol{\nu}.$$

In this way we obtain a p.d.f. depending solely on the parameters of interest $\boldsymbol{\theta}$. The corresponding likelihood function of $\boldsymbol{\theta}$ is

$$L_{\theta}(\boldsymbol{\theta}|\mathbf{x}) = \int L(\boldsymbol{\theta}, \boldsymbol{\nu}|\mathbf{x}) \pi(\boldsymbol{\nu}) d\boldsymbol{\nu} = \int f(\mathbf{x}|\boldsymbol{\theta}, \boldsymbol{\nu}) \pi(\boldsymbol{\nu}) d\boldsymbol{\nu}. \quad (6.32)$$

Example 98. Nuisance parameter: measurement of a Poisson rate with a digital clock

An automatic monitoring device measures a Poisson rate θ with a digital clock with a least count of Δ . For n observed reactions within a time interval ν the p.d.f. is given by the Poisson distribution $P(n|\theta\nu)$. If we consider both, the rate parameter θ and the length of the time interval ν as unknown parameters, the corresponding likelihood function is

$$L(\theta, \nu) = \frac{e^{-\theta\nu} [\theta\nu]^n}{n!}.$$

For a clock reading t_0 , the true measurement time is contained in the time interval $t_0 \pm \Delta/2$. We can assume that all times ν within that interval are equally probable and thus the prior of ν is $\pi(\nu) = 1/\Delta$ for ν in the interval $[t_0 - \Delta/2, t_0 + \Delta/2]$ and equal to zero elsewhere. We eliminate constant factors, and, integrating over ν ,

$$L_{\theta}(\theta) = \int_{t_0 - \Delta/2}^{t_0 + \Delta/2} e^{-\theta\nu} [\theta\nu]^n d\nu,$$

we get rid of the nuisance parameter. The integral can be evaluated numerically.

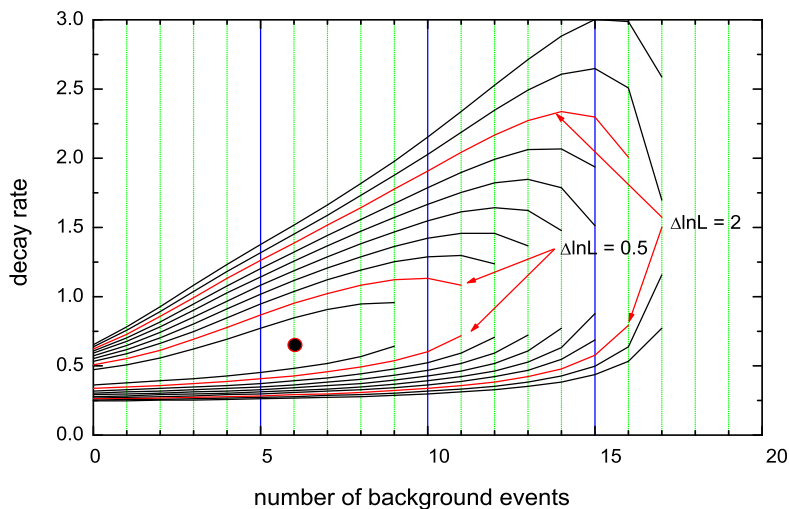


Fig. 6.15. Log-likelihood contour as a function of decay rate and number of background events. For better visualization the discrete values of the event numbers are connected.

Let us resume the problem discussed in the introduction. We now assume that we have prior information on the amount of background: The background expectation had been determined in an independent experiment to be 10 with sufficient precision to neglect its uncertainty. The actual number of background events follows a Poisson distribution. The likelihood function is

$$L(\gamma) = \sum_{N=0}^{\infty} \frac{e^{-10} 10^N}{N!} \prod_{i=1}^{20} \left[\left(1 - \frac{N}{20} \right) \gamma e^{-\gamma t_i} + \frac{N}{20} 0.2 e^{-0.2 t_i} \right] .$$

Since our nuisance parameter is discrete we have replaced the integration in (6.32) by a sum.

6.9.2 Factorizing the Likelihood Function

Very easy is the elimination of the nuisance parameter if the p.d.f. is of the form

$$f(\mathbf{x}|\theta, \nu) = f_{\theta}(\mathbf{x}|\theta) f_{\nu}(\mathbf{x}|\nu) , \quad (6.33)$$

i.e. only the first term f_{θ} depends on θ . Then we can write the likelihood as a product

$$L(\theta, \nu) = L_{\theta}(\theta) L_{\nu}(\nu)$$

with

$$L_{\theta} = \prod f_{\theta}(\mathbf{x}_i|\theta) ,$$

independent of the nuisance parameter ν .

Example 99. Elimination of a nuisance parameter by factorization of a two-dimensional normal distribution

A sample of space points (x_i, y_i) , $i = 1, \dots, N$ follow a normal distribution

$$\begin{aligned} f(x, y|\theta, \nu) &= \frac{ab}{2\pi} \exp\left(-\frac{1}{2} [a^2(x - \theta)^2 + b^2(y - \nu)^2]\right) \\ &= \frac{ab}{2\pi} \exp\left(-\frac{a^2}{2}(x - \theta)^2\right) \exp\left(-\frac{b^2}{2}(y - \nu)^2\right). \end{aligned}$$

with θ the parameter which we are interested in. The normalized x distribution depends only on θ . Whatever value ν takes, the shape of this distribution remains always the same. Therefore we can estimate θ independently of ν . The likelihood function is proportional to a normal distribution of θ ,

$$L(\theta) \sim \exp\left(-\frac{a^2}{2}(\theta - \hat{\theta})^2\right),$$

with the estimate $\hat{\theta} = \bar{x} = \sum x_i/N$.

6.9.3 Parameter Transformation, Restructuring

Sometimes we manage by means of a parameter transformation $\nu' = \nu'(\theta, \nu)$ to bring the p.d.f. into the desired form (6.33) where the p.d.f. factorizes into two parts which depend separately on the parameters θ and ν' . We have already sketched an example in Sect. 4.3.6: When we are interested in the slope θ and not in the intersection ν with the y-axis of a straight line $y = \theta x + \nu$ which should pass through measured points, then we are able to eliminate the correlation between the two parameters. To this end we express the equation of the straight line by the slope and the ordinate at the center of gravity, see Example 114 in Chap. 7.

A simple transformation $\nu' = c_1\nu + c_2\theta$ also helps to disentangle correlated parameters of a Gaussian likelihood

$$L(\theta, \nu) \sim \exp\left(-\frac{a^2(\theta - \hat{\theta})^2 - 2ab\rho(\theta - \hat{\theta})(\nu - \hat{\nu}) + b^2(\nu - \hat{\nu})^2}{2(1 - \rho^2)}\right),$$

With suitable chosen constants c_1, c_2 it produces a likelihood function that factorizes in the new parameter pair θ, ν' . In the notation where the quantities $\hat{\theta}, \hat{\nu}$ maximize the likelihood function, the transformation produces the result

$$L_\theta(\theta) \sim \exp\left(-\frac{a^2}{2}(\theta - \hat{\theta})^2\right).$$

We set the proof of this assertion aside.

It turns out that this procedure yields the same result as simply integrating out the nuisance parameter and as the profile likelihood method which we will discuss below. This result is interesting in the following respect: In many situations the likelihood function is nearly of Gaussian shape. As is shown in Appendix 13.3, the likelihood function approaches a Gaussian with increasing number of observations. Therefore, integrating out the nuisance parameter, or better to apply the profile likelihood method, is a sensible approach in many practical situations. Thus nuisance parameters are a problem only if the sample size is small.

The following example is frequently discussed in the literature [16].

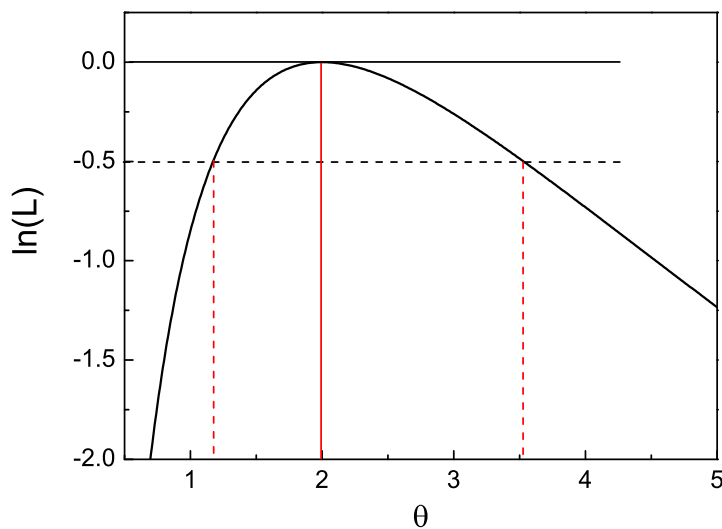


Fig. 6.16. Log-likelihood function of an absorption factor.

Example 100. Elimination of a nuisance parameter by restructuring: absorption measurement

The absorption factor θ for radioactive radiation by a plate is determined from the numbers of events r_1 and r_2 , which are observed with and without the absorber within the same time intervals. The numbers r_1, r_2 follow Poisson distributions with mean values ρ_1 and ρ_2 :

$$f_1(r_1|\rho_1) = \frac{e^{-\rho_1} \rho_1^{r_1}}{r_1!},$$

$$f_2(r_2|\rho_2) = \frac{e^{-\rho_2} \rho_2^{r_2}}{r_2!}.$$

The interesting parameter is the expected absorption $\theta = \rho_2/\rho_1$. In first approximation we can use the estimates r_1, r_2 of the two independent parameters ρ_1 and ρ_2 and their errors to calculate in the usual way through error propagation θ and its uncertainty:

$$\hat{\theta} = \frac{r_2}{r_1},$$

$$\frac{(\delta\hat{\theta})^2}{\hat{\theta}^2} = \frac{1}{r_1} + \frac{1}{r_2}.$$

For large numbers r_1, r_2 this method is justified but the correct way is to transform the parameters ρ_1, ρ_2 of the combined distribution

$$f(r_1, r_2|\rho_1, \rho_2) = \frac{e^{-(\rho_1+\rho_2)} \rho_1^{r_1} \rho_2^{r_2}}{r_1! r_2!}$$

into the independent parameters $\theta = \rho_2/\rho_1$ and $\nu = \rho_1 + \rho_2$. The transformation yields:

$$\tilde{f}(r_1, r_2 | \theta, \nu) = e^{-\nu} \nu^{r_1+r_2} \frac{(1+1/\theta)^{-r_1} (1+\theta)^{-r_2}}{r_1! r_2!},$$

$$L(\theta, \nu | r_1, r_2) = L_\nu(\nu | r_1, r_2) L_\theta(\theta | r_1, r_2).$$

Thus the log-likelihood function of θ is

$$\ln L_\theta(\theta | r_1, r_2) = -r_1 \ln(1+1/\theta) - r_2 \ln(1+\theta).$$

It is presented in Fig. 6.16 for the specific values $r_1 = 10$, $r_2 = 20$. The maximum is located at $\hat{\theta} = r_2/r_1$, as obtained with the simple estimation above. However the errors are asymmetric.

Instead of a parameter transformation it is sometimes possible to eliminate the nuisance parameter if we can find a statistic y which is independent of the nuisance parameter ν , i.e. ancillary with respect to ν , but dependent on the parameter θ . Then we can use the p.d.f. of y , $f(y|\theta)$, which per definition is independent of ν , to estimate θ . Of course, we may lose information because y is not necessarily a sufficient statistic relative to θ . The following example illustrates this method. In this case there is no loss of information.

Example 101. Eliminating a nuisance parameter by restructuring: Slope of a straight line with the y-axis intercept as nuisance parameter

We come back to one of our standard examples which can, as we have indicated, be solved by a parameter transformation. Now we solve it in a simpler way. Points (x_i, y_i) are distributed along a straight line. The x coordinates are exactly known, the y coordinates are the variates. The p.d.f. $f(y_1, \dots, y_n | \theta, \nu)$ contains the slope parameter θ and the uninteresting intercept ν of the line with the y axis. It is easy to recognize that the statistic $\{\hat{y}_1 = y_1 - y_n, \hat{y}_2 = y_2 - y_n, \dots, \hat{y}_{n-1} = y_{n-1} - y_n\}$ is independent of ν . In this specific case the new statistic is also sufficient relative to the slope θ which clearly depends only on the differences of the ordinates. We leave the details of the solution to the reader.

Further examples for the elimination of a nuisance parameter by restructuring have been given already in Sect. 6.5.2, Examples 79 and 80.

6.9.4 Profile Likelihood

We now turn to approximate solutions.

Some scientists propose to replace the nuisance parameter by its estimate. This corresponds to a delta function for the prior of the nuisance parameter and is for that reason quite exotic and dangerous. It leads to an illegitimate reduction of the error limits whenever the nuisance parameter and the interesting parameter are correlated. Remark, that a correlation always exists unless a factorization is possible. In the extreme case of full correlation the error would shrink to zero.

A much more sensible approach to eliminate the nuisance parameter uses the so-called profile likelihood [33]. To explain it, we give an example with a single nuisance parameter.

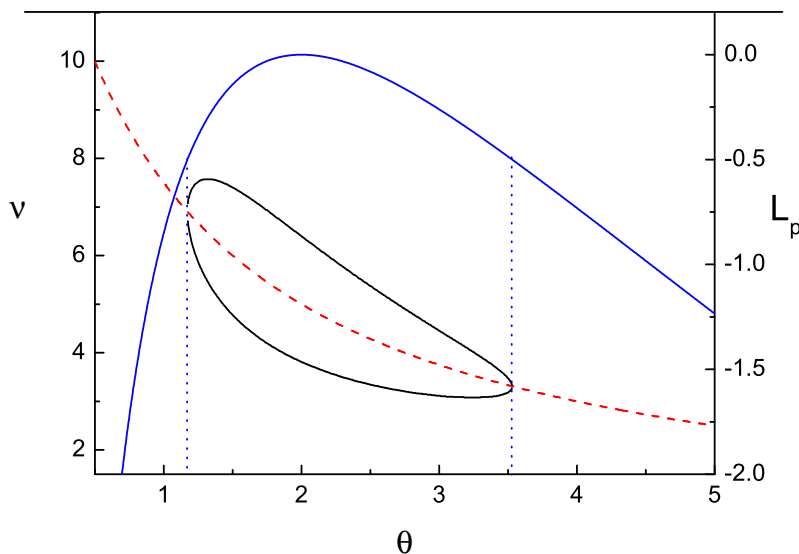


Fig. 6.17. Profile likelihood (solid curve, right hand scale) and $\Delta \ln L(\theta, \nu) = 1/2$ contour (left hand scale). The dashed curve is $\hat{\nu}(\theta)$.

The likelihood function is maximized with respect to the nuisance parameter ν as a function of the wanted parameter θ . The function $\hat{\nu}(\theta)$ which maximizes L then satisfies the relation

$$\left. \frac{\partial L(\theta, \nu | x)}{\partial \nu} \right|_{\hat{\nu}} = 0 \rightarrow \hat{\nu}(\theta) .$$

It is inserted into the likelihood function and provides the profile likelihood L_p ,

$$L_p = L(\theta, \hat{\nu}(\theta) | x) ,$$

which depends solely on θ .

This method has the great advantage that only the likelihood function enters and no assumptions about priors have to be made. It also takes correlations into account. Graphically we can visualize the error interval of the profile likelihood $\Delta \ln L_p(\theta, \nu) = 1/2$ by drawing the tangents of the curve $\Delta \ln L = 1/2$ parallel to the ν axis. These tangents include the error interval.

An example of a profile likelihood is depicted in Fig. 6.17. We have used the absorption example from above to compute the curves. The nuisance parameter is ρ_1 . The result is the same as with the exact factorization.

In the literature we find methods which orthogonalize the parameters at the maximum of the likelihood function [35] which means to diagonalize a more dimensional Gaussian. The result is similar to that of the profile likelihood approach.

In the limit of a large number of observations where the likelihood function approaches the shape of a normal distribution, the profile likelihood method is identical to restructuring and factorizing the likelihood.

6.9.5 Integrating out the Nuisance Parameter

If the methods fail which we have discussed so far, we are left with only two possibilities: Either we give up the elimination of the nuisance parameter or we integrate it out. The simple integration

$$L_{\theta}(\theta|x) = \int_{-\infty}^{\infty} L(\theta, \nu|x) d\nu$$

implicitly contains the assumption of a uniform prior of ν and therefore depends to some extent on the validity of this condition. However, in most cases it is a reasonable approximation. The effect of varying the prior is usually negligible, except when the likelihood function is very asymmetric. Also a linear term in the prior does usually not matter. It is interesting to notice that in most cases integrating out the nuisance parameter leads to the same result as restructuring the problem.

6.9.6 Explicit Declaration of the Parameter Dependence

It is not always possible to eliminate the nuisance parameter in such a way that the influence of the method on the result can be neglected. When the likelihood function has a complex structure, we are obliged to document the full likelihood function. In many cases it is possible to indicate the dependence of the estimate θ and its error limits θ_1, θ_2 on the nuisance parameter ν explicitly by a simple linear function

$$\begin{aligned}\hat{\theta} &= \hat{\theta}_0 + c(\nu - \hat{\nu}) , \\ \theta_{1,2} &= \theta_{1,2} + c(\nu - \hat{\nu}) .\end{aligned}$$

Usually the error limits will show the same dependence as the MLE which means that the width of the interval is independent of ν .

However, publishing a dependence of the parameter of interest on the nuisance parameter is useful only if ν corresponds to a physical constant and not to an internal parameter of an experiment like *efficiency* or *background*.

6.9.7 Advice

If it is impossible to eliminate the nuisance parameter explicitly and if the shape of the likelihood function does not differ dramatically from that of a Gaussian, the profile likelihood approach should be used for the parameter and interval estimation. In case the deviation from a Gaussian is considerable, we will try to avoid problems by an explicit declaration of the dependence of the likelihood estimate and limits on the nuisance parameter. If also this fails, we abstain from the elimination of the nuisance parameter and publish the full likelihood function.

Parameter Inference II

7.1 Likelihood and Information

7.1.1 Sufficiency

In a previous example, we have seen that the likelihood function for a sample of exponentially distributed decay times is a function only of the sample mean. In fact, in many cases, the i.i.d. individual elements of a sample $\{x_1, \dots, x_N\}$ can be combined to fewer quantities, ideally to a single one without affecting the estimation of the interesting parameters. The set of these quantities which are functions of the observations is called a *sufficient statistic*. The sample itself is of course a sufficient, while uninteresting statistic.

According to R. A. Fisher, a statistic is sufficient for one or several parameters, if by addition of arbitrary other statistics of the same data sample, the parameter estimation cannot be improved. More precise is the following definition [1]: A statistic $\mathbf{t}(x_1, \dots, x_N) \equiv \{t_1(x_1, \dots, x_N), \dots, t_M(x_1, \dots, x_N)\}$ is sufficient for a parameter set $\boldsymbol{\theta}$, if the distribution of a sample $\{x_1, \dots, x_N\}$, given $\mathbf{t}$, does not depend on $\boldsymbol{\theta}$:

$$f(x_1, \dots, x_N | \boldsymbol{\theta}) = g(\mathbf{t}_1, \dots, \mathbf{t}_M | \boldsymbol{\theta}) h(x_1, \dots, x_N). \quad (7.1)$$

The distribution $g(\mathbf{t} | \boldsymbol{\theta})$ then contains all the information which is relevant for the parameter estimation. This means that for the estimation process we can replace the sample by the sufficient statistic. In this way we may reduce the amount of data considerably. In the standard situation where all parameter components are constraint by the data, the dimension of $\mathbf{t}$ must be larger or equal to the dimension of the parameter vector $\boldsymbol{\theta}$. Every set of uniquely invertible functions of $\mathbf{t}$ is also a sufficient statistic.

The relevance of sufficiency is expressed in a different way in the so-called *sufficiency principle*:

If two different sets of observations have the same values of a sufficient statistic, then the inference about the unknown parameter should be the same.

Of special interest is a *minimal sufficient statistic*. It consists of a minimal number of components, ideally only of one element per parameter.

In what follows, we consider the case of a one-dimensional sufficient statistic $t(x_1, \dots, x_N)$ and a single parameter θ . The likelihood function can according to

(7.1) be written in the following way:

$$L = L_1(\theta|t(\mathbf{x})) \cdot L_2(\mathbf{x}) . \quad (7.2)$$

It is easy to realize that the second factor L_2 which is independent of θ^1 , has no bearing on the likelihood ratios of different values of θ . We obtain a data reduction of N to 1. This means that all samples of size N which have the same value of the statistic t lead to the same likelihood function and thus to the same MLE and the same likelihood ratio interval.

If a minimal sufficient statistic of one element per parameter exists, then the MLE itself is a minimal sufficient statistic and the MLE together with the sample size N fix the likelihood function up to an irrelevant factor. (For the Cauchy distribution the full sample is a minimal sufficient statistic. No further reduction in size is possible. Thus its MLE is not sufficient.)

If in the general situation with P parameters a minimal sufficient statistic $\mathbf{t}$ of P components exists, the data reduction is N to P and the MLE for the P parameters will be a unique function of $\mathbf{t}$ and is therefore itself a sufficient statistic.

Once we have found a sufficient statistic it is mostly easy to compute the MLE.

Example 102. Sufficient statistic and expected value of a normal distribution

Let $x_1, \dots, x_N$ be N normally distributed observations with width σ . The parameter of interest be the expected value μ of the distribution. The likelihood function is

$$\begin{aligned} L(\mu|x_1, \dots, x_N) &= c \prod_{i=1}^N \exp[-(x_i - \mu)^2/(2\sigma^2)] \\ &= c \exp\left[-\sum_{i=1}^N (x_i - \mu)^2/(2\sigma^2)\right] , \end{aligned}$$

with $c = (\sqrt{2\pi}\sigma)^{-N}$. The exponent can be expressed in the following way:

$$\sum_{i=1}^N (x_i - \mu)^2/(2\sigma^2) = N(\overline{x^2} - 2\overline{x}\mu + \mu^2)/(2\sigma^2) .$$

Now the likelihood L factorizes:

$$L(\mu, \sigma|x_1, \dots, x_N) = \exp[-N(-2\overline{x}\mu + \mu^2)/(2\sigma^2)] \cdot c \exp[-N\overline{x^2}/(2\sigma^2)] . \quad (7.3)$$

Only the first factor depends on μ . Consequently the experimental quantity $\overline{x}$ contains the full information on μ and thus is a one-dimensional sufficient statistic. Setting the derivative of the first factor equal to zero, we obtain the MLE $\hat{\mu} = \overline{x}$.

In the following example we show that a sufficient two-dimensional statistic can be found when besides the expectation value also the width σ is to be estimated.

Example 103. Sufficient statistic for mean value and width of a normal distribution

¹Note, that also the domain of x has to be independent of θ .

Let $x_1, \dots, x_N$ be N normally distributed observations. The mean value μ and the width σ be the parameters of interest. From (7.3) we deduce that $\bar{x}$ and $\overline{x^2}$ together form a sufficient statistic $\{\bar{x}, \overline{x^2}\}$. Alternatively, also $\bar{x}$ and $v^2 = \sum (x_i - \bar{x})^2$ form a sufficient statistic. The MLE in the two-dimensional parameter space μ, σ^2 is

$$\hat{\mu} = \bar{x}, \quad \hat{\sigma}^2 = \frac{1}{N}(\overline{x^2} - \bar{x}^2).$$

There is no one-dimensional sufficient statistic for σ alone, as has been the case for μ .

Remark: In the examples which we have discussed, the likelihood function is fixed up to an irrelevant multiplicative factor if we consider the sample size N as a constant. In case N is also a random variable, then N is part of the sufficient statistic, e.g. in the last example it is $\{\bar{x}, \overline{x^2}, N\}$. Usually N is given and is then an *ancillary statistic*.

Definition: A statistic y is called *ancillary*, if $f(y|\theta) = f(y)$, i.e. the p.d.f. of y is independent of the parameter of interest².

The value of the ancillary statistic has no influence on the MLE but is relevant for the shape of the likelihood function and thus for the precision of the estimation. The sample size is in most cases an ancillary statistic and responsible for the accuracy of the estimation.

7.1.2 The Conditionality Principle

Imagine that a measurement is performed either with the precise device A or with the imprecise device B . The device is selected by a stochastic process. After the measurement has been realized, we know the device which had been selected. Let us assume this was device B . The *conditionality principle* tells us that for the parameter inference we are allowed to use this information which means that we may act as if device A had not existed. The analysis is not “blind”. Stochastic results influence the way we evaluate the parameters.

More generally, the conditionality principle states:

If an experiment concerning the inference about θ is chosen from a collection of possible random experiments, independently of θ , then any experiment not chosen is irrelevant to the inference.

Example 104. Conditionality

We measure the position coordinate of the trajectory of an ionizing particle passing a drift chamber. A certain wire responds. Its position provides a rough coordinate. In 90 % of all cases a drift time is registered and we obtain a much more precise value of the coordinate. The conditionality principle tells us that in this case we are allowed to use the drift time information without considering the worse resolution of a possible but not realized failure of the time measurement.

²Note that the combination of two ancillary statistics is not necessarily ancillary.

The conditionality principle seems to be trivial. Nevertheless the belief in its validity is not shared by all statisticians because it leads to the *likelihood principle* with its far reaching consequences which are not always intuitively obvious.

7.1.3 The Likelihood Principle

We now discuss a principle which concerns the foundations of statistical inference and which plays a central role in Bayesian statistics.

The likelihood principle (LP) states the following:

Given a p.d.f. $f(x|\theta)$ containing an unknown parameter of interest θ and an observation x , all information relevant for the estimation of the parameter θ is contained in the likelihood function $L(\theta|x) = f(x|\theta)$.

Furthermore, two likelihood functions which are proportional to each other, contain the same information about θ . The general form of the p.d.f. is considered as irrelevant. The p.d.f. at variate values which have not been observed has no bearing for the parameter inference.

Correspondingly, for discrete hypotheses H_i the full experimental information relevant for discriminating between them is contained in the likelihoods L_i .

The following examples are intended to make plausible the LP which we have implicitly used in Chap. 6.

Example 105. Likelihood principle, dice

We have a bag of biased dice of type A and B . Dice A produces the numbers 1 to 6 with probabilities $1/12, 1/6, 1/6, 1/6, 1/6, 3/12$. The corresponding probabilities for dice B are $3/12, 1/6, 1/6, 1/6, 1/6, 1/12$. The result of an experiment where one of the dice is selected randomly is “3”. We are asked to bet for A or B . We are unable to draw a conclusion from the observed result because both dice produce this number with the same probability, the likelihood ratio is equal to *one*. The LP tells us – what intuitively is clear – that for a decision the additional information, i.e. the probabilities of the two dice to yield values different from “3”, are irrelevant.

Example 106. Likelihood principle, $V - A$

We come back to an example which we had discussed already in Sect. 6.3. An experiment investigates $\tau^- \rightarrow \mu^- \nu_\tau \bar{\nu}_\mu$, $\mu^- \rightarrow e^- \nu_\mu \bar{\nu}_e$ decays and measures the slope $\hat{\alpha}$ of the cosine of the electron direction with respect to the muon direction in the muon center-of-mass. The parameter α depends on the $\tau - \mu$ coupling. Is the τ decay proceeding through $V - A$ or $V + A$ coupling? The LP implies that the probabilities $f_-(\alpha)$, $f_+(\alpha)$ of the two hypotheses to produce values α different from the observed value $\hat{\alpha}$ do not matter.

When we now allow that the decay proceeds through a mixture $r = g_V/g_A$ of V and A interaction, the inference of the ratio r is based solely on the observed value $\hat{\alpha}$, i.e. on $L(r|\hat{\alpha})$.

The LP follows inevitably from the sufficiency principle and the conditioning principle. It goes back to Fisher, has been reformulated and derived several times

[36, 37, 38, 39]. Some of the early promoters (Barnard, Birnbaum) of the LP later came close to rejecting it or to restrict its applicability. The reason for the refusal of the LP has probably its origin in its incompatibility with some concepts of the classical statistics. A frequently expressed argument against the LP is that the confidence intervals of the frequentist statistics cannot be derived from the likelihood function alone and thus contradict the LP. But this fact merely shows that certain statistical methods do not use the full information content of a measurement or / and use irrelevant information. Another reason lies in problems applying the LP outside the natural sciences like in medicine or biology. There it is often not possible to parameterize the empirical models in a stringent way. But uncertainties in the model prohibit the application of the LP. The exact validity of the model is a basic requirement for the application of the LP.

In the literature examples are presented which are pretended to contradict the LP. These examples are not really convincing and rather strengthen the LP. Anyway, they often contain quite exotic distributions which are irrelevant in physics applications and which lead when treated in a frequentist way to unacceptable results [39].

We abstain from a reproduction of the rather abstract proof of the LP and limit us to present a simple and transparent illustration of it:

The quantity which contains all the information we have on θ after the measurement is the p.d.f. of θ ,

$$g(\theta) = \frac{L(\theta|x)\pi(\theta)}{\int L(\theta|x)\pi(\theta)d\theta} .$$

It is derived from the prior density and the likelihood function. The prior does not depend on the data, thus the complete information that can be extracted *from the data*, and which is relevant for $g(\theta)$, must be contained in the likelihood function.

A direct consequence of the LP is that in the absence of prior information, optimal parameter inference has to be based solely upon the likelihood function. It is then logical to select for the estimate the value of the parameter which maximizes the likelihood function and to choose the error interval such that the likelihood is constant at the border, i.e. is smaller everywhere outside than inside. (see Chap. 8). All approaches which are not based on the likelihood function are inferior to the likelihood method or at best equivalent to it.

7.1.4 Bias of Maximum Likelihood Results

In frequentist statistics an important aspect of point estimation is the evaluation of the bias of an estimator. The bias b of an estimate $\hat{\theta}$ is the deviation of its expectation value from the true value θ of the parameter:

$$b = E(\hat{\theta}) - \theta .$$

Example 107. Bias of the estimate of a decay parameter

We estimate the decay parameter γ from 5 observed decays of an unstable particle. We have seen in a previous example that the MLE $\hat{\gamma}$ is the inverse of the average of the individual decay times, $\hat{\gamma} = 1/\bar{t}$. The mean value $\bar{t}$ follows a gamma distribution (see Sect. 3.6.8).

$$f(\bar{t}|\gamma) = \frac{(5\gamma)^5}{4!} \bar{t}^4 \exp(-5\gamma\bar{t}) ,$$

and thus the expectation value $E(\hat{\gamma})$ of $\hat{\gamma}$ is

$$\begin{aligned} E(\hat{\gamma}) &= \int_0^\infty \hat{\gamma} f(\bar{t}|\gamma) d\bar{t} \\ &= \int_0^\infty \frac{(5\gamma)^5 \bar{t}^3}{4!} \exp(-5\gamma\bar{t}) d\bar{t} = \frac{5}{4}\gamma. \end{aligned}$$

When in a large number of similar experiments with 5 observed events the MLE of the decay time is determined then the arithmetic mean differs from the true value by 25 %, the bias of the MLE is $b = E(\hat{\gamma}) - \gamma = \gamma/4$. For a single decay the bias is infinite.

With increasing number of observations the bias decreases, the MLE is a consistent estimator (see Appendix 13.2.2).

Biases occur quite frequently at small samples. The word *bias* somehow suggests that something is wrong and thus it appears quite disturbing at first sight that estimates may be systematically biased. In fact in most of the conventional statistical literature it is recommended to correct for the bias. However, there is no obvious reason for a correction and a closer study reveals that bias corrections lead to difficulties when we combine different measurements $\hat{\theta}_i$ in the usual way, weighting the results by the inverse covariance matrix, or in the one dimensional case according to (4.7) simply by the inverse error squared

$$\bar{\theta} = \frac{\sum \hat{\theta}_i / \delta_i^2}{\sum 1 / \delta_i^2}.$$

Since usually the estimated errors depend on the value of the MLE, the weighting introduces a bias which often partially compensates a bias of the MLE.

Let us resume our last example and assume that many experiments measure the decay rate from sample sizes of $N = 5$. The estimates $\hat{\gamma}_i$ will vary from experiment to experiment. Each experiment will apart from the estimate evaluate the error δ_i which will turn out to be proportional to $\hat{\gamma}_i$, namely $\delta_i = \hat{\gamma}_i / \sqrt{5}$. Averaging without bias correction according to our prescription, we will obtain $E(\bar{\gamma}) = 5/6 \gamma$, thus the bias is reduced, while averaging the bias corrected estimates would lead to $E(\bar{\gamma}) = 2/3 \gamma$, a result which is considerably worse. Anyway, the bias correction is rather unsatisfactory for a small number of observations, for $N = 1$ the bias is infinite and thus the corrected value would be zero!

As we have seen, likelihood estimates have the desirable property that they are invariant against parameter transformations. For instance, the estimates for the rate $\hat{\gamma}$ and the mean life $\hat{\tau}$ obey the same relation as the true values, namely $\hat{\gamma} = 1/\hat{\tau}$ while the corresponding relation is not valid for the bias corrected quantities. Generally, the bias is not invariant against parameter transformations and thus its correction is problematic. The requirement of an unbiased estimate is in conflict with the parametrization invariance.

Contrary to many colleagues, we argue that a systematic bias correction of the estimate is baneful. It is in contradiction with the LP and should be applied only in well-founded situations. It is much more sensible to present the full likelihood function or to specify asymmetric errors since the exact procedure to combine measurements is to add the log-likelihoods. We will come back to this point in Chap. 8 and discuss the virtue of bias corrections again in Appendix 13.6.

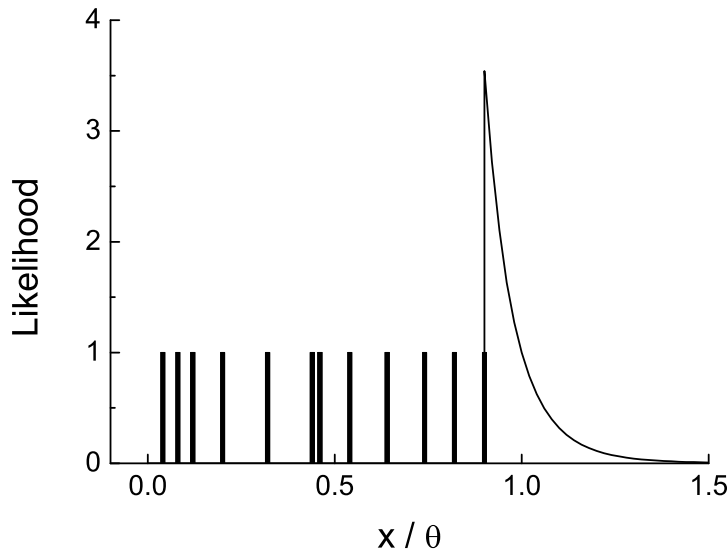


Fig. 7.1. Likelihood function of the width of a uniform distribution for 12 observations.

For a similar reason it is dangerous to interpret the likelihood function as p.d.f. of the parameter. The following example illustrates this fact.

Example 108. Bias of the estimate of a Poisson rate with observation zero

We search for a rare decay but we do not observe any. The likelihood for the mean rate λ is according to the Poisson statistic

$$L(\lambda) = \frac{e^{-\lambda} \lambda^0}{0!} = e^{-\lambda}.$$

When we normalize the likelihood function to obtain the Bayesian p.d.f. with a uniform prior, we obtain the expectation value $\langle \lambda \rangle = 1$ while the value $\hat{\lambda} = 0$ corresponds to the maximum of the likelihood function.

(It may seem astonishing, that an expectation value *one* follows from a null-measurement. This result is a consequence of the assumption of a uniform distribution of the prior which is not unreasonable because had we not anticipated the possibility of a decay, we would not have performed the measurement. Since also mean rates different from zero may lead to the observation *zero* it is natural that the expectation value of λ is different from *zero*.)

Now if none of 10 similar experiments would observe a decay, a naive averaging of the expected values alone would again result in a mean of *one*, a crazy value. Strictly speaking, the likelihoods of the individual experiments should be multiplied, or, equivalently the null rate would have to be normalized to ten times the original time with the Bayesian result $1/10$.

We study a further example.

Example 109. Bias of the measurement of the width of a uniform distribution

Let $x_1, \dots, x_N$ be N observations of a sample following a uniform distribution $f(x) = 1/\theta$ with $0 < x < \theta$. We search for the value of the parameter θ . Figure 7.1 shows the observations and the likelihood function for $N = 12$. The likelihood function is

$$\begin{aligned} L &= 0 \quad \text{for } \theta < x_{\max}, \\ &= \frac{1}{\theta^N} \quad \text{for } \theta \geq x_{\max}. \end{aligned}$$

Obviously, the likelihood has its maximum when θ coincides with the largest observation $x_{\max}$ of the sample: $\hat{\theta} = x_{\max}$. (Here $x_{\max}$ is a sufficient statistic.) At smaller values of x , the likelihood is zero. The estimate is biased towards small values. Given a sample size of N , we obtain $N + 1$ gaps between the observations and the borders $[0, \theta]$. The average distance of the largest observation from θ thus is $\theta/(N + 1)$. The bias is $-\hat{\theta}/N$. There is no reason to correct for the bias. We rather prefer to present the biased result with a one-sided error

$$\theta = x_{\max} \begin{smallmatrix} +x_{\max}/N \\ -0 \end{smallmatrix}$$

or, alternatively, the full likelihood function.

A further, more general discussion of the bias problem is given in Appendix 13.6.

7.1.5 Stopping Rules

An experiment searches for a rare reaction. Just after the first successful observation at time t the experiment is stopped. Is the resulting rate $1/t$ systematically biased? Do we have to consider the stopping rule in the inference process? The answer is “no” but many people have a different opinion. This is the reason why we find the expression *stopping rule paradox* in the literature.

The possibility to stop an experiment without compromising the data analysis, for instance because a detector failed, no money was left or because the desired precision has been reached, means a considerable simplification of the data analysis.

The validity of the independence of the parameter estimation on a large class of stopping conditions is a direct consequence of the LP because the likelihood function of a parameter determined from sequentially obtained data does not depend on stopping rules.

In this context we examine a simple example.

Example 110. Stopping rule: four decays in a time interval

In two similar experiments the lifetime of the same instable particle is measured. In experiment *A* the time interval t is fixed and 4 decays are observed. In experiment *B* the time t is measured which is required to observe 4 decays. Accidentally the two times coincide. Thus in both experiments 4 decays are registered in the time interval t but in experiment *A* the number n of decays is the random variable whereas in experiment *B* it is the time t . Do both experiments find the same rate, namely $\tau = t/4$ and the same error interval? We could think “no” because in the first experiment the fourth decay has happened earlier than in the second. The likelihood functions for the two

situations are deduced for experiment A from the Poisson distribution and for experiment B from the exponential time distribution:

$$\begin{aligned} L_A(\theta|n) &= P(n|\theta t) \\ &= \frac{e^{-\theta t}(\theta t)^4}{4!} \sim \theta^4 e^{-\theta t}, \\ L_B(\theta|t) &= \theta^4 e^{-\theta t} \sim L_A(\theta|n). \end{aligned}$$

The likelihood functions are equal up to an irrelevant factor and consequently also the results are the same. The stopping rule does not influence the analysis. The only relevant data are the number of decays and the time interval.

The fact that an arbitrary sequential stopping rule does not change the expectation value is illustrated with an example given in Fig. 7.2. A rate is determined. The measurement is stopped if a sequence of 3 decays occurs within a short time interval of only one second. It is probable that the observed rate is higher than the true one, the estimate is too high in most cases. However, if we perform many such experiments one after the other, their combination is equivalent to a single very long experiment where the stopping rule does not influence the result and from which we can estimate the mean value of the rate with high precision. Since the log-likelihood of the long experiment is equal to the sum of the log-likelihoods of the short experiments, the log-likelihoods of the short experiments obviously represent correctly the measurements. The stopping rule does not enter into the likelihood function and therefore must be irrelevant.

Why does the fact that neglecting the stopping rule is justified contradict our intuition? Well, most of the sequences indeed lead to too high rates but when we combine measurements the few long sequences get a higher weight and they tend to produce lower rates, and the average is correct. On the other hand, one might argue that the LP ignores the information that in most cases the true value of the rate is lower than the MLE. This information clearly matters if we would bet on this property, but it is irrelevant for estimating the probability density of the parameter value. A bias correction would improve the not very precise estimate somewhat for small sequences but be very unfavorable for the fewer but more precise long sequences and for a single experiment we cannot know whether we have a short or a long sequence. In fact, in exploratory experiments like those in particle physics or astrophysics we are always concerned with a single experiment. (see also Appendix 13.6).

7.2 Further Methods of Parameter Inference

In the previous sections we have discussed in some detail the maximum likelihood method. Another very popular method is the *least square method* which we can derive from the maximum likelihood method. A further approach to estimate unknown parameters of a distribution is the *moments method* which we will treat briefly in the following section.

7.2.1 The Moments Method

The moments of a distribution which depends on a parameter θ usually also depend on θ :

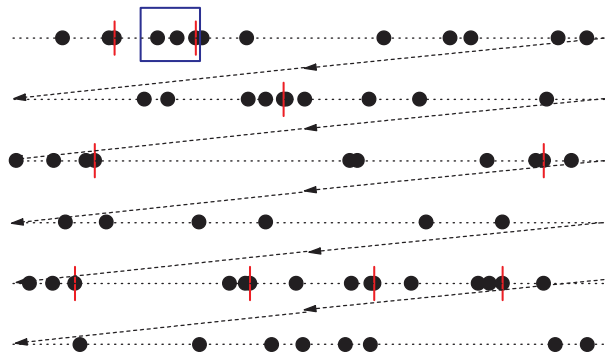


Fig. 7.2. An experiment is stopped when 3 observations are registered within a short time interval (indicated by a box). An arbitrarily long experiment can be subdivided into many such sub-experiments following the stopping rule.

$$\mu_n(\theta) = \int x^n f(x|\theta) dx . \quad (7.4)$$

The empirical moments

$$\hat{\mu}_n = \frac{1}{N} \sum_i x_i^n ,$$

e.g. the sample mean or the mean of squares, which we can extract trivially for a sample, are estimators of the moments of the distribution. From the inverse function μ^{-1} we obtain a consistent estimate of the parameter,

$$\hat{\theta} = \mu^{-1}(\hat{\mu}) ,$$

because according to the law of large numbers we have (see Appendix 13.1)

$$\lim_{N \rightarrow \infty} P\{|\hat{\mu} - \mu| > \varepsilon\} = 0 .$$

It is clear that any function $u(x)$ for which expected value and variance exist, and where $\langle u \rangle$ is an invertible function of θ , can be used instead of x^n . Therefore the method is somewhat more general than suggested by its name.

If the distribution has several parameters to be estimated, we must use several moments or expected values, approximate them by empirical averages, and solve the resulting system of – in general non-linear – equations for the unknown parameters.

The estimators derived from the lower moments are usually more precise than those computed from the higher ones. Parameter estimation from the moments is usually inferior to that of the ML method. Only if the moments used form a sufficient statistic, the two approaches produce the same result.

The uncertainties of the fitted parameters have to be estimated from the covariance matrix of the corresponding moments and subsequently by error propagation or alternatively by a Monte Carlo simulation, generating the measurement several

times. Also the *bootstrap method* which will be introduced in Chap. 12, can be employed. Sometimes the error calculation is a bit annoying and reproduces the ML error intervals only in the large sample limit.

Example 111. Moments method: mean and variance of the normal distribution

We come back to the example from Sect. 6.5.2 For a sample $\{x_1, \dots, x_N\}$ following the distribution

$$f(x|\mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-(x - \mu)^2 / (2\sigma^2) \right] .$$

We determine independently the parameters μ and σ . We use the abbreviations $\bar{x}$ for the sample mean and $\overline{x^2}$ for the mean of the squares and $v^2 = \overline{(x - \bar{x})^2} = \overline{x^2} - \bar{x}^2$ for the empirical variance.

The relation between the moment μ_1 and μ is simply $\mu_1 = \mu$, therefore

$$\hat{\mu} = \hat{\mu}_1 = \bar{x} .$$

In Chap. 3, we have derived the relation (3.15) $\langle v^2 \rangle = \sigma^2(N-1)/N$ between the expectation of the empirical variance and the variance of the distribution; inverting it, we get

$$\hat{\sigma} = v \sqrt{\frac{N}{N-1}} .$$

The two estimates are uncorrelated. The error of $\hat{\mu}$ is derived from the estimated variance

$$\delta_\mu = \frac{\hat{\sigma}}{\sqrt{N}} ,$$

and the error of $\hat{\sigma}$ is determined from the expected variance of v . We omit the calculation, the result is:

$$\delta_\sigma = \frac{\hat{\sigma}}{\sqrt{2(N-1)}} .$$

In the special case of the normal distribution, the independent point estimates of μ and σ of the moments method are identical to those of the maximum likelihood method. The errors differ for small samples but coincide in the limit $N \rightarrow \infty$.

The moments method has the advantage of being very simple, especially in the case of distributions which depend linearly on the parameters – see the next example below:

Example 112. Moments method: asymmetry of an angular distribution

Suppose we have to determine from a sample the asymmetry parameter α of a distribution $f(x) = (1 + \alpha x)/2$ linear in $x = \cos \beta$. The first moment of the distribution is $\mu_1 = \alpha/3$. Thus we can compute the parameter from the sample mean $\bar{x} = \sum x_i/N$:

$$\hat{\alpha} = 3 \bar{x} .$$

The mean square error from an individual measurement x is proportional to the variance of the distribution:

$$\text{var}(\hat{\alpha}) = 9 \text{var}(x) = 3 - \alpha^2 . \quad (7.5)$$

Using instead of α its estimate, we get

$$\delta_{\hat{\alpha}} = 3 \delta_{\bar{x}} = \sqrt{\frac{3 - 9\bar{x}^2}{N}} .$$

A likelihood fit, according to the likelihood principle, is more accurate and reflects much better the result of the experiment which, because of the kinematical limits³ $|\alpha| < 1$, cannot be described very well by symmetric errors; especially when the sample size is small and the estimate happens to lie near the boundary. In this case the maximum likelihood method should be applied.

An indication of the relative precision of the estimate moments method as compared to the MLE is provided by a comparison of the asymptotic efficiencies. In the asymptotic limit $N \rightarrow \infty$ the variance of the moments estimate $\hat{\alpha}$ does not approach the limiting value given by the Cramer–Rao inequality, see (13.6) in Appendix 13.2, which can be shown to be achieved by the MLE:

$$\text{var}(\hat{\alpha}_{ML}) \approx \frac{\alpha^2}{N} \left[\frac{1}{2\alpha} \ln\left(\frac{1+\alpha}{1-\alpha}\right) - 1 \right]^{-1} .$$

A comparison with (7.5) shows that the asymptotic efficiency of the moments method, defined as

$$\varepsilon = \frac{\text{var}(\hat{\alpha}_{ML})}{\text{var}(\hat{\alpha})} ,$$

is unity only for $\alpha = 0$. It is 0.92 for $\alpha = 0.5$ and drops to 0.73 for $\alpha = 0.8$. (At the boundary, $|\alpha| = 1$ the Cramer–Rao relation cannot be applied.)

Note that the p.d.f. of our example is a special case of the usual expansion of an angular distribution into Legendre polynomials $P_l(\cos \beta)$:

$$f(x|\boldsymbol{\theta}) = (1 + \sum_{l=1}^L \theta_l P_l(x))/2 .$$

From the orthogonality of the P_l with the usual normalization

$$\int_{-1}^1 P_l(x) P_m(x) dx = \frac{2}{2l+1} \delta_{l,m}$$

it is easy to see that $\theta_l = (2l+1)\langle P_l \rangle$. In the case $l = 1$, $P_1 = x$, this is the first moment of the distribution and we reproduce $\mu_1 = \alpha/3$.

The moments method can also be used to provide start values for approximations of likelihood estimators which we discussed in Sect. 6.8.

³These limits stem from the obvious requirement that the p.d.f. has to be positive.

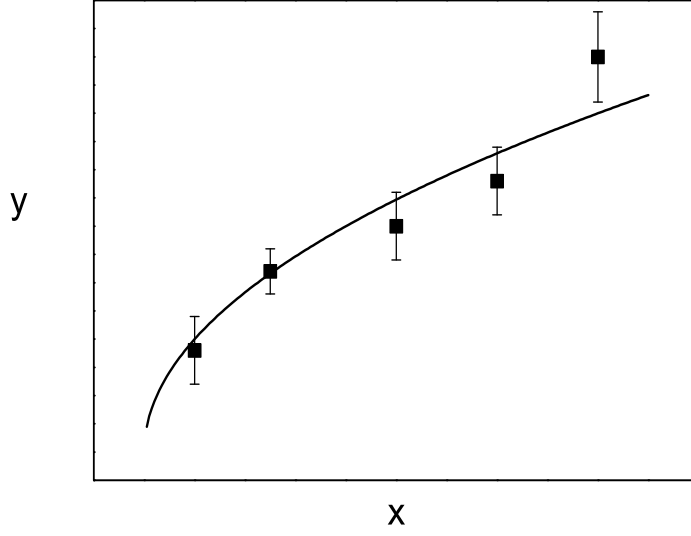


Fig. 7.3. Fit of a curve to measurements.

7.2.2 The Least Square Method

A frequently occurring problem is that measured points and error margins are given through which we want to fit a curve as shown in Fig. 7.3. The curve $y = t(x, \lambda)$ be measured at N positions x_i , yielding the values $y_i(x_i)$. The standard solution of this so-called regression problem is provided by the least square method which fixes parameters of a given function by minimizing the sum of the normalized square deviations of the function from measured points.

Given N measured points $x_i, y_i \pm \delta_i$, and a function $t(x, \theta)$, known up to some free parameters θ , the latter are determined such that

$$\chi^2 = \sum_{i=1}^N \frac{(y_i - t(x_i, \theta))^2}{\delta_i^2} \quad (7.6)$$

takes its minimal value.

The least square method goes back to Gauss. Historically it has successfully been applied to astronomical problems and is still the best method we have to adjust parameters of a curve to measured points if only the variance of the error distribution is known. It is closely related to the likelihood method when the errors are normally distributed. Then we can write the p.d.f. of the measurements in the following way:

$$f(y_1, \dots, y_N | \theta) \propto \exp \left[- \sum_{i,j=1}^N \frac{(y_i - t_i(x_i, \theta))^2}{2\delta_i^2} \right],$$

and the log-likelihood is

$$\ln L(\theta | \mathbf{y}) = -\frac{1}{2} \sum_{i,j=1}^N \frac{(y_i - t_i(x_i, \theta))^2}{\delta_i^2},$$

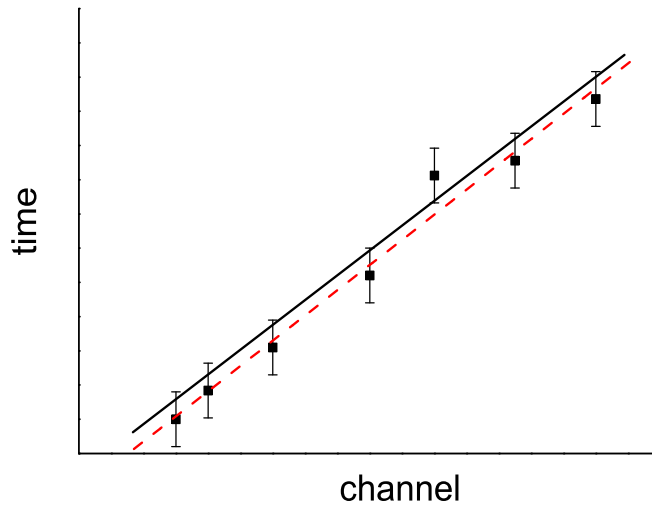


Fig. 7.4. χ^2 -Fit (dashed) of straight line through digital measurements.

$$= -\frac{1}{2}\chi^2. \quad (7.7)$$

Thus minimizing χ^2 is equivalent to maximizing the likelihood if the errors are normally distributed, a condition which frequently is at least approximately satisfied. From (7.7) we conclude that the standard deviation errors of the parameters in a least square fit correspond to one unit, $\Delta\chi^2 = 1$, twice the value $1/2$ of the maximum likelihood method. In Sect. 3.6.7 we have seen that χ^2 follows a χ^2 distribution of $f = N - Z$ (Z is the number of free parameters) degrees of freedom, provided the normality of the errors is satisfied. Thus we expect χ^2 to be of the order of f , large values indicate possible problems with the data or their description. We will investigate this in chapter 10.

That the least square method can lead to false results if the condition of Gaussian uncertainties is not fulfilled, is illustrated in the following example.

Example 113. Counter example to the least square method: gauging a digital clock

A digital clock has to be gauged. Fig. 7.4 shows the time channel as a function of the true time and a least square fit by a straight line. The error bars in the figure are not error bars in the usual sense but indicate the channel width. The fit fails to meet the allowed range of the fifth point and therefore is not compatible with the data. All straight lines which meet all “error bars” have the same likelihood. One correct solution is indicated in the figure.

We can easily generalize the expression (7.6) to the case of correlated errors. Then we have

$$\chi^2 = \sum_{i,j=1}^N (y_i - t_i)V_{ij}(y_j - t_j)$$

where V , the weight matrix, is the inverse of the covariance matrix. The quantity χ^2 is up to a factor *two* equal to the negative log-likelihood of a multivariate normal

distribution,

$$f(y_1, \dots, y_N | \boldsymbol{\theta}) \propto \exp \left[-\frac{1}{2} \sum_{i,j=1}^N (y_i - t_i) V_{ij} (y_j - t_j) \right],$$

see Sect. 6.5.6. Maximizing the likelihood is again equivalent to minimizing χ^2 if the errors are normally distributed.

The sum χ^2 is not invariant against a non-linear variable transformation $y'(y)$. The least square method is also used when the error distribution is unknown. In this situation we do not dispose of a better method.

Example 114. Least square method: fit of a straight line

We fit the parameters a, b of the straight line

$$y(x) = ax + b \quad (7.8)$$

to a sample of points (x_i, y_i) with uncertainties δ_i of the ordinates. We minimize χ^2 :

$$\begin{aligned} \chi^2 &= \sum_i \frac{(y_i - ax_i - b)^2}{\delta_i^2}, \\ \frac{\partial \chi^2}{\partial a} &= \sum_i \frac{(-y_i + ax_i + b)2x_i}{\delta_i^2}, \\ \frac{\partial \chi^2}{\partial b} &= \sum_i \frac{(-y_i + ax_i + b)2}{\delta_i^2}. \end{aligned}$$

We set the derivatives to zero and introduce the following abbreviations. (In parentheses we put the expressions for the special case where all uncertainties are equal, $\delta_i = \delta$):

$$\begin{aligned} \bar{x} &= \sum_i \frac{x_i}{\delta_i^2} / \sum_i \frac{1}{\delta_i^2} \quad \left(\sum_i x_i / N \right), \\ \bar{y} &= \sum_i \frac{y_i}{\delta_i^2} / \sum_i \frac{1}{\delta_i^2} \quad \left(\sum_i y_i / N \right), \\ \overline{x^2} &= \sum_i \frac{x_i^2}{\delta_i^2} / \sum_i \frac{1}{\delta_i^2} \quad \left(\sum_i x_i^2 / N \right), \\ \overline{xy} &= \sum_i \frac{x_i y_i}{\delta_i^2} / \sum_i \frac{1}{\delta_i^2} \quad \left(\sum_i x_i y_i / N \right). \end{aligned}$$

We obtain

$$\begin{aligned} \hat{b} &= \bar{y} - \hat{a} \bar{x}, \\ \overline{xy} - \hat{a} \overline{x^2} - \hat{b} \bar{x} &= 0, \end{aligned}$$

and

$$\begin{aligned} \hat{a} &= \frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2}, \\ \hat{b} &= \frac{\overline{x^2} \bar{y} - \bar{x} \overline{xy}}{\overline{x^2} - \bar{x}^2}. \end{aligned}$$

The problem is simplified when we put the origin of the abscissa at the center of gravity $\bar{x}$:

$$x' = x - \bar{x} ,$$

$$\hat{a}' = \frac{\overline{x'y}}{\overline{x'^2}} ,$$

$$\hat{b}' = \bar{y} .$$

Now the equation of the straight line reads

$$y = \hat{a}'(x - \bar{x}) + \hat{b}' . \quad (7.9)$$

We gain an additional advantage, the errors of the estimated parameters are no longer correlated.

$$\delta^2(\hat{a}') = 1 / \sum_i \frac{x_i^2}{\delta_i^2} ,$$

$$\delta^2(\hat{b}') = 1 / \sum_i \frac{1}{\delta_i^2} .$$

We recommend to use always the form (7.9) instead of (7.8).

7.2.3 Linear Regression

If the prediction depends only linearly on the parameters, we can compute the parameters which minimize χ^2 analytically. We put

$$\mathbf{y}_t(\boldsymbol{\theta}) = \mathbf{a} + \mathbf{T}\boldsymbol{\theta} . \quad (7.10)$$

Here $\boldsymbol{\theta}$ is the P -dimensional parameter vector, $\mathbf{a}$ is a given N -dimensional vector, $\mathbf{y}_t$ is the N -dimensional vector of predictions. $\mathbf{T}$, also called the design matrix, is a rectangular matrix of given elements with P columns and N rows.

The straight line fit discussed in Example 114 is a special case of (7.10) with $y_t = \theta_1 x + \theta_2$, $\mathbf{a} = 0$, and

$$\mathbf{T} = \begin{pmatrix} x_1 & \cdots & x_N \\ 1 & \cdots & 1 \end{pmatrix}^T .$$

We have to find the minimum of

$$\chi^2 = (\mathbf{y} - \mathbf{a} - \mathbf{T}\boldsymbol{\theta})^T \mathbf{V} (\mathbf{y} - \mathbf{a} - \mathbf{T}\boldsymbol{\theta})$$

where, as usual, $\mathbf{V}$ is the weight matrix, the inverse of the covariance matrix: $\mathbf{V} = \mathbf{C}^{-1}$. In our case it is a diagonal $N \times N$ matrix with elements $1/\delta_i^2$. To simplify the notation we transform the observations

$$\mathbf{y}' = \mathbf{y} - \mathbf{a} ,$$

derive

$$\chi^2 = (\mathbf{y}' - \mathbf{T}\boldsymbol{\theta})^T \mathbf{V} (\mathbf{y}' - \mathbf{T}\boldsymbol{\theta})$$

with respect to the parameters $\boldsymbol{\theta}$ and set the derivatives equal to zero:

$$\frac{1}{2} \frac{\partial \chi^2}{\partial \boldsymbol{\theta}} \Big|_{\hat{\boldsymbol{\theta}}} = \mathbf{0} = -\mathbf{T}^T \mathbf{V} (\mathbf{y}' - \mathbf{T}\hat{\boldsymbol{\theta}}) . \quad (7.11)$$

From these so-called normal equations we get the estimate for the P parameters $\hat{\theta}$:

$$\hat{\theta} = (\mathbf{T}^T \mathbf{V} \mathbf{T})^{-1} \mathbf{T}^T \mathbf{V} \mathbf{y}' . \quad (7.12)$$

Note that $\mathbf{T}^T \mathbf{V} \mathbf{T}$ is a symmetric $P \times P$ matrix with unique inverse which turns out to be the error (i.e. covariance) matrix of θ : With the usual propagation of errors we obtain (see Sect. 4.3):

$$\begin{aligned} \mathbf{E}_{\theta} &= \mathbf{D} \mathbf{C} \mathbf{D}^T , \\ &= (\mathbf{T}^T \mathbf{V} \mathbf{T})^{-1} , \end{aligned}$$

substituting for $\mathbf{D}$ the matrix on the right hand side of (7.12), after some simplifications.

Linear regression provides an optimal solution only for normally distributed errors. If the distribution of errors is not known up to their variance⁴, it is according to the Gauss–Markov theorem optimal in the following restricted sense: If the error distribution has mean value zero, the estimator is unbiased and among all other unbiased estimators which are linear in the observations its estimator has minimal variance. In fact, the result then coincides with that of a likelihood fit if normally distributed errors are assumed.

Linear problems are rare. When the prediction is a non-linear function of the parameters, the problem can be linearized by a Taylor expansion as a first rough approximation. By iteration the precision can be improved.

The importance of non-linear parameter inference by iterative linear regression has decreased considerably. The minimum searching routines which we find in all computer libraries are more efficient and easier to apply. Some basic minimum searching approaches are presented in Appendix 13.9.

7.3 Comparison of Estimation Methods

The following table contains an evaluation of the virtues and properties of the estimation approaches which we have been discussing.

Whenever possible, the likelihood method should be applied. It requires a sample of observations and a p.d.f. in analytic or well defined numerical form and is very sensitive to wrongly assigned observations in the sample. When the theoretical description of the data is given in form of a simulated histogram, the Poisson likelihood adjustment of the simulation to the bin content should be chosen. When we have to fit a function to measured data points, we use the least square method. If computing time is a limitation like in some on-line applications, the moments method lends itself. In many situations all three methods are equivalent.

All methods are sensitive to spurious background. Especially robust methods have been invented to solve this problem. An introduction and references are given in Appendix 13.15. For completeness we present in Appendix 13.3.1 some frequentist criteria of point and interval estimation which are relevant when parameters of many objects of the same type, for instance particle tracks, are measured. In the Appendix 13.6 we discuss the virtues of different point and interval inference approaches. Algorithms for minimum search are sketched in Appendix 13.9.

⁴Usually, when the variance is known, we have also an idea about the the shape of the distribution.

Table 7.1. Virtues and caveats of different methods of parameter estimation.

	moments	χ^2	max. likelihood
simplicity	++	+	–
precision	–	+	++
individual observations	+	–	+
measured points	–	+	–
histograms	+	+	+
upper and lower limits	–	–	+
external constraints	–	+	+
background included	+	+	–
error assignment	from error propagation	$\chi_{\min}^2 + 1$	$\ln L_{\max} - 0.5$
requirement	full p.d.f.	only variance	full p.d.f.

Interval Estimation

8.1 Introduction

In Chap. 4 we had presented a short introduction into error calculus. It was based on probability theory. In principle, error estimation is an essential part of statistics and of similar importance as parameter estimation. Measurements result from point estimation of one or several parameters, measurement errors from interval¹ estimation. These two parts form an ensemble and have to be defined in an consistent way.

As we have already mentioned, the notation *measurement error* used by scientists is somewhat misleading, more precise is the term *measurement uncertainty*. In the field of statistics the common term is *confidence intervals*, an expression which often is restricted to the specific frequentist intervals as introduced by Neyman which we sketch in the Appendix.

It is in no way obvious how we ought define error or confidence intervals and this is why statisticians have very different opinions on this subject. There are various conventions in different fields of physics, and particle physicists have not yet adopted a common solution.

Let us start with a wish list which summarizes the properties in the single parameter case which we would like to realize. The extrapolation to several parameters is straight forward.

1. Error intervals should contain the wanted true parameter with a fixed probability.
2. For a given probability, the interval should be as short as possible.
3. The error interval should represent the mean square spread of measurements around the true parameter value. In allusion to the corresponding probability term we talk about standard deviation errors.
4. The definition has to be consistent, i.e. observations containing identical information about the parameters should lead to identical intervals. More precise measurements should have shorter intervals than less precise ones. The error interval has to contain the point estimate.
5. Error intervals should be invariant under transformation of the estimated parameter.

¹The term *interval* is not restricted to a single dimension. In n dimensions it describes a n -dimensional volume.

6. The computation of the intervals should be free from subjective, e.g. more or less arbitrary model depending assumptions.
7. A consistent method for the combination of measurements and for error propagation has to exist.
8. The approach has to be simple and transparent.

Unfortunately it is absolutely impossible to fulfil simultaneously all these conditions which partially contradict each other. We will have to set priorities and sometimes we will have to use ad hoc solutions which are justified only from experience and common sense. Under all circumstances, we will satisfy point 4, i.e. consistency. As far as possible, we will follow the likelihood principle and derive the interval limits solely from the likelihood function.

It turns out that not always the same procedure is optimum for the interval estimation. For instance, if we measure the size or the weight of an object, *precision* is the dominant requirement, i.e. properties denoting the reliability or reproducibility of the data. Here, a quantity like the variance corresponding to the mean quadratic deviation is appropriate to describe the *error* or *uncertainty intervals*. Contrary, limits, for instance of the mass of a hypothetical particle like the Higgs particle, will serve to verify theoretical predictions. Here the dominant aspect is *probability* and we talk about *confidence* or *credibility intervals*². Confidence intervals are usually defined such that they contain a parameter with high probability, e.g. 90% or 95% while error intervals comprise one standard deviation or something equivalent. The exact calculation of the standard deviation as well as that of the probability that a parameter is contained inside an interval require the knowledge of its p.d.f. which depends not only on the likelihood function but in addition on the prior density which in most cases is unknown. To introduce a subjective prior, however, is something which we want to avoid.

First we treat situations where the aspect *precision* dominates. There, as far as possible, we will base our considerations on the likelihood function only. Then we will discuss cases where the probability aspect is important. These will deal mainly with limits on hypothetical quantities, like masses of SUSY particles. There we will be obliged to include prior densities.

8.2 Error Intervals

The purpose of error intervals is to document the precision of a measurement. They are indispensable when we combine measurements. The combination of measurements permits us to improve continuously the precision of a parameter estimate with increasing number of measurements.

If the prior density is known with sufficient precision, we determine the probability density of the parameter(s) and subsequently the moments. But this condition is so rarely fulfilled that we need not discuss it. Normally, we are left with the likelihood function only.

In what follows we will always assume that the likelihood function is of simple shape, differentiable, with only one maximum and decreasing continuously to all

²The term *credibility interval* is used for Bayesian intervals.

sides. This condition is realized in most cases. In the remaining ones where it is of complicated shape we have to renounce the simple parametrization by point and interval estimates and present the full likelihood function.

The width of the likelihood function indicates how precise a measurement is. The standard error limits, as introduced in Sect. 6.5.1 – decrease by a factor $e^{1/2}$ from the maximum – rely on the likelihood ratio. These limits have the positive property of being independent of the parameter metric: This means in the one-dimensional case that for a parameter $\lambda(\theta)$ which is a monotonic function of θ that the limits $\lambda_1, \lambda_2, \theta_1, \theta_2$ fulfill the relations $\lambda_1 = \lambda(\theta_1)$ and $\lambda_2 = \lambda(\theta_2)$. It does not matter whether we write the likelihood as a function of θ or of λ .

In large experiments usually there are many different effects which influence the final result and consequently also many different independent sources of uncertainty, most of which are of the systematic type. Systematic errors (see Sect. 4.2.3) such as calibration uncertainties can only be treated in the Bayesian formalism. We have to estimate their p.d.f. or at least a mean value and a standard deviation.

8.2.1 Parabolic Approximation

The error assignment is problematic only for small samples. As is shown in Appendix 13.3, the likelihood function approaches a Gaussian with increasing size of the sample. At the same time its width decreases with increasing sample size, and we can neglect possible variations of the prior density in the region where the likelihood is significant. Under this condition we obtain a normally distributed p.d.f. for the parameter(s) with a standard deviation error interval given by the $e^{1/2}$ decrease from the maximum. It includes the parameter with probability 68.3 % (see Sect. 4.5) The log-likelihood then is parabolic and the error interval corresponds to the region within which it decreases from its maximum by a value of $1/2$ as we had fixed it already previously. This situation is certainly realized for the large majority of all measurements which are published in the Particle Data Book [24].

In the parabolic approximation the MLE and the expectation value coincide, as well as the likelihood ratio error squared and the variance. Thus we can also derive the standard deviation $\delta\theta$ from the curvature of the likelihood function at its maximum. For a single parameter we can approximate the likelihood function by the expression

$$-\ln L_p = \frac{1}{2}V(\theta - \hat{\theta})^2 + \text{const.} . \quad (8.1)$$

Consequently, a change of $\ln L_p$ by $1/2$ corresponds to the second derivative of L at $\hat{\theta}$:

$$(\delta\theta)^2 = V^{-1} = - \left(\frac{d^2 \ln L}{d\theta^2} \Big|_{\hat{\theta}} \right)^{-1} .$$

For several parameters the parabolic approximation can be expressed by

$$-\ln L = \frac{1}{2} \sum_{i,j} (\theta_i - \hat{\theta}_i) V_{ij} (\theta_j - \hat{\theta}_j) + \text{const.} .$$

We obtain the symmetric weight matrix³ V from the derivatives

³It is also called Fisher-information.

$$V_{ij} = -\frac{\partial^2 \ln L}{\partial \theta_i \partial \theta_j} \Big|_{\hat{\theta}}$$

and the covariance or error matrix from its inverse $C = V^{-1}$.

If we are interested only in part of the parameters, we can eliminate the remaining nuisance parameters simply forgetting about the part of the matrix which contains the corresponding elements. This is a consequence of the considerations from Sect. 6.9.

In most cases the likelihood function is not known analytically. Usually, we have a computer program which delivers the likelihood function for arbitrary values of the parameters. Once we have determined the maximum, we are able to estimate the second derivative and the weight matrix V computing the likelihood function at parameter points close to the MLE. To ensure that the parabolic approximation is valid, we should increase the distance of the points and check whether the result remains consistent.

In the literature we find frequently statements like “The measurement excludes the theoretical prediction by four standard deviations.” These kind of statements have to be interpreted with caution. Their validity relies on the assumption that the log-likelihood is parabolic over a very wide parameter range. Neglecting tails can lead to completely wrong conclusions. We have also to remember that for a given number of standard deviations the probability decreases with the number of dimensions (see Tab. 4.2 in Sect. 4.5).

In the following section we address more problematic situations which usually occur with small data samples where the asymptotic solutions are not appropriate. Fortunately, they are rather the exception. We keep in mind that a relatively rough estimate of the error often is sufficient such that approximate methods in most cases are justified.

8.2.2 General Situation

As above, we again use the likelihood ratio to define the error limits which now usually are asymmetric. In the one-dimensional case the two errors δ_- and δ_+ satisfy

$$\ln L(\hat{\theta}) - \ln L(\hat{\theta} - \delta_-) = \ln L(\hat{\theta}) - \ln L(\hat{\theta} + \delta_+) = 1/2. \quad (8.2)$$

If the log-likelihood function deviates considerably from a parabola it makes sense to supplement the one standard deviation limits $\Delta \ln L = -1/2$ with the two standard deviation limits $\Delta \ln L = -2$ to provide a better documentation of the shape of the likelihood function. This complication can be avoided if we can obtain an approximately parabolic likelihood function by an appropriate parameter transformation. In some situations it is useful to document in addition to the mode of the likelihood function and the asymmetric errors, if available, also the mean and the standard deviation which are relevant, for instance, in some cases of error propagation which we will discuss below.

Example 115. Error of a lifetime measurement

To determine the mean lifetime τ of a particle from a sample of observed decay times, we use the likelihood function

$$L_\tau = \prod_{i=1}^N \frac{1}{\tau} e^{-t_i/\tau} = \frac{1}{\tau^N} e^{-N\bar{t}/\tau}. \quad (8.3)$$

The corresponding likelihood for the decay rate is

$$L_\lambda = \prod_{i=1}^N \lambda e^{-\lambda t_i} = \lambda^N e^{-N\bar{t}\lambda}.$$

The values of the functions are equal at equivalent values of the two parameters τ and λ , i.e. for $\lambda = 1/\tau$:

$$L_\lambda(\lambda) = L_\tau(\tau).$$

Fig. 8.1 shows the two log-likelihoods for a small sample of ten events with mean value $\bar{t} = 0.5$. The lower curves for the parameter τ are strongly asymmetric. This is also visible in the limits for changes of the log-likelihood by 0.5 or 2 units which are indicated on the right hand cut-outs. The likelihood with the decay rate as parameter (upper figures) is much more symmetric than that of the mean life. This means that the decay rate is the more appropriate parameter to document the shape of the likelihood function, to average different measurement and to perform error propagation, see below. On the other hand, we can of course transform the maximum likelihood estimates and errors of the two parameters into each other without knowing the likelihood function itself.

Generally, it does not matter whether we use one or the other parameter to present the result but for further applications it is always simpler and more precise to work with approximately symmetric limits. For this reason usually $1/p$ (p is the absolute value of the momentum) instead of p is used as parameter when charged particle trajectories are fitted to the measured hits in a magnetic spectrometer.

In the general case we satisfy the conditions 4 to 7 of our wish list but the first three are only approximately valid. We neither can associate an exact probability content to the intervals nor do the limits correspond to moments of a p.d.f..

8.3 Error Propagation

In many situations we have to evaluate a quantity which depends on one or several measurements with individual uncertainties. We thus have a problem of point estimation and of interval estimation. We look for the parameter which is best supported by the different measurements and for its uncertainty. Ideally, we are able to construct the likelihood function. In most cases this is not necessary and approximate procedures are adequate.

8.3.1 Averaging Measurements

In Chap. 4 we have shown that the mean of measurements with Gaussian errors δ_i which are independent of the measurement, is given by the weighted sum of the individual measurements (4.7) with weights proportional to the inverse errors squared

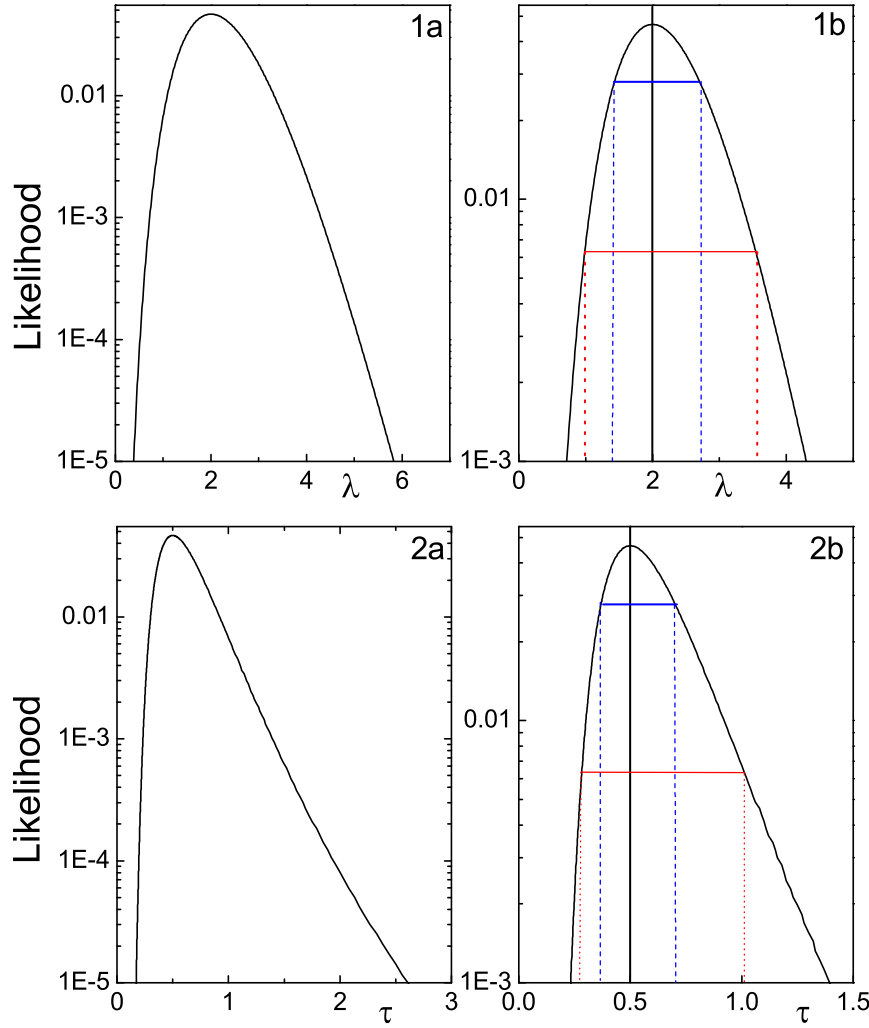


Fig. 8.1. Likelihood functions for the parameters decay rate (top) and lifetime (below). The standard deviation limits are shown in the cut-outs on the right hand side.

$1/\delta_i^2$. In case the errors are correlated with the measurements which occurs frequently with small event numbers, this procedure introduces a bias (see Example 56 in Chap. 4) From (6.6) we conclude that the exact method is to add the log-likelihoods of the individual measurements. Adding the log-likelihoods is equivalent to combining the raw data as if they were obtained in a single experiment. There is no loss of information and the method is not restricted to specific error conditions.

Example 116. Averaging lifetime measurements

N experiments quote lifetimes $\hat{\tau}_i \pm \delta_i$ of the same unstable particle. The estimates and their errors are computed from the individual measurements t_{ij} of the i -th experiment according to $\hat{\tau}_i = \sum_{j=1}^{n_i} t_{ij}/n_i$, respectively $\delta_i = \hat{\tau}_i/\sqrt{n_i}$ where n_i is the number of observed decays. We can reconstruct the individual log-likelihood functions and their sum $\ln L$, with $n, n = \sum_{i=1}^N n_i$, the overall event number:

$$\begin{aligned}\ln L(\tau) &= \sum_{i=1}^N -n_i(\ln \tau + \hat{\tau}_i/\tau) \\ &= -n \ln \tau - \sum \frac{n_i \hat{\tau}_i}{\tau}\end{aligned}$$

with the maximum at

$$\hat{\tau} = \frac{\sum n_i \hat{\tau}_i}{n}$$

and its error

$$\delta = \frac{\hat{\tau}}{\sqrt{n}}.$$

The individual measurements are weighted by their event numbers, instead of weights proportional to $1/\delta_i^2$. As the errors are correlated with the measurements, the standard weighted mean (4.7) with weights proportional to $1/\delta_i^2$ would be biased. In our specific example the correlation of the errors and the parameter values is known and we could use weights proportional to $(\tau_i/\delta_i)^2$.

Example 117. Averaging ratios of Poisson distributed numbers

In absorption measurements and many other situations we are interested in a parameter which is the ratio of two numbers which follow the Poisson distribution. Averaging naively these ratios $\hat{\theta}_i = m_i/n_i$ using the weighted mean (4.7) can lead to strongly biased results. Instead we add the log-likelihood functions which we have derived in Sect. 6.9.3

$$\begin{aligned}\ln L &= \sum [n_i \ln(1 + 1/\theta) - m_i \ln(1 + \theta)] \\ &= n \ln(1 + 1/\theta) - m \ln(1 + \theta)\end{aligned}$$

with $m = \sum m_i$ and $n = \sum n_i$. The MLE is $\hat{\theta} = m/n$ and the error limits have to be computed numerically in the usual way or for not too small n, m by linear error propagation, $\delta_\theta^2/\theta^2 = 1/n + 1/m$.

In the standard situation where we do not know the full likelihood function but only the MLE and the error limits we have to be content with an approximate procedure. If the likelihood functions which have been used to extract the error limits are parabolic, then the standard weighted mean (4.7) is exactly equal to the result which we obtain when we add the log-likelihood functions and extract then the estimate and the error.

Proof: A sum of terms of the form (8.1) can be written in the following way:

$$\frac{1}{2} \sum V_i(\theta - \theta_i)^2 = \frac{1}{2} \tilde{V}(\theta - \tilde{\theta})^2 + \text{const.}$$

Since the right hand side is the most general form of a polynomial of second order, a comparison of the coefficients of θ^2 and θ yields

$$\begin{aligned}\tilde{V} &= \sum V_i, \\ \tilde{V}\tilde{\theta} &= \sum V_i\theta_i,\end{aligned}$$

that is just the weighted mean including its error. Consequently, we should aim at approximately parabolic log-likelihood functions when we present experimental results. Sometimes this is possible by a suitable choice of the parameter. For example, we are free to quote either the estimate of the mass or of the mass squared.

8.3.2 Approximating the Likelihood Function

We also need a method to average statistical data with asymmetric errors, a method which works without knowing the exact shape of the likelihood function. To this end we try to reconstruct the log-likelihood functions approximately, add them, and extract the parameter which maximizes the sum and the likelihood ratio errors. The approximation has to satisfy the constraints that the derivative at the MLE is zero and the error relation (8.2).

The simplest parametrization uses two different parabola branches

$$-\ln L(\theta) = \frac{1}{2}(\theta - \hat{\theta})^2 / \delta_{\pm}^2 \quad (8.4)$$

with

$$\delta_{\pm} = \frac{1}{2}\delta_+[1 + \text{sgn}(\theta - \hat{\theta})] + \frac{1}{2}\delta_-[1 - \text{sgn}(\theta - \hat{\theta})],$$

i.e. the parabolas meet at the maximum and obey (8.2). Adding functions of this type produces again a piecewise parabolic function which fixes the mean value and its asymmetric errors. The solution for both the mean value and the limits is unique.

Parameterizations [43] varying the width σ of a parabola linearly or quadratically with the parameter are usually superior to the simple two branch approximation. We set

$$-\ln L(\theta) = \frac{1}{2} \left[(\theta - \hat{\theta}) / \sigma(\theta) \right]^2$$

with

$$\sigma(\theta) = \frac{2\delta_+\delta_-}{\delta_+ + \delta_-} + \frac{\delta_+ - \delta_-}{\delta_+ + \delta_-}(\theta - \hat{\theta}) \quad (8.5)$$

or

$$(\sigma(\theta))^2 = \delta_+\delta_- + (\delta_+ - \delta_-)(\theta - \hat{\theta}), \quad (8.6)$$

respectively. The log-likelihood function has poles at locations of θ where the width becomes zero, $\sigma(\theta) = 0$. Thus our approximations are justified only in the range of θ which excludes the corresponding parameter values.

In Fig. 8.2 we present four typical examples of asymmetric likelihood functions. The log-likelihood function of the mean life of four exponentially distributed times is shown in 8.2 a. Fig. 8.2 b is the corresponding log-likelihood function of the decay time⁴. Figs. 8.2 c, d have been derived by a parameter transformation from normally distributed observations where in one case the new parameter is one over the mean and the square root of the mean⁵ in the other case. A method which is optimum for all cases does not exist. All three approximations fit very well inside the one standard deviation limits. Outside, the two parameterizations (8.5) and (8.6) are superior to the two-parabola approximation.

We propose to use one of two parameterizations (8.5, 8.6) but to be careful if $\sigma(\theta)$ becomes small.

⁴The likelihood function of a Poisson mean has the same shape.

⁵An example of such a situation is a fit of a particle mass from normally distributed mass squared observations.

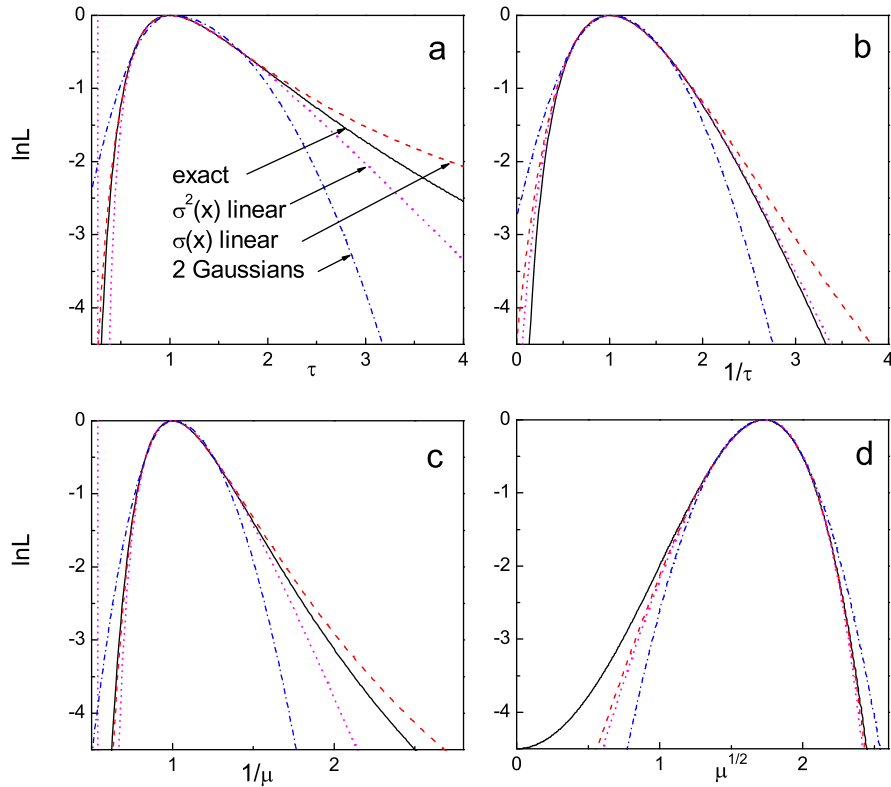


Fig. 8.2. Asymmetric likelihood functions and parametrizations.

8.3.3 Incompatible Measurements

Before we rely on a mean value computed from the results of different experiments we should make sure that the various input data are statistically compatible. What we mean with *compatible* is not obvious at this point. It will become clearer in Chap. 10, where we discuss significance tests which lead to the following plausible procedure that has proven to be quite useful in particle physics [24].

We compute the weighted mean value $\tilde{\theta}$ of the N results and form the sum of the quadratic deviations of the individual measurements from their average, normalized to their expected errors squared:

$$\chi^2 = \sum (\theta_i - \tilde{\theta})^2 / \delta_i^2.$$

The expectation value of this quantity is $N - 1$ if the deviations are normally distributed with variances δ_i^2 . If χ^2 is sizably (e.g. by 50%) higher than $N - 1$, then we can suspect that at least one of the experiments has published a wrong value, or what is more likely, has underestimated the error, for instance when systematic errors have not been detected. Under the premise that none of the experiments can be discarded a priori, we scale-up all declared errors by a common scaling factor

$S = \chi^2/(N - 1)$ and publish this factor together with mean value and the scaled error. Large scaling factors indicate problems in one or several experiments.

A similar procedure is applied if the errors are asymmetric even though the condition of normality then obviously is violated. We form

$$\chi^2 = \sum (\theta_i - \tilde{\theta})^2 / \delta_{i\pm}^2 ,$$

where δ_{i+} and δ_{i-} , respectively, are valid for $\theta_i < \tilde{\theta}$ and $\theta_i > \tilde{\theta}$.

8.3.4 Error Propagation for a Scalar Function of a Single Parameter

If we have to propagate the MLE and its error limits of a parameter θ to another parameter $\theta' = \theta'(\theta)$, we should apply the direct functional relation which is equivalent to a transformation of the likelihood function:

$$\begin{aligned} \hat{\theta}' &= \theta'(\hat{\theta}) , \\ \hat{\theta}' + \delta'_+ &= \theta'(\hat{\theta} + \delta_+) , \\ \hat{\theta}' - \delta'_- &= \theta'(\hat{\theta} - \delta_-) . \end{aligned}$$

Here we have assumed that $\theta'(\theta)$ is monotonically increasing. If it is decreasing, the arguments of θ' have to be interchanged.

The errors of the output quantity are asymmetric either because the input errors are asymmetric or because the functional dependence is non-linear. For instance an angular measurement $\alpha = 87^\circ \pm 1^\circ$ would transform into $\sin \alpha = 0.9986^{+0.0008}_{-0.0010}$.

8.3.5 Error Propagation for a Function of Several Parameters

A difficult problem is the determination of the error of a scalar quantity $\theta(\boldsymbol{\mu})$ which depends on several measured input parameters $\boldsymbol{\mu}$ with asymmetric errors. We have to eliminate nuisance parameters.

If the complete likelihood function $\ln L(\boldsymbol{\mu})$ of the input parameters is available, we derive the error limits from the profile likelihood function of θ as proposed in Sect. 6.5.1.

The MLE of θ is simply $\hat{\theta} = \theta(\hat{\boldsymbol{\mu}})$. The profile likelihood of θ has to fulfil the relation $\Delta \ln L(\theta) = \ln L(\hat{\theta}) - \ln L(\theta) = \ln L(\hat{\boldsymbol{\mu}}) - \ln L(\boldsymbol{\mu})$. To find the two values of θ for the given $\Delta \ln L$, we have to find the maximum and the minimum of θ fulfilling the constraint. The one standard deviation limits are the two extreme values of θ located on the $\Delta \ln L(\theta) = \ln L(\hat{\boldsymbol{\mu}}) - \ln L(\boldsymbol{\mu}) = 1/2$ surface⁶.

There are various numerical methods to compute these limits. Constraint problems are usually solved with the help of Lagrange multipliers. A simpler method is the one which has been followed when we discussed constrained fits (see Sect. 6.6): With an extremum finding program, we minimize

$$\theta(\boldsymbol{\mu}) + c [\ln L(\hat{\boldsymbol{\mu}}) - \ln L(\boldsymbol{\mu}) - 1/2]^2$$

⁶Since we assumed likelihood functions with a single maximum, this is a closed surface, in two dimensions a line of constant height.

where c is a number which has to be large compared to the absolute change of θ within the $\Delta \ln L = 1/2$ region. We get μ_{low} and $\theta_{low} = \theta(\mu_{low})$ and maximizing

$$\theta(\mu) - c [\ln L(\hat{\mu}) - \ln L(\mu) - 1/2]^2$$

we get θ_{up} .

If the likelihood functions are not known, the only practical way is to resort to a Bayesian treatment, i.e. to make assumptions about the p.d.f.s of the input parameters. In many cases part of the input parameters have systematic uncertainties. Then, anyway, the p.d.f.s of those parameters have to be constructed. Once we have established the complete p.d.f. $f(\mu)$, we can also determine the distribution of θ . The analytic parameter transformation and reduction described in Chap. 3 will fail in most cases and we will adopt the simple Monte Carlo solution where we generate a sample of events distributed according to $f(\mu)$ and where $\theta(\mu)$ provides the θ distribution in form of a histogram and the uncertainty of this parameter. To remain consistent with our previously adopted definitions we would then interpret this p.d.f. of θ as a likelihood function and derive from it the MLE $\hat{\theta}$ and the likelihood ratio error limits.

We will not discuss the general scheme in more detail but add a few remarks related to special situations and discuss two simple examples.

Sum of Many Measurements

If the output parameter $\theta = \sum \xi_i$ is a sum of “many” input quantities ξ_i with variances σ_i^2 of similar size and their mean values and variances are known, then due to the central limit theorem we have

$$\begin{aligned}\hat{\theta} &= \langle \theta \rangle = \Sigma \langle \xi_i \rangle, \\ \delta_\theta^2 &= \sigma_\theta^2 \approx \Sigma \sigma_i^2\end{aligned}$$

independent of the shape of the distributions of the input parameters and the error of θ is normally distributed. This situation occurs in experiments where many systematic uncertainties of similar magnitude enter in a measurement.

Product of Many Measurements

If the output parameter $\theta = \prod \xi_i$ is a product of “many” positive input quantities ξ_i with relative uncertainties σ_i/ξ_i of similar size then due to the central limit theorem

$$\begin{aligned}\langle \ln \theta \rangle &= \Sigma \langle \ln \xi_i \rangle, \\ \sigma_{\ln \theta} &\approx \sqrt{\Sigma \sigma_{\ln \xi_i}^2}\end{aligned}$$

independent of the shape of the distributions of the input parameters and the error of $\ln \theta$ is normally distributed which means that θ follows a log-normal distribution (see Sect. 3.6.10). Such a situation may be realized if several multiplicative efficiencies

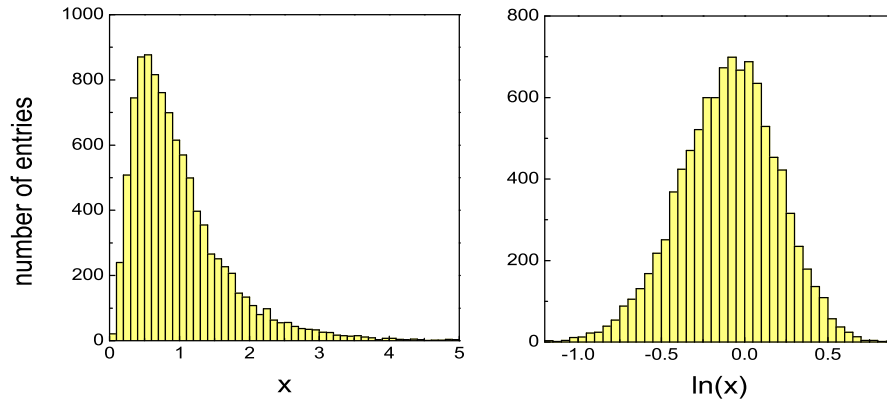


Fig. 8.3. Distribution of the product of 10 variates with mean 1 and standard deviation 0.2.

with similar uncertainties enter into a measurement. The distribution of θ is fully specified only once we know the quantities $\langle \ln \xi_i \rangle$ and $\sigma_{\ln \xi_i}$. The latter condition will usually not be fulfilled and $\langle \ln \xi_i \rangle, \sigma_{\ln \xi_i}$ have to be set by some educated guess. In most cases, however, the approximations $\langle \theta \rangle = \prod \langle \xi_i \rangle$ and $\delta_\theta^2 / \theta^2 = \sum \delta_i^2 / \xi_i^2$ may be adequate. These two quantities fix the log-normal distribution from which we can derive the maximum and the asymmetric errors. If the relative errors are sufficiently small, the log-normal distribution approaches a normal distribution and we can simply use the standard linear error propagation with symmetric errors. As always, it is useful to check approximations by a simulation.

Example 118. Distribution of a product of measurements

We simulate the distribution of $\theta = \prod \xi_i$, of 10 measured quantities with mean equal to 1 and standard deviation of 0.2, all normally distributed. The result is very different from a Gaussian and is well described by a log-normal distribution as is shown in Fig. 8.3. The mean is compatible with $\langle \theta \rangle = 1$ and the standard deviation is 0.69, slightly larger than the prediction from simple error propagation of 0.63. These results remain the same when we replace the Gaussian errors by uniform ones with the same standard deviation. Thus details of the distributions of the input parameters are not important.

Sum of Weighted Poisson Numbers

If $\theta = \sum w_i \eta_i$ is a sum of Poisson numbers η_i weighted with w_i then we can apply the simple linear error propagation rule:

$$\begin{aligned}\hat{\theta} &= \sum w_i \eta_i, \\ \delta_\theta^2 &= \sum w_i^2 \eta_i.\end{aligned}$$

The reason for this simple relation is founded on the fact that a sum of weighted Poisson numbers can be approximated by the Poisson distribution of the equivalent number of events (see Sect. 3.6.3). A condition for the validity of this approximation

is that the number of equivalent events is large enough to use symmetric errors. If this number is low we derive the limits from the Poisson distribution of the equivalent number of events which then will be asymmetric.

Example 119. Sum of weighted Poisson numbers

Particles are detected in three detectors with efficiencies $\varepsilon_1 = 0.7$, $\varepsilon_2 = 0.5$, $\varepsilon_3 = 0.9$. The observed event counts are $n_1 = 10$, $n_2 = 12$, $n_3 = 8$. A background contribution is estimated in a separate counting experiment as $b = 9$ with a reduction factor of $r = 2$. The estimate $\hat{n}$ for total number of particles which traverse the detectors is $\hat{n} = \sum n_i/\varepsilon_i - b/r = 43$. From linear error propagation we obtain the uncertainty $\delta_n = 9$. A more precise calculation based on the Poisson distribution of the equivalent number of events would yield asymmetric errors, $\hat{n} = 43^{+10}_{-8}$.

Averaging Correlated Measurements

The following example is a warning that naive linear error propagation may lead to false results.

Example 120. Average of correlated cross section measurements, Peelle's pertinent puzzle

The results of a cross section measurements is ξ_1 with uncertainties due to the event count, δ_{10} , and to the beam flux. The latter leads to an error $\delta_f \xi$ which is proportional to the cross section ξ . The two contributions are independent and thus the estimated error squared in the Gaussian approximation is $\delta_1^2 = \delta_{10}^2 + \delta_f^2 \xi_1^2$. A second measurement ξ_2 with different statistics but the same uncertainty on the flux has an uncertainty $\delta_2^2 = \delta_{20}^2 + \delta_f^2 \xi_2^2$. Combining the two measurements we have to take into account the correlation of the errors. In the literature [45] the following covariance matrix is discussed:

$$C = \begin{pmatrix} \delta_{10}^2 + \delta_f^2 \xi_1^2 & \delta_f^2 \xi_1 \xi_2 \\ \delta_f^2 \xi_1 \xi_2 & \delta_{20}^2 + \delta_f^2 \xi_2^2 \end{pmatrix}.$$

It can lead to the strange result that the least square estimate $\hat{\xi}$ of the two cross sections is located outside the range defined by the individual results [46], e.g. $\hat{\xi} < \xi_1, \xi_2$. This anomaly is known as *Peelle's Pertinent Puzzle* [44]. Its reason is that the normalization error is proportional to the true cross section and not to the observed one and thus has to be the same for the two measurements, i.e. in first approximation proportional to the estimate $\hat{\xi}$ of the true cross section. The correct covariance matrix is

$$C = \begin{pmatrix} \delta_{10}^2 + \delta_f^2 \hat{\xi}^2 & \delta_f^2 \hat{\xi}^2 \\ \delta_f^2 \hat{\xi}^2 & \delta_{20}^2 + \delta_f^2 \hat{\xi}^2 \end{pmatrix}. \quad (8.7)$$

Since the best estimate of ξ cannot depend on the common scaling error it is given by the weighted mean

$$\hat{\xi} = \frac{\delta_{10}^{-2} \xi_1 + \delta_{20}^{-2} \xi_2}{\delta_{10}^{-2} + \delta_{20}^{-2}}. \quad (8.8)$$

The error δ is obtained by the usual linear error propagation,

$$\delta^2 = \frac{1}{\delta_{10}^{-2} + \delta_{20}^{-2}} + \delta_f^2 \hat{\xi}^2. \quad (8.9)$$

Proof: The weighted mean for $\hat{\xi}$ is defined as the combination

$$\hat{\xi} = w_1 \xi_1 + w_2 \xi_2$$

which, under the condition $w_1 + w_2 = 1$, has minimal variance (see Sect. 4.3):

$$\text{var}(\hat{\xi}) = w_1^2 C_{11} + w_2^2 C_{22} + 2w_1 w_2 C_{12} = \min.$$

Using the correct C (8.7), this can be written as

$$\text{var}(\hat{\xi}) = w_1^2 \delta_{10}^2 + (1 - w_1)^2 \delta_{20}^2 + \delta_f^2 \hat{\xi}^2.$$

Setting the derivative with respect to w_1 to zero, we get the usual result

$$w_1 = \frac{\delta_{10}^{-2}}{\delta_{10}^{-2} + \delta_{20}^{-2}}, \quad w_2 = 1 - w_1,$$

$$\delta^2 = \min[\text{var}(\hat{\xi})] = \frac{\delta_{10}^{-2}}{(\delta_{10}^{-2} + \delta_{20}^{-2})^2} + \frac{\delta_{20}^{-2}}{(\delta_{10}^{-2} + \delta_{20}^{-2})^2} + \delta_f^2 \hat{\xi}^2,$$

proving the above relations (8.8), (8.9).

8.4 One-sided Confidence Limits

8.4.1 General Case

Frequently, we cannot achieve the precision which is necessary to resolve a small physical quantity. If we do not obtain a value which is significantly different from zero, we usually present an upper limit. A typical example is the measurement of the lifetime of a very short-lived particle which cannot be resolved by the measurement. The result of such a measurement is then quoted by a phrase like “The lifetime of the particle is smaller than ... with 90 % confidence.” Upper limits are often quoted for rates of rare reactions if no reaction has been observed or the observation is compatible with background. For masses of hypothetical particles postulated by theory but not observed with the limited energy of present accelerators, experiments provide lower limits.

In this situation we are interested in probabilities. Thus we have to introduce prior densities or to remain with likelihood ratio limits. The latter are not very popular. As a standard, we fix the prior to be constant in order to achieve a uniform procedure allowing to compare and to combine measurements from different experiments. This means that a priori all values of the parameter are considered as equally likely. As a consequence, the results of such a procedure depend on the choice of the variable. For instance lower limits of a mass u_m and a mass squared u_{m^2} , respectively, would not obey the relation $u_{m^2} = (u_m)^2$. Unfortunately we cannot avoid this unattractive property when we want to present probabilities. Knowing that a

uniform prior has been applied, the reader of a publication can interpret the limit as a sensible parametrization of the experimental result and draw his own conclusions. Of course, it is also useful to present the likelihood function which fully documents the result.

To obtain the p.d.f. of the parameter of interest, we just have to normalize the likelihood function⁷ to the allowed range of the parameter θ . The probability $P\{\theta < \theta_0\}$ computed from this density is the confidence level C for the upper limit θ_0 :

$$C(\theta_0) = \frac{\int_{-\infty}^{\theta_0} L(\theta) d\theta}{\int_{-\infty}^{\infty} L(\theta) d\theta}. \quad (8.10)$$

Lower limits are computed in an analogous way:

$$C_{low}(\theta_0) = \frac{\int_{\theta_0}^{\infty} L(\theta) d\theta}{\int_{-\infty}^{\infty} L(\theta) d\theta}. \quad (8.11)$$

Here the confidence level C is given and the relations (8.10), (8.11) have to be solved for θ_0 .

8.4.2 Upper Poisson Limits, Simple Case

When, in an experimental search for a certain reaction, we do not find the corresponding events, we quote an upper limit for its existence. Similarly in some cases where an experiment records one or two candidate events but where strong theoretical reasons speak against accepting those as real, it is common practice not to quote a rate but rather an upper limit. The result is then expressed in the following way: The rate for the reaction x is less than λ_0 with 90 % confidence.

The upper limit is again obtained as above by integration of the normalized likelihood function.

For k observed events, we want to determine an upper limit μ_0 with $C = 90\%$ confidence for the expectation value of the Poisson rate. The normalization integral over the parameter μ of the Poisson distribution $P(k|\mu) = e^{-\mu} \mu^k / k!$ is equal to one. Thus we obtain:

$$\begin{aligned} C &= \int_0^{\mu_0} P(k|\mu) d\mu \\ &= \frac{\int_0^{\mu_0} e^{-\mu} \mu^k d\mu}{k!}. \end{aligned} \quad (8.12)$$

The integral is solved by partial integration,

$$\begin{aligned} C &= 1 - \sum_{i=0}^k \frac{e^{-\mu_0} \mu_0^i}{i!} \\ &= 1 - \sum_{i=0}^k P(i|\mu_0). \end{aligned}$$

⁷In case the likelihood function cannot be normalized, we have to renounce to produce a p.d.f. and present only the likelihood function.

However, the sum over the Poisson probabilities cannot be solved analytically for μ_0 . It has to be solved numerically, or (8.12) is evaluated with the help of tables of the incomplete gamma function.

A special role plays the case $k = 0$, e.g. when no event has been observed. The integral simplifies to:

$$C = 1 - e^{-\mu_0} , \\ \mu_0 = -\ln(1 - C) .$$

For $C = 0.9$ this relation is fulfilled for $\mu_0 \approx 2.3$.

Remark that for Poisson limits of rates without background the frequentist statistics (see Appendix 13.5) and the Bayesian statistics with uniform prior give the same results. For the following more general situations, this does not hold anymore.

8.4.3 Poisson Limit for Data with Background

When we find in an experiment events which can be explained by a background reaction with expected mean number b , we have to modify (8.12) correspondingly. The expectation value of k is then $\mu + b$ and the confidence C is

$$C = \frac{\int_0^{\mu_0} P(k|\mu + b) d\mu}{\int_0^{\infty} P(k|\mu + b) d\mu} .$$

Again the integrals can be replaced by sums:

$$C = 1 - \frac{\sum_{i=0}^k P(i|\mu_0 + b)}{\sum_{i=0}^k P(i|b)} .$$

Example 121. Upper limit for a Poisson rate with background

Expected are two background events and observed are also two events. Thus the mean signal rate μ is certainly small. We obtain an upper limit μ_0 for the signal with 90% confidence by solving numerically the equation

$$0.9 = 1 - \frac{\sum_{i=0}^2 P(i|\mu_0 + 2)}{\sum_{i=0}^2 P(i|2)} .$$

We find $\mu_0 = 3.88$. The Bayesian probability that the mean rate μ is larger than 3.88 is 10 %. Fig. 8.4 shows the likelihood functions for the two cases $b = 2$ and $b = 0$ together with the limits. For comparison are also given the likelihood ratio limits which correspond to a decrease from the maximum by e^{-2} . (For a normal distribution this would be equivalent to two standard deviations).

We now investigate the more general case that both the acceptance ε and the background are not perfectly known, and that the p.d.f.s of the background and the acceptance f_b , f_ε are given. For a mean Poisson signal μ the probability to observe k events is

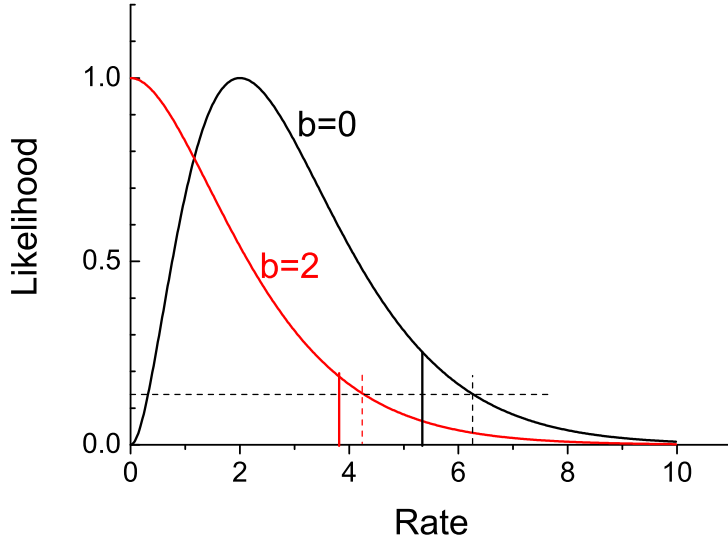


Fig. 8.4. Upper limits for poisson rates. The dashed lines are likelihood ratio limits (decrease by e^2).

$$g(k|\mu) = \int db \int d\varepsilon P(k|\varepsilon\mu + b) f_b(b) f_\varepsilon(\varepsilon) = L(\mu|k) .$$

For k observations this is also the likelihood function of μ . According to our scheme, we obtain the upper limit μ_0 by normalization and integration,

$$C = \frac{\int_0^{\mu_0} L(\mu|k) d\mu}{\int_0^\infty L(\mu|k) d\mu}$$

which is solved numerically for μ_0 .

Example 122. Upper limit for a Poisson rate with uncertainty in background and acceptance

Observed are 2 events, expected are background events following a normal distribution $N(b|2.0, 0.5)$ with mean value $b_0 = 2$ and standard deviation $\sigma_b = 0.5$. The acceptance is assumed to follow also a normal distribution with mean $\varepsilon_0 = 0.5$ and standard deviation $\sigma_\varepsilon = 0.1$. The likelihood function is

$$L(\mu|2) = \int d\varepsilon \int db P(2|\varepsilon\mu + b) N(\varepsilon|0.5, 0.1) N(b|2.0, 0.5) .$$

We solve this integral numerically for values of μ in the range of $\mu_{\min} = 0$ to $\mu_{\max} = 20$, in which the likelihood function is noticeable different from zero (see Fig. 8.5). Subsequently we determine μ_0 such that the fraction $C = 0.9$ of the normalized likelihood function is located left of μ_0 . Since negative values of the normal distributions are unphysical, we cut these distributions and renormalize them. The computation in our case yields the upper limit $\mu_0 = 7.7$. In the figure we also indicate the e^{-2} likelihood ratio limit.

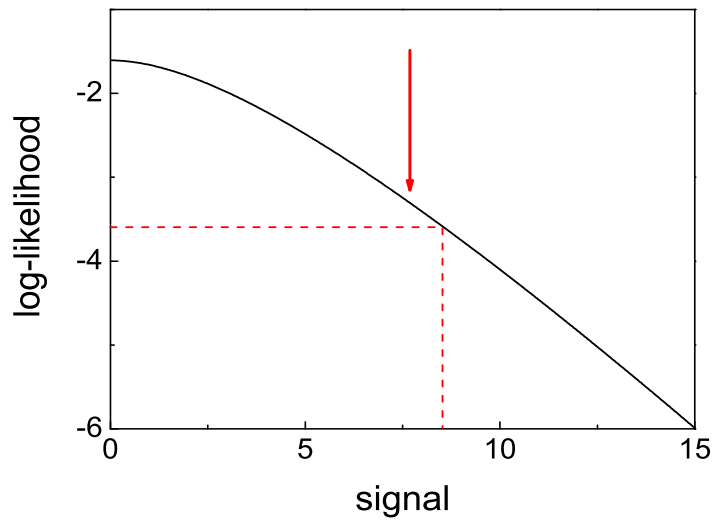


Fig. 8.5. Log-likelihood function for a Poisson signal with uncertainty in background and acceptance. The arrow indicates the upper 90% limit. Also shown is the likelihood ratio limit (decrease by e^2 , dashed lines).

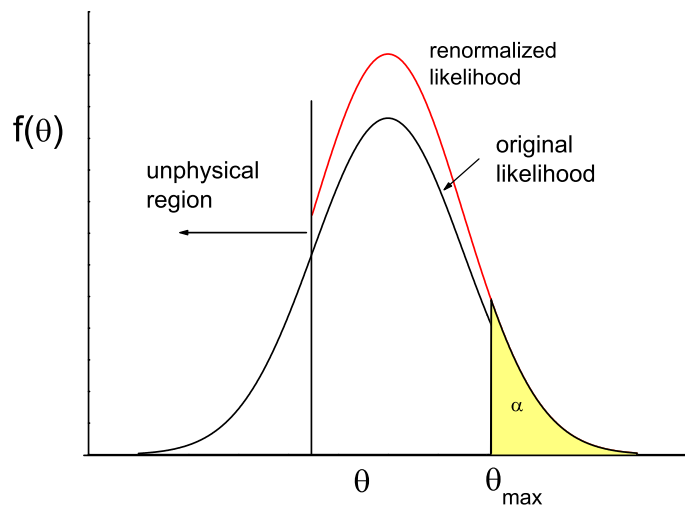


Fig. 8.6. Renormalized likelihood function and upper limit.

8.4.4 Unphysical Parameter Values

Sometimes the allowed range of a parameter is restricted by physical or mathematical boundaries, for instance it may happen that we infer from the experimental data a negative mass. In these circumstances the parameter range will be cut and the likelihood function will be normalized to the allowed region. This is illustrated in Fig. 8.6. The integral of the likelihood in the physical region is one. The shaded area is equal to α . The parameter θ is less than $\theta_{\max}$ with confidence $C = 1 - \alpha$.

We have to treat observations which are outside the allowed physical region with caution and check whether the errors have been estimated correctly and no systematic uncertainties have been neglected.

8.5 Summary

Measurements are described by the likelihood function.

- The standard likelihood ratio limits are used to represent the precision of the measurement.
- If the log-likelihood function is parabolic and the prior can be approximated by a constant, e.g. the likelihood function is very narrow, the likelihood function is proportional to the p.d.f. of the parameter, error limits represent one standard deviation and a 68.3 % probability interval.
- If the likelihood function is asymmetric, we derive asymmetric errors from the likelihood ratio. The variance of the measurement or probabilities can only be derived if the prior is known or if additional assumptions are made. The likelihood function should be published.
- Nuisance parameters are eliminated by the methods described in Chap. 7, usually using the profile likelihood.
- Error propagation is performed using the direct functional dependence of the parameters.
- Confidence intervals, upper and lower limits are computed from the normalized likelihood function, i.e. using a flat prior. These intervals usually correspond to 90% or 95% probability.
- In many cases it is not possible to assign errors or confidence intervals to parameters without making assumptions which are not uniquely based on experimental data. Then the results have to be presented such that the reader of a publication is able to insert his own assumptions and the procedure used by the author has to be documented.

Deconvolution

9.1 Introduction

9.1.1 The Problem

In many experiments statistical samples are distorted by limited acceptance, sensitivity, or resolution of the detectors. Is it possible to reconstruct from this distorted sample the original distribution from which the undistorted sample has been drawn?

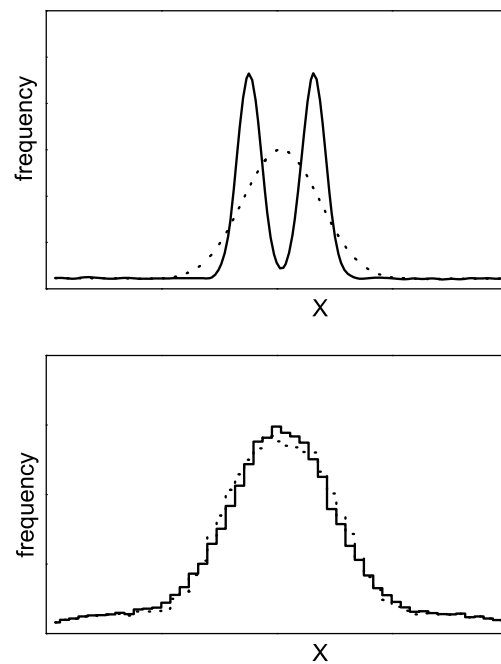


Fig. 9.1. Convolution of two different distributions. The distributions (top) differ hardly after the convolution (bottom).

The correction of losses is straight forward, but the deconvolution¹ of effects caused by the limited resolution is difficult because it implies a priori assumptions about the solution. Therefore, we should ask ourselves, whether a deconvolution is really a necessary step in our analysis. If we want to verify a theoretical prediction for a distribution $f(\mathbf{x})$, it is much easier and more accurate to convolute f with the known resolution and to compare then the smeared prediction and the experimental distributions with the methods discussed in Chaps. 6 and 10. If a prediction contains interesting parameters, also those should be estimated by comparing the convoluted distribution with the observed data. When we study, for instance, a sharp resonance peak on a slowly varying background, we adapt the resonance parameters of a theoretical distribution and the background of the true distribution in such a way that after the convolution it agrees with the observed distribution within the expected uncertainties. However, in situations where a reliable theoretical description is missing, or where the measurement is to be compared with a distribution obtained in another experiment with different experimental conditions, a deconvolution of the data cannot be avoided.

In the following we will call the original distribution from which the sample is drawn the *true distribution*. The *convoluted (smeared, observed) distribution* describes the observed sample consisting of the smeared events. We will also use the terms *coordinates* and *points* to describe our variables.

The observed sample is taken from a convoluted distribution $f'(\mathbf{x}')$ which is obtained from the true distribution $f(\mathbf{x})$ via the convolution integral

$$f'(\mathbf{x}') = \int_{-\infty}^{\infty} t(\mathbf{x}, \mathbf{x}') f(\mathbf{x}) d\mathbf{x} . \quad (9.1)$$

Supposed to be known is the convolution or transfer function² $t(\mathbf{x}, \mathbf{x}')$, which for a sample element located at $\mathbf{x}$ leads with the probability $t(\mathbf{x}, \mathbf{x}') d\mathbf{x}'$ to an observation in the interval $d\mathbf{x}'$ at $\mathbf{x}'$. In many cases it has the form of a normal distribution with the argument $\mathbf{x} - \mathbf{x}'$. In any case it should approach zero for large values of $|\mathbf{x} - \mathbf{x}'|$.

To determine $f(\mathbf{x})$ given $t(\mathbf{x}, \mathbf{x}')$ and $f'(\mathbf{x}')$ is called an inverse problem in numerical mathematics. Solving the (9.1), a Fredholms integral equation of the first kind, in the presence of statistical fluctuations is a difficult task because small variations of the input f' can cause large changes in f . Because of this instability, the problem is qualified by mathematicians as “ill-posed” and can only be solved by introducing additional input.

More specifically, our task is to reconstruct the statistical distribution $f(\mathbf{x})$ from a sample $\{\mathbf{x}'_1, \dots, \mathbf{x}'_N\}$ drawn from an distribution $f'(\mathbf{x}')$. Can this be accomplished if nothing is known about $f(\mathbf{x})$?

Unfortunately, the answer to this question is negative.

Methods for solving our problem are offered by numerical mathematics. However, they do not always optimally take into account the statistical fluctuations. The precision of the reconstruction of f depends on the size of the sample as well as on the accuracy with which we know the convolution function. In nuclear- and particle physics the sample size is often the limiting factor, in other sciences, like optics, the difficulties more often are related to a limited knowledge of the measurement resolution. In any case we have to take into account that the convolution leads to an

¹In the literature frequently the term *unfolding* is used instead of *deconvolution*.

²Other names are resolution function, point spread function, or kernel.

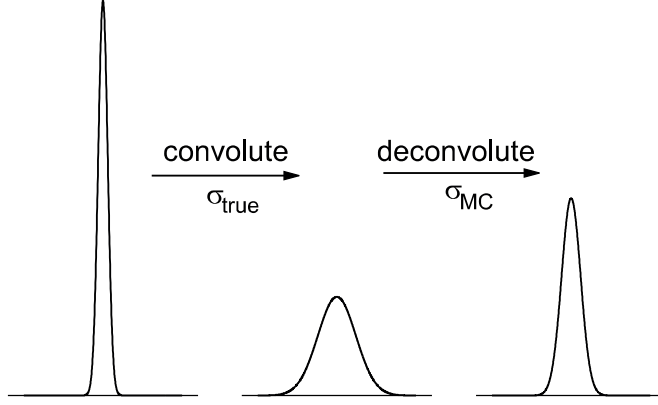


Fig. 9.2. Effect of deconvolution with a resolution wrong by 10%.

information loss. Since in realistic situations neither the sample size is infinite nor the convolution function t is known exactly, to obtain a stable result, we are forced to assume that the solution $f(\mathbf{x})$ is smooth. The various deconvolution methods differ in how they accomplish the smoothing which is called *regularization*.

Fig. 9.1 shows two different original distributions and the corresponding distributions smeared with a Gaussian. In spite of extremely different original distributions, the distributions of the samples are very similar. This demonstrates the sizeable information loss, especially in the case of the distribution with two adjacent peaks where the convolution function is broader than the structure. Sharp structures are washed out. In case of the two peak distribution it will be difficult to reconstruct the true distribution.

Fig. 9.2 shows the effect of using a wrong resolution function. The distribution in the middle is produced from that on the left hand side by convolution with a Gaussian with width σ_f . The deconvolution produces the distribution on the right hand side, where the applied width σ'_f was taken too low by 10%. For a relative error δ ,

$$\delta = \frac{|\sigma_f - \sigma'_f|}{\sigma_f},$$

we obtain an artificial broadening of a Gaussian line after deconvolution by

$$\begin{aligned}\sigma_{art}^2 &= |\sigma_f^2 - \sigma_f'^2|, \\ \sigma_{art} &= \sigma_f(2\delta - \delta^2)^{1/2} \approx \sqrt{2\delta}\sigma_f,\end{aligned}$$

where σ_{art}^2 has to be added to the squared width of the original line. Thus a Dirac δ -function becomes a normal distribution of width σ_{art} . From Table 9.1 we find that even small deviations in the resolution – e.g. estimates wrong by only 1 % – can lead to substantial artificial broadening of sharp structures. In the following parts of this chapter we will assume that the resolution is known exactly. Eventually, the corresponding uncertainties have to be evaluated by a Monte Carlo simulation.

Since statistical fluctuations prevent narrow structures in the true distribution from being resolved, we find it, vice versa, impossible to exclude artificially strongly oscillating solutions of the deconvolution problem without additional restrictions.

Table 9.1. Effect of different smearing in convolution and deconvolution.

δ	σ_{art}/σ_f
0.50	0.87
0.20	0.60
0.10	0.44
0.05	0.31
0.01	0.14

In the following we first discuss the common case that the observed distribution is given in the form of a histogram, and, accordingly, that a histogram should be produced by the deconvolution. Later we will consider also binning-free methods. We will assume that the convolution function also includes acceptance losses, since this does not lead to additional complications.

9.1.2 Deconvolution by Matrix Inversion

We denote the content of the bin i in the observed histogram by d_i , and correspondingly as θ_i in the true histogram. The relation 9.1 reads now

$$d_i = \sum_j T_{ij} \theta_j, \quad (9.2)$$

where the matrix element T_{ij} gives the probability to find an event in bin i of the observed histogram which was produced in bin j of the true histogram. Of course, the relation 9.2 is valid only for expected values, the values observed in a specific case $\hat{d}_i$ fluctuate according to the Poisson distribution:

$$P(\hat{d}_i) = \frac{e^{-d_i} d_i^{\hat{d}_i}}{\hat{d}_i!}.$$

In order to simplify the notation we consider in the following a one-dimensional histogram and combine the bin contents to a vector $\mathbf{d}$.

If we know the histogram, i.e. the vector $\mathbf{d}$, and the transfer matrix $\mathbf{T}$, we get the true histogram $\boldsymbol{\theta}$ by solving a set of linear equations. If both histograms have the same number of bins, that means if the matrix $\mathbf{T}$ is quadratic, we have simply to invert the matrix $\mathbf{T}$.

$$\begin{aligned} \mathbf{d} &= \mathbf{T} \boldsymbol{\theta}, \\ \boldsymbol{\theta} &= \mathbf{T}^{-1} \mathbf{d}. \end{aligned} \quad (9.3)$$

Here we have to assume that $\mathbf{T}^{-1}$ exists. Without smearing and equal bin sizes in both distributions, $\mathbf{T}$ and $\mathbf{T}^{-1}$ are diagonal and describe acceptance losses. Often the number of bins in the true histogram is chosen smaller than in the observed one. We will come back to this slightly more complicated situation at the end of this section.

We obtain an estimate $\hat{\boldsymbol{\theta}}$ for the original distribution if we substitute in (9.3) the expected value $\mathbf{d}$ by the observation $\hat{\mathbf{d}}$

$$\hat{\boldsymbol{\theta}} = \mathbf{T}^{-1} \hat{\mathbf{d}}.$$

By the usual propagation of errors we find the error matrix

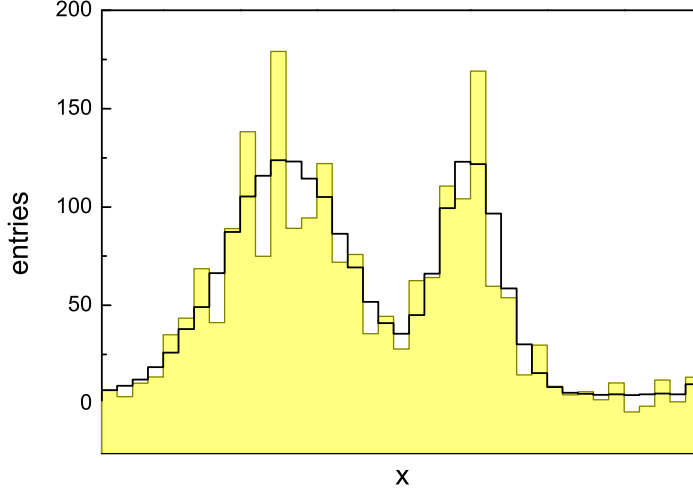


Fig. 9.3. Original histogram (open) and result of deconvolution by matrix inversion (shaded).

$$C_{ik} = \sum_j (\mathbf{T}^{-1})_{ij} (\mathbf{T}^{-1})_{kj} d_j .$$

Remark: For the error estimate it is not required to take into account explicitly also the fluctuations of the bin counts in the true histogram. They are contained already in the Poisson fluctuations $(\delta d_j)^2 = d_j$ of the observed event numbers.

The non-diagonal elements of the error matrix can be sizeable when the resolution is bad and correspondingly $\mathbf{T}$ differs considerably from a diagonal matrix.

The inversion method provides an exact solution of the deconvolution problem, but is not really appreciated by the users, because the event numbers of neighboring bins in the deconvoluted histogram often differ considerably, even negative bin contents are possible, which makes the result incompatible with our intuitive expectation. This is seen in the example of Fig. 9.3. The reason for this effect is obvious: The smeared distributions from two different true distributions cannot be distinguished if the latter are similar globally, but very different on a microscopic scale which is below the experimental resolution. Even though the deconvoluted distribution is very different from the true distribution it is compatible with it within the statistical errors which are strongly (anti-)correlated for neighboring bins.

In spite of the large fluctuations, the solution by matrix inversion is well suited for a comparison with theoretical models if the complete error matrix is taken into account. For histograms with many bins, however, the publication of the full error matrix with important non-diagonal elements – due to strong correlations of neighboring bins – is hardly practicable. Also the visual comparison with other data or theories is disturbed by the oscillations. For this reason methods have been developed which suppress non-significant fluctuations. Before turning to these methods, we come back once again to the transfer matrix.

9.1.3 The Transfer Matrix

Until now we have assumed that we know exactly the probability T_{ij} for observing elements in bin i which originally were produced in bin j . This is, at least in principle, impossible, as we have to average the true distribution $f(x)$ – which we do not know – over the respective bin interval

$$T_{ij} = \frac{\int_{Bin_i} \left[\int_{Bin_j} t(x, x') f(x) dx \right] dx'}{\int_{Bin_j} f(x) dx} . \quad (9.4)$$

Therefore T depends on f . Only if f can be approximated by constants in all bins the dependence is negligible. This condition is satisfied if the width of the transfer function, i.e. the smearing, is large compared to the bin width in the true histogram. On the other hand, small bins mean strong oscillations and correlations between neighboring bins. The matrix inversion method implies oscillations. Suppressing this feature by using wide bins can lead to inconsistent results.

All methods working with histograms have to derive the transfer matrix from a smooth distribution, which is not too different from the true one. For methods using regularization (see below), an iteration is required, if the deconvoluted distribution differs markedly from that used for the determination of T .

Remark: The deconvolution produces wrong results, if not all values of x' which are used as input in the deconvolution process are covered by x values of the considered true distribution. The range in x thus has to be larger than that of x' . The safest thing is not to restrict it at all even if some regions suffer from low statistics and thus will be reconstructed with marginal precision. We can eliminate these regions after the deconvolution.

In practice, the transfer matrix is usually obtained not analytically from (9.4), but by Monte Carlo simulation. In this case, the statistical fluctuations of the simulation eventually have to be taken into account. This leads to multinomial errors of the transfer matrix elements. The correct treatment of these errors is rather involved, thus, if possible, one should generate a number of simulated observations which is much larger than the experimental sample such that the fluctuations can be neglected. A rough estimate shows that for a factor of ten more simulated observations the contribution of the simulation to the statistical error of the result is only about 5% and then certainly tolerable.

9.1.4 Regularization Methods

From the above discussions it follows that it makes sense to suppress insignificant oscillations in the deconvoluted distribution. The various methods developed to achieve this, can be classified roughly as follows:

1. The deconvoluted function is processed by one of the usual smoothing procedures.
2. The true distribution is parameterized as a histogram. The parameters, i.e. the bin contents, are fitted to the data. A regularization term R_{regu} is subtracted from the purely statistical log-likelihood, respectively added to the squared deviation χ^2 :

$$\begin{aligned}\chi^2 &= \chi_{stat}^2 + R_{regu} , \\ \ln L &= \ln L_{stat} - R_{regu} .\end{aligned}$$

This term is constructed in such a way that smooth functions are favored. It has to get larger for stronger fluctuations of the solution.

3. The deconvolution is performed iteratively, starting from a smooth zeroth approximation of the true distribution. The iteration is stopped, once insignificant oscillations appear.
4. Insignificant eigenvalues of the transfer matrix are suppressed.

Of course, the size of the regularization must be limited such that the deconvoluted function is still compatible with the observation. This condition is used to fix the regularization parameter.

We will restrict our considerations to the methods 2 to 4. The accuracy of method 1 is hard to estimate. Reviews and comparisons of some deconvolution methods can be found in [47, 48, 50, 51, 52].

9.2 Deconvolution of Histograms

9.2.1 Fitting the Bin Content

The Likelihood Function

The expected values of the bin contents are parameters to be estimated. They have to be chosen such that the likelihood for the given observation is maximum. Since the observed event numbers are Poisson distributed, the log-likelihood for an observed bin is, according to (6.15) and the discussions in Sect. 6.5.6, given by

$$\begin{aligned}\ln L_i &= \hat{d}_i \ln d_i - d_i \\ &= \hat{d}_i \ln \left(\sum_j T_{ij} \theta_j \right) - \sum_j T_{ij} \theta_j ,\end{aligned}$$

and for the whole histogram it is

$$\ln L_{stat} = \sum_i \left[\hat{d}_i \ln \left(\sum_j T_{ij} \theta_j \right) - \sum_j T_{ij} \theta_j \right] . \quad (9.5)$$

The estimates of the parameters θ_j and their errors are obtained in the usual way. Of course, also this solution will oscillate. But contrary to the method of matrix inversion, it cannot produce negative values. As discussed above, we suppress the oscillations by adding a regularization term R_{regu} to the purely statistical term L_{stat} :

$$\ln L(\boldsymbol{\theta}) = \ln L_{stat}(\boldsymbol{\theta}) - R_{regu}(\boldsymbol{\theta}) . \quad (9.6)$$

For the likelihood method the number of bins in the true and in the observed histogram may be different. In case they are equal, the likelihood solution without

regularization and the solution by simple matrix inversion coincide, provided the latter does not yield negative parameter values.

For large event numbers we may use instead of the maximum likelihood adjustment a χ^2 fit:

$$\chi_{stat}^2 = \sum_{i=1}^M \sum_{j=1}^N \frac{(T_{ij}\theta_j - \hat{d}_i)^2}{T_{ij}\theta_j},$$

$$\chi^2(\boldsymbol{\theta}) = \chi_{stat}^2(\boldsymbol{\theta}) + R_{regu}(\boldsymbol{\theta}).$$

Curvature Regularization

An often applied regularization function R_{regu} is,

$$R_{regu}(x) = r \left(\frac{d^2 f}{dx^2} \right)^2. \quad (9.7)$$

It increases with the curvature of f . It penalizes strong fluctuations and favors a linear variation of the unfolded distribution. The regularization constant r determines the power of the regularization.

For a histogram of N bins with constant bin width we approximate (9.7) by

$$R_{regu} = r \sum_{i=2}^{N-1} (2\theta_i - \theta_{i-1} - \theta_{i+1})^2. \quad (9.8)$$

This function becomes zero for a linear distribution. It is not difficult to adapt (9.8) to variable bin widths.

The terms of the sum (9.8) in general become large for strongly populated bins. In less populated regions of the histogram, where the relative statistical fluctuations are large, the regularization by (9.8) will not be very effective. Therefore the terms should be weighted according to their statistical significance:

$$R_{regu} = r \sum_{i=2}^{N-1} \frac{(2\theta_i - \theta_{i-1} - \theta_{i+1})^2}{\delta_i^2}. \quad (9.9)$$

Here δ_i^2 is the variance of the numerator. For Poisson distributed bin contents, it is

$$\delta_i^2 = 4\theta_i + \theta_{i-1} + \theta_{i+1}.$$

Usually the deconvolution also corrects for acceptance losses. Then this error estimate has to be modified. We leave the trivial calculation to the reader.

For higher dimensional histograms a regularization term can be introduced analogously by penalizing the deviation of each bin content from the mean value of its neighbors as in (9.8).

There may be good reasons to use regularization functions other than (9.8), e.g. when it is known that the function which we try to reconstruct is strongly non linear and when its shape is approximately known. Instead of suppressing the curvature, we may penalize the deviation from the expected shape of the histogram. We accomplished this with the transformation

$$\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \boldsymbol{\tau} ,$$

where $\boldsymbol{\theta}_0$ refers to the expectation and the new parameter vector $\boldsymbol{\tau}$ is fitted and regularized.

Entropy Regularization

The entropy S of a discrete distribution with probabilities p_i , $i = 1, \dots, N$ is defined through

$$S = - \sum_{i=1}^N p_i \ln p_i .$$

We borrow the entropy concept from thermodynamics, where the entropy S measures the randomness of a state and the maximum of S corresponds to the equilibrium state which is the state with the highest probability. It has also been introduced into information theory and into Bayesian statistics to fix prior probabilities.

For our deconvolution problem we consider a histogram to be constructed with N bins containing $\theta_1, \dots, \theta_N$ events. The probability for one of the n events to fall into true bin i is given by θ_i/n . Therefore the entropy of the distribution is

$$S = - \sum_{i=1}^N \frac{\theta_i}{n} \ln \frac{\theta_i}{n} .$$

The maximum of the entropy corresponds to an uniform population of the bins, i.e. $\theta_i = \text{const.} = n/N$, and equals $S_{\max} = \ln N$, while its minimum $S_{\min} = 0$ is found for the one-point distribution (all events in the same bin) $\theta_i = n\delta_{i,j}$. Thus $R_{\text{regu}} \propto -S$ can be used to smoothen a distribution. We minimize

$$\chi^2 = \chi_{\text{stat}}^2 - rS ,$$

or equivalently maximize

$$\ln L = \ln L_{\text{stat}} + rS$$

where r determines again the strength of the regularization. For further details see [55, 54, 56]. We do not recommend this method for the usual applications in particle physics, see Sect. 9.4 below. It is included here because it is popular in astronomy.

Regularization Strength

As we have seen in Chap. 6, Sect. 6.5.6, for sufficiently large statistics the negative log-likelihood is equivalent to the χ^2 statistic:

$$\chi^2 \approx -2 \ln L + \text{const} .$$

The regularized solution has to be statistically compatible with the observation. The expectation of χ^2 is equal to the number of degrees of freedom, $N - 1 \approx N$, and its standard deviation is $\sqrt{2(N - 1)} \approx \sqrt{2N}$, see Chap. 3. Therefore, the addition of the regularization term should not increase the minimum of χ^2 by more than about $\sqrt{2N}$, the square root of twice the number of bins. The corresponding tolerable decrease of the maximal log-likelihood is $\sqrt{N/2}$:

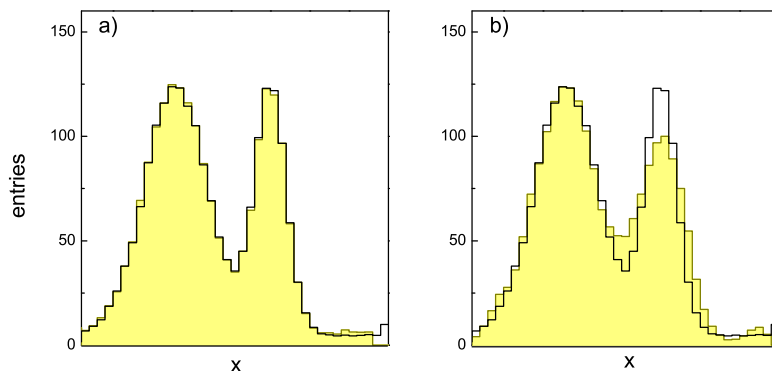


Fig. 9.4. Comparison of the deconvoluted histogram (shaded) with the original histogram. a) normal regularization strength, b) too strong regularization. (without regularization see Fig. 9.3.)

$$\min(\chi^2) = \min(\chi_{stat}^2) + \sqrt{2N} ,$$

$$\min(-\ln L) = \min(-\ln L_{stat}) + \sqrt{N/2} .$$

Depending on the concrete problem, also a weaker or stronger regularization may be adequate. In many cases it makes sense to tolerate some spurious fluctuations and to conserve in this way possible fine structures in the spectra. If it is known that the distribution is rather smooth, e.g. for structure functions, a stronger regularization might make sense.

In Fig. 9.4a we compare the result of a deconvolution with regularization with the true histogram. The agreement is very good. The initial true distribution is the same as in Fig. 9.3, where the result of the deconvolution without regularization was shown. Figure 9.4b indicates the effect of a too strong regularization: The narrow peak is sizably broadened, because the regularization suppresses the curvature at its top.

Choice of Bin Width

The bin width of the true histogram which is finally presented as the result should be adapted to the experimental resolution. It does not make sense to choose it much smaller than the f.w.h.m. of the transfer function. Too small intervals suggest a measurement precision which does not really exist. On the other hand the binning of the transfer matrix must be fine enough to justify the approximations discussed above. Thus it may be not unreasonable to perform the deconvolution with relatively small bins but to merge neighboring bins afterwards.

There is no reason, however, to choose a small number of bins in the observed histogram. It is recommended to choose it larger than in the deconvoluted histogram. Then the number of parameters is smaller than the number of measurements, and the result is constrained. This has the advantage that we are able to apply goodness-of-fit tests.

9.2.2 Iterative Deconvolution

We have seen above that small variations in the observed distribution may produce large changes in the reconstructed distribution, if we do not introduce additional restrictions. To avoid this kind of numerical difficulties, an iterative procedure for matrix inversion has been developed [49] which is also suited for the deconvolution of histograms [50]. The stepwise inversion corresponds to a stepwise deconvolution. We start with a first guess of the true distribution and modify it in such a way that the difference between left and right hand side in (9.2) is reduced. Then we iterate this procedure.

It is possible to show that it converges to the result of matrix inversion, if the latter is positive in all bins. Similar to the maximum likelihood solution negative bin contents are avoided. This represents some progress but the unpleasant oscillations are still there. But since we start with a smooth initial distribution, the artifacts occur only after a certain number of iterations. The regularization is then performed simply by interrupting the iteration sequence.

To define an appropriate stopping rule, two ways offer themselves:

1. We calculate after each step χ^2 . The sequence is interrupted if χ^2 differs by N , the number of bins, from its minimal value which is reached in the limit of infinitely many iterations.
2. We introduce an explicit regularization term, like (9.8), and calculate in each step the regularized χ^2 according to (9.6). The iteration is stopped at the maximum of the log-likelihood.

We prefer the simpler first method.

We will now look at this method more closely and formulate the relevant relations.

A simple example for the first iteration is sketched in Fig. 9.5. The start distribution in 9.5a for the true histogram has been chosen as uniform. It consists of three bins. Application of the transfer matrix produces 9.5b, a first prediction for the data. The different shadings indicate the origin of the entries in the five observed bins. Now the distribution 9.5b is compared with the observed distribution 9.5c. The agreement is bad, for instance the second observed bin differs from the prediction by a factor two. All contributions of the prediction to this bin are now multiplied by two. Finally the scaled contributions are summed up according to their origin to a new true distribution 9.5d. If there would be only losses, but no migration of entries, this first iteration would lead to a complete correction.

This procedure corresponds to the following equations: The prediction $\mathbf{d}^{(k)}$ of the iteration k is obtained from the true vector $\boldsymbol{\theta}^{(k)}$:

$$d_i^{(k)} = \sum_j T_{ij} \theta_j^{(k)} .$$

The components $T_{ij} \theta_j^{(k)}$ are scaled with $\hat{d}_i / d_i^{(k)}$ and added up into the bin j of the true distribution from which it originated:

$$\theta_j^{(k+1)} = \sum_i T_{ij} \theta_j^{(k)} \frac{\hat{d}_i}{d_i^{(k)}}$$

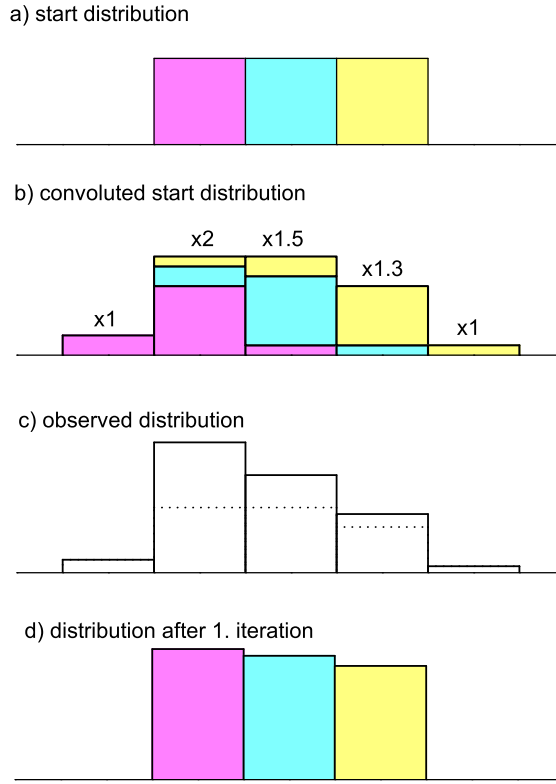


Fig. 9.5. Iterative deconvolution.

$$= \sum_i T_{ij} \theta_j^{(k)} \frac{\hat{d}_i}{\sum_l T_{il} \theta_l^{(k)}}.$$

The result of the iteration method for the same example that has been deconvoluted with a likelihood fit (Fig. 9.4) is shown in Fig. 9.6. After the first iteration the agreement is still not optimal. After the eighth iteration we realize some obvious oscillations. The sequence should be interrupted after the second iteration.

To control the regularization it may be necessary to have finer iteration steps. This is achieved simply by multiplying the scaling factor by a constant smaller than one.

9.2.3 Regularization of the Transfer Matrix

We transform the relation $\mathbf{d} = \mathbf{T}\boldsymbol{\theta}$ by a unitary transformation³ $\mathbf{U}$ to a basis in which the transformed matrix $\mathbf{D}$ is diagonal:

$$\mathbf{D} = \mathbf{U}\mathbf{T}\mathbf{U}^{-1},$$

³The matrix $\mathbf{U}$ is obtained by the usual methods, solving the characteristic equation $\det(\mathbf{T} - \lambda \mathbf{I}) = 0$, giving the roots λ and determining the respective orthogonal and normalized eigenvectors which form the columns of $\mathbf{U}$.

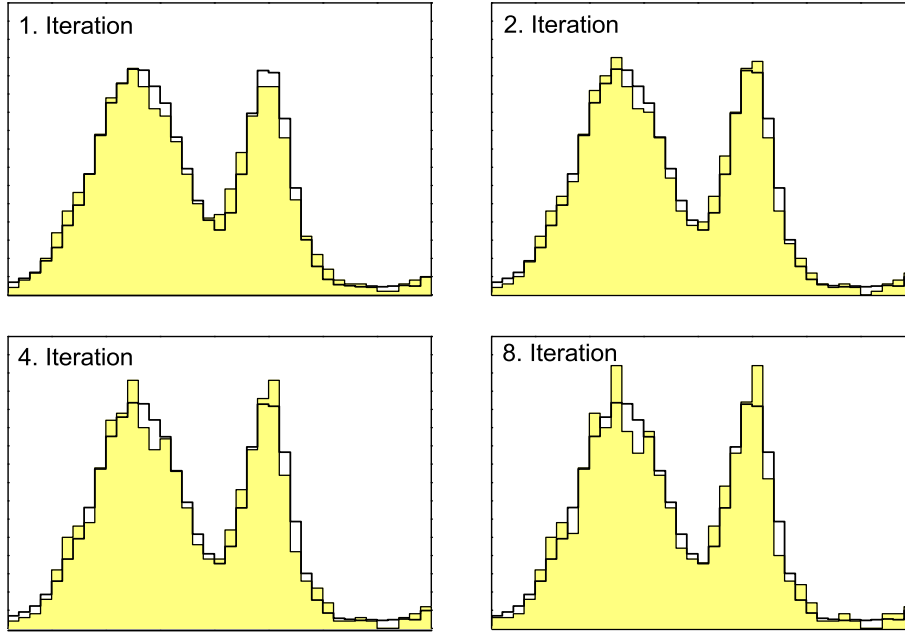


Fig. 9.6. Iterative deconvolution (shaded histograms) with different number of iterations. The original distribution corresponds to the open histogram.

$$\begin{aligned} \mathbf{U}\mathbf{d} &= \mathbf{U}\mathbf{T}\mathbf{U}^{-1}\mathbf{U}\boldsymbol{\theta} \\ &= \mathbf{D}\mathbf{U}\boldsymbol{\theta} . \end{aligned}$$

The diagonal elements of $\mathbf{D}$ are the eigenvalues λ_i . In the transformed system the observed histogram is found by multiplying the true histogram $\mathbf{U}\boldsymbol{\theta}$ with the eigenvalues:

$$(\mathbf{U}\mathbf{d})_i = \lambda_i(\mathbf{U}\boldsymbol{\theta})_i .$$

This new relation can be solved for the components of the transformed true vector:

$$(\mathbf{U}\boldsymbol{\theta})_i = \frac{(\mathbf{U}\mathbf{d})_i}{\lambda_i} .$$

This equation is valid for the expected values. If we use it to calculate the true histogram $\boldsymbol{\theta}$ from the estimate $\hat{\mathbf{d}}$ which deviates by $\boldsymbol{\varepsilon}$ from the expected value,

$$\hat{\mathbf{d}} = \mathbf{d} + \boldsymbol{\varepsilon} ,$$

we find

$$(\widehat{\mathbf{U}\boldsymbol{\theta}})_i = \frac{(\mathbf{U}\hat{\mathbf{d}})_i}{\lambda_i} = \frac{(\mathbf{U}(\mathbf{d} + \boldsymbol{\varepsilon}))_i}{\lambda_i} ,$$

where the fluctuations $\boldsymbol{\varepsilon}$ are weighted especially when the eigenvalues are small. This is the origin of our problem!

We may now regularize the transfer matrix by suppressing small eigenvalues $\lambda_i < \lambda_{\min}$ in $\mathbf{D}$. This means, the corresponding components of the vector

$$(\widehat{\mathbf{U}\boldsymbol{\theta}})_i = \frac{(\mathbf{U}\hat{\mathbf{d}})_i}{\lambda_i}$$

are set to zero. We then get the estimate $\hat{\boldsymbol{\theta}}$ by the inverse transformation:

$$\hat{\boldsymbol{\theta}} = \mathbf{U}^{-1}(\widehat{\mathbf{U}\boldsymbol{\theta}}) .$$

Instead of cutting abruptly the eigenvalues which are smaller than $\lambda_{\min}$, it has been proposed to damp their contributions continuously. An advantage of such a procedure is not evident, however.

The choice of the regularization parameter $\lambda_{\min}$ follows the same scheme as for the other methods. The solutions with and without regularization are convoluted and compared with the observed histogram and in both cases χ^2 is calculated. The value of χ^2 of the regularized solution should exceed the not regularized one not more than allowed statistically.

Often the same bin width is chosen for the true and the observed histogram. It is, however, often appropriate to add at least some marginal bins in the observed histogram which is broader due to the finite resolution. The transfer matrix then becomes rectangular. We have to recast (9.3) somewhat to obtain a quadratic matrix $\mathbf{T}^T \mathbf{T}$ which can be inverted:

$$\begin{aligned} \mathbf{T}^T \mathbf{d} &= \mathbf{T}^T \mathbf{T} \boldsymbol{\theta} , \\ \boldsymbol{\theta} &= (\mathbf{T}^T \mathbf{T})^{-1} \mathbf{T}^T \mathbf{d} . \end{aligned}$$

To shorten the notation we introduce the matrix $\tilde{\mathbf{T}}$ and the vector $\tilde{\mathbf{d}}$:

$$\begin{aligned} \tilde{\mathbf{T}} &= \mathbf{T}^T \mathbf{T} , \\ \tilde{\mathbf{d}} &= \mathbf{T}^T \mathbf{d} . \end{aligned}$$

After this transformation the treatment is as above.

9.3 Binning-free Methods

We now present binning-free methods where the observations need not be combined into bins. The deconvolution produces again a sample. The advantage of this approach is that arbitrary histograms under various selection criteria can be constructed afterwards. It is especially suited for low statistics distributions in high dimensional spaces where histogramming methods fail.

9.3.1 Iterative Deconvolution

We can realize the iterative weighting method described in Sect. 9.2.2 also in a similar way without binning [52].

We start with a Monte Carlo sample of events, each event being defined by the true coordinate x and the observation x' . During the iteration process we modify at each step a weight which we associate to the events such that the densities in the observation space of simulated and real events approach each other. Initially all weights are equal to one. At the end of the procedure we have a sample of weighted events which corresponds to the deconvoluted distribution.

To this end, we estimate a local density $d'(x'_i)$ in the vicinity of any point x'_i in the observation space. (For simplicity, we restrict ourselves again to a one-dimensional space since the generalization to several dimensions is trivial.) The following density estimation methods (see Chap. 12) lend themselves:

1. The density is taken as the number of observations within a certain fixed region around x'_i , divided by the length of the region. The length should correspond roughly to the resolution, if the region contains a sufficient number of entries.
2. The density is chosen proportional to the inverse length of that interval which contains the K nearest neighbors, where K should be not less than about 10 and should be adjusted by the user to the available resolution and statistics.

We denote by $t(x)$ the simulated density in the true space at location x , by $t'(x')$ the observed simulated density at x' and the corresponding data density be $d'(x')$. The density $d'(x')$ is estimated from the length of the interval containing K events, $t'(x')$ from the number of simulated events $M(x')$ in the same interval. The simulated densities are updated in each iteration step k . We associate a preliminary weight

$$w_i^{(1)} = \frac{d'(x'_i)}{t^{(0)}(x'_i)} = \frac{K}{M(x')}$$

to the Monte Carlo event i . The weighted events in the vicinity of x represent a new density $t^{(1)}(x)$ in the true space. We now associate a true weight w_i to the event which is just the average over the preliminary weights of all K events in the neighborhood of x_i , $w_i = \sum_j w'_j / K$. With the smoothed weight w_i a new observed simulated density $t^{(1)}$ is computed. In the k 'th iteration the preliminary weight is given by

$$w_i^{(k+1)} = \frac{d'(x'_i)}{t^{(k)}(x'_i)} w_i^{(k)}.$$

The weight will remain constant once the densities t' and d' agree. As result we obtain a discrete distribution of coordinates x_i with appropriate weights w_i , which represents the deconvoluted distribution. The degree of regularization depends on the parameters K used for the density estimation.

The method is obviously not restricted to one-dimensional distributions, and is indeed useful in multi-dimensional cases, where histogram bins suffer from small numbers of entries. We have to replace x_i , x'_i by $\mathbf{x}_i$, $\mathbf{x}'_i$, and the regions for the density estimation are multi-dimensional.

9.3.2 The Satellite Method

The basic idea of this method [53] is the following: We generate a Monte Carlo sample of the same size as the experimental data sample. We let the Monte Carlo events migrate until the convolution of their position is compatible with the observed data.

With the help of a test variable ϕ , which could for example be the negative log likelihood and which we will specify later, we have the possibility to judge quantitatively the compatibility. When the process has converged, i.e. ϕ has reached its minimum, the Monte Carlo sample represents the deconvoluted distribution.

We proceed as follows:

We denote by $\{\mathbf{x}'_1, \dots, \mathbf{x}'_N\}$ the locations of the points of the experimental sample and by $\{\mathbf{y}_1, \dots, \mathbf{y}_N\}$ those of the Monte Carlo sample. The observed density of the simulation is $f(\mathbf{y}') = \sum_i t(\mathbf{y}_i, \mathbf{y}')$, where t is the resolution or transfer function. The test variable $\phi[\mathbf{x}'_1, \dots, \mathbf{x}'_N; f(\mathbf{y}')] is a function of the sample coordinates $\mathbf{x}_i$ and the density expected for the simulation. We execute the following steps:$

1. The points of the experimental sample $\{\mathbf{x}'_1, \dots, \mathbf{x}'_N\}$ are used as a first approximation to the true locations $\mathbf{y}_1 = \mathbf{x}'_1, \dots, \mathbf{y}_N = \mathbf{x}'_N$.
2. We compute the test quantity ϕ of the system.
3. We select randomly a Monte Carlo event and let it migrate by a random amount $\Delta\mathbf{y}_i$ into a randomly chosen direction, $\mathbf{y}_i \rightarrow \mathbf{y}_i + \Delta\mathbf{y}_i$.
4. We recompute ϕ . If ϕ has decreased, we keep the move, otherwise we reject it. If ϕ has reached its minimum, we stop, if not, we return to step 3.

The resolution or smearing function t is normally not a simple analytic function, but only numerically available through a Monte Carlo simulation. Thus we associate to each true Monte Carlo point i a set of K generated observations $\{\mathbf{y}'_{i1}, \dots, \mathbf{y}'_{iK}\}$, which we call satellites and which move together with $\mathbf{y}_i$. The test quantity ϕ is now a function of the N experimental positions and the $N \times K$ smeared Monte Carlo positions.

Choices of the test variable ϕ are presented in Chap. 10. We recommend to use the variable *energy*.

The migration distances $\Delta\mathbf{y}_i$ should be taken from a distribution with a width somewhat larger than the measurement resolution, while the exact shape of the distribution is not relevant. We therefore recommend to use a uniform distribution, for which the generation of random numbers is faster than for a normal or other distributions. The result of the deconvolution is independent from these choices, but the number of iteration steps can raise appreciably for a bad choice of parameters.

Example 123. Deconvolution of a blurred picture

Figure 9.7 shows a two-dimensional application. The observed picture consisted of lines and points which are convoluted with a two-dimensional normal distribution. In the Monte Carlo simulation for each true point $K = 25$ satellites have been generated. The energy ϕ is minimized. The resolution of the lines in the deconvoluted figure on the right hand side is restricted by the low experimental statistics. For the eyes the restriction is predominantly due to the low Monte Carlo factor K . Each eye has $N = 60$ points. The maximal resolution for a point measured N times is obtained for measurement error σ_f as

$$\Delta x = \sigma_f \sqrt{\frac{1}{N} + \frac{1}{K}}$$

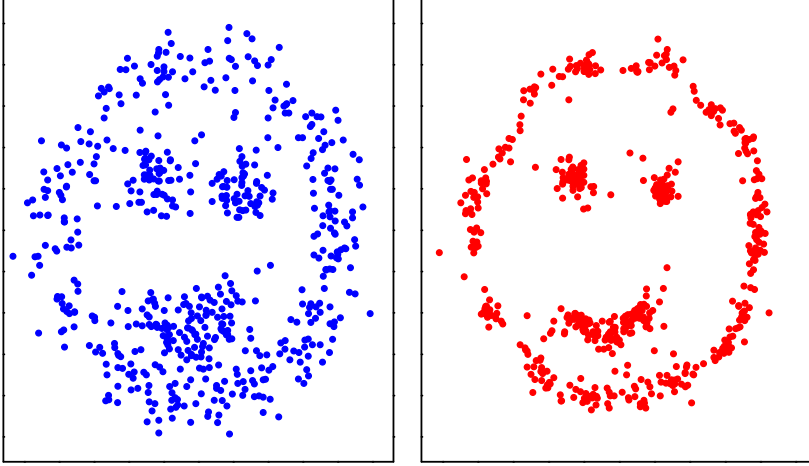


Fig. 9.7. Deconvolution of a blurred picture with the satellite method.

$$= \sigma_f \sqrt{\frac{1}{60} + \frac{1}{25}} = 0.24 \sigma_f .$$

Measurement resolution and acceptance should stay approximately constant in the region in which the events migrate. When we start with a reasonably good approximation of the true distribution, this condition is usually satisfied. In exceptional cases it would be necessary to update the distribution of the satellites y'_{ik} after each move, i.e. to simulate or correct them once again. It is more efficient, however, to perform the adaptation for all elements periodically after a certain number of migration steps.

The number K determines the maximal resolution after the deconvolution, it has therefore a regularization effect; e.g. for a measurement resolution σ_f and $K = 16$ the minimal sampling interval is $\sigma_T = \sigma_f / \sqrt{K} = \sigma_f / 4$.

If the true p.d.f. has several maxima, we may find several relative minima of the energy. In this case a new stochastic element has to be introduced in the minimization (see Sect. 5.2.7). In this case a move towards a position with smaller energy is not performed automatically, but only preferred statistically.

We have not yet explained how acceptance losses are taken into account. The simplest possibility is the following: If there are acceptance losses, we need $K_{i0} > K$ trials to generate the K satellites of the event y_i . Consequently, we relate a weight $w_i = K_{i0}/K$ to the element y_i . After the deconvolution we then obtain a weighted sample.

A more detailed description of the satellite method is found in [53].

9.3.3 The Maximum Likelihood Method

In the rare cases where the transfer function $t(x, x')$ is known analytically or easily calculable otherwise, we can maximize the likelihood where the parameters are the locations of the true points. Neglecting acceptance losses, the p.d.f. for an observation x' , with the true values $x_1, \dots, x_N$ as parameters is

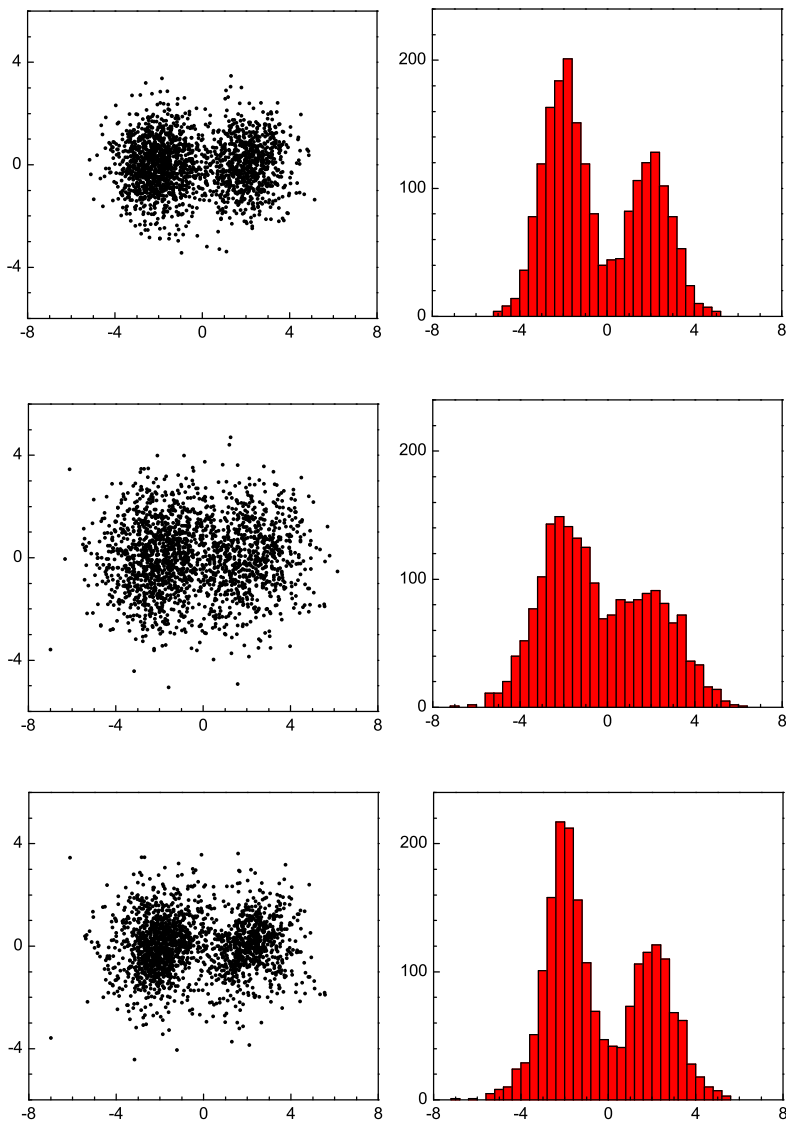


Fig. 9.8. Deconvolution of point locations. The left hand side shows from top to bottom the true distribution, the smeared distribution and the deconvoluted distribution. The right hand side shows the corresponding projections onto the x axis in form of histograms.

$$f_N(\mathbf{x}'|\mathbf{x}_1, \dots, \mathbf{x}_N) = \frac{1}{N} \sum_{i=1}^N t(\mathbf{x}_i, \mathbf{x}')$$

where t is assumed to be normalized with respect to $\mathbf{x}'$. The log likelihood then is given, up to a constant, by

$$\ln L(\mathbf{x}|\mathbf{x}') = \sum_{k=1}^N \ln \sum_{i=1}^N t(\mathbf{x}_i, \mathbf{x}_k) .$$

The maximum can either be found using the well known minimum searching procedures or the migration method which we have described above and which is not restricted to low event numbers. Of course maximizing the likelihood leads to the same artifacts as observed in the histogram based methods. The true points form clusters which, eventually, degenerate into discrete distributions. A smooth result is obtained by stopping the maximizing process before the maximum has been reached. For definiteness, similar to the case of histogram deconvolution, a fixed difference of the likelihood from its maximum value should be chosen to stop the maximization process. Similarly to the histogram case, this difference should be of the order of $\Delta L \approx \sqrt{NDF/2}$ where the number of degrees of freedom NDF is equal to the number of points times the dimension of the space.

Example 124. : Deconvolution by fitting the true event locations

Fig. 9.8 top shows 2000 points randomly generated according to the superposition of two normal distributions denoted as $N(x', y'|\mu_x, \mu_y, \sigma_x, \sigma_y)$:

$$f(x', y') = 0.6 N(x', y'|-2, 0, 1, 1) + 0.4 N(x', y'|+2, 0, 1, 1) .$$

The transfer function again is a normal distribution centered at the true points with symmetric standard deviations of one unit. It is used to convolute the original distribution with the result shown in Fig. 9.8 middle. The starting values of the parameters $\hat{x}_i, \hat{y}_i$ are set equal to the observed locations x'_i, y'_i . Randomly selected points are then moved within squares of size 4×4 units and moves that improve the likelihood are kept. After 5000 successful moves the procedure is stopped to avoid clustering of the true points. The result is shown in the lower plot of Fig. 9.8. On the right hand side of the same figure the projections of the distribution onto the x axis in form of histograms are presented.

Generally, as with other deconvolution methods, we have to find a sensible compromise between smoothness and resolution and to choose the corresponding regularization strength. In astronomy and optics, often the signals originate from point sources. In this case it is reasonable to completely omit the regularization, and to determine the locations and strength of the sources by maximizing the likelihood. Then, eventually, after inspection of the result, the number of sources might be fixed and the parameters then could be determined in a standard likelihood fit.

9.4 Comparison of the Methods

We have discussed three different methods for deconvolution of histograms, and three binning-free methods:

1. Likelihood fit of the true histogram with curvature sensitive or entropy regularization
2. Multiplication of the observed histogram vector with the inverted, regularized transfer matrix
3. Iterative deconvolution
4. Iterative binning-free deconvolution
5. The satellite method
6. The binning-free likelihood method

The first method is more transparent than the others. The user has the possibility to adapt the regularization function to his specific needs. With curvature regularization he may, for instance, choose a different regularization for different regions of the histogram, or for the different dimensions in a higher-dimensional histogram. He may also regularize with respect to an assumed shape of the resulting histogram. The statistical accuracy in different parts of the histogram can be taken into account. Regularization with the entropy approach is technically simpler but it is not suited for applications in particle physics, because it favors a globally uniform distribution while the local smearing urges for a local smoothing. It has, however, been successfully applied in astronomy and been further adjusted to specific problems there. An overview with critical remarks is given in [56].

The second method is independent from the shape of the distribution to be deconvoluted. It depends on the transfer matrix only. This has the advantage to be independent from subjective influences of the user. A disadvantage is that regions of the true histogram with high statistics are treated not differently from those with only a few entries. A refined version which has successfully been applied in several experiments is presented in [8].

The third procedure is technically the simplest. It can be shown that it is very similar to the second method. It also suppresses small eigenvalues of the transfer matrix.

The binning-free, iterative method has the disadvantage that the user has to choose some parameters. It requires sufficiently high statistics in all regions of the observation space. An advantage is that there are no approximations related to the binning. The deconvolution produces again single points in the observation space which can be subjected to selection criteria and collected into arbitrary histograms, while methods working with histograms have to decide on the corresponding parameters before the deconvolution is performed.

The satellite method has the same advantages. Important parameters must not be chosen, however. It is especially well suited for small samples and multidimensional distributions, where other methods have difficulties. For large samples it is rather slow even on large computers.

The binning-free likelihood method requires an analytic transfer function. It is much faster than the satellite method, and is especially well suited for the deconvolution of narrow structures like point sources.

An qualitative comparison of the different methods does not show big differences in the results. In the majority of problems the deconvolution of histograms with the fitting method and curvature regularization is the preferred solution.

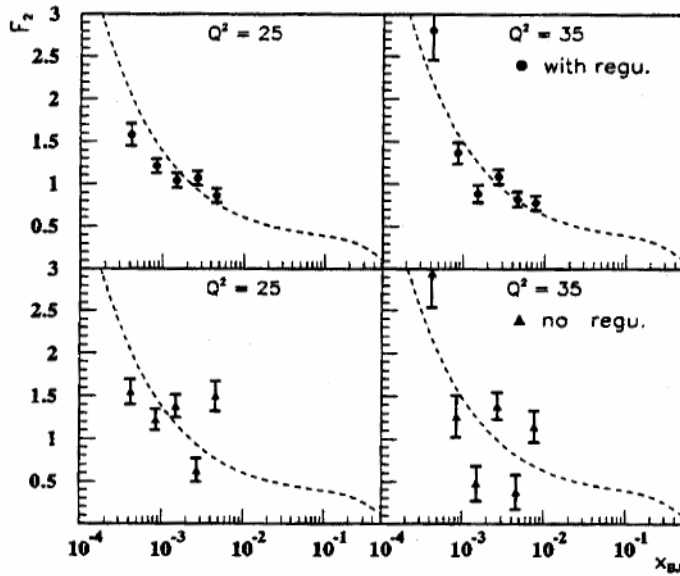


Fig. 9.9. Result of a deconvolution with strong (top) and weak (bottom) regularization. The errors depend on the regularization strength.

As stated above, whenever the possibility exists to parametrize the true distribution, the deconvolution process should be avoided and replaced by a standard fit.

9.5 Error Estimation for the Deconvoluted Distribution

The fitting methods produce error estimates automatically, for the other methods the uncertainties can be obtained by the usual error propagation. But we run into another unavoidable difficulty connected to the regularization: The size of the errors depends on the strength of the regularization which on the other hand is unrelated to the statistical accuracy of the data. This is illustrated in Fig. 9.9 where the deconvolution of a structure function with different regularization strengths is shown. A strongly regularized distribution may exhibit smaller errors than the distribution before the convolution. This is unsatisfactory, as we lose information by the smearing. We should present errors which do not depend on data manipulations.

As described above, the errors of neighboring bins of a histogram are negatively correlated. The goodness-of-fit changes only slightly if we, for instance, enhance a bin content and accordingly reduce both the two neighboring bin contents or vice versa. The regularization has the effect to minimize the difference while keeping the sum of entries nearly unchanged, as can be seen in the example of Fig. 9.9. The effect of the regularization is sketched in Fig. 9.10. Even a soft regularization will reduce the area of the error ellipse considerably.

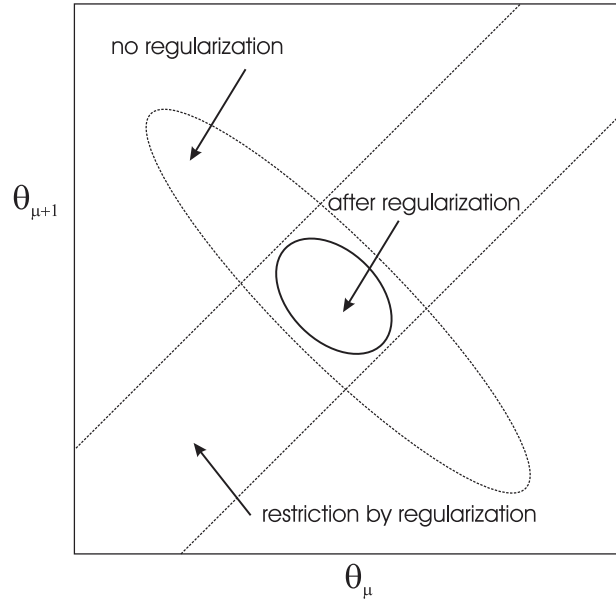


Fig. 9.10. Effect of regularization. The correlation of neighboring bins is reduced.

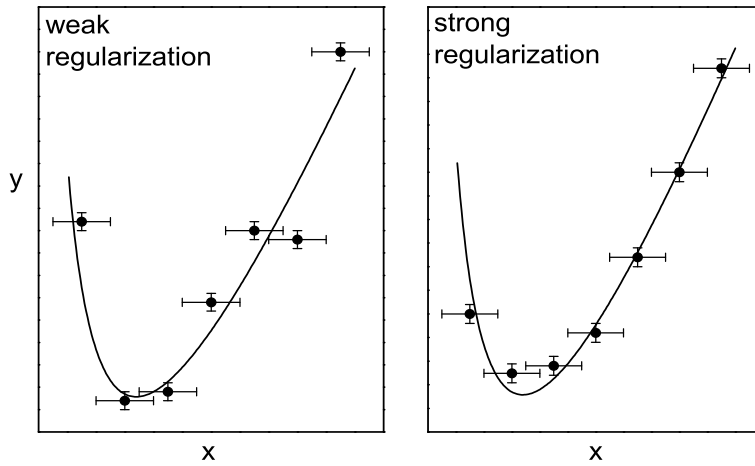


Fig. 9.11. Deconvoluted distribution with strong and weak regularization. The horizontal error bars represent the experimental resolution.

A sensible presentation of the result of a deconvolution where the values but not the errors depend on the regularization is the following: For each deconvoluted point we calculate its statistical error $\delta\theta_j$ neglecting uncertainties due to a possible wrong association of observations to the corresponding bin. This means essentially, that the relative error is equal to one over the square root of the number of observed events associated to the true bin.

The number of observed events belonging to bin j is $\sum_i T_{ij}\theta_j$. Thus we get for the relative bin error

$$\frac{\delta\theta_j}{\theta_j} = \frac{1}{\sqrt{\sum_i T_{ij}\theta_j}} .$$

Besides the pure Poisson fluctuations the graphical presentation should also show the measurement resolution. We represent it by a horizontal bar for each point. Fig. 9.11 shows a deconvolution result for two regularizations of different strength. The curve represents the true distribution. Contrary to the representation of Fig. 9.9, the vertical error bars are now independent of the regularization strength. The horizontal bars indicate the resolution. With the strong regularization the valley is somewhat filled up due to the suppression of curvature and the points are following the curve better than expected from the error bars. The experienced scientist is able to judge also for the weak regularization that the curve is compatible with the data.

In multi-dimensional and in binning-free applications a graphical representation of the resolution is difficult but the transfer function has to be documented in some way.

Hypothesis Tests

10.1 Introduction

So far we treated problems where a data sample was used to discriminate between completely fixed competing hypotheses or to estimate free parameters of a given distribution. Now we turn to the case where we would like to find out whether a single hypothesis, without a completely defined alternative, is true or not. Some of the questions which we would like to address are the following:

1. Track parameters are fitted to some measured points. Are the deviations of the coordinate from the adjusted track compatible with statistical fluctuations or should we reject the hypothesis that they are related to a particle trajectory?
2. Can we describe a sample of e^+e^- reactions by quantum electrodynamics?
3. Do two samples obtained at different times in an experiment significantly differ from each other?
4. Is a signal in a spectral distribution significant?
5. Can we describe a background distribution significantly better by a linear or by a higher order polynomial.

To answer this kind of questions, we will have to set up a test procedure which quantifies the compatibility of a given data sample with a hypothesis. The test has to provide a quantitative result which is used to judge how plausible or unlikely a hypothesis is, definite judgements – right or wrong – are outside the possibilities of statistics. A test can never prove the validity of a hypothesis, it can only indicate problems with it.

A scientist who chooses a certain test procedure has to fix all parameters of the test before looking at the data¹. Under no circumstances is it allowed to base the selection of a test on the data which one wants to analyze, to optimize a test on the bases of the data or to terminate the running time of an experiment as a function of the output of a test. This would bias the result. Obviously, it is allowed to optimize a test with a part of the data which is excluded in the final analysis.

¹Scientists often call this a *blind* analysis.

Usually a test is associated with a decision: *accept* or *reject*. We will not always attempt a decision but confine ourselves to fix the parameters which form the bases for a possible decision.

As mentioned, we will primarily deal with a part of test theory which is especially important in natural sciences and also in many other empirical research areas, namely that only one hypothesis, we call it the null hypothesis H_0 , is tested while the admitted alternative is so vague or general that it cannot be parameterized. The alternative hypothesis H_1 is simply “ H_0 is false”. The question is whether the sample at hand is in agreement with H_0 or whether it deviates significantly from it. The corresponding tests are called *goodness-of-fit (GOF) tests*.

Strongly related to GOF tests are *two-sample tests* which check whether two samples belong to the same population.

At the end of this chapter we will treat another case in which we have a partially specified alternative and which plays an important role in physics. There the goal is to investigate whether a small signal is significant or explainable by a fluctuation of a background distribution corresponding to H_0 . We call this procedure *signal test*.

10.2 Some Definitions

Before addressing GOF tests, we introduce some notations.

10.2.1 Single and Composite Hypotheses

We distinguish between *simple* and *composite* hypotheses. The former fix the population uniquely. Thus H_0 : “The sample is drawn from a normal distribution with mean zero and variance one, i.e. $N(0, 1)$.” is a simple hypothesis. If the alternative is also simple, e.g. H_1 : “ $N(5, 1)$ ”, then we have the task to decide between two simple hypotheses which we have already treated in Chap. 6, Sect. 6.3. In this simple case there exists an optimum test, the likelihood ratio test.

Composite hypotheses are characterized by free parameters, like H_0 : “The sample is drawn from a normal distribution.” The user will adjust mean and variance of the normal distribution and test whether the adjusted Gaussian is compatible with the data.

The hypothesis that we want to test is always H_0 , the null hypothesis, and the alternative H_1 is in most cases the hypothesis that H_0 does not apply. H_1 then represents an infinite number of specified hypotheses.

10.2.2 Test Statistic, Critical Region and Significance Level

After we have fixed the null hypothesis and the admitted alternative H_1 , we must choose a *test statistic* $t(\mathbf{x})$, which is a function of the sample values $\mathbf{x} \equiv \{x_1, \dots, x_N\}$, possibly in such a way that the difference between the distribution $f(t|H_0)$ and distributions belonging to H_1 are as large as possible. To simplify the notation, we consider one-dimensional distributions. The generalization to multi-dimensional observations is trivial.

When we test, for instance, the hypothesis that a coordinate is distributed according to $N(0, 1)$, then for a sample consisting of a single measurement x , a reasonable test statistic is the absolute value $|x|$. We assume that if H_0 is wrong then $|x|$ would be large. A typical test statistic is the χ^2 deviation of a histogram from a prediction. Large values of χ^2 indicate that something might be wrong with the prediction.

Before we apply the test we have to fix a *critical region* K which leads to the rejection of H_0 if t is located inside of it. Under the condition that H_0 is true, the probability of rejecting H_0 is α , $P\{t \in K|H_0\} = \alpha$ where $\alpha \in [0, 1]$ normally is a small quantity (e.g. 5 %). It is called *significance level* or *size* of the test. For a test based on the χ^2 statistic, the critical region is defined by $\chi^2 > \chi_{\max}^2(\alpha)$ where the parameter $\chi_{\max}^2$ is a function of the significance level α . It fixes the range of the critical region.

To compute rejection probabilities we have to compute the p.d.f. $f(t)$ of the test statistic. In some cases it is known as we will see below, but in other cases it has to be obtained by Monte Carlo simulation. The distribution f has to include all experimental conditions under which t is determined, e.g. the measurement uncertainties of t .

10.2.3 Errors of the First and Second Kind, Power of a Test

After the test parameters are selected, we can apply the test to our data. If the actually obtained value of t is outside the critical region, $t \notin K$, then we accept H_0 , otherwise we reject it. This procedure implies four different outcomes with the following a priori probabilities:

1. $H_0 \cap t \in K$, $P\{t \in K|H_0\} = \alpha$: *error of the first kind*. (H_0 is true but rejected.),
2. $H_0 \cap t \notin K$, $P\{t \notin K|H_0\} = 1 - \alpha$ (H_0 is true and accepted.),
3. $H_1 \cap t \in K$, $P\{t \in K|H_1\} = 1 - \beta$ (H_0 is false and rejected.),
4. $H_1 \cap t \notin K$, $P\{t \notin K|H_1\} = \beta$: *error of the second kind* (H_0 is false but accepted.).

When we apply the test to a large number of data sets or events, then the rate α , the error of the first kind, is the inefficiency in the selection of H_0 events, while the rate β , the error of the second kind, represents the background with which the selected events are contaminated with H_1 events. Of course, for α given, we would like to have β as small as possible. Given the rejection region K which depends on α , also β is fixed for given H_1 . For a reasonable test we expect that β is monotonically decreasing with α increasing: With $\alpha \rightarrow 0$ also the critical region K is shrinking, while the *power* $1 - \beta$ must decrease, and the background is less suppressed. For fixed α , the *power* indicates the quality of a test, i.e. how well alternatives to H_0 can be rejected.

The power is a function, the *power function*, of the significance level α . Tests which provide maximum power $1 - \beta$ with respect to H_1 for all values of α are called *Uniformly Most Powerful (UMP) tests*. Only in rare cases where H_1 is restricted in some way, there exists an optimum, i.e. UMP test. If both hypotheses are simple then as already mentioned in Chap. 6, Sect. 6.3, according to a lemma of Neyman and E. S. Pearson, the likelihood ratio can be used as test statistic to discriminate between H_0 and H_1 and provides a uniformly most powerful test.

The interpretation of α and β as error rates makes sense when many experiments or data sets of the same type are investigated. In a search experiment where we want to find out whether a certain physical process or a phenomenon exists or in an isolated GOF test they refer to virtual experiments and it is not obvious which conclusions we can draw from their values.

10.2.4 P-Values

Strictly speaking, the result of a test is that a hypothesis is “accepted” or “rejected”. In most practical situations it is useful to replace this digital answer by a continuous parameter, the so called *p-value* which is a function $p(t)$ of the test statistic t and which measures the compatibility of the sample with the null hypothesis, a small value of p casting some doubt on the validity of H_0 . For an observed value t_{obs} of the test statistic, p is the probability to obtain a value $t \geq t_{obs}$ under the null hypothesis:

$$p = P\{t \geq t_{obs} | H_0\}.$$

To simplify its definition, we assume that the test statistic t is confined to values between zero and infinity with a critical region $t > t_c$ ². Its distribution under H_0 be $f_0(t)$. Then we have

$$p(t) = 1 - \int_0^t f_0(t') dt' = 1 - F_0(t). \quad (10.1)$$

Since p is a unique monotonic function of t , we can consider p as a normalized test statistic which is completely equivalent to t .

The relationship between the different quantities which we have introduced is illustrated in Fig. 10.1. The upper graph represents the p.d.f of the test statistic under H_0 . The critical region extends from t_c to infinity. The a priori rejection probability for a sample under H_0 is α , equal to the integral of the distribution of the test statistic over the critical region. The lower graph shows the p-value function. It starts at one and is continuously decreasing to zero at infinity. The smaller the test statistic is – think of χ^2 – the higher is the p-value. At $t = t_c$ the p-value is equal to the significance level α . The condition $p < \alpha$ leads to rejection of H_0 . Due to its construction, the p.d.f. of the p-value under H_0 is uniform. The name p-value is derived from the word probability, but its experimentally observed value does not represent the probability that H_0 is true. This is obvious from the fact that the p-value is a function of the selected test statistic. We will come back to this point when we discuss goodness-of-fit.

10.2.5 Consistency and Bias of Tests

A test is called *consistent* if its power tends to unity as the sample size tends to infinity. In other words: If we have an infinitely large data sample, we should always be able to decide between H_0 and the alternative H_1 .

We also want that independent of α the rejection probability for H_1 is higher than for H_0 , i.e. $\alpha < 1 - \beta$. Tests that violate this condition are called *biased*. Consistent tests are asymptotically unbiased.

²This condition can always be realized for one-sided tests. Two-sided tests are rare – see Example 129.

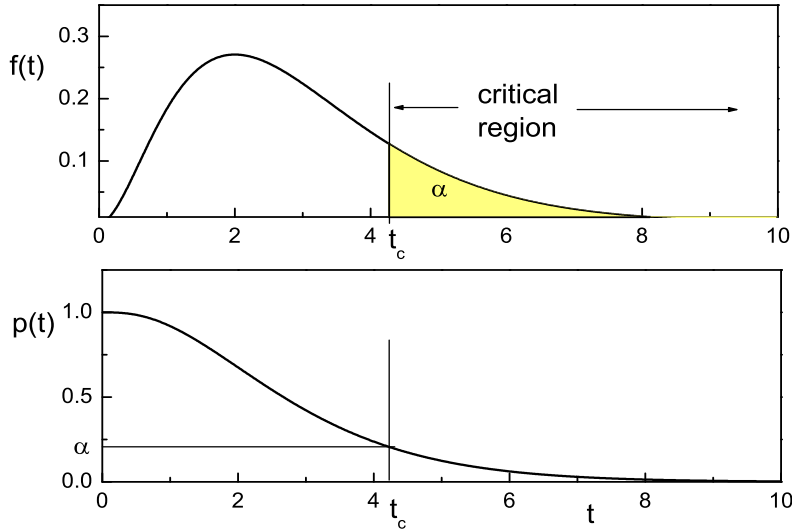


Fig. 10.1. Distribution of a test statistic and corresponding p-value curve.

When H_1 represents a family of distributions, consistency and non-biasedness are valid only if they apply to all members of the family. Thus in case that the alternative H_1 is not specified, a test is biased if there is an arbitrary hypothesis different from H_0 with rejection probability less than α and it is inconsistent if we can find a hypothesis different from H_0 which is not rejected with power unity in the large sample limit.

Example 125. Bias and inconsistency of a test

Assume, we select in an experiment events of the type $K^0 \rightarrow \pi^+\pi^-$. The invariant mass $m_{\pi\pi}$ of the pion pairs has to match the K^0 mass. Due to the finite experimental resolution the experimental masses of the pairs are normally distributed around the kaon mass m_K with variance σ^2 . With the null hypothesis H_0 that we observe $K^0 \rightarrow \pi^+\pi^-$ decays, we may apply to our sample a test with the test quantity $t = (m_{\pi\pi} - m_K)^2/\sigma^2$, the normalized mean quadratic difference between the observed masses of N pairs and the nominal K^0 mass. Our sample is accepted if it satisfies $t < t_0$ where t_0 is the critical quantity which determines the error of the first kind α and the acceptance $1 - \alpha$. The distribution of Nt under H_0 is a χ^2 distribution with N degrees of freedom. Clearly, the test is biased, because we can imagine mass distributions with acceptance larger than $1 - \alpha$, for instance a uniform distribution in the range $t \leq t_0$. This test is also inconsistent, because it would favor this specific realization of H_1 also for infinitely large samples. Nevertheless it is not unreasonable for very small samples in the considered case and for $N = 1$ there is no alternative. The situation is different for large samples where more powerful tests exist which take into account the Gaussian shape of the expected distribution under H_0 .

While consistency is a necessary condition for a sensible test applied to a large sample, bias and inconsistency of a test applied to a small sample cannot always be avoided and are tolerable under certain circumstances.

10.3 Goodness-of-Fit Tests

10.3.1 General Remarks

Goodness-of-fit (GOF) tests check whether a sample is compatible with a given distribution. Even though this is not possible in principle without a well defined alternative, this works quite well in practice, the reason being that the choice of the test statistic is influenced by speculations about the behavior of alternatives, speculations which are based on our experience. Our presumptions depend on the specific problem to be solved and therefore very different testing procedures are on the market.

In the empirical research outside the exact sciences, questions like “Is a certain drogue effective?”, “Have girls less mathematical ability than boys?”, “Does the IQ follow a normal distribution?” are to be answered. In the natural sciences, GOF tests usually serve to detect unknown systematic errors in experimental results. When we measure the mean life of an unstable particle, we know that the lifetime distribution is exponential but to apply a GOF test is informative, because a low p-value may indicate a contamination of the events by background or problems with the experimental equipment. But there are also situations where we accept or reject hypotheses as a result of a test. Examples are event selection (e.g. B-quark production), particle track selection on the bases of the quality of reconstruction and particle identification, (e.g. electron identification based on calorimeter or Cerenkov information). Typical for these examples is that we examine a number of similar objects and accept a certain error rate α , while when we consider the p-value of the final result of an experiment, discussing an error rate does not make sense.

An experienced scientist has a quite good feeling for deviations between two distributions just by looking at a plot. For instance, when we examine the statistical distribution of Fig. 10.2, we will realize that its description by an exponential distribution is rather unsatisfactory. The question is: How can we quantify the disagreement? Without a concrete alternative it is rather difficult to make a judgement.

Let us discuss a different example: Throwing a dice produces “1” ten times in sequence. Is this result compatible with the assumption H_0 that the dice is unbiased? Well, such a sequence does not occur frequently and the guess that something is wrong with the dice is well justified. On the other hand, the sequence of ten times “1” is not less probable than any other sequence, namely $(1/6)^{10} = 1.7 \cdot 10^{-8}$. Our doubt relies on our experience: We have an alternative to H_0 in mind, namely asymmetric dice. We can imagine asymmetric dice but not dice that produce with high probability a sequence like “4,5,1,6,3,3,6,2,5,2”. As a consequence we would choose a test which is sensitive to deviations from a uniform distribution. When we test a random number generator we would be interested, for example, in a periodicity of the results or a correlation between subsequent numbers and we would choose a different test. In GOF tests, we cannot specify H_1 precisely, but we need to have an idea of it which then enters in the selection of the test. We search for test parameters where we suppose that they discriminate between the null hypothesis and possible alternatives.

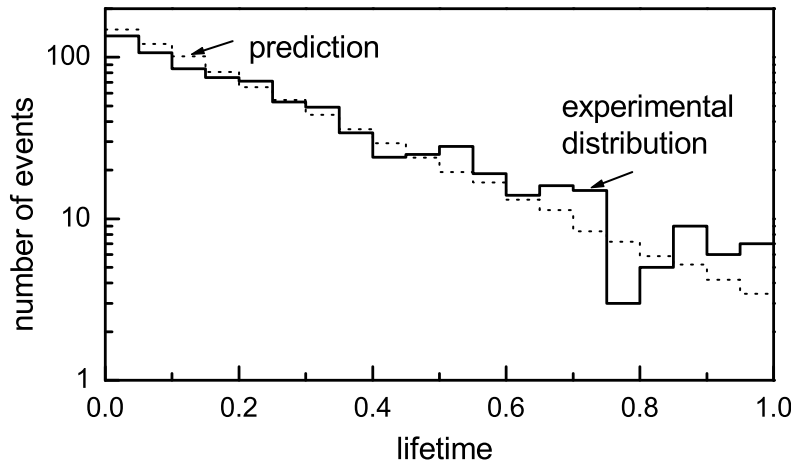


Fig. 10.2. Comparison of an experimental distribution to a prediction.

However, there is not such a thing as a best test quantity as long as the alternative is not completely specified.

A typical test quantity is the χ^2 -variable which we have introduced to adjust parameters of functions to experimental histograms or measured points with known error distributions. In the least square method of parameter inference, see Chap. 7, the parameters are fixed such that the sum χ^2 of the normalized quadratic deviations is minimum. Deviating parameter values produce larger values of χ^2 , consequently we expect the same effect when we compare the data to a wrong hypothesis. If χ^2 is abnormally large, it is likely that the null hypothesis is not correct.

Unfortunately, physicists use almost exclusively the χ^2 test, even though for many applications more powerful tests are available. Scientists also often overestimate the significance of the χ^2 test results. Other tests like the *Kolmogorov–Smirnov Test* and tests of the *Cramer–von Mises family* avoid the always somewhat arbitrary binning of histograms in the χ^2 test. These tests are restricted to univariate distributions, however. Other binning-free methods can also be applied to multivariate distributions.

Sometimes students think that a good test statistic would be the likelihood L_0 of the null hypothesis, i.e. for H_0 with single event distribution $f_0(x)$ the product $\prod f_0(x_i)$. That this is not a good idea is illustrated in Fig. 10.3 where the null hypothesis is represented by a fully specified normal distribution. From the two samples, the narrow one clearly fits the distribution worse but it has the higher likelihood. A sample where all observations are located at the center would per definition maximize the likelihood but such a sample would certainly not support the null hypothesis.

While the indicated methods are *distribution-free*, i.e. applicable to arbitrary distributions specified by H_0 , there are procedures to check the agreement of data with specific distributions like normal, uniform or exponential distributions. These methods are of inferior importance for physics applications. We will deal only with *distribution-free* methods.

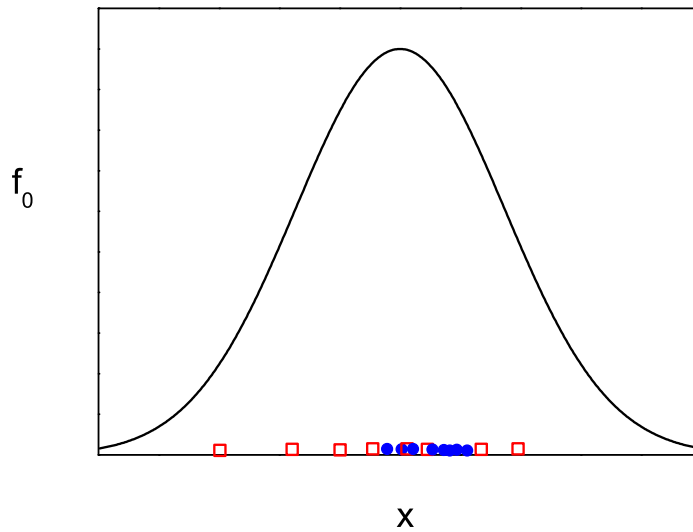


Fig. 10.3. Two different samples and a hypothesis.

We will also exclude tests based on order statistics from our discussion. These tests are mainly used to test properties of time series and are not very powerful in most of our applications.

At the end of this section we want to stress that parameter inference with a valid hypothesis and GOF test which doubt the validity of a hypothesis touch two completely different problems. Whenever possible deviations can be parameterized it is always appropriate to determine the likelihood function of the parameter and use the likelihood ratio to discriminate between different parameter values.

A good review of GOF tests can be found in [57], in which, however, more recent developments are missing.

10.3.2 P-Values

Interpretation and Use of P-Values

We have introduced p-values p in order to dispose of a quantity which measures the agreement between a sample and a distribution $f_0(t)$ of the test statistic t . Small p-values should indicate a bad agreement. Since the distribution of p under H_0 is uniform in the interval $[0, 1]$, all values of p in this interval are equally probable. When we reject a hypothesis under the condition $p < 0.1$ we have a probability of 10% to reject H_0 . The rejection probability would be the same for a rejection region $p > 0.9$. The reason for cutting at low p-values is the expectation that distributions of H_1 would produce low p-values.

The p-value is not the probability that the hypothesis under test is true. It is the probability under H_0 to obtain a p-value which is smaller than the one actually observed. A p-value between zero and p is expected to occur in the fraction p of experiments if H_0 is true.

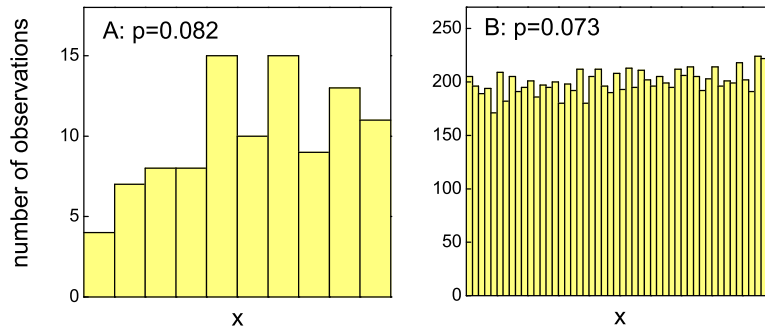


Fig. 10.4. Comparison of two experimental histograms to a uniform distribution.

Example 126. The p-value and the probability of a hypothesis

In Fig. 10.4 we have histogrammed two distributions from two simulated experiments *A* and *B*. Are these uniform distributions? For experiment *B* with 10000 observations this is conceivable, while for experiment *A* with only 100 observations it is difficult to guess the shape of the distribution. Alternatives like strongly rising distributions are more strongly excluded in *B* than in *A*. We would therefore attribute a higher probability for the validity of the hypothesis of a uniform distribution for *B* than for *A*, but the p-values based on the χ^2 test are very similar in both cases, namely $p \approx 0.08$. Thus the deviations from a uniform distribution would have in both cases the same significance

We learn from this example also that *the p-value is more sensitive to deviations in large samples than in small samples*. Since in practice small unknown systematic errors can rarely be excluded, we should not be astonished that in high statistics experiments often small p-values occur. The systematic uncertainties which usually are not considered in the null hypothesis then dominate the purely statistical fluctuation.

Even though we cannot transform significant deviations into probabilities for the validity of a hypothesis, they provide useful hints for hidden measurement errors or contamination with background. In our example a linearly rising distribution has been added to uniform distributions. The fractions were 45% in experiment *A* and 5% in experiment *B*.

In some experimental situations we are able to compare many replicates of measurements to the same hypothesis. In particle physics experiments usually a huge number of tracks has to be reconstructed. The track parameters are adjusted by a χ^2 fit to measured points assuming normally distributed uncertainties. The χ^2 value of each fit can be used as a test statistic and transformed into a p-value, often called χ^2 probability. Histograms of p-values obtained in such a way are very instructive. They often look like the one shown in Fig. 10.5. The plot has two interesting features: It is slightly rising with increasing p-value which indicates that the errors have been slightly overestimated. The peak at low p-values is due to fake tracks which do not correspond to particle trajectories and which we would eliminate almost completely by a cut at about $p_c = 0.05$. We would have to pay for it by a loss of good tracks of

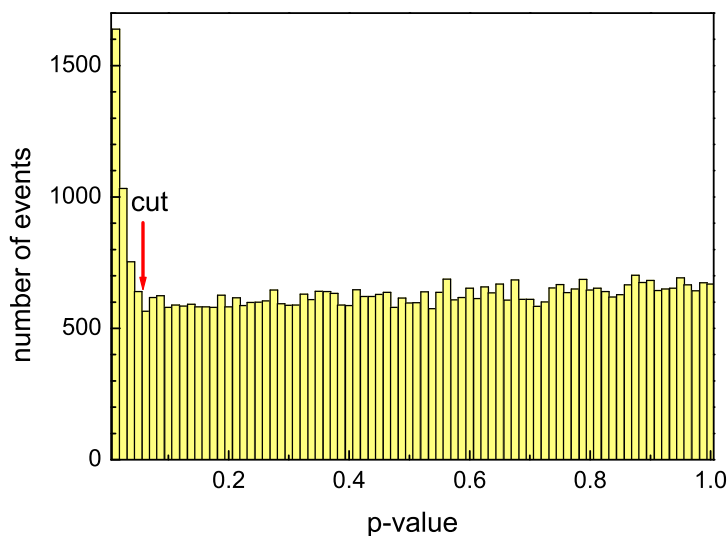


Fig. 10.5. Experimental distribution of p-values.

somewhat less than 5 %. A more precise estimate of the loss can be obtained by an extrapolation of the smooth part of the p-value distribution to $p = 0$.

Combination of p-Values

If two p-values p_1, p_2 which have been derived from independent test statistics t_1, t_2 are available, we would like to combine them to a single p-value p . The at first sight obvious idea to set $p = p_1 p_2$ suffers from the fact that the distribution of p will not be uniform. A popular but arbitrary choice is

$$p = p_1 p_2 [1 - \ln(p_1 p_2)] \quad (10.2)$$

which can be shown to be uniformly distributed [65]. This choice has the unpleasant feature that the combination of the p-values is not associative, i.e. $p[(p_1, p_2), p_3] \neq p[p_1, (p_2, p_3)]$. There is no satisfactory way to combine p-values.

We propose, if possible, not to use (10.2) but to go back to the original test statistics and construct from them a combined statistic t and the corresponding p-distribution. For instance, the obvious combination of two χ^2 statistics would be $t = \chi_1^2 + \chi_2^2$.

10.3.3 The χ^2 Test in Generalized Form

The Idea of the χ^2 Comparison

We consider a sample of N observations which are characterized by the values x_i of a variable x and a prediction $f_0(x)$ of their distribution. We subdivide the range of x into B intervals to which we attach sequence numbers k . The prediction p_k for the probability that an observation is contained in interval k is:

$$p_k = \int_k f_0(x) dx ,$$

with $\sum p_k = 1$. The integration extends over the interval k . The number of sample observations d_k found in this bin has to be compared with the expectation value Np_k . To interpret the deviation $d_k - Np_k$, we have to evaluate the expected mean quadratic deviation δ_k^2 under the condition that the prediction is correct. Since the distribution of the observations into bins follows a binomial distribution, we have

$$\delta_k^2 = Np_k(1 - p_k) .$$

Usually the observations are distributed into typically 10 to 50 bins. Thus the probabilities p_k are small compared to unity and the expression in brackets can be omitted. This is the Poisson approximation of the binomial distribution. The mean quadratic deviation is equal to the number of expected observations in the bin:

$$\delta_k^2 = Np_k .$$

We now normalize the observed to the expected mean quadratic deviation,

$$\chi_k^2 = \frac{(d_k - Np_k)^2}{Np_k} ,$$

and sum over all B bins:

$$\chi^2 = \sum_{k=1}^B \frac{(d_k - Np_k)^2}{Np_k} . \quad (10.3)$$

By construction we have:

$$\begin{aligned} \langle \chi_k^2 \rangle &\approx 1 , \\ \langle \chi^2 \rangle &\approx B . \end{aligned}$$

If the quantity χ^2 is considerably larger than the number of bins, then obviously the measurement deviates significantly from the prediction.

A significant deviation to small values $\chi^2 \ll B$ even though considered as unlikely is tolerated, because we know that alternative hypotheses do not produce smaller $\langle \chi^2 \rangle$ than H_0 .

The χ^2 Distribution and the χ^2 Test

We now want to be more quantitative. If H_0 is valid, the distribution of χ^2 follows to a very good approximation the χ^2 distribution which we have introduced in Sect. 3.6.7 and which is displayed in Fig. 3.20. The approximation relies on the approximation of the distribution of observations per bin by a normal distribution, a condition which in most applications is sufficiently good if the expected number of entries per bin is larger than about 10. The parameter *number of degrees of freedom (NDF)* f of the χ^2 distribution is equal to the expectation value and to the number of bins minus one:

$$\langle \chi^2 \rangle = f = B - 1 . \quad (10.4)$$

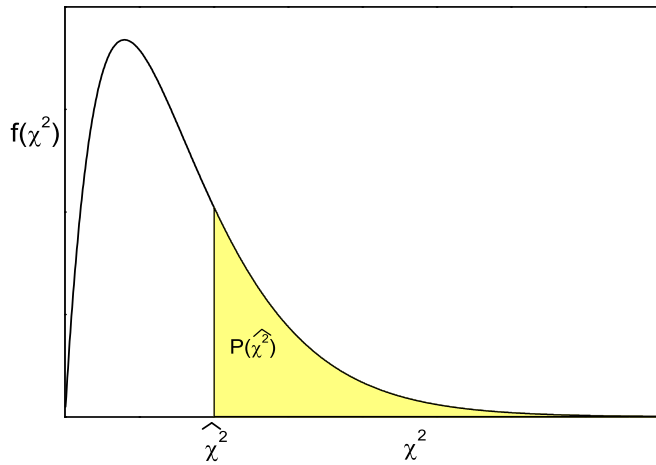


Fig. 10.6. P-value for the obseration $\hat{\chi}^2$.

Originally we had set $\langle \chi^2 \rangle \approx B$ but this relation overestimates χ^2 slightly. The smaller value $B - 1$ is plausible because the individual deviations are somewhat smaller than one – remember, we had approximated the binomial distribution by a Poisson distribution. For instance, in the limit of a single bin, the mean deviation is not *one* but *zero*. We will come back to this point below.

In some cases we have not only a prediction of the shape of the distribution but also a prediction N_0 of the total number of observations. Then the number of entries in each bin should follow a Poisson distribution with mean $N_0 p_k$, (10.3) has to be replaced by

$$\chi^2 = \sum_{k=1}^B \frac{(d_k - N_0 p_k)^2}{N_0 p_k}. \quad (10.5)$$

and we have $f = B = \langle \chi^2 \rangle$.

In experiments with low statistics the approximation that the distribution of the number of entries in each bin follows a normal distribution is sometimes not justified and the distribution of the χ^2 quantity as defined by (10.3) or (10.5) is not very well described by a χ^2 distribution. Then we have the possibility to determine the distribution of our χ^2 variable under H_0 by a Monte Carlo simulation³.

In Fig. 10.6 we illustrate how we can deduce the p-value or χ^2 probability from the distribution and the experimental value $\hat{\chi}^2$ of our test statistic χ^2 . The experimental value $\hat{\chi}^2$ divides the χ^2 distribution, which is fixed through the number of degrees of freedom, and which is independent of the data, into two parts. According to its definition (10.1), the p-value $p(\hat{\chi}^2)$ is equal to the area of the right hand part. It is the fraction of many imagined experiments where χ^2 is larger than the experimentally observed value $\hat{\chi}^2$ – always assuming that H_0 is correct. As mentioned above, high values of χ^2 and correspondingly low values of p indicate that the theoretical description is inadequate to describe the data. The reason is in most cases found in experimental problems.

³We have to be especially careful when α is small.

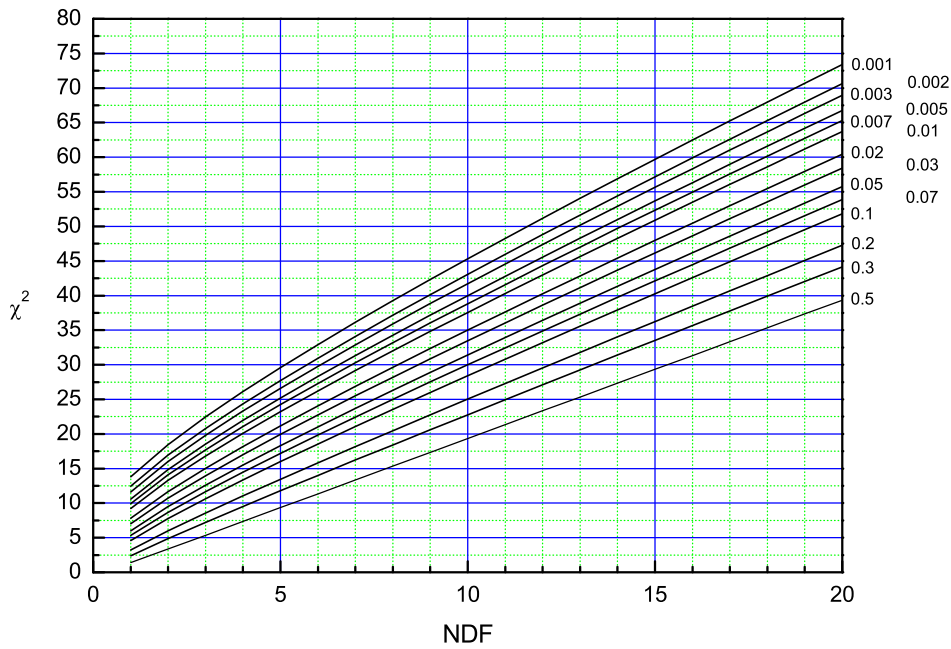


Fig. 10.7. Critical χ^2 values as a function of the number of degrees of freedom with the significance level as parameter.

The χ^2 comparison becomes a test, if we accept the theoretical description of the data only if the p-value exceeds a critical value, the significance level α , and reject it for $p < \alpha$. The χ^2 test is also called *Pearson test* after the statistician Karl Pearson who has introduced it already in 1900.

Figure 10.7 gives the critical values of χ^2 , as a function of the number of degrees of freedom with the significance level as parameter. To simplify the presentation we have replaced the discrete points by curves. The p-value as a function of χ^2 with NDF as parameter is available in the form of tables or in graphical form in many books. For large f , about $f > 20$, the χ^2 distribution can be approximated sufficiently well by a normal distribution with mean value $x_0 = f$ and variance $s^2 = 2f$. We are then able to compute the p-values from integrals over the normal distribution. Tables can be found in the literature or alternatively, the computation can be performed with computer programs like Mathematica or Maple.

The Choice of Binning

There is no general rule for the choice of the number and width of the histogram bins for the χ^2 comparison but we note that the χ^2 test loses significance when the number of bins becomes too large.

To estimate the effect of fine binning for a smooth deviation, we consider a systematic deviation which is constant over a certain region with a total number of entries N_0 and which produces an excess of εN_0 events. Partitioning the region into B bins would add to the statistical χ^2 in each single bin the contribution:

Table 10.1. P-values for χ^2 and EDF statistic.

test	p value
χ^2 , 50 Bins	0.10
χ^2 , 50 Bins, e.p.	0.05
χ^2 , 20 Bins	0.08
χ^2 , 20 Bins, e.p.	0.07
χ^2 , 10 Bins	0.06
χ^2 , 10 Bins, e.p.	0.11
χ^2 , 5 Bins	0.004
χ^2 , 5 Bins, e.p.	0.01
D_{max}	0.005
W^2	0.001
A^2	0.0005

$$\chi_s^2 = \frac{(\varepsilon N_0/B)^2}{N_0/B} = \frac{\varepsilon^2 N_0}{B}.$$

For B bins we increase χ^2 by $\varepsilon^2 N_0$ which is to be compared to the purely statistical contribution χ_0^2 which is in average equal to B . The significance S , i.e. the systematic deviation in units of the expected fluctuation $\sqrt{2B}$ is

$$S = \varepsilon^2 \frac{N_0}{\sqrt{2B}}.$$

It decreases with the square root of the number of bins.

We recommend a fine binning only if deviations are considered which are restricted to narrow regions. This could be for instance pick-up spikes. These are pretty rare in our applications. Rather we have systematic deviations produced by non-linearity of measurement devices or by background and which extend over a large region. Then wide intervals are to be preferred.

In [58] it is proposed to choose the number of bins according to the formula $B = 2N^{2/5}$ as a function of the sample size N .

Example 127. Comparison of different tests for background under an exponential distribution

In Fig. 10.2 a histogrammed sample is compared to an exponential. The sample contains, besides observations following this distribution, a small contribution of uniformly distributed events. From Table 10.1 we recognize that this defect expresses itself by small p-values and that the corresponding decrease becomes more pronounced with decreasing number of bins.

Some statisticians propose to adjust the bin parameters such that the number of events is the same in all bins. In our table this partitioning is denoted by e.p. (equal probability). In the present example this does not improve the significance.

The value of χ^2 is independent of the signs of the deviations. However, if several adjacent bins show an excess (or lack) of events like in the left hand histogram of Fig. 10.8 this indicates a systematic discrepancy which one would not expect at the same level for the central histogram which produces the same value for χ^2 . Because correlations between neighboring bins do not enter in the test, a visual inspection is often more effective than the mathematical test. Sometimes it is helpful to present

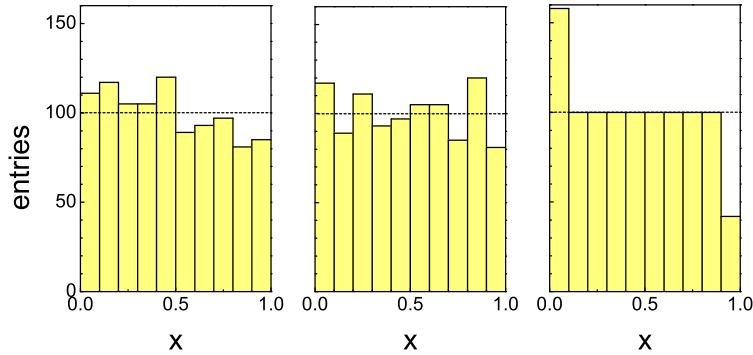


Fig. 10.8. The left hand and the central histogram produce the same χ^2 p-value, the left hand and the right hand histograms produce the same Kolmogorov p-value.

Table 10.2. χ^2 values with sign of deviation for a two-dimensional histogram.

i \ j	1	2	3	4	5	6	7	8
1	0.1	-0.5	1.3	-0.3	1.6	-1.1	2.0	1.2
2		-1.9	0.5	-0.4	0.1	-1.2	1.3	1.5
3			-1.2	-0.8	0.2	0.1	1.3	1.9
4				0.2	0.7	-0.6	1.1	2.2

for every bin the value of χ^2 multiplied by the sign of the deviation either graphically or in form of a table.

Example 128. χ^2 comparison for a two-dimensional histogram

In Table 10.2 for a two-dimensional histogram the values of χ^2 accompanied with the sign are presented. The absolute values are well confined in the range of our expectation but near the right hand border we observe an accumulation of positive deviations which point to a systematic effect.

Generalization to Arbitrary Measurements

The Pearson method can be generalized to arbitrary measurements y_k with mean square errors δ_k^2 . For theoretical predictions t_k we compute χ^2 ,

$$\chi^2 = \sum_{k=1}^N \frac{(y_k - t_k)^2}{\delta_k^2},$$

where χ^2 follows a χ^2 distribution of $f = N$ degrees of freedom. A necessary condition for the validity of the χ^2 distribution is that the uncertainties follow a normal distribution. For a large number N of individual measurements the central limit theorem applies and we can relax the condition of normality. Then χ^2 is approximately normally distributed with mean N and variance $2N$.

A further generalization is given in the Appendix 13.8 where weighted events and statistical errors of the theoretical predictions, resulting from the usual Monte Carlo calculation, are considered.

Remark: The quantity δ^2 has to be calculated under the assumption that the theoretical description which is to be tested is correct. This means, that normally the raw measurement error cannot be inserted. For example, instead of ascribing to a measured quantity an error δ'_k which is proportional to its value y_k , a corrected error

$$\delta_k = \delta'_k \frac{t_k}{y_k}$$

should be used.

Sometimes extremely small values of χ^2 are presented. The reason is in most cases an overestimation of the errors.

The variable χ^2 is frequently used to separate signal events from background. To this end, the experimental distribution of χ^2 is transformed into a p-value distribution like the one presented in Fig. 10.5. In this situation it is not required that χ^2 follows the χ^2 distribution. It is only necessary that it is a discriminative test variable.

The χ^2 Test for Composite Hypotheses

In most cases measurements do not serve to verify a fixed theory but to estimate one or more parameters. The method of least squares for parameter estimation has been discussed in Sect. 7.2. To fit a curve $y = t(x, \theta)$ to measured points y_i with Gaussian errors σ_i , $i = 1, \dots, N$, we minimize the quantity

$$\chi^2 = \sum_{i=1}^N \frac{(y_i - t(x_i, \theta_1, \dots, \theta_Z))^2}{\sigma_i^2}, \quad (10.6)$$

with respect to the Z free parameters θ_k .

It is plausible that with increasing number of parameters, which are adjusted, the description of the data improves, χ^2 decreases. In the extreme case where the number of parameters is equal to the number N of measured points or histogram bins it becomes zero. The distribution of χ^2 in the general case where Z parameters are adjusted follows under conditions to be discussed below a χ^2 distribution of $f = N - Z$ degrees of freedom.

Setting in (10.6) $z_i = (y_i - t(x_i, \theta))/\sigma_i$ we may interpret $\chi^2 = \sum_1^N z_i^2$ as the (squared) distance of a point $\mathbf{z}$ with normally distributed components from the origin in an N -dimensional space. If all parameters are fixed except one, say θ_1 , which is left free and adjusted to the data by minimizing χ^2 , we have to set the derivative with respect to θ_1 equal to zero:

$$-\frac{1}{2} \frac{\partial \chi^2}{\partial \theta_1} = \sum_{i=1}^N z_i \frac{\partial t}{\partial \theta_1} / \sigma_i = 0.$$

If t is a linear function of the parameters, an assumption which is often justified at least approximately⁴, the derivatives are constants, and we get a linear relation (constraint) of the form $c_1 z_1 + \dots + c_N z_N = 0$. It defines a $(N - 1)$ -dimensional subspace,

⁴Note that also the σ_i have to be independent of the parameters.

a hyperplane containing the origin, of the N -dimensional z -space. Consequently, the distance in z -space is confined to this subspace and derived from $N - 1$ components. For Z free parameters we get Z constraints and a $(N - Z)$ -dimensional subspace. The independent components (dimensions) of this subspace are called degrees of freedom. The number of degrees of freedom is $f = N - Z$ as pretended above. Obviously, the sum of f squared components will follow a χ^2 distribution with f degrees of freedom.

In the case of fitting a normalized distribution to a histogram with B bins which we have considered above, we had to set (see Sect. 3.6.7) $f = B - 1$. This is explained by a constraint of the form $z_1 + \dots + z_B = 0$ which is valid due to the equality of the normalization for data and theory.

The χ^2 Test for Small Samples

When the number of entries per histogram bin is small, the approximation that the variations are normally distributed is not justified. Consequently, the χ^2 distribution should no longer be used to calculate the p-value.

Nevertheless we can use in this situation the sum of quadratic deviations χ^2 as test statistic. The distribution $f_0(\chi^2)$ has then to be determined by a Monte Carlo simulation. The test then is slightly biased but the method still works pretty well.

Warning

The assumption that the distribution of the test statistic under H_0 is described by a χ^2 distribution relies on the following assumptions: 1. The entries in all bins of the histogram are normally distributed. 2. The expected number of entries depends linearly on the free parameters in the considered parameter range. An indication for a non-linearity are asymmetric errors of the adjusted parameters. 3. The estimated uncertainties σ_i in the denominators of the summands of χ^2 are independent of the parameters. Deviations from these conditions affect mostly the distribution at large values of χ^2 and thus the estimation of small p-values. Corresponding conditions have to be satisfied when we test the GOF of a curve to measured points. Whenever we are not convinced about their validity we have to generate the distribution of χ^2 by a Monte Carlo simulation.

10.3.4 The Likelihood Ratio Test

General Form

The likelihood ratio test compares H_0 to a parameter dependent alternative H_1 which includes H_0 as a special case. The two hypothesis are defined through the p.d.f.s $f(\mathbf{x}|\boldsymbol{\theta})$ and $f(\mathbf{x}|\boldsymbol{\theta}_0)$ where the parameter set $\boldsymbol{\theta}_0$ is a subset of $\boldsymbol{\theta}$, often just a fixed value of $\boldsymbol{\theta}$. The test statistic is the likelihood ratio λ , the ratio of the likelihood of H_0 and the likelihood of H_1 where the parameters are chosen such that they maximize the likelihoods for the given observations $\mathbf{x}$. It is given by the expression

$$\lambda = \frac{\sup L(\boldsymbol{\theta}_0|\mathbf{x})}{\sup L(\boldsymbol{\theta}|\mathbf{x})}, \quad (10.7)$$

or equivalently by

$$\ln \lambda = \sup \ln L(\boldsymbol{\theta}_0|\mathbf{x}) - \sup \ln L(\boldsymbol{\theta}|\mathbf{x}) .$$

If $\boldsymbol{\theta}_0$ is a fixed value, this expression simplifies to $\ln \lambda = \ln L(\boldsymbol{\theta}_0|\mathbf{x}) - \sup \ln L(\boldsymbol{\theta}|\mathbf{x})$.

From the definition (10.7) follows that λ always obeys $\lambda \leq 1$.

Example 129. Likelihood ratio test for a Poisson count

Let us assume that H_0 predicts $\mu_0 = 10$ decays in an hour, observed are 8. The likelihood to observe 8 for the Poisson distribution is $L_0 = P(8|10) = e^{-10}10^8/8!$. The likelihood is maximal for $\mu = 8$, it is $L = P(8|8) = e^{-8}8^8/8!$. Thus the likelihood ratio is $\lambda = P(8|10)/P(8|8) = e^{-2}(5/4)^8 = 0.807$. The probability to observe a ratio smaller than or equal to 0.807 is

$$p = \sum_k P(k|10) \quad \text{for } k \text{ with } P(k|10) \leq 0.807 P(10|10) .$$

Relevant numerical values of $\lambda(k, \mu_0) = P(k|\mu_0)/P(k|k)$ and $P(k|\mu_0)$ for $\mu_0 = 10$ are given in Table 10.3 It is seen, that the sum over k runs over

Table 10.3. Values of λ and P , see text.

k	8	9	10	11	12	13
λ	0.807	0.950	1.000	0.953	0.829	0.663
P	0.113	0.125	0.125	0.114	0.095	0.073

all k , except $k = 9, 10, 11, 12$: $p = \sum_{k=0}^8 P(k|10) + \sum_{k=13}^{\infty} P(k|10) = 1 - \sum_{k=9}^{12} P(k|10) = 0.541$ which is certainly acceptable. This is an example for the p-value of a two-sided test.

The likelihood ratio test in this general form is useful to discriminate between a specific and a more general hypothesis, a problem which we will study in Sect. 10.5.2. To apply it as a goodness-of-fit test, we have to histogram the data.

The Likelihood Ratio Test for Histograms

We have shown that the likelihood $L_0 = \prod_i f_0(x_i)$ of a sample cannot be used as a test statistic, but when we combine the data into bins, a likelihood ratio can be defined for the histogram and used as test quantity. The test variable is the ratio of the likelihood for the hypothesis that the bin content is predicted by H_0 and the likelihood for the hypothesis that maximizes the likelihood for the given sample. The latter is the likelihood for the hypothesis where the prediction for the bin coincides with its content. If H_0 is not simple, we take the ratio of the maximum likelihood allowed by H_0 and the unconstrained maximum of L .

For a bin with content d , prediction t and p.d.f. $f(d|t)$ this ratio is $\lambda = f(d|t)/f(d|d)$ since at $t = d$ the likelihood is maximal. For the histogram we have to multiply the ratios of the B individual bins. Instead we change to the log-likelihoods and use as test statistic

$$V = \ln \lambda = \sum_{i=1}^B [\ln f(d_i|t_i) - \ln f(d_i|d_i)] .$$

If the bin content follows the Poisson statistics we get (see Chap. 6, Sect. 6.5.6)

$$\begin{aligned} V &= \sum_{i=1}^B [-t_i + d_i \ln t_i - \ln(d_i!) + d_i - d_i \ln d_i + \ln(d_i!)] \\ &= \sum_{i=1}^B [d_i - t_i + d_i \ln(t_i/d_i)] . \end{aligned}$$

The distribution of the test statistic V is not universal, i.e. not independent of the distribution to be tested as in the case of χ^2 . It has to be determined by a Monte Carlo simulation. In case parameters of the prediction have been adjusted to data, the parameter adjustment has to be included in the simulation.

The method can be extended to weighted events and to the case of Monte Carlo generated predictions with corresponding statistical errors, see Appendix 13.8.

Asymptotically, $N \rightarrow \infty$, the test statistic V approaches $-\chi^2/2$ as is seen from the expansion of the logarithm, $\ln(1+x) \approx x - x^2/2$. After introducing $x_i = (d_i - t_i)/t_i$ which, according to the law of large numbers, becomes small for large d_i , we find

$$\begin{aligned} V &= \sum_{i=1}^B [t_i x_i - t_i(1 + x_i) \ln(1 + x_i)] \\ &\approx \sum_{i=1}^B t_i \left[x_i - (1 + x_i) \left(x_i - \frac{1}{2} x_i^2 \right) \right] \\ &\approx \sum_{i=1}^B t_i \left(-\frac{1}{2} x_i^2 \right) = -\frac{1}{2} \sum_{i=1}^B \left(\frac{(d_i - t_i)^2}{t_i} \right) = -\frac{1}{2} \chi_B^2 , \end{aligned}$$

and thus $-2V$ is distributed according to a χ^2 distribution with B degrees of freedom, but then we may also use directly the χ^2 test.

If the prediction is normalized to the data, we have to replace the Poisson distribution by the multinomial distribution. We omit the calculation and present the result:

$$V = \sum_{i=1}^B d_i \ln(t_i/d_i) .$$

In this case, V approaches asymptotically the χ^2 distribution with $B - 1$ degrees of freedom.

10.3.5 The Kolmogorov–Smirnov Test

The subdivision of a sample into intervals is arbitrary and thus subjective. Unfortunately some experimenters use the freedom to choose histogram bins such that the data agree as well as possible with the theoretical description in which they believe. This problem is excluded in binning-free tests which have the additional advantage that they are also applicable to small samples.

The Kolmogorov–Smirnov test compares the distribution function

$$F_0(x) = \int_{-\infty}^x f_0(x) dx$$

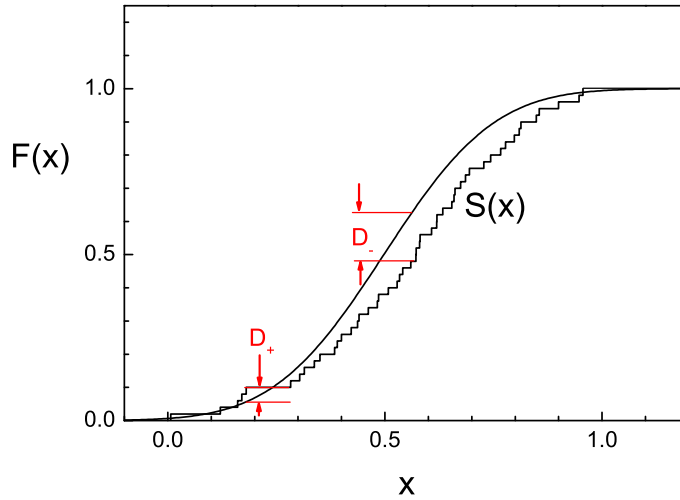


Fig. 10.9. Comparison of the empirical distribution function $S(x)$ with the theoretical distribution function $F(x)$.

with the corresponding experimental quantity S ,

$$S(x) = \frac{\text{Number of observations with } x_i < x}{\text{Total number}} .$$

The test statistic is the maximum difference D between the two functions:

$$\begin{aligned} D &= \sup |F(x) - S(x)| \\ &= \sup(D_+, D_-) . \end{aligned}$$

The quantities D_+, D_- denote the maximum positive and negative difference, respectively. $S(x)$ is a step function, an experimental approximation of the distribution function and is called *Empirical Distribution Function (EDF)*. It is depicted in Fig. 10.9 for an example and compared to the distribution function $F(x)$ of H_0 . To calculate $S(x)$ we sort all N elements in ascending order of their values, $x_i < x_{i+1}$ and add $1/N$ at each location x_i to $S(x)$. Then $S(x_i)$ is the fraction of observations with x values less or equal to x_i ,

$$\begin{aligned} S(x_i) &= \frac{i}{N} , \\ S(x_N) &= 1 . \end{aligned}$$

As in the χ^2 test we can determine the expected distribution of D , which will depend on N and transform the experimental value of D into a p-value. To get rid of the N dependence of the theoretical D distribution we use $D^* = \sqrt{N}D$. Its distribution under H_0 is for not too small N ($N > \approx 100$) independent of N and available in form of tables and graphs. For event numbers larger than 20 the approximation $D^* = D(\sqrt{N} + 0.12 + 0.11/\sqrt{N})$ is still a very good approximation⁵. The function $p(D^*)$ is displayed in Fig. 10.10.

⁵ D does not scale exactly with $\sqrt{N}$ because S increases in discrete steps.

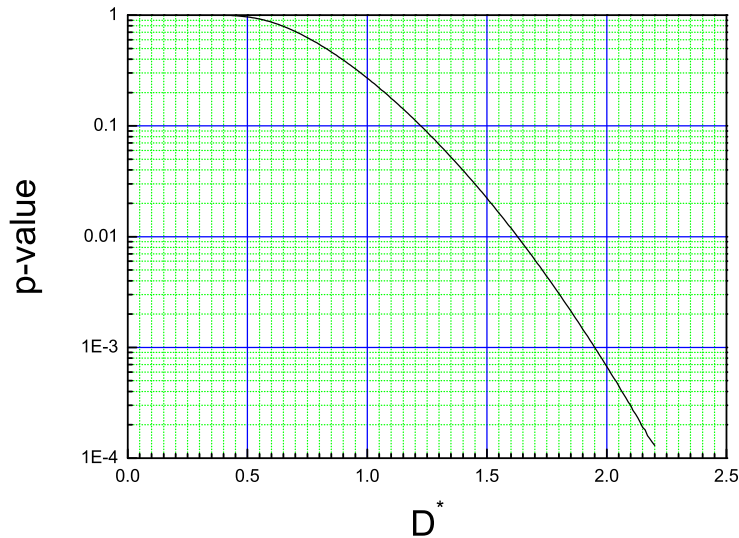


Fig. 10.10. P-value as a function of the Kolmogorov test statistic D^* .

The Kolmogorov–Smirnov test emphasizes more the center of the distribution than the tails because there the distribution function is tied to the values *zero* and *one* and thus is little sensitive to deviations at the borders. Since it is based on the distribution function, deviations are integrated over a certain range. Therefore it is not very sensitive to deviations which are localized in a narrow region. In Fig. 10.8 the left hand and the right hand histograms have the same excess of entries in the region left of the center. The Kolmogorov–Smirnov test produces in both cases approximately the same value of the test statistic, even though we would think that the distribution of the right hand histogram is harder to explain by a statistical fluctuation of a uniform distribution. This shows again, that the power of a test depends strongly on the alternatives to H_0 . The deviations of the left hand histogram are well detected by the Kolmogorov–Smirnov test, those of the right hand histogram much better by the Anderson–Darling test which we will present below.

There exist other EDF tests [57], which in most situations are more effective than the simple Kolmogorov–Smirnov test.

10.3.6 Tests of the Kolmogorov–Smirnov – and Cramer–von Mises Families

In the Kuiper test one uses as the test statistic the sum $V = D_+ + D_-$ of the two deviations of the empirical distribution function S from F . This quantity is designed for distributions “on the circle”. This are distributions where the beginning and the end of the distributed quantity are arbitrary, like the distribution of the azimuthal angle which can be presented with equal justification in all intervals $[\varphi_0, \varphi_0 + 2\pi]$ with arbitrary φ_0 .

The tests of the Cramer–von Mises family are based on the quadratic difference between F and S . The simple Cramer–von Mises test employs the test statistic

$$W^2 = \int_{-\infty}^{\infty} [(F(x) - S(x))^2] dF .$$

In most situations the *Anderson–Darling* test with the test statistic A^2 and the test of *Watson* with the test statistic U^2

$$A^2 = N \int_{-\infty}^{\infty} \frac{[S(x) - F(x)]^2}{F(x)[1 - F(x)]} dF ,$$

$$U^2 = N \int_{-\infty}^{\infty} \left\{ S(x) - F(x) - \int_{-\infty}^{\infty} [S(x) - F(x)] dF \right\}^2 dF ,$$

are superior to the Kolmogorov–Smirnow test.

The test of Anderson emphasizes especially the tails of the distribution while Watson’s test has been developed for distributions on the circle. The formulas above look quite complicated at first sight. They simplify considerably when we perform a *probability integral transformation* (PIT). This term stands for a simple transformation of the variate x into the variate $z = F_0(x)$, which is uniformly distributed in the interval $[0, 1]$ and which has the simple distribution function $H_0(z) = z$. With the transformed step distribution $S^*(z)$ of the sample we get

$$A^2 = N \int_{-\infty}^{\infty} \frac{[S^*(z) - z]^2}{z(1 - z)} dz ,$$

$$U^2 = N \int_{-\infty}^{\infty} \left\{ S^*(z) - z - \int_{-\infty}^{\infty} [S^*(z) - z] dz \right\}^2 dz .$$

In the Appendix 13.7 we show how to compute the test statistics. There also the asymptotic distributions are collected.

10.3.7 Neyman’s Smooth Test

This test [59] is different from those discussed so far in that it parameterizes the alternative hypothesis. Neyman introduced the smooth test in 1937 (for a discussion by E. S. Pearson see [60]) as an alternative to the χ^2 test, in that it is insensitive to deviations from H_0 which are positive (or negative) in several consecutive bins. He insisted that in hypothesis testing the investigator has to bear in mind which departures from H_0 are possible and thus to fix partially the p.d.f. of the alternative. The test is called “smooth” because, contrary to the χ^2 test, the alternative hypothesis approaches H_0 smoothly for vanishing parameter values. The hypothesis under test H_0 is again that the sample after the PIT, $z_i = F_0(x_i)$, follows a uniform distribution in the interval $[0, 1]$.

The smooth test excludes alternative distributions of the form

$$g_k(z) = \sum_{i=0}^k \theta_i \pi_i(z), \quad (10.8)$$

where θ_i are parameters and the functions $\pi_i(z)$ are modified orthogonal Legendre polynomials that are normalized to the interval $[0, 1]$ and symmetric or antisymmetric with respect to $z = 1/2$:

$$\begin{aligned}
\pi_0(z) &\equiv 1, \\
\pi_1(z) &= \sqrt{3}(2z-1), \\
&\dots\dots\dots \\
\pi_i(z) &= \sqrt{2i+1}P_i(2z-1).
\end{aligned}$$

Here $P_i(x)$ is the Legendre polynomial in the usual form. The first parameter θ_0 is fixed, $\theta_0 = 1$, and the other parameters are restricted such that g_k is positive. The user has to choose the parameter k which limits the degree of the polynomials. If the alternative hypothesis is suspected to contain narrow structures, we have to admit large k . The test with $k = 1$ rejects a linear contribution, $k = 2$ in addition a quadratic component and so on. Obviously, the null hypothesis H_0 corresponds to $\theta_1 = \dots = \theta_k = 0$, or equivalently to $\sum_{i=1}^k \theta_i^2 = 0$. We have to look for a test statistic which increases with the value of this sum.

For a sample of size N the test statistic proposed by Neyman is

$$r_k^2 = \frac{1}{N} \sum_{i=1}^k t_i^2 = \frac{1}{N} \sum_{i=1}^k \left(\sum_{j=1}^N \pi_i(z_j) \right)^2. \quad (10.9)$$

This choice is plausible, because a large absolute value of t_i is due to a strong contribution of the polynomial π_i to the observed distribution and thus also to a large value of θ_i^2 , while under H_0 we have for $i \geq 1$

$$\langle t_i \rangle = N \langle \pi_i(z) \rangle = 0,$$

because

$$\int_0^1 \pi_i(z) dz = 0.$$

Asymptotically, $N \rightarrow \infty$, under H_0 the test statistic r_k^2 follows a χ^2 distribution with k degrees of freedom (see 3.6.7). This is a consequence of the orthonormality of the polynomials π_i and the central limit theorem: We have

$$\text{var}(t_i) = \langle t_i^2 \rangle = N \int_0^1 \pi_i^2(z) dz = N$$

and as a sum of N random variables the statistic $t_i/\sqrt{N}$ is normally distributed for large N , with expectation value zero and variance one. Due to the orthogonality of the π_i , the t_i are uncorrelated. For small N the distribution of the test statistic r_k^2 has to be obtained by a Monte Carlo simulation.

In any case, large values of r_k^2 indicate bad agreement of the data with H_0 , but for a fixed value of k the smooth test is not consistent⁶. Its power approaches unity for $N \rightarrow \infty$ only for the class of alternatives H_k having a PIT which is represented by an expansion in Legendre polynomials up to order k . Hence with respect to these, while usually uninteresting, restricted alternatives it is consistent. Thus for large samples and especially for the exclusion of narrow structures k should not be chosen too small. The value of k in the smooth test corresponds roughly to the number of bins in the χ^2 -test.

The smooth test is in most cases superior to the χ^2 test. This can be understood in the following way: The smooth test scrutinizes not only for structures of a fixed

⁶For $k = 1$, for instance, the test cannot exclude distributions concentrated near $z = 1/2$.

frequency but for all frequencies up to k while the χ^2 test with $B \gg 1$ bins is rather insensitive to low frequency variations.

Remark: The alternative distribution quoted by Neyman was the exponential

$$g_k(z) = C \exp \left(\sum_{i=0}^k \theta_i \pi_i(z) \right) \quad (10.10)$$

where $C(\boldsymbol{\theta})$ is a normalization constant. Neyman probably chose the exponential form, because it guaranties positivity without further restrictions of the parameters θ_i . Moreover, with this class of alternatives, it has been shown by E. S. Pearson [60] that the smooth test can be interpreted as a likelihood ratio test. Anyway, (10.8) or (10.10) serve only as a motivation to choose the test statistic (10.9) which is the relevant quantity.

10.3.8 The L_2 Test

The binning-free tests discussed so far are restricted to one dimension, i.e. to univariate distributions. We now turn to multivariate tests.

A very obvious way to express the difference between two distributions f and f_0 is the integrated quadratic difference

$$L_2 = \int [f(\mathbf{r}) - f_0(\mathbf{r})]^2 d\mathbf{r}. \quad (10.11)$$

Unfortunately, we cannot use this expression for the comparison of a sample $\{\mathbf{r}_1, \dots, \mathbf{r}_N\}$ with a continuous function f_0 . Though we can try to derive from our sample an approximation of f . Such a procedure is called *probability density estimation (PDE)*. A common approach (see Chap. 12) is the Gaussian smearing or smoothing. The N discrete observations at the locations $\mathbf{r}_i$ are transformed into the function

$$f_G(\mathbf{r}) = \frac{1}{N} \sum e^{-\alpha(\mathbf{r}_i - \mathbf{r})^2}.$$

The smearing produces a broadening which has also to be applied to f_0 :

$$f_{0G}(\mathbf{r}) = \int f_0(\mathbf{r}') e^{-\alpha(\mathbf{r} - \mathbf{r}')^2} d\mathbf{r}'.$$

We now obtain a useful test statistic L_{2G} ,

$$L_{2G} = \int [f_G(\mathbf{r}) - f_{0G}(\mathbf{r})]^2 d\mathbf{r}.$$

So far the L_2 test [61] has not found as much attention as it deserves because the calculation of the integral is tedious. However its Monte Carlo version is pretty simple. It offers the possibility to adjust the width of the smearing function to the density f_0 . Where we expect large distances of observations, the Gaussian width should be large, $\alpha \sim f_0^2$.

A more sophisticated version of the L_2 test is presented in [61]. The Monte Carlo version is included in Sect. 10.3.11, see below.

10.3.9 Comparing a Data Sample to a Monte Carlo Sample and the Metric

We now turn to tests where we compare our sample not to an analytic distribution but to a Monte Carlo simulation of f_0 . This is not a serious restriction because anyhow acceptance and resolution effects have to be taken into account in the majority of all experiments. Thus the null hypothesis is usually represented by a simulation sample.

To compare two samples we have to construct a relation between observations of the samples which in the multi-dimensional case has to depend in some way on the distance between them. We can define the distance in the usual way using the standard Euclidian metric but since the different dimensions often represent completely different physical quantities, e.g. spatial distance, time, mass etc., we have considerable freedom in the choice of the metric and we will try to adjust the metric such that the test is powerful.

We usually want that all coordinates enter with about equal weight into the test. If, for example, the distribution is very narrow in x but wide in y , then the distance r of points is almost independent of y and it is reasonable to stretch the distribution in the x direction before we apply a test. Therefore we propose for the general case to scale linearly all coordinates such that the empirical variances of the sample are the same in all dimensions. In addition we may want to get rid of correlations when for instance a distribution is concentrated in a narrow band along the x - y diagonal.

Instead of transforming the coordinates we can use the *Mahalanobis distance*⁷ in order to normalize distances between observations $(\mathbf{x}_1, \dots, \mathbf{x}_N)$ with sample mean $\bar{\mathbf{x}}$. (The bold-face symbols here denote P -dimensional vectors describing different features measured on each of the N sampled objects.)

The Mahalanobis distance d_M of two observations $\mathbf{x}$ and $\mathbf{x}'$ is

$$d_M = \sqrt{(\mathbf{x} - \mathbf{x}')^T \mathbf{C}^{-1} (\mathbf{x} - \mathbf{x}')},$$

with

$$C_{ij} = \sum_{n=1}^N (x_{ni} - \bar{x}_i)(x_{nj} - \bar{x}_j) / N.$$

It is equivalent to the Euclidian distance after a linear transformation of the vector components which produces a sample with unity covariance matrix. If the covariance matrix is diagonal, then the resulting distance is the normalized Euclidean distance in the P -dimensional space:

$$d_M = \sqrt{\sum_{p=1}^P \frac{(x_p - x'_p)^2}{\sigma_p^2}}.$$

In the following tests the choice of the metric is up to the user. In many situations it is reasonable to use the Mahalanobis distance, even though moderate variations of the metric normally have little influence on the power of a test.

⁷This is a distance measure introduced by P. C. Mahalanobis in 1936.

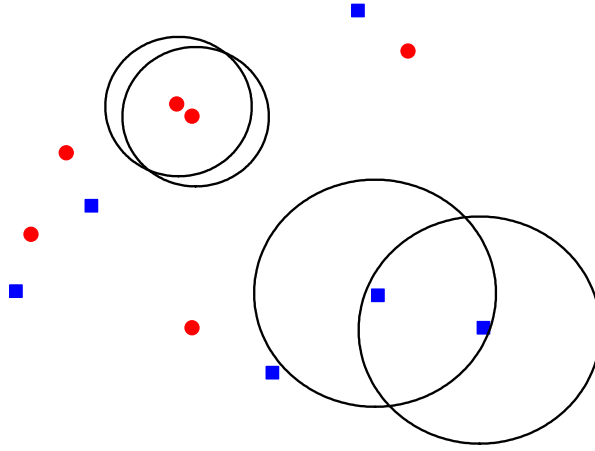


Fig. 10.11. K-nearest neighbor test.

10.3.10 The k-Nearest Neighbor Test

We consider two samples, one generated by a Monte Carlo simulation of a null distribution f_0 and the experimental sample. The test statistic is the number $n(k)$ of observations of the mixed sample where all of its k nearest neighbors belong to the same sample as the observation itself. This is illustrated in Fig. 10.11 for an unrealistically simple configuration. We find $n(1) = 4$ and $n(2) = 4$. The parameter k is a small number to be chosen by the user, in most cases it is one, two or three.

Of course we expect n to be large if the two parent distributions are very different. The k-nearest neighbor test is very popular and quite powerful. It has one caveat: We would like to have the number M of Monte Carlo observations much larger than the number N of experimental observations. In the situation with $M \gg N$ each observation tends to have as next neighbor a Monte Carlo observation and the test becomes less significant.

10.3.11 The Energy Test

A very general expression that measures the difference between two distributions $f(\mathbf{r})$ and $f_0(\mathbf{r})$ in an n dimensional space is

$$\phi = \frac{1}{2} \int d\mathbf{r} \int d\mathbf{r}' [f(\mathbf{r}) - f_0(\mathbf{r})] [f(\mathbf{r}') - f_0(\mathbf{r}')] R(\mathbf{r}, \mathbf{r}') . \quad (10.12)$$

Here we call R the distance function. The factor $1/2$ is introduced to simplify formulas which we derive later. The special case $R = \delta(\mathbf{r} - \mathbf{r}')$ leads to the simple integrated quadratic deviation (10.11) of the L_2 test

$$\phi = \frac{1}{2} \int d\mathbf{r} [f(\mathbf{r}) - f_0(\mathbf{r})]^2 . \quad (10.13)$$

However, we do not intend to compare two distributions but rather two samples $A \{\mathbf{r}_1, \dots, \mathbf{r}_N\}$, $B \{\mathbf{r}_{01}, \dots, \mathbf{r}_{0M}\}$, which are extracted from the distributions f and f_0 , respectively. For this purpose we start with the more general expression (10.12) which connects points at different locations. We restrict the function R in such a way that it is a function of the distance $|\mathbf{r} - \mathbf{r}'|$ only and that ϕ is minimum for $f \equiv f_0$.

The function (10.12) with $R = 1/|\mathbf{r} - \mathbf{r}'|$ describes the electrostatic energy of the sum of two charge densities f and f_0 with equal total charge but different sign of the charge. In electrostatics the energy reaches a minimum if the charge is zero everywhere, i.e. the two charge densities are equal up to the sign. Because of this analogy we refer to ϕ as *energy*.

For our purposes the logarithmic function $R(r) = -\ln(r)$ and the bell function $R(r) \sim \exp(-cr^2)$ are more suitable than $1/r$.

We multiply the expressions in brackets in (10.12) and obtain

$$\phi = \frac{1}{2} \int d\mathbf{r} \int d\mathbf{r}' [f(\mathbf{r})f(\mathbf{r}') + f_0(\mathbf{r})f_0(\mathbf{r}') - 2f(\mathbf{r})f_0(\mathbf{r}')] R(|\mathbf{r} - \mathbf{r}'|). \quad (10.14)$$

A Monte Carlo integration of this expression is obtained when we generate M random points $\{\mathbf{r}_{01} \dots \mathbf{r}_{0M}\}$ of the distribution $f_0(\mathbf{r})$ and N random points $\{\mathbf{r}_1, \dots, \mathbf{r}_N\}$ of the distribution $f(\mathbf{r})$ and weight each combination of points with the corresponding distance function. The Monte Carlo approximation is:

$$\begin{aligned} \phi &\approx \frac{1}{N(N-1)} \sum_i \sum_{j>i} R(|\mathbf{r}_i - \mathbf{r}_j|) - \frac{1}{NM} \sum_i \sum_j R(|\mathbf{r}_i - \mathbf{r}_{0j}|) + \\ &+ \frac{1}{M(M-1)} \sum_i \sum_{j>i} R(|\mathbf{r}_{0i} - \mathbf{r}_{0j}|) \\ &\approx \frac{1}{N^2} \sum_i \sum_{j>i} R(|\mathbf{r}_i - \mathbf{r}_j|) - \frac{1}{NM} \sum_i \sum_j R(|\mathbf{r}_i - \mathbf{r}_{0j}|) + \\ &+ \frac{1}{M^2} \sum_i \sum_{j>i} R(|\mathbf{r}_{0i} - \mathbf{r}_{0j}|). \end{aligned} \quad (10.15)$$

This is the energy of a configuration of discrete charges. Alternatively we can understand this result as the sum of three expectation values which are estimated by the two samples. The value of ϕ from (10.15) thus is the estimate of the energy of two samples that are drawn from the distributions f_0 and f and that have the total charge zero.

We can use the expression (10.15) as test statistic when we compare the experimental sample to a Monte Carlo sample, the null sample representing the null distribution f_0 . Small energies signify a good, large ones a bad agreement of the experimental sample with H_0 . To be independent of statistical fluctuations of the simulated sample, we choose M large compared to N , typically $M \approx 10N$.

The test statistic energy ϕ is composed of three terms ϕ_1, ϕ_2, ϕ_3 which correspond to the interaction of the experimental sample with itself, to its interaction with the null sample and with the interaction of the null sample with itself:

$$\phi = \phi_1 - \phi_2 + \phi_3, \quad (10.16)$$

$$\phi_1 = \frac{1}{N(N-1)} \sum_{i<j} R(|\mathbf{r}_i - \mathbf{r}_j|), \quad (10.17)$$

$$\phi_2 = \frac{1}{NM} \sum_{i,j} R(|\mathbf{r}_i - \mathbf{r}_{0j}|) , \quad (10.18)$$

$$\phi_3 = \frac{1}{M(M-1)} \sum_{i < j} R(|\mathbf{r}_{0i} - \mathbf{r}_{0j}|) . \quad (10.19)$$

The term ϕ_3 is independent of the data and can be omitted but is normally included to reduce statistical fluctuations.

The distance function R relates sample points and simulated points of the null hypothesis to each other. Proven useful have the functions

$$R_l = -\ln(r + \varepsilon) , \quad (10.20)$$

$$R_s = e^{-r^2/(2s^2)} . \quad (10.21)$$

The small positive constant ε suppresses the pole of the logarithmic distance function. Its value should be chosen approximately equal to the experimental resolution⁸ but variations of ε by a large factor have no sizable influence on the result. With the function $R_l = 1/r$ we get the special case of electrostatics. With the Gaussian distance function R_s the test is very similar to the χ^2 test with bin width $2s$ but avoids the arbitrary binning of the latter. For slowly varying functions we propose to use the logarithmic distance function.

Empirical studies have shown that the test statistic follows to a very good approximation a distribution from the family of the extreme value distributions,

$$f(\phi) = \frac{1}{\sigma} \left(1 + \xi \frac{\phi - \mu}{\sigma} \right)^{-1/\xi - 1} \exp \left\{ - \left(1 + \xi \frac{\phi - \mu}{\sigma} \right)^{-1/\xi} \right\} ,$$

with parameters μ, σ, ξ which we have discussed in Sect. 3.6.12. We have to compute this distribution $f_0(\phi)$ by a Monte Carlo simulation. To get a sufficient precision for small p-values we have to generate a very large sample. We gain computing time when we extract the first three moments of a relatively modest Monte Carlo sample and compute from those the parameters of the extreme value distribution.

The energy test is consistent [63]. It is quite powerful in many situations and has the advantage that it is not required to sort the sample elements.

The energy test with Gaussian distance function is completely equivalent to the L_2 test. It is more general than the latter in that it allows to use various distance functions.

10.3.12 Tests Designed for Specific Problems

The power of tests depends on the alternatives. If we have an idea of it, even if it is crude, we can design a GOF test which is especially sensitive to the deviations from H_0 which we have in mind. The distribution of the test statistic has to be produced by a Monte Carlo simulation.

Example 130. Designed test: three region test

⁸Distances between two points that are smaller than the resolution are accidental and thus insignificant.

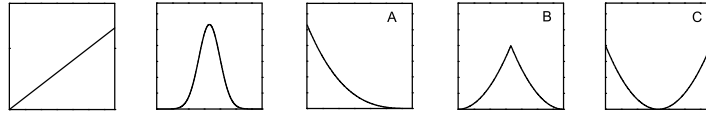


Fig. 10.12. Different admixtures to a uniform distribution.

Experimental distributions often show a local excess of observations which are either just a statistical fluctuation or stem from a physical process. To check whether an experimental sample is compatible with the absence of a bump caused by a physical process, we propose the following *three region test*. We subdivide the domain of the variable in three regions with expected numbers of observations n_{10}, n_{20}, n_{30} and look for differences to the corresponding experimental numbers n_1, n_2, n_3 . The subdivision is chosen such that the sum of the differences is maximum. The test statistic R_3 is

$$R_3 = \sup_{n_1, n_2} [(n_1 - n_{10}) + (n_2 - n_{20}) + (n_3 - n_{30})] .$$

Notice, that $n_3 = N - n_1 - n_2$ is a function of n_1 and n_2 . The generalization to more than three regions is trivial. Like in the χ^2 test we could also divide the individual squared differences by their expected value:

$$R'_3 = \sup_{n_1, n_2} \left[\frac{(n_1 - n_{10})^2}{n_{10}} + \frac{(n_2 - n_{20})^2}{n_{20}} + \frac{(n_3 - n_{30})^2}{n_{30}} \right] .$$

10.3.13 Comparison of Tests

Univariate Distributions

Whether a test is able to detect deviations from H_0 depends on the distribution f_0 and on the kind of distortion. Thus there is no test which is most powerful in all situations.

To get an idea of the power of different tests, we consider six different admixtures to a uniform distribution and compute the fraction of cases in which the distortion of the uniform distribution is detected at a significance level of 5%. For each distribution constructed in this way, we generate stochastically 1000 mixtures with 100 observations each. The distributions which we add are depicted in Fig. 10.12. One of them is linear, two are normal with different widths, and three are parabolic. The χ^2 test was performed with 12 bins following the prescription of Ref. [58], the parameter of Neyman's smooth test was $k = 2$ and the width of the Gaussian of the energy test was $s = 1/8$. The sensitivity of different tests is presented in Fig. 10.13.

The histogram of Fig. 10.13 shows that none of the tests is optimum in all cases. The χ^2 test performs only mediocly. Probably a lower bin number would improve the result. The tests of Neyman, Anderson–Darling and Kolmogorov–Smirnov are sensitive to a shift of the mean value while the Anderson–Darling test reacts especially to changes at the borders of the distribution. The tests of Watson and Kuiper detect preferentially variations of the variance. Neyman's test and the energy test with logarithmic distance function are rather efficient in most cases.

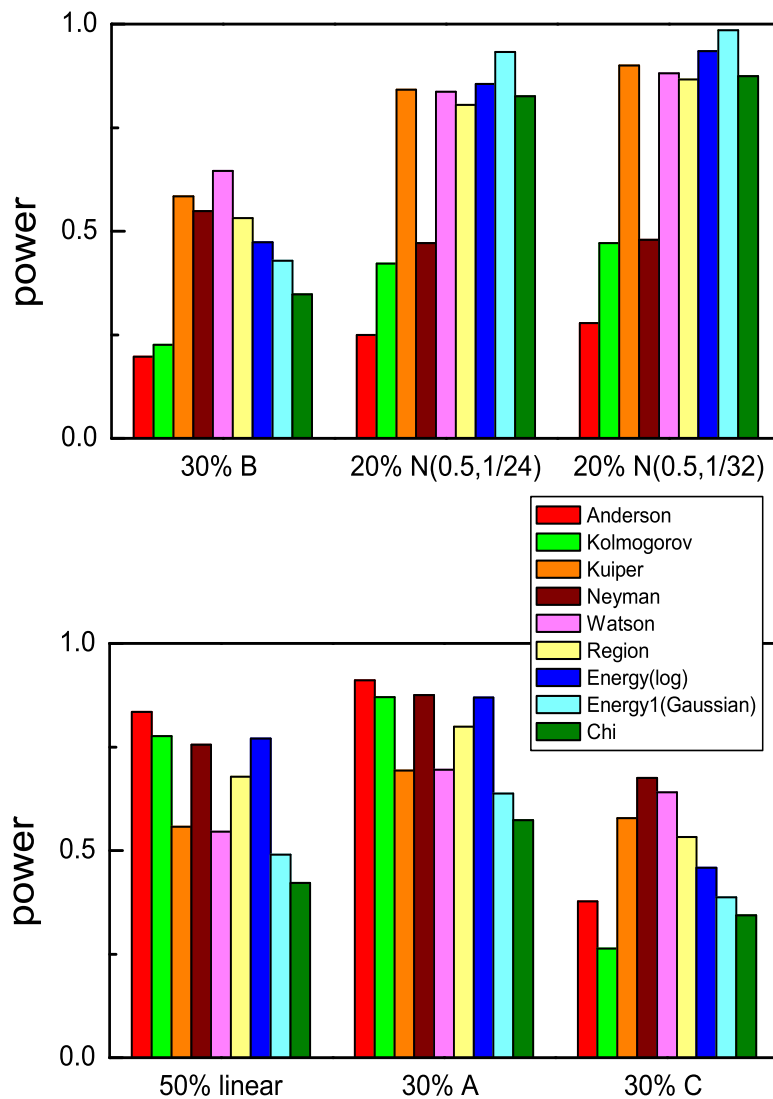


Fig. 10.13. Power (fraction of identified distortions) of different tests.

Multivariate Distributions

The goodness-of-fit of multivariate distributions cannot be tested very well with simple tests. The χ^2 test often suffers from the small number of entries per bin. Here the k-nearest neighbor test and the energy test with the long range logarithmic distance function are much more efficient.

Example 131. GOF test of a two-dimensional sample

Figure 10.14 shows a comparison of a sample H_1 with a two-dimensional normal distribution (H_0). H_1 corresponds to the distribution of H_0 but contains an admixture of a linear distribution. The p-value of the energy test is 2%.

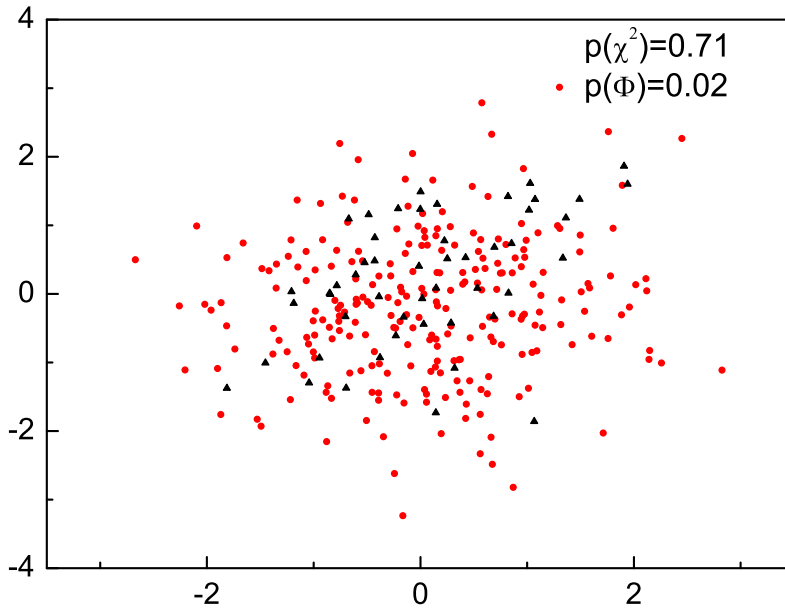


Fig. 10.14. Comparison of a normally distributed sample (circles) from H_0 with a linear admixture (triangles) with the normal distribution of H_0 .

With a χ^2 test with 9 bins we obtain a p-value of 71%. It is unable to identify the deformation of f_0 .

More detailed investigations can be found in [62].

10.4 Two-Sample Tests

10.4.1 The Problem

A standard situation in particle physics is that H_0 cannot be compared directly to the data but has first to be transformed to a Monte Carlo sample, to take into account acceptance losses and resolution effects. We have to compare two samples, a procedure which we had already applied in the *energy test*. Here the distribution of the test statistic needed to compute p-values can be generated by a simple Monte Carlo program.

In other sciences, a frequently occurring problem is that the effectiveness of two or several procedures have to be compared. This may concern drugs, animal feed or the quality of fabrication methods. A similar problem is to test whether a certain process is stable or whether its results have changed during time. Also in the natural sciences we frequently come across the problem that we observe an interesting phenomenon in one data sample which apparently has disappeared in another sample taken at a later time. It is important to investigate whether the two data samples are compatible with each other. Sometimes it is also of interest to investigate whether a Monte Carlo sample and an experimental sample are compatible. Thus we are interested in

a statistical procedure which tells us whether two samples A and B are compatible, i.e. drawn from the same parent distribution. Thereby we assume that the parent distribution itself is unknown. If it were known, we could apply one of the GOF tests which we have discussed above. We have to invent procedures to generate the distribution of the test statistic. In some cases this is trivial. In the remaining cases, we have to use combinatorial methods.

10.4.2 The χ^2 Test

To test whether two samples are compatible, we can apply the χ^2 test or the Kolmogorov–Smirnov test with minor modifications.

When we calculate the χ^2 statistic we have to normalize the two samples A and B of sizes N and M to each other. For a_i and b_i entries in bin i , $a_i/N - b_i/M$ should be compatible with zero. With the usual error propagation we obtain an estimate $a_i/N^2 + b_i/M^2$ of the quadratic error of this quantity and

$$\chi^2 = \sum_{i=1}^B \frac{(a_i/N - b_i/M)^2}{(a_i/N^2 + b_i/M^2)} . \quad (10.22)$$

It follows approximately a χ^2 distribution of $B - 1$ degrees of freedom, but not exactly, as we had to replace the expected values by the observed numbers in the error estimation. We have to be careful if the number of observations per bin is small.

10.4.3 The Likelihood Ratio Test

The likelihood ratio test is less vulnerable to low event numbers than the χ^2 test.

Setting $r = M/N$ we compute the likelihood that we observe in a single bin a entries with expectation λ and b entries with expectation $\rho\lambda$, where the hypothesis H_0 is characterized by $\rho = r$:

$$L(\lambda, \rho | a, b) = \frac{e^{-\lambda} \lambda^a}{a!} \frac{e^{-\rho\lambda} (\rho\lambda)^b}{b!} .$$

Leaving out constant factors the log-likelihood is

$$\ln L = -\lambda(1 + \rho) + (a + b) \ln \lambda + b \ln \rho .$$

We determine the conditional maximum likelihood value of λ under $\rho = r$ and the corresponding log-likelihood:

$$\begin{aligned} 1 + r &= (a + b) \frac{1}{\hat{\lambda}_c} , \\ \hat{\lambda}_c &= \frac{a + b}{1 + r} , \\ \ln L_{cmax} &= (a + b) \left[-1 + \ln \frac{a + b}{1 + r} \right] + b \ln r . \end{aligned}$$

The unconditional maximum of the likelihood is found for $\hat{\lambda} = a$ and $\hat{\rho} = b/a$:

$$\ln L_{umax} = -(a + b) + a \ln a + b \ln b .$$

Our test statistic is V_{AB} , the logarithm of the likelihood ratio, now summed over all bins:

$$\begin{aligned} V_{AB} &= \ln L_{cmax} - \ln L_{umax} \\ &= \sum_i \left[(a_i + b_i) \ln \frac{a_i + b_i}{1 + r} - a_i \ln a_i - b_i \ln b_i + b_i \ln r \right] . \end{aligned}$$

Note that $V_{AB}(r) = V_{BA}(1/r)$, as it should.

Now we need a method to determine the expected distribution of the test statistic V_{AB} under the assumption that both samples originate from the same population.

To generate a distribution from a sample, the so-called *bootstrap* method [64] (see Chap. 12.2) has been developed. In our situation a variant of it, a simple permutation method is appropriate.

We combine the two samples to a new sample with $M + N$ elements and form new pairs of samples, the bootstrap samples, with M and N elements by permutation: We draw randomly M elements from the combined sample and associate them to A and the remaining elements to B . Computationally this is easier than to use systematically all individual possibilities. For each generated pair i we determine the statistic V_i . This procedure is repeated many times and the values V_i form the reference distribution. Our experimental p-value is equal to the fraction of generated V_i which are larger than V_{AB} :

$$p = \frac{\text{Number of permutations with } V_i > V_{AB}}{\text{Total number of permutations}} .$$

10.4.4 The Kolmogorov–Smirnov Test

Also the Kolmogorov–Smirnov test can easily be adapted to a comparison of two samples. We construct the test statistic in an analogous way as above. The test statistic is $D^* = D\sqrt{N_{eff}}$, where D is the maximum difference between the two empirical distribution functions S_A, S_B , and N_{eff} is the effective or equivalent number of events, which is computed from the relation:

$$\frac{1}{N_{eff}} = \frac{1}{N} + \frac{1}{M} .$$

In a similar way other EDF multi-dimensional tests which we have discussed above can be adjusted.

10.4.5 The Energy Test

For a binning-free comparison of two samples A and B with M and N observations we can again use the energy test [63] which in the multi-dimensional case has only few competitors.

We compute the energy ϕ_{AB} in the same way as above, replacing the Monte Carlo sample by one of the experimental samples. The expected distribution of the test statistic ϕ_{AB} is computed in the same way as for the likelihood ratio test from

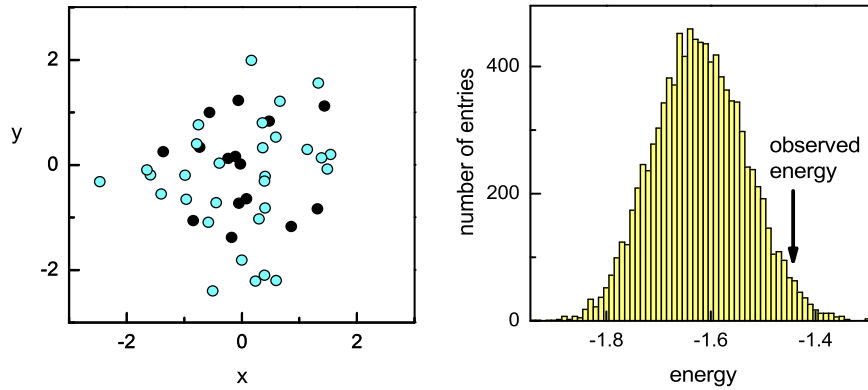


Fig. 10.15. Two-sample test. Left hand: the samples which are to be compared. Right hand: distribution of test statistic and actual value.

the combined sample using the bootstrap permutation technique. Our experimental p-value is equal to the fraction of generated ϕ_i from the bootstrap sample which are larger than ϕ_{AB} :

$$p = \frac{\text{Number of permutations with } \phi_i > \phi_{AB}}{\text{Total number of permutations}}.$$

Example 132. Comparison of two samples

We compare two two-dimensional samples with 15 and 30 observations with the energy test. The two samples are depicted in a scatter plot at the left hand side of Fig. 10.15. The energy of the system is $\phi_{AB} = -1.480$ (The negative value arises because we have omitted the term ϕ_3). From the mixed sample 10000 sample combinations have been selected at random. Its energy distribution is shown as a histogram in the figure. The arrow indicates the location of ϕ_{AB} . It corresponds to a p-value of 0.06. We can estimate the error of the p-value p computing it from many permutation sets each with a smaller number of permutations. From the variation of p from 100 times 100 permutations we find $\delta p = 0.02$. The p-value is small, indicating that the samples belong to different populations. Indeed they have been drawn from different distributions, a uniform distribution, $-1.5 < x, y < 1.5$ and a normal distribution with standard deviations $\sigma_x = \sigma_y = 1$.

10.4.6 The k-Nearest Neighbor Test

The k-nearest neighbor test is per construction a two-sample test. The distribution of the test statistic is obtained in exactly the same way as in the two-sample energy test which we have discussed in the previous section.

The performance of the k-nearest neighbor test is similar to that of the energy test. The energy test (and the L^2 test which is automatically included in the former) is more flexible than the k-nearest neighbor test and includes all observation of the

sample in the continuous distance function. The k-nearest neighbor test on the other hand is less sensitive to variations of the density which are problematic for the energy test with the Gaussian distance function of constant width.

10.5 Significance of Signals

10.5.1 Introduction

Tests for signals are closely related to goodness-of-fit tests but their aim is different. We are not interested to verify that H_0 is compatible with a sample but we intend to quantify the evidence of signals which are possibly present in a sample which consists mainly of uninteresting background. Here not only the distribution of the background has to be known but in addition we must be able to parameterize the alternative which we search for. The null hypothesis H_0 corresponds to the absence of deviations from the background. The alternative H_s is not fully specified, otherwise it would be sufficient to compute the simple likelihood ratio which we have discussed in Chap. 6.

Signal tests are applied when we search for rare decays or reactions like neutrino oscillations. Another frequently occurring problem is that we want to interpret a line in a spectrum as indication for a resonance or a new particle. To establish the evidence of a signal, we usually require a very significant deviation from the null hypothesis, i.e. the sum of background and signal has to describe the data much better than the background alone because particle physicists look in hundreds of histograms for more or less wide lines and thus always find candidates⁹ which in most cases are just background fluctuations. For this reason, signals are only accepted by the community if they have a significance of at least four or five standard deviations. In cases where we search more specifically for a certain phenomenon a smaller significance may be sufficient. A high significance for a signal corresponds to a low p-value of the null hypothesis.

To quote the p-value instead of the significance as expressed by the number of standard deviations by which the signal exceeds the background expectation is to be preferred because it is a measure which is independent of the form of the distribution. However, the standard deviation scale is better suited to illustrate the significance than the p-values scale where very small values dominate. For this reason it has become customary to transform the p-value p into the number of Gaussian standard deviations s_G which are related through

$$p = 1/\sqrt{2\pi} \int_{s_G}^{\infty} \exp(-x^2/2) dx \quad (10.23)$$

$$= [1 - \operatorname{erf}(s_G/\sqrt{2})] / 2. \quad (10.24)$$

The function $s_G(p)$ is given in Fig. 10.16. Relations (10.23), (10.24) refer to one-sided tests. For two-sided tests, p has to be multiplied by a factor two.

When we require very low p-values for H_0 to establish signals, we have to be especially careful in modeling the distribution of the test statistic. Often the distribution corresponding to H_0 is approximated by a polynomial and/or a signal by a Gaussian

⁹This is the so-called *look-else-where effect*.

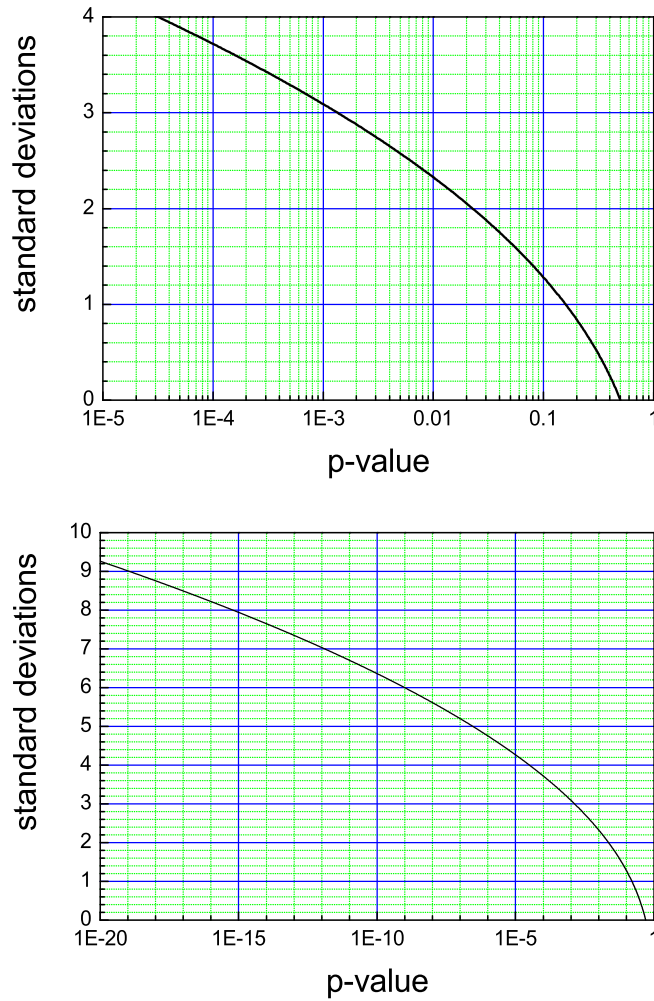


Fig. 10.16. Transformation of p-values to one-sided number of standard deviations.

with some uncertainties in the parameters and assumptions which are difficult to implement in the test procedure. We then have to be especially conservative. It is better to underestimate the significance of a signal than to present evidence for a new phenomenon based on a doubtful number.

To illustrate this problem we return to our standard example where we search for a line in a one-dimensional spectrum. Usually, the background under an observed bump is estimated from the number of events outside but near the bump in the so-called side bands. If the side bands are chosen too close to the signal they are affected by the tails of the signal, if they are chosen too far away, the extrapolation into the signal region is sensitive to the assumed shape of the background distribution which often is approximated by a linear or quadratic function. This makes it difficult to estimate the size and the uncertainty of the expected background with sufficient accuracy to establish the p-value for a large (>4 st. dev.) signal. As numerical example let

us consider an expectation of 1000 background events which is estimated by the experimenter too low by 2%, i.e. equal to 980. Then a 4.3 st. dev. excess would be claimed by him as a 5 st. dev. effect and he would find too low a p-value by a factor of 28. We also have to be careful with numerical approximations, for instance when we approximate a Poisson distribution by a Gaussian.

Usually, the likelihood ratio, i.e. the ratio of the likelihood which maximizes H_s and the maximum likelihood for H_0 is the most powerful test statistic. In some situations a relevant parameter which characterizes the signal strength is more informative.

10.5.2 The Likelihood Ratio Test

Definition

An obvious candidate for the test statistic is the likelihood ratio (LR) which we have introduced and used in Sect. 10.3 to test goodness-of-fit of histograms, and in Sect. 10.4 as a two-sample test. We repeat here its general definition:

$$\lambda = \frac{\sup [L_0(\boldsymbol{\theta}_0|\mathbf{x})]}{\sup [L_s(\boldsymbol{\theta}_s|\mathbf{x})]},$$

$$\ln \lambda = \ln \sup [L_0(\boldsymbol{\theta}_0|\mathbf{x})] - \ln \sup [L_s(\boldsymbol{\theta}_s|\mathbf{x})]$$

where L_0, L_s are the likelihoods under the null hypothesis and the signal hypothesis, respectively. The supremum is to be evaluated relative to the parameters, i.e. the likelihoods are to be taken at the MLEs of the parameters. The vector $\mathbf{x}$ represents the sample of the N observations $x_1, \dots, x_N$ of a one-dimensional geometric space. The extension to a multi-dimensional space is trivial but complicates the writing of the formulas. The parameter space of H_0 is assumed to be a subset of that of H_s . Therefore λ will be smaller or equal to *one*.

For example, we may want to find out whether a background distribution is described significantly better by a cubic than by a linear distribution:

$$\begin{aligned} f_0 &= \alpha_0 + \alpha_1 x, \\ f_s &= \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \alpha_3 x^3. \end{aligned} \quad (10.25)$$

We would fit separately the parameters of the two functions to the observed data and then take the ratio of the corresponding maximized likelihoods.

Frequently the data sample is so large that we better analyze it in form of a histogram. Then the distribution of the number of events y_i in bin i , $i = 1, \dots, B$ can be approximated by normal distributions around the parameter dependent predictions $t_i(\boldsymbol{\theta})$. As we have seen in Chap. 6, Sect. 6.5.6 we then get the log-likelihood

$$\ln L = -\frac{1}{2} \sum_{i=1}^B \frac{[y_i - t_i]^2}{t_i} + \text{const.}$$

which is equivalent to the χ^2 statistic, $\chi^2 \approx -2 \ln L$. In this limit the likelihood ratio statistic is equivalent to the χ^2 difference, $\Delta\chi^2 = \min \chi_0^2 - \min \chi_s^2$, of the χ^2 deviations, $\min \chi_0^2$ with the parameters adjusted to the null hypothesis H_0 , and $\min \chi_s^2$ with its parameters adjusted to the alternative hypothesis H_s , background plus signal:

$$\ln \lambda = \ln \sup [L_0(\boldsymbol{\theta}_0|\mathbf{y})] - \ln \sup [L_s(\boldsymbol{\theta}_s|\mathbf{y})] \quad (10.26)$$

$$\approx -\frac{1}{2}(\min \chi_0^2 - \min \chi_s^2) . \quad (10.27)$$

The p-value derived from the LR statistic does not take into account that a simple hypothesis is a priori more attractive than a composite one which contains free parameters. Another point of criticism is that the LR is evaluated only at the parameters that maximize the likelihood while the parameters suffer from uncertainties. Thus conclusions should not be based on the p-value only.

A Bayesian approach applies so-called *Bayes factors* to correct for the mentioned effects but is not very popular because it has other caveats. Its essentials are presented in the Appendix 13.14

Distribution of the Test Statistic

The distribution of λ under H_0 in the general case is not known analytically; however, if the approximation (10.27) is justified, the distribution of $-2 \ln \lambda$ under certain additional regularity conditions and the conditions mentioned at the end of Sect. 10.3.3 will be described by a χ^2 distribution. In the example corresponding to relations (10.25) this would be a χ^2 distribution of 2 degrees of freedom since f_s compared to f_0 has 2 additional free parameters. Knowing the distribution of the test statistic reduces the computational effort required for the numerical evaluation of p-values considerably.

Let us look at a specific problem: We want to check whether an observed bump above a continuous background can be described by a fluctuation or whether it corresponds to a resonance. The two hypotheses may be described by the distributions

$$\begin{aligned} f_0 &= \alpha_0 + \alpha_1 x + \alpha_2 x^2, \\ f_s &= \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \alpha_3 N(x|\mu, \sigma) , \end{aligned} \quad (10.28)$$

and we can again use $\ln \lambda$ or $\Delta\chi^2$ as test statistic. Since we have to define the test before looking at the data, μ and σ will be free parameters in the fit of f_s to the data. Unfortunately, now $\Delta\chi^2$ no longer follows a χ^2 distribution of 3 degrees of freedom and has a significantly larger expectation value than expected from the χ^2 distribution. The reason for this dilemma is that for $\alpha_3 = 0$ which corresponds to H_0 the other parameters μ and σ are undefined and thus part of the χ^2 fluctuation in the fit to f_s is unrelated to the difference between f_s and f_0 .

More generally, only if the following conditions are satisfied, $\Delta\chi^2$ follows in the large number limit a χ^2 distribution with the number of degrees of freedom given by the difference of the number of free parameters of the null and the alternative hypotheses:

1. The distribution f_0 of H_0 has to be a special realization of the distribution f_s of H_s .
2. The fitted parameters have to be inside the region, i.e. off the boundary, allowed by the hypotheses. For example, the MLE of the location of a Gaussian should not be outside the range covered by the data.
3. All parameters of H_s have to be defined under H_0 .

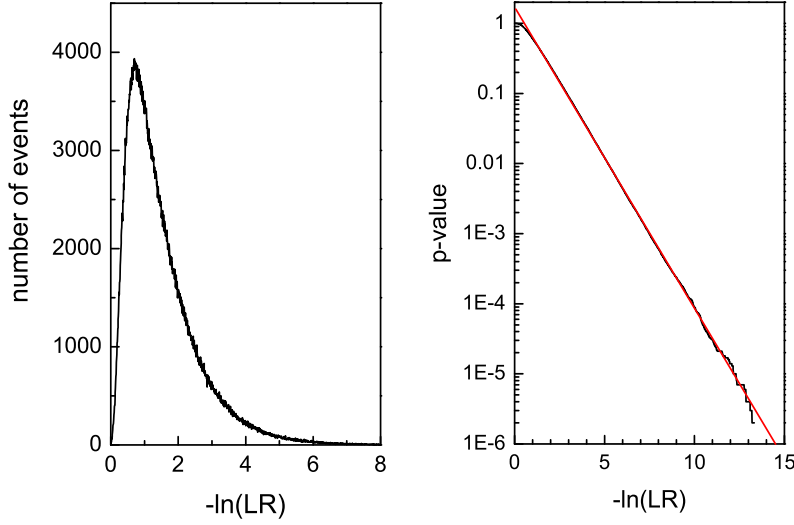


Fig. 10.17. Distributions of the test statistic under H_0 and p-value as a function of the test statistic.

If one of these conditions is not satisfied, the distribution of the test statistic has to be obtained via a Monte Carlo simulation. This means that we generate many fictive experiments of H_0 and count how many of those have values of the test statistic that exceed the one which has actually been observed. The corresponding fraction is the p-value for H_0 . This is a fairly involved procedure because each simulation includes fitting of the free parameters of the two hypotheses. In Ref. [65] it is shown that the asymptotic behavior of the distribution can be described by an analytical function. In this way the amount of simulation can be reduced.

Example 133. Distribution of the likelihood ratio statistic

We consider a uniform distribution (H_0) of 1000 events in the interval $[0, 1]$ and as alternative a resonance with Gaussian width, $\sigma = 0.05$, and arbitrary location μ in the range $0.2 \leq \mu \leq 0.8$ superposed to a uniform distribution. The free parameters are ε , the fraction of resonance events and μ . The logarithm of the likelihood ratio statistic is

$$\begin{aligned} \ln \lambda &= \ln \sup [L_0(\boldsymbol{\theta}_0|\mathbf{x})] - \ln \sup [L_s(\boldsymbol{\theta}_s|\mathbf{x})] \\ &= \sum_{i=1}^{1000} \ln(1) - \sum_{i=1}^{1000} \ln \left[1 - \hat{\varepsilon} + \frac{\hat{\varepsilon}}{\sqrt{2\pi}\sigma} \exp \left(-\frac{(x_i - \hat{\mu})^2}{2\sigma^2} \right) \right] \\ &= - \sum_{i=1}^{1000} \ln \left[1 - \hat{\varepsilon} + \frac{\hat{\varepsilon}}{\sqrt{2\pi}\sigma} \exp \left(-\frac{(x_i - \hat{\mu})^2}{2\sigma^2} \right) \right], \end{aligned}$$

essentially the negative logarithm of the likelihood of the MLE. Fig. 10.17 shows the results from a million simulated experiments. The distribution of $-\ln \lambda$ under H_0 has a mean value of -1.502 which corresponds to $\langle \Delta\chi^2 \rangle = 3.004$. The p-value as a function of $-\ln \lambda$ follows asymptotically an exponential as is illustrated in the right hand plot of Fig. 10.17. Thus it is possible to extrapolate the function to smaller p-values which is necessary to

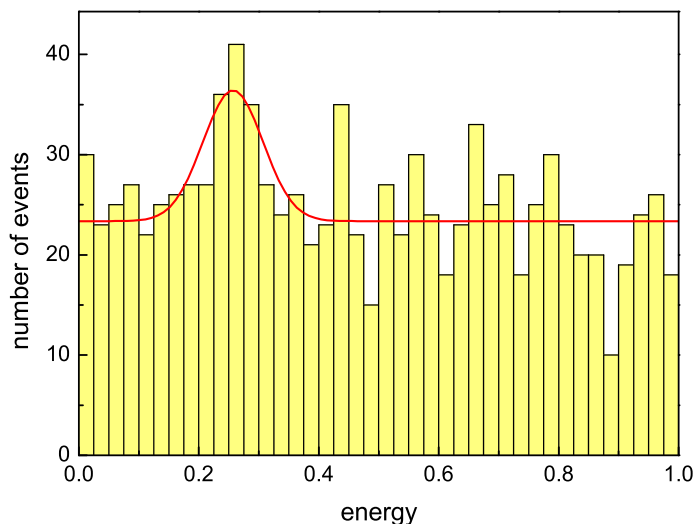


Fig. 10.18. Histogram of event sample used for the likelihood ratio test. The curve is an unbinned likelihood fit to the data.

claim large effects. Figure 10.18 displays the result of an experiment where a likelihood fit finds a resonance at the energy 0.257. It contains a fraction of 0.0653 of the events. The logarithm of the likelihood ratio is 9.277. The corresponding p-value for H_0 is $p_{LR} = 1.8 \cdot 10^{-4}$. Hence it is likely that the observed bump is a resonance. In fact it had been generated as a 7 % contribution of a Gaussian distribution $N(x|0.25, 0.05)$ to a uniform distribution.

We have to remember though that the p-value is not the probability that H_0 is true, it is the probability that H_0 simulates the resonance of the type seen in the data. In a Bayesian treatment, see Appendix 13.14, we find betting odds in favor of H_0 of about 2% which is much less impressive. The two numbers refer to different issues but nonetheless we have to face the fact that the two different statistical approaches lead to different conclusions about how evident the existence of a bump really is.

In experiments with a large number of events, the computation of the p-value distribution based on the unbinned likelihood ratio becomes excessively slow and we have to turn to histograms and to compute the likelihood ratio of H_0 and H_s from the histogram. Figure 10.19 displays some results from the simulation of 10^6 experiments of the same type as above but with 10000 events distributed over 100 bins.

In the figure also the distribution of the signal fraction under H_0 and for experiments with 1.5% resonance added is shown. The large spread of the signal distributions reflects the fact that identical experiments by chance may observe a very significant signal or just a slight indication of a resonance.

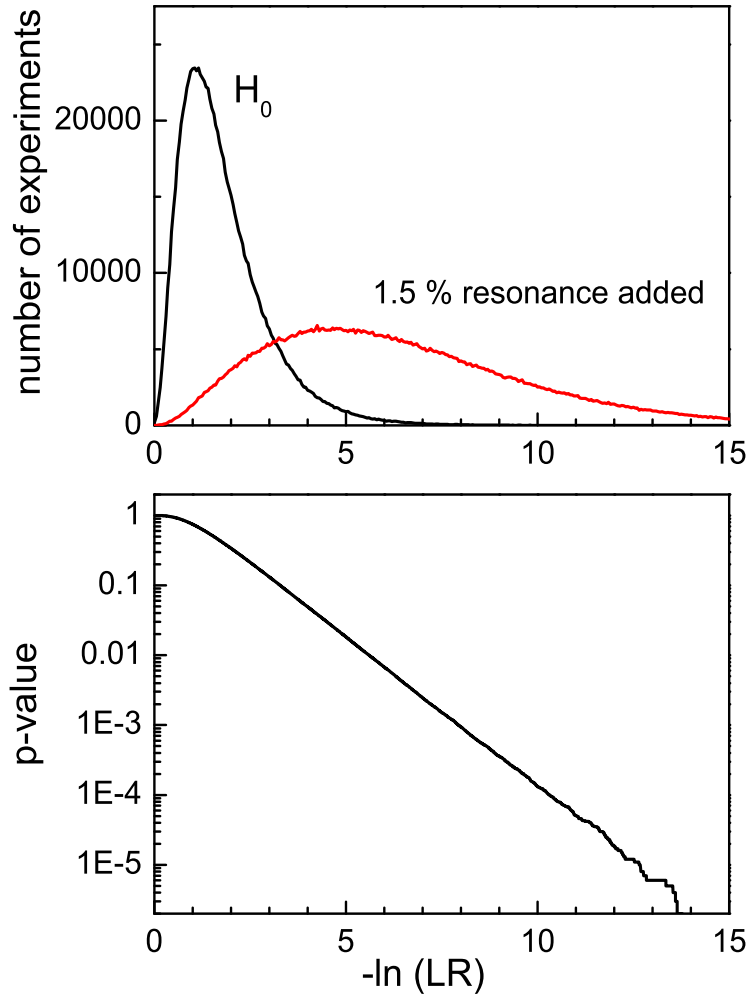


Fig. 10.19. Distributions of the test statistic under H_0 and p-value as a function of the test statistic. In the upper graph also the distribution for experiments with a resonance contribution is shown.

General Multi-Channel Case

It is easy to extend the likelihood ratio test to the multi-channel case. We assume that the observations $\mathbf{x}_k$ of the channels $k = 1, \dots, K$ are independent of each other. The overall likelihood is the product of the individual likelihoods. For the log-likelihood ratio we then have to replace (10.26) by

$$\ln \lambda = \sum_{k=1}^K \{ \ln \sup [L_{0k}(\boldsymbol{\theta}_{0k} | \mathbf{x}_k)] - \ln \sup [L_{sk}(\boldsymbol{\theta}_{sk} | \mathbf{x}_k)] \} .$$

As an example, we consider an experiment where we observe bumps at the same mass in K different decay channels, bumps which are associated to the same phenomenon, i.e. a particle decaying into different secondaries.

When we denote the decay contribution into channel k by ε_k , the p.d.f. of the decay distribution by $f_k(\mathbf{x}_k|\boldsymbol{\theta}_k)$ and the corresponding background distributions by $f_{0k}(\mathbf{x}_k|\boldsymbol{\theta}_{0k})$, the distribution under H_0 is

$$f_0(\mathbf{x}_1, \dots, \mathbf{x}_K|\boldsymbol{\theta}_{01}, \dots, \boldsymbol{\theta}_{0K}) = \prod_{k=1}^K f_{0k}(\mathbf{x}_k|\boldsymbol{\theta}_{0k})$$

and the alternative signal distribution is

$$f_s(\mathbf{x}_1, \dots, \mathbf{x}_K|\boldsymbol{\theta}_{01}, \dots, \boldsymbol{\theta}_{0K}; \boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_K; \varepsilon_1, \dots, \varepsilon_K) = \prod_{k=1}^K [(1 - \varepsilon_k)f_{0k}(\mathbf{x}_k|\boldsymbol{\theta}_{0k}) + \varepsilon_k f_k(\mathbf{x}_k|\boldsymbol{\theta}_k)] .$$

The likelihood ratio is then

$$\ln \lambda = \sum_{k=1}^K \left\{ \ln f_{0k}(\mathbf{x}_k|\hat{\boldsymbol{\theta}}_{0k}) - \ln [(1 - \hat{\varepsilon}_k)f_{0k}(\mathbf{x}_k|\hat{\boldsymbol{\theta}}'_{0k}) + \hat{\varepsilon}_k f_k(\mathbf{x}_k|\hat{\boldsymbol{\theta}}_k)] \right\} .$$

Remark, that the MLEs of the parameters $\boldsymbol{\theta}_{0k}$ depend on the hypothesis. They are different for the null and the signal hypotheses and, for this reason, have been marked by an apostrophe in the latter.

10.5.3 Tests Based on the Signal Strength

Instead of using the LR statistic it is often preferable to use a parameter of H_s as test statistic. In the simple example of (10.25) the test statistic $t = \alpha_3$ would be a sensible choice. When we want to estimate the significance of a line in a background distribution, instead of the likelihood ratio the number of events which we associate to the line (or the parameter α_3 in our example (10.28)) is a reasonable test statistic. Compared to the LR statistic it has the advantage to represent a physical parameter but usually the corresponding test is less powerful.

Example 134. Example 133 continued

Using the fitted fraction of resonance events as test statistic, the p-value for H_0 is $p_f = 2.2 \cdot 10^{-4}$, slightly less stringent than that obtained from the LR. Often physicists compare the number of observed events directly to the prediction from H_0 . In our example we have 243 events within two standard deviations around the fitted energy of the resonance compared to the expectation of 200 from a uniform distribution. The probability to observe ≥ 243 for a Poisson distribution with mean 200 is $p_p = 7.3 \cdot 10^{-4}$. This number cannot be compared directly with p_{LR} and p_f because the latter two values include the *look-else-where effect*, i.e. that the simulated resonance may be located at an arbitrary energy. A lower number for p_p is obtained if the background is estimated from the side bands, but then the computation becomes more involved because the error on the expectation has to be included. Primitive methods are only useful for a first crude estimate.

We learn from this example that the LR statistic provides the most powerful test among the considered alternatives. It does not only take into account the excess of events of a signal but also its expected shape. For this reason p_{LR} is smaller than p_f .

Often the significance of a signal s is stated in units of standard deviations σ :

$$s = \frac{N_s}{\sqrt{N_0 + \delta_0^2}} .$$

Here N_s is the number of events associated to the signal, N_0 is the number of events in the signal region expected from H_0 and δ_0 its uncertainty. In the Gaussian approximation it can be transformed into a p-value via (10.23). Unless N_0 is very large and δ_0 is very well known, this p-value has to be considered as a lower limit or a rough guess.

Statistical Learning

11.1 Introduction

In the process of its mental evolution a child learns to classify objects, persons, animals, and plants. This process partially proceeds through explanations by parents and teachers (supervised learning), but partially also by cognition of the similarities of different objects (unsupervised learning). But the process of learning – of children and adults – is not restricted to the development of the ability merely to classify but it includes also the realization of relations between similar objects, which leads to ordering and quantifying physical quantities, like size, time, temperature, etc.. This is relatively easy, when the laws of nature governing a specific relation have been discovered. If this is not the case, we have to rely on approximations, like inter- or extrapolations.

Also computers, when appropriately programmed, can perform learning processes in a similar way, though to a rather modest degree. The achievements of the so-called artificial intelligence are still rather moderate in most areas, however a substantial progress has been achieved in the fields of supervised learning and classification and there computers profit from their ability to handle a large amount of data in a short time and to provide precise quantitative solutions to well defined specific questions. The techniques and programs that allow computers to learn and to classify are summarized in the literature under the term *machine learning*.

Let us now specify the type of problems which we discuss in this chapter: For an input vector $\mathbf{x}$ we want to find an output $\hat{\mathbf{y}}$. The input is also called *predictor*, the output *response*. Usually, each input consists of several components (*attributes*, *properties*), and is written therefore in boldface letters. Normally, it is a metric (quantifiable) quantity but it could also be a categorical quantity like a color or a particle type. The output can also contain several components or consists of a single real or discrete (*Yes* or *No*) variable. Like a human being, a computer program learns from past experience. The teaching process, called training, uses a training sample $\{(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2) \dots (\mathbf{x}_N, \mathbf{y}_N)\}$, where for each input vector $\mathbf{x}_i$ the response $\mathbf{y}_i$ is known. When we ask for the response to an arbitrary continuous input $\mathbf{x}$, usually its estimate $\hat{\mathbf{y}}(\mathbf{x})$ will be more accurate when the distance to the nearest input vector of the training sample is small than when it is far away. Consequently, the training sample should be as large as possible or affordable. The region of interest should be covered with input vectors homogeneously, and we should be aware that the accuracy of the estimate decreases at the boundary of this region.

Learning which exceeds simple memorizing relies on the existence of more or less simple relations between input and response: Similar input corresponds to similar response. In our approach this translates into the requirement that the responses are similar for input vectors which are close. We can not learn much from erratic distributions.

Example 135. Simple empirical relations

The resistance R of a wire is used for a measurement of the temperature T . In the teaching process which here is called calibration, a sample of corresponding values R_i, T_i is acquired. In the application we want to find for a given input R an estimate of T . Usually a simple interpolation will solve this problem.

For more complicated relations, approximations with polynomials, higher spline functions or orthogonal functions are useful.

Example 136. Search for common properties

A certain class of molecules has a positive medical effect. The structure, physical and chemical properties $\mathbf{x}$ of these molecules are known. In order to find out which combination of the properties is relevant, the distribution of all attributes of the molecules which represent the training objects is investigated. A linear method for the solution of this task is the *principal component analysis*.

Example 137. Two-class classification, SPAM mails

A sizable fraction of electronic mails are of no interest to the addressee and considered by him as a nuisance. Many mailing systems use filter programs to eliminate these undesired so-called SPAM¹ mails. After evaluation of a training sample where the classification into *Yes* or *No* (accept or reject) is done by the user, the programs are able to take over the classification job. They identify certain characteristic words, like *Viagra*, *sex*, *profit*, *advantage*, *meeting*, *experiment*, *university* and other attributes like *large letters*, *colors* to distinguish between SPAM and serious mails. This kind of problem is efficiently solved by *decision trees* and *artificial neural networks*.

The attributes are here categorical variables. In the following we will restrict ourselves mainly to continuous variables.

Example 138. Multi-class classification, pattern recognition

Hand-written letters or figures have to be recognized. Again a sample for which the relation between the written pixels and the letters is known, is used to train the program. Also this problem can be treated by *decision trees*, *artificial neural networks*, and by *kernel methods*. Here the attributes are the pixel coordinates.

As we have observed also previously, multivariate applications suffer from the *curse of dimensionality*. There are two reasons: i) With increasing number d of dimensions, the distance between the input vectors increases and ii) the surface effects

¹SPAM is an artificial nonsense word borrowed from a sketch of a British comedy series of Monty Python's Flying Circus where in a cafe every meal contains SPAM.

are enhanced. When a fixed number of points is uniformly distributed over a hyper-cube of dimension d , the mean distance between the points is proportional to $\sqrt{d}$. The higher the dimension, the more empty is the space. At the same time the region where estimates become less accurate due to surface effects increases. The fraction of the volume taken by a hyper-sphere inscribed into a hyper-cube is only 5.2% for $d = 5$, and the fraction of the volume within a distance to the surface less than 10% of the edge length increases like $1 - 0.8^d$, this means from 20% for $d = 1$ to 67% for $d = 5$.

Example 139. Curse of dimensionality

A training sample of 1000 five-dimensional inputs is uniformly distributed over a hyper-cube of edge length a . To estimate the function value at the center of the region we take all sample elements within a distance of $a/4$ from the center. These are on average *one to two* only ($1000 \times 0.052 \times 0.5^5 = 1.6$), while in one dimension 500 elements would contribute.

In the following, we will first discuss the approximation of measurements afflicted with errors by analytic functions and the interpolation by smoothing techniques. Next we introduce the factor analysis, including the so-called principal component analysis. The last section deals with classification methods, based on artificial neural networks, kernel algorithms, and decision trees. In recent years we observed a fast progress in this field due to new developments, i.e. *support vector machines*, *boosting*, and the availability of powerful general computer algorithms. This book can only introduce these methods, without claim of completeness. A nice review of the whole field is given in [13].

11.2 Smoothing of Measurements and Approximation by Analytic Functions

We start with two simple examples, which illustrate applications:

i) In a sequence of measurements the gas amplification of an ionization chamber as a function of the applied voltage has been determined. We would like to describe the dependence in form of a smooth curve.

ii) With optical probes it is possible to scan a surface profile point-wise. The objects may be workpieces, tools, or human bodies. The measurements can be used by milling machines or cutting devices to produce replicates or clothes. To steer these machines, a complete surface profile of the objects is needed. The discrete points have to be approximated by a continuous function. When the surface is sufficiently smooth, this may be achieved by means of a spline approximation.

More generally, we are given a number N of measurements y_i with uncertainties δ_i at fixed locations x_i , the independent variables, but are interested in the values of the dependent or response variable y at different values of x , that is, we search for a function $f(x)$ which approximates the measurements, improves their precision and inter- and extrapolates in x . The simplest way to achieve this is to smooth the polygon connecting the data points.

More efficient is the approximation of the measurement by a parameter dependent analytic function $f(x, \theta)$. We then determine the parameters by a least square fit, i.e.

minimize the sum over the squared and normalized residuals $\sum [(y_i - f(x_i, \boldsymbol{\theta}))^2 / \delta_i^2]$ with respect to $\boldsymbol{\theta}$. The approximation should be compatible with the measurements within their statistical errors but the number of free parameters should be as small as possible. The accuracy of the measurements has a decisive influence on the number of free parameters which we permit in the fit. For large errors we allow also for large deviations of the approximation from the measurements. As a criterion for the number of free parameters, we use statistical tests like the χ^2 test. The value of χ^2 should then be compatible with the number of constraints, i.e. the number of measured points minus the number of fitted parameters. Too low a number of parameters leads to a bias of the predictions, while too many parameters reduce the accuracy, since we profit less from constraints.

Both approaches rely on the presumption that the true function is simple and smooth. Experience tells us that these conditions are justified in most cases.

The approximation of measurements which all have the same uncertainty by analytic functions is called regression analysis. Linear regression had been described in Chap. 7.2.3. In this section we treat the general non-linear case with arbitrary errors.

In principle, the independent variable may also be multi-dimensional. Since then the treatment is essentially the same as in the one-dimensional situation, we will mainly discuss the latter.

11.2.1 Smoothing Methods

We use the measured points in the neighborhood of $\mathbf{x}$ to get an estimate of the value of $\mathbf{y}(\mathbf{x})$. We denote the uncertainties of the output vectors of the training sample by δ_j for the component j of $\mathbf{y}$. When the points of the training sample have large errors, we average over a larger region than in the case of small errors. The better accuracy of the average for a larger region has to be paid for by a larger bias, due to the possibility of larger fluctuations of the true function in this region. Weighting methods work properly if the function is approximately linear. Difficulties arise in regions with lot of structure and at the boundaries of the region if there the function is not approximately constant.

k-Nearest Neighbors

The simplest method for a function approximation is similar to the density estimation which we treat in Chap. 9 and which uses the nearest neighbors in the training sample. We define a distance $d_i = |\mathbf{x} - \mathbf{x}_i|$ and sort the elements of the training sample in the order of their distances $d_i < d_{i+1}$. We choose a number K of nearest neighbors and average over the corresponding output vectors:

$$\hat{\mathbf{y}}(\mathbf{x}) = \frac{1}{K} \sum_{i=1}^K \mathbf{y}_i .$$

This relation holds for constant errors. Otherwise for the component j of $\mathbf{y}$ the corresponding weighted mean

$$\hat{y}_j(\mathbf{x}) = \frac{\sum_{i=1}^K y_{ij} / \delta_{ij}^2}{\sum_{i=1}^K 1 / \delta_{ij}^2}$$

has to be used. The choice of K depends on the density of points in the training sample and the expected variation of the true function $\mathbf{y}(\mathbf{x})$.

If all individual points in the projection j have mean square errors δ_j^2 , the error of the prediction δy_j is given by

$$(\delta y_j)^2 = \frac{\delta_j^2}{K} + \langle y_j(\mathbf{x}) - \hat{y}_j(\mathbf{x}) \rangle^2 . \quad (11.1)$$

The first term is the statistical fluctuation of the mean value. The second term is the bias which is equal to the systematic shift squared, and which is usually difficult to evaluate. There is the usual trade-off between the two error components: with increasing K the statistical term decreases, but the bias increases by an amount depending on the size of the fluctuations of the true function within the averaging region.

k-Nearest Neighbors with Linear Approximation

The simple average suffers from the drawback that at the boundary of the variable space the measurements contributing to the average are distributed asymmetrically with respect to the point of interest $\mathbf{x}$. If, for instance, the function falls strongly toward the left-hand boundary of a one-dimensional space, averaging over points which are predominantly located at the right hand side of x leads to too large a result. (See also the example at the end of this section). This problem can be avoided by fitting a linear function through the K neighboring points instead of using the mean value of $\mathbf{y}$.

Gaussian Kernels

To take all K-nearest neighbors into account with the same weight independent of their distance to $\mathbf{x}$ is certainly not optimal. Furthermore, its output function is piecewise constant (or linear) and thus discontinuous. Better should be a weighting procedure, where the weights become smaller with increasing distances. An often used weighting or kernel function² is the Gaussian. The sum is now taken over all N training inputs:

$$\hat{\mathbf{y}}(\mathbf{x}) = \frac{\sum_{i=1}^N \mathbf{y}_i e^{-\alpha|\mathbf{x}-\mathbf{x}_i|^2}}{\sum_{i=1}^N e^{-\alpha|\mathbf{x}-\mathbf{x}_i|^2}} .$$

The constant α determines the range of the correlation. Therefore the width $s = 1/\sqrt{2\alpha}$ of the Gaussian has to be adjusted to the density of the points and to the curvature of the function. If computing time has to be economized, the sum may of course be truncated and restricted to the neighborhood of $\mathbf{x}$, for instance to the distance $2s$. According to (11.1) the mean squared error becomes³:

$$(\delta y_j)^2 = \delta_j^2 \frac{\sum e^{-2\alpha|\mathbf{x}-\mathbf{x}_i|^2}}{[\sum e^{-\alpha|\mathbf{x}-\mathbf{x}_i|^2}]^2} + \langle y_j(\mathbf{x}) - \hat{y}_j(\mathbf{x}) \rangle^2 .$$

²The denotation *kernel* will be justified later, when we introduce classification methods.

³This relation has to be modified if not all errors are equal.

11.2.2 Approximation by Orthogonal Functions

Complete orthogonal function systems offer three attractive features: i) The fitted function coefficients are uncorrelated, ii) The function systems are complete and thus able to approximate any well behaved, i.e. square integrable, function, iii) They are naturally ordered with increasing oscillation frequency⁴. The function system to be used depends on the specific problem, i.e. on the domain of the variable and the asymptotic behavior of the function. Since the standard orthogonal functions are well known to physicists, we will be very brief and omit all mathematical details, they can be looked-up in mathematical handbooks.

Complete normalized orthogonal function systems $\{u_i(x)\}$ defined on the finite or infinite interval $[a, b]$ fulfil the orthogonality and the completeness relations. To simplify the notation, we introduce the inner product (g, h)

$$(g, h) \equiv \int_a^b g^*(x)h(x)dx$$

and have

$$(u_i, u_j) = \delta_{ij},$$

$$\sum_i u_i^*(x)u_i(x') = \delta(x - x').$$

For instance, the functions of the well known Fourier system for the interval $[a, b] = [-L/2, L/2]$ are $u_n(x) = \frac{1}{\sqrt{L}} \exp(i2\pi nx/L)$.

Every square integrable function can be represented by the series

$$f(x) = \sum_{i=0}^{\infty} a_i u_i(x), \text{ with } a_i = (u_i, f)$$

in the sense that the squared difference converges to zero with increasing number of terms⁵:

$$\lim_{K \rightarrow \infty} \left[f(x) - \sum_{i=0}^K a_i u_i(x) \right]^2 = 0. \quad (11.2)$$

The coefficients a_i become small for large i , if $f(x)$ is smooth as compared to the $u_i(x)$, which oscillate faster and faster for increasing i . Truncation of the series therefore causes some smoothing of the function.

The approximation of measurements by orthogonal functions works quite well for very smooth data. When the measurements show strong short range variations, sharp peaks or valleys, then a large number of functions is required to describe the data. Neglecting individually insignificant contributions may lead to a poor approximation. Typically, their truncation may produce spurious oscillations (“ringing”) in regions near to the peaks, where the true function is already flat.

For large data sets with equidistant points and equal errors the *Fast Fourier Transform*, FFT, plays an important role, especially for data smoothing and image processing. Besides the trigonometric functions, other orthogonal systems are

⁴We use the term *frequency* also for spatial dimensions.

⁵At eventual discontinuities, $f(x)$ should be taken as $[f(x+0) + f(x-0)]/2$.

Table 11.1. Characteristics of orthogonal polynomials.

Polynomial	Domain	Weight function
Legendre, $P_i(x)$	$[-1, +1]$	$w(x) = 1$
Hermite, $H_i(x)$	$(-\infty, +\infty)$	$w(x) = \exp(-x^2)$
Laguerre, $L_i(x)$	$[0, \infty)$	$w(x) = \exp(-x)$

useful, some of which are displayed in Table 11.2.2. The orthogonal functions are proportional to polynomials $p_i(x)$ of degree i multiplied by the square root of a weight function $w(x)$, $u_i(x) = p_i(x)\sqrt{w(x)}$. Specifying the domain $[a, b]$ and w , and requiring orthogonality for $u_{i,j}$,

$$(u_i, u_j) = c_i \delta_{ij} ,$$

fixes the polynomials up to the somewhat conventional normalization factors $\sqrt{c_i}$.

The most familiar orthogonal functions are the trigonometric functions used in the Fourier series mentioned above. From electrodynamics and quantum mechanics we are also familiar with Legendre polynomials and spherical harmonics. These functions are useful for data depending on variables defined on the circle or on the sphere, e.g. angular distributions. For example, the distribution of the intensity of the microwave background radiation which contains information about the curvature of the space, the baryon density and the amount of dark matter in the universe, is usually described as a function of the solid angle by a superposition of spherical harmonics. In particle physics the angular distributions of scattered or produced particles can be described by Legendre polynomials or spherical harmonics. Functions extending to $\pm\infty$ are often approximated by the eigenfunctions of the harmonic oscillator consisting of Hermite polynomials multiplied by the exponential $\exp(-x^2/2)$ and functions bounded to $x \geq 0$ by Laguerre polynomials multiplied by $e^{-x/2}$.

In order to approximate a given measurement by one of the orthogonal function systems, one usually has to shift and scale the independent variable x .

Polynomial Approximation

The simplest function approximation is achieved with a simple polynomial $f(x) = \sum a_k x^k$ or more generally by $f(x) = \sum a_k u_k$ where u_k is a polynomial of order k . Given data y_ν with uncertainties δ_ν at locations x_ν we minimize

$$\chi^2 = \sum_{\nu=1}^N \frac{1}{\delta_\nu^2} \left[y_\nu - \sum_{k=0}^K a_k u_k(x_\nu) \right]^2 , \quad (11.3)$$

in order to determine the coefficients a_k . To constrain the coefficients, their number $K+1$ has to be smaller than the number N of measurements. All polynomial systems of the same order describe the data equally well but differ in the degree to which the coefficients are correlated. The power of the polynomial is increased until it is compatible within statistics with the data. The decision is based on a χ^2 criterion.

The purpose of this section is to show how we can select polynomials with uncorrelated coefficients. In principle, these polynomials and their coefficients can be computed through diagonalization of the error matrix but they can also be obtained

directly with the Gram–Schmidt method. This method has the additional advantage that the polynomials and their coefficients are given by simple algebraic relations.

For a given sample of measured points $y_\nu = f(x_\nu)$ with errors δ_ν , we fix the weights in the usual way

$$w_\nu = w(x_\nu) = \frac{1}{\delta_\nu^2} / \sum_j \frac{1}{\delta_j^2} ,$$

and now define the inner product of two functions $g(x), h(x)$ by

$$(g, h) = \sum_\nu w_\nu g(x_\nu) h(x_\nu)$$

with the requirement

$$(u_i, u_j) = \delta_{ij} .$$

Minimizing χ^2 is equivalent to minimizing

$$X^2 = \sum_{\nu=1}^N w_\nu \left[y_\nu - \sum_{k=0}^K a_k u_k(x_\nu) \right]^2 .$$

For $K = N - 1$ the square bracket at the minimum of X^2 is zero,

$$y_\nu - \sum_{k=0}^{N-1} a_k u_k(x_\nu) = 0$$

for all ν , and forming the inner product with u_j we get

$$(y, u_j) = a_j . \quad (11.4)$$

This relation produces the coefficients also in the interesting case $K < N - 1$.

To construct the orthogonal polynomials, we set $v_0 = 1$,

$$u_i = \frac{v_i}{\sqrt{(v_i, v_i)}} , \quad (11.5)$$

$$v_{i+1} = x^{i+1} - \sum_{j=0}^i (u_j, x^{i+1}) u_j . \quad (11.6)$$

The first two terms in the corresponding expansion, $a_0 u_0$ and $a_1 u_1$, are easily calculated. From (11.5), (11.6), (11.4) and the following definition of the moments of the weighted sample

$$\bar{x} = \sum_\nu w_\nu x_\nu , \quad s_x^2 = \sum_\nu w_\nu (x_\nu - \bar{x})^2 , \quad s_{xy} = \sum_\nu w_\nu (x_\nu y_\nu - \bar{x} \bar{y})$$

we find the coefficients and functions which fix the polynomial expansion of y :

$$y = \bar{y} + \frac{s_{xy}}{s_x^2} (x - \bar{x}) .$$

We recover the well known result for the best fit by a straight line in the form with independent coefficients: This is of course no surprise, as the functions that are

minimized are identical, namely χ^2 in both cases, see Example 114 in Chap. 7. The calculation of higher order terms is straight forward but tedious. The uncertainties δa_i of the coefficients are all equal independent of i and given by the simple relation

$$(\delta a_i)^2 = 1 / \sum_{\nu=1}^N \frac{1}{\delta_\nu^2} .$$

The derivation of this formula is given in the Appendix 13.11 together with formulas for the polynomials in the special case where all measurements have equal errors and are uniformly distributed in x .

The Gram–Charlier Series

The following example for the application of Hermite functions, strictly speaking, does not concern the approximation of measurements by a function but the approximation of an empirical p.d.f. (see Sect. 12.1.1 in the following Chapter). We discuss it here since it is mathematically closely related to the subject of this section.

The Gram–Charlier series is used to approximate empirical distributions which do not differ very much from the normal distribution. It expresses the quotient of an empirical p.d.f. $f(x)$ to the standard normal distribution $N(x|0, 1)$ as an expansion in the slightly modified Hermite polynomials $\tilde{H}_i(x)$ in the form

$$f(x) = N(x) \sum_{i=0}^{\infty} a_i \tilde{H}_i(x) . \quad (11.7)$$

Here, $N(x) \equiv (2\pi)^{-1/2} \exp(-x^2/2)$, the standard normal distribution, differs somewhat from the weight function $\exp(-x^2)$ used in the definition of the Hermite polynomials $H(x)$ given above in Table 11.2.2. The two definitions of the polynomials are related by

$$\tilde{H}_i(x) = \frac{1}{\sqrt{2^i}} H_i \left(\frac{x}{\sqrt{2}} \right) .$$

The orthogonality relation of the modified polynomials is

$$(\tilde{H}_i, \tilde{H}_j) = \int_{-\infty}^{+\infty} N(x) \tilde{H}_i(x) \tilde{H}_j(x) dx = i! \delta_{ij} , \quad (11.8)$$

and their explicit form can be obtained by the simple recursion relation:

$$\tilde{H}_{i+1} = x \tilde{H}_i - i \tilde{H}_{i-1} .$$

With $\tilde{H}_0 = 1$, $\tilde{H}_1 = x$ we get

$$\begin{aligned} \tilde{H}_2 &= x^2 - 1 , \\ \tilde{H}_3 &= x^3 - 3x , \\ \tilde{H}_4 &= x^4 - 6x^2 + 3 , \end{aligned}$$

and so on.

When we multiply both sides of (11.7) with $\tilde{H}_j(x)$ and integrate, we find, according to (11.8), the coefficients a_i from

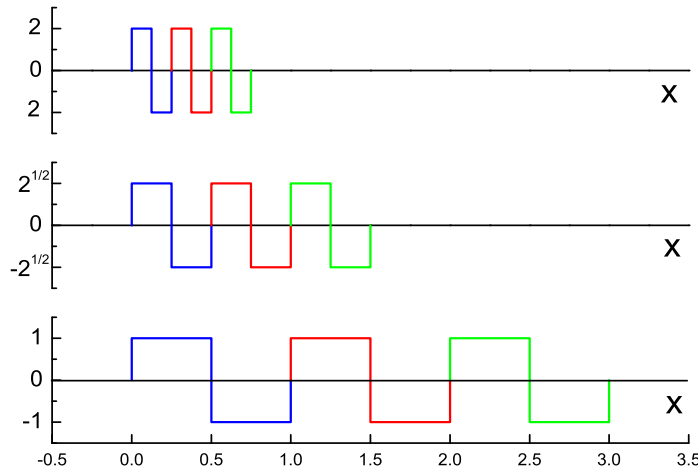


Fig. 11.1. Nine orthonormalized wavelets with three different frequencies..

$$a_i = \frac{1}{i!} \int f(x) \tilde{H}_i(x) dx .$$

These integrals can be expressed as combinations of moments of $f(x)$, which are to be approximated by the sample moments of the experimental distribution. First, the sample mean and the sample variance are used to shift and scale the experimental distribution such that the transformed mean and variance equal 0 and 1, respectively. Then $a_{1,2} = 0$, and the empirical skewness and excess of the normalized sample $\gamma_{1,2}$ as defined in Sect. 3.2 are proportional to the parameters $a_{3,4}$. The approximation to this order is

$$f(x) \approx N(x) \left(1 + \frac{1}{3!} \gamma_1 \tilde{H}_3(x) + \frac{1}{4!} \gamma_2 \tilde{H}_4(x) \right) .$$

As mentioned, this approximation is well suited to describe distributions which are close to normal distributions. This is realized, for instance, when the variate is a sum of independent variates such that the central limit theorem applies. It is advisable to check the convergence of the corresponding Gram–Charlier series and not to truncate the series too early [2].

11.2.3 Wavelets

The trigonometric functions used in the Fourier series are discrete in the frequency domain, but extend from minus infinity to plus infinity in the spatial domain and thus are not very well suited to describe strongly localized function variations. To handle this kind of problems, the wavelet system has been invented. Wavelets are able to describe pulse signals and spikes like those generated in electrocardiograms, nuclear magnetic resonance (NMR) records or seismic records, in data transmission, and for the coding of images and hand-written text. For data reduction and storage they have become an indispensable tool.

The simplest orthogonal system with the desired properties are the *Haar wavelets* shown in Fig. 11.1. The lowest row shows three wavelets which are orthogonal, because they have no overlap. The next higher row contains again three wavelets with one half the length of those below. They are orthogonal to each other and to the wavelets in the lower row. In the same way the higher frequency wavelets in the following row are constructed. We label them with two indices j, k indicating length and position. We define a *mother function* $\psi(x)$, the bottom left wavelet function of Fig. 11.1.

$$\psi(x) = \begin{cases} 1, & \text{if } 0 \leq x < \frac{1}{2} \\ -1, & \text{if } \frac{1}{2} \leq x < 1 \\ 0, & \text{else} \end{cases}$$

and set $W_{00} = \psi(x)$. The remaining wavelets are then obtained by translations and dilatations in discrete steps from the mother function $\psi(x)$:

$$W_{jk}(x) = 2^{j/2} \psi(2^j x - k) .$$

The factor $2^{j/2}$ provides the normalization in the orthonormality relation⁶.

$$\int_{-\infty}^{+\infty} W_{ik}^*(x) W_{jl}(x) dx = \delta_{ij} \delta_{kl} . \quad (11.9)$$

It is evident that wavelets are much better suited to fit local structures than the sine and cosine functions of the Fourier expansion, since the wavelet expansion coefficients c_{jk} contain information on frequency *and* location of a signal.

The simple Haar wavelets shown in Fig. 11.1 which we have introduced to demonstrate the principal properties of wavelets are rarely used in applications as functions with infinitely sharp edges are usually absent in a realistic phenomenon. More common are the smoother wavelets

$$\psi(x) = \frac{1}{\sqrt{2\pi}\sigma^3} e^{-x^2/(2\sigma^2)} \left(1 - \frac{x^2}{\sigma^2}\right) \quad (\text{Mexican Hat}) , \quad (11.10)$$

$$\psi(x) = (e^{ix} - c) e^{-x^2/(2\sigma^2)} \quad (\text{Morlet-Wavelet}) , \quad (11.11)$$

and many others. The first function, the Mexican hat, is the second derivative of the Gaussian function, Fig. 11.2. The second, the Morlet function, is a complex monochromatic wave, modulated by a Gaussian. The constant $c = \exp(-\sigma^2/2)$ in the Morlet function can usually be neglected by choosing a wide lowest order function, $\sigma \gtrsim 5$. In both functions σ defines the width of the window.

The mother function ψ has to fulfil apart from the trivial normalization property 11.9, also the relation

$$\int \psi(x) dx = 0 .$$

Any square integrable function $f(x)$ fulfilling $\int f(x) dx = 0$ can be expanded in the discrete wavelet series,

$$f(x) = \sum_{j,k} c_{jk} W_{jk}(x) .$$

As usual, in order to regularize the function $f(x)$, the expansion is truncated when the coefficients become insignificant with increasing j , corresponding to small details or large frequencies.

⁶The Haar wavelets are real, but some types of wavelets are complex.

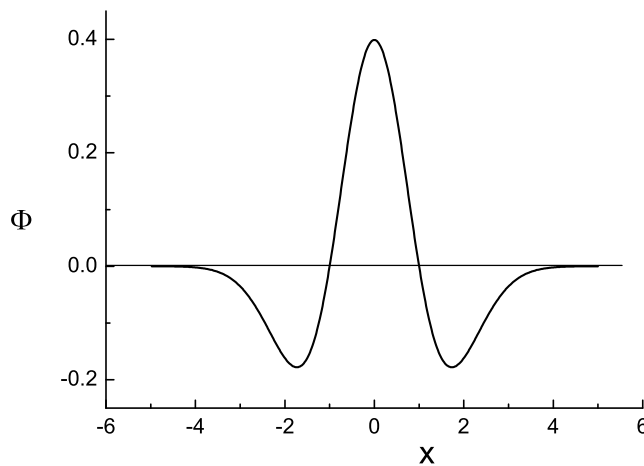


Fig. 11.2. Mexican hat wavelet.

The calculation of the coefficients c_{jk} is in principle analogous to the calculation of Fourier coefficients by convolution of f with the wavelets⁷. For given measurements a least square fit can be applied. The success of the wavelet applications was promoted by the appearance of fast numerical algorithms, like the multi-scale analysis. They work on a function f which need not integrate to zero, sampled at equidistant points, similarly to the fast Fourier transform (FFT).

An elementary introduction to the wavelet analysis is found in [66]. Programs are available in program libraries and in the internet.

11.2.4 Spline Approximation

The mathematical and numerical treatment of polynomials is especially simple and effective. Therefore, they are often chosen for the approximation of experimental data. A disadvantage of polynomials is however that they tend to infinity for large absolute values of the independent variable. This difficulty is resolved by using piecewise polynomial functions, the splines.

According to the degree of the polynomials used, we distinguish between linear, quadratic, cubic etc. splines.

The simplest spline approximation is the linear one, consisting of a polygon. The steps in the independent variable x between the knots are constant (Fig. 11.3). The lower the chosen number of knots and the spline order are, the larger will be on average the deviations of the points from the fitted curve. A sensible choice should take into account the mean squared dispersion of the points, i.e. the χ^2 -sum should be of the order of the number of degrees of freedom. When the response values y are exact and equidistant, the points are simply connected by a polygon.

A smoother approximation with no kinks is obtained with quadratic splines. A curve with continuous derivatives up to order n is produced with splines of degree

⁷The wavelets (11.10), (11.11) are not orthogonal. Thus the coefficients are correlated.

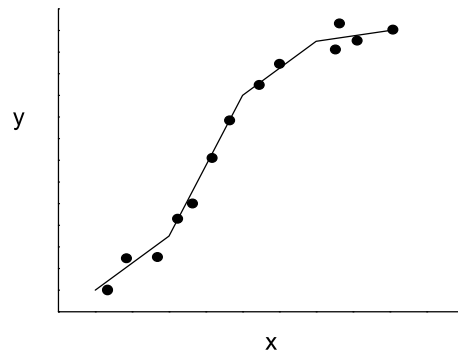


Fig. 11.3. Linear spline approximation.

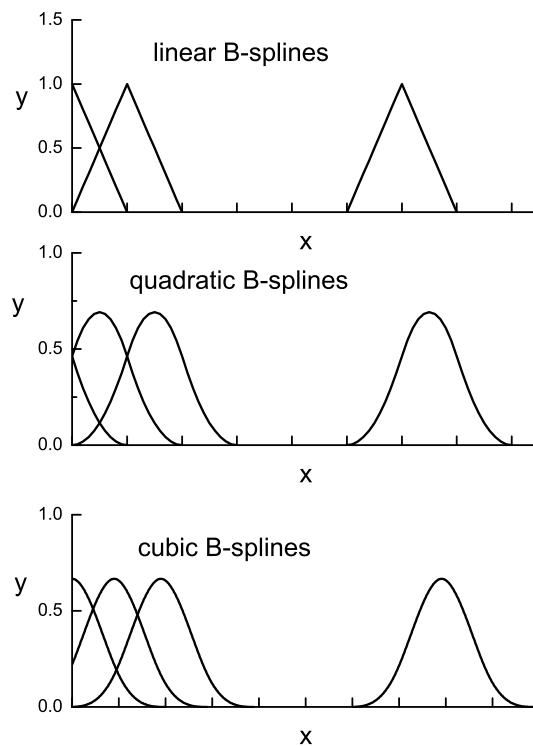


Fig. 11.4. Linear, quadratic and cubic B-splines.

$\geq n+1$. Since a curve with continuous second derivatives looks smooth to the human eye, splines of degree higher than cubic are rarely used.

Spline approximations are widely used in technical disciplines. They have also been successfully applied to the deconvolution problem [8] (Chap. 9). Instead of adapting a histogram to the true distribution, the amplitudes of spline functions can be fitted. This has the advantage that we obtain a continuous function which incorporates the desired degree of regularization.

For the numerical computations the so called *B-splines* (basis splines) are especially useful. Linear, quadratic and cubic B-splines are shown in Fig. 11.4. The

superposition of B-splines fulfils the continuity conditions at the knots. The superposition of the triangular linear B-splines produces a polygon, that of quadratic and cubic B-splines a curve with continuous slope and curvature, respectively.

A B-spline of given degree is determined by the step width b and the position x_0 of its center. Their explicit mathematical expressions are given in Appendix 13.12.

The function is approximated by

$$\hat{f}(x) = \sum_{k=0}^K a_k B_k(x) . \quad (11.12)$$

The amplitudes a_k can be obtained from a least squares fit. For values of the response function y_i and errors δy_i at the input points $x_i, i = 1, \dots, N$, we minimize

$$\chi^2 = \sum_{i=1}^N \frac{\left[y_i - \sum_{k=0}^K a_k B_k(x_i) \right]^2}{(\delta y_i)^2} . \quad (11.13)$$

Of course, the number N of input values has to be at least equal to the number K of splines. Otherwise the number of degrees of freedom would become negative and the approximation under-determined.

Spline Approximation in Higher Dimensions

In principle, the spline approximation can be generalized to higher dimensions. However, there the difficulty is that a grid of intervals (knots) destroys the rotation symmetry. It is again advantageous to work with B-splines. Their definition becomes more complicated: In two dimensions we have instead of triangular functions pyramids and for quadratic splines also mixed terms $\propto x_1 x_2$ have to be taken into account. In higher dimensions the number of mixed terms explodes, another example of the curse of dimensionality.

11.2.5 Approximation by a Combination of Simple Functions

There is no general recipe for function approximation. An experienced scientist would try, first of all, to find functions which describe the asymptotic behavior and the rough distribution of the data, and then add further functions to describe the details. This approach is more tedious than using programs from libraries but will usually produce results superior to those of the general methods described above.

Besides polynomials, $a_0 + a_1 x + a_2 x^2 + \dots$, rational functions can be used, i.e. quotients of two polynomials (Padé approximation), the exponential function $e^{\alpha x}$, the logarithm $^b \log x$, the Gaussian e^{-ax^2} , and combinations like $x^a e^{-bx}$. In many cases a simple polynomial will do. The results usually improve when the original data are transformed by translation and dilatation $x \rightarrow a(x + b)$ to a normalized form.

11.2.6 Example

In order to compare different methods, we have generated at equidistant values x_i of the variable x , function values y_i according to the function $x e^{-x}$ with superimposed

Gaussian fluctuations. The measurements generated in this way are smoothed, respectively fitted by different functions. The results are shown in Fig. 10.9. All eight panels show the original function and the measured points connected by a polygon.

In the upper two panels smoothing has been performed by weighting. Typical for both methods are that structures are washed-out and strong deviations at the borders. The Gaussian weighting in the left hand panel performs better than the method of nearest neighbors on the right hand side which also shows spurious short range fluctuations which are typical for this method.

As expected, also the linear spline approximation is not satisfactory but the edges are reproduced better than with the weighting methods. Both quadratic and cubic splines with 10 free parameters describe the measurement points adequately, but the cubic splines show some unwanted oscillations. The structure of the spline intervals is clearly seen. Reducing the number of free parameters to 5 suppresses the spurious fluctuations but then the spline functions cannot follow any more the steep rise at small x . There is only a marginal difference between the quadratic and the cubic spline approximations.

The approximation by a simple polynomial of fourth order, i.e. with 5 free parameters, works excellently. By the way, it differs substantially from the Taylor expansion of the true function. The polynomial can adapt itself much better to regions of different curvature than the splines with their fixed step width.

To summarize: The physicist will usually prefer to construct a clever parametrization with simple analytic functions to describe his data and avoid the more general standard methods available in program libraries. Those are useful to get a fast overview and for the parametrization of a large amount of data.

As we have already mentioned, the approximation of measurements by the standard set of orthogonal functions works quite well for very smooth functions where sharp peaks and valleys are absent. Peaks and bumps are better described with wavelets than with the conventional orthogonal functions. Smoothing results of measurements with the primitive kernel methods which we have discussed are usually unsatisfactory. A better performance is obtained with kernels with variable width and corrections for a possible boundary bias. The reader is referred to the literature [86]. Spline approximations are useful when the user has no idea about the shape of the function and when the measurements are able to constrain the function sufficiently to suppress fake oscillations.

11.3 Linear Factor Analysis and Principal Components

Factor analysis and *principal component analysis* (PCA) both reduce a multi-dimensional variate space to lower dimensions. In the literature there is no clear distinction between the two techniques.

Often several features of an object are correlated or redundant, and we want to express them by a few uncorrelated components with the hope to gain deeper insight into latent relations. One would like to reduce the number of features to as low a number of components, called factors, as possible.

Let us imagine that for 20 cuboids we have determined 6 geometrical and physical quantities: volume, surface, basis area, sum of edge lengths, mass, and principal moments of inertia. We submit these data which may be represented by a 6-dimensional

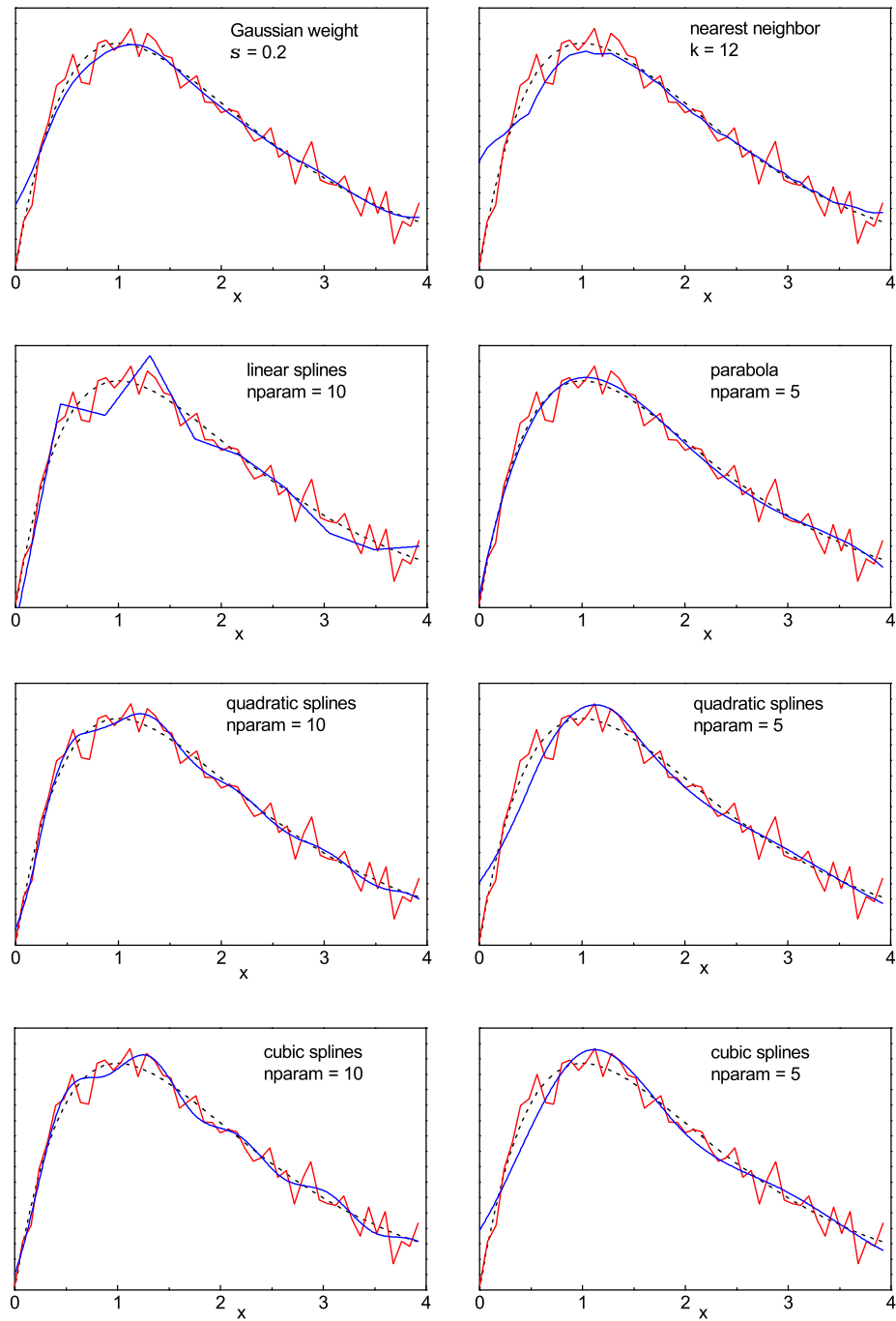


Fig. 11.5. Smoothing and function approximation. The measurements are connected by a polygon. The curve corresponding to the original function is dashed.

vector to a colleague without further information. He will look for similarities and correlations, and he might guess that these data can be derived for each cuboid from only 4 parameters, namely length, width, height, and density. The search for these basis parameters, the components or factors is called *factor analysis* [67, 68].

A general solution for this problem cannot be given, without an ansatz for the functional relationship between the feature matrix, in our example build from the 20 six-dimensional data vectors and its errors. Our example indicates though, in which direction we might search for a solution of this problem. Each body is represented by a point in a six-dimensional feature space. The points are, however, restricted to a four-dimensional subspace, the component space. The problem is to find this subspace. This is relatively simple if it is linear.

In general, and in our example, the subspace is not linear, but a linear approximation might be justified if the cuboids are very similar such that the components depend approximately linearly on the deviations of the input vectors from a center of gravity. Certainly in the general situation it is reasonable to look first for a linear relationship between features and parameters. Then the subspace is a linear vector space and easy to identify. In the special situation where only one component exists, all points lie approximately on a straight line, deviations being due to measurement errors and non-linearity. To identify the multi-dimensional plane, we have to investigate the correlation matrix. Its transformation into diagonal form delivers the *principal components* – linear combinations of the feature vectors in the direction of the principal axes. The principal components are the eigenvectors of the correlation matrix ordered according to decreasing eigenvalues. When we ignore the principal components with small eigenvalues, the remaining components form the planar subspace.

Factor analysis or PCA has been developed in psychology, but it is widely used also in other descriptive fields, and there are numerous applications in chemistry and biology. Its moderate computing requirements which are at the expense of the restriction to linear relations, are certainly one of the historical reasons for its popularity. We sketch it below, because it is still in use, and because it helps to get a quick idea of hidden structures in multi-dimensional data. When no dominant components are found, it may help to disprove expected relations between different observations.

A typical application is the search for factors explaining similar properties between different objects: Different chemical compounds may act similarly, e.g. decrease the surface tension of water. The compounds may differ in various features, as molecular size and weight, electrical dipole moment, and others. We want to know which parameter or combination of parameters is relevant for the interesting property. Another application is the search for decisive factors for a similar curing effect of different drugs. The knowledge of the principal factors helps to find new drugs with the same positive effect.

In physics factor analysis does not play a central role, mainly because its results are often difficult to interpret and, as we will see below, not unambiguous. It is not easy, therefore, to find examples from our discipline. Here we illustrate the method with an artificially constructed example taken from astronomy.

Example 140. Principal component analysis

Galaxies show the well known red-shift of the spectrum which is due to their escape velocity. Besides the measurement value or feature *red-shift* x_1 we know the *brightness* x_2 of the galaxies. To be independent of scales and

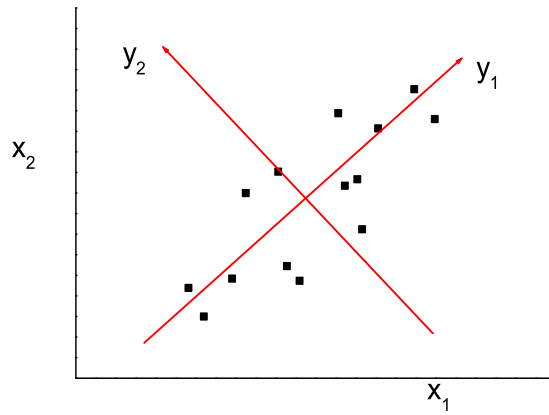


Fig. 11.6. Scatter diagram of two attributes of 11 measured objects.

mean values, we transform these quantities in such a way that sample mean and variance are zero, respectively unity. To demonstrate the concept, we have invented some data which are shown in Fig. 11.6. The two coordinates are strongly correlated. The correlation is eliminated in a rotated coordinate system where the objects have coordinates y_1 and y_2 which are linear combinations of red-shift and brightness in the directions of the principal axes of the correlation matrix. Now we consider as important those directions, where the observed objects show the largest differences. In our case this is the direction of y_1 , while y_2 has apparently only a minor influence on both features. We may conclude that red-shift and brightness have mainly one and the same cause which determines the value of y_1 . In our example, we know that this is the distance, both brightness and red shift depend on it. Since, apparently, the distance determines y_1 , we can use it, after a suitable calibration, as a measure for the distance.

We will now put these ideas into concrete terms.

The input data for the factor analysis are given in the form of a matrix X of N rows and P columns. The element x_{np} is the measured value of the feature p of the object n , thus X is a rectangular matrix. In a first step we determine the correlations between the P input attributes. By a simple transformation, we obtain uncorrelated linear combinations of the features. The hope is that there are few dominant combinations and that the others can be neglected. Then the data can be described by a small number of $Q < P$ linear combinations, the principal components.

We first transform the data X_{np} into standardized form where the sample mean and variance are zero, respectively unity. We get the normalized variables⁸

$$x_{np} = \frac{X_{np} - \bar{X}_p}{\delta_p}$$

with

$$\bar{X}_p = \frac{1}{N} \sum_{n=1}^N X_{np} ,$$

⁸The normalization (division by δ_p) is not always required.

$$\delta_p^2 = \frac{1}{N-1} \sum_{n=1}^N (X_{np} - \bar{X}_p)^2 .$$

The quantity x_{np} is the normalized deviation of the measurement value of type p for the object n from the average over all objects for this measurement.

In the same way as in Chap. 4 we construct the correlation matrix for our sample by averaging the $P \times P$ products of the components $x_{n1} \dots x_{nP}$ over all N objects:

$$\begin{aligned} \mathbf{C} &= \frac{1}{N-1} \mathbf{X}^T \mathbf{X} , \\ C_{pq} &= \frac{1}{N-1} \sum_{n=1}^N x_{np} x_{nq} . \end{aligned}$$

It is a symmetric positive definite $P \times P$ matrix. Due to the normalization the diagonal elements are equal to unity.

Then this matrix is brought into diagonal form by an orthogonal transformation corresponding to a rotation in the P -dimensional feature space.

$$\mathbf{C} \rightarrow \mathbf{V}^T \mathbf{C} \mathbf{V} = \text{diag}(\lambda_1 \dots \lambda_P) .$$

The uncorrelated feature vectors in the rotated space $\mathbf{y}_n = \{y_{n1}, \dots, y_{nP}\}$ are given by

$$\mathbf{y}_n = \mathbf{V}^T \mathbf{x}_n , \quad \mathbf{x}_n = \mathbf{V} \mathbf{y}_n .$$

To obtain eigenvalues and -vectors we solve the linear equation system

$$(\mathbf{C} - \lambda_p \mathbf{I}) \mathbf{v}_p = 0 , \quad (11.14)$$

where λ_p is the eigenvalue belonging to the eigenvector $\mathbf{v}_p$:

$$\mathbf{C} \mathbf{v}_p = \lambda_p \mathbf{v}_p .$$

The P eigenvalues are found as the solutions of the characteristic equation

$$\det(\mathbf{C} - \lambda \mathbf{I}) = 0 .$$

In the simple case described above of only two features, this is a quadratic equation

$$\begin{vmatrix} C_{11} - \lambda & C_{12} \\ C_{21} & C_{22} - \lambda \end{vmatrix} = 0 ,$$

that fixes the two eigenvalues. The eigenvectors are calculated from (11.14) after substituting the respective eigenvalue. As they are fixed only up to an arbitrary factor, they are usually normalized. The rotation matrix $\mathbf{V}$ is constructed by taking the eigenvectors $\mathbf{v}_p$ as its columns: $v_{qp} = (\mathbf{v}_p)_q$.

Since the eigenvalues are the diagonal elements in the rotated, diagonal correlation matrix, they correspond to the variances of the data distribution with respect to the principal axes. A small eigenvalue means that the projection of the data on this axis has a narrow distribution. The respective component is then, presumably, only of small influence on the data, and may perhaps be ignored in a model of the data. Large eigenvalues belong to the important principal components.

Factors f_{np} are obtained by standardization of the transformed variables y_{np} by division by the square root of the eigenvalues λ_p :

$$f_{np} = \frac{y_{np}}{\sqrt{\lambda_p}} .$$

By construction, these factors represent variates with zero mean and unit variance. In most cases they are assumed to be normally distributed. Their relation to the original data x_{np} is given by a linear (not orthogonal) transformation with a matrix $\mathbf{A}$, the elements of which are called factor *loadings*. Its definition is

$$\mathbf{x}_n = \mathbf{A} \mathbf{f}_n , \text{ or } \mathbf{X}^T = \mathbf{A} \mathbf{F}^T . \quad (11.15)$$

Its components show, how strongly the input data are influenced by certain factors.

In the classical factor analysis, the idea is to reduce the number of factors such that the description of the data is still satisfactory within tolerable deviations ε :

$$\begin{aligned} x_1 &= a_{11}f_1 + \cdots + a_{1Q}f_Q + \varepsilon_1 \\ x_2 &= a_{21}f_1 + \cdots + a_{2Q}f_Q + \varepsilon_2 \\ &\vdots \\ x_P &= a_{P1}f_1 + \cdots + a_{PQ}f_Q + \varepsilon_P \end{aligned}$$

with $Q < P$, where the “factors” (latent random variables) $f_1, \dots, f_Q$ are considered as uncorrelated and distributed according to $N(0, 1)$, plus uncorrelated zero-mean Gaussian variables ε_p , with variances σ_p^2 , representing the residual statistical fluctuations not described by the linear combinations. As a first guess, Q is taken as the index of the smallest eigenvalue λ_Q which is considered to be still significant. In the ideal case $Q = 1$ only one decisive factor would be the dominant element able to describe the data.

Generally, the aim is to estimate the loadings a_{pq} , the eigenvalues λ_p , and the variances σ_p^2 from the sampling data, in order to reduce the number of relevant quantities responsible for their description.

The same results as we have found above by the traditional method by solving the eigenvalue problem for the correlation matrix⁹ can be obtained directly by using the *singular value decomposition* (SVD) of the matrix $\mathbf{X}$ (remember that it has N rows and P columns):

$$\mathbf{X} = \mathbf{U} \mathbf{D} \mathbf{V}^T ,$$

where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal. $\mathbf{U}$ is not a square matrix, nevertheless $\mathbf{U}^T \mathbf{U} = \mathbf{I}$, where the unit matrix $\mathbf{I}$ has dimension P . $\mathbf{D}$ is a diagonal matrix with elements $\sqrt{\lambda_p}$, ordered according to decreasing values, and called here singular values. The decomposition (11.15) is obtained by setting $\mathbf{F} = \mathbf{U}$ and $\mathbf{A} = \mathbf{V} \mathbf{D}$.

The decomposition (11.15) is not unique: If we multiply both $\mathbf{F}$ and $\mathbf{A}$ with a rotation matrix $\mathbf{R}$ from the right we get an equivalent decomposition:

$$\mathbf{X} = \tilde{\mathbf{F}} \tilde{\mathbf{A}}^T = \mathbf{F} \mathbf{R} (\mathbf{A} \mathbf{R})^T = \mathbf{F} \mathbf{R} \mathbf{R}^T \mathbf{A}^T = \mathbf{U} \mathbf{D} \mathbf{V}^T , \quad (11.16)$$

which is the same as (11.15), with factors and loadings being rotated.

There exist program packages which perform the numerical calculation of principal components and factors.

Remarks:

⁹Physicists may find the method familiar from the discussion of the inertial momentum tensor and many similar problems.

1. The transformation of the correlation matrix to diagonal form makes sense, as we obtain in this way uncorrelated inputs. The new variables help to understand better the relations between the various measurements.
2. The silent assumption that the principal components with larger eigenvalues are the more important ones is not always convincing, since starting with uncorrelated measurements, due to the scaling procedure, would result in eigenvalues which are all identical. An additional difficulty for interpreting the data comes from the ambiguity (11.16) concerning rotations of factors and loadings.

11.4 Classification

We have come across classification already when we have treated goodness-of-fit. There the problem was either to accept or to reject a hypothesis without a clear alternative. Now we consider a situation where we dispose of information of two or more classes of events.

The assignment of an object according to some quality to a class or category is described by a so-called categorical variable. For two categories we can label the two possibilities by discrete numbers; usually the values ± 1 or 1 and 0 are chosen. In most cases it makes sense, to give as a result instead of a discrete classification a continuous variable as a measure for the correctness of the classification. The classification into more than two cases can be performed sequentially by first combining classes such that we have a two class system and then splitting them further.

Classification is indispensable in data analysis in many areas. Examples in particle physics are the identification of particles from shower profiles or from Cerenkov ring images, beauty, top or Higgs particles from kinematics and secondaries and the separation of rare interactions from frequent ones. In astronomy the classification of galaxies and other stellar objects is of interest. But classification is also a precondition for decisions in many scientific fields and in everyday life.

We start with an example: A patient suffers from certain symptoms: stomach-ache, diarrhoea, temperature, head-ache. The doctor has to give a diagnosis. He will consider further factors, as age, sex, earlier diseases, possibility of infection, duration of the illness, etc.. The diagnosis is based on the experience and education of the doctor.

A computer program which is supposed to help the doctor in this matter should be able to learn from past cases, and to compare new inputs in a sensible way with the stored data. Of course, as opposed to most problems in science, it is not possible here to provide a functional, parametric relation. Hence there is a need for suitable methods which interpolate or extrapolate in the space of the input variables. If these quantities cannot be ordered, e.g. sex, color, shape, they have to be classified. In a broad sense, all this problems may be considered as variants of function approximation.

The most important methods for this kind of problems are the *discriminant analysis*, *artificial neural nets*, *kernel* or *weighting* methods, and *decision trees*. In the last years, remarkable progress in these fields could be realized with the development of *support vector machines*, *boosted decision trees*, and *random forests* classifiers.

Before discussing these methods in more detail let us consider a further example.

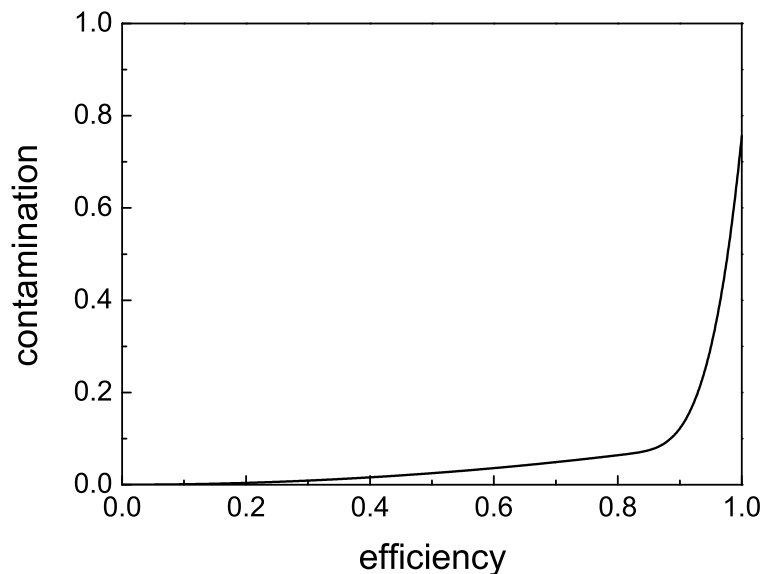


Fig. 11.7. Fraction of wrongly assigned events as a function of the efficiency.

Example 141. Classification of particles with calorimeters

The interactions of electrons and hadrons in calorimeter detectors of particle physics differ in a many parameters. Calorimeters consist of a large number of detector elements, for which the signal heights are evaluated and recorded. The system should learn from a training sample obtained from test measurements with known particle beams to classify electrons and hadrons with minimal error rates.

An optimal classification is possible if the likelihood ratio is available which then is used as a cut variable. The goal of intelligent classification methods is to approximate the likelihood ratio or an equivalent variable which is a unique function of the likelihood ratio. The relation itself need not be known.

When we optimize a given method, it is not only the percentage of right decisions which is of interest, but we will also consider the consequences of the various kinds of errors. It is less serious if a SPAM has not been detected, as compared to the loss of an important message. In statistics this is taken into account by a *loss function* which has to be defined by the user. In the standard situation where we want to select a certain class of events, we have to consider *efficiency* and *contamination*¹⁰. The larger the efficiency, the larger is also the relative contamination by wrongly assigned events. A typical curve showing this relation is plotted in Fig. 11.7. The user will select the value of his cut variable on the bases of this curve.

The loss has to be evaluated from the training sample. It is recommended to use a part of the training sample to develop the method and to reserve a certain fraction to validate it. As statistics is nearly always too small, also more economic methods

¹⁰One minus the contamination is called purity.

of validation, *cross validation* and *bootstrap* (see Sect.12.2), have been developed which permit to use the full sample to adjust the parameters of the method. In an n -fold cross validation the whole sample is randomly divided into n equal parts of N/n events each. In turn one of the parts is used for the validation of the training result from the other $n - 1$ parts. All n validation results are then averaged. Typical choices are n equal to 5 or 10.

11.4.1 The Discriminant Analysis

The classical discriminant analysis as developed by Fisher is a special case of the classification method that we introduce in the following. We follow our discussions of Chap. 6, Sect. 6.3.

If we know the p.d.f.s $f_1(\mathbf{x})$ and $f_2(\mathbf{x})$ for two classes of events it is easy to assign an observation $\mathbf{x}$ to one of the two classes in such a way that the error rate is minimal (case 1):

$$\begin{aligned}\mathbf{x} &\rightarrow \text{class 1, if } f_1(\mathbf{x}) > f_2(\mathbf{x}) , \\ \mathbf{x} &\rightarrow \text{class 2, if } f_1(\mathbf{x}) < f_2(\mathbf{x}) .\end{aligned}$$

Normally we will get a different number of wrong assignments for the two classes: observations originating from the broader distribution will be miss-assigned more often, see Fig. 11.8) than those of the narrower distribution. In most cases it will matter whether an input from class 1 or from class 2 is wrongly assigned. An optimal classification is then reached using an appropriately adjusted likelihood ratio:

$$\begin{aligned}\mathbf{x} &\rightarrow \text{class 1, if } f_1(\mathbf{x})/f_2(\mathbf{x}) > c , \\ \mathbf{x} &\rightarrow \text{class 2, if } f_1(\mathbf{x})/f_2(\mathbf{x}) < c .\end{aligned}$$

If we want to have the same error rates (case 2), we must choose the constant c such that the integrals over the densities in the selected regions are equal:

$$\int_{f_1/f_2 > c} f_1(\mathbf{x})d\mathbf{x} = \int_{f_1/f_2 < c} f_2(\mathbf{x})d\mathbf{x} . \quad (11.17)$$

This assignment has again a minimal error rate, but now under the constraint (11.17). We illustrate the two possibilities in Fig. 11.8 for univariate functions.

For normal distributions we can formulate the condition for the classification explicitly: For case 2 we choose that class for which the observation $\mathbf{x}$ has the smallest distance to the mean measured in standard deviations. This condition can then be written as a function of the exponents. With the usual notations we get

$$\begin{aligned}(\mathbf{x} - \boldsymbol{\mu}_1)^T \mathbf{V}_1 (\mathbf{x} - \boldsymbol{\mu}_1) - (\mathbf{x} - \boldsymbol{\mu}_2)^T \mathbf{V}_2 (\mathbf{x} - \boldsymbol{\mu}_2) < 0 &\rightarrow \text{class 1} , \\ (\mathbf{x} - \boldsymbol{\mu}_1)^T \mathbf{V}_1 (\mathbf{x} - \boldsymbol{\mu}_1) - (\mathbf{x} - \boldsymbol{\mu}_2)^T \mathbf{V}_2 (\mathbf{x} - \boldsymbol{\mu}_2) > 0 &\rightarrow \text{class 2} .\end{aligned}$$

This condition can easily be generalized to more than two classes; the assignment according to the standardized distances will then, however, no longer lead to equal error rates for all classes.

The classical discriminant analysis sets $\mathbf{V}_1 = \mathbf{V}_2$. The left-hand side in the above relations becomes a linear combination of the x_p . The quadratic terms cancel. Equating it to zero defines a hyperplane which separates the two classes. The sign of this

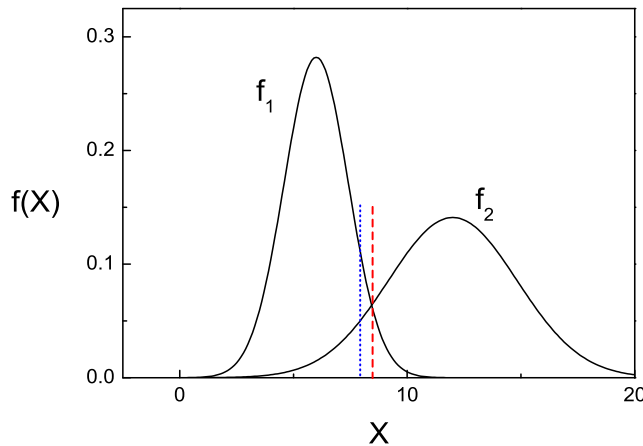


Fig. 11.8. Separation of two classes. The dashed line separates the events such that the error rate is minimal, the dotted line such that the wrongly assigned events are the same in both classes.

linear combination thus determines the class membership. Note that the separating hyperplane is cutting the line connecting the distribution centers under a right angle only for spherical symmetric distributions.

If the distributions are only known empirically from representative samples, we approximate them by continuous distributions, usually by a normal distribution, and fix their parameters to reproduce the empirical moments. In situations where the empirical distributions strongly overlap, for instance when a narrow distribution is located at the center of a broad one, the simple discriminant analysis does no longer work. The classification methods introduced in the following sections have been developed for this and other more complicated situations and where the only source of information on the population is a training sample. The various approaches are all based on the continuity assumption that observations with similar attributes lead to similar outputs.

11.4.2 Artificial Neural Networks

Introduction

The application of *artificial neural networks*, ANN, has seen a remarkable boom in the last decades, parallel to the exploding computing capacities. From its many variants, in science the most popular are the relatively simple forward ANNs with back-propagation, to which we will restrict our discussion. The interested reader should consult the broad specialized literature on this subject, where fascinating self organizing networks are described which certainly will play a role also in science, in the more distant future. It could e.g. be imagined that a self-organizing ANN would be able to classify a data set of events produced at an accelerator without human intervention and thus would be able to discover new reactions and particles.

The species considered here has a comparably more modest aim: The network is trained in a first step to ascribe a certain output (response) to the inputs. In this supervised learning scheme, the response is compared with the target response, and then the network parameters are modified to improve the agreement. After a training phase the network is able to classify new data.

ANNs are used in many fields for a broad variety of problems. Examples are pattern recognition, e.g. for hand-written letters or figures, or the forecast of stock prices. They are successful in situations where the relations between many parameters are too complex for an analytical treatment. In particle physics they have been used, among other applications, to distinguish electron from hadron cascades and to identify reactions with heavy quarks.

With ANNs, many independent computing steps have to be performed. Therefore specialized computers have been developed which are able to evaluate the required functions very fast by parallel processing.

Primarily, the net approximates an algebraic function which transforms the input vector $\mathbf{x}$ into the response vector $\mathbf{y}$,

$$\mathbf{y} = f(\mathbf{x}|w) .$$

Here w symbolizes a large set of parameters, typically there are, depending on the application, $10^3 - 10^4$ parameters. The training process corresponds to a fitting problem. The parameters are adjusted such that the response agrees within the uncertainties with a target vector $\mathbf{y}_t$ which is known for the events of the training sample.

There are two different applications of neural nets, simple function approximation¹¹ and classification. The net could be trained, for instance, to estimate the energy of a hadronic showers from the energies deposited in different cells of the detector. The net could also be trained to separate electron from hadron showers. Then it should produce a number close to 1 for electrons and close to 0 for hadrons.

With the large number of parameters it is evident that the solution is not always unique. Networks with different parameters can perform the same function within the desired accuracy.

For the fitting of the large number of parameters minimizing programs like simplex (see Appendix 13.9) are not suited. The gradient descent method is much more practicable here. It is able to handle a large number of parameters and to process the input data sequentially.

A simple, more detailed introduction to the field of ANN is given in [69]

Network Structure

Our network consists of two layers of knots (neurons), see Fig. 11.9. Each component x_k of the n -component input vector $\mathbf{x}$ is transmitted to all knots, labeled $i = 1, \dots, m$, of the first layer. Each individual data line $k \rightarrow i$ is ascribed a weight $W_{ik}^{(1)}$. In each unit the weighted sum $u_i = \sum_k W_{ik}^{(1)} x_k$ of the data components connected

¹¹Here function approximation is used to perform calculations. In the previous section its purpose was to parametrize data.

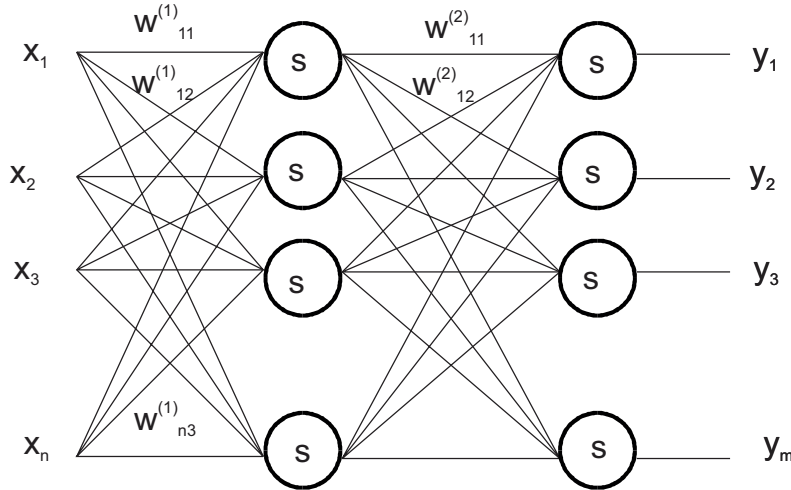


Fig. 11.9. Back propagation. At each knot the sigmoid function of the sum of the weighted inputs is computed.

to it is calculated. Each knot symbolizes a non-linear so-called activation function $x'_i = s(u_i)$, which is identical for all units. The first layer produces a new data vector $\mathbf{x}'$. The second layer, with m' knots, acts analogously on the outputs of the first one. We call the corresponding $m \times m'$ weight matrix $W^{(2)}$. It produces the output vector $\mathbf{y}$. The first layer is called hidden layer, since its output is not observed directly. In principle, additional hidden layers could be implemented but experience shows that this does not improve the performance of the net.

The net executes the following functions:

$$x'_j = s \left(\sum_k W_{jk}^{(1)} x_k \right),$$

$$y_i = s \left(\sum_j W_{ij}^{(2)} x'_j \right).$$

This leads to the final result:

$$y_i = s \left\{ \sum_j W_{ij}^{(2)} s \left(\sum_k W_{jk}^{(1)} x_k \right) \right\}. \quad (11.18)$$

Sometimes it is appropriate to shift the input of each unit in the first layer by a constant amount (bias). This is easily realized by specifying an artificial additional input component $x_0 \equiv 1$.

The number of weights (the parameters to be fitted) is, when we include the component x_0 , $(n+1) \times m + mm'$.

Activation Function

The activation function $s(x)$ has to be non-linear, in order to achieve that the superposition (11.18) is able to approximate widely arbitrary functions. It is plausible

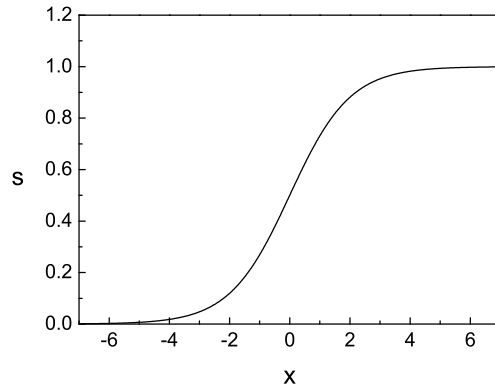


Fig. 11.10. Sigmoid function.

that it should be more sensitive to variations of the arguments near zero than for very large absolute values. The input bias x_0 helps to shift input parameters which have a large mean value into the sensitive region. The activation function is usually standardized to vary between zero and one. The most popular activation function is the sigmoid function

$$s(u) = \frac{1}{e^{-u} + 1},$$

which is similar to the Fermi function. It is shown in Fig. 11.10.

The Training Process

In the training phase the weights will be adapted after each new input object. Each time the output vector of the network $\mathbf{y}$ is compared with the target vector $\mathbf{y}_t$. We define again the loss function E :

$$E = (\mathbf{y} - \mathbf{y}_t)^2, \quad (11.19)$$

which measures for each training object the deviation of the response from the expected one.

To reduce the error E we walk downwards in the weight space. This means, we change each weight component by ΔW , proportional to the sensitivity $\partial E / \partial W$ of E to changes of W :

$$\begin{aligned} \Delta W &= -\frac{1}{2}\alpha \frac{\partial E}{\partial W} \\ &= -\alpha(\mathbf{y} - \mathbf{y}_t) \cdot \frac{\partial \mathbf{y}}{\partial W}. \end{aligned}$$

The proportionality constant α , the learning rate, determines the step width.

We now have to find the derivatives. Let us start with s :

$$\frac{ds}{du} = \frac{e^{-u}}{(e^{-u} + 1)^2} = s(1 - s). \quad (11.20)$$

From (11.18) and (11.20) we compute the derivatives with respect to the weight components of the first and the second layer,

$$\frac{\partial y_i}{\partial W_{ij}^{(2)}} = y_i(1 - y_i)x'_j,$$

and

$$\frac{\partial y_i}{\partial W_{jk}^{(1)}} = y_i(1 - y_i)x'_j W_{ij}^{(2)}(1 - x'_j)x_k.$$

It is seen that the derivatives depend on the same quantities which have already been calculated for the determination of $\mathbf{y}$ (the forward run through the net). Now we run backwards, change first the matrix $W^{(2)}$ and then with already computed quantities also $W^{(1)}$. This is the reason why this process is called *back propagation*. The weights are changed in the following way:

$$\begin{aligned} W_{jk}^{(1)} &\rightarrow W_{jk}^{(1)} - \alpha(\mathbf{y} - \mathbf{y}_t) \sum_i y_i(1 - y_i)x'_j W_{ij}^{(2)}(1 - x'_j)x_k, \\ W_{ij}^{(2)} &\rightarrow W_{ij}^{(2)} - \alpha(\mathbf{y} - \mathbf{y}_t)y_i(1 - y_i)x'_j. \end{aligned}$$

Testing and Interpreting

The gradient descending minimum search has not necessarily reached the minimum after processing the training sample a single time, especially when the available sample is small. Then the should be used several times (e.g. 10 or 100 times). On the other hand it may happen for too small a training sample that the net performs correctly for the training data, but produces wrong results for new data. The network has, so to say, learned the training data by heart. Similar to other minimizing concepts, the net interpolates and extrapolates the training data. When the number of fitted parameters (here the weights) become of the same order as the number of constraints from the training data, the net will occasionally, after sufficient training time, describe the training data exactly but fail for new input data. This effect is called *over-fitting* and is common to all fitting schemes when too many parameters are adjusted.

It is therefore indispensable to validate the network function after the optimization, with data not used in the training phase or to perform a cross validation. If in the training phase simulated data are used, it is easy to generate new data for testing. If only experimental data are available with no possibility to enlarge the sample size, usually a certain fraction of the data is reserved for testing. If the validation result is not satisfactory, one should try to solve the problem by reducing the number of network parameters or the number of repetitions of the training runs with the same data set.

The neural network generates from the input data the response through the fairly complicated function (11.18). It is impossible by an internal analysis of this function to gain some understanding of the relation between input and resulting response. Nevertheless, it is not necessary to regard the ANN as a “black box”. We have the possibility to display graphically correlations between input quantities and the result, and all functional relations. In this way we gain some insight into possible connections. If, for instance, a physicist would have the idea to train a net with an experimental data sample to predict for a certain gas the volume from the pressure and the temperature, he would be able to reproduce, with a certain accuracy, the results of the van-der-Waals equation. He could display the relations between the

three quantities graphically. Of course the analytic form of the equation and its interpretation cannot be delivered by the network.

Often a study of the optimized weights makes it possible to simplify the net. Very small weights can be set to zero, i.e. the corresponding connections between knots are cut. We can check whether switching off certain neurons has a sizable influence on the response. If this is not the case, these neurons can be eliminated. Of course, the modified network has to be trained again.

Practical Hints for the Application

Computer Programs for ANNs with back-propagation are relatively simple and available at many places but the effort to write an ANN program is also not very large. The number of input vector components n and the number of knots m and m' are parameters to be chosen by the user, thus the program is universal, only the loss function has to be adapted to the specific problem.

- The number of units in each layer should more or less match the number of input components. Some experts plead for a higher number. The user should try to find the optimal number.
- The sigmoid function has values only between zero and unity. Therefore the output or the target value has to be appropriately scaled by the user.
- The raw input components are usually correlated. The net is more efficient if the user orthogonalizes them. Then often some of the new components have negligible effect on the output and can be discarded.
- The weights have to be initialized at the beginning of the training phase. This can be done by a random number generator or they can be set to fixed values.
- The loss function E (11.19) has to be adjusted to the problem to be solved.
- The learning rate α should be chosen relatively high at the beginning of a training phase, e.g. $\alpha = 10$. In the course of fitting it should be reduced to avoid oscillations.
- The convergence of minimizing process is slow if the gradient is small. If this is the case, and the fit is still bad, it is recommended to increase the learning constant for a certain number of iterations.
- In order to check whether a minimum is only local, one should train the net with different start values of the weights.
- Other possibilities for the improvement of the convergence and the elimination of local minima can be found in the substantial literature. An ANN program package that proceeds automatically along many of the proposed steps is described in [70].

Example: Čerenkov circles

Charged, relativistic particles can emit photons by the Čerenkov effect. The photons hit a detector plane at points located on a circle. Of interest are radius and center of this circle, since they provide information on direction and velocity of the emitting particle. The number of photons and the coordinates where they hit the detector

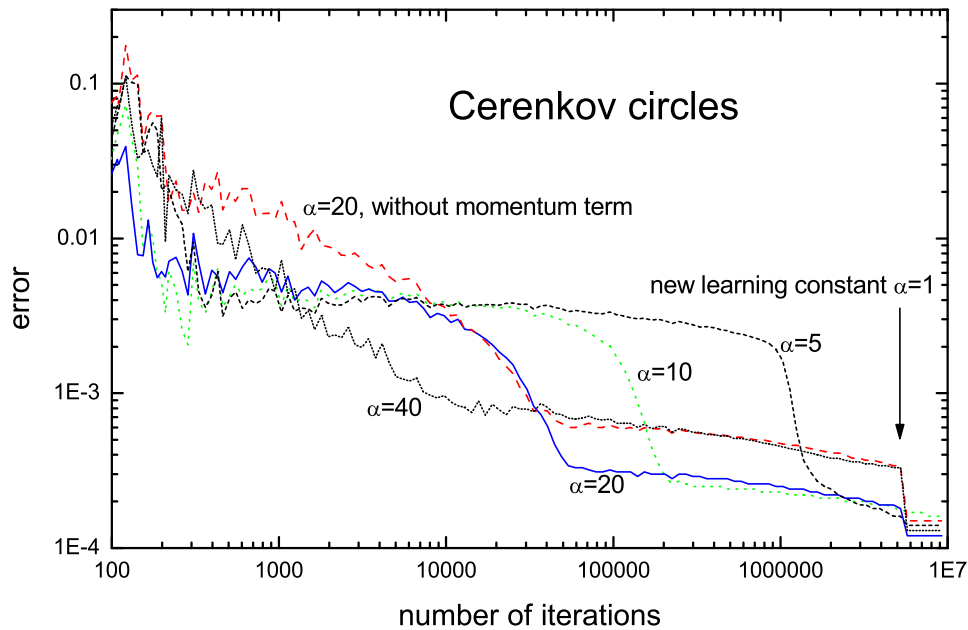


Fig. 11.11. Reconstruction of radii of circles through 5 points by means of an ANN with different sequences of learning constants α .

fluctuate statistically and are disturbed by spurious noise signals. It has turned out that ANNs can reconstruct the parameters of interest from the available coordinates with good efficiency and accuracy.

We study this problem by a Monte Carlo simulation. In a simplified model, we assume that exactly 5 photons are emitted by a particle and that the coordinate pairs are located on a circle and registered. The center, the radii, and the hit coordinates are generated stochastically. The input vector of the net thus consists of 10 components, the 5 coordinate pairs. The output is a single value, the radius R . The loss function is $(R - R_{true})^2$, where the true value R_{true} is known from the simulation.

The relative accuracy of the reconstruction as a function of the iteration step is shown in Fig. 11.11. Different sequences of the learning rate have been tried. Typically, the process is running by steps, where after a flat phase follows a rather abrupt improvement. The number of iterations required to reach the minimum is quite large.

Hardware Realization

The structure of back propagation network can be implemented by a hardware network. The weights are stored locally at the units which are realized by rather simple microprocessors. Each microprocessor performs the knot function, e.g. the sigmoid function. A trained net can then calculate the fitted function very fast, since all processors are working in parallel. Such processors can be employed for the triggering

in experiments where a quick decision is required, whether to accept an event and to store the corresponding data.

11.4.3 Weighting Methods

For the decision whether to assign an observation at the location $\mathbf{x}$ to a certain class, an obvious option is to do this according to the classification of neighboring objects of the training sample. One possibility is to consider a certain region around $\mathbf{x}$ and to take a “majority vote” of the training objects inside this region to decide about the class membership of the input. The region to be considered here can be chosen in different ways; it can be a fixed volume around $\mathbf{x}$, or a variable volume defined by requiring that it contains a fixed number of observations, or an infinite volume, introducing weights for the training objects which decrease with their distance from $\mathbf{x}$.

In any case we need a metric to define the distance. The choice of a metric in multi-dimensional applications is often a rather intricate problem, especially if some of the input components are physically of very different nature. A way-out seems to be to normalize the different quantities to equal variance and to eliminate global correlations by a linear variable transformation. This corresponds to the transformation to principal components discussed above (see Sect. 11.3) with subsequent scaling of the principal components. An alternative but equivalent possibility is to use a direction dependent weighting. The same result is achieved when we apply the Mahalanobis metric, which we have introduced in Sect. 10.3.9.

For a large training sample the calculation of all distances is expensive in computing time. A drastic reduction of the number of distances to be calculated is in many cases possible by the so-called *support vector machines* which we will discuss below. Those are not machines, but programs which reduce the training sample to a few, but decisive inputs, without impairing the results.

K-Nearest Neighbors

We choose a number K which of course will depend on the size of the training sample and the overlap of the classes. For an input $\mathbf{x}$ we determine the K nearest neighbors and the numbers $k_1, k_2 = K - k_1$, of observations that belong to class I and II, respectively. For a ratio k_1/k_2 greater than α , we assign the new observation to class I, in the opposite case to class II:

$$\begin{aligned} k_1/k_2 > \alpha &\implies \text{class I,} \\ k_1/k_2 < \alpha &\implies \text{class II.} \end{aligned}$$

The choice of α depends on the loss function. When the loss function treats all classes alike, then α will be unity and we get a simple majority vote. To find the optimal value of K we minimize the average of the loss function computed for all observations of the training sample.

Distance Dependent Weighting

Instead of treating all training vector inputs $\mathbf{x}'$ within a given region in the same way, one should attribute a larger weight to those located nearer to the input $\mathbf{x}$. A

sensible choice is again a Gaussian kernel,

$$K(\mathbf{x}, \mathbf{x}') \sim \exp\left(-\frac{(\mathbf{x} - \mathbf{x}')^2}{2s^2}\right).$$

With this choice we obtain for the class β the weight w_β ,

$$w_\beta = \sum_i K(\mathbf{x}, \mathbf{x}_{\beta i}), \quad (11.21)$$

where $\mathbf{x}_{\beta i}$ are the locations of the training vectors of the class β .

If there are only two classes, writing the training sample as

$$\{\mathbf{x}_1, y_1 \dots \mathbf{x}_N, y_N\}$$

with the response vector $y_i = \pm 1$, the classification of a new input $\mathbf{x}$ is done according to the value ± 1 of the classifier $\hat{y}(\mathbf{x})$, given by

$$\hat{y}(\mathbf{x}) = \text{sign}\left(\sum_{y_i=+1} K(\mathbf{x}, \mathbf{x}_i) - \sum_{y_i=-1} K(\mathbf{x}, \mathbf{x}_i)\right) = \text{sign}\left(\sum_i y_i K(\mathbf{x}, \mathbf{x}_i)\right). \quad (11.22)$$

For a direction dependent density of the training sample, we can use a direction dependent kernel, eventually in the Mahalanobis form mentioned above:

$$K(\mathbf{x}, \mathbf{x}') \sim \exp\left[-\frac{1}{2}(\mathbf{x} - \mathbf{x}')^T \mathbf{V}(\mathbf{x} - \mathbf{x}')\right].$$

with the weight matrix $\mathbf{V}$. When we first normalize the sample, this complication is not necessary. The parameter s of the matrix $\mathbf{V}$, which determines the width of the kernel function, again is optimized by minimizing the loss for the training sample.

Support Vector Machines

Support vector machines (SVMs) produce similar results as ordinary distance depending weighting methods, but they require less memory for the storage of learning data and the classification is extremely fast. Therefore, they are especially useful in on-line applications.

The class assignment usually is the same for all elements in large connected regions of the variable $\mathbf{x}$. Very often, in a two case classification, there are only two regions separated by a hypersurface. For short range kernels it is obvious then that for the classification of observations, the knowledge of only those input vectors of the training sample is essential which are located in the vicinity of the hypersurface. These input vectors are called *support vectors* [73]. SVMs are programs which try to determine them, respectively their weights, in an optimal way, setting the weights of all other inputs vectors to zero.

In the one-dimensional case with non-overlapping classes it is sufficient to know those inputs of each class which are located nearest to the dividing limit between the classes. Sums like (11.21) are then running over one element only. This, of course, makes the calculation extremely fast.

In higher dimensional spaces with overlapping classes and for more than two classes the problem to determine support vectors is of course more complicated. But

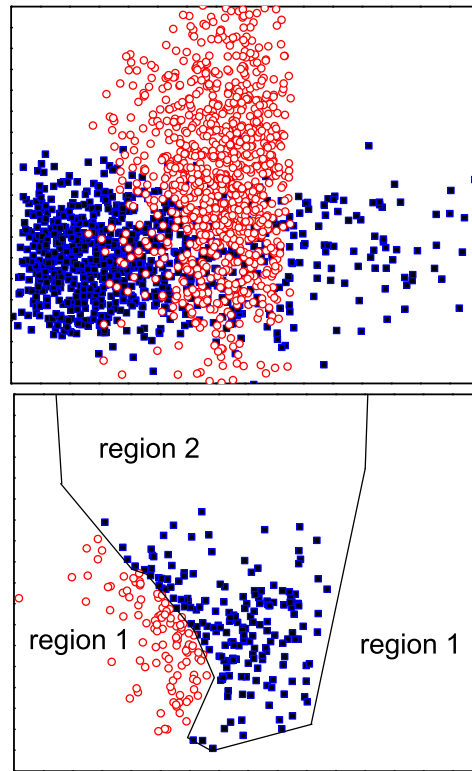


Fig. 11.12. Separation of two classes. Top: learning sample, bottom: wrongly assigned events of a test sample.

also in these circumstances the number of relevant training inputs can be reduced drastically. The success of SVMs is based on the so-called *kernel trick*, by which non-linear problems in the input space are treated as linear problems in some higher-dimensional space by well known optimization algorithms. For the corresponding algorithms and proofs we refer to the literature, e.g. [13, 72]. A short introduction is given in Appendix 13.13.

Example and Discussion

In Fig. 11.12 are shown in the top panel two overlapping training samples of 500 inputs each. The loss function is the number of wrong assignments independent of the respective class. Since the distributions are quite similar in both coordinates we do not change the metric. We use a Gaussian kernel. The optimization of the parameter s by means of the training sample shows only a small change of the error rate for a change of s by a factor four. The lower panel displays the result of the classification for a test sample of the same size (500 inputs per class). Only the wrong assignments are shown.

We realize that wrongly assigned training observations occur in two separate, non overlapping regions which can be separated by a curve or a polygon chain as indicated

in the figure. Obviously all new observations would be assigned to the class corresponding to the region in which they are located. If we would have used instead of the distance-depending weighting the K-nearest neighbors method, the result would have been almost identical. In spite of the opposite expectation, this more primitive method is more expensive in both the programming and in the calculation, when compared to the weighting with a distance dependent kernel.

Since for the classification only the separation curve between the classes is required, it must be sufficient to know the class assignment for those training observations which lie near this curve. They would define the support vectors of a SVM. Thus the number of inputs needed for the assignment of new observations would be drastically reduced. However, for a number of assignments below about 10^6 the effort to determine support vectors usually does not pay. The SVMs are useful for large event numbers in applications where computing time is relevant.

11.4.4 Decision Trees

Simple Trees

We consider the simple case, the two class classification, i.e. the assignment of inputs to one of two classes I and II , and N observations with P features $x_1, x_2, \dots, x_P$, which we consider, as before, as the components of an input vector.

In the first step we consider the first component $x_{11}, x_{21}, \dots, x_{N1}$ for all N input vectors of the training sample. We search for a value x_{c1} which optimally divides the two classes and obtain a division of the training sample into two parts A and B . Each of these parts which belong to two different subspaces, will now be further treated separately. Next we take the subspace A , look at the feature x_2 , and divide it, in the same way as before the full space, again into two parts. Analogously we treat the subspace B . Now we can switch to the next feature or return to feature 1 and perform further splittings. The sequence of divisions leads to smaller and smaller subspaces, each of them assigned to a certain class. This subdivision process can be regarded as the development of a decision tree for input vectors for which the class membership is to be determined. The growing of the tree is stopped by a pruning rule. The final partitions are called leaves.

In Fig. 11.13 we show schematically the subdivision into subspaces and the corresponding decision tree for a training sample of 32 elements with only two features. The training sample which determines the decisions is indicated. At the end of the tree (here at the bottom) the decision about the class membership is taken.

It is not obvious, how one should optimize the sequence of partitions and the position of cuts, and also not, under which circumstances the procedure should be stopped.

For the optimization of splits we must again define a loss function which will depend on the given problem. A simple possibility in the case of two classes is, to maximize for each splitting the difference $\Delta N = N_r - N_f$ between right and wrong assignments. We used this in our example Fig. 11.13. For the first division this quantity was equal to $20 - 12 = 8$. To some extent the position of the splitting hyperplane is still arbitrary, the loss function changes its value only when it hits the nearest input. It could, for example, be put at the center between the two nearest

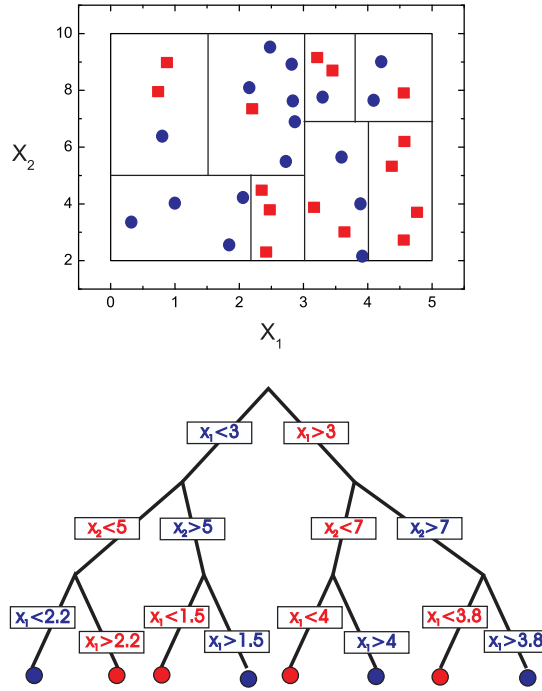


Fig. 11.13. Decision tree (bottom) corresponding to the classification shown below.

points. Often the importance of efficiency and purity is different for the two classes. Then we would choose an asymmetric loss function.

Very popular is the following, slightly more complicated criterion: We define the impurity P_I of class I

$$P_I = \frac{N_I}{N_I + N_{II}}, \quad (11.23)$$

which for optimal classification would be 1 or 0. The quantity

$$G = P_I(1 - P_I) + P_{II}(1 - P_{II}) \quad (11.24)$$

the *Gini-index*, should be as small as possible. For each separation of a parent node E with Gini index G_E into two children nodes A , B with G_A , respectively G_B , we minimize the sum $G_A + G_B$.

The difference

$$D = G_E - G_A - G_B$$

is taken as stopping or pruning parameter. The quantity D measures the increase in purity, it is large for a parent node with large G and two children nodes with small G . When D becomes less than a certain critical value D_c the branch will not be split further and ends at a leaf. The leaf is assigned to the class which has the majority in it.

Besides the Gini index, also other measures for the purity or impurity are used [13]. An interesting quantity is entropy $S = -P_I \ln P_I - P_{II} \ln P_{II}$, a well known measure of disorder, i.e. of impurity.

The purity parameter, e.g. G , is also used to organize the splitting sequence. We choose always that input vector component in which the splitting produces the most significant separation.

A further possibility would be to generalize the orthogonal splitting by allowing also non-orthogonal planes to reach better separations. But in the standard case all components are treated independently.

Unfortunately, the classification by decision trees is usually not perfect. The discontinuity at the boundaries and the fixed splitting sequence impair the accuracy. On the other hand, they are simple, transparent and the corresponding computer programs are extremely fast.

Boosted Decision Trees

Boosting [75] is based on a simple idea: By a weighted superposition of many moderately effective classifiers it should be possible to reach a fairly precise assignment. Instead of only one decision tree, many different trees are grown. Each time, before the development of a new tree is started, wrongly assigned training inputs are *boosted* to higher weights in order to lower their probability of being wrongly classified in the following tree. The final class assignment is then done by averaging the decisions from all trees. Obviously, the computing effort for these *boosted decision trees* is increased, but the precision is significantly enhanced. The results of boosted decision trees are usually as good as those of ANNs. Their algorithm is very well suited for parallel processing. There are first applications in particle physics [76].

Before the first run, all training inputs have the weight 1. In the following run each input gets a weight w_i , determined by a certain boosting algorithm (see below) which depends on the particular method. The definition of the node impurity P for calculating the loss function, see (11.23), (11.24), is changed accordingly to

$$P = \frac{\sum_I w_i}{\sum_I w_i + \sum_{II} w_i} ,$$

where the sums $\sum_I$, $\sum_{II}$ run over all events in class I or II , respectively. Again the weights will be boosted and the next run started. Typically $M \approx 1000$ trees are generated in this way.

If we indicate the decision of a tree m for the input $\mathbf{x}_i$ by $T_m(\mathbf{x}_i) = 1$ (for class I) and $= -1$ (for class II), the final result will be given by the sign of the weighted sum over the results from all trees

$$T_M(\mathbf{x}_i) = \text{sign} \left(\sum_{m=1}^M \alpha_m T_m(\mathbf{x}_i) \right) .$$

We proceed in the following way: To the first tree we assign a weight $\alpha_1 = 1$. The weights of the wrongly assigned input vectors are increased. The weight¹² α_2 of the second tree $T_2(\mathbf{x})$ is chosen such that the overall loss from all input vectors of the training sample is minimal for the combination $[\alpha_1 T_1(\mathbf{x}) + \alpha_2 T_2(\mathbf{x})] / [\alpha_1 + \alpha_2]$. We continue in the same way and add further trees. For tree i the weight α_i is optimized such that the existing trees are complemented in an optimal way. How this is done depends of course on the loss function.

¹²We have two kinds of weight, weights of input vectors (w_i) and weights of trees (α_m).

A well tested recipe for the choice of weights is AdaBoost [75]. The training algorithm proceeds as follows:

- The i -th input $\mathbf{x}_i$ gets the weight $w_i = 1$ and the value $y_i = 1$, ($= -1$), if it belongs to class I , (II).
- $T_m(\mathbf{x}_i) = 1$ ($= -1$), if the input ends in a leave belonging to class I (II).
 $S_m(\mathbf{x}_i) = (1 - y_i T_m(\mathbf{x}_i))/2 = 1$ ($= 0$), if the assignment is wrong (right).
- The fraction of the weighted wrong assignments ε_m is used to change the weights for the next iteration:

$$\begin{aligned}\varepsilon_m &= \sum_i w_i S_m(\mathbf{x}_i) / \sum_i w_i , \\ \alpha_m &= \ln \frac{1 - \varepsilon_m}{\varepsilon_m} , \\ w_i &\rightarrow w_i e^{\alpha_m S_m} .\end{aligned}$$

Weights of correctly assigned training inputs thus remain unchanged. For example, for $\varepsilon_m = 0.1$, wrongly assigned inputs will be boosted by a factor $0.9/0.1 = 9$. Note that $\alpha_m > 0$ if $\varepsilon < 0.5$; this is required because otherwise the replacement $T_m(\mathbf{x}_i) \rightarrow -T_m(\mathbf{x}_i)$ would produce a better decision tree.

- The response for a new input which is to be classified is

$$T_M(\mathbf{x}_i) = \text{sign} \left(\sum_{m=1}^M \alpha_m T_m(\mathbf{x}_i) \right) .$$

For $\varepsilon_m = 0.1$ the weight of the tree is $\alpha_m = \ln 9 \approx 2.20$. For certain applications it may be useful to reduce the weight factors α_m somewhat, for instance $\alpha_m = 0.5 \ln((1 - \varepsilon_m)/\varepsilon_m)$ [76].

11.4.5 Bagging and Random Forest

Bagging

The concept of bagging was first introduced by Breiman [83]. He has shown that the performance of unstable classifiers can be improved considerably by training many classifiers with bootstrap replicates and then using a majority vote of those: From a training sample containing N input vectors, N vectors are drawn at random with replacement. Some vectors will be contained several times. This bootstrap¹³ sample is used to train a classifier. Many, 100 or 1000 classifiers are produced in this way. New inputs are run through all trees and each tree “votes” for a certain classification. The classification receiving the majority of votes is chosen. In a study of real data [83] a reduction of error rates by bagging between 20% and 47% was found. There the bagging concept had been applied to simple decision trees, however, the bagging concept is quite general and can be adopted also to other classifiers.

¹³We will discuss bootstrap methods in the following chapter.

Random Forest

Another new development [84] which includes the bootstrap idea, is the extension of the decision tree concept to the *random forest* classifier.

Many trees are generated from bootstrap samples of the training sample, but now part of the input vector components are suppressed. A tree is constructed in the following way: First m out of the M components or attributes of the input vectors are selected at random. The tree is grown in a m -dimensional subspace of the full input vector space. It is not obvious how m is to be chosen, but the author proposes $m \ll M$ and says that the results show little dependence on m . With large m the individual trees are powerful but strongly correlated. The value of m is the same for all trees.

From the N truncated bootstrap vectors, N_b are separated, put into a *bag* and reserved for testing. A fraction $f = N_b/N \approx 1/3$ is proposed. The remaining ones are used to generate the tree. For each split that attribute out of the m available attributes is chosen which gives the smallest number of wrong classifications. Each leave contains only elements of a single class. There is no pruning.

Following the bagging concept, the classification of new input vectors is obtained by the majority vote of all trees.

The out-of-the-bag (oob) data are used to estimate the error rate. To this end, each oob-vector of the k -th sample is run through the k -th tree and classified. The fraction of wrong classifications from all oob vectors is the error rate. (For T trees there are in total $T \times N_b$ oob vectors.) The oob data can also be used to optimize the constant m .

The random forest classifier has received quite some interest. The concept is simple and seems to be similarly powerful as that of other classifiers. It is especially well suited for large data sets in high dimensions.

11.4.6 Comparison of the Methods

We have discussed various methods for classification. Each of them has its advantages and its drawbacks. It depends on the special problem, which one is the most suitable.

The *discriminant analysis* offers itself for one- or two dimensional continuous distributions (preferably Gaussians or other unimodal distributions). It is useful for event selection in simple situations.

Kernel methods are relatively easy to apply. They work well if the division line between classes is sufficiently smooth and transitions between different classes are continuous. Categorical variables cannot be treated. The variant with support vectors reduces computing time and the memory space for the storage of the training sample. In standard cases with not too extensive statistics one should avoid this additional complication. Kernel methods can perform event selection in more complicated environments than is possible with the primitive discriminant analysis. For the better performance the possibility of interpreting the results is diminished, however.

Artificial neural networks are, due to the enormous number of free parameters, able to solve any problem in an optimal way. They suffer from the disadvantage that the user usually has to intervene to guide the minimizing process to a correct minimum. The user has to check and improve the result by changing the network

structure, the learning constant and the start values of the weights. New program packages are able to partially take over these tasks. ANN are able to separate classes in very involved situations and extract very rare events from large samples.

Decision trees are a very attractive alternative to ANN. One should use boosted decision trees, random forest or apply bagging though, since those discriminate much better than simple trees. The advantage of simple trees is that they are very transparent and that they can be displayed graphically. Like ANN, decision trees can, with some modifications, also be applied to categorical variables.

At present, there is lack of theoretical framework and experimental information on some of the new developments. We would like to know to what extent the different classifiers are equivalent and which classifier should be selected in a given situation. There will certainly be answers to these questions in the near future.

Auxiliary Methods

12.1 Probability Density Estimation

12.1.1 Introduction

In the subsection *function approximation* we have considered measurements $\mathbf{y}$ at fixed locations $\mathbf{x}$ where $\mathbf{y}$ due to statistical fluctuations deviates from an unknown function. Now we start from a sample $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ which follow an unknown statistical distribution which we want to approximate. We have to estimate the density $\hat{f}(\mathbf{x})$ at the location $\mathbf{x}$ from the frequency of observations $\mathbf{x}_i$ in the vicinity of $\mathbf{x}$. The corresponding technique, *probability density estimation* (PDE), is strongly correlated with function approximation. Both problems are often treated together under the title *smoothing methods*. In this section we discuss only non-parametric approaches; a parametric method, where parameters are adjusted to approximate Gaussian like distributions has been described in Sect. 11.2.2. We will essentially present results and omit the derivations. For details the reader has to consult the specialized literature.

PDE serves mainly to visualize an empirical frequency distribution. Visualization of data is an important tool of scientific research. It can lead to new discoveries and often constitutes the basis of experimental decisions. PDE also helps to classify data and sometimes the density which has been estimated from some ancillary measurement is used in subsequent Monte Carlo simulations of experiments. However, to solve certain problems like the estimation of moments and other characteristic properties of a distribution, it is preferable to deal directly with the sample instead of performing a PDE. This path is followed by the bootstrap method which we will discuss in a subsequent section. When we have some knowledge about the shape of a distribution, then PDE can improve the precision of the bootstrap estimates. For instance there may exist good reasons to assume that the distribution has only one maximum and/or it may be known that the random variable is restricted to a certain region with known boundary.

The PDE $\hat{f}(\mathbf{x})$ of the true density $f(\mathbf{x})$ is obtained by a smoothing procedure applied to the discrete experimental distribution of observations. This means, that some kind of averaging is done which introduces a bias which is especially large if the distribution $f(\mathbf{x})$ varies strongly in the vicinity of $\mathbf{x}$.

The simplest and most common way to measure the quality of the PDE is to evaluate the integrated square error (ISE) L_2

$$L_2 = \int_{-\infty}^{\infty} [\hat{f}(\mathbf{x}) - f(\mathbf{x})]^2 dx$$

and its expectation value $E(L_2)$, the *mean integrated square error* (*MISE*)¹. The mean quadratic difference $E([\hat{f}(\mathbf{x}) - f(\mathbf{x})]^2)$ has two components, according to the usual decomposition:

$$E([\hat{f}(\mathbf{x}) - f(\mathbf{x})]^2) = \text{var}(\hat{f}(\mathbf{x})) + \text{bias}^2(\hat{f}(\mathbf{x})) .$$

The first term, the variance, caused by statistical fluctuations, decreases with increasing smoothing and the second term, the bias squared, decreases with decreasing smoothing. The challenge is to find the optimal balance between these two contributions.

We will give a short introduction to PDE mainly restricted to one-dimensional distributions. The generalization of the simpler methods to multi-dimensional distributions is straight forward but for the more sophisticated ones this is more involved. A rather complete and comprehensive overview can be found in the books by J.S. Simonoff [85], A.W. Bowman and A. Azzalini [86], D. W. Scott [87] and W. Härdle et al. [89]. A summary is presented in an article by D. W. Scott and S. R. Sain [88].

12.1.2 Fixed Interval Methods

Histogram Approximation

The simplest and most popular method of density estimation is *histogramming* with fixed bin width. For the number ν_k of N events falling into bin B_k and bin width h the estimated density is

$$\hat{f}(x) = \frac{\nu_k}{Nh} \text{ for } x \in B_k .$$

It is easy to construct, transparent, does not contain hidden parameters which often are included in other more sophisticated methods and indicates quite well which distributions are compatible with the data. However it has, as we have repeatedly stated, the disadvantage of the rather arbitrary binning and its discontinuity. Fine binning provides a good resolution of structures and low bias but has to be paid for by large fluctuations. Histograms with wide bins have the advantage of small statistical errors but are biased. A sensible choice for the bin width h is derived from the requirement that the mean squared integrated error should be as small as possible. The mean integrated square error, *MISE*, for a histogram is

$$MISE = \frac{1}{Nh} + \frac{1}{12}h^2 \int f'(x)^2 dx + O\left(\frac{h^4}{N}\right) . \quad (12.1)$$

The integral $\int f'(x)^2 dx = R(f')$ is called *roughness*. For a normal density with variance σ^2 it is $R = (4\sqrt{\pi}\sigma^3)^{-1}$. Neglecting the small terms ($h \rightarrow 0$) we can derive [88] the optimal bin width h^* and the corresponding *asymptotic mean integrated square error* *AMISE*:

¹The estimate $\hat{f}(\mathbf{x})$ is a function of the set $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ of random variables and thus also a random variable.

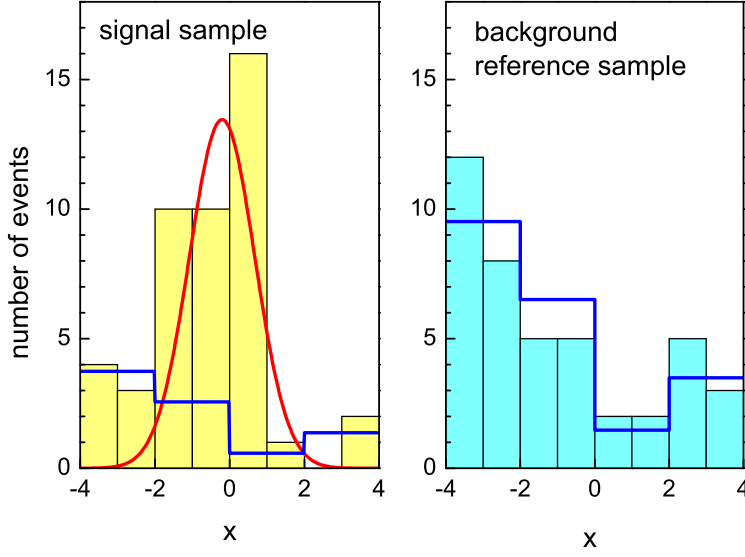


Fig. 12.1. Experimental signal with some background (left) and background reference sample with two times longer exposure time (right). The fitted signal and background functions are indicated.

$$h^* \approx \left[\frac{6}{N \int f'(x)^2 dx} \right]^{1/3} \approx 3.5\sigma N^{-1/3}, \quad (12.2)$$

$$AMISE \approx \left[\frac{9 \int f'(x)^2 dx}{16N^2} \right]^{1/3} \approx 0.43N^{-2/3}/\sigma.$$

The second part of relation (12.2) holds for a Gaussian p.d.f. with variance σ^2 and is a reasonable approximation for a distribution with typical σ . Even though the derivative f' and the bandwidth² σ are not precisely known, they can be estimated from the data. As expected, the optimal bin width is proportional to the band width, whereas its $N^{-1/3}$ dependence on the sample size N is less obvious.

In d dimensions similar relations hold. Of course the N -dependence has to be modified. For d -dimensional cubical bins the optimal bin width scales with $N^{-1/(d+2)}$ and the mean square error scales with $N^{-2/(d+2)}$.

Example 142. PDE of a background distribution and signal fit

We analyze a signal sample containing a Gaussian signal $N(x|\mu, \sigma)$ with unknown location and scale parameters μ, σ containing some unknown background. In addition, we dispose of a reference sample containing only background. From the data taking times and fluxes we know that the background in the signal sample should nominally be half ($r = 0.5$) of that in the reference sample. In Fig. 12.1 we show the two experimental distributions. From the shape of the experimental background distribution we estimate the slope

²Contrary to what is understood usually under bandwidth, in PDE this term is used to describe the typical width of structures. For a Gaussian it equals the standard deviation.

$y' = 0.05$ of its p.d.f. $h(x)$, and, using relation 12.2, find a bin width of 2 units. The heights $\beta_1, \beta_2, \beta_3, \beta_4$ of the 4 equally wide bins of the histogram distribution are left as free parameters in the fit. Because of the normalization, we have $\beta_4 = 1 - \beta_1 - \beta_2 - \beta_3$. Further parameters are the expected rate of background events in the reference sample ρ and the fraction ϕ of true signal events in the signal sample. These 7 parameters are to be determined in a likelihood fit.

The log-likelihood function $\ln L = \ln L_1 + \ln L_2 + \ln L_3 + \ln L_4$ comprises 4 terms, with:

1. L_1 , the likelihood of the n_s events in the signal sample (superposition of signal and background distribution):

$$\ln L_1(\mu, \sigma, \phi, \beta_1, \beta_2, \beta_3) = \sum_{i=1}^{n_s} \ln [\phi N(x_i|\mu, \sigma) + (1 - \phi)h(x_i|\beta_1, \beta_2, \beta_3)] .$$

2. L_2 , the likelihood of the n_r events of the reference sample (background distribution):

$$\ln L_2(\beta_1, \beta_2, \beta_3) = \sum_{i=1}^{n_r} \ln h(x_i|\beta_1, \beta_2, \beta_3) .$$

3. L_3 , the likelihood to observe n_r reference events where ρ are expected (Poisson distribution):

$$\ln L_3 = -\rho + n_r \ln \rho - \ln n_r! .$$

4. L_4 , the likelihood to get $n_s(1 - \phi)$ background events³ in the signal sample where $r\rho$ are expected (Poisson distribution):

$$\ln L_4 = -r\rho + n_s(1 - \phi) \ln(r\rho) - \ln \{[n_s(1 - \phi)]!\} .$$

The results of the fit are indicated in the Fig. 12.1 which is a histogram of the observed events. The MLE of the interesting parameters are $\mu = -0.18 \pm 0.32$, $\sigma = 0.85 \pm 0.22$, $\phi = 0.60_{-0.09}^{+0.05}$, the correlation coefficients are of the order of 0.3. We abstain from showing the full error matrix. The samples have been generated with the nominal parameter values $\mu_0 = 0$, $\sigma_0 = 1$, $\phi = 0.6$. To check the influence of the background parametrization, we repeat the fit with only two bins. The results change very little to $\mu = -0.12 \pm 0.34$, $\sigma = 0.85 \pm 0.22$, $\phi = 0.57_{-0.11}^{+0.07}$. When we represent the background p.d.f. by a polygon (see next chapter) instead of a histogram, the result again remains stable. We then get $\mu = -0.20 \pm 0.30$, $\sigma = 0.82 \pm 0.21$, $\phi = 0.60_{-0.09}^{+0.05}$. The method which we have applied in the present example is more precise than that of Sect. 6.5.10 but depends to a certain degree on the presumed shape of the background distribution.

Linear and Higher Order Parabolic Approximation

In the previous chapter we had adjusted spline functions to measurements with errors. Similarly, we can use them to approximate the probability density. We will consider

³The fit will favor a natural number. It is recommended to replace here $n!$ by $\Gamma(n + 1)$

here only the linear approximation by a polygon but it is obvious that the method can be extended to higher order parabolic functions. The discontinuity corresponding to the steps between bins is avoided when we transform the histogram into a polygon. We just have to connect the points corresponding to the histogram functions at the center of the bins. It can be shown that this reduces the *MISE* considerably, especially for large samples. The optimum bin width in the one-dimensional case now depends on the average second derivative f'' of the p.d.f. and is much wider than for a histogram and the error is smaller [88] than in the corresponding histogram case:

$$h^* \approx 1.6 \left[\frac{1}{N \int f''(x)^2 dx} \right]^{1/5},$$

$$MISE^* \approx 0.5 \left[\frac{\int f''(x)^2 dx}{N^4} \right]^{1/5}.$$

In d dimensions the optimal bin width for polygon bins scales with $N^{-1/(d+4)}$ and the mean square error scales with $N^{-4/(d+4)}$.

12.1.3 Fixed Number and Fixed Volume Methods

To estimate the density at a point $\mathbf{x}$ an obvious procedure is to divide the number k of observations in the neighborhood of $\mathbf{x}$ by the volume V which they occupy, $\hat{f}(\mathbf{x}) = k/V$. Either we can fix k and compute the corresponding volume $V(\mathbf{x})$ or we can choose V and count the number of observations contained in that volume. The quadratic uncertainty is $\sigma^2 = k + bias^2$, hence the former emphasizes fixed statistical uncertainty and the latter rather aims at small variations of the bias.

The k -nearest neighbor method avoids large fluctuations in regions where the density is low. We obtain a constant statistical error if we estimate the density from the spherical volume V taken by the k -nearest neighbors of point $\mathbf{x}$:

$$\hat{f}(\mathbf{x}) = \frac{k}{V_k(\mathbf{x})}. \quad (12.3)$$

As many other PDE methods, the k -nearest neighbor method is problematic in regions with large curvature of f and at boundaries of $\mathbf{x}$.

Instead of fixing the number of observations k in relation (12.3) we can fix the volume V and determine k . Strong variations of the bias in the k -nearest neighbor method are somewhat reduced but both methods suffer from the same deficiencies, the boundary bias and a loss of precision due to the sharp cut-off due to either fixing k or V . Furthermore it is not guaranteed that the estimated density is normalized to one. Hence a renormalization has to be performed.

The main advantage of fixed number and fixed volume methods is their simplicity.

12.1.4 Kernel Methods

We now generalize the fixed volume method and replace (12.3) by

$$\hat{f}(\mathbf{x}) = \frac{1}{NV} \sum K(\mathbf{x} - \mathbf{x}_i)$$

where the kernel K is equal to 1 if $\mathbf{x}_i$ is inside the sphere of volume V centered at $\mathbf{x}$ and 0 otherwise. Obviously, smooth kernel functions are more attractive than the uniform kernel of the fixed volume method. An obvious candidate for the kernel function $K(u)$ in the one-dimensional case is the Gaussian $\propto \exp(-u^2/2h^2)$. A very popular candidate is also the parabolically shaped Epanechnikov kernel $\propto (1 - c^2u^2)$, for $|cu| \leq 1$, and else zero. Here c is a scaling constant to be adjusted to the bandwidth of f . Under very general conditions the Epanechnikov kernel minimizes the asymptotic mean integrated square error $AMISE$ obtained in the limit where the effective binwidth tends to zero, but other kernels perform nearly as well. The $AMISE$ of the Gaussian kernel is only 5% larger and that of the uniform kernel by 8% [85].

The optimal bandwidth of the kernel function obviously depends on the true density. For example for a Gaussian true density $f(x)$ with variance σ^2 the optimal bandwidth h of a Gaussian kernel is $h_G \approx 1.06\sigma N^{-1/5}$ [85] and the corresponding constant c of the Epanechnikov kernel is $c \approx 2.2/(2h_G)$. In practice, we will have to replace the Gaussian σ in the relation for h_0 by some estimate depending on the structure of the observed data. $AMISE$ of the kernel PDE is converging at the rate $N^{-4/5}$ while this rate was only $N^{-2/3}$ for the histogram.

12.1.5 Problems and Discussion

The simple PDE methods sketched above suffer from several problems, some of which are unavoidable:

1. The boundary bias: When the variable x is bounded, say $x < a$, then $\hat{f}(x)$ is biased downwards unless $f(a) = 0$ in case the averaging process includes the region $x > a$ where we have no data. When the averaging is restricted to the region $x < a$, the bias is positive (negative) for a distribution decreasing (increasing) towards the boundary. In both cases the size of the bias can be estimated and corrected for, using so-called boundary kernels.
2. Many smoothing methods do not guarantee normalization of the estimated probability density. While this effect can be corrected for easily by renormalizing $\hat{f}$, it indicates some problem of the method.
3. Fixed bandwidth methods over-smooth in regions where the density is high and tend to produce fake bumps in regions where the density is low. Variable bandwidth kernels are able to avoid this effect partially. Their bandwidth is chosen inversely proportional to the square root of the density, $h(x_i) = h_0 f(x_i)^{-1/2}$. Since the true density is not known, f must be replaced by a first estimate obtained for instance with a fixed bandwidth kernel.
4. Kernel smoothing corresponds to a convolution of the discrete data distribution with a smearing function and thus unavoidably tends to flatten peaks and to fill-up valleys. This is especially pronounced where the distribution shows strong structure, that is where the second derivative f'' is large. Convolution and thus also PDE implies a loss of some information contained in the original data. This defect may be acceptable if we gain sufficiently due to knowledge about f that we put into the smoothing program. In the simplest case this is only the fact that the distribution is continuous and differentiable but in some situations also the asymptotic behavior of f may be given, or we may know that it is unimodal. Then we will try to implement this information into the smoothing method.

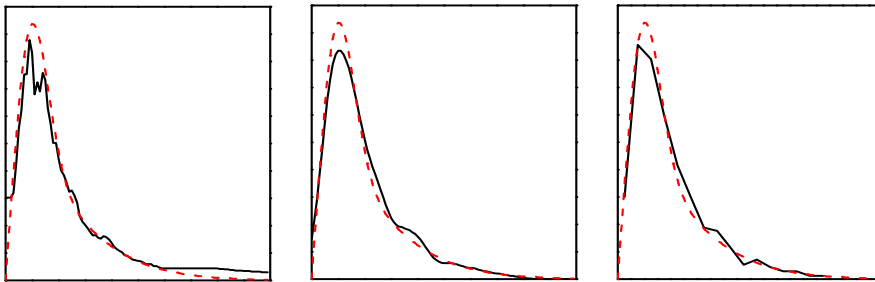


Fig. 12.2. Estimated probability density. Left hand: Nearest neighbor, center: Gaussian kernel, right hand: Polygon.

Some of the remedies for the difficulties mentioned above use estimates of f and its derivatives. Thus iterative procedures seem to be the solution. However, the iteration process usually does not converge and thus has to be supervised and stopped before artifacts appear.

In Fig. 12.2 three simple smoothing methods are compared. A sample of 1000 events has been generated from the function shown as a dashed curve in the figure. A k -nearest neighbor approximation of the p.d.f. of a sample is shown in the left hand graph of Fig. 12.2. The value of k chosen was 100 which is too small to produce enough smoothing but too large to follow the distribution at the left hand border. The result of the PDE with a Gaussian kernel with fixed width is presented in the central graph and a polygon approximation is shown in the right-hand graph. All three graphs show the typical defects of simple smoothing methods, broadening of the peak and fake structures in the region where the statistics is low.

Alternatively to the standard smoothing methods, complementary approaches often produce better results than the former. The typical smoothing problems can partially be avoided when the p.d.f. is parameterized and adjusted to the data sample in a likelihood fit. A simple parametrization is the superposition of normal distributions,

$$f(\mathbf{x}) = \sum \alpha_i N(\mathbf{x}; \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i),$$

with the free parameters, weights α_i , mean values $\boldsymbol{\mu}_i$ and covariance matrixes $\boldsymbol{\Sigma}_i$.

If information about the shape of the distribution is available, more specific parameterizations which describe the asymptotic behavior can be applied. Distributions which resemble a Gaussian should be approximated by the Gram-Charlier series (see last paragraph of Sect. 11.2.2). If the data sample is sufficiently large and the distribution is unimodal with known asymptotic behavior the construction of the p.d.f. from the moments as described in [74] is quite efficient.

Physicists use PDE mainly for the visualization of the data. Here, in one dimension, histogramming is the standard method. When the estimated distribution is used to simulate an experiment, frequency polygons are to be preferred. Whenever a useful parametrization is at hand, then PDE should be replaced by an adjustment of the corresponding parameters in a likelihood fit. Only in rare situations it pays to

construct complicated kernels. For a quick qualitative illustration of a distribution off-the-shelf programs may do. For most quantitative evaluations of moments and parameters of the unknown distribution we recommend to use the bootstrap method which is discussed in the following section.

12.2 Resampling Techniques

12.2.1 Introduction

In the previous section we have discussed a method to construct a distribution approximately starting from a sample drawn from it. Knowing the distribution allows us to calculate certain parameters like moments or quantiles. In most cases, however, it is preferable to determine the wanted quantities directly from the sample. A trivial example for this approach is the estimation of the mean value and the variance from a series of measurements as we have discussed in Chap. 4 treating error calculation where we had used the sample mean $\bar{x} = \sum x_i / N$ and the empirical variance $s^2 = [\sum (x_i - \bar{x})^2] / (N - 1)$. In a similar way we can also determine higher moments, correlations, quantiles and other statistical parameters. However, the analytical derivation of the corresponding expressions is often not as simple as that of the mean value or the variance. Here the bootstrap concept, which to a large extent has been developed by Efron, helps. A comprehensive presentation of the method is given in Ref. [64], which has served as bases for this section.

The name *Bootstrap* goes back to the famous book of Erich Raspe in which he narrates the adventures of the lying Baron von Münchhausen. Münchhausen had pretended to have saved himself out of a swamp by pulling himself up with his own bootstraps⁴. In statistics, the expression bootstrap is used because from a small sample the quasi complete distribution is generated. There is not quite as much lying as in Münchhausen's stories.

The bootstrap concept is based upon a simple idea: The sample itself replaces the unknown distribution. The sample *is* the distribution from which we draw individual observations.

As already mentioned, the bootstrap method permits us, apart from error estimation, to compute p-values for significance tests and the error rate in classifications. It relies, as will be shown in subsequent examples, on the combination of randomly selected observations.

A subvariant of the bootstrap technique is called *jackknife* which is mainly used to estimate biases from subsets of the data.

In Chap. 10 where we had evaluated the distribution of the energy test statistic in two-sample problems, we have used another resampling technique. We had reshuffled the elements of two partitions applying *random permutations*. Whereas in the bootstrap method, elements are drawn with replacement, permutations generate samples where every element occurs only a single time.

The reason for not using all possible permutations is simply that their number is in most cases excessively large and a finite random sample provides sufficiently precise results. While bootstrap techniques are used mainly to extract parameters

⁴In the original version he is pulling himself with his hair.

of an unknown distribution from a single sample, randomization methods serve to compare two or more samples taken under different conditions.

Remark: We may ask with some right whether resampling makes sense since randomly choosing elements from a sample does not seem to be as efficient as a systematic evaluation of the complete sample. Indeed, it should always be optimal to evaluate the interesting parameter directly using all elements of the sample with the same weight, either analytically or numerically; but, as in Monte Carlo simulations, the big advantage of parameter estimation by randomly selecting elements relies on the simplicity of the approach. Nowadays, lack of computing capacity is not a problem and in the limit of an infinite number of drawn combinations of observations the complete available information is exhausted.

12.2.2 Definition of Bootstrap and Simple Examples

We sort the N observations of a given data sample $\{x_1, x_2, \dots, x_N\}$ according to their value, $x_i \leq x_{i+1}$, and associate to each of them the probability $1/N$. We call this discrete distribution $P_0(x_i) = 1/N$ the *sample distribution*. We obtain a bootstrap sample $\{x_1^*, x_2^*, \dots, x_M^*\}$ by generating M observations following P_0 . Bootstrap observations are marked with a star "*". A bootstrap sample may contain the same observation several times. The evaluation of interesting quantities follows closely that which is used in Monte Carlo simulations. The p.d.f. used for event generation in Monte Carlo procedures is replaced by the sample distribution.

Example 143. Bootstrap evaluation of the accuracy of the estimated mean value of a distribution

We dispose already of an efficient method to estimate the variance. Here we present an alternative approach in order to introduce and illustrate the bootstrap method. Given be the sample of $N = 10$ observations $\{0.83, 0.79, 0.31, 0.09, 0.72, 2.31, 0.11, 0.32, 1.11, 0.75\}$. The estimate of the mean value is obviously $\hat{\mu} = \bar{x} = \sum x_i/N = 0.74$. We have also derived in Sect. 3.2 a formula to estimate the uncertainty, $\delta_\mu = s/\sqrt{N-1} = 0.21$. When we treat the sample as representative of the distribution, we are able to produce an empirical distribution of the mean value: We draw from the complete sample sequentially N observations (*drawing with replacement*) and get for instance the bootstrap sample $\{0.72, 0.32, 0.79, 0.32, 0.11, 2.31, 0.83, 0.83, 0.72, 1.11\}$. We compute the sample mean and repeat this procedure B times and obtain in this way B mean values μ_k^* . The number of bootstrap replicates should be large compared to N , for example B typically equal to 100 or 1000 for $N = 10$. From the distribution of the values μ_b , $b = 1, \dots, B$ we can compute again the mean value $\hat{\mu}^*$ and its uncertainty δ_μ :

$$\begin{aligned}\hat{\mu}^* &= \frac{1}{B} \sum \mu_b^*, \\ \delta_\mu^{*2} &= \frac{1}{B} \sum (\mu_b^* - \hat{\mu}^*)^2.\end{aligned}$$

Fig. 12.3 shows the sample distribution corresponding to the 10 observations and the bootstrap distribution of the mean values. The bootstrap estimates $\hat{\mu}^* = 0.74$, $\delta_\mu^* = 0.19$, agree reasonably well with the directly obtained values. The larger value of δ_μ compared to δ_μ^* is due to the bias correction in

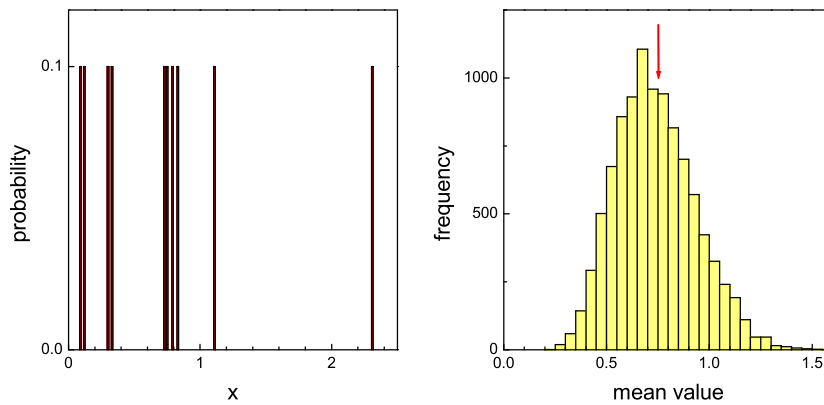


Fig. 12.3. Sample distribution (left) and distribution of bootstrap sample mean values (right).

its evaluation. The bootstrap values correspond to the maximum likelihood estimates. The distribution of Fig. 12.3 contains further information. We realize that the distribution is asymmetric, the reason being that the sample was drawn from an exponential. We could, for example, also derive the skewness or the frequency that the mean value exceeds 1.0 from the bootstrap distribution.

While we know the exact solution for the estimation of the mean value and mean squared error of an arbitrary function $u(x)$, it is difficult to compute the same quantities for more complicated functions like the median or for correlations.

Example 144. Error of mean distance of stochastically distributed points in a square

Fig. 12.4 shows 20 points drawn from an unknown p.d.f. distributed in a square. The mean distance is 0.55. We determine the standard deviation for this quantity from 10^4 bootstrap samples and obtain the value 0.10. This example is rather abstract and has been chosen because it is simple and demonstrates that the bootstrap method is able to solve problems which are hardly accessible with other methods.

Example 145. Acceptance of weighted events

We resume an example from Sect. 4.3.6 where we had presented an analytic solution. Now we propose a simpler solution: We dispose of a sample of N Monte Carlo generated events with weights w_i , $i = 1, \dots, N$, where we know for each of them whether it is accepted, $\varepsilon_i = 1$, or not, $\varepsilon_i = 0$. The mean acceptance is $\varepsilon = \sum w_i \varepsilon_i / \sum w_i$. Now we draw from the sample randomly B new bootstrap samples $\{(w_1^*, \varepsilon_1^*), \dots, (w_N^*, \varepsilon_N^*)\}$ and compute in each case ε^* . The empirical variance σ^2 of the distribution of ε^* is the bootstrap estimate of the error squared, $\delta_\varepsilon^2 = \sigma^2$, of the acceptance ε .

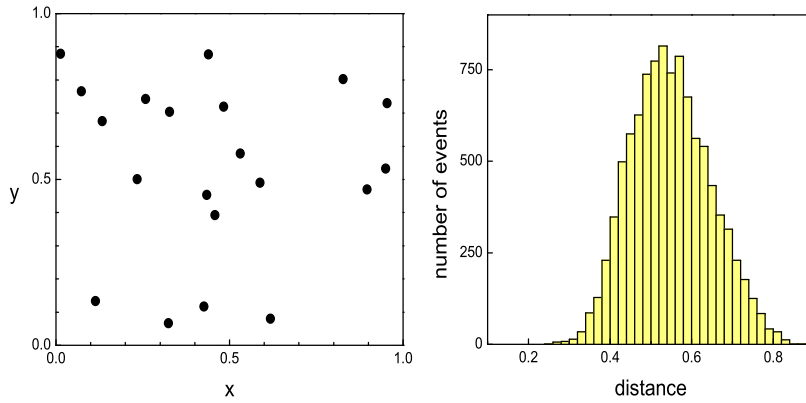


Fig. 12.4. Distribution of points in a unit square. The right hand graph shows the bootstrap distribution of the mean distance of the points.

12.2.3 Precision of the Error Estimate

Usually we are not interested in the uncertainty σ_δ of the error estimate δ . This is a higher order effect, yet we want to know how many bootstrap samples are required to avoid additional error contributions related to the method.

The standard deviation has two components, σ_t , which depends on the shape of the true distribution and the sample size N , and, σ_B , which depends on the number B of bootstrap replicates. Since the two causes are independent and of purely statistical nature, we can put

$$\begin{aligned}\sigma_\delta^2 &= \sigma_t^2 + \sigma_B^2, \\ \frac{\sigma_\delta^2}{\delta^2} &= \frac{c_1}{N} + \frac{c_2}{B}.\end{aligned}$$

We can only influence the second term, N being given. Obviously it is sufficient to choose the number B of bootstrap replicates large compared to the number N of experimental observations. For a normal distribution the two constants c_1, c_2 are both equal to $1/2$. (A derivation is given in [64].) For distributions with long tails, i.e. large excess γ_2 , they are larger. (Remember: $\gamma_2 = 0$ for the normal distribution.) The value of c_2 is in the general case given by [64]:

$$c_2 = \frac{\gamma_2 + 2}{4}.$$

An estimate for γ_2 can be derived from the empirical fourth moment of the sample. Since error estimates are rather crude anyway, we are satisfied with the choice $B \gg N$.

12.2.4 Confidence Limits

To compute confidence limits or the p-value of a parameter we generate its distribution from bootstrap samples. In a preceding example where we computed the mean

distance of random points, we extract from the distribution of Fig. 12.4 that the probability to find a distance less than 0.4 is approximately equal to 10%. Exact confidence intervals can only be derived from the exact distribution.

12.2.5 Precision of Classifiers

Classifiers like decision trees and ANNs usually subdivide the learning sample in two parts, one part is used to train the classifier and a smaller part is reserved to test the classifier. The precision can be enhanced considerably by using bootstrap samples for both training and testing.

12.2.6 Random Permutations

In Chap. 10 we have treated the two-sample problem: “Do two experimental distributions belong to the same population?” In one of the tests, the energy test, we had used permutations of the observations to determine the distribution of the test quantity. The same method can be applied to an arbitrary test quantity which is able to discriminate between samples.

Example 146. Two-sample test with a decision tree

Let us assume that we want to test whether the two samples $\{\mathbf{x}_1, \dots, \mathbf{x}_{N_1}\}$, $\{\mathbf{x}_1, \dots, \mathbf{x}_{N_2}\}$ of sizes N_1 and N_2 belong to the same population. This is our null hypothesis. Instead of using one of the established two-sample methods we may train a decision tree to separate the two samples. As a test quantity serves the number of misclassifications $\tilde{N}$ which of course is smaller than $(N_1 + N_2)/2$, half the size of the total sample. Now we combine the two samples, draw from the combined sample two new random samples of sizes N_1 and N_2 , train again a decision tree to identify for each element the sample index and count the number of misclassifications. We repeat this procedure many, say 1000, times and obtain this way the distribution of the test statistic under the null hypothesis. The fraction of cases where the random selection yields a smaller number of misclassifications than the original samples is equal to the p-value of the null hypothesis.

Instead of a decision tree we can use any other classifier, for instance a ANN. The corresponding tests are potentially very powerful but also quite involved. Even with nowadays computer facilities, training of some 1000 decision trees or artificial neural nets is quite an effort.

12.2.7 Jackknife and Bias Correction

Jackknife is mainly used for bias removal. It is assumed that in first order the bias decreases with N , i.e. that it is proportional to $1/N$. How jackknife works, is explained in the following simple example where we know the exact result from Chap. 3.

When we estimate the mean squared error of a quantity x from a sample of size N , using the formula

$$\delta_N^2 = \sum (x_i - \bar{x})^2 / N$$

then δ_N^2 is biased. We assume that the expected bias decreases linearly with N . Thus it is possible to estimate the bias by changing N . We remove one observation at a time and compute each time the mean squared error $\delta_{N-1,i}^2$ and average the results:

$$\delta_{N-1}^2 = \frac{1}{N} \sum_{i=1}^N \delta_{N-1,i}^2 .$$

A linear dependence of the average bias b_N on $1/N$ implies

$$\begin{aligned} b_N N &= b_{N-1} (N-1) \\ [E(\delta_N^2) - \sigma^2] N &= [E(\delta_{N-1}^2) - \sigma^2] (N-1) , \\ \sigma^2 &= N E(\delta_N^2) - (N-1) E(\delta_{N-1}^2) . \end{aligned} \tag{12.4}$$

Thus an improved estimate is $\delta_c^2 = N \delta_N^2 - (N-1) \delta_{N-1}^2$.

Inserting the known expectation values (see Chap. 3),

$$E(\delta_N^2) = \sigma^2 \frac{N-1}{N} , \quad E(\delta_{N-1}^2) = \sigma^2 \frac{N-2}{N-1} ,$$

we confirm the bias corrected result $E(\delta_c^2) = \sigma^2$.

This result can be generalized to any estimate t with a dominantly linear bias, $b \propto 1/N$.

$$t_c = N t_N - (N-1) t_{N-1}$$

The remaining bias is zero, of order $1/N^2$, or of higher order.

Appendix

13.1 Large Number Theorems

13.1.1 The Chebyshev Inequality and the Law of Large Numbers

For a probability density $f(x)$ with expected value μ , finite variance σ and arbitrary given positive δ , the following inequality, known as *Chebyshev inequality*, is valid:

$$P\{|x - \mu| \geq \delta\} \leq \frac{\sigma^2}{\delta^2} . \quad (13.1)$$

This very general theorem says that a given, fixed deviation from the expected value becomes less probable when the variance becomes smaller. It is also valid for discrete distributions.

To prove the inequality, we use the definition

$$P_I \equiv P\{x \in I\} = \int_I f(x) dx ,$$

where the domain of integration I is given by $1 \leq |x - \mu|/\delta$. The assertion follows from the following inequalities for the integrals:

$$\int_I f(x) dx \leq \int_I \left(\frac{x - \mu}{\delta} \right)^2 f(x) dx \leq \int_{-\infty}^{\infty} \left(\frac{x - \mu}{\delta} \right)^2 f(x) dx = \sigma^2 / \delta^2 .$$

Applying (13.1) to the arithmetic mean $\bar{x}$ from N independent identical distributed random variables $x_1, \dots, x_N$ results in one of the so-called laws of large numbers:

$$P\{|\bar{x} - \langle x \rangle| \geq \delta\} \leq \text{var}(x) / (N\delta^2) , \quad (13.2)$$

with the relations $\langle \bar{x} \rangle = \langle x \rangle$, $\text{var}(\bar{x}) = \text{var}(x)/N$ obtained in Sects. 3.2.2 and 3.2.3. The right-hand side disappears for $N \rightarrow \infty$, thus in this limit the *probability* to observe an arithmetic mean value outside an arbitrary interval centered at the expected value approaches zero. We talk about stochastic convergence or convergence in probability, here of the arithmetic mean against the expected value.

We now apply (13.2) to the indicator function $\mathcal{I}_I(x) = 1$ for $x \in I$, else 0. The sample mean $\bar{\mathcal{I}}_I = \sum \mathcal{I}_I(x_i)/N$ is the observed relative frequency of events $x \in I$ in the sample. The expected value and the variance are

$$\begin{aligned}
\langle \mathcal{I}_I \rangle &= \int \mathcal{I}_I(x) f(x) dx \\
&= P_I, \\
\text{var}(\mathcal{I}_I) &= \int \mathcal{I}_I^2(x) f(x) dx - \langle \mathcal{I}_I \rangle^2 \\
&= P_I(1 - P_I) \leq 1/4,
\end{aligned}$$

where, as above, P_I is the probability $P\{x \in I\}$ to find x in the set I . When we insert these results into (13.2), we obtain

$$P\{|\overline{\mathcal{I}_I} - P_I| \geq \delta\} \leq 1/(4N\delta^2). \quad (13.3)$$

The relative frequency of events of a certain type in a sample converges with increasing N stochastically to the probability to observe an event of that type¹.

13.1.2 The Central Limit Theorem

The central limit theorem states that the distribution of the sample mean $\bar{x}$,

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i,$$

of N i.i.d. variables x_i with finite variance σ^2 in the limit $N \rightarrow \infty$ will approach a normal distribution with variance σ^2/N independent of the form of the distribution $f(x)$. The following proof assumes that its characteristic function exists.

To simplify the notation, we transform the variable to $y = (x - \mu)/(\sqrt{N}\sigma)$, where μ is the mean of x . The characteristic function of a p.d.f. with mean zero and variance $1/N$ and thus also the p.d.f. of y is of the form

$$\phi(t) = 1 - \frac{t^2}{2N} + c \frac{t^3}{N^{3/2}} + \dots$$

The characteristic function of the sum $z = \sum_{i=1}^N y_i$ is given by the product

$$\phi_z = \left[1 - \frac{t^2}{2N} + c \frac{t^3}{N^{3/2}} + \dots \right]^N$$

which in the limit $N \rightarrow \infty$, where only the first two terms survive, approaches the characteristic function of the standard normal distribution $N(0, 1)$:

$$\lim_{N \rightarrow \infty} \phi_z = \lim_{N \rightarrow \infty} \left[1 - \frac{t^2}{2N} \right]^N = e^{-t^2/2}.$$

It can be shown that the convergence of characteristic functions implies the convergence of the distributions. The distribution of $\bar{x}$ for large N is then approximately

$$f(\bar{x}) \approx \frac{\sqrt{N}}{\sqrt{2\pi}\sigma} \exp \left[-\frac{N(\bar{x} - \mu)^2}{\sigma^2} \right].$$

Remark: The law of large numbers and the central limit theorem can be generalized to sums of independent but not identically distributed variates. The convergence is relatively fast when the variances of all variates are of similar size.

¹This theorem was derived by the Dutch-Swiss mathematician Jakob I. Bernoulli (1654-1705).

13.2 Consistency, Bias and Efficiency of Estimators

The following estimator properties are essential in frequentist statistics. We will discuss their relevance in Appendix 13.6.

Throughout this chapter we assume that samples consist of N i.i.d. variables x_i , the true parameter value is θ_0 , estimates are $\hat{\theta}$ or t .

13.2.1 Consistency

We expect from an useful estimator that it becomes more accurate with increasing size of the sample, i.e. larger deviations from the true value should become more and more improbable.

A sequence of estimators t_N of a parameter θ is called consistent, if their p.d.f.s for $N \rightarrow \infty$ are shrinking towards a central value equal to the true parameter value θ_0 , or, expressing it mathematically, if

$$\lim_{N \rightarrow \infty} P\{|t_N - \theta_0| > \varepsilon\} = 0 \quad (13.4)$$

is valid for arbitrary ε . A sufficient condition for consistency which can be easier checked than (13.4), is the combination of the two requirements

$$\lim_{N \rightarrow \infty} \langle t_N \rangle = \theta_0, \quad \lim_{N \rightarrow \infty} \text{var}(t_N) = 0,$$

where of course the existence of mean value and variance for the estimator t_N has to be assumed.

For instance, as implied by the law of large numbers, the sample moments

$$t_m = \frac{1}{N} \sum_{i=1}^N x_i^m$$

are consistent estimators for the respective m -th moments μ_m of $f(x)$ if this moments exist.

13.2.2 Bias of Estimates

The bias of an estimate has been already introduced in Sect. 7.1.4: An estimate t_N for θ is unbiased if already for finite N (eventually $N > N_0$) and all parameter values considered, the estimator satisfies the condition

$$\langle t_N \rangle = \theta.$$

The bias of an estimate is defined as:

$$b = \langle t_N \rangle - \theta.$$

Obviously, consistent estimators are asymptotically unbiased:

$$\lim_{N \rightarrow \infty} b(N) = 0.$$

The bias of a consistent estimator can be removed without affecting the consistency by multiplying the estimate with a factor like $(N+a)/(N+b)$ which approaches unity for $N \rightarrow \infty$.

13.2.3 Efficiency

An important characteristics is of course the accuracy of the statistical estimate. A useful measure for accuracy is the mean square deviation $\langle (t - \theta_0)^2 \rangle$ of the estimate from the true value of the parameter. According to (3.11) it is related to the variance of the estimator and the bias by

$$\langle (t - \theta_0)^2 \rangle = \text{var}(t) + b^2. \quad (13.5)$$

Definition: An estimator t is (asymptotically) efficient for the parameter θ if for all permitted parameter values it fulfils the following conditions for $N \rightarrow \infty$:

1. $\sqrt{N}(t - \theta)$ approaches a normal distribution of constant width and mean equal to zero.
2. $\text{var}(t) \leq \text{var}(t')$ for any other estimator t' which satisfies condition 1.

In other words, an efficient estimate is asymptotically normally distributed and has minimal variance. According to condition 1, its variance decreases with $1/N$. An efficient estimator therefore reaches the same accuracy as a competing one with a smaller sample size N , and is therefore economically superior. Not in all situations an efficient estimator exists.

Example 147. Efficiency of different estimates of the location parameter of a Gaussian [77]

Let us compare three methods to estimate the expected value μ of a Gaussian $N(x|\mu, \sigma)$ with given width σ from a sample $\{x_i\}$, $i = 1, \dots, N$. For large N we obtain for $\text{var}(t)$:

Method 1: sample mean	σ^2/N
Method 2: sample median	$\sigma^2/N \cdot \pi/2$
Method 3: $(x_{\min} + x_{\max})/2$	$\sigma^2/N \cdot N\pi^2/(12 \ln N)$

Obviously methods 2 and 3 are not efficient. Especially the third method, taking the mean of the two extremal values found in the sample, performs badly here. For other distributions, different results will be found. For the rather exotic two-sided exponential distribution (an exponential distribution of the absolute value of the variate, also called Laplace distribution) method 2 would be efficient and equal to the MLE. For a uniform distribution the estimator of method 3 would be efficient and also equal to the MLE.

While it is of interest to find the estimator which provides the smallest variance, it is not obvious how we could prove this property, since a comparison with all thinkable methods is of course impossible. Here a useful tool is the Cramer–Rao inequality. It provides a lower bound of the variance of an estimator. If we reach this minimum, we can be sure that the optimal accuracy is obtained.

The *Cramer–Rao inequality* states:

$$\text{var}(t) \geq \frac{[1 + (db/d\theta)]^2}{N \langle (\partial \ln f / \partial \theta)^2 \rangle}. \quad (13.6)$$

The denominator of the right-hand side is also called, after R. A. Fisher, the information² about the parameter θ from a sample of size N of i.i.d. variates.

²This is a special use of the word *information* as a technical term.

To prove this inequality, we define the random variable $y = \sum y_i$ with

$$y_i = \frac{\partial \ln f_i}{\partial \theta}, \quad f_i \equiv f(x_i|\theta). \quad (13.7)$$

It has the expected value

$$\begin{aligned} \langle y_i \rangle &= \int \frac{1}{f_i} \frac{\partial f_i}{\partial \theta} f_i \, dx_i \\ &= \int \frac{\partial f_i}{\partial \theta} \, dx_i \\ &= \frac{\partial}{\partial \theta} \int f_i \, dx_i \\ &= \frac{\partial}{\partial \theta} 1 = 0. \end{aligned} \quad (13.8)$$

Because of the independence of the y_i we have $\langle y_i y_j \rangle = \langle y_i \rangle \langle y_j \rangle = 0$ and

$$\text{var}(y) = N \langle y_i^2 \rangle = N \left\langle \left(\frac{\partial \ln f}{\partial \theta} \right)^2 \right\rangle. \quad (13.9)$$

Using the definition $L = \prod f_i$, we find for $\text{cov}(ty) = \langle (t - \langle t \rangle)(y - \langle y \rangle) \rangle$:

$$\begin{aligned} \text{cov}(ty) &= \int t \frac{\partial \ln L}{\partial \theta} L \, dx_1 \cdots dx_N \\ &= \int t \frac{\partial}{\partial \theta} L \, dx_1 \cdots dx_N \\ &= \frac{\partial}{\partial \theta} \langle t \rangle \\ &= 1 + \frac{db}{d\theta}. \end{aligned} \quad (13.10)$$

From the Cauchy–Schwarz inequality

$$[\text{cov}(ty)]^2 \leq \text{var}(t) \text{var}(y)$$

and (13.9), (13.10) follows (13.6).

The equality sign in (13.6) is valid if and only if the two factors t, y in the covariance are proportional to each other. In this case t is called a Minimum Variance Bound (MVB) estimator. It can be shown to be also minimal sufficient.

In most of the literature efficiency is defined by the stronger condition: An estimator is called efficient, if it is bias-free and if it satisfies the MVB.

13.3 Properties of the Maximum Likelihood Estimator

13.3.1 Consistency

The maximum likelihood estimator (MLE) is consistent under mild assumptions.

To prove this, we consider the expected value of

$$\ln L(\theta|\mathbf{x}) = \sum_{i=1}^N \ln f(x_i|\theta) \quad (13.11)$$

which is to be calculated by integration over the variables³ $\mathbf{x}$ using the true p.d.f. (with the true parameter θ_0). First we prove the inequality

$$\langle \ln L(\theta|\mathbf{x}) \rangle < \langle \ln L(\theta_0|\mathbf{x}) \rangle, \quad (13.12)$$

for $\theta \neq \theta_0$: Since the logarithm is a strongly convex function, there is always $\langle \ln(\dots) \rangle < \ln \langle \dots \rangle$, hence

$$\left\langle \ln \frac{L(\theta|\mathbf{x})}{L(\theta_0|\mathbf{x})} \right\rangle < \ln \left\langle \frac{L(\theta|\mathbf{x})}{L(\theta_0|\mathbf{x})} \right\rangle = \ln \int \frac{L(\theta|\mathbf{x})}{L(\theta_0|\mathbf{x})} L(\theta_0|\mathbf{x}) d\mathbf{x} = \ln 1 = 0.$$

In the last step we used

$$\int L(\theta|\mathbf{x}) d\mathbf{x} = \int \prod f(x_i|\theta) dx_1 \cdots dx_N = 1.$$

Since $\ln L(\theta|\mathbf{x})/N = \sum \ln f(x_i|\theta)/N$ is an arithmetic sample mean which, according to the law of large numbers (13.2), converges stochastically to the expected value for $N \rightarrow \infty$, we have also (in the sense of stochastic convergence)

$$\ln L(\theta|\mathbf{x})/N \rightarrow \langle \ln f(x|\theta) \rangle = \sum \langle \ln f(x_i|\theta) \rangle / N = \langle \ln L(\theta|\mathbf{x}) \rangle / N,$$

and from (13.12)

$$\lim_{N \rightarrow \infty} P\{\ln L(\theta|\mathbf{x}) < \ln L(\theta_0|\mathbf{x})\} = 1, \quad \theta \neq \theta_0. \quad (13.13)$$

On the other hand, the MLE $\hat{\theta}$ is defined by its extremal condition

$$\ln L(\hat{\theta}|\mathbf{x}) \geq \ln L(\theta_0|\mathbf{x}).$$

A contradiction to (13.13) can be avoided only, if also

$$\lim_{N \rightarrow \infty} P\{|\hat{\theta} - \theta_0| < \varepsilon\} = 1$$

is valid. This means consistency of the MLE.

13.3.2 Efficiency

Since the MLE is consistent, it is unbiased asymptotically for $N \rightarrow \infty$. Under certain assumptions in addition to the usually required regularity⁴ the MLE is also efficient asymptotically.

Proof:

³We keep the form of the argument list of L , although now $\mathbf{x}$ is not considered as fixed to the experimentally sampled values, but as a random vector with given p.d.f..

⁴The boundaries of the domain of x must not depend on θ and the maximum of L should not be reached at the boundary of the range of θ .

With the notations of the last paragraph with $L = \prod f_i$ and using (13.8), the expected value and variance of $y = \sum y_i = \partial \ln L / \partial \theta$ are given by the following expressions:

$$\langle y \rangle = \int \frac{\partial \ln L}{\partial \theta} L \, d\mathbf{x} = 0, \quad (13.14)$$

$$\sigma_y^2 = \text{var}(y) = \left\langle \left(\frac{\partial \ln L}{\partial \theta} \right)^2 \right\rangle = - \left\langle \frac{\partial^2}{\partial \theta^2} \ln L \right\rangle. \quad (13.15)$$

The last relation follows after further differentiation of (13.14) and from the relation

$$\int \frac{\partial^2 \ln L}{\partial \theta^2} L \, d\mathbf{x} = - \int \frac{\partial \ln L}{\partial \theta} \frac{\partial L}{\partial \theta} \, d\mathbf{x} = - \int \frac{\partial \ln L}{\partial \theta} \frac{\partial \ln L}{\partial \theta} L \, d\mathbf{x}.$$

From the Taylor expansion of $\partial \ln L / \partial \theta|_{\theta=\hat{\theta}}$ which is zero by definition and with (13.15) we find

$$\begin{aligned} 0 &= \frac{\partial \ln L}{\partial \theta} \Big|_{\theta=\hat{\theta}} \approx \frac{\partial \ln L}{\partial \theta} \Big|_{\theta=\theta_0} + (\hat{\theta} - \theta_0) \frac{\partial^2 \ln L}{\partial \theta^2} \Big|_{\theta=\theta_0} \\ &\approx y - (\hat{\theta} - \theta_0) \sigma_y^2, \end{aligned} \quad (13.16)$$

where the consistency of the MLE guaranties the validity of this approximation in the sense of stochastic convergence. Following the central limit theorem, y/σ_y being the sum of i.i.d. variables, is asymptotically normally distributed with mean zero and variance unity. The same is then true for $(\hat{\theta} - \theta_0)\sigma_y$, i.e. $\hat{\theta}$ follows asymptotically a normal distribution with mean θ_0 and asymptotically vanishing variance $1/\sigma_y^2 \sim 1/N$, as seen from (13.9).

13.3.3 Asymptotic Form of the Likelihood Function

A similar result as derived in the last paragraph for the p.d.f. of the MLE $\hat{\theta}$ can be derived for the likelihood function itself.

If one considers the Taylor expansion of $y = \partial \ln L / \partial \theta$ around the MLE $\hat{\theta}$, we get with $y(\hat{\theta}) = 0$

$$y(\theta) \approx (\theta - \hat{\theta}) y'(\hat{\theta}). \quad (13.17)$$

As discussed in the last paragraph, we have for $N \rightarrow \infty$

$$y'(\hat{\theta}) \rightarrow y'(\theta_0) \rightarrow \langle y' \rangle = -\sigma_y^2 = \text{const}.$$

Thus $y'(\hat{\theta})$ is independent of $\hat{\theta}$ and higher derivatives disappear. After integration of (13.17) over θ we obtain a parabolic form for $\ln L$:

$$\ln L(\theta) = \ln L(\hat{\theta}) - \frac{1}{2} \sigma_y^2 (\theta - \hat{\theta})^2,$$

where the width of the parabola decreases with $\sigma_y^{-2} \sim 1/N$ (13.9). Up to the missing normalization, the likelihood function has the same form as the distribution of the MLE with $\hat{\theta} - \theta_0$ replaced by $\theta - \hat{\theta}$.

13.3.4 Properties of the Maximum Likelihood Estimate for Small Samples

The criterium of asymptotic efficiency, fulfilled by the MLE for large samples, is usually extended to small samples, where the normal approximation of the sampling distribution does not apply, in the following way: A bias-free estimate t is called a *minimum variance* (MV) estimate if $\text{var}(t) \leq \text{var}(t')$ for any other bias-free estimate t' . If, moreover, the Cramer–Rao inequality (13.6) is fulfilled as an equality, one speaks of a *minimum variance bound* (MVB) estimate, often also called efficient or most efficient, estimate (not to be confused with the asymptotic efficiency which we have considered before in Appendix 13.2). The latter, however, exists only for a certain function $\tau(\theta)$ of the parameter θ if it has a one-dimensional sufficient statistic (see 7.1.1). It can be shown [2] that under exactly this condition the MLE for τ will be this MVB estimate, and therefore bias-free for any N . The MLE for any non-linear function of τ will in general be biased, but still optimal in the following sense: if bias-corrected, it becomes an MV estimate, i.e. it will have the smallest variance among all unbiased estimates.

Example 148. : Efficiency of small sample MLEs

The MLE for the variance σ^2 of a normal distribution with known mean μ ,

$$\widehat{\sigma^2} = \frac{1}{N} \sum (x_i - \mu)^2 ,$$

is unbiased and efficient, reaching the MVB for all N . The MLE for σ is of course

$$\hat{\sigma} = \sqrt{\widehat{\sigma^2}} ,$$

according to the relation between σ and σ^2 . It is biased and thus not efficient in the sense of the above definition.

A bias-corrected estimator for σ is (see for instance [90])

$$\hat{\sigma}_{corr} = \sqrt{\frac{N}{2} \frac{\Gamma(\frac{N}{2})}{\Gamma(\frac{N+1}{2})}} \hat{\sigma} .$$

This estimator can be shown to have the smallest variance of all unbiased estimators, independent of the sample size N .

In the above example a one-dimensional sufficient statistic exists. If this is not the case, the question of optimality of the MLE for small samples has – from the frequentist point of view – no general answer.

In summary, also for finite N the MLE for a certain parameter achieves the optimal – from the frequentist point of view – properties of an MVB estimator, if the latter does exist. Of course these properties cannot be preserved for other parameterizations, since variance and bias are not invariant properties.

13.4 Error of Background-Contaminated Parameter Estimates

In order to calculate the additional uncertainty of a parameter estimate due to the presence of background, if the latter is taken from a reference experiment in the

way described in Sect. 6.5.10, we consider the general definition of the pseudo log-likelihood

$$\ln \tilde{L} = \sum_{i=1}^N \ln f(x_i|\theta) - r \sum_{i=1}^M \ln f(x'_i|\theta) ,$$

restricting ourselves at first to a single parameter θ , see (6.20). The generalization to multi-dimensional parameter spaces is straight forward, and will be indicated later. From $\partial \ln \tilde{L} / \partial \theta|_{\hat{\theta}} = 0$, we find

$$\left[\sum_{i=1}^S \frac{\partial \ln f(x_i^{(S)}|\theta)}{\partial \theta} + \sum_{i=1}^B \frac{\partial \ln f(x_i^{(B)}|\theta)}{\partial \theta} - r \sum_{i=1}^M \frac{\partial \ln f(x'_i|\theta)}{\partial \theta} \right]_{\hat{\theta}} = 0 .$$

This formula defines the background-corrected estimate $\hat{\theta}$. It differs from the “ideal” estimate $\hat{\theta}^{(S)}$ which would be obtained in the absence of background, i.e. by equating to zero the first sum on the left hand side. Writing $\hat{\theta} = \hat{\theta}^{(S)} + \Delta\hat{\theta}$ in the first sum, and Taylor expanding it up to the first order, we get

$$\sum_{i=1}^S \frac{\partial^2 \ln f(x_i^{(S)}|\theta)}{\partial \theta^2} \Big|_{\hat{\theta}^{(S)}} \Delta\hat{\theta} + \left[\sum_{i=1}^B \frac{\partial \ln f(x_i^{(B)}|\theta)}{\partial \theta} - r \sum_{i=1}^M \frac{\partial \ln f(x'_i|\theta)}{\partial \theta} \right]_{\hat{\theta}} = 0 . \quad (13.18)$$

The first sum, if taken with a minus sign, is the Fisher information of the signal sample on $\theta^{(S)}$, and equals $-1/\text{var}(\hat{\theta}^{(S)})$, asymptotically. The approximation relies on the assumption that $\sum \ln f(x_i|\theta)$ is parabolic in the region $\hat{\theta}^{(S)} \pm \Delta\hat{\theta}$. Then we have

$$\Delta\hat{\theta} \approx \text{var}(\hat{\theta}^{(S)}) \left[\sum_{i=1}^B \frac{\partial \ln f(x_i^{(B)}|\theta)}{\partial \theta} - r \sum_{i=1}^M \frac{\partial \ln f(x'_i|\theta)}{\partial \theta} \right]_{\hat{\theta}} . \quad (13.19)$$

We take the expected value with respect to the background distribution and obtain

$$\langle \Delta\hat{\theta} \rangle = \text{var}(\hat{\theta}^{(S)}) \langle B - rM \rangle \left\langle \frac{\partial \ln f(x|\theta)}{\partial \theta} \right\rangle_{\hat{\theta}} .$$

Since $\langle B - rM \rangle = 0$, the background correction is asymptotically bias-free.

Squaring (13.19), and writing the summands in short hand as y_i, y'_i , we get

$$\begin{aligned} (\Delta\hat{\theta})^2 &= (\text{var}(\hat{\theta}^{(S)}))^2 \left[\sum_{i=1}^B y_i - r \sum_{i=1}^M y'_i \right]^2 , \\ [\dots]^2 &= \sum_i^B \sum_j^B y_i y_j + r^2 \sum_i^M \sum_j^M y'_i y'_j - 2r \sum_i^B \sum_j^M y_i y'_j \\ &= \sum_i^B y_i^2 + r^2 \sum_i^M y_i'^2 + \sum_{j \neq i}^B y_i y_j + r^2 \sum_{j \neq i}^M y'_i y'_j - 2r \sum_i^B \sum_j^M y_i y'_j , \\ &= (\text{var}(\hat{\theta}^{(S)}))^2 \left[\langle B + r^2 M \rangle \langle y^2 \rangle - \langle y \rangle^2 + \langle (B - rM)^2 \rangle \langle y \rangle^2 \right] . \end{aligned} \quad (13.20)$$

In physics experiments, the event numbers M, B, S are independently fluctuating according to Poisson distributions with expected values $\langle M \rangle = \langle B \rangle / r$, and $\langle S \rangle$. Then $\langle B + r^2 M \rangle = \langle B \rangle (1 + r)$ and

$$\langle (B - rM)^2 \rangle = \langle B^2 \rangle + \langle r^2 M^2 \rangle - 2r \langle B \rangle \langle M \rangle = \langle B \rangle + r^2 \langle M \rangle = \langle B \rangle (1 + r) .$$

Adding the contribution from the uncontaminated estimate, $\text{var}(\hat{\theta}^{(S)})$, to (13.20) leads to the final result

$$\begin{aligned}\text{var}(\hat{\theta}) &= \text{var}(\hat{\theta}^{(S)}) + \langle (\Delta\hat{\theta})^2 \rangle \\ &= \text{var}(\hat{\theta}^{(S)}) + (1+r)(\text{var}(\hat{\theta}^{(S)}))^2 \langle B \rangle \langle y^2 \rangle \\ &= \text{var}(\hat{\theta}^{(S)}) + r(1+r)(\text{var}(\hat{\theta}^{(S)}))^2 \langle M \rangle \langle y^2 \rangle .\end{aligned}\quad (13.21)$$

But $\langle M \rangle$, the expected number of background events in both the main- and the control sample, is not known and can only be estimated by the empirical value M . In the same way we have to use empirical estimates for the expected values $\langle y \rangle$ and $\langle y^2 \rangle$, since the distribution of background events is unknown. Thus we replace

$$\langle M \rangle \rightarrow M, \quad \langle y \rangle \rightarrow \sum_{i=1}^M y_i / M, \quad \langle y^2 \rangle \rightarrow \sum_{i=1}^M y_i^2 / M$$

where $y_i = \partial \ln f(x'_i | \theta) / \partial \theta$. As usual in error calculation, the dependence of y_i on the true value of θ has to be approximated by a dependence on the estimated value $\hat{\theta}$. Similarly, we approximate $\text{var}(\hat{\theta}^{(S)})$:

$$\begin{aligned}-1/\text{var}(\hat{\theta}^{(S)}) &= \sum_{i=1}^S \frac{\partial^2 \ln f(x_i | \theta)}{\partial \theta^2} \Big|_{\hat{\theta}^{(S)}} \\ &\approx \left[\sum_{i=1}^N \frac{\partial^2 \ln f(x_i | \theta)}{\partial \theta^2} - r \sum_{i=1}^M \frac{\partial^2 \ln f(x'_i | \theta)}{\partial \theta^2} \right]_{\hat{\theta}} .\end{aligned}$$

We realize from (13.21) that it is advantageous to take a large reference sample, i.e. r small. The variance $\langle (\Delta\hat{\theta})^2 \rangle$ increases with the square of the error of the uncontaminated sample. Via the quantity $\langle y^2 \rangle$ it depends also on the shape of the background distribution.

For P -dimensional parameter space $\boldsymbol{\theta}$ we see from (13.18) that the first sum is given by the weight matrix $V^{(S)}$ of the estimated parameters in the absence of background

$$-\sum_{l=1}^P \sum_{i=1}^S \frac{\partial^2 \ln f(x_i^{(S)} | \boldsymbol{\theta})}{\partial \theta_k \partial \theta_l} \Big|_{\hat{\boldsymbol{\theta}}^{(S)}} \Delta \hat{\theta}_l = \sum_{l=1}^P (V^{(S)})_{kl} \Delta \hat{\theta}_l .$$

Solving the linear equation system for $\Delta \hat{\boldsymbol{\theta}}$ and constructing from its components the error matrix E , we find in close analogy to the one-dimensional case

$$E = C^{(S)} Y C^{(S)} ,$$

with $C^{(S)} = V^{(S)-1}$ being the covariance matrix of the background-free estimates and Y defined as

$$Y_{kl} = r(1+r) \langle M \rangle \langle y_k y_l \rangle ,$$

with $y_k = y_k(x_i)$ short hand for $\partial \ln f(x'_i | \boldsymbol{\theta}) / \partial \theta_k$. As in the one-dimensional case, the total covariance matrix of the estimated parameters is the sum

$$\text{cov}(\hat{\theta}_k, \hat{\theta}_l) = C_{kl}^{(S)} + E_{kl} .$$

The following example illustrates the error due to background contamination for the above estimation method.

Example 149. Parameter uncertainty for background contaminated signals

We investigate how well our asymptotic error formula works in a specific example. To this end, we consider a Gaussian signal distribution with width unity and mean zero over a background modeled by an exponential distribution with decay constant $\gamma = 0.2$ of the form $c \exp[-\gamma(x + 4)]$ where both distributions are restricted to the range $[-4, 4]$. The numbers of signal events S , background events B and reference events M follow Poisson distributions with mean values $\langle S \rangle = 60$, $\langle B \rangle = 40$ and $\langle M \rangle = 100$. This implies a correction factor $r = \langle B \rangle / \langle M \rangle = 0.4$ for the reference experiment. From 10^4 MC experiments we obtain a distribution of $\hat{\mu}$, with mean value and width 0.019 and 0.34, respectively. The pure signal $\hat{\mu}^{(S)}$ has mean and width 0.001 and 0.13 ($= 1/\sqrt{60}$). From our asymptotic error formula (13.21) we derive an error of 0.31, slightly smaller than the MC result. The discrepancy will be larger for lower statistics. It is typical for Poisson fluctuations.

13.5 Frequentist Confidence Intervals

We associate error intervals to measurements to indicate that the parameter of interest has a reasonably high probability to be located inside the interval. However to compute the probability a prior probability has to be introduced with the problem which we have discussed in Sect. 6.1. To circumvent this problem, J. Neyman has proposed a method to construct intervals without using prior probabilities. Unfortunately, as it is often the case, one problem is traded for another one.

Neyman's confidence intervals have the following defining property: *The true parameter lies in the interval on the average in the fraction C of intervals of confidence level C .* In other words: Given a true value θ , a measurement t will include it in its associated confidence interval $[t_1, t_2]$ – “cover” it – with probability C . (Remark that this does not necessarily imply that given a certain confidence interval the true value is included in it with probability C .)

Traditionally chosen values for the confidence level are 68.3%, 90%, 95% – the former corresponds to the standard error interval of the normal distribution.

Confidence intervals are constructed in the following way:

For each parameter value θ a probability interval $[t_1(\theta), t_2(\theta)]$ is defined, such that the probability that the observed value t of θ is located in the interval is equal to the confidence level C :

$$P\{t_1(\theta) \leq t \leq t_2(\theta)\} = \int_{t_1}^{t_2} f(t|\theta) dt = C. \quad (13.22)$$

Of course the p.d.f. $f(t|\theta)$ or error distribution of the estimator t must be known. To fix the interval completely, an additional condition is applied. In the univariate case, a common procedure is to choose central intervals,

$$P\{t < t_1\} = P\{t > t_2\} = \frac{1 - C}{2}.$$

Other conventions are minimum length and equal probability intervals defined by $f(t_1) = f(t_2)$. The confidence interval consists of those parameter values which include the measurement $\hat{t}$ within their probability intervals. Somewhat simplified: Parameter values are accepted, if the observation is compatible with them.

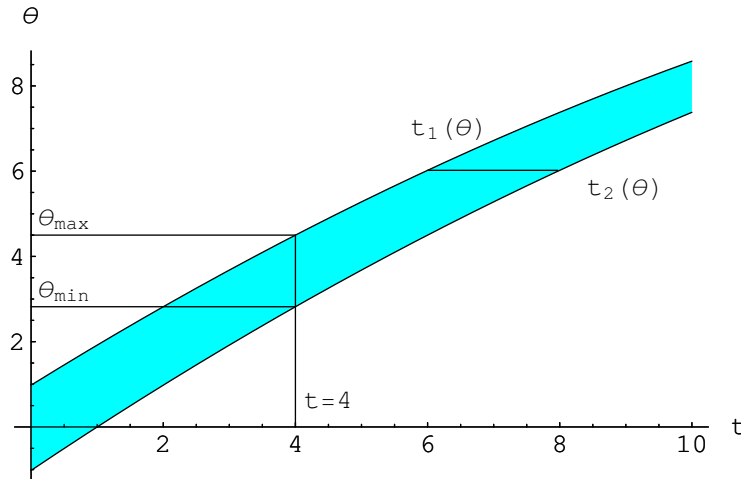


Fig. 13.1. Confidence belt. The shaded area is the confidence belt, consisting of the probability intervals $[t_1(\theta), t_2(\theta)]$ for the estimator t . The observation $t = 4$ leads to the confidence interval $[\theta_{\min}, \theta_{\max}]$.

The one-dimensional case is illustrated in Fig. 13.1. The pair of curves $t = t_1(\theta)$, $t = t_2(\theta)$ in the (t, θ) -plane comprise the so-called *confidence belt*. To the measurement $\hat{t} = 4$ then corresponds the confidence interval $[\theta_{\min}, \theta_{\max}]$ obtained by inverting the relations $t_{1,2}(\theta_{\max, \min}) = \hat{t}$, i.e. the section of the straight line $t = \hat{t}$ parallel to the θ axis.

The construction shown in Fig. 13.1 is not always feasible: It has to be assumed that $t_{1,2}(\theta)$ are monotone functions. If the curve $t_1(\theta)$ has a maximum say at $\theta = \theta_0$, then the relation $t_1(\theta) = \hat{t}$ cannot always be inverted: For $\hat{t} > t_1(\theta_0)$ the confidence belt *degenerates* into a region bounded from below, while for $\hat{t} < t_1(\theta_0)$ there is no unique solution. In the first case one usually declares a lower confidence bound as an infinite interval bounded from below. In the second case one could construct a set of disconnected intervals, some of which may be excluded by other arguments.

The construction of the confidence contour in the two-parameter case is illustrated in Fig. 13.2 where for simplicity the parameter and the observation space are chosen such that they coincide. For each point θ_1, θ_2 in the parameter space we fix a *probability contour* which contains a measurement of the parameters with probability C . Those parameter points with probability contours passing through the actual measurement $\hat{\theta}_1, \hat{\theta}_2$ are located at the *confidence contour*. All parameter pairs located inside the shaded area contain the measurement in their probability region.

Frequentist statistics avoids prior probabilities. This feature, while desirable in general, can have negative consequences if prior information exists. This is the case if the parameter space is constrained by mathematical or physical conditions. In frequentist statistics it is not possible to exclude un-physical parameter values without introducing additional complications. Thus, for instance, a measurement could lead for a mass to a 90% confidence interval which is situated completely in the negative region, or for an angle to a complex angular region. The problem is mitigated somewhat by a newer method [78], but not without introducing other complications [79], [80].

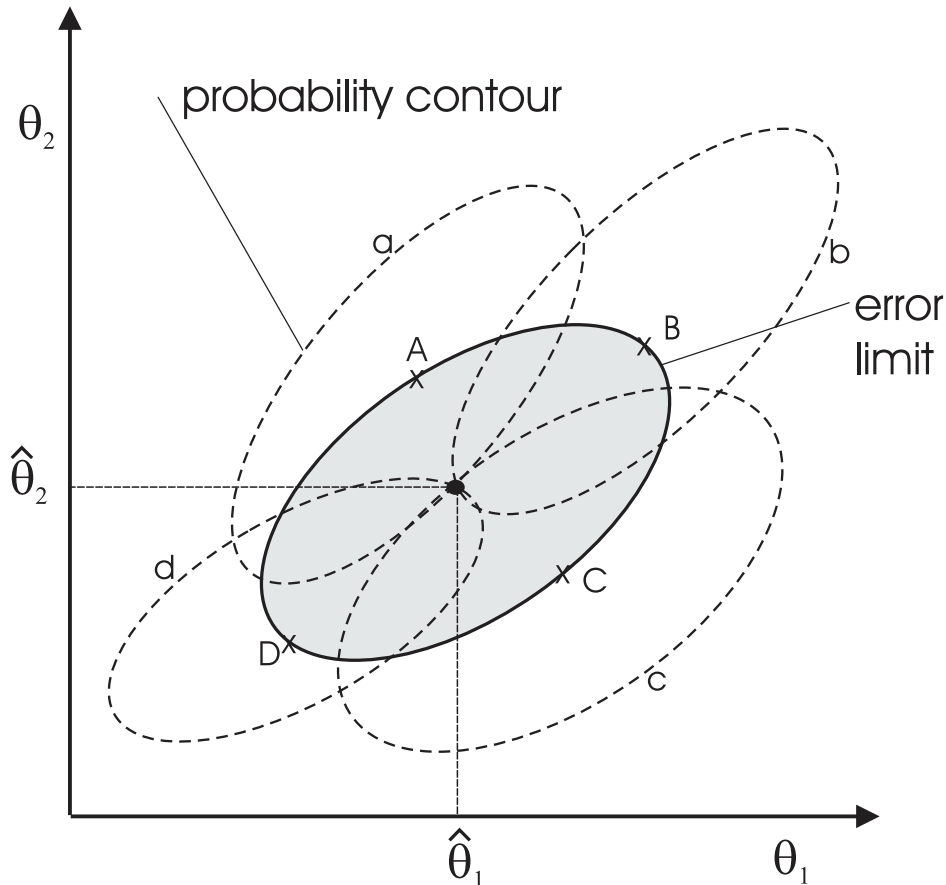


Fig. 13.2. Confidence interval. The shaded area is the confidence region for the two-dimensional measurement $(\hat{\theta}_1, \hat{\theta}_2)$. The dashed curves indicate probability regions associated to the locations denoted by capital letters.

13.6 Comparison of Different Inference Methods

13.6.1 A Few Examples

Before we compare the different statistical philosophies let us look at a few examples.

Example 150. Coverage: performance of magnets

A company produces magnets which have to satisfy the specified field strength within certain tolerances. The various measurements performed by the company are fed into a fitting procedure producing 99% confidence intervals which are used to accept (if they are inside the tolerances) or reject the product before sending it off. The client is able to repeat the measurement with high precision and accepts only magnets within the agreed specification. To calculate the price the company must rely on the condition that the confidence interval in fact covers the true value with the presumed confidence level.

Example 151. Bias in the measurements for a mass determination

The mass of a particle is determined from the momenta and the angular configuration of its decay products. The mean value of the masses from many events is reported. The momenta of the charged particles are measured by means of a spectrometer consisting of a magnet and tracking chambers. In this configuration, the χ^2 fit of the absolute values of track momenta and consequently also the mass estimates are biased. This bias, which can be shown to be positive, propagates into the mean value. Here a bias in the momentum fit has to be corrected for, because it would lead to a systematic shift of the resulting average of the mass values.

Example 152. Inference with known prior

We repeat an example presented in Sect. 6.2.2. In the reconstruction of a specific, very interesting event, for instance providing experimental evidence for a new particle, we have to infer the distance θ between the production and decay vertices of an unstable particle produced in the reaction. From its momentum and its known mean life we calculate its expected decay length λ . The prior density for the actual decay length θ is $\pi(\theta) = \exp(-\theta/\lambda)/\lambda$. The experimental distance measurement which follows a Gaussian with standard deviation s yields d . According to (6.2.2), the p.d.f. for the actual distance is given by

$$f(\theta|d) = \frac{e^{-(d-\theta)^2/(2s^2)}e^{-\theta/\lambda}}{\int_0^\infty e^{-(d-\theta)^2/(2s^2)}e^{-\theta/\lambda}d\theta}.$$

This is an ideal situation. We can determine the mean value and the standard deviation or the mode of the θ distribution and an asymmetric error interval with well defined probability content, for instance 68.3%. The confidence level is of no interest and due to the application of the prior the estimate of θ is biased, but this is irrelevant.

Example 153. Bias introduced by a prior

We now modify and extend our example. Instead of the decay length we discuss the lifetime of the particle. The reasoning is the same, we can apply the prior and determine an estimate and an error interval. We now study N decays, to improve our knowledge of the mean lifetime τ of the particle species. For each individual decay we use a prior with an estimate of τ as known from previous experiments, determine each time the lifetime $\hat{t}_i$ and the mean value $\bar{t} = \sum \hat{t}_i/N$ from all measurements. Even though the individual time estimates are improved by applying the prior the average $\bar{t}$ is a very bad estimate of τ because the $\hat{t}_i$ are biased towards low values and consequently also their mean value is shifted. (Remark that in this and in the third example we have two types of parameters which we have to distinguish. We discuss the effect of a bias afflicting the primary parameter set, i.e. λ , respectively τ).

Example 154. Comparing predictions with strongly differing accuracies: Earth quake

Two theories H_1, H_2 predict the time θ of an earth quake. The predictions differ in the expected values as well as in the size of the Gaussian errors:

$$\begin{aligned} H_1 : \theta_1 &= (7.50 \pm 2.25) \text{ h} , \\ H_2 : \theta_2 &= (50 \pm 100) \text{ h} . \end{aligned}$$

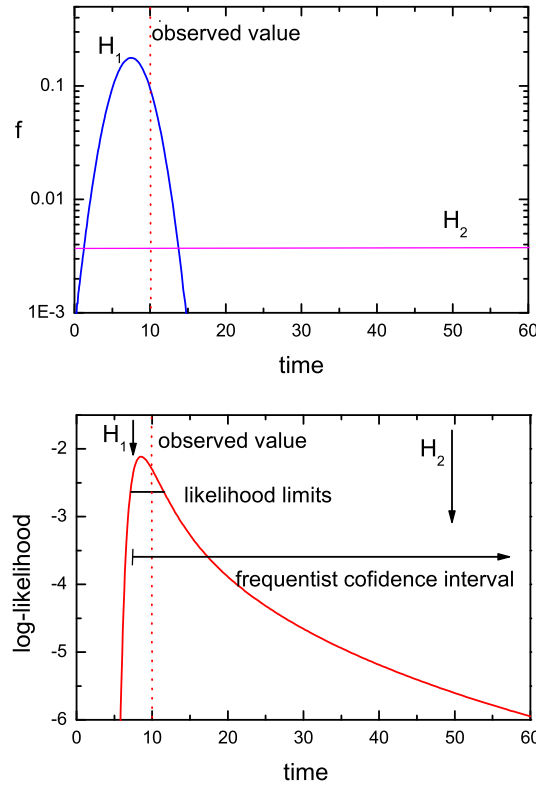


Fig. 13.3. Two hypotheses compared to an observation. The likelihood ratio supports hypothesis 1 while the distance in units of st.dev. supports hypothesis 2.

To keep the discussion simple, we do not exclude negative times t . The earthquake then takes place at time $t = 10$ h. In Fig. 13.3 are shown both hypothetical distributions in logarithmic form together with the actually observed time. The first prediction H_1 differs by more than one standard deviation from the observation, prediction H_2 by less than one standard deviation. Is then H_2 the more probable theory? Well, we cannot attribute probabilities to the theories but the likelihood ratio R which here has the value $R = 26$, strongly supports hypothesis H_1 . We could, however, also consider both hypotheses as special cases of a third general theory with the parametrization

$$f(t) = \frac{25}{\sqrt{2\pi}\theta^2} \exp \left[-\frac{625(t - \theta)^2}{2\theta^4} \right]$$

and now try to infer the parameter θ and its error interval. The observation produces the likelihood function shown in the lower part of Fig. 13.3. The usual likelihood ratio interval contains the parameter θ_1 and excludes θ_2 while the frequentist standard confidence interval $[7.66, \infty]$ would lead to the reverse conclusion which contradicts the likelihood ratio result and also our intuitive conclusions.

The presented examples indicate that depending on the kind of problem, different statistical methods are to be applied.

13.6.2 The Frequentist approach

The frequentist approach emphasizes efficiency, i.e. minimal variance combined with unbiasedness, and coverage. These quantities are defined through *the expected fluctuations of the estimated parameter of interest given its true value*. To compute these quantities we need to know the full p.d.f.. Variance and bias are related to point estimation. A bias has to be avoided whenever we average over several estimates like in the second and the fourth example. Frequentist interval estimation guarantees coverage⁵. A producer of technical items has to guarantee that a certain fraction of a sample fulfils given tolerances. He will choose, for example, a 99 % confidence interval for the lifetime of a bulb and then be sure that complaints will occur only in 1 % of the cases. Insurance companies want to estimate their risk and thus have to know, how frequently damage claims will occur.

Common to frequentist parameter inference (examples 1, 2 and 4) is that we are interested in the *properties of a set of parameter values*. The parameters are associated to many different objects, events or accidents, e.g. field strengths of various magnets, momenta of different events, or individual lifetimes. Here coverage and unbiasedness are essential, and efficiency is an important quality. As seen in the fourth example, the application of prior information – even when it is known exactly – would be destructive. Physicists usually impose transformation invariance to important parameters (The estimates of the lifetime $\hat{\tau}$ and the decay rate $\hat{\gamma}$ of a particle should satisfy $\hat{\gamma} = 1/\hat{\tau}$ but only one of these two parameters can be unbiased.) but in many situations the fact that bias and efficiency are not invariant under parameter transformations does not matter. In a business contract in the bulb example an agreement would be on the lifetime and the decay rate would be of no interest. The combination of the results from different measurements is difficult but mostly of minor interest.

13.6.3 The Bayesian Approach

The Bayesian statistics defines probability or credibility intervals. The interest is directed towards *the true value given the observed data*. As the probability of data that are not observed is irrelevant, the p.d.f. is not needed, the likelihood principle applies, only the prior and the likelihood function are relevant. The Bayesian approach is justified if we are interested in *a constant of nature, a particle mass, a coupling constant or in a parameter describing a unique event* like in examples three and five. In these situations we have to associate to the measurement an error interval in a consistent way. In the Bayesian approach the interval has a well defined probability content. Point and interval estimation cannot be treated independently. Coverage and bias are of no importance – in fact it does not make much sense to state that a certain fraction of physical constants are covered by their error intervals and it is of no use to know that out of 10 measurements of a particle mass one has to expect that about 7 contain the true value within their error intervals. Systematic errors and nuisance parameters for which no p.d.f. is available can only be treated in the Bayesian framework.

The drawback of the Bayesian method is the need to invent a prior probability. In example three the prior is known but this is one of the rare cases. In the fifth example,

⁵In the frequentist statistics point and interval estimation are unrelated.

like in many other situations, a uniform prior would be acceptable to most scientists and then the Bayesian interval would coincide with a likelihood ratio interval.

13.6.4 The Likelihood Ratio Approach

To avoid the introduction of prior probabilities, physicists are usually satisfied with the information contained in the likelihood function. In most cases the MLE and the likelihood ratio error interval are sufficient to summarize the result. Contrary to the frequentist confidence interval this concept is compatible with the maximum likelihood point estimation as well as with the likelihood ratio comparison of discrete hypotheses and allows to combine results in a consistent way. Parameter transformation invariance holds. However, there is no coverage guarantee and an interpretation in terms of probability is possible only for small error intervals, where prior densities can be assumed to be constant within the accuracy of the measurement.

13.6.5 Conclusion

The choice of the statistical method has to be adapted to the concrete application. The frequentist reasoning is relevant in rare situations like event selection, where coverage could be of some importance or when secondary statistics is performed with estimated parameters. In some situations Bayesian tools are required to proceed to sensible results. In all other cases the presentation of the likelihood function, or, as a summary of it, the MLE and a likelihood ratio interval are the best choice to document the inference result.

13.7 P-Values for EDF-Statistics

The formulas reviewed here are taken from the book of D'Agostino and Stephens [81] and generalized to include the case of the two-sample comparison.

Calculation of the Test Statistics

The calculation of the supremum statistics D and of $V = D_+ + D_-$ is simple enough, so we will skip a further discussion.

The quadratic statistics W^2 , U^2 and A^2 are calculated after a probability integral transformation (PIT). The PIT transforms the expected theoretical distribution of x into a uniform distribution. The new variate z is found from the relation $z = F(x)$, whereby F is the integral distribution function of x .

With the transformed observations z_i , ordered according to increasing values, we get for W^2 , U^2 and A^2 :

$$W^2 = \frac{1}{12N} + \sum_{i=1}^N \left(z_i - \frac{2i-1}{2N} \right)^2, \quad (13.23)$$

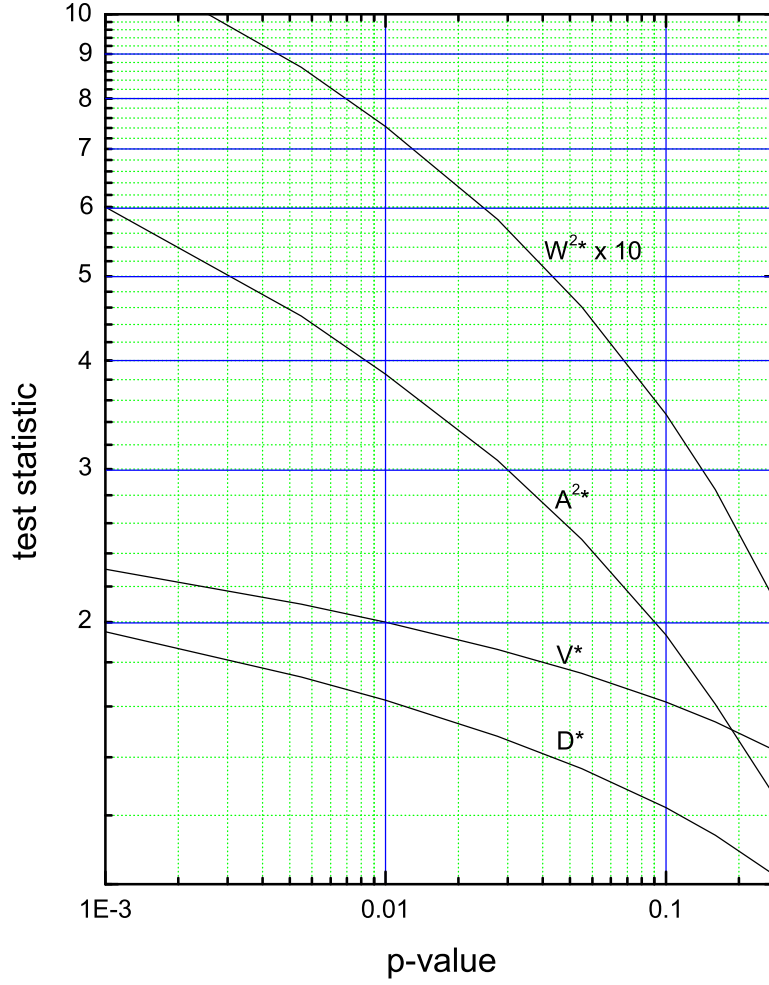


Fig. 13.4. P-values for empirical test statistics.

$$U^2 = \frac{\sum_{i=2}^N (z_i - z_{i-1})^2}{\sum_{i=1}^N z_i^2},$$

$$A^2 = -N + \sum_{i=1}^{N-1} (z_i - 1) (\ln z_i + \ln(1 - z_{N+1-i})) . \quad (13.24)$$

If we know the distribution function not analytically but only from a Monte-Carlo simulation, the z -value for an observation x is approximately $z \approx (\text{number of Monte-Carlo observations with } x_{MC} < x) / (\text{total number of Monte Carlo observations})$. (Somewhat more accurate is an interpolation). For the comparison with a simulated distribution is N to be taken as the equivalent number of observations

$$\frac{1}{N} = \frac{1}{N_{\text{exp}}} + \frac{1}{N_{MC}} .$$

Here N_{exp} and N_{MC} are the experimental respectively the simulated sample sizes.

Calculation of p-Values

After normalizing the test variables with appropriate powers of N they follow p.d.f.s which are independent of N . The test statistics' D^* , W^{2*} , A^{2*} modified in this way are defined by the following empirical relations

$$D^* = D_{\max}(\sqrt{N} + 0.12 + \frac{0.11}{\sqrt{N}}) , \quad (13.25)$$

$$W^{2*} = (W^2 - \frac{0.4}{N} + \frac{0.6}{N^2})(1.0 + \frac{1.0}{N}) , \quad (13.26)$$

$$A^{2*} = A^2 . \quad (13.27)$$

The relation between these modified statistics and the p-values is given in Fig. 13.7.

13.8 Comparison of two Histograms, Goodness-of-Fit and Parameter Estimation

In the main text we have treated goodness-of-fit and parameter estimation from a comparison of two histograms in the simple situation where the statistical errors of one of the histograms (generated by Monte Carlo simulation) was negligible compared to the uncertainties of the other histogram. Here we take the errors of both histograms into account and also permit that the histogram bins contain weighted entries.

13.8.1 Comparison of two Poisson Numbers with Different Normalization

We compare $c_n n$ with $c_m m$ where the normalization constants c_n, c_m are known and n, m are Poisson distributed. The null hypothesis H_0 is that n is drawn from a distribution with mean λ/c_n and m from a distribution with mean λ/c_m . We form a χ^2 expression

$$\chi^2 = \frac{(c_n n - c_m m)^2}{\delta^2} \quad (13.28)$$

where the denominator δ^2 is the expected variance of the parenthesis in the numerator under the null hypothesis.

The log-likelihood of λ is

$$\ln L(\lambda) = n \ln \frac{\lambda}{c_n} - \frac{\lambda}{c_n} + m \ln \frac{\lambda}{c_m} - \frac{\lambda}{c_m} + \text{const.}$$

with the MLE

$$\hat{\lambda} = \frac{n + m}{1/c_n + 1/c_m} = c_n c_m \frac{n + m}{c_n + c_m} . \quad (13.29)$$

We compute δ^2 under the assumption that n is distributed according to a Poisson distribution with mean $\hat{n} = \hat{\lambda}/c_n$ and respectively, mean $\hat{m} = \hat{\lambda}/c_m$.

$$\begin{aligned}\delta^2 &= c_n^2 \hat{n} + c_m^2 \hat{m} \\ &= (c_n + c_m) \hat{\lambda} \\ &= c_n c_m (n + m)\end{aligned}$$

and inserting the result into (13.28), we obtain

$$\chi^2 = \frac{1}{c_n c_m} \frac{(c_n n - c_m m)^2}{n + m}. \quad (13.30)$$

Notice that the normalization constants are defined up to a common factor, only the relative normalization c_n/c_m is relevant.

13.8.2 Comparison of Weighted Sums

When we compare experimental data to a Monte Carlo simulation, the simulated events frequently are weighted. We generalize our result to the situation where both numbers n, m consist of a sum of weights. Now the equivalent numbers of events $\tilde{n}$ and $\tilde{m}$ are approximately Poisson distributed. We simply have to replace (13.30) by

$$\chi^2 = \frac{1}{\tilde{c}_n \tilde{c}_m} \frac{(\tilde{c}_n \tilde{n} - \tilde{c}_m \tilde{m})^2}{\tilde{n} + \tilde{m}} \quad (13.31)$$

where now $\tilde{c}_n, \tilde{c}_m$ are the relative normalization constants for the equivalent numbers of events. We summarize in short the relevant relations, assuming that $n = \sum v_k$ is composed of a sum of individual events with weights v_k and correspondingly, $m = \sum w_k$ and as before $c_n n$ is supposed to agree with $c_m m$ as before. As discussed in 3.6.3 we set

$$\tilde{n} = \frac{\left[\sum v_k\right]^2}{\sum v_k^2}, \quad \tilde{m} = \frac{\left[\sum w_k\right]^2}{\sum w_k^2}$$

and find with $\tilde{c}_n \tilde{n} = c_n n$, $\tilde{c}_m \tilde{m} = c_m m$:

$$\tilde{c}_n = c_n \frac{\sum v_k \sum v_k^2}{\left[\sum v_k\right]^2}, \quad \tilde{c}_m = c_m \frac{\sum w_k \sum w_k^2}{\left[\sum w_k\right]^2}.$$

13.8.3 χ^2 of Histograms

We have to evaluate the expression (13.31) for each bin and sum over all B bins

$$\chi^2 = \sum_{i=1}^B \left[\frac{1}{\tilde{c}_n \tilde{c}_m} \frac{(\tilde{c}_n \tilde{n} - \tilde{c}_m \tilde{m})^2}{\tilde{n} + \tilde{m}} \right]_i \quad (13.32)$$

where the prescription indicated by the index i means that all quantities in the bracket have to be evaluated for bin i . In case the entries are not weighted the tilde is obsolete. The c_n, c_m usually are overall normalization constants and equal for all bins of the corresponding histogram. If the histograms are normalized with respect to each other, we have $c_n \Sigma n_i = c_m \Sigma m_i$ and we can set $c_n = \Sigma m_i = M$ and $c_m = \Sigma n_i = N$.

χ^2 Goodness-of-Fit Test

This expression can be used for goodness-of-fit tests. In case the normalization constants are given externally, for instance through the luminosity, χ^2 follows approximately a χ^2 distribution of B degrees of freedom. Frequently the histograms are normalized with respect to each other. Then we have one degree of freedom less, i.e. $B - 1$. If P parameters have been adjusted in addition, then we have $B - P - 1$ degrees of freedom.

Likelihood Ratio Test

In Chap. 10, Sect. 10.3.4 we have introduced the likelihood ratio test for histograms. For a pair of Poisson numbers n, m the likelihood ratio is the ratio of the maximal likelihood under the condition that the two numbers are drawn from the same distribution to the unconditioned maximum of the likelihood for the observation of n . The corresponding difference of the logarithms is our test statistic V (see likelihood ratio test for histograms):

$$\begin{aligned} V &= n \ln \frac{\lambda}{c_n} - \frac{\lambda}{c_n} - \ln n! + m \ln \frac{\lambda}{c_m} - \frac{\lambda}{c_m} - \ln m! - [n \ln n - n - \ln n!] \\ &= n \ln \frac{\lambda}{c_n} - \frac{\lambda}{c_n} + m \ln \frac{\lambda}{c_m} - \frac{\lambda}{c_m} - \ln m! - n \ln n + n . \end{aligned}$$

We now turn to weighted events and perform the same replacements as above:

$$V = \tilde{n} \ln \frac{\tilde{\lambda}}{\tilde{c}_n} - \frac{\tilde{\lambda}}{\tilde{c}_n} + \tilde{m} \ln \frac{\tilde{\lambda}}{\tilde{c}_m} - \frac{\tilde{\lambda}}{\tilde{c}_m} - \ln \tilde{m}! - \tilde{n} \ln \tilde{n} + \tilde{n} .$$

Here the parameter $\tilde{\lambda}$ is the MLE corresponding to (13.29) for weighted events:

$$\tilde{\lambda} = \tilde{c}_n \tilde{c}_m \frac{\tilde{n} + \tilde{m}}{\tilde{c}_n + \tilde{c}_m} .$$

The test statistic of the full histogram is the sum of the contributions from all bins:

$$V = \sum_{i=1}^B \left[\tilde{n} \ln \frac{\tilde{\lambda}}{\tilde{c}_n} - \frac{\tilde{\lambda}}{\tilde{c}_n} + \tilde{m} \ln \frac{\tilde{\lambda}}{\tilde{c}_m} - \frac{\tilde{\lambda}}{\tilde{c}_m} - \ln \tilde{m}! - \tilde{n} \ln \tilde{n} + \tilde{n} \right]_i .$$

The distribution of the test statistic under H_0 for large event number follows approximately a χ^2 distribution of $B - 1$ degrees of freedom, for small numbers it has to be obtained by simulation.

13.8.4 Parameter Estimation

When we compare experimental data to a parameter dependent Monte Carlo simulation, one of the histograms depends on the parameter, e.g. $m(\theta)$ and the comparison is used to determine the parameter. During the fitting procedure, the parameter is modified and this implies a change of the weights of the Monte Carlo events.

χ^2 Parameter Estimation

In principle all weights $[w_k]_i$ of all bins might depend on θ . We have to minimize (13.32) and recompute all weights for each minimization step. The standard situation is however, that the experimental events are not weighted and that the weights of the Monte Carlo events are equal for all events of the same bin. Then (13.32) simplifies with $\tilde{n}_i = n_i$, $\tilde{c}_n = c_n$, $\tilde{c}_m = c_m w$ and $\tilde{m}_i$ is just the number of unweighted Monte Carlo events in bin i :

$$\chi^2 = \sum_{i=1}^B \left[\frac{1}{c_n c_m w} \frac{(c_n n - c_m w \tilde{m})^2}{(n + \tilde{m})} \right]_i$$

with

$$c_n = M = \sum_{i=1}^B w_i \tilde{m}_i, \quad c_m = N.$$

Maximum Likelihood Method

We generalize the binned likelihood fit to the situation with weighted events. We have to establish the likelihood to observe $\tilde{n}, \tilde{m}$ equivalent events given $\hat{\lambda}$ from (13.29). We start with a single bin. The probability to observe $\tilde{n}$ ($\tilde{m}$) is given by the Poisson distribution with mean $\tilde{\lambda}/c_n$, ($\tilde{\lambda}/c_m$). The corresponding likelihood function is

$$\ln L = \tilde{n} \ln(\tilde{\lambda}/\tilde{c}_n) - (\tilde{\lambda}/\tilde{c}_n) + \tilde{m} \ln(\tilde{\lambda}/\tilde{c}_m) - (\tilde{\lambda}/\tilde{c}_m) - \ln \tilde{m}! + \text{const.}$$

We omit the constant term $\ln n!$, sum over all bins, and obtain

$$\ln L = \sum_{i=1}^B \left[\tilde{n} \ln \frac{\tilde{\lambda}}{\tilde{c}_n} - \frac{\tilde{\lambda}}{\tilde{c}_n} + \tilde{m} \ln \frac{\tilde{\lambda}}{\tilde{c}_m} - \frac{\tilde{\lambda}}{\tilde{c}_m} - \ln \tilde{m}! \right]_i$$

which we have to maximize with respect to the parameters entering into the Monte Carlo simulation hidden in $\tilde{\lambda}, \tilde{c}_m, \tilde{m}$.

For the standard case which we have treated in the χ^2 fit, we can apply here the same simplifications:

$$\ln L = \sum_{i=1}^B \left[n \ln \frac{\hat{\lambda}}{c_n} - \frac{\hat{\lambda}}{c_n} + \tilde{m} \ln \frac{\hat{\lambda}}{c_m w} - \frac{\hat{\lambda}}{c_m w} - \ln \tilde{m}! \right]_i.$$

13.9 Extremum Search

If we apply the maximum-likelihood method for parameter estimation, we have to find the maximum of a function in the parameter space. This is, as a rule, not possible without numerical tools. An analogous problem is posed by the method of least squares. Minimum and maximum search are principally not different problems, since we can invert the sign of the function. We restrict ourselves to the minimum search.

Before we engage off-the-shelf computer programs, we should obtain some rough idea of their function. The best way in most cases is a graphical presentation. It is not important for the user to know the details of the programs, but some knowledge of their underlying principles is helpful.

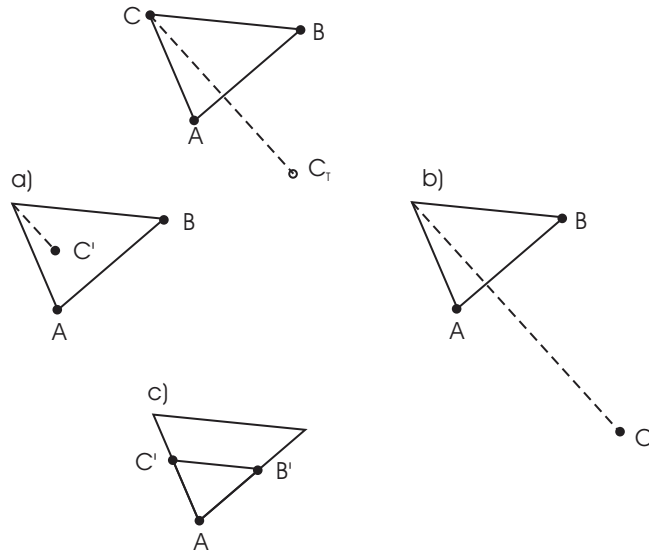


Fig. 13.5. Simplex algorithm.

13.9.1 Monte Carlo Search

In order to obtain a rough impression of the function to be investigated, and of the approximate location of its minimum, we may sample the parameter stochastically. A starting region has to be selected. Usual programs will then further restrict the parameter space in dependence of the search results. An advantage of this method is that the probability to end up in a relative minimum is rather small. In the literature this rather simple and not very effective method is sometimes sold under the somewhat pretentious name *genetic algorithm*. Since it is fairly inefficient, it should be used only for the first step of a minimization procedure.

13.9.2 The Simplex Algorithm

Simplex is a quite slow but robust algorithm, as it needs no derivatives. For an n -dimensional parameter space $n + 1$ starting points are selected, and for each point the function values calculated. The point which delivers the largest function value is rejected and replaced by a new point. How this point is found is demonstrated in two dimensions.

Fig. 13.5 shows in the upper picture three points. let us assume that A has the lowest function value and point C the largest $f(x_C, y_C)$. We want to eliminate C and to replace it by a superior point C' . We take its mirror image with respect to the center of gravity of points A, B and obtain the test point C_T . If $f(x_{C_T}, y_{C_T}) < f(x_C, y_C)$ we did find a better point, thus we replace C by C_T and continue with the new triplet. In the opposite case we double the step width (13.5b) with respect to the center of gravity and find C' . Again we accept C' , if it is superior to C . If not, we compare it with the test point C_T and if $f(x_{C_T}, y_{C_T}) < f(x_{C'}, y_{C'})$ holds, the step width is halved and reversed in direction (13.5a). The point C' now moves to the inner region of the simplex triangle. If it is superior to C it replaces C as above. In all other cases the original simplex is shrunk by a factor two in the direction of

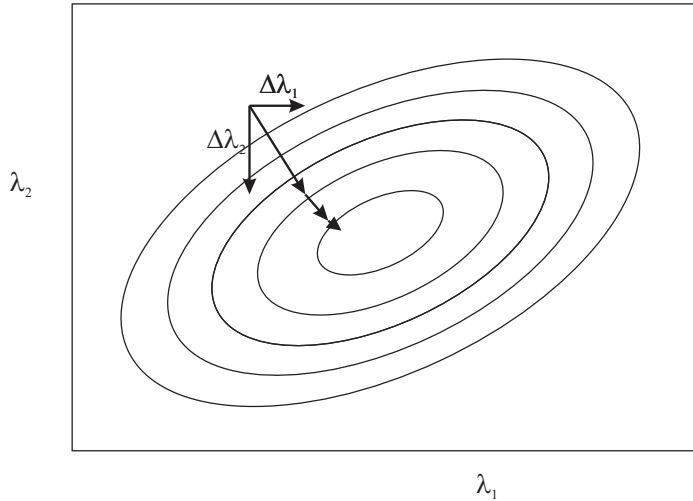


Fig. 13.6. Method of steepest descent.

the best point A (13.5c). In each case one of the four configurations is chosen and the iteration continued.

13.9.3 The Parabola Method

Again we begin with starting points in parameter space. In the one-dimensional case we choose 3 points and put a parabola through them. The point with the largest function value is dropped and replaced by the minimum of the parabola and a new parabola is computed. In the general situation of an n -dimensional space, $2n + 1$ points are selected which determine a paraboloid. Again the worst point is replaced by the vertex of the paraboloid. The iteration converges for functions which are convex in the search region.

13.9.4 The Method of Steepest Descent

A traveler, walking in a landscape unknown to him, who wants to find a lake, will chose a direction down-hill perpendicular to the curves of equal height (if there are no insurmountable obstacles). The same method is applied when searching for a minimum by the method of steepest descent. We consider this local method in more detail, as in some cases it has to be programmed by the user himself.

We start from a certain point λ_0 in the parameter space, calculate the gradient $\nabla_{\lambda} f(\lambda)$ of the function $f(\lambda)$ which we want to minimize and move by $\Delta\lambda$ downhill.

$$\Delta\lambda = -\alpha \nabla_{\lambda} f(\lambda) .$$

The step length depends on the learning constant α which is chosen by the user. This process is iterated until the function remains essentially constant. The method is sketched in Fig. 13.6.

The method of steepest descent has advantages as well as drawbacks:

- The decisive advantage is its simplicity which permits to handle a large number of parameters at the same time. If convenient, for the calculation of the gradient rough approximations can be used. Important is only that the function decreases with each step. As opposed to the simplex and parabola methods its complexity increases only linear with the number of parameters. Therefore problems with huge parameter sets can be handled.
- It is possible to evaluate a sample sequentially, element by element, which is especially useful for the back-propagation algorithm of neural networks.
- Unsatisfactory is that the learning constant is not dimensionless. In other words, the method is not independent of the parameter scales. For a space-time parameter set the gradient path will depend, for instance, on the choice whether to measure the parameters in meters or millimeters, respectively hours or seconds.
- In regions with flat parameter space the convergency is slow. In a narrow valley oscillations may appear. For too large values of α oscillations will make exact minimizing difficult.

The last mentioned problems can be reduced by various measures where the step length and direction partially depend on results of previous steps. When the function change is small and similar in successive steps α is increased. Oscillations in a valley can be avoided by adding to the gradient in step i a fraction of the gradient of step $i - 1$:

$$\Delta \lambda_i = \alpha (\nabla_{\lambda} f(\lambda_i) + 0.5 \nabla_{\lambda} f(\lambda_{i-1})) .$$

Oscillations near the minimum are easily recognized and removed by decreasing α .

The method of steepest descent is applied in ANN and useful in the updating alignment of tracking detectors [82].

13.9.5 Stochastic Elements in Minimum Search

A physical system which is cooled down to the absolute zero point will principally occupy an energetic minimum. When cooled down fast it may, though, be captured in a local (relative) minimum. An example is a particle in a potential wall. For somewhat higher temperature it may leave the local minimum, thanks to the statistical energy distribution (Fig. 13.7). This is used for instance in the stimulated annealing of defects in solid matter.

This principle can be used for minimum search in general. A step in the wrong direction, where the function increases by Δf , can be accepted, when using the method of steepest descent, e.g. with a probability

$$P(\Delta f) = \frac{1}{1 + e^{\Delta f/T}} .$$

The scale factor T ("temperature") steers the strength of the effect. It has been shown that for successively decreasing T the absolute minimum will be reached.

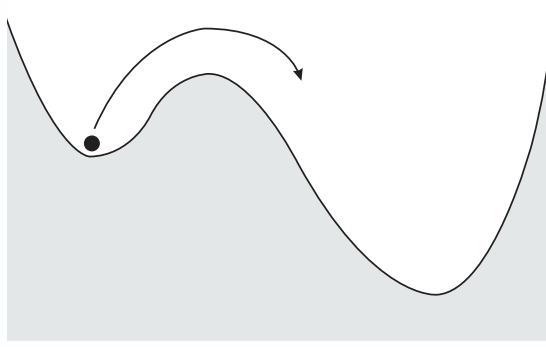


Fig. 13.7. Stochastic annealing. A local minimum can be left with a certain probability.

13.10 Linear Regression with Constraints

We consider N measurements $\mathbf{y}$ at known locations $\mathbf{x}$, with a $N \times N$ covariance matrix $\mathbf{C}_N$ and a corresponding weight matrix $\mathbf{V}_N = \mathbf{C}_N^{-1}$. (We indicate the dimensions of quadratic matrices with an index).

In the linear model the measurements are described by $P < N$ parameters $\boldsymbol{\theta}$ in form of linear relations

$$\mathbf{y} = \mathbf{T}(\mathbf{x})\boldsymbol{\theta} ,$$

with the rectangular $N \times P$ “design” matrix $\mathbf{T}$.

In 7.2.3 we have found that the corresponding χ^2 expression is minimized by

$$\hat{\boldsymbol{\theta}} = (\mathbf{T}^T \mathbf{V}_N \mathbf{T})^{-1} \mathbf{T}^T \mathbf{V}_N \mathbf{y} .$$

We now include constraints between the parameters, expressed by $K < P$ linear relations:

$$\mathbf{H}\boldsymbol{\theta} = \boldsymbol{\rho} ,$$

with $\mathbf{H}(\mathbf{x})$ a given rectangular $K \times P$ matrix and $\boldsymbol{\rho}$ a K -dimensional vector.

This problem is solved by introducing K Lagrange multipliers $\boldsymbol{\alpha}$ and looking for a *stationary* point of the lagrangian

$$\Lambda = (\mathbf{y} - \mathbf{T}\boldsymbol{\theta})^T \mathbf{V}_N (\mathbf{y} - \mathbf{T}\boldsymbol{\theta}) + 2\boldsymbol{\alpha}^T (\mathbf{H}\boldsymbol{\theta} - \boldsymbol{\rho}) .$$

Differentiating with respect to $\boldsymbol{\theta}$ and $\boldsymbol{\alpha}$ gives the normal equations

$$\mathbf{T}^T \mathbf{V}_N \mathbf{T} \boldsymbol{\theta} + \mathbf{H} \boldsymbol{\alpha} = \mathbf{T}^T \mathbf{V}_N \mathbf{y} , \quad (13.33)$$

$$\mathbf{H} \boldsymbol{\theta} = \boldsymbol{\rho} \quad (13.34)$$

to be solved for $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\alpha}}$. Note that Λ is minimized only with respect to $\boldsymbol{\theta}$, but maximized with respect to $\boldsymbol{\alpha}$: The stationary point is a saddle point, which complicates a direct extremum search. Solving (13.33) for $\boldsymbol{\theta}$ and inserting it into (13.34), we find

$$\hat{\boldsymbol{\alpha}} = \mathbf{C}_K^{-1} (\mathbf{H} \mathbf{C}_P \mathbf{T}^T \mathbf{V}_N \mathbf{y} - \boldsymbol{\rho})$$

and, re-inserting the estimates into (13.33), we obtain

$$\hat{\boldsymbol{\theta}} = \mathbf{C}_P [\mathbf{T}^T \mathbf{V}_N \mathbf{y} - \mathbf{H}^T \mathbf{C}_K^{-1} (\mathbf{H} \mathbf{C}_P \mathbf{T}^T \mathbf{V}_N \mathbf{y} - \boldsymbol{\rho})],$$

where the abbreviations $\mathbf{C}_P = (\mathbf{T}^T \mathbf{V}_N \mathbf{T})^{-1}$, $\mathbf{C}_K = \mathbf{H} \mathbf{C}_P \mathbf{H}^T$ have been used.

The covariance matrix is found from linear error propagation, after a somewhat lengthy calculation, as

$$\text{cov}(\hat{\boldsymbol{\theta}}) = \mathbf{D} \mathbf{C}_N \mathbf{D}^T = (\mathbf{I}_P - \mathbf{C}_P \mathbf{H}^T \mathbf{C}_K^{-1} \mathbf{H}) \mathbf{C}_P,$$

where

$$\mathbf{D} = \mathbf{C}_P (\mathbf{I}_P - \mathbf{H}^T \mathbf{C}_K^{-1} \mathbf{H} \mathbf{C}_P) \mathbf{T}^T \mathbf{V}_N$$

has been used. The covariance matrix is symmetric positive definite. Without constraints, it equals $\mathbf{C}_P$, the negative term is absent. Of course, the introduction of constraints reduces the errors and thus improves the parameter estimation.

13.11 Formulas Related to the Polynomial Approximation

Errors of the Expansion Coefficients

In Sect. 11.2.2 we have discussed the approximation of measurements by orthogonal polynomials and given the following formula for the error of the expansion coefficients a_k ,

$$\text{var}(a_k) = 1 / \sum_{\nu=1}^N \frac{1}{\delta_\nu^2}$$

which is valid for all $k = 1, \dots, K$. Thus all errors are equal to the error of the weighted mean of the measurements y_ν .

Proof: from linear error propagation we have, for independent measurements y_ν ,

$$\begin{aligned} \text{var}(a_k) &= \text{var} \left(\sum_{\nu} w_{\nu} u_k(x_{\nu}) y_{\nu} \right) \\ &= \sum_{\nu} w_{\nu}^2 (u_k(x_{\nu}))^2 \delta_{\nu}^2 \\ &= \sum_{\nu} w_{\nu} u_k^2(x_{\nu}) / \sum_{\nu} \frac{1}{\delta_{\nu}^2} \\ &= 1 / \sum_{\nu} \frac{1}{\delta_{\nu}^2}, \end{aligned}$$

where in the third step we used the definition of the weights, and in the last step the normalization of the polynomials u_k .

Polynomials for Data with Uniform Errors

If the errors $\delta_1, \dots, \delta_N$ are uniform, the weights become equal to $1/N$, and for certain patterns of the locations $x_1, \dots, x_N$, for instance for an equidistant distribution, the orthogonalized polynomials $u_k(x)$ can be calculated. They are given in mathematical handbooks, for instance in Ref. [94]. Although the general expression is quite involved, we reproduce it here for the convenience of the reader. For

x defined in the domain $[-1, 1]$ (eventually after some linear transformation and shift), and $N = 2M + 1$ equidistant (with distance $\Delta x = 1/M$) measured points $x_\nu = \nu/M$, $\nu = 0, \pm 1, \dots, \pm M$, they are given by

$$u_k(x) = \left(\frac{(2M+1)(2k+1)[(2M)!]^2}{(2M+k+1)!(2M-k)!} \right)^{1/2} \sum_{i=0}^k (-1)^{i+k} \frac{(i+k)^{[2i]}(M+t)^{[i]}}{(i!)^2(2M)^{[i]}} ,$$

for $k = 0, 1, 2, \dots, 2M$, where we used the notation $t = x/\Delta x = xM$ and the definitions

$$\begin{aligned} z^{[i]} &= z(z-1)(z-2)\cdots(z-i+1) \\ z^{[0]} &= 1, \quad z \geq 0, \quad 0^{[i]} = 0, \quad i = 1, 2, \dots \end{aligned}$$

13.12 Formulas for B-Spline Functions

13.12.1 Linear B-Splines

Linear B-splines cover an interval $2b$ and overlap with both neighbors:

$$\begin{aligned} B(x; x_0) &= 2 \frac{x - x_0 - b}{b} \quad \text{for } x_0 - b \leq x \leq x_0 , \\ &= 2 \frac{-x - x_0 + b}{b} \quad \text{for } x_0 \leq x \leq x_0 + b , \\ &= 0 \quad \text{else .} \end{aligned}$$

They are normalized to unit area. Since the central values are equidistant, we fix them by the lower limit $x_{\min}$ of the x -interval and count them as $x_0(k) = x_{\min} + kb$, with the index k running from $k_{\min} = 0$ to $k_{\max} = (x_{\max} - x_{\min})/b = K$.

At the borders only half of a spline is used.

Remark: The border splines are defined in the same way as the other splines. After the fit the part of the function outside of its original domain is ignored. In the literature the definition of the border splines is often different.

13.12.2 Quadratic B-Splines

The definition of quadratic splines is analogous:

$$\begin{aligned} B(x; x_0) &= \frac{1}{2b} \left(\frac{x - x_0 + 3/2b}{b} \right)^2 \quad \text{for } x_0 - 3b/2 \leq x \leq x_0 - b/2 , \\ &= \frac{1}{2b} \left[\frac{3}{2} - 2 \left(\frac{x - x_0}{b} \right)^2 \right] \quad \text{for } x_0 - b/2 \leq x \leq x_0 + b/2 , \\ &= \frac{1}{2b} \left(\frac{x - x_0 - 3/2b}{b} \right)^2 \quad \text{for } x_0 + b/2 \leq x \leq x_0 + 3b/2 , \\ &= 0 \quad \text{else .} \end{aligned}$$

The supporting points $x_0 = x_{\min} + (k - 1/2)b$ lie now partly outside of the x -domain. The index k runs from 0 to $k_{\max} = (x_{\max} - x_{\min})/b + 2$. Thus, the number K of splines is by two higher than the number of intervals. The relations (11.13) and (11.12) are valid as before.

13.12.3 Cubic B-Splines

Cubic B-splines are defined as follows:

$$\begin{aligned}
 B(x; x_0) &= \frac{1}{6b} \left(2 + \frac{x - x_0}{b} \right)^3 \quad \text{for } x_0 - 2b \leq x \leq x_0 - b, \\
 &= \frac{1}{6b} \left[-3 \left(\frac{x - x_0}{b} \right)^3 - 6 \left(\frac{x - x_0}{b} \right)^2 + 4 \right] \quad \text{for } x_0 - b \leq x \leq x_0, \\
 &= \frac{1}{6b} \left[3 \left(\frac{x - x_0}{b} \right)^3 - 6 \left(\frac{x - x_0}{b} \right)^2 + 4 \right] \quad \text{for } x_0 \leq x \leq x_0 + b, \\
 &= \frac{1}{6b} \left(2 - \frac{x - x_0}{b} \right)^3 \quad \text{for } x_0 + b \leq x \leq x_0 + 2b, \\
 &= 0 \quad \text{else.}
 \end{aligned}$$

The shift of the center of the spline is performed as before: $x_0 = x_{\min} + (k - 1)b$. The index k runs from 0 to $k_{\max} = (x_{\max} - x_{\min})/b + 3$. The number $k_{\max} + 1$ of splines is equal to the number of intervals plus 3.

13.13 Support Vector Classifiers

Support vector machines are described some detail in Refs. [13, 71, 72, 73].

13.13.1 Linear Classifiers

Linear classifiers⁶ separate the two training samples by a hyperplane. Let us initially assume that in this way a complete separation is possible. Then an optimal hyperplane is the plane which divides the two samples with the largest margin. This is shown in Fig. 13.8. The hyperplane can be constructed in the following way: The shortest connection Δ between the convex hulls⁷ of the two non-overlapping classes determines the direction $\mathbf{w}/|\mathbf{w}|$ of the normal $\mathbf{w}$ of this plane which cuts the distance at its center. We represent the hyperplane in the form

$$\mathbf{w} \cdot \mathbf{x} + b = 0, \quad (13.35)$$

where b fixes its distance from the origin. Note that $\mathbf{w}$ is not normalized, a common factor in $\mathbf{w}$ and b does not change condition (13.35). Once we have found the hyperplane $\{\mathbf{w}, b\}$ which separates the two classes $y_i = \pm 1$ of the training sample $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$ we can use it to classify new input:

$$\hat{y} = f(\mathbf{x}) = \text{sign}(\mathbf{w} \cdot \mathbf{x} + b). \quad (13.36)$$

To find the optimal hyperplane which divides Δ into equal parts, we define the two marginal planes which touch the hulls:

⁶A linear classification scheme was already introduced in Sect. 11.4.1.

⁷The convex hull is the smallest polyhedron which contains all points and their connecting straight lines.

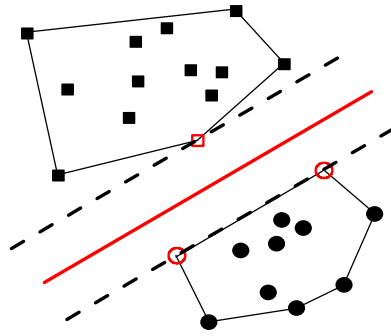


Fig. 13.8. The central hyperplane separates squares from circles. Shown are the convex hulls and the support vectors (open symbols).

$$\mathbf{w} \cdot \mathbf{x} + b = \pm 1 .$$

If $\mathbf{x}^+, \mathbf{x}^-$ are located at the two marginal hyperplanes, the following relations hold which also fix the norm of $\mathbf{w}$:

$$\mathbf{w} \cdot (\mathbf{x}^+ - \mathbf{x}^-) = 2 \Rightarrow \Delta = \frac{\mathbf{w}}{|\mathbf{w}|} \cdot (\mathbf{x}^+ - \mathbf{x}^-) = \frac{2}{|\mathbf{w}|} .$$

The optimal hyperplane is now found by solving a constrained quadratic optimization problem

$$|\mathbf{w}|^2 = \text{minimum} , \text{ subject to } y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 , i = 1, \dots, N .$$

For the solution, only the constraints with equals sign are relevant. The vectors corresponding to points on the marginal planes form the so-called active set and are called support vectors (see Fig. 13.8). The optimal solution can be written as

$$\mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i$$

with $\alpha_i > 0$ for the active set, else $\alpha_i = 0$, and furthermore $\sum \alpha_i y_i = 0$. The last condition ensures translation invariance: $\mathbf{w}(\mathbf{x}_i - \mathbf{a}) = \mathbf{w}(\mathbf{x}_i)$. Together with the active constraints, after substituting the above expression for $\mathbf{w}$, it provides just the required number of linear equations to fix α_i and b . Of course, the main problem is to find the active set. For realistic cases this requires the solution of a large quadratic optimization problem, subject to linear inequalities. For this purpose an extended literature as well as program libraries exist.

This picture can be generalized to the case of overlapping classes. Assuming that the optimal separation is still given by a hyperplane, the picture remains essentially the same, but the optimization process is substantially more complex. The standard way is to introduce so called *soft margin classifiers*. There some points on the wrong side of their marginal plane are tolerated, but with a certain penalty in the optimization process. It is chosen proportional to the sum of their distances or their square distance from their own territory. The proportionality constant is adjusted to the given problem.

13.13.2 General Kernel Classifiers

All quantities determining the linear classifier $\hat{y}$ (13.36) depend only on inner products of vectors of the input space. This concerns not only the dividing hyperplane, given by (13.35), but also the expressions for $\mathbf{w}$, b and the factors α_i associated to the support vectors. The inner product $\mathbf{x} \cdot \mathbf{x}'$ which is a bilinear symmetric scalar function of two vectors, is now replaced by another scalar function $K(\mathbf{x}, \mathbf{x}')$ of two vectors, the kernel, which need not be bilinear, but should also be symmetric, and is usually required to be positive definite. In this way a linear problem in an inner product space is mapped into a very non-linear problem in the original input space where the kernel is defined. We then are able to separate the classes by a hyperplane in the inner product space that may correspond to a very complicated hypersurface in the input space. This is the so-called *kernel trick*.

To illustrate how a non-linear surface can be mapped into a hyperplane, we consider a simple example. In order to work with a linear cut, i.e. with a dividing hyperplane, we transform our input variables $\mathbf{x}$ into new variables: $\mathbf{x} \rightarrow \mathbf{X}(\mathbf{x})$. For instance, if x_1, x_2, x_3 are momentum components and a cut in energy, $x_1^2 + x_2^2 + x_3^2 < r^2$, is to be applied, we could transform the momentum space into a space

$$\mathbf{X} = \{x_1^2, x_2^2, x_3^2, \dots\}.$$

where the cut corresponds to the hyperplane $X_1 + X_2 + X_3 = r^2$. Such a mapping can be realized by substituting the inner product by a kernel:

$$\mathbf{x} \cdot \mathbf{x}' \rightarrow K(\mathbf{x}, \mathbf{x}') = \mathbf{X}(\mathbf{x}) \cdot \mathbf{X}(\mathbf{x}').$$

In our example a kernel of the so-called monomial form is appropriate:

$$\begin{aligned} K(\mathbf{x}, \mathbf{x}') &= (\mathbf{x} \cdot \mathbf{x}')^d \text{ with } d = 2, \\ (\mathbf{x} \cdot \mathbf{x}')^2 &= (x_1x'_1 + x_2x'_2 + x_3x'_3)^2 = \mathbf{X}(\mathbf{x}) \cdot \mathbf{X}(\mathbf{x}') \end{aligned} \quad (13.37)$$

with

$$\mathbf{X}(\mathbf{x}) = \{x_1^2, x_2^2, x_3^2, \sqrt{2}x_1x_2, \sqrt{2}x_1x_3, \sqrt{2}x_2x_3\}.$$

The sphere $x_1^2 + x_2^2 + x_3^2 = r^2$ in x -space is mapped into the 5-dimensional hyperplane $X_1 + X_2 + X_3 = r^2$ in 6-dimensional X -space. (A kernel inducing instead of monomials of order d (13.37), polynomials of all orders, up to order d is $K(\mathbf{x}, \mathbf{x}') = (1 + \mathbf{x} \cdot \mathbf{x}')^d$.)

The most common kernel used for classification is the Gaussian (see Sect. 11.2.1):

$$K(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{(\mathbf{x} - \mathbf{x}')^2}{2s^2}\right).$$

It can be shown that it induces a mapping into a space of infinite dimensions [71] and that nevertheless the training vectors can in most cases be replaced by a relatively small number of support vectors. The only free parameter is the penalty constant which regulates the degree of overlap of the two classes. A high value leads to a very irregular shape of the hypersurface separating the training samples of the two classes to a high degree in the original space whereas for a low value its shape is much smoother and more minority observations are tolerated.

In practice, this mapping into the inner product space is not performed explicitly, in fact it is even rarely known. All calculations are performed in x -space, especially the determination of the support vectors and their weights α . The kernel trick merely

serves to prove that a classification with support vectors is feasible. The classification of new input then proceeds with the kernel K and the support vectors directly:

$$\hat{y} = \text{sign} \left(\sum_{y_i=+1} \alpha_i K(\mathbf{x}, \mathbf{x}_i) - \sum_{y_i=-1} \alpha_i K(\mathbf{x}, \mathbf{x}_i) \right).$$

The use of a relatively small number of support vectors (typically only about 5% of all α_i are different from zero) drastically reduces the storage requirement and the computing time for the classification. Remark that the result of the support vector classifier is not identical to that of the original kernel classifier but very similar.

13.14 The Bayes Factor

In Chap. 6 we have introduced the likelihood ratio to discriminate between simple hypotheses. For two composite hypotheses H_1 and H_2 with free parameters, in the Bayesian approach the simple ratio is to be replaced by the so-called Bayes factors.

Let us assume for a moment that H_1 applies. Then the actual parameters will follow a p.d.f. proportional to $L_1(\boldsymbol{\theta}_1|\mathbf{x})\pi_1(\boldsymbol{\theta}_1)$ where $L_1(\boldsymbol{\theta}_1|\mathbf{x})$ is the likelihood function and $\pi_1(\boldsymbol{\theta}_1)$ the prior density of the parameters. The same reasoning is valid for H_2 . The probability that H_1 (H_2) is true is proportional to the integral over the parameter space, $\int L_1(\boldsymbol{\theta}_1|\mathbf{x})\pi_1(\boldsymbol{\theta}_1)d\boldsymbol{\theta}_1$ ($\int L_2(\boldsymbol{\theta}_2|\mathbf{x})\pi_2(\boldsymbol{\theta}_2)d\boldsymbol{\theta}_2$). The relative betting odds thus are given by the Bayes factor B ,

$$B = \frac{\int L_1(\boldsymbol{\theta}_1|\mathbf{x})\pi_1(\boldsymbol{\theta}_1)d\boldsymbol{\theta}_1}{\int L_2(\boldsymbol{\theta}_2|\mathbf{x})\pi_2(\boldsymbol{\theta}_2)d\boldsymbol{\theta}_2}.$$

In the case with no free parameters, B reduces to the simple likelihood ratio L_1/L_2 .

The two terms forming the ratio are called *marginal likelihoods*. The integration automatically introduces a penalty for additional parameters and related overfitting: The higher the dimensionality of the parameter space is, the larger is in average the contribution of low likelihood regions to the integral. In this way the concept follows the philosophy of *Ockham's razor*⁸ which in short states that *from different competing theories, the one with the fewest assumptions, i.e. the simplest, should be preferred*.

The Bayes factor is intended to replace the p-value of frequentist statistics.

H. Jeffreys [19] has suggested a classification of Bayes factors into different categories ranging from < 3 (barely worth mentioning) to > 100 (decisive).

For the example of Chap. 10 Sect. 10.5, Fig.10.18 with a resonance above a uniform background for uniform prior densities in the signal fraction t , $0 \leq t \leq 0.5$ and the location μ , $0.2 \leq \mu \leq 0.8$ the Bayes factor is $B = 54$ which is considered as very significant. This result is inversely proportional to the range in μ as is expected because the probability to find a fake signal in a flat background is proportional to its range. In the cited example we had found a likelihood ratio of $1.1 \cdot 10^4$ taken at the MLE. The corresponding p-value was $p = 1.8 \cdot 10^{-4}$ for the hypothesis of a flat background, much smaller than the betting odds of $1/54$ for this hypothesis. While

⁸Postulated by William of Ockham, English logician in the 14th century.

the Bayes factor takes into account the uncertainty of the parameter estimate, the uncertainty is completely neglected in the the p-value derived from the likelihood ratio taken simply at the MLE. On the other hand, for the calculation of the Bayes factor an at least partially subjective prior probability has to be included.

For the final rating the Bayes factor has to be multiplied by the prior factors of the competing hypotheses:

$$R = B \frac{\pi_{H1}}{\pi_{H2}} = \frac{\int L_1(\boldsymbol{\theta}_1|\mathbf{x})\pi_1(\boldsymbol{\theta}_1)d\boldsymbol{\theta}_1}{\int L_2(\boldsymbol{\theta}_2|\mathbf{x})\pi_2(\boldsymbol{\theta}_2)d\boldsymbol{\theta}_2} \frac{\pi_{H1}}{\pi_{H2}}.$$

The posterior rating is equal to the prior rating times the Bayes factor.

The Bayes factor is a very reasonable and conceptually attractive concept which requires little computational effort. It is to be preferred to the frequentist p-value approach in decision making. However, for the documentation of a measurement it has the typical Bayesian drawback that it depends on prior densities and unfortunately there is no objective way to fix those.

13.15 Robust Fitting Methods

13.15.1 Introduction

If one or a few observations in a sample are separated from the bulk of the data, we speak of outliers. The reasons for their existence range from trivial mistakes or detector failures to important physical effects. In any case, the assumed statistical model has to be questioned if one is not willing to admit that a large and very improbable fluctuation did occur.

Outliers are quite disturbing: They can change parameter estimates by large amounts and increase their errors drastically.

Frequently outliers can be detected simply by inspection of appropriate plots. It goes without saying, that simply dropping them is not a good advice. In any case at least a complete documentation of such an event is required. Clearly, objective methods for their detection and treatment are preferable.

In the following, we restrict our treatment to the simple one-dimensional case of Gaussian-like distributions, where outliers are located far from the average, and where we are interested in the mean value. If a possible outlier is contained in the allowed variate range of the distribution – which is always true for a Gaussian – a statistical fluctuation cannot be excluded as a logical possibility. Since the outliers are removed on the basis of a statistical procedure, the corresponding modification of results due to the possible removal of correct observations can be evaluated.

We distinguish three cases:

1. The standard deviations of the measured points are known.
2. The standard deviations of the measured points are unknown but known to be the same for all points.
3. The standard deviations are unknown and different.

It is obvious that case 3 of unknown and unequal standard deviations cannot be treated.

The treatment of outliers, especially in situations like case 2, within the LS formalism is not really satisfying. If the data are of bad quality we may expect a sizeable fraction of outliers with large deviations. These may distort the LS fit to such an extent that outliers become difficult to define (*masking* of outliers). This kind of fragility of the LS method, and the fact that in higher dimensions the outlier detection becomes even more critical, has lead statisticians to look for estimators which are less disturbed by data not obeying the assumed statistical model (typical are deviations from the assumed normal distribution), even when the efficiency suffers. In a second – not robust – fit procedure with cleaned data it is always possible to optimize the efficiency.

In particle physics, a typical problem is the reconstruction of particle tracks from hits in wire or silicon detectors. Here outliers due to other tracks or noise are a common difficulty, and for a first rough estimate of the track parameters and the associated hit selection for the pattern recognition, robust methods are useful.

13.15.2 Robust Methods

Truncated Least Square Fits

The simplest method to remove outliers is to eliminate those measurements which contribute excessively to the χ^2 of a least square (LS) fit. In this truncated least square fit (LST) all observations that deviate by more than a certain number of standard deviations from the mean are excluded. Reasonable values lie between 1.5 and 2 standard deviations, corresponding to a χ^2 cut $\chi_{\max}^2 = 2.25$ to 4. The optimal value of this cut depends on the expected amount of background or false measurements and the number of observations. In case 2 the variance has to be estimated from the data and the estimated variance $\hat{\delta}^2$ is, according to Chap. 3.2.3, given by

$$\hat{\delta}^2 = \sum (y_i - \hat{\mu})^2 / (N - 1) .$$

This method can be improved by removing outliers sequentially (LSTS). In a first step we use all measurements $y_1, \dots, y_N$, with standard deviations $\delta_1, \dots, \delta_N$ to determine the mean value $\hat{\mu}$ which in our case is just the weighted mean. Then we compute the normalized residuals, also called pulls, $r_i = (y_i - \hat{\mu})/\delta_i$ and select the measurement with the largest value of r_i^2 . The value of χ^2 is computed with respect to the mean and variance of the remaining observations and the measurement is excluded if it exceeds the parameter $\chi_{\max}^2$ ⁹. The fit is repeated until all measurements are within the margin. In case that all measurements are genuine Gaussian measurements, this procedure only marginally reduces the precision of the fit.

In both methods LST and LSTS a minimum fraction of measurements has to be retained. A reasonable value is 50 % but depending on the problem other values may be appropriate.

⁹If the variance has to be estimated from the data its value is biased towards smaller values because for a genuine Gaussian distribution eliminating the measurement with the largest pull reduces the expected variance.

The Sample Median

A first step (already proposed by Laplace) in the direction to estimators more robust than the sample mean is the introduction of the sample median as estimator for location parameters. While the former follows to an extremely outlying observation up to $\pm\infty$, the latter stays nearly unchanged in this case. This change can be expressed as a change of the object function, i.e. the function to be minimized with respect to μ , from $\sum_i (y_i - \mu)^2$ to $\sum_i |y_i - \mu|$ which is indeed minimized if $\hat{\mu}$ coincides with the sample median in case of N odd. For even N , $\hat{\mu}$ is the mean of the two innermost points. Besides the slightly more involved computation (sorting instead of summing), the median is not an optimal estimator for a pure Gaussian distribution:

$$\text{var}(\text{median}) = \frac{\pi}{2} \text{var}(\text{mean}) = 1.571 \text{var}(\text{mean}) ,$$

but it weights large residuals less and therefore performs better than the arithmetic mean for distributions which have longer tails than the Gaussian. Indeed for large N we find for the Cauchy distribution $\text{var}(\text{median}) = \pi^2/(4N)$, while $\text{var}(\text{mean}) = \infty$ (see 3.6.9), and for the two-sided exponential (Laplace) distribution $\text{var}(\text{median}) = \text{var}(\text{mean})/2$.

M-Estimators

The object function of the LS approach can be generalized to

$$\sum_i \rho \left(\frac{y_i - t(x_i, \boldsymbol{\theta})}{\delta_i} \right) \quad (13.38)$$

with $\rho(z) = z^2$ for the LS method which is optimal for Gaussian errors. For the Laplace distribution mentioned above the optimal object function is based on $\rho(z) = |z|$, derived from the likelihood analog which suggests $\rho \propto \ln f$. To obtain a more robust estimation the function ρ can be modified in various ways but we have to retain the symmetry, $\rho(z) = \rho(-z)$ and to require a single minimum at $z = 0$. This kind of estimators with object functions ρ different from z^2 are called M-estimators, “M” reminding maximum likelihood. The best known example is due to Huber, [92]. His proposal is a kind of mixture of the appropriate functions of the Gauss and the Laplace cases:

$$\rho(z) = \begin{cases} z^2/2 & \text{if } |z| \leq c \\ c(|z| - c/2) & \text{if } |z| > c . \end{cases}$$

The constant c has to be adapted to the given problem. For a normal population the estimate is of course not efficient. For example with $c = 1.345$ the inverse of the variance is reduced to 95% of the standard value. Obviously, the fitted object function (13.38) no longer follows a χ^2 distribution with appropriate degrees of freedom.

Estimators with High Breakdown Point

In order to compare different estimators with respect to their robustness, the concept of the *breakdown point* has been introduced. It is the smallest fraction ε of corrupted data points which can lead the fitted values to differ by an arbitrary large amount

from the correct ones. For LS, ε approaches zero, but for M-estimators or truncated fits, changing a single point would be not sufficient to shift the fitted parameter by a large amount. The maximal value of ε is smaller than 50% if the outliers are the minority. It is not difficult to construct estimators which approach this limit, see [93]. This is achieved, for instance, by ordering the residuals according to their absolute value (or ordering the squared residuals, resulting in the same ranking) and retaining only a certain fraction, at least 50%, for the minimization. This so-called *least trimmed squares* (LTS) fit is to be distinguished from truncated least square fit (LST, LSTS) with a fixed cut against large residuals.

An other method relying on rank order statistics is the so-called *least median of squares* (LMS) method. It is defined as follows: Instead of minimizing with respect to the parameters μ the *sum* of squared residuals, $\sum_i r_i^2$, one searches the minimum of the *sample median* of the squared residuals:

$$\text{minimize}_{\mu} \{ \text{median}(r_i^2(\mu)) \} .$$

This definition implies that for N data points, $N/2 + 1$ points enter for N even and $(N + 1)/2$ for N odd. Assuming equal errors, this definition can be illustrated geometrically in the one-dimensional case considered here: $\hat{\mu}$ is the center of the smallest interval (vertical strip in Fig. 13.9) which covers half of all x values. The width 2Δ of this strip can be used as an estimate of the error. Many variations are of course possible: Instead of requiring 50% of the observations to be covered, a larger fraction can be chosen. Usually, in a second step, a LS fit is performed with the retained observations, thus using the LMS only for outlier detection. This procedure is chosen, since it can be shown that, at least in the case of normal distributions, ranking methods are statistically inferior as compared to LS fits.

Example 155. Fitting a mean value in presence of outliers

In Fig. 13.9 a simple example is presented. Three data points, representing the outliers, are taken from $N(3, 1)$ and seven from $N(10, 1)$. The LS fit (7.7 ± 1.1) is quite disturbed by the outliers. The sample median is here initially 9.5, and becomes 10.2 after excluding the outliers. It is less disturbed by the outliers. The LMS fit corresponds to the thick line, and the minimal strip of width 2Δ to the dashed lines. It prefers the region with largest point density and is therefore a kind of mode estimator. While the median is a location estimate which is robust against large symmetric tails, the mode is also robust against asymmetric tails, i.e. skew distributions of outliers.

A more quantitative comparison of different fitting methods is presented in Table 13.1. We have generated 100000 samples with a 7 point signal given by $N(10, 1)$ and 3 points of background, a) asymmetric: $N(3, 1)$ (the same parameters as used in the example before), b) uniform in the interval $[5, 15]$, and in addition c) pure normally distributed points following $N(10, 1)$ without background. The table contains the root mean squared deviation of the mean values from the nominal value of 10. To make the comparison fair, as in the LMS method also in the trimmed LS fit 6 points have been retained and in the sequentially truncated LS fit a minimum of 6 points was used.

With the asymmetric background, the first three methods lead to biased mean values (7.90 for the simple LS, 9.44 for the median and 9.57 for the trimmed LS) and thus the corresponding r.m.s. values are relatively large. As expected the median suffers much less from the background than a stan-

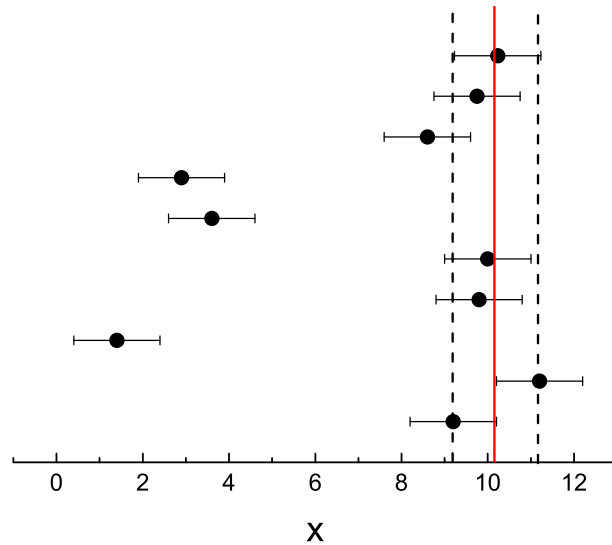


Fig. 13.9. Estimates of the location parameter for a sample with three outliers.

Table 13.1. Root mean squared deviation of different estimates from the nominal value.

method	background		
	asymm.	uniform	none
simple LS	2.12	0.57	0.32
median	0.72	0.49	0.37
LS trimmed	0.60	0.52	0.37
LS sequentially truncated	0.56	0.62	0.53
least median of squares	0.55	0.66	0.59

dard LS fit. The results of the other two methods, LMS and LS sequentially truncated perform reasonable in this situation, they succeed to eliminate the background completely without biasing the result but are rather weak when little or no background is present. The result of LMS is not improved in our example when a least square fit is performed with the retained data.

The methods can be generalized to the multi-parameter case. Essentially, the r.m.s deviation is replaced by χ^2 . In the least square fits, truncated or trimmed, the measurements with the largest χ^2 values are excluded. The LMS method searches for the parameter set where the median of the χ^2 values is minimal.

More information than presented in this short and simplified introduction into the field of robust methods can be found in the literature cited above and the newer book of R. Maronna, D. Martin and V. Yohai [91].

References

1. M. G. Kendall and W. R. Buckland, *A Dictionary of Statistical Terms*, Longman, London (1982).
2. M. G. Kendall and A. Stuart, *The Advanced Theory of Statistics*, Griffin, London (1979).
3. S. Brandt, *Data Analysis*, Springer, Berlin (1999).
4. A. G. Frodesen, O. Skjeggstad, M. Tofte, *Probability and Statistics in Particle Physics*, Universitetsforlaget, Bergen (1979).
5. L. Lyons, *Statistics for Nuclear and Particle Physicists*, Cambridge University Press (1992).
6. W. T. Eadie et al., *Statistical Methods in Experimental Physics*, North-Holland, Amsterdam, (1982).
7. F. James, *Statistical Methods in Experimental Physics*, World Scientific Publishing, Singapore (2007).
8. V. Blobel, E. Lohrmann, *Statistische und numerische Methoden der Datenanalyse*, Teubner, Stuttgart (1998).
9. B. P. Roe, *Probability and Statistics in Experimental Physics*, Springer, Berlin (2001).
10. R. Barlow, *Statistics*, Wiley, Chichester (1989).
11. G. Cowan, *Statistical Data Analysis*, Clarendon Press, Oxford (1998).
12. G. D'Agostini, *Bayesian Reasoning in Data Analysis: A Critical Introduction*, World Scientific Pub., Singapore (2003).
13. T. Hastie, R. Tibshirani, J. H. Friedman, *The Elements of Statistical Learning*, Springer, Berlin (2001).
14. G. E. P. Box and G. C. Tiao, *Bayesian Inference in Statistical Analysis*, Addison-Wesley, Reading (1973).
15. R. A. Fisher, *Statistical Methods, Experimental Design and Scientific Inference*, Oxford University Press (1990). (First publication 1925).
16. A. W. F. Edwards, *Likelihood*, The John Hopkins University Press, Baltimore (1992).
17. I. J. Good, *Good Thinking, The Foundations of Probability and its Applications*, Univ. of Minnesota Press, Minneapolis (1983).
18. L. J. Savage, *The Writings of Leonard Jimmie Savage - A Memorial Selection*, ed. American Statistical Association, Washington (1981).
19. H. Jeffreys, *Theory of Probability*, Clarendon Press, Oxford (1983).
20. L. J. Savage, *The Foundation of Statistical Inference*, Dover Publ., New York (1972).
21. Proceedings of PHYSTAT03, *Statistical Problems in Particle Physics, Astrophysics and Cosmology* ed. L. Lyons et al., SLAC, Stanford (2003) Proceedings of PHYSTAT05, *Statistical Problems in Particle Physics, Astrophysics and Cosmology* ed. L. Lyons et al., Imperial College Press, Oxford (2005).
22. M. Abramowitz, I.A. Stegun, *Handbook of Mathematical Functions*, Dover Publications, inc., New York (1970).
23. Bureau International de Poids et Mesures, *Rapport du Groupe de travail sur l'expression des incertitudes*, P.V. du CIPM (1981) 49, P. Giacomo, *On the expression of uncertainties in quantum metrology and fundamental physical constants*, ed. P. H. Cutler and A.

- A. Lucas, Plenum Press (1983), International Organization for Standardization (ISO), *Guide to the expression of uncertainty in measurement*, Geneva (1993).
24. A. B. Balantekin et al., *Review of Particle Physics*, J. of Phys. G 33 (2006), 1.
25. P. Sinervo, *Definition and Treatment of Systematic Uncertainties in High Energy Physics and Astrophysics*, Proceedings of PHYSTAT2003, P123, ed. L. Lyons, R. Mount, R. Reitmeyer, Stanford, Ca (2003).
26. R. J. Barlow, *Systematic Errors, Fact and Fiction*, hep-ex/0207026 (2002).
27. R.M. Neal *Probabilistic Inference Using Markov Chain Monte Carlo Methods*, University of Toronto, Department of Computer Science, Tech. Rep. CRG-TR-9 3-1 (1993).
28. N. Metropolis, A.W. Rosenbluth, M.N. Rosenbluth, A.H. Teller, and E. Teller *Equation of state calculations by fast computing machines*, J. Chem. Phys. 21 (1953) 1087.
29. R. J. Barlow, *Extended maximum likelihood*, Nucl. Instr. and Meth. A 297 (1990) 496.
30. G. Zech, *Reduction of Variables in Parameter Inference*, Proceedings of PHYSTAT2005, ed. L. Lyons, M. K. Unel, Oxford (2005).
31. G. Zech, *A Monte Carlo Method for the Analysis of Low Statistic Experiments*, Nucl. Instr. and Meth. 137 (1978) 551.
32. V. S. Kurbatov and A. A. Tyapkin, in Russian edition of W. T. Eadie et al., *Statistical Methods in Experimental Physics*, Atomisdat, Moscow (1976).
33. O. E. Barndorff-Nielsen, *On a formula for the distribution of a maximum likelihood estimator*, Biometrika 70 (1983), 343.
34. P.E. Condon and P.L. Cowell, *Channel likelihood: An extension of maximum likelihood to multibody final states*, Phys. Rev. D 9 (1974) 2558.
35. D. R. Cox and N. Reid, *Parameter Orthogonality and Approximate Conditional Inference*, J. R. Statist. Soc. B 49, No 1, (1987) 1, D. Fraser and N. Reid, *Likelihood inference with nuisance parameters*, Proc. of PHYSTAT2003, ed. L. Lyons, R. Mount, R. Reitmeyer, SLAC, Stanford (2003) 265.
36. G. A. Barnard, G.M. Jenkins and C.B. Winstein, *Likelihood inference and time series*, J. Roy. Statist. Soc. A 125 (1962).
37. A. Birnbaum, *More on the concepts of statistical evidence*, J. Amer. Statist. Assoc. 67 (1972), 858.
38. D. Basu, *Statistical Information and Likelihood*, Lecture Notes in Statistics 45, ed. J. K. Ghosh, Springer, Berlin (1988).
39. J. O. Berger and R. L. Wolpert, *The Likelihood Principle*, Lecture Notes of Inst. of Math. Stat., Hayward, Ca, ed. S. S. Gupta (1984).
40. L. G. Savage, *The Foundations of Statistics Reconsidered*, Proceedings of the forth Berkeley Symposium on Mathematical Statistics and Probability, ed. J. Neyman (1961) 575.
41. C. Stein, *A remark on the likelihood principle*, J. Roy. Statist. Soc. A 125 (1962) 565.
42. M. Diehl and O. Nachtmann, *Optimal observables for the measurement of the three gauge boson couplings in $e^+e^- \rightarrow W^+W^-$* , Z. f. Phys. C 62, (1994), 397.
43. R. Barlow, *Asymmetric Errors*, arXiv:physics/0401042v1 (2004), Proceedings of PHYSTAT2005, ed. L. Lyons, M. K. Unel, Oxford (2005), 56.
44. R. W. Peelle, *Peelle's Pertinent Puzzle*, Informal memorandum dated October 13, 1987, ORNL, USA (1987)
45. S. Chiba and D. L. Smith, *Impacts of data transformations on least-squares solutions and their significance in data analysis and evaluation*, J. Nucl. Sci. Technol. 31 (1994) 770.
46. H. J. Behrendt et al., *Determination of α_s , and $\sin^2\theta$ from measurements of the total hadronic cross section in e^+e^- annihilation at PETRA*, Phys. Lett. 183B (1987) 400.
47. V. B. Anykeyev, A. A. Spiridonov and V. P. Zhigunov, *Comparative investigation of unfolding methods*, Nucl. Instr. and Meth. A303 (1991) 350.
48. G. Zech, *Comparing statistical data to Monte Carlo simulation - parameter fitting and unfolding*, Desy 95-113 (1995).
49. G. I. Marchuk, *Methods of Numerical Mathematics*, Springer, Berlin (1975).
50. Y. Vardi and D. Lee, *From image deblurring to optimal investments: Maximum likelihood solution for positive linear inverse problems (with discussion)*, J. R. Sttist. Soc.

- B55, 569 (1993), L. A. Shepp and Y. Vardi, *Maximum likelihood reconstruction for emission tomography*, IEEE trans. Med. Imaging MI-1 (1982) 113, Y. Vardi, L. A. Shepp and L. Kaufmann, *A statistical model for positron emission tomography*, J. Am. Stat. Assoc. (1985) 8, A. Kondor, *Method of converging weights - an iterative procedure for solving Fredholm's integral equations of the first kind*, Nucl. Instr. and Meth. 216 (1983) 177, H. N. Mülthei and B. Schorr, *On an iterative method for the unfolding of spectra*, Nucl. Instr. and Meth. A257 (1986) 371.
51. D.M. Titterton, *Some aspects of statistical image modeling and restoration*, Proceedings of PHYSTAT 05, ed. L. Lyons and M.K. Ünel, Oxford (2005).
 52. L. Lindemann and G. Zech, *Unfolding by weighting Monte Carlo events*, Nucl. Instr. and Meth. A354 (1994) 516.
 53. B. Aslan and G. Zech, *Statistical energy as a tool for binning-free, multivariate goodness-of-fit tests, two-sample comparison and unfolding*, Nucl. Instr. and Meth. A 537 (2005) 626.
 54. M. Schmelling, *The method of reduced cross-entropy - a general approach to unfold probability distributions*, Nucl. Instr. and Meth. A340 (1994) 400.
 55. R. Narayan, R. Nityananda, *Maximum entropy image restoration in astronomy*, Ann. Rev. Astrophys. 24 (1986) 127.
 56. P. Magan, F. Courbin and S. Sohy, *Deconvolution with correct sampling*, Astrophys. J. 494 (1998) 472.
 57. R. B. d'Agostino and M. A. Stephens (editors), *Goodness of Fit Techniques*, M. Dekker, New York (1986).
 58. D. S. Moore, *Tests of Chi-Squared Type in Goodness of Fit Techniques*, ed. R. B. d'Agostino and M. A. Stephens, M. Dekker, New York (1986).
 59. J. Neyman, *"Smooth test" for goodness of fit*, Scandinavisk Aktuaristidskrift 11 (1937), 149.
 60. E. S. Pearson, *The Probability Integral Transformation for Testing Goodness of Fit and Combining Independent Tests of Significance*, Biometrika 30 (1938), 134.
 61. A. W. Bowman, *Density based tests for goodness-of-fit*, J. Statist. Comput. Simul. 40 (1992) 1.
 62. B. Aslan, *The concept of energy in nonparametric statistical inference - goodness-of-fit tests and unfolding*, Dissertation, Siegen (2004).
 63. B. Aslan and G. Zech, *New Test for the Multivariate Two-Sample Problem based on the concept of Minimum Energy*, J. Statist. Comput. Simul. 75, 2 (2004), 109.
 64. B. Efron and R. T. Tibshirani, *An Introduction to the Bootstrap*, Chapman & Hall, London (1993).
 65. L. Demortier, *P Values: What They Are and How to Use Them*, CDF/MEMO/STATISTICS/PUBLIC (2006).
 66. S. G. Mallat, *A Wavelet Tour of Signal Processing*, Academic Press, New York (1999), A. Graps, *An introduction to wavelets*, IEEE, Computational Science and Engineering, 2 (1995) 50 und www.amara.com/IEEEwave/IEEEwavelet.html.
 67. I.T. Jolliffe, *Principal Component Analysis*, Springer, Berlin (2002).
 68. P.M. Schulze, *Beschreibende Statistik*, Oldenburg Verlag (2003).
 69. R. Rojas, *Theorie der neuronalen Netze*, Springer, Berlin (1991).
 70. M. Feindt and U. Kerzel, *The NeuroBayes neural network package*, Nucl. Instr. and Meth A 559 (2006) 190.
 71. J. H. Friedman, *Recent Advances in Predictive (Machine) Learning* Proceedings of PHYSTAT03, *Statistical Problems in Particle Physics, Astrophysics and Cosmology* ed. L. Lyons et al., SLAC, Stanford (2003).
 72. B. Schölkopf, *Statistical Learning and Kernel Methods*, <http://research.microsoft.com>.
 73. V. Vapnik, *The Nature of Statistical Learning Theory*, Springer, Berlin (1995), V. Vapnik, *Statistical Learning Theory*, Wiley, New York (1998), B. Schölkopf and A. Smola, *Learning with Kernels*, MIT press, Cambridge, MA (2002).
 74. E. L. Lehmann, *Elements of Large-Sample Theory*, Springer, Berlin (1999).
 75. Y. Freund and R. E. Schapire, *Experiments with a new boosting algorithm*, Proc. COLT, Academic Press, New York (1996) 209.

76. B.P. Roe et. al., *Boosted decision trees as an alternative to artificial neural networks for particle identification*, Nucl. Instr. and Meth. A543 (2005) 577.
77. T. A. Bancroft and Chien-Pai Han, *Statistical Theory and Inference in Research*, ed. D. B. Owen, Dekker, New York (1981).
78. G. J. Feldman, R. D. Cousins, *Unified approach to the classical statistical analysis of small signals*. Phys. Rev. D 57 (1998) 3873.
79. B. Efron, *Why isn't Everyone a Bayesian?* Am. Stat. 40 (1986) 1,
R. D. Cousins, *Why Isn't Every Physicist a Bayesian?* Am. J. Phys. 63 (1995) 398,
D. V. Lindley, *Wald Lectures: Bayesian Statistics*, Statistical Science, 5 (1990) 44.
80. G. Zech, *Frequentist and Bayesian confidence intervals*, EPJdirect C12 (2002) 1.
81. M. A. Stephens, *Tests based on EDF Statistics*, in Goodness of Fit Techniques, ed. R. B. d'Agostino and M. A. Stephens, Dekker, New York (1986).
82. G. Zech, *A Simple Iterative Alignment Method using Gradient Descending Minimum Search*, Proceedings of PHYSTAT03, *Statistical Problems in Particle Physics, Astrophysics and Cosmology*, ed. L. Lyons et al., SLAC, Stanford (2003), 226.
83. L. Breiman, *Bagging predictors*, Technical Report No. 421, Department of Statistics, University of California, Berkeley, Ca, (1994).
84. L. Breiman, *Random Forests*, Technical Report, Department of Statistics, University of California, Berkeley, Ca (2001).
85. J. S. Simonoff, *Smoothering Methods in Statistics*, Springer, Berlin (1996).
86. A. W. Bowman and A. Azzalini, *Applied Smoothing Techniques for Data Analysis: The Kernel Approach with S-Plus Illustration*, Oxford University Press (1997).
87. D. W. Scott, *Multivariate Density Estimation: Theory, Practice, and Visualization*, John Wiley, New York (1992).
88. D. W. Scott, and S. R. Sain, *Multi-Dimensional Density Estimation*, in Handbook of Statistics, Vol 24: Data Mining and Computational Statistics, ed. C.R. Rao and E.J. Wegman, Elsevier, Amsterdam (2004).
89. W. Härdle, M. Müller, S. Sperlich, A. Werwatz, *Nonparametric and Semiparametric Models*, Springer, Berlin (2004).
90. M. Fisz, *Probability Theorie and Mathematical Statistics*, R.E. Krieger Publishing Company, Malabre, Florida (1980) 464.
91. R. Maronna, D. Martin and V. Yohai, *Robust Statistics – Theory and Methods*, John Wiley, New York (2006).
92. P. J. Huber, *Robust Statistics*, John Wiley, New York (1981).
93. P. J. Rousseeuw, *Robust Regression and Outlier Detection*, John Wiley, New York (1987).
94. G. A. Korn and Th. M. Korn, *Mathematical Handbook for Scientists and Engineers*, McGraw-Hill, New York (1961)
95. Some useful internet links
 - *Statistics Glossary* (V. J. Easton and J. H. McColl):
http://www.stats.gla.ac.uk/steps/glossary/basic_definitions.html
 - *Useful Statistics Links for Particle Physicists*: <http://www.nu.to.infn.it/Statistics/>
 - *Electronic Textbook Statsoft*: <http://www.statsoft.com/textbook/stathome.html>
 - *Electronic Statistics Book*: <http://wiki.stat.ucla.edu/socr/index.php/EBook>
 - *Life and Work of Statisticians* (University of York, Dept. of Mathematics):
<http://www.york.ac.uk/depts/maths/histstat/lifework.htm>
 - *Definition of Measurement Uncertainty Terms* (D. L. Deardorff, University of North Carolina, Dept. of Physics and Astronomy):
<http://www.physics.unc.edu/~deardorff/uncertainty/definitions.html>
 - *Some enhancements of Decision Tree Bagging* (P. Geurts):
<http://www.montefiore.ulg.ac.be/services/stochastic/pubs/2000/Geu00/geurts-pkdd2000-bagging.pdf>
 - *Wavelet script* (in German): <http://www.math.ethz.ch/~blatter/Waveletsvortrag.pdf>
 - *A Tutorial on Support Vector Machines for Pattern Recognition* (Ch. J. C. Burges):
<http://www.umi.acs.umd.edu/~joseph/support-vector-machines4.pdf>
 - *Tutorial on SVM*: <http://www.dimap.ufrn.br/~marcilio/AM/SVM-01.ppt>
 - *Recent Advances in Predictive (Machine) Learning* (J. B. Friedman) <http://www-stat.stanford.edu/~jhf/ftp/machine.pdf>

Table of Symbols

<i>Symbol</i>	<i>Explanation</i>
A, B	Events
$\overline{A}$	Negation of A
$\Omega / \emptyset$	Certain / impossible event
$A \cup B, A \cap B, A \subset B$	A OR B , A AND B , A implies B etc.
$P\{A\}$	Probability of A
$P\{A B\}$	Conditional probability of A (for given B)
$x, y, z; k, l, m$	(Continuous; discrete) random variable (variate)
θ, μ, σ	Parameter of distributions
$f(x), f(x \theta)$	Probability density function
$F(x), F(x \theta)$	Integral (probability-) distribution function (for parameter value θ , respectively)(p. 16)
$f(\mathbf{x}), f(\mathbf{x} \boldsymbol{\theta})$	Respective multidimensional generalizations (p. 42)
$L(\theta), L(\theta x_1, \dots, x_N)$	Likelihood function (p. 137)
$L(\boldsymbol{\theta} x_1, \dots, x_N)$	Generalization to more dimensional parameter space
$\hat{\theta}$	Statistical estimate of the parameter θ (p. 143)
$E(u(x)), \langle u(x) \rangle$	Expected value of $u(x)$
$\overline{u(x)}$	Arithmetic sample mean, average (p. 21)
δx	Measurement error of x (p. 83)
σ_x	Standard deviation of x
$\sigma_x^2, \text{var}(x)$	Variance (dispersion) of x (p. 23)
$\text{cov}(x, y), \sigma_{xy}$	Covariance (p. 45)
ρ_{xy}	Correlation coefficient
μ_i	Moment of order i with respect to origin 0, initial moment
μ'_i	Central moment (p. 32)
μ_{ij}, μ'_{ij} etc.	Two-dimensional generalizations (p. 44)
$\gamma_1, \beta_2, \gamma_2$	Skewness, kurtosis, excess (p. 26)
κ_i	Semiinvariants (cumulants) of order i , (p. 35)

Index

- acceptance fluctuations, 57
- activation function, 314
- AdaBoost, 325
- AMISE, *see* asymptotic mean integrated square error
- ancillary statistic, 185
- Anderson–Darling statistic, 359
- Anderson–Darling test, 266
- angular distribution, 53
 - generation, 113
- ANN, *see* artificial neural network
- approximation of functions, *see* function approximation
- artificial neural network, *see* neural network
- asymptotic mean integrated square error
 - of histogram approximation, 330
- attributes, 289
- averaging measurements, 205

- B-splines, 301
- back propagation of ANN, 316
- bagging, 325
- Bayes factor, 282, **374**
- Bayes’ postulat, 5
- Bayes’ probability, 5
- Bayes’ theorem, **11**, 133, 135
 - for probability densities, 43
- Bayesian statistics, 3
- Bernoulli distribution, 56
- bias, 187
 - of estimate, **345**, 358
 - of measurement, 103
- binomial distribution, 55
 - Poisson limit, 60
 - weighted observations, 57
- boosting, 324
- bootstrap, 277, **336**
 - confidence limits, 339
 - estimation of variance, 337
 - jackknife, 340
 - precision, 339
 - two-sample test, 340
- breakdown point, 377
- Breit-Wigner distribution, 74
 - generation, 113
- brownian motion, 30

- categorical variables, 309
- Cauchy distribution, 74
 - generation, 113
- central limit theorem, **66**, 73, 344
- characteristic function, 33
 - of binomial distribution, 56
 - of Cauchy distribution, 74
 - of exponential distribution, 37
 - of extremal value distribution, 78
 - of normal distribution, 66
 - of Poisson distribution, 36
 - of uniform distribution, 65
- Chebyshev inequality, 343
- chi-square, 151, **196**
 - of histograms, 153
 - histogram of weighted events, 364
 - of histograms of weighted events, 362
- chi-square distribution, 41, **70**
- chi-square probability, 253
- chi-square test, 254
 - binning, 257
 - composite hypothesis, 260
 - generalized, 259
 - small samples, 261
 - two-sample, 276
- CL, *see* confidence level
- classification, 290, **309**
 - decision tree, 322
 - k-nearest neighbors, 319
 - kernel methods, **319**
 - support vector machines, 320
 - weighting, 319
- classifiers
 - training and testing, 340
- combining measurements, **151**, 205

- conditional probability, 11
- conditionality principle, 185
- confidence belt, 354
- confidence interval, 104, **202**
 - classical, 353
 - unphysical parameter values, 219
 - upper limits, 214
- confidence level, 353
- confidence region, 354
- consistency
 - of estimate, **345**, 358
 - of test, 248
- constraints, 163, 368
- convolution function, 222
- convolution integral, **48**, 222
- convolution matrix, 226
- correlation, 45, 52
 - coefficient, **45**, 96
- covariance, 45
- covariance matrix, 97
- coverage probability, 353
- Cramer–Rao inequality, 346
- Cramer–von Mises test, 265
- Cramer–von-Mises statistic, 359
- credibility interval, 202
- critical region, 247
- cross validation, 311
- cumulants, 35
- curse of dimensionality, 290
- decision tree, 290, **322**, 327
 - boosted, 324
- deconvolution, 221
 - binning of histograms, 230
 - binning-free, 234
 - by matrix inversion, 224
 - error estimation, 241
 - iterative, 231
 - migration method, 236
 - of histograms, 227
 - regularization, **226**, 229
 - regularization of transfer matrix, 232
- degree of belief, 3
- degrees of freedom, 71, **72**, 255
- diffusion, 30
- digital measurements, 31
- direct probability, 138
- discriminant analysis, 311
- distribution
 - angular, 53
 - continuous, 16
 - discrete, 16
 - multivariate, 51
 - sample width, 71
- distribution function, 16
- EDF, *see* empirical distribution function
- EDF statistics, 359
- efficiency
 - of estimators, 346
- efficiency fluctuations, 57
- efficiency of estimate, 358
- empirical distribution function, 264
- empirical moments, 87
- energy test, 270
 - distance function, 270, **272**
 - two-sample, 277
- entropy regularization, 229
- Epanechnikov kernel, 334
- equivalent number of events, 63
- error, **81**, 201
 - declaration of, 82
 - definition, 83
 - definition of, 204
 - determination of, 85
 - of a product, 211
 - of a sum, 211, 212
 - of average, 98
 - of correlated measurements, 98
 - of empirical variance, 87
 - of error, 87
 - of ratio, 207
 - of weighted sum, 102
 - one-sided, 214
 - parabolic, 203
 - propagation of, 94, **94**, 205, 210
 - relative, 82
 - several variables, 100
 - statistical, 84
 - systematic, 88, 90
 - types of, 84
 - unphysical parameter values, 219
 - verification of, 82
- error ellipsoid, 97
- error interval, 202
- error matrix, 97
- error of the first kind, 247
- error of the second kind, 247
- error propagation, 94, **94**, 205, 210
- estimate, 3
- estimator
 - minimum variance bound, 347
- event, **2**, 9
- excess, 26
- expected value, 20
 - definition, 21
- exponential distribution, 69
 - generation, 112
 - generation from uniform distribution, 42
- extended likelihood, 154
- extreme value distribution
 - generation, 113
- extreme value distributions, 77

- extremum search, 364
 - method of steepest descent, 366
 - Monte Carlo methods, 365
 - parabola method, 366
 - simplex algorithm, 365
 - stochastic, 367
- f.w.h.m., *see* full width at half maximum
- factor analysis, 303
- Fisher information, 346
- Fisher's spherical distribution, 55
- Fisher–Tippett distribution, 78
- frequentist confidence intervals, 353
- frequentist statistics, 3
- full width at half maximum, 28
- function approximation, 291
 - adapted functions, 302
 - Gaussian weighting, 293
 - k-nearest neighbors, 292
 - orthogonal functions, 294
 - polynomial, **295**, 369
 - splines, 300
 - wavelets, 298
 - weighting methods, 292
- gamma distribution, 72
- Gauss distribution, 65
- Gauss–Markov theorem, 199
- Gini-index, 323
- GOF test, *see* goodness of fit test
- goodness-of-fit test, 250, 363
- Gram–Charlier series, 297
- Gram–Schmidt method, 296
- Gumbel distribution, 78
- Haar wavelet, 299
- Hermite polynomial, 295
- histogram, comparison of, 361
- hypothesis
 - composite, 246
 - simple, 246
- hypothesis test, 245
 - multivariate, 268
- i.i.d. variables, *see* independent, identically distributed variables
- importance sampling, 115
- incompatible measurements, 209
- independence, 52
- independence of variates, 46
- independent, identically distributed variables, 52
- information, 186
- input vector, 289
- integrated square error, 329
- interval estimation, **201**, 355
- inverse probability, 137
- ISE, *see* integrated square error
- iterative deconvolution, 231
- jackknife, 340
- k-nearest neighbor test, 278
 - two-sample, 270
- k-nearest neighbors, **292**, 319
- kernel method, 326
- kernel methods, 290, 333
 - classification, 319
- kernel trick, 373
- kinematical fit, 165
- Kolmogorov–Smirnov test, **263**, 277
- Kuiper test, 265
- kurtosis, 26
 - coefficient of, 26
- L2 test, 268
- Laguerre polynomial, 295
- law of large numbers, 73, 343
- learning, 289
- least median of squares, 378
- least square fit, 195
 - truncated, 376
- least square method, 195
 - counter example, 196
- least trimmed squares, 378
- Legendre polynomial, 295
- lifetime distribution
 - moments of, 28
 - Monte Carlo adjustment, 158
- likelihood, 137
 - definition, 137
 - extended, 154
 - histogram of weighted events, 364
 - histograms, 152
 - histograms with background, 153
 - map, 155
- likelihood function, 137
 - approximation, 208
 - asymptotic form, 349
 - parametrization, 208
 - transformation invariance, 142
- likelihood principle, 186
- likelihood ratio, 137, **137**, 140
 - examples, 140
- likelihood ratio test, **261**, 281
 - for histograms, 262, 363
 - two-samples, 276
- linear distribution
 - generation, 112
- linear regression, 198, 292
 - with constraints, 368
- literature, 6
- LMS, *see* least median of squares

- loadings, 308
- location parameter, 27
- log-likelihood, 138
- log-normal distribution, **75**, 211
- log-Weibull distribution, 78
 - generation, 113
- look-else-where effect, **279**, 286
- Lorentz distribution, 74
 - generation, 113
- loss function
 - decision tree, 324
- LP, *see* likelihood principle
- LS, *see* least squares
- LST, *see* least squares truncated
- LTS, *see* least trimmed squares

- machine learning, 289
- Mahalanobis distance, 269
- marginal distribution, 43
- marginal likelihood, 374
- Markov chain Monte Carlo, 120
- maximum likelihood estimate, 142
 - bias of, 187
 - consistency, 347
 - efficiency, 348
 - small sample properties, 350
- maximum likelihood method, 142
 - examples, 144
 - recipe, 143
 - several parameters, 148
 - signal with background, 150
- MCMC, *see* Markov chain Monte Carlo
- mean integrated square error, **330**, 333
 - of histogram approximation, 330
 - of linear spline approximation, 333
- mean value, 22
- measurement, 2
 - average, 205
 - bias, 103
 - combination of correlated results, 98
 - combining, 98, **151**, 202, 205
- measurement error, *see* error
- measurement uncertainty, *see* error
- median, **27**, 377
- method of steepest descent, 366
- Mexican hat wavelet, 299
- minimal sufficient statistic, 183
- minimum search, 364
- minimum variance bound estimate, 350
- minimum variance estimate, 350
- MISE, *see* mean integrated square error
- MLE, *see* maximum likelihood estimate
- mode, 27
- moments, 32
 - exponential distribution, 38
 - higher-dimensional distributions, 44
 - of Poisson distribution, 36
- Monte Carlo integration, 123
 - accuracy, 57
 - advantages of, 129
 - expected values, 128
 - importance sampling, 126
 - selection method, 123
 - stratified sampling, 128
 - subtraction method, 127
 - weighting method, 127
 - with improved selection, 125
- Monte Carlo search, 365
- Monte Carlo simulation, 107
 - additive distributions, 118
 - by variate transformation, 110
 - discrete distributions, 114
 - generation of distributions, 109
 - histogram distributions, 114
 - importance sampling, 115
 - Markov chain Monte Carlo, 120
 - Metropolis algorithm, 120
 - parameter inference, 155, 157
 - Planck distribution, 117
 - selection method, 115
 - with weights, 119
- Morlet wavelet, 299
- multinomial distribution, 58
- multivariate distributions
 - correlation, 52
 - correlation matrix, 52
 - covariance matrix, 52
 - expected values, 52
 - independence, 52
 - transformation, 52
- MV estimate, *see* minimum variance estimate
- MVB estimate, *see* minimum variance bound estimate

- neural network, 290, **312**, 326
 - activation function, 314
 - loss function, 315
 - testing, 316
 - training, 315
- Neyman's smooth test, 266
- normal distribution, 65
 - generation, 113
 - generation from uniform p.d.f., 51
 - in polar coordinates, 47
 - two-dimensional, 66
 - two-dimensional rotation, 68
- nuisance parameter, 174
 - dependence on, 181
 - elimination, 174
 - elimination by factorization, 176
 - elimination by integration, 181

- elimination by restructuring, 177
- profile likelihood, 179
- null hypothesis, 246, **246**
- number of degrees of freedom, 71, **72**, 255
- observation, 2
- Ockham's razor, 374
- optimal variable method, 171
- orthogonal functions, 294
- p-value, **248**, 252
 - combination of, 254
- p.d.f., *see* probability density function
- parameter inference, 131
 - approximated likelihood estimator, 171
 - least square method, 195
 - moments method, 191
 - Monte Carlo simulation, 155
 - optimal variable method, 171
 - reduction of number of variates, 168
 - weighted Monte Carlo, 157
 - with constraints, 163
 - with given prior, 133
- PCA, *see* principal component analysis
- PDE, *see* probability density estimation
- Pearson test, 257
- Peelle's pertinent puzzle, 213
- PIT, **266**, 359
- Planck distribution
 - generation, 117
- point spread function, 222
- Poisson distribution, 58
 - weighted observations, 61
- Poisson numbers
 - weighted, 63
- polynomial approximation, 295
- population, 3
- power law distribution
 - generation, 112
- principal component analysis, 290, **303**
- principal components, 306
- prior probability, **134**, 136
 - for particle mass, 5
- probability, 3
 - assignment of, 4
 - axioms, 10
 - conditional, 11
 - independent, 11
- probability density
 - conditional, 43
 - two-dimensional, 42
- probability density estimation, 268, **329**
 - by Gram–Charlier series, 297
 - fixed volume, 333
 - histogram approximation, 330
 - k-nearest neighbors, 333
 - kernel methods, 333
 - linear spline approximation, 332
- probability density function, 16
- probability integral transformation, **266**, 359
- probability of causes, 137
- profile likelihood, 179
- propagation of errors, 94, **94**
 - linear, 94
 - several variables, 95
- pseudo random number, 109
- quantile, 28
- random event, **2**, 9
- random forest, 325
- random number, 109
- random variable, 10
- random walk, 30
- reduction of number of variables, 47
- regression, 195
- regression analysis, 292
- regularization, **226**, 229, 241
 - minimize curvature, 228
 - of the transfer matrix, 232
- regularization function, 228
- resampling techniques, 336
- response, 289
- robust fitting methods, 375
 - breakdown point, 377
 - least median of squares, 378
 - least trimmed squares, 378
 - M-estimator, 377
 - sample median, 377
 - truncated least square fit, 376
- sample, 1
- sample mean, 22
- sample width, 25, 71
 - relation to variance, 25
- scale parameter, 27
- shape parameter, 27
- sigmoid function, 315
- signal test, 246
 - multi-channel, 285
- signal with background, 61
- significance, 63
- significance level, 247
- significance test, 245
 - small signals, 279
- simplex, 365
- singular value decomposition, 308
- skewness, 26
 - coefficient of, 26
- soft margin classifier, 372
- solid angle, 55

- spline approximation, 300
- spline functions, 370
 - cubic, 371
 - linear, 370
 - normalized, 301
 - quadratic, 370
- stability, 37
- standard deviation, 23
- statistic, 144
 - ancillary, 185
 - minimal sufficient, 183
 - sufficient, 183
- statistical error
 - definition, 91
- statistical learning, 289
- statistics
 - Bayesian, 3
 - frequentist, 3
 - goal of, 1
- stimulated annealing, 367
- stopping rule paradox, 190
- stopping rules, 190
- straight line fit, 179, **197**
- Student's t distribution, 75
- sufficiency, 145, **183**
- sufficiency principle, 183
- sufficient statistic, 183
- support vector, 322
- support vector machine, 291, **320**, 371
- SVD, *see* singular value decomposition
- SVM, *see* support vector machine
- systematic error, 88, 90
 - definition, 91
 - detection of, 92
 - examples, 91
- test, 245
 - bias, 248
 - comparison, 273
 - consistency, 248
 - distribution-free, 251
 - goodness-of-fit, 250, 363
 - power, 247
 - significance, 245
 - size, 247
 - uniformly most powerful, 247
- test statistic, 246
- training sample, 289
- transfer function, 222
- transfer matrix, 226
- transformation of variables, 38
 - multivariate, 46
 - transformation function, 50
- truncated least square fit, 376
- two-point distribution, 56
- two-sample test, 246, **275**
 - chi-square test, 276
 - energy test, 277
 - k-nearest neighbor test, 278
 - Kolmogorov–Smirnov test, 277
 - likelihood ratio, 276
- UMP test, *see* test, uniformly most powerful
- unfolding, *see* deconvolution
- uniform distribution, 31, **65**
- upper limit, 214
 - Poisson statistics with background, 216
 - Poisson statistics, 215
- v. Mises distribution, 53
- variables
 - independent, identically distributed, 52
- variance, 23
 - estimation by bootstrap, 337
 - of a sum, 23
 - of a sum of distributions, 25
 - of sample mean, 24
- variate, 10
 - transformation, 41
- Venn diagram, **10**, 134
- Watson statistic, 359
- Watson test, 266
- wavelets, 298
- Weibull distribution, 78
- weight matrix, 69
- weighted observations, 61
 - statistics of, 61
- width of sample, 25
 - relation to variance, 25

List of Examples

Chapter 1

1. Uniform prior for a particle mass

Chapter 2

2. Card game, independent events
3. Random coincidences, measuring the efficiency of a counter
4. Bayes' theorem, fraction of women among students
5. Bayes' theorem, beauty filter

Chapter 3

6. Discrete probability distribution (dice)
7. Probability density of an exponential distribution
8. Probability density of the normal distribution
9. Relation between the expected values of the track momentum and of its curvature
10. Variance of the convolution of two distributions
11. Expected values, dice
12. Expected values, lifetime distribution
13. Mean value of the volume of a sphere with a normally distributed radius
14. Playing poker until the bitter end
15. Diffusion
16. Mean kinetic energy of a gas molecule
17. Reading accuracy of a digital clock
18. Efficiency fluctuations of a detector
19. Characteristic function of the Poisson distribution
20. Distribution of a sum of independent, Poisson distributed variates
21. Characteristic function and moments of the exponential distribution
22. Calculation of the p.d.f. for the volume of a sphere from the p.d.f. of the radius

- 23. Distribution of the quadratic deviation
- 24. Distribution of kinetic energy in the one-dimensional ideal gas
- 25. Generation of an exponential distribution starting from a uniform distribution
- 26. Superposition of two two-dimensional normal distributions
- 27. Correlated variates
- 28. Dependent variates with correlation coefficient zero
- 29. Transformation of a normal distribution from cartesian into polar coordinates
- 30. Distribution of the difference of two digitally measured times
- 31. Distribution of the transverse momentum squared of particle tracks
- 32. Quotient of two normally distributed variates
- 33. Generation of a two-dimensional normal distribution starting from uniform distributions
- 34. The v. Mises distribution
- 35. Fisher's spherical distribution
- 36. Efficiency fluctuations of a Geiger counter
- 37. Accuracy of a Monte Carlo integration
- 38. Acceptance fluctuations for weighted events
- 39. Poisson limit of the binomial distribution
- 40. Fluctuation of a counting rate minus background
- 41. Distribution of weighted, Poisson distributed observations
- 42. Distribution of the mean value of decay times

Chapter 4

- 43. Scaling error
- 44. Low decay rate
- 45. Poisson distributed rate
- 46. Digital measurement (uniform distribution)
- 47. Efficiency of a detector (binomial distribution)
- 48. Calorimetric energy measurement (normal distribution)
- 49. Average from 5 measurements
- 50. Average of measurements with common off-set error
- 51. Average outside the range defined by the individual measurements
- 52. Error propagation: velocity of a sprinter
- 53. Error propagation: area of a rectangular table
- 54. Straight line through two measured points
- 55. Error of a sum of weighted measurements

- 56. Bias in averaging measurements
- 57. Confidence levels for the mean of normally distributed measurements

Chapter 5

- 58. Area of a circle of diameter d
- 59. Volume of the intersection of a cone and a torus
- 60. Correction of decay times
- 61. Efficiency of particle detection
- 62. Measurement of a cross section in a collider experiment
- 63. Reaction rates of gas mixtures
- 64. Importance sampling
- 65. Generation of the Planck distribution
- 66. Generation of an exponential distribution with constant background
- 67. Mean distance of gas molecules
- 68. Photon-yield for a particle crossing a scintillating fiber
- 69. Determination of π

Chapter 6

- 70. Bayes' theorem: pion- or kaon decay?
- 71. Time of a decay with exponential prior
- 72. Likelihood ratio: $V + A$ or $V - A$ reaction?
- 73. Likelihood ratio of Poisson frequencies
- 74. Likelihood ratio of normal distributions
- 75. Likelihood ratio for two decay time distributions
- 76. MLE of the mean life of an unstable particle
- 77. MLE of the mean value of a normal distribution with known width (case Ia)
- 78. MLE of the width of a normal distribution with given mean (case Ib)
- 79. MLE of the mean of a normal distribution with unknown width (case IIa)
- 80. MLE of the width of a normal distribution with unknown mean (case IIb)
- 81. MLEs of the mean value and the width of a normal distribution
- 82. Determination of the axis of a given distribution of directions
- 83. Likelihood analysis for a signal with background
- 84. Adjustment of a linear distribution to a histogram
- 85. Fit of the slope of a linear distribution with Monte Carlo correction
- 86. Fit of a lifetime with Monte Carlo correction
- 87. Signal over background with background reference sample
- 88. Fit with constraint: two pieces of a rope

- 89. Fit of the particle composition of an event sample
- 90. Kinematical fit with constraints: eliminating parameters
- 91. Example 88 continued
- 92. Example 90 continued
- 93. Example 88 continued
- 94. Reduction of the variate space
- 95. Approximated likelihood estimator: lifetime fit from a distorted distribution
- 96. Approximated likelihood estimator: linear and quadratic distributions
- 97. Nuisance parameter: decay distribution with background
- 98. Nuisance parameter: measurement of a Poisson rate with a digital clock
- 99. Elimination of a nuisance parameter by factorization of a two-dimensional normal distribution
- 100. Elimination of a nuisance parameter by restructuring: absorption measurement
- 101. Eliminating a nuisance parameter by restructuring: slope of a straight line with the y-axis intercept as nuisance parameter

Chapter 7

- 102. Sufficient statistic and expected value of a normal distribution
- 103. Sufficient statistic for mean value and width of a normal distribution
- 104. Conditionality
- 105. Likelihood principle, dice
- 106. Likelihood principle, $V - A$
- 107. Bias of the estimate of a decay parameter
- 108. Bias of the estimate of a Poisson rate with observation zero
- 109. Bias of the measurement of the width of a uniform distribution
- 110. Stopping rule: four decays in a time interval
- 111. Moments method: mean and variance of the normal distribution
- 112. Moments method: asymmetry of an angular distribution
- 113. Counter example to the least square method: gauging a digital clock
- 114. Least square method: Fit of a straight line

Chapter 8

- 115. Error of a lifetime measurement
- 116. Averaging lifetime measurements
- 117. Averaging ratios of Poisson distributed numbers
- 118. Distribution of a product of measurements
- 119. Sum of weighted Poisson numbers
- 120. Average of correlated cross section measurements, Peelle's pertinent puzzle

- 121. Upper limit for a Poisson rate with background
- 122. Upper limit for a Poisson rate with uncertainty in background and acceptance

Chapter 9

- 123. Deconvolution of a blurred picture
- 124. Deconvolution by fitting the true event locations

Chapter 10

- 125. Bias and inconsistency of a test
- 126. The p-value and the probability of a hypothesis
- 127. Comparison of different tests for background under an exponential distribution
- 128. χ^2 comparison for a two-dimensional histogram
- 129. Likelihood ratio test for a Poisson count
- 130. Designed test: three region test
- 131. GOF test of a two-dimensional sample
- 132. Comparison of two samples
- 133. Significance test: signal over background, distribution of the likelihood ratio statistic
- 134. Example 133 continued

Chapter 11

- 135. Simple empirical relations
- 136. Search for common properties
- 137. Two-class classification, SPAM mails
- 138. Multiclass classification, pattern recognition
- 139. Curse of dimensionality
- 140. Principal component analysis
- 141. Calorimeters in particle physics

Chapter 12

- 142. PDE of a background distribution and signal fit
- 143. Bootstrap evaluation of the accuracy of the estimated mean value of a distribution
- 144. Error of mean distance of stochastically distributed points in a square
- 145. Acceptance of weighted events
- 146. Two-sample test with a decision tree

Appendix

- 147. Efficiencies of different estimates of the location parameter of a Gaussian [77]
- 148. Efficiency of small sample MLEs
- 149. Parameter uncertainty for background contaminated signals

- 150. Coverage: performance of magnets
- 151. Bias in the measurements for a mass determination
- 152. Inference with known prior
- 153. Bias introduced by a prior
- 154. Comparing predictions with strongly differing accuracies: Earth quake
- 155. Fitting a mean value in presence of outliers